review questions

INSTRUCTIONS Each of the questions or incomplete statements that follow is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

Hematopoiesis

1. What is the first type of cell produced by the developing embryo?
   A. Erythrocyte
   B. Granulocyte
   C. Lymphocyte
   D. Thrombocyte

2. What percentage of tissue located in the bone marrow cavities of adults is fat?
   A. 10%
   B. 25%
   C. 50%
   D. 75%

3. Which of the following is *not* characteristic of pluripotent hematopoietic stem cells?
   A. Possess self-renewal ability
   B. Produce progenitor cells committed to a single cell lineage
   C. Express the stem cell marker CD13
   D. Are morphologically unrecognizable

4. In an adult, what are the two best areas for obtaining active bone marrow by aspiration?
   A. Vertebra, tibia
   B. Sternum, vertebra
   C. Anterior iliac crest, tibia
   D. Posterior iliac crest, sternum

5. What is the normal ratio of myeloid to erythroid precursors in bone marrow (M:E ratio)?
   A. 1:1
   B. 1:3
   C. 4:1
   D. 8:1

6. Which of the following does *not* accurately describe hematopoietic growth factors?
   A. Bind to target cell receptors to express activity
   B. Action of majority is lineage restricted
   C. May promote or suppress cell death
   D. Can stimulate or inhibit cell proliferation

7. In the third month of gestation, what is the primary site of hematopoiesis?
   A. Liver
   B. Marrow of long bones
   C. Spleen
   D. Yolk sac

8. The mechanism that relays information about tissue oxygen levels to erythropoietin-producing sites is located in the
   A. Brain
   B. Kidney
   C. Liver
   D. Spleen

9. Antigen-independent lymphopoiesis occurs in primary lymphoid tissue located in the
   A. Liver and kidney
   B. Spleen and lymph nodes
   C. Peyer's patches and spleen
   D. Thymus and bone marrow

10. Programmed cell death is called
    A. Necrosis
    B. Apoptosis
    C. Cellular senescence
    D. Terminal differentiation

11. In what area of the bone marrow does hematopoiesis take place?
    A. Cords
    B. Endosteum
    C. Endothelium
    D. Sinuses

12. Bone marrow cellularity refers to the ratio of
    A. Red cell precursors to white cell precursors
    B. Hematopoietic tissue to adipose tissue
    C. Granulocytic cells to erythrocytic cells
    D. Extravascular tissue to intravascular tissue

13. Interleukins and colony stimulating factors are cytokines produced by
    A. B lymphocytes and erythrocytes
    B. Erythrocytes and thrombocytes
    C. Monocytes and T lymphocytes
    D. Neutrophils and monocytes

14. What is the approximate total blood volume in an adult?
    A. 1 L
    B. 2 L
    C. 6 L
    D. 12 L

15. The myeloid progenitor cell can produce cells committed to
    A. Granulocytic, erythrocytic, monocytic, or megakaryocytic lineages
    B. Granulocytic, monocytic, lymphocytic, or megakaryocytic lineages
    C. Erythrocytic, granulocytic, monocytic, or lymphocytic lineages
    D. Erythrocytic, granulocytic, lymphocytic, or megakaryocytic lineages

16. The largest hematopoietic cells in normal bone marrow are
    A. Osteoblasts
    B. Osteoclasts
    C. Megakaryocytes
    D. Plasma cells

17. When evaluating a bone marrow aspirate smear, which finding is considered abnormal?
    A. A predominance of granulocyte precursors as compared to nucleated red cells
    B. Detection of stainable iron in macrophages and erythroid precursors with Prussian blue
    C. An average of three megakaryocytes seen per low power (10×) field
    D. The presence of 10% myeloblasts on the cell differential count

18. As most blood cell lines mature, which of the following is characteristic?
    A. Cell diameter increases
    B. Nucleus to cytoplasm ratio (N:C) decreases
    C. Nuclear chromatin becomes less condensed
    D. Basophilia of the cytoplasm increases

19. Which of the following describes thrombopoietin (TPO)?
    A. Renal hormone that regulates marrow red cell production
    B. Marrow hormone secreted by developing megakaryoblasts
    C. Hormone produced by the liver that stimulates megakaryopoiesis
    D. Pituitary hormone that controls platelet sequestration by the spleen

20. When the hepatic phase of fetal life is reactivated in an adult, hematopoiesis can be termed
    A. Myeloid or medullary
    B. Myeloid metaplasia or extramedullary
    C. Myelophthisis or myelodysplasia
    D. Mesoblastic or mesenchymal

Erythrocytes

21. What is the average life span of a normal red blood cell?
    A. 1 day
    B. 10 days
    C. 60 days
    D. 120 days

22. The Na^+–K^+ cation pump is an important mechanism in keeping the red blood cell intact. Its function is to maintain a high level of
    A. Intracellular Na^+
    B. Intracellular K^+
    C. Plasma Na^+
    D. Plasma K^+

23. Which of the following depicts the structure of the hemoglobin molecule?
    A. Two heme groups, two globin chains
    B. Four heme groups, two globin chains
    C. Two heme groups, four globin chains
    D. Four heme groups, four globin chains

24. Which of the following describes the process known as *culling?*
    A. Release of red cells from the bone marrow
    B. Binding of free hemoglobin by transport proteins
    C. Incorporation of iron into protoporphyrin IX
    D. Removal of abnormal red cells by the spleen

25. Hemoglobin forms that are incapable of oxygen transport include
    A. Deoxyhemoglobin and oxyhemoglobin
    B. Oxyhemoglobin and carboxyhemoglobin
    C. Carboxyhemoglobin and methemoglobin
    D. Methemoglobin and deoxyhemoglobin

26. The majority of iron found in an adult is a constituent of
    A. Ferritin
    B. Myoglobin
    C. Hemoglobin
    D. Peroxidase

27. A *senescent* red blood cell is one that has
    A. Been hemolyzed
    B. Lived its life span
    C. Become deformed
    D. Lost its mitochondria

28. What red cell morphologic abnormality is described by the term "poikilocytosis"?
    A. Variations in size
    B. Deviations from normal shape
    C. Presence of inclusions
    D. Alterations in hemoglobin concentration

29. Howell-Jolly bodies are composed of
    A. DNA
    B. Iron
    C. Reticulum
    D. RNA

30. When spherocytes are reported, what is observed on the peripheral blood smear?
    A. Red cells without a central pallor
    B. Red cells with blunt projections
    C. Red cells with sharp projections
    D. Red cells with intracellular rod-shaped crystals

31. The red cells found in lead poisoning characteristically exhibit coarse granules composed of _____ that are reported as _____.
    A. Precipitated hemoglobin; Pappenheimer bodies
    B. Aggregated ribosomes; basophilic stippling
    C. Nuclear fragments; Pappenheimer bodies
    D. Excess iron deposits; basophilic stippling

32. Rouleaux of red blood cells when seen in the monolayer of a blood smear is characteristic of
    A. Hypersplenism
    B. Hypogammaglobulinemia
    C. Cold hemagglutinin disease
    D. Multiple myeloma

33. Which of the following is most frequently associated with the inclusion bodies seen in Color Plate 1■?
    A. Iron overload state
    B. Post-transfusion
    C. Post-splenectomy
    D. Iron-deficient state

34. Which of the following statements about iron absorption is *true?*
    A. Absorption occurs in the ileum.
    B. The mucosal cell always absorbs the correct amount of iron to meet needs.
    C. Absorption increases when erythropoietic activity increases.
    D. Alkaline pH favors absorption.

35. What term describes a mature red blood cell that contains iron granules or deposits?
    A. Siderosome
    B. Sideroblast
    C. Ringed sideroblast
    D. Siderocyte

36. Which of the following is associated with a "shift to the left" in the oxygen dissociation curve of hemoglobin?
    A. Decreased pH and elevated temperature
    B. Decreased oxygen affinity
    C. Decreased oxygen release
    D. Presence of 2,3-bisphosphoglycerate (2,3-BPG)

37. Which of the following statements does *not* characterize erythropoietin (EPO)?
    A. Transforms the CFU-E into the earliest recognizable RBC precursor
    B. Increases the rate of red blood cell production by the bone marrow
    C. Shortens the maturation time of developing erythroid precursors
    D. Decreases stimulation of erythropoiesis when cellular hypoxia increases

38. Which of the following factors will result in an immediate increase in oxygen delivery to the tissues?
    A. Increased pH
    B. High altitudes
    C. Increased hemoglobin binding of 2,3-BPG
    D. Increased renal release of erythropoietin

39. Periods of intense erythropoietin activity cause premature release of marrow reticulocytes into the blood. Which of the following is *not* true of these early reticulocytes?
    A. Loss of residual RNA occurs immediately upon marrow release
    B. Circulate longer than usual before reaching maturity
    C. May be termed shift or stress reticulocytes
    D. Show diffuse basophilia with Wright's stain

40. Which of the following inclusions is *only* visible with supravital staining?
    A. Basophilic stippling
    B. Cabot rings
    C. Heinz bodies
    D. Pappenheimer bodies

41. The presence of schistocytes on the peripheral blood smear is commonly associated with
    A. Increased iron mobilization
    B. Increased red cell destruction
    C. Decreased erythropoietin activity
    D. Decreased red cell proliferation

42. Which of the following may be a sign of accelerated bone marrow erythropoiesis?
    A. Hypercellular marrow with a decreased number of RBC precursors
    B. Bone marrow M:E ratio of 6:1
    C. Nucleated red cells in the peripheral circulation
    D. Low erythrocyte, hemoglobin, and hematocrit levels

43. Microcytic, hypochromic red cells are most often associated with impaired
    A. DNA synthesis
    B. RNA metabolism
    C. Hemoglobin synthesis
    D. Enzyme metabolism

44. When in bone marrow, the nucleated red cells present in Color Plate 2■ would be staged as
    A. Basophilic normoblasts
    B. Polychromatophilic normoblasts
    C. Orthochromic normoblasts
    D. Pronormoblasts

45. When acanthocytes are found on the blood smear, it is usually the result of
    A. Abnormal membrane permeability
    B. Altered membrane lipids
    C. Mechanical trauma
    D. Polymerization of hemoglobin molecules

46. Which erythrocyte metabolic pathway generates adenosine triphosphate (ATP) via glycolysis?
    A. Embden-Meyerhof
    B. Hexose monophosphate
    C. Rapoport-Luebering
    D. Methemoglobin reductase

47. Which of the following red blood cell precursors is the last stage to undergo mitosis?
    A. Pronormoblast
    B. Basophilic normoblast
    C. Polychromatophilic normoblast
    D. Orthochromic normoblast

48. The major adult hemoglobin requires the synthesis of alpha-globin chains and
    A. Beta-globin chains
    B. Delta-globin chains
    C. Epsilon-globin chains
    D. Gamma-globin chains

49. Defective nuclear maturation commonly results in the production of red cells that are
    A. Normocytic
    B. Hypochromic
    C. Macrocytic
    D. Microcytic

50. The major storage form of iron is
    A. Ferritin
    B. Transferrin
    C. Hemosiderin
    D. Hemachromatin

51. The red cells observed on a peripheral blood smear show extreme anisocytosis with an equal number of macrocytes and microcytes. Which of the following values correlate with this finding?
    A. MCV 108.0 fL, RDW 14.0%
    B. MCV 90.0 fL, RDW 25.0%
    C. MCV 75.0 fL, RDW 16.0%
    D. MCV 88.0 fL, RDW 12.0%

52. Excessive extravascular red cell destruction is associated with
    A. Hemoglobinemia
    B. Bilirubinemia
    C. Hemoglobinuria
    D. Hemosiderinuria

53. Which protein is primarily responsible for transport of hemoglobin dimers resulting from intravascular hemolysis?
    A. Hemopexin
    B. Albumin
    C. Hemosiderin
    D. Haptoglobin

54. The morphologic abnormality characteristically found in hemoglobinopathies is
    A. Elliptocytes
    B. Dacryocytes
    C. Codocytes
    D. Discocytes

55. Where do the early and late stages of heme synthesis occur?
    A. On ribosomes
    B. In mitochondria
    C. In cytoplasm
    D. In nucleoli

56. Spectrin is a protein that occupies a major role in
    A. Red cell membrane structure
    B. Reducing ferric iron
    C. Red cell transport and removal of CO_2
    D. Iron recovery during hemoglobin degradation

57. What is the function of reduced glutathione (GSH) in the red blood cell?
    A. Promotes Kreb's cycle activity
    B. Maintains anion balance during the "chloride shift"
    C. Neutralizes intracellular oxidants that accumulate
    D. Prevents oxygen uptake by hemoglobin

58. What does measuring the total iron-binding capacity (TIBC) represent?
    A. Amount of free iron in serum
    B. Circulating protein-bound iron
    C. Amount of iron that transferrin can bind
    D. Indirect measurement of iron stores

59. Serum ferritin is a good indicator of the amount of
    A. Cytochrome iron
    B. Storage iron
    C. Hemoglobin iron
    D. Transferrin saturation

60. Fetal hemoglobin differs from adult hemoglobin in that hemoglobin F
    A. Has a lower oxygen affinity
    B. Resists elution from red cells with acid solutions
    C. Is no longer synthesized after birth in a normal individual
    D. Has four gamma-globin chains

Erythrocyte Disorders

61. Impaired DNA metabolism is characteristic of
    A. Hemoglobin C disease
    B. Iron-deficiency anemia
    C. Sideroblastic anemia
    D. Megaloblastic anemia

62. Which of the following is associated with glucose-6-phosphate dehydrogenase (G6PD) deficiency?
    A. G6PD gene is located on the X chromosome.
    B. Ongoing intravascular hemolysis occurs.
    C. All circulating red cells, including reticulocytes, lack enzyme activity.
    D. Splenectomy can relieve the rate of red cell destruction.

63. In regard to variant hemoglobin E, $\alpha_2 \beta_2^{\ 26\ Glu \rightarrow Lys}$, which of the following statements is *false?*
    A. There are two normal alpha chains.
    B. Glutamic acid replaces lysine on position 26 of the beta chains.
    C. Hemoglobin E is the second most common hemoglobin variant known.
    D. Glutamic acid is normally found at position 26 of the beta chain.

64. Color Plate 3■ shows the peripheral blood of a 16-year-old female with a sporadic history of dizzy spells, fainting, and jaundice. This patient also had a history of periodic abdominal pain related to gallstones. Upon physical examination, she exhibited mild splenomegaly. Her hemoglobin was 107 g/L (10.7 g/dL), hematocrit was 0.32 L/L (32%), red cell indices were normal, and the direct antiglobulin test was negative. Based on history and peripheral blood morphology, which of the following statements is most likely *true?*
    A. Hemoglobin S will be revealed by electrophoresis.
    B. Tests to confirm iron deficiency should be ordered.
    C. An intrinsic hereditary defect of red cells should be suspected.
    D. The anemia is secondary to spleen and gallbladder disorders.

65. A 9-month-old male was seen in the Emergency Department with a femur fracture that had occurred from a fall down the stairs. Upon physical examination, the physician noted hepatosplenomegaly, extreme pallor, and a slight arrhythmia. A complete blood count revealed the following:

WBC	12.2×10^9/L ($12.2 \times 10^3/\mu L$)
RBC	3.05×10^{12}/L ($3.05 \times 10^6/\mu L$)
Hemoglobin	61 g/L (6.1 g/dL)
Hematocrit	0.20 L/L (20%)
MCV	65.5 fL
MCH	20 pg
MCHC	305 g/L (30.5 g/dL)
RDW	25%

The Wright's stained blood smear showed the findings seen in Color Plate 4■. Hemoglobin electrophoresis was ordered with results as follows:

Hgb A	0%
Hgb A_2	3%
Hgb F	97%

Which condition is most likely causing the hematologic abnormalities?
A. Alpha-thalassemia major
B. Cooley beta-thalassemia major
C. Hemoglobin H disease
D. Hereditary persistence of hemoglobin F

66. A 14-year-old African-American male was seen in the clinic for abdominal pain. A complete blood count revealed the following:

WBC	7.0×10^9/L (7.0×10^3/μL)
RBC	2.90×10^{12}/L (2.90×10^6/μL)
Hemoglobin	85 g/L (8.5 g/dL)
Hematocrit	0.25 L/L (25%)
MCV	86.2 fL
MCH	29.3 pg
MCHC	340 g/L (34.0 g/dL)
RDW	21%

The peripheral smear showed the red blood cell morphology seen in Color Plate 5■. What condition is suggested by these findings?

A. Hemoglobin E disease
B. Hemoglobin S disease
C. Hemoglobin SC disease
D. Hemoglobin C disease

67. Pica is most commonly associated with which of the following conditions?

A. Pyridoxine deficiency
B. Lack of erythrocyte folate
C. Iron deficiency
D. Porphyrias

68. Of the following, the leading cause of folate deficiency is

A. Increased requirements
B. Dietary insufficiency
C. Drug inhibition
D. Malabsorption

69. Which of the following statements about sickle cell syndromes is *false?*

A. Asplenism may result from repeated sickling crises in the homozygous state.
B. Heterozygous persons may be partly protected from infection by falciparum malaria.
C. Hemoglobin S is more soluble in dithionite than is normal hemoglobin.
D. Trait conditions are generally asymptomatic with no sickle cell formation.

70. The findings seen in Color Plate 6■ can be found in patients with microangiopathic hemolytic anemia (MAHA). Which of the following conditions could *not* be responsible for this type of red cell destruction?

A. Disseminated intravascular coagulation (DIC)
B. Hemolytic uremic syndrome (HUS)
C. Thrombotic thrombocytopenic purpura (TTP)
D. Idiopathic thrombocytopenic purpura (ITP)

71. Which of the following blood findings does *not* correlate with the presence of ringed sideroblasts in the bone marrow?

A. Pappenheimer bodies
B. Basophilic stippling
C. Increased total iron-binding capacity
D. Increased percent transferrin saturation

72. Which of the following conditions is *not* usually associated with marked reticulocytosis?

A. Four days after a major hemorrhage
B. Drug-induced autoimmune hemolytic anemia
C. Sickle cell anemia
D. Pernicious anemia

73. Hereditary stomatocytosis is manifested physiologically by changes in
    A. Hemoglobin oxygen affinity
    B. Membrane cation permeability
    C. Efficiency of hemoglobin reduction
    D. Glycolytic ATP production

74. In addition to an increase in red blood cells, which of the following is characteristic of polycythemia vera?
    A. Decreased platelets, decreased granulocytes, decreased erythropoietin level
    B. Decreased platelets, decreased granulocytes, increased erythropoietin level
    C. Increased platelets, increased granulocytes, increased erythropoietin level
    D. Increased platelets, increased granulocytes, decreased erythropoietin level

75. Which of the following is *not* characteristic of aplastic anemia?
    A. Extramedullary hematopoiesis
    B. Bone marrow hypoplasia
    C. Absolute reticulocytopenia
    D. Blood findings of pancytopenia

76. What values would you expect to obtain on hemoglobin and hematocrit determinations done immediately after a major hemorrhage, if hemoglobin and hematocrit values were normal prior to the hemorrhage?
    A. Both normal
    B. Both decreased
    C. Hemoglobin decreased, hematocrit normal
    D. Hemoglobin normal, hematocrit decreased

77. Results from a 1-day-old infant include a hemoglobin of 201 g/L (20.1 g/dL), hematocrit of 0.60 L/L (60.0%), MCV of 110.2 fL, and 4 nucleated red cells/100 WBCs. How should these results be interpreted?
    A. The elevated hemoglobin and hematocrit values indicate possible dehydration.
    B. The nucleated red cells suggest accelerated erythropoiesis due to a hemolytic process.
    C. Testing should be done to identify the cause of the macrocytosis.
    D. No further testing is indicated.

78. When viewing Color Plate 7■, the red blood cells with a single elongated projection are known as _____ and may be seen in _____
    A. Acanthocytes; liver disease
    B. Echinocytes; liver disease
    C. Drepanocytes; myelofibrosis
    D. Dacryocytes; myelofibrosis

79. A patient with normocytic, normochromic anemia secondary to small cell carcinoma may be exhibiting an anemia designated as
    A. Hemolytic
    B. Megaloblastic
    C. Myelophthisic
    D. Sideroblastic

80. Idiopathic aplastic anemia is best defined as a form of anemia that
    A. Has no identifiable cause
    B. Is caused by a physician's treatment
    C. Follows exposure to ionizing radiation
    D. Develops after a viral infection

81. Which of the following is a true red blood cell aplasia?
    A. Marrow replacement anemia
    B. Fanconi anemia
    C. Diamond-Blackfan anemia
    D. Donath-Landsteiner anemia

82. Which of the following is *not* a cause of absolute secondary erythrocytosis?
    A. Defective cardiac or pulmonary function
    B. High-altitude adjustment
    C. Dehydration secondary to diuretic use
    D. Hemoglobins with increased oxygen affinity

83. A cellulose acetate hemoglobin electrophoresis (alkaline pH), performed on the blood of a stillborn infant, revealed a single band that migrated farther toward the anode than did the Hb A control. What is the most likely composition of the stillborn infant's hemoglobin?
    A. Four beta chains
    B. Four gamma chains
    C. Two alpha and two beta chains
    D. Two alpha and two gamma chains

84. The most likely cause of the stillborn infant's condition in question 83 is
    A. Erythroblastosis fetalis
    B. Rh hemolytic disease of the fetus
    C. Hydrops fetalis
    D. ABO hemolytic disease of the newborn

85. Which of the following conditions show similar CBC and blood smear findings?
    A. Beta-thalassemia major and minor
    B. Folic acid and vitamin B_{12} deficiencies
    C. Acute and chronic blood loss
    D. Sickle cell disease and trait

86. Which of the following would be useful in identifying the cause of the blood profile seen in Color Plate 8■?
    A. Osmotic fragility test
    B. Reticulocyte count
    C. Direct antiglobulin test
    D. Urine urobilinogen level

87. Which of the following conditions is *not* associated with the presence of schistocytes and spherocytes?
    A. Clostridial septicemia
    B. Prosthetic heart valves
    C. Severe thermal burns
    D. Aplastic anemia

88. A 30-year-old woman who has been vomiting for 3 days has a hemoglobin value of 180 g/L (18.0 g/dL) and a hematocrit of 0.54 L/L (54.0%). Her results suggest the presence of
    A. Absolute erythrocytosis
    B. Primary polycythemia
    C. Secondary polycythemia
    D. Relative polycythemia

89. An excessive accumulation of iron in body tissues is called
    A. Hemochromatosis
    B. Erythroblastosis
    C. Megaloblastosis
    D. Acrocyanosis

90. Abetalipoproteinemia is characterized by mild anemia and numerous _____ on the peripheral blood smear.
    A. Acanthocytes
    B. Elliptocytes
    C. Echinocytes
    D. Stomatocytes

91. What is the most common cause of iron deficiency?
    A. Bleeding
    B. Gastrectomy
    C. Inadequate diet
    D. Intestinal malabsorption

92. Which of the following does *not* characterize beta-thalassemia major?
    A. Transfusion-dependent anemia
    B. Decreased alpha chains result in excess beta chains.
    C. Iron chelation therapy is necessary.
    D. Common in persons of Mediterranean ancestry

93. In the anemia of chronic disease, what are the usual serum iron and transferrin levels?
    A. Serum iron decreased, transferrin decreased
    B. Serum iron decreased, transferrin increased
    C. Serum iron normal, transferrin normal
    D. Serum iron increased, transferrin increased

94. In children, the most important effect of lead poisoning is on the
    A. Liver
    B. Kidney
    C. Neurologic system
    D. Development of erythrocytes

95. Which of the following would *not* result in the dual population of red cells represented in Color Plate 9■?
    A. Blood transfusion
    B. Oral iron therapy
    C. Spleen removal
    D. Coexisting deficiencies

96. What is the most likely genetic defect in the hemoglobin of cells seen in Color Plate 10■?
    A. Substitution of valine for glutamic acid in position 6 of the alpha-globin chain
    B. Substitution of valine for glutamic acid in position 6 of the beta-globin chain
    C. Substitution of lysine for glutamic acid in position 6 of the alpha-globin chain
    D. Substitution of lysine for glutamic acid in position 6 of the beta-globin chain

97. On what is the classification of sickle cell trait versus sickle cell disease based?
    A. Severity of the clinical symptoms
    B. Number of irreversibly sickled cells (ISCs)
    C. Level of compensatory hemoglobin F
    D. Percentage of hemoglobin S on electrophoresis

98. Which of the following is the most appropriate treatment for sickle cell anemia?
    A. Hydroxyurea
    B. Supportive therapy
    C. Hyperbaric oxygen
    D. Iron

99. Which of the following values can be used to indicate the presence of a hemolytic anemia?
    A. Hemoglobin level
    B. Hematocrit level
    C. Erythrocyte count
    D. Reticulocyte count

100. A pre-operative, 20-year-old female has a mild microcytic anemia, with target cells and stippled red cells observed on the blood smear. Her hemoglobin A_2 level is quantified at 5%. What do these findings suggest?
    A. Iron-deficiency anemia
    B. Heterozygous alpha-thalassemia
    C. Heterozygous beta-thalassemia
    D. Hemoglobin S/beta-thalassemia

101. What causes the hemolytic process in glucose-6-phosphate dehydrogenase deficiency following oxidant exposure?
    A. Coating of red cells by antibody
    B. Osmotic pressure changes
    C. Complement attachment
    D. Precipitation of denatured hemoglobin

102. In clinically severe hereditary spherocytosis, which of the following findings would *not* be found post-splenectomy?
   A. Rise in the red cell count and hemoglobin level
   B. Higher number of circulating reticulocytes
   C. Increased number of Howell-Jolly bodies
   D. Transient elevation in the platelet count

103. Which of the following laboratory results is *not* consistent with accelerated red cell destruction?
   A. Increased serum bilirubin
   B. Increased plasma hemoglobin
   C. Increased serum lactate dehydrogenase (LD)
   D. Increased serum haptoglobin

104. Acquired hemolytic anemias are usually due to
   A. Extracorpuscular factors
   B. Defects within the bone marrow
   C. Intracellular factors
   D. Changes in hemoglobin stability

105. The antibody associated with paroxysmal cold hemoglobinuria shows specificity for
   A. ABO antigens
   B. I antigens
   C. P antigens
   D. Rh antigens

106. A 69-year-old male is admitted with pallor, mild tachycardia, and difficulty walking because of numbness in the extremities. His CBC reveals a hemoglobin of 78 g/L (7.8 g/dL), a hematocrit of 0.25 L/L (25.0%), and MCV of 118.5 fL. This patient's symptoms and the blood findings seen in Color Plate 11■ are most suggestive of anemia due to a lack of
   A. Folic acid
   B. Vitamin B_{12}
   C. Vitamin B_6
   D. Ascorbic acid

107. A clinical laboratory scientist examined a Wright's stained peripheral smear and saw what appeared to be small, dark-staining granules in the mature erythrocytes. A second smear was stained with Prussian blue and a positive result was obtained. Based on this information, which of the following would you expect to be abnormal?
   A. Plasma hemoglobin level
   B. Serum ferritin level
   C. Hemoglobin electrophoresis
   D. Test for parietal cell antibodies

108. Hemoglobinopathies are characterized by
   A. Absent or reduced rate of globin-chain synthesis
   B. Inability to transport and release oxygen to the tissues
   C. Inhibition of iron chelation needed for heme biosynthesis
   D. Production of structurally abnormal hemoglobin variants

109. Which of the following statements about hereditary spherocytosis is *true?*
   A. Abnormally shaped cells are produced in the bone marrow.
   B. Cells have a decreased mean cell hemoglobin concentration (MCHC).
   C. Membrane loss and red cell trapping occur in the splenic microcirculation.
   D. Red cell osmotic fragility is decreased.

110. Which of the following statements about hereditary elliptocytosis (HE) is *true?*
   A. Characteristic oval shape occurs in mature erythrocytes.
   B. Heterogeneous group of disorders linked to Rh-null individuals.
   C. Cellular defect involves the lipid composition of the membrane.
   D. HE cells are abnormally permeable to calcium.

111. Which of the following disorders is *not* commonly linked to the development of anemia of chronic disease?
   A. Persistent infections
   B. Noninfectious inflammatory disorders
   C. Chronic gastrointestinal blood loss
   D. Malignancy

112. Which of the following statements about hemoglobin C disease is *false?*
   A. Electrophoresis shows approximately 60% hemoglobin A and 40% hemoglobin C.
   B. Target cells are frequently seen on peripheral smears.
   C. Red cells may contain bar-shaped intracellular crystals.
   D. The disorder is less severe than sickle cell disease.

113. Which of the following is associated with sickle cells?
   A. Increased oxygen tension promotes sickling.
   B. There is decreased mechanical fragility.
   C. There is increased deformability.
   D. Increased sickling occludes vessels.

114. A bone marrow M:E ratio of 4:1 would be an expected finding for
   A. Sickle cell anemia
   B. Aplastic anemia
   C. Beta-thalassemia major
   D. Megaloblastic anemia

115. An elderly man with a 10-year history of chronic lymphocytic leukemia presented with jaundice and fatigue that was attributed to a recent 3-gram drop in his hemoglobin. Many spherocytes and polychromatophilic red cells were found on his Wright's stained blood smear. Which type of immune hemolytic anemia is most likely?
   A. Idiopathic warm autoimmune hemolytic anemia
   B. Secondary warm autoimmune hemolytic anemia
   C. Primary cold hemagglutinin disease
   D. Paroxysmal cold hemoglobinuria

116. A moderately anemic patient with suspected pernicious anemia (PA) shows intrinsic factor antibodies and a low cobalamin level. Which of the following would *not* support the diagnosis of PA?
   A. Gastric atrophy and achlorhydria
   B. Oval macrocytes and Howell-Jolly bodies
   C. Bone marrow erythroid precursors exhibit normoblastic maturation.
   D. Elevated serum lactate dehydrogenase (LD) and bilirubin levels

117. A cellulose acetate electrophoresis revealed a large band of hemoglobin in the hemoglobin S position. This band quantified at 95%. The peripheral smear revealed 70% target cells, and the solubility test was negative. Based on this information, what is the hemoglobin?
   A. Hemoglobin C
   B. Hemoglobin D
   C. Hemoglobin E
   D. Hemoglobin S

118. A previously healthy man experiences weakness and hemoglobinuria after taking the antimalarial agent primaquine. This hemolytic attack most likely occurred because of a deficiency of
   A. Pyruvate kinase
   B. Glucose-6-phosphate dehydrogenase
   C. 2,3-Bisphosphoglycerate
   D. Methemoglobin reductase

119. Which of the following is an acquired red cell membrane defect that results in increased sensitivity to complement binding?
   A. March hemoglobinuria
   B. Paroxysmal nocturnal hemoglobinuria
   C. Paroxysmal cold hemoglobinuria
   D. Methemoglobinemia

120. Which of the following is *not* associated with acquired reversible sideroblastic anemias?
   A. Methotrexate therapy
   B. Lead intoxication
   C. Isoniazid treatment for tuberculosis
   D. Acute alcohol ingestion

121. Which of the following statements about the relative anemia of pregnancy is *false?*
   A. It is due to a reduction in the number of erythrocytes.
   B. It is normocytic and normochromic.
   C. It does not produce an oxygen deficit for the fetus.
   D. It is associated with an increase in plasma volume.

122. The anemia found in chronic renal failure is most likely caused by
   A. Loss of erythropoietin synthesis
   B. Lack of cellular oxygen demand
   C. Defective iron absorption
   D. Destruction of red cells by uremic metabolites

123. Which of the following phrases about aplastic anemia is *false?*
   A. Stem cell disorder
   B. Risk of life-threatening infection
   C. Frequent bleeding complications
   D. Reduced red cell survival

124. The fish tapeworm *Diphyllobothrium latum* is associated with the development of
   A. Microcytic anemia
   B. Macrocytic anemia
   C. Hemolytic anemia
   D. Hypoproliferative anemia

125. An increase in erythropoietin is *not* a normal compensating mechanism in which of the following conditions?
   A. Renal tumors
   B. Heavy smoking
   C. Cardiovascular disease
   D. Pulmonary disease

126. Thalassemias are the result of a
   A. Structural defect in the heme portion of hemoglobin
   B. Quantitative defect in globin-chain synthesis
   C. Qualitative defect in globin-chain structure
   D. Change in hemoglobin solubility properties

127. Which of the following characterizes iron-deficiency anemia?
   A. Decreased serum iron, decreased transferrin saturation, normal ferritin
   B. Decreased serum transferrin, decreased transferrin saturation, decreased ferritin
   C. Increased serum transferrin, decreased transferrin saturation, decreased ferritin
   D. Increased serum transferrin, increased transferrin saturation, decreased serum iron

128. Clinical manifestations of a homozygous mutation involving the beta-globin gene will most likely appear
   A. During embryonic development
   B. In the neonate at birth
   C. No later than 3 weeks after birth
   D. By 6 months of age

129. The hemolysis associated with infection by malaria organisms is due to the
   A. Release of merozoites from erythrocytes
   B. Invasion of erythrocytes by merozoites
   C. Host's immunologic response to infected erythrocytes
   D. Toxins produced by the malarial organism

130. A clinical laboratory scientist received a 5 mL EDTA tube that contained 0.5 mL of anticoagulated blood. A smear was prepared and stained with Wright's stain. When examined microscopically, the majority of cells appeared to have many evenly distributed, blunt spicules on the surface. How should this cellular appearance be interpreted?
   A. An anemic condition requiring further testing
   B. Spur cells caused by using incorrect technique during slide preparation
   C. Artifact caused by a dirty spreader slide
   D. Crenated cells caused by incorrect blood to anticoagulant ratio

131. A failure to generate sufficient ATP is characteristic of red blood cells with
   A. Pyruvate kinase deficiency
   B. Glucose-6-phosphate dehydrogenase deficiency
   C. Lipoprotein deficiency
   D. Hexokinase deficiency

132. When iron use exceeds absorption, which of the following occurs *first?*
   A. Hemoglobin level decreases.
   B. Iron stores are depleted.
   C. Transferrin synthesis increases.
   D. Excretion of iron decreases.

133. The major mechanism responsible for the anemia of chronic disease is
   A. Impaired release of storage iron because of increased hepcidin levels
   B. Damaged bone marrow stem cells
   C. Immune destruction caused by red cell autoantibodies
   D. Increased erythropoietin response by committed red cell progenitor cells

134. Which of the following is *not* a characteristic of the idiopathic type of sideroblastic anemia?
   A. Refractory to treatment
   B. Blocks in heme synthesis are unknown
   C. Reversible with intramuscular vitamin B_{12} injections
   D. Subtype of myelodysplastic syndromes

135. Thinning of bones and deformation of facial bone structure seen in homozygous beta-thalassemia is a
   A. Consequence of disturbances in calcium metabolism
   B. Result of hyperplastic marrow activity
   C. Secondary disorder due to immunologic response
   D. Result of increased fibroclast activity

136. Which of the following does *not* accurately describe cold autoimmune hemolytic anemia?
   A. Red cell agglutination in extremities induces Raynaud's phenomenon.
   B. It may occur secondary to *Mycoplasma* pneumonia.
   C. Hemolysis is complement-mediated or via removal of coated cells.
   D. The autoantibody is usually an IgG type directed against Rh antigens.

137. Which of the following represents an anemia that would have a high red cell distribution width (RDW)?
   A. Sickle cell disease during crisis
   B. Thalassemia minor
   C. Aplastic anemia
   D. Anemia of chronic disorders

138. In which of the following disorders would splenomegaly *not* be a common finding?
   A. Homozygous beta-thalassemia
   B. Hereditary spherocytosis
   C. Hemoglobin SC disease
   D. Folic acid deficiency

Leukocytes

139. Functionally, white blood cells are divided into
   A. Granulocytes, nongranulocytes
   B. Polymorphonuclears, mononuclears
   C. Phagocytes, immunocytes
   D. Granulocytes, lymphocytes

140. What is the largest white blood cell normally found in the peripheral blood?
   A. Eosinophil
   B. Neutrophil
   C. Lymphocyte
   D. Monocyte

141. What is the approximate amount of time a granulocyte spends in the circulation before migrating into the tissues?
   A. Less than 1 day
   B. About 3 days
   C. Up to 5 days
   D. More than 10 days

142. What percentage of neutrophils in the peripheral blood constitutes the circulating pool?
   A. 100%
   B. 80%
   C. 50%
   D. 30%

143. What is the major phagocytic cell involved in the initial defense against bacterial pathogens?
   A. Neutrophil
   B. Eosinophil
   C. Basophil
   D. Monocyte

144. What is the growth factor that is primarily responsible for regulating granulocyte and monocyte production?
   A. Erythropoietin
   B. Colony stimulating factor
   C. Interleukin
   D. Thrombopoietin

145. What does the granulocyte mitotic pool in the bone marrow contain?
   A. Myeloblasts and promyelocytes
   B. Band and segmented forms
   C. The majority of marrow granulocytes
   D. Myelocytes and metamyelocytes

146. A "shift to the left," when used to describe a cell population, refers to
   A. Increased cells in the blood due to a redistribution of blood pools
   B. An increase in immature blood cells following release of bone marrow pools
   C. A cell production "hiatus" or gap
   D. A higher percentage of lymphocytes than neutrophils

147. Which of the following is characteristic of agranulocytosis?
   A. Neutrophils without granules
   B. Decreased numbers of granulocytes, red cells, and platelets
   C. Immature granulocytes in the peripheral blood
   D. Decreased numbers of granulocytes

148. Which of the following is *not* a characteristic of T lymphocytes?
   A. Secrete cytokines
   B. Synthesize antibody
   C. Comprise majority of cells in the blood lymphocyte pool
   D. Regulate the immune response

149. An adult has a total white blood cell count of 4.0×10^9/L (4.0×10^3/μL). The differential count is as follows: polymorphonuclear neutrophils (PMNs) 25%, bands 5%, lymphocytes 65%, and monocytes 5%. The absolute value reference range for lymphocytes is $1.0–4.0 \times 10^9$/L. Which of the following is *true?*
   A. The percentage of lymphocytes is normal.
   B. There is an absolute lymphocytosis.
   C. There is a relative lymphocytosis.
   D. There is both an absolute and a relative lymphocytosis.

150. Which of the following statements is *correct?*
   A. Hypersegmented neutrophils have four nuclear lobes.
   B. Auer rods are composed of fused primary granules.
   C. Toxic granules are prominent secondary granules.
   D. Döhle bodies are agranular patches of DNA.

151. Which of the following factors is *not* associated with variations in the total white blood cell count?
   A. Age
   B. Exercise
   C. Emotional stress
   D. Sex

152. Of the following, an absolute neutrophil count of 1.0×10^9/L would be associated with
   A. Shortness of breath
   B. Bleeding tendencies
   C. Risk of infection
   D. No clinical symptoms

153. Which of the following statements about basophils is *false?*
   A. Morphologically, basophils resemble tissue mast cells.
   B. Membrane receptors bind IgG, initiating anaphylactic reactions.
   C. Basophilic granules contain heparin and histamine.
   D. Granules are water soluble.

154. The most mature granulocyte precursor that can undergo mitosis is the
   A. Myeloblast
   B. Promyelocyte
   C. Myelocyte
   D. Metamyelocyte

155. Production of primary granules ceases and production of secondary granules commences with what cell stage?
   A. Myeloblast
   B. Promyelocyte
   C. Myelocyte
   D. Metamyelocyte

156. Which of the following statements about eosinophils is *false?*
   A. They contain a type of peroxidase that is distinct from that of neutrophils.
   B. Eosinophilic granules contain lysozyme.
   C. Eosinophils are an important line of defense against parasites.
   D. Major basic protein is a component of eosinophil granules.

157. Which of the following is characteristic of primary granules?
   A. Coated with a phospholipid membrane
   B. Called azurophilic or specific granules
   C. Contain myeloperoxidase and lactoferrin
   D. Present in the promyelocyte stage only

158. Which of the following are indicators of a neutrophilic response to tissue damage or inflammatory stimuli?
   A. Toxic granules and Döhle bodies in the neutrophils
   B. Vacuoles and Barr bodies in the neutrophils
   C. Hypersegmented neutrophils and Auer rods
   D. Pyknotic neutrophils and Russell bodies

159. What is the term for cell movement through blood vessels to a tissue site?
   A. Diapedesis
   B. Opsonization
   C. Margination
   D. Chemotaxis

160. Vasodilation and bronchoconstriction are the result of degranulation by which of the following blood cells?
   A. Eosinophils
   B. Monocytes
   C. Neutrophils
   D. Basophils

161. On what basis can B and T lymphocytes be distinguished?
   A. Differences in nuclear shape
   B. Monoclonal antibody reactions to surface and cytoplasmic antigens
   C. Cytoplasmic granularity and overall cell size
   D. Chromatin pattern in the nucleus

162. Cells that produce immunoglobulins in response to antigenic stimulation are designated
   A. Natural killer cells
   B. Plasma cells
   C. Virocytes
   D. Thymocytes

163. Which of the following statements about neutrophils is *false?*
   A. Suppress allergic reactions caused by basophils
   B. Have surface receptors for IgG and complement components
   C. Contain alkaline phosphatase and muramidase
   D. Act in nonspecific phagocytosis and are destined to die

164. Which of the following characteristics would be *least* likely to distinguish reactive lymphocytes from monocytes?
   A. Sharp indentation of the cytoplasmic margin by adjacent red blood cells
   B. Presence of large azurophilic granules
   C. Irregular, indented nuclear shape
   D. Abundant, deeply basophilic cytoplasm

165. Which of the following can differentiate metamyelocytes from other stages of granulocyte maturation?
   A. Presence of specific granules
   B. Indentation of nucleus
   C. Absence of nucleoli
   D. Color of cytoplasm

166. Lymphocyte concentrations in the peripheral blood are greatest during what age interval?
   A. 1 to 4 years
   B. 4 to 15 years
   C. 16 to 40 years
   D. 40 to 70 years

167. Which of the following is the *least* likely to be expressed by early B cell precursors?
A. SIgM, a surface membrane immunoglobulin
B. CD34, a hematopoietic stem cell marker
C. TdT (terminal deoxynucleotidyl transferase), a nuclear enzyme
D. CD10 (CALLA), a surface antigen

168. Which of the following statements about macrophages is *incorrect?*
A. They are mature tissue forms of blood monocytes.
B. They serve as antigen-presenting cells to the immune system.
C. Their quantity of lysosomes and acid hydrolases decreases during maturation.
D. They remove damaged or dying cells and cellular debris.

169. Antigen-dependent lymphopoiesis occurs in secondary lymphoid tissue located in the
A. Liver and kidney
B. Spleen and lymph nodes
C. Lungs and Peyer's patches
D. Thymus and bone marrow

170. Which of the following is *not* produced by neutrophils during the respiratory burst?
A. Hydroxyl radicals (OH^-)
B. Hydrogen peroxide (H_2O_2)
C. Superoxide anion (O_2^-)
D. Myeloperoxidase

Leukocyte Disorders

171. In patients with infectious mononucleosis, which blood cells are infected by the causative agent?
A. Monocytes
B. T lymphocytes
C. B lymphocytes
D. Histiocytes

172. Which of the following statements about hairy cell leukemia is *true?*
A. It is an acute disease, primarily affecting young adults.
B. Splenomegaly is an unusual finding.
C. Hairy cells contain tartrate-resistant acid phosphatase.
D. Hairy cells are abnormal T lymphocytes.

173. Based on the WHO classification system, B cell ALL (FAB type L3) and _____ represent different clinical presentations of the same disease entity.
A. Burkitt lymphoma
B. Hodgkin lymphoma
C. Mycosis fungoides
D. Small lymphocytic lymphoma

174. The presence of both immature neutrophils and nucleated erythrocytes in the peripheral blood is most accurately called a
A. Neutrophilic left shift
B. Regenerative left shift
C. Neutrophilic leukemoid reaction
D. Leukoerythroblastic reaction

175. In which anomaly is a failure of granulocytes to divide beyond the band or two-lobed stage observed?
A. Pelger-Huët
B. May-Hegglin
C. Alder-Reilly
D. Chédiak-Higashi

176. In which of the following are eosinophils *not* increased?
A. Cushing syndrome
B. Allergic disorders
C. Skin disorders
D. Parasitic infection

177. Which of the following represents the principal defect in chronic granulomatous disease (CGD)?
A. Chemotactic migration
B. Phagocytosis
C. Lysosomal formation and function
D. Oxidative respiratory burst

178. The blood shown in Color Plate 11■ is from a leukemia patient following treatment. These findings are most suggestive of therapy with
A. Corticosteroids (e.g., prednisone)
B. A folate antagonist (e.g., methotrexate)
C. Recombinant erythropoietin
D. Chloramphenicol

179. A patient with normal hemoglobin and WBC count values, a persistently elevated platelet count (over 1000×10^9/L), increased marrow megakaryocytes, and a history of frequent bleeding and clotting episodes most likely has
A. Polycythemia vera
B. Chronic myelofibrosis
C. Essential thrombocythemia
D. Chronic myelogenous leukemia

180. An adult patient with massive splenomegaly has mild anemia, a slightly elevated WBC count, and an LAP score of 170. The blood smear shows teardrop erythrocytes and leukoerythroblastosis. These findings are most consistent with
A. Chronic myelogenous leukemia
B. Idiopathic myelofibrosis
C. Primary polycythemia
D. Primary thrombocythemia

181. Which of the following infections does *not* reveal a blood picture as seen in Color Plate 12■?
A. Epstein-Barr virus (EBV)
B. *Bordetella pertussis* (whooping cough)
C. Cytomegalovirus (CMV)
D. *Toxoplasma gondii* (toxoplasmosis)

182. The most common type of chronic lymphocytic leukemia (CLL) in the United States involves the
A. B cell
B. NK cell
C. T cell
D. Plasma cell

183. Which of the following are characteristic findings in Waldenström disease?
A. Increased IgA and hepatosplenomegaly
B. Increased IgE and renal failure
C. Increased IgG and hypercalcemia
D. Increased IgM and blood hyperviscosity

184. Which of the following would *not* cause a total WBC count of 62.2×10^9/L ($62.2 \times 10^3/\mu$L) and the blood findings seen in Color Plate 13■?
A. Treatment with myeloid growth factors
B. Gram-negative septicemia
C. Human immunodeficiency virus (HIV)
D. Systemic fungal infection

185. The peripheral blood shown in Color Plate 14■ is from a 69-year-old female. Her WBC count was 83.0×10^9 cells/L ($83.0 \times 10^3/\mu$L) and her platelet count was normal. Based on the cell morphology and this information, what is the most likely diagnosis?
A. Acute lymphoblastic leukemia
B. Chronic lymphocytic leukemia
C. Waldenström macroglobulinemia
D. Viral infection

186. In which of the following is progression to acute leukemia *least* likely?
A. Chronic myelogenous leukemia (CML)
B. Refractory anemia with excess blasts (RAEB)
C. Refractory anemia with ringed sideroblasts (RARS)
D. Chronic lymphocytic leukemia (CLL)

187. A Gaucher cell is best described as a macrophage with
   A. "Wrinkled" cytoplasm due to an accumulation of glucocerebroside
   B. "Foamy" cytoplasm filled with unmetabolized sphingomyelin
   C. Pronounced vacuolization and deposits of cholesterol
   D. Abundant cytoplasm containing storage iron and cellular remnants

188. Which of the following suggests a diagnosis of Hodgkin disease rather than other lymphoproliferative disorders?
   A. Presence of a monoclonal population of large lymphoid cells
   B. Predominance of immature B cells with irregular nuclear clefts
   C. Circulating T cells with a convoluted, cerebriform nucleus
   D. Presence of giant binucleated Reed-Sternberg cells with prominent nucleoli

189. In a patient with fever of unknown origin, which of the following findings is *not* consistent with an inflammatory process?
   A. Increased C-reactive protein
   B. Increased albumin level
   C. Increased fibrinogen level
   D. Increased erythrocyte sedimentation rate

190. The presence of the chromosomal abnormality t(15;17) and a high incidence of disseminated intravascular coagulation (DIC) is diagnostic of
   A. Acute myeloblastic leukemia without maturation (FAB type M1)
   B. Acute myeloblastic leukemia with maturation (FAB type M2)
   C. Acute promyelocytic leukemia (FAB type M3)
   D. Acute myelomonocytic leukemia (FAB type M4)

191. Which of the following is *not* commonly found in acute myelogenous leukemias?
   A. Neutropenia
   B. Thrombocytopenia
   C. Hepatosplenomegaly
   D. Lymphadenopathy

192. The child whose blast cells are shown in Color Plate 15■ has acute lymphoblastic leukemia that is precursor B cell type and CALLA positive. Analysis by flow cytometry would likely show cells that immunophenotype for
   A. CD2, CD7
   B. CD10, CD19
   C. CD13, CD33
   D. CD14, CD34

193. The patient whose bone marrow is shown in Color Plate 16■ most likely has a(n)
   A. Acute leukemia
   B. Chronic leukemia
   C. Myelodysplastic syndrome
   D. Aplastic anemia

194. Multiple myeloma is characterized by the presence in urine of large amounts of
   A. Cryoglobulins
   B. IgG heavy chains
   C. IgG light chains
   D. Beta microglobulins

195. Which of the following is *not* classified as a myeloproliferative disorder?
   A. Polycythemia vera
   B. Essential thrombocythemia
   C. Multiple myeloma
   D. Chronic myelogenous leukemia

196. Which of the following gene mutations correlates with the t(9;22) that is present in Philadelphia chromosome positive chronic myelogenous leukemia?
   A. MYC/IGH
   B. BCR/ABL
   C. PML/RARA
   D. JAK2

197. Which of the following statements does *not* correctly describe the WHO (World Health Organization) classification of hematopoietic neoplasms?
   A. Acute leukemia is defined as the presence of at least 20% bone marrow blasts.
   B. Diagnosis is based on cellular morphology and cytochemistry.
   C. It groups lymphoid disorders into B cell, T/NK cell, and Hodgkin lymphoma.
   D. Diagnostic criteria include morphologic, cytochemical, immunologic, cytogenetic, and molecular features.

198. Which of the following would be *least* helpful in distinguishing chronic myelogenous leukemia (CML) from a neutrophilic leukemoid reaction?
   A. An extreme leukocytosis with increased neutrophilic bands, metamyelocytes, and myelocytes
   B. Leukocyte alkaline phosphatase score
   C. Presence of marked splenomegaly
   D. Neutrophils with Döhle bodies and toxic granulation

199. The cytoplasmic inclusion present in the cell shown in Color Plate 17■
   A. Excludes a diagnosis of acute myelogenous leukemia
   B. Stains positive with leukocyte alkaline phosphatase (LAP)
   C. Stains positive with myeloperoxidase (MPO)
   D. Identifies the cell as a malignant lymphoblast

200. Which of the following is a typical finding in chronic leukemias at onset?
   A. Symptoms of infection and bleeding
   B. Significant thrombocytopenia
   C. Severe anemia
   D. Elevated leukocyte count

201. In what condition would an LAP score of 10 most likely be found?
   A. Bacterial septicemia
   B. Late pregnancy
   C. Polycythemia vera
   D. Chronic myelogenous leukemia

202. Which of the following is *not* associated with neutrophilia?
   A. Staphylococcal pneumonia
   B. Crushing injury
   C. Infectious hepatitis
   D. Neoplasms (tumors)

203. In which of the following would an absolute monocytosis *not* be seen?
   A. Tuberculosis
   B. Recovery stage of acute bacterial infection
   C. Collagen disorders
   D. Infectious mononucleosis

204. Coarse PAS positivity may be found in the leukemic cells of
   A. Acute myeloblastic leukemia (FAB type M1)
   B. Acute lymphoblastic leukemia (FAB type L1)
   C. Acute myelomonocytic leukemia (FAB type M4)
   D. Acute monocytic leukemia (FAB type M5)

205. Which of the following is *not* among the diagnostic criteria used for classifying the myelodysplastic syndromes?
   A. Unexplained anemia refractory to treatment
   B. Hypogranular and hyposegmented neutrophils
   C. Abnormal platelet size and granulation
   D. Hypocellular bone marrow with 25% blasts

206. Naphthol AS-D chloroacetate esterase (specific) is usually positive in _____ cells, and alpha-naphthyl acetate esterase (nonspecific) is useful for identifying blast cells of _____ lineage.
   A. Granulocytic; monocytic
   B. Monocytic; granulocytic
   C. Granulocytic; lymphocytic
   D. Monocytic; lymphocytic

207. The familial disorder featuring pseudo-Döhle bodies, thrombocytopenia, and large platelets is called
   A. May-Hegglin anomaly
   B. Chédiak-Higashi syndrome
   C. Pelger-Huët anomaly
   D. Alder-Reilly anomaly

208. Alder-Reilly anomaly is an abnormality of
   A. Lysosomal fusion
   B. Nuclear maturation
   C. Oxidative metabolism
   D. Mucopolysaccharide metabolism

209. What is the initial laboratory technique for the diagnosis of monoclonal gammopathies?
   A. Immunologic markers of marrow biopsy cells
   B. Cytochemical staining of marrow and peripheral blood cells
   C. Serum and urine protein electrophoresis
   D. Cytogenetic analysis of marrow cells

210. Which of the following statements about Hodgkin disease is *false?*
   A. Peak incidence occurs in young adults.
   B. Staging determines extent of disease and treatment course.
   C. Stage IV has the best prognosis.
   D. Almost a 2:1 male predominance over females is characteristic.

211. The blast cells shown in Color Plate 18■ are CD14 and CD33 positive, Sudan black B positive, specific esterase positive, and nonspecific esterase positive. Which type of acute leukemia is most consistent with the immunophenotyping and cytochemical staining results?
   A. Acute lymphoblastic leukemia, T cell type
   B. Acute erythroleukemia
   C. Acute myelomonocytic leukemia
   D. Acute monocytic leukemia

212. Which type of leukemia is associated with the best prognosis for a cure?
   A. Chronic lymphocytic leukemia in the elderly
   B. Acute lymphoblastic leukemia in children
   C. Acute myelogenous leukemia in children
   D. Chronic myelogenous leukemia in young adults

213. What is the key diagnostic test for Hodgkin lymphoma?
   A. Bone marrow biopsy
   B. Lymph node biopsy
   C. Spinal tap
   D. Skin biopsy

214. A bone marrow with 90% cellularity and myeloid:erythroid (M:E) ratio of 10:1 is most characteristic of
   A. Chronic myelogenous leukemia
   B. Primary polycythemia
   C. Beta-thalassemia major
   D. Aplastic anemia

215. A 60-year-old patient presents with extreme fatigue. Her blood and bone marrow findings are as follows: severe anemia with a dual RBC population, 3% marrow blasts, and numerous ringed sideroblasts. This information is most consistent with
   A. Refractory anemia (RA)
   B. Refractory anemia with ringed sideroblasts (RARS)
   C. Refractory anemia with excess blasts (RAEB)
   D. Chronic myelomonocytic leukemia (CMML)

216. Which of the following is *not* a mechanism by which neutropenia may be produced?
   A. Hypersplenism
   B. Marrow injury or replacement
   C. Recent strenuous exercise
   D. Drug-induced antibodies

217. Which of the following is *not* a characteristic finding in polycythemia vera?
   A. Blood pancytosis
   B. Increased red cell mass
   C. Increased erythropoietin level
   D. Increased blood viscosity

218. In what disorder is significant basophilia most commonly seen?
   A. Hairy cell leukemia
   B. Plasma cell leukemia
   C. Acute lymphoblastic leukemia
   D. Chronic myelogenous leukemia

219. Acute erythroleukemia (FAB type M6) is characterized by increased
   A. Promyelocytes and lysozyme activity
   B. Marrow megakaryocytes and thrombocytosis
   C. Marrow erythroblasts and multinucleated red cells
   D. Marrow monoblasts and immature monocytes

220. The blood findings present in Color Plate 20■ are from a patient with complaints of fatigue and severe lower back pain. Which of the following would *not* be typical of this disease?
   A. Bone tumors of plasma cells
   B. Hypercalcemia
   C. Progressive renal impairment
   D. Normal sedimentation rate

221. Myeloid metaplasia refers to
   A. Displacement of normal marrow cells by fibrous tissue
   B. Hematopoietic failure
   C. Extramedullary hematopoiesis
   D. Tumors (neoplasms) of the bone marrow

222. Which of the following statements about non-Hodgkin types of lymphoma is *true?*
   A. Lymphadenopathy is the most common presenting symptom.
   B. Initially, they present as a systemic disease rather than a localized tumor.
   C. They are often associated with multiple bone lesions.
   D. They are characterized by proliferation of malignant cells primarily involving the bone marrow.

Methodology

223. What combination of reagents is used to measure hemoglobin?
   A. Hydrochloric acid and *p*-dimethylaminobenzaldehyde
   B. Potassium ferricyanide and potassium cyanide
   C. Sodium bisulfite and sodium metabisulfite
   D. Sodium citrate and hydrogen peroxide

224. The slowest-moving hemoglobin(s) on an alkaline electrophoresis at pH 8.6 is(are)
   A. A
   B. A_2, C, E, and O
   C. F
   D. S, D, and G

225. A patient with suspected sickle cell trait has negative solubility test results, but hemoglobin electrophoresis at pH 8.6 shows an apparent A-S pattern. What is the most likely explanation?
   A. Patient has hemoglobin AS, and the solubility test is incorrect.
   B. Patient has hemoglobin AA, and the electrophoresis is incorrect.
   C. Patient has hemoglobin AD or AG, and both procedures are correct.
   D. Tests need to be repeated; impossible to determine which procedure is correct.

226. Which of the following is an *incorrect* statement about the solubility test for Hemoglobin S?
   A. Hemoglobin S polymerizes when deoxygenated.
   B. Testing performed on a 2-day-old infant can result in a false negative result.
   C. Sickle cell trait can be differentiated from sickle cell anemia with this test.
   D. The test is positive in hemoglobin C_{Harlem}.

227. Which of the following is *not* associated with causing a falsely low ESR?
   A. Column used is slanted.
   B. EDTA tube is clotted.
   C. EDTA tube is one-third full.
   D. EDTA specimen is 24 hours old.

228. A platelet count is performed on an automated instrument from an EDTA blood sample. Smear evaluation reveals the presence of platelet clumps. The specimen is redrawn using sodium citrate as the anticoagulant, and a count of 300×10^9/L is obtained. What is the correct platelet count to report?
   A. 270×10^9/L
   B. 300×10^9/L
   C. 330×10^9/L
   D. 360×10^9/L

229. To best preserve cellular morphology, differential smears from an EDTA specimen should be made no more than _____ hour(s) after collection.
   A. 1
   B. 5
   C. 12
   D. 24

230. The blood smear made on a patient with polycythemia vera is too short. What should be done to correct this problem?
   A. Decrease the angle of the spreader slide.
   B. Increase the angle of the spreader slide.
   C. Adjust the angle of the spreader slide to 45 degrees.
   D. Use a smaller drop of blood.

231. The components of Wright's stain include
   A. Crystal violet and safranin
   B. Brilliant green and neutral red
   C. New methylene blue and carbolfuchsin
   D. Methylene blue and eosin

232. What is the reason for red blood cells to be bright red and the WBC nuclei to be poorly stained when using Wright's stain?
   A. The staining time is too long.
   B. The stain or buffer is too alkaline.
   C. The stain or buffer is too acidic.
   D. The smear was not washed long enough.

233. If 60 reticulocytes are counted in 1000 red blood cells, what is the reticulocyte count?
   A. 0.06%
   B. 0.6%
   C. 6.0%
   D. 60.0%

234. Using the percent reticulocyte from question 233 and an RBC count of 3.00×10^{12}/L ($3.00 \times 10^6/\mu L$), the calculated absolute reticulocyte count reported in SI units is
   A. 1.8×10^9/L
   B. 18×10^9/L
   C. 180×10^9/L
   D. $180 \times 10^3/\mu L$

235. The Sudan black B stain shown in Color Plate 19■ is a stain for
A. Glycogen
B. Lipids
C. Myeloperoxidase
D. Acid phosphatase

236. The following numbers were obtained in evaluating leukocyte alkaline phosphatase (LAP) activity in neutrophils. What is the score?

0	1	2	3	4
15	20	30	20	15

A. 100
B. 115
C. 200
D. 215

237. Perl's Prussian blue is a stain used to detect
A. DNA
B. RNA
C. Iron
D. Glycogen

238. Which of the following red cell inclusions stain with *both* Perl's Prussian blue and Wright's stain?
A. Howell-Jolly bodies
B. Basophilic stippling
C. Pappenheimer bodies
D. Heinz bodies

239. What is the depth between the counting platform and the coverslip on a hemacytometer?
A. 0.01 mm
B. 0.10 mm
C. 1.00 mm
D. 0.1 cm

240. A WBC count is performed on a hemacytometer using a 1:20 dilution. 308 cells are seen in a total area of 8 mm^2. What is the WBC count?
A. 3.8×10^9/L
B. 7.7×10^9/L
C. 15.4×10^9/L
D. 38.5×10^9/L

241. Which set of results indicates that an error in measurement has occurred?

	RBC $\times 10^{12}$/L	Hgb (g/dL)	Hct (%)
A.	2.50	7.6	22.9
B.	2.75	9.5	24.8
C.	3.40	10.0	31.0
D.	3.75	11.1	34.0

242. Which of the following would *not* be the cause of a falsely high MCHC of 38.3 g/dL on an automated instrument?
A. Hereditary spherocytosis
B. Lipemia
C. Presence of a cold agglutinin
D. Instrument sampling or mixing error

243. What is the principle of automated impedance cell counters?
A. Angle of laser beam scatter by cells
B. Amplification of an electrical current by cells
C. Interruption of an electrical current by cells
D. Change in optical density of the solution containing cells

244. A clinically significant difference between two electronic cell counts is indicated when the standard deviation is greater than
A. ±1.0
B. ±1.5
C. ±2.0
D. ±3.0

245. Side angle scatter in a laser-based cell counting system is used to measure
A. Cell size
B. Cytoplasmic granularity
C. Cell number
D. Immunologic (antigenic) identification

246. A white blood cell count is done on an automated impedance cell counter from a patient with the blood picture seen in Color Plate 4■. The WBC count is most likely
A. Falsely increased because of nRBCs
B. Falsely increased because of red cell fragments
C. Falsely decreased because of nRBCs
D. Accurate; no error with this methodology

247. The hemoglobin A_2 quantification using anion exchange chromatography will be valid in
A. Hemoglobin C disease
B. Hemoglobin E trait
C. Hemoglobin O trait
D. Beta-thalassemia minor

248. Which of the following is *not* associated with an increased osmotic fragility and a decreased surface area-to-volume ratio?
A. Beta-thalassemia major
B. Hereditary spherocytosis
C. Warm autoimmune hemolytic anemia
D. Burn victims

249. A clotted EDTA tube can be used to perform a(n)
A. Erythrocyte sedimentation rate
B. Solubility test for hemoglobin S
C. Hematocrit
D. Platelet count

250. The test value range that includes 95% of the normal population is the
A. Reference interval
B. Linearity limit
C. Reportable range
D. Critical range

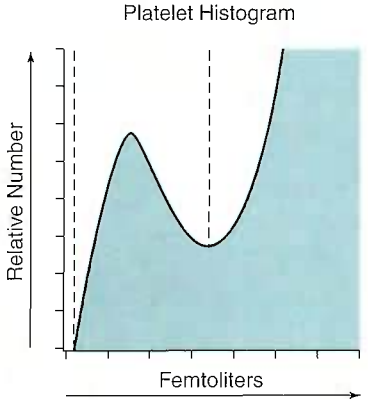
251. To establish a standard curve for reading hemoglobin concentration,
A. A commercial control material is used.
B. A wavelength of 640 nm is employed.
C. Certified standards are used.
D. A patient blood sample of known hemoglobin concentration is used.

252. Which of the following is *not* a source of error when measuring hemoglobin by the cyanmethemoglobin method?
A. Excessive anticoagulant
B. White blood cell count that exceeds linearity limits
C. Lipemic plasma
D. Scratched or dirty hemoglobin measuring cell

253. Which of the following statements about microhematocrits is *false?*
A. Excessive centrifugation causes falsely low results.
B. A tube less than half full causes falsely low results.
C. Hemolysis causes falsely low results.
D. Trapped plasma causes falsely high results.

254. The erythrocyte sedimentation rate (ESR) is influenced by the red cell phenomenon seen in Color Plate 20■. Which of the following factors will neither contribute to this phenomenon nor affect the ESR?
A. Size of the red blood cells
B. Shape of the red blood cells
C. Hemoglobin content of the red blood cells
D. Composition of the plasma

255. An EDTA blood sample run on an automated impedance cell counter has generated a warning flag at the upper region of the platelet histogram illustrated below. Which of the following would *not* be a cause of this warning flag?
A. Nucleated RBCs
B. Microcytic RBCs
C. EDTA-dependent platelet agglutinins
D. Giant platelets



256. To evaluate normal platelet numbers in an appropriate area of a blood smear, approximately how many platelets should be observed per oil immersion field?
A. 1–4
B. 4–10
C. 8–20
D. 20–50

257. Which of the following statements about manual reticulocyte counts is *false?*
A. The blood/stain mixture is incubated for 5–10 minutes.
B. New methylene blue, a supravital stain, is used.
C. RBC inclusions can result in falsely elevated counts.
D. An erythrocyte must have at least 4 blue particles to be counted as a reticulocyte.

258. When are automated cell counters required to have a calibration check performed?
A. At least every 3 months
B. After replacement of any major part
C. After performing monthly maintenance
D. When the control values are greater than 2 standard deviations from the mean

259. A blood sample was run through an automated cell counter and the following results were obtained: WBC 6.9×10^9/L (6.9×10^3/μL), RBC 3.52×10^{12}/L (3.52×10^6/μL), Hgb 120 g/L (12.0 g/dL), Hct 0.32 L/L (32.0%), MCH 34.1 pg, MCHC 37.5 g/dL. Which of the troubleshooting steps that follows should be performed to obtain reportable results?
A. Perform a saline replacement procedure.
B. Warm the specimen to 37°C and rerun.
C. Perform a microhematocrit.
D. None; the results are reportable.

260. Which of the following tests could be performed on a hemolyzed blood sample?
A. Hemoglobin only
B. Hemoglobin and platelet count
C. RBC count and hematocrit
D. No results would be reportable.

261. For which of the following procedures would heparin be a recommended anticoagulant?
A. Platelet count
B. Coagulation tests
C. Smear-based red cell morphology
D. Osmotic fragility

262. In the platelet count procedure using phase microscopy,
   A. Platelets appear dark against a light background.
   B. The entire ruled counting surface of the hemacytometer is used.
   C. Ammonium oxalate will lyse the WBCs.
   D. Platelets should be counted immediately after plating the hemacytometer.

263. What is the quality control term used to describe the reproducibility of a test?
   A. Accuracy
   B. Precision
   C. Standard deviation
   D. Specificity

Case Histories

Use the following information to answer questions 264–268.

The peripheral blood shown in Color Plate 4■ is from a 10-month-old Greek boy with the following results on an automated impedance counter: WBC 35.0×10^9/L ($35.0 \times 10^3/\mu$L); RBC 2.50×10^{12}/L ($2.50 \times 10^6/\mu$L); hemoglobin 45 g/L (4.5 g/dL); hematocrit 0.16 L/L (16%); platelet count 250×10^9/L (250,000/μL); reticulocyte count 8.0%; 110 nucleated red blood cells/100 WBCs and many targets are seen. Other laboratory results are as follows: serum iron elevated; total iron-binding capacity (TIBC) decreased; serum ferritin elevated.

264. What is the corrected white blood cell count expressed in SI units of $\times 10^9$/L?
   A. 4.6
   B. 12.5
   C. 16.7
   D. 18.4

265. What would be the appearance of the child's red blood cells on a peripheral smear?
   A. Microcytic, hypochromic
   B. Normocytic, hypochromic
   C. Normocytic, normochromic
   D. Microcytic, normochromic

266. The CBC, serum iron, total iron-binding capacity, and serum ferritin levels are most characteristic of
   A. Beta-thalassemia minor
   B. Iron-deficiency anemia
   C. Alpha-thalassemia minor
   D. Beta-thalassemia major

267. What type(s) of hemoglobin will be detected on this child using hemoglobin electrophoresis?
   A. A only
   B. A and F
   C. A, increased A_2, F
   D. F only

268. Why is it difficult to diagnose this disorder in a newborn?
   A. The liver is immature.
   B. The beta chains are not fully developed at birth.
   C. It is similar to hemolytic disease of the newborn (HDN) because of ABO incompatibility.
   D. There are normally many erythrocyte precursors in the peripheral blood.

Use the following information to answer questions 269–271.

A 75-year-old man with rheumatoid arthritis complains to his physician of pain and fatigue. His CBC results are as follows: WBC 6.8×10^9/L (6.8×10^3/μL); RBC 3.49×10^{12}/L (3.49×10^6/μL); hemoglobin 97 g/L (9.7 g/dL); hematocrit 0.29 L/L (29%); MCV 83 fL; MCHC 33.9 g/dL. Other laboratory results are as follows: serum iron and total iron-binding capacity (TIBC) both decreased, serum ferritin slightly elevated.

269. If the serum iron is 22 μg/dL and the TIBC is 150 μg/dL, what is the percent transferrin?
   A. 7%
   B. 10%
   C. 12%
   D. 15%

270. The results of the CBC and iron studies in this case are most characteristic of
   A. Beta-thalassemia minor
   B. Iron deficiency
   C. Sideroblastic anemia
   D. Anemia of chronic disease

271. Which of the following is *not* associated with the anemia described in question 270?
   A. Chronic gastrointestinal blood loss
   B. Hodgkin lymphoma
   C. Tuberculosis
   D. Systemic lupus erythematosus

Use the following information to answer questions 272–274.

The peripheral blood shown in Color Plate 11■ is from a 19-year-old female college student who has been living primarily on tea, beer, and cereal for the past 9 months because she finds dining hall food distasteful. She visits student health complaining of fatigue. Her CBC results are as follows: WBC 2.5×10^9/L (2.5×10^3/μL); RBC 2.10×10^{12}/L (2.10×10^6/μL); hemoglobin 85 g/L (8.5 g/dL); hematocrit 0.24 L/L (24%); platelet count 110×10^9/L (110,000/μL); MCV 114 fL; MCHC 35.0 g/dL; reticulocyte count 0.8%.

272. What test(s) should be done *first* to determine a diagnosis in this patient?
   A. Vitamin B_{12} and folate levels
   B. Iron studies
   C. Bone marrow examination
   D. Osmotic fragility

273. In the absence of neurological symptoms, the anemia in this patient is most likely caused by a lack of
   A. An enzyme
   B. Iron
   C. Folic acid
   D. Intrinsic factor

274. Which of the following is *not* a laboratory finding in this general classification of anemia?
   A. Target cells and schistocytes
   B. Teardrop cells and macro-ovalocytes
   C. Howell-Jolly bodies and Cabot rings
   D. Elevated serum LD and iron levels

Use the following information to answer questions 275–277.

A 45-year-old Scandinavian woman with white hair appears older than her age. She complains to her physician of weakness, a tingling sensation in her lower extremities, and shortness of breath. Her CBC results are as follows: WBC 3.4×10^9/L ($3.4 \times 10^3/\mu$L); RBC 1.90×10^{12}/L ($1.90 \times 10^6/\mu$L); hemoglobin level 86 g/L (8.6 g/dL); hematocrit 0.25 L/L (25%); MCV 132 fL; MCHC 34.4 g/dL; platelet count 100×10^9/L (100,000/μL). Cabot rings are noted on the peripheral smear.

275. The clinical and laboratory findings are most consistent with
   A. Liver disease
   B. Pernicious anemia
   C. Folic acid deficiency
   D. Aplastic anemia

276. Which of the following is *not* associated with this disorder?
   A. Alcoholism
   B. Antibodies to intrinsic factor or parietal cells
   C. *Diphyllobothrium latum* infection
   D. Achlorhydria

277. Which of the following statements about megaloblastic anemia is *true?*
   A. Oral folate therapy reverses the neurologic symptoms of PA.
   B. Intramuscular injections of vitamin B_{12} will reverse the neurologic symptoms of PA.
   C. Methotrexate (chemotherapeutic agent) is a vitamin B_{12} antagonist.
   D. Folate deficiency takes years to develop.

Use the following information to answer questions 278–280.

A 32-year-old African-American traveling to Africa on business had been healthy until he began taking primaquine for prevention of malaria. He went to his physician because he felt faint and his urine was black. His CBC results are as follows: WBC 6.5×10^9/L ($6.5 \times 10^3/\mu$L); RBC 1.67×10^{12}/L ($1.67 \times 10^6/\mu$L); hemoglobin level 50 g/L (5.0 g/dL); hematocrit 0.15 L/L (15%); MCV 89.8 fL; MCHC 33.3 g/dL; platelet count 175×10^9/L (175,000/μL); reticulocyte 25.0%.

278. The most likely cause of this hemolytic episode is
   A. G6PD deficiency
   B. Hereditary spherocytosis
   C. Sickle cell disease
   D. Pyruvate kinase deficiency

279. The defect in this disorder is caused by an
   A. Amino acid substitution
   B. Intrinsic red blood cell membrane defect
   C. Enzyme deficiency in the hexose monophosphate shunt
   D. Enzyme deficiency in the Embden-Meyerhof pathway

280. Inclusions that form when the patient is oxidatively challenged are composed of
   A. RNA
   B. Denatured hemoglobin
   C. DNA
   D. Iron

Use the following information to answer questions 281–283.

A 15-month-old malnourished child is brought to the clinic for a routine examination. Her CBC results are as follows: WBC 9.5×10^9/L (9.5×10^3/μL); RBC 2.70×10^{12}/L (2.70×10^6/μL); hemoglobin 67 g/L (6.7 g/dL); hematocrit 0.25 L/L (25%); MCV 73.5 fL; MCHC 26.8 g/dL; reticulocyte 0.2%; RDW 19%. Abnormal RBC morphology present included pencil forms and target cells.

281. What is this toddler's most probable diagnosis?
   A. Folic acid deficiency
   B. Hereditary spherocytosis
   C. Iron deficiency
   D. Erythroblastosis fetalis

282. The earliest indicator of this disease state is
   A. Decreased folic acid
   B. Decreased serum iron
   C. Decreased serum ferritin
   D. Increased bilirubin

283. What is the toddler's absolute reticulocyte count?
   A. 0.05×10^9/L
   B. 0.5×10^9/L
   C. 5×10^9/L
   D. 50×10^9/L

Use the following information to answer questions 284–288.

An 8-year-old girl is seen by the family physician. On physical examination, the physician notes fever, sore throat, bruising, petechiae, and pallor. A CBC is drawn and the results are as follows: WBC 110×10^9/L (110×10^3/μL); RBC 1.70×10^{12}/L (1.70×10^6/μL); hemoglobin 55 g/L (5.5 g/dL); hematocrit 0.16 L/L (16%); differential count shows 93% blasts and 7% lymphocytes. A bone marrow examination is performed and reveals 85% blasts. All of the blasts are small with no variation in their appearance.

284. Which of the following would you expect to most accurately reflect the child's platelet count?
   A. 10×10^9/L
   B. 100×10^9/L
   C. 200×10^9/L
   D. 400×10^9/L

285. What is this child's most probable diagnosis?
   A. Acute lymphoblastic leukemia
   B. Acute myelogenous leukemia
   C. Hairy cell leukemia
   D. Myelodysplastic syndrome

286. Which of the following cytochemical stains would most likely be positive in the blast cells of this patient?
   A. Myeloperoxidase
   B. Leukocyte alkaline phosphatase
   C. Periodic acid–Schiff
   D. Nonspecific esterase

287. Terminal deoxyribonucleotidyl transferase (TdT) is present in
A. Precursor B and precursor T lymphoid cells
B. Mature B and T lymphocytes
C. Precursor B cells and mature B lymphocytes
D. Precursor T cells and mature T lymphocytes

288. The presence of CD2, CD5, CD7 and the *absence* of CD10 (CALLA) are associated with
A. B lymphocytes
B. T lymphocytes
C. Myeloid cells
D. Monocytic cells

Use the following information to answer questions 289–292.

The peripheral blood smear in Color Plate 17 ■ and the Sudan black B stain in Color Plate 19 ■ are from a 90-year-old man complaining of fatigue and nosebleeds. The physician noted the patient was febrile and had petechiae. CBC results were as follows: WBC 20.0×10^9/L ($20.0 \times 10^3/\mu L$); RBC 2.58×10^{12}/L ($2.58 \times 10^6/\mu L$); hemoglobin 77 g/L (7.7 g/dL); hematocrit 0.24 L/L (24%); platelet count 32×10^9/L (32,000/μL); differential count shows 75% blasts, 20% lymphocytes, and 5% segmented neutrophils. A bone marrow examination revealed 80% cellularity with 80% blasts. The blasts were myeloperoxidase and specific esterase positive; nonspecific esterase and PAS negative.

289. What is this patient's most likely diagnosis?
A. Acute myelogenous leukemia without maturation (FAB type M1)
B. Acute myelogenous leukemia with maturation (FAB type M2)
C. Acute monocytic leukemia (FAB type M5)
D. Myelodysplastic syndrome

290. Cytogenetic studies would most likely show which of the following chromosome abnormalities?
A. t(8;21)
B. t(8;14)
C. t(9;22)
D. t(15;17)

291. Using World Health Organization (WHO) criteria for the diagnosis of acute leukemia, the percentage of bone marrow blasts must be at least
A. 5
B. 20
C. 30
D. 50

292. Which of the following is *not* considered an underlying condition that predisposes a patient to acute leukemia?
   A. Viral infections
   B. Bacterial infections
   C. Chronic bone marrow dysfunction
   D. Congenital chromosome abnormalities

Use the following information to answer questions 293–296.

An 83-year-old woman is seen in the emergency department complaining of fatigue and recent weight loss. Her CBC results are as follows: WBC 2.6×10^9/L ($2.6 \times 10^3/\mu$L); RBC 2.79×10^{12}/L ($2.79 \times 10^6/\mu$L); hemoglobin 92 g/L (9.2 g/dL); hematocrit 0.28 L/L (28%); MCV 100.0 fL; RDW 23.5%; platelet count 42×10^9/L (42,000/μL); differential count shows 42% segmented neutrophils, 45% band neutrophils, 3% lymphocytes, 3% metamyelocytes, 4% myelocytes, 3% blasts, and 4 nRBC/100 WBC. Morphologic changes noted on the differential smear include poor granulation and hyposegmentation of the neutrophils, giant platelets that display poor granulation, oval macrocytes, basophilic stippling, Cabot rings, Pappenheimer bodies, and Howell-Jolly bodies. Three micromegakaryocytes are seen per 100 WBCs. Serum B_{12} and folate levels are normal.

293. The peripheral blood findings are most consistent with
   A. Myelodysplastic syndrome
   B. Degenerative left shift
   C. Megaloblastic anemia
   D. Chronic myelogenous leukemia

294. The expected bone marrow findings in this disorder using WHO criteria are
   A. Hypocellular; blasts $\geq 20\%$
   B. Hypocellular; blasts $< 20\%$
   C. Hypercellular; blasts $\geq 20\%$
   D. Hypercellular; blasts $< 20\%$

295. If the bone marrow in this patient had 18% blasts, the most likely disorder would be
   A. Chronic myelomonocytic leukemia (CMML)
   B. Chronic myelogenous leukemia (CML)
   C. Refractory anemia with ringed sideroblasts (RARS)
   D. Refractory anemia with excess blasts (RAEB)

296. Which of the following is a *false* statement about myelodysplastic syndromes?
   A. MDS is "preleukemic" and frequently terminates in acute leukemia.
   B. Treatment for MDS is only supportive and not a cure.
   C. Median survival for all types of MDS is 5 years.
   D. The lower the blast percent, the longer is the survival rate.

Use the following information to answer questions 297–300.

A 53-year-old man reported to the laboratory for routine blood work as part of a yearly physical. He had been feeling tired for the last few months. Physical examination revealed splenomegaly. His CBC results are as follows: WBC 80.0×10^9/L ($80.0 \times 10^3/\mu L$); RBC 4.10×10^{12}/L ($4.10 \times 10^6/\mu L$); hemoglobin 123 g/L (12.3 g/dL); hematocrit 0.37 L/L (37.0%); platelet count 650×10^9/L (650,000/μL); differential count shows 40% polymorphonuclear neutrophils, 18% bands, 5% metamyelocytes, 7% myelocyte, 28% lymphocytes, and 2% monocytes. No RBC or WBC morphologic abnormalities are seen.

297. The peripheral blood findings are most consistent with a diagnosis of
   A. Neutrophilic leukemoid reaction
   B. Chronic myelogenous leukemia
   C. Acute myelogenous leukemia
   D. Regenerative left shift

298. Which of the following would yield the most diagnostic information for this patient?
   A. Sudan black B (SBB)
   B. Periodic acid–Schiff (PAS)
   C. Tartrate-resistant acid phosphatase (TRAP)
   D. Leukocyte alkaline phosphatase (LAP)

299. Which of the following myeloproliferative disorders is characterized by the presence of a t(9;22) chromosome abnormality and the BCR/ABL oncogene?
   A. Polycythemia vera
   B. Essential thrombocythemia
   C. Chronic myelogenous leukemia
   D. Chronic idiopathic myelofibrosis

300. How does the presence of this chromosome abnormality affect the prognosis?
   A. It is not a prognostic indicator.
   B. The prognosis is better when the abnormality is present.
   C. The prognosis is better when the abnormality is not present.
   D. Progression to acute lymphoblastic leukemia occurs more often when the abnormality is present.

answers & rationales

Hematopoiesis

1.

A. The need for oxygen delivery to developing tissues results in the production of erythrocytes before other blood cells. Erythropoiesis commences in the yolk sac as early as the fourteenth day of embryonic development. These primitive red cells produce embryonic hemoglobins that temporarily serve oxygen needs of the fetus. Myelopoietic and lymphopoietic activities begin when the liver and spleen become sites of production at 6–9 weeks of gestation; however, erythropoiesis still predominates. At this time, the red cells produce hemoglobin F, which is the chief oxygen carrier during fetal life.

2.

C. In the infant, there is an increased demand for blood formation because of the rate of growth. At birth, all bone marrow cavities are filled with hematopoietic tissue (active red marrow). As the growth rate slows, there is less need for active marrow. Fatty infiltration of the marrow becomes noticeable at about 4 years of age as cell production diminishes within the shafts of the long bones and is filled with yellow inactive tissue. Fat comprises 50% of the total marrow space in the adult. Except for lymphopoiesis, hematopoiesis is confined to the flat bones and pelvic area by the age of 25 years.

3.

C. Hematopoietic stem cells can make copies of themselves to maintain the stem cell pool and possess the ability to generate cells of all lineages (pluripotential). These stem cells give rise to multipotential myeloid and lymphoid progenitor cells, which ultimately produce progenitor cells that are restricted to a specific cell lineage. With appropriate cytokine stimulus, the committed progenitor cells undergo proliferation to recognizable precursors that produce an amplified number of mature end-stage cells. Stem cells and progenitor cells cannot be morphologically distinguished (look similar to small lymphocytes) but can be identified phenotypically by markers such as the stem cell marker CD34. CD34 expression is lost as antigens for a specific cell lineage are expressed. CD13 is a marker expressed by myeloid precursors.

4.

D. Unlike the infant, in which all bone marrow is capable of forming blood cells, the active marrow in an adult is confined to the flat bones of the skeleton such as the sternum and posterior iliac crest. Although the spinous processes of the vertebrae contain active marrow, these sites are rarely used for aspiration in adults because of the danger of damage to the spinal cord. Sternal puncture also presents a possibility of serious damage to underlying structures, but this site may be used because of easy accessibility or if the aspirate is a "dry tap" in the iliac crest. To obtain both a bone marrow aspirate and core biopsy, most marrow specimens are taken from the posterior iliac crest. The anterior iliac crest may occasionally be used in adults and sometimes the tibia in children less than 2 years of age.

5.

C. The ratio between all granulocytes and their precursors and all nucleated red cell precursors represents the myeloid-to-erythroid ratio. Myeloid precursors outnumber erythroid precursors by about 3 or 4 to 1 in the normal bone marrow. Although there are many more red blood cells in the peripheral blood than granulocytes, red blood cells have a much longer life span in circulation (120 days) as compared to granulocytes (about 8 hours). Granulocytes, therefore, require a more continual production than erythrocytes and are the most numerous marrow precursors. Alterations in the M:E ratio, such as 1:1 or 8:1, may indicate erythroid hyperplasia or granulocytic hyperplasia, respectively.

6.

B. A diverse group of growth factors (cytokines) regulate and maintain hematopoiesis in a steady state. Most hematopoietic growth factors are not lineage restricted but can act on more than one cell type and have multiple functions. For example, interleukins (IL-3) and colony stimulating factors (GM-CSF) affect multiple cell lines; whereas erythropoietin action is limited to erythroid cells. Cytokines are glycoproteins that usually express activity by binding to specific receptors on target cells. The action of growth factors on hematopoietic progenitor and precursor cells can stimulate or inhibit cell proliferation and differentiation as well as promote or suppress cell death. Growth factors may act alone or together to exert a positive or negative influence on hematopoiesis as well as on the function of mature cells. A determining factor for controlling the rate of cell production is cytokine stimulation in response to physiologic need.

7.

A. The liver of the fetus assumes responsibility for hematopoiesis about the second month of gestation. From 3 to 6 months of fetal development, the spleen, thymus, and lymph nodes are also involved, but the principal site of hematopoiesis remains the liver. By the seventh gestational month, the bone marrow becomes the primary hematopoietic site. Around birth, the liver and spleen have ceased hematopoiesis (except for splenic lymphopoiesis) but maintain the potential for reactivation of hematopoiesis.

8.

B. Erythropoietin (EPO) is a hormone that stimulates red cell production in the bone marrow by its action on the committed RBC progenitor cells. To maintain optimal erythrocyte mass for tissue oxygenation, the body's mechanism for sensing tissue oxygen levels is located in the kidney. Erythropoietin production increases when hypoxia is detected by renal oxygen sensors, with 90% being synthesized in the kidney and 10% in the liver. EPO levels in the blood vary according to the oxygen carrying capacity of the blood (e.g., EPO levels rise in anemia and fall when tissue oxygen levels return to normal).

9.

D. The marrow-derived common lymphoid progenitor cell ultimately gives rise to lymphocytes of T, B, or NK (presumably) cell lineages. Antigen-independent lymphopoiesis occurs in primary lymphoid tissue located in the thymus and bone marrow. The formation of immunocompetent T and B cells from precursor cells is influenced by environment (thymus, bone marrow) and several interleukins. Antigen-dependent lymphopoiesis occurs in secondary lymphoid tissue (spleen, lymph nodes, Peyer's patches of the gastrointestinal tract) and begins with antigenic stimulation of immunocompetent cells.

10.

B. Apoptosis is physiological cell death that can be induced by deprivation of growth factors or prevented by growth-promoting cytokines. Apoptosis plays an important role in the regulation of cell number and is deregulated in certain malignancies. Necrosis is accidental cell death by phagocytic cells and is associated with lethal physical damage. Cellular senescence describes cells that have lived their life span and will die of old age. Terminal differentiation refers to mature end-stage cells that are no longer capable of replication.

11.

A. Bone marrow consists of vessels, nerves, hematopoietic cells at various levels of maturation, and stromal cells encased in a membrane lining called the endosteum. The vascular system empties into a system of sinuses (venous sinusoids). A layer of endothelium lines these sinusoids. Blood cell formation occurs in hematopoietic cords located outside of the sinusoids and between the trabeculae of spongy bone. The bone marrow stroma (macrophages, adipocytes, fibroblasts, endothelial cells) forms an optimal microenvironment for developing cells by providing support and secreting cytokines. Mature differentiated cells can deform to penetrate the vessel wall and enter the sinuses and blood circulation.

12.

B. Bone marrow cellularity in the normal adult is approximately 50% hematopoietic tissue and 50% adipose tissue (fat), with a range of 30–70% cellularity being normocellular. Marrow cellularity is usually estimated from the core biopsy. An intact bone marrow can respond to demand by increasing its activity several times the normal rate if sufficient supplies and growth factors are available. The marrow becomes hypercellular when inactive fatty tissue is replaced by active hematopoietic marrow. In contrast, bone marrow failure may result in hypocellularity or aplasia with increased fat and a reduced number of hematopoietic cells.

13.

C. Interleukins and colony stimulating factors are cytokines produced by a variety of cells, including monocytes/macrophages, T lymphocytes, fibroblasts, and endothelial cells. It is essential that cytokines are continuously supplied by cells present in the bone marrow microenvironment during hematopoietic cell development, or cells will die. Erythropoietin functions as a true hormone because it is produced by the kidney, released into the blood, and carried to the bone marrow, where it stimulates red cell production.

14.

C. In a normal adult, the total blood volume is approximately 12 pints or 6 liters. Cells account for about 45% (44% is red cell mass) and plasma accounts for 55%. Alterations in red cell mass or plasma volume are reflected in the RBC count and in measurements of hemoglobin and hematocrit. True anemia or polycythemia is due to a decrease or increase in total RBC mass, respectively. A reduction in plasma volume with a normal RBC mass may cause relative (pseudo) polycythemia. Conversely, an increase in plasma volume with normal RBC mass may cause relative (pseudo) anemia.

15.

A. The pluripotent hematopoietic stem cell gives rise to lymphoid and myeloid progenitor cells. The lymphoid progenitor produces cells destined to become lymphocytic cells, whereas the myeloid progenitor cell produces progenitors committed to differentiation into granulocytic, erythrocytic, monocytic, or megakaryocytic lineages with appropriate stimulus. The cells produced by progenitor cells can be demonstrated using *in vitro* culture techniques; thus, the myeloid progenitor cell is termed CFU (colony forming unit)-GEMM based on the cell colonies formed.

16.

C. The mature megakaryocyte, the largest hematopoietic cell in normal bone marrow, has a multilobed nucleus and abundant, granular cytoplasm. Plasma cells are characterized by a round, eccentric nucleus and intensely blue cytoplasm. Osteoblasts and osteoclasts are nonhematopoietic cells that may be present in normal bone marrow. Osteoblasts are cells involved in bone formation that resemble plasma cells but are larger and often found in groups. Osteoclasts reabsorb bone and are similar to megakaryocytes in size but are multinucleated.

17.

D. In normal adult marrow, about 50% is fat, 40% is myeloid (granulocytic) cells, and 10% is erythroid cells. The M:E ratio is determined by performing a differential count of marrow precursor cells. The presence of 10% myeloblasts is an abnormal finding (reference range 0–2%), and a hematologic disease is likely. Megakaryocytes should be seen when scanning and are usually reported as normal, increased, or decreased in number. Marrow iron is assessed with Perl's Prussian blue stain, and it is normal to see stainable iron in macrophages, as well as iron granules in the cytoplasm of developing red cell precursors.

18.

B. The nucleus-to-cytoplasm ratio decreases as blood cell lines mature. With maturation, cells generally become smaller, the nuclear chromatin becomes clumpy and condensed, nucleoli disappear, and the cytoplasm loses its deep blue basophilia when stained with Wright's stain. Exceptions include megakaryocytes (because of endomitosis they grow larger as cytoplasm accumulates) and plasma cells (increased RNA and protein synthesis produces a deep basophilia).

19.

C. Thrombopoietin (TPO) is the major regulator of platelet production in the bone marrow by its action on committed progenitor and precursor cells of the megakaryocytic line. It is primarily produced by hepatocytes and possibly by the kidney. After marrow release, about 70% of platelets are in the blood circulation and 30% are sequestered in the spleen. Unlike erythropoietin, which is manufactured for routine therapeutic use, recombinant TPO is still being evaluated.

20.

B. Hematopoiesis within the medulla or inner part of the bone marrow is termed medullary or myeloid. Hematopoiesis that occurs in the liver and spleen (reactivation of fetal life) is called extramedullary or myeloid metaplasia (organs may enlarge). Cell production outside of the marrow space takes place when the bone marrow is unable to meet its production demands. This may occur in severe hemolytic anemias when the maximal capacity of the bone marrow to increase activity is exceeded. Myeloid metaplasia may also be an extension of a disease process such as myelofibrosis. Myelophthisis refers to the replacement of normal marrow hematopoietic tissue by fibrotic tissue or cancer cells, whereas myelodysplasia describes abnormal maturation of erythrocytic, granulocytic, and/or megakaryocytic cell lines. The period of intrauterine life when cell production occurs in the yolk sac may be termed mesoblastic.

Erythrocytes

21.

D. Normal red blood cells survive about 4 months, or 120 days. The entire life span of the red cell is spent inside the vascular tree, making it easier to determine the rate of production and destruction. Red cell survival depends upon an intact RBC membrane, sufficient cellular energy, and normal hemoglobin function. As red cells circulate for 120 days, enzymes are depleted and the ability to deform decreases. Under normal conditions, red cell loss due to aging (~1%) is equal to daily replacement. Most destruction of aged red cells occurs extravascularly by macrophages of the reticuloendothelial system (spleen, liver).

22.

B. The erythrocyte has a semipermeable membrane that allows water and some anions, such as chloride (Cl^-) and bicarbonate HCO_3^-, to enter the cell rapidly. Sodium ions (Na^+) enter the cell and potassium ions (K^+) leave the cell slowly but continuously. In order to maintain a high intracellular K^+ concentration and remove excess Na^+, ATP-dependent cationic pumps expel Na^+ and take in K^+. This regulation of intracellular cations allows the red cell to control its volume and water content.

23.

D. A molecule of hemoglobin is composed of four globular, protein subunits, and each subunit contains a heme group bound within a convoluted globin chain. Heme groups are identical and consist of protoporphyrin IX with a central iron atom, made largely in the mitochondria. Amino acids are sequenced on ribosomes to produce four types of globin chains (alpha, beta, delta, and gamma) that combine in identical pairs. A normal hemoglobin molecule consists of two alpha-globin chains and two non-alpha-globin chains, each of which binds a heme group. The different globin chains determine the hemoglobin type (A, A_2, or F).

24.

D. Culling is the process of removing aged or abnormal red blood cells from the circulation by the spleen. Red cells (7 μm) enter the spleen through the splenic artery and must squeeze back into active circulation through 2- to 4-μm clefts in the venous sinusoids. Aged or abnormally shaped red cells with impaired membrane flexibility are trapped in the splenic microcirculation and ingested by macrophages. The spleen is the largest filter of blood in the body and has an essential role in the "quality control" of red cells.

25.

C. Each hemoglobin molecule has four heme groups located at its surface, and oxygen binds to the central ferrous iron (Fe^{2+}) in heme. Deoxyhemoglobin (not carrying O_2) and oxyhemoglobin (carrying up to four O_2) are normal physiologic forms of hemoglobin with iron in the ferrous state. Hemoglobin in which the ferrous iron (Fe^{2+}) has been oxidized to the ferric state (Fe^{3+}) is known as methemoglobin and is unable to carry O_2. Carboxyhemoglobin is hemoglobin with carbon monoxide (CO) attached to ferrous iron rather than O_2. Both methemoglobin and carboxyhemoglobin are reversible.

26.

C. Of the total body iron present in a normal adult, approximately 70% is contained in hemoglobin (in red cells of the blood and marrow). Most of the remainder, ~25%, is found in storage sites as ferritin or hemosiderin. A much smaller amount of iron is contained in muscle myoglobin (4%) and respiratory enzymes such as peroxidase (1%). The structures of hemoglobin and myoglobin are similar (both consist of globin and heme), but myoglobin functions as an oxygen trap in the tissues.

27.

B. A senescent red blood cell is one that has lived its life span. Repeated passes through the spleen deplete the cells of glucose and decrease their surface area as membrane lipids are lost. The red cells are removed from the circulation by splenic macrophages that recognize subtle abnormalities in these cells, sequester them, and destroy them.

28.

B. "Poikilocytosis" is a general term that refers to deviations from the normal red cell shape (biconcave, discoid). "Anisocytosis" is the term used when differences in the sizes of red cells are described. Color in red cells is designated as normochromic (normal) or hypochromic (indicating a decreased hemoglobin concentration). Abnormally shaped red cells and red cell inclusions are associated with rigid red cells that have reduced deformability and shortened survival.

29.

A. Howell-Jolly bodies are nuclear (DNA) remnants that remain in the red cell after the nucleus has been extruded and may represent nuclear instability. These inclusions are associated with the defective nuclear maturation found in megaloblastic anemias and the rapid cell division that occurs in severe hemolytic anemias. Under normal circumstances, the spleen effectively pits these bodies from the cell. Pitting is a process that removes inclusions while leaving the rest of the red cell intact. It may be that the pitting mechanism is overwhelmed and cannot keep pace with inclusion formation in hemolytic anemias. Howell-Jolly bodies can also be seen in individuals after splenectomy who lack the normal pitting function.

30.

A. Spherocytes appear smaller and more densely staining than normal red cells and lack a central pallor area. Because they are the result of membrane loss, their surface area-to-volume ratio is decreased. Spherocytes should be distinguished from acanthocytes, which also lack a pallor area but have sharp, irregular projections. Echinocytes have a central pallor area and blunt, short projections. Red cells with intracellular rod- or bar-shaped crystals contain hemoglobin C crystals.

31.

B. The presence of lead causes an inhibition of several of the enzymes important in heme synthesis. Among these is pyrimidine 5′-nucleotidase, which is normally responsible for degradation of ribosomal ribonucleic acid (RNA). The lack of this enzyme apparently allows aggregates of incompletely degraded RNA to remain in the cell cytoplasm. It is this ribosomal material that appears on Wright's stain as punctate basophilic stippling. Precipitated hemoglobin forms Heinz bodies (not visible with Wright's stain), nuclear fragments are called Howell-Jolly bodies, and iron deposits are Pappenheimer bodies.

32.

D. The presence of red cell rouleaux (coining pattern) is a characteristic finding in multiple myeloma because of the increased concentration of immunoglobulins in the blood plasma (hypergammaglobulinemia). The excessive immunoglobulins are produced by malignant plasma cells. Cold hemagglutinin disease is characterized by red cell agglutination or clumping in a nonspecific pattern. "Hypersplenism" refers to an enlarged, overactive spleen that destroys both normal and abnormal cells, possibly causing pancytopenia.

33.

C. In viewing Color Plate 1 ■, the inclusions in the red blood cells are Howell-Jolly bodies. During passage through the microvessels of the spleen, the red cell is examined for intracellular inclusions or membrane-bound antibodies, which, if present, are removed. Abnormal red cells circulate longer, and inclusions such as Howell-Jolly bodies or Pappenheimer bodies will be seen post-splenectomy (or in conditions with splenic atrophy). The phagocytic removal of abnormal red cells is assumed by the liver, but the liver is not as efficient as the spleen. Howell-Jolly bodies are not associated with iron deficient or iron overload states.

34.

C. One of the reasons for increased intestinal absorption of iron is an accelerated rate of erythropoiesis (another is depletion of iron stores). Although the mucosal cell does act as a barrier in normal circumstances, this function is not absolute and controls break down in the presence of large amounts of iron, causing an excess to be absorbed. An acid pH is required for iron absorption, and sites of maximal absorption are the duodenum and upper jejunum. The body has no effective means for iron excretion.

35.

D. The presence of iron granules or deposits can be detected with Perl's Prussian blue iron stain. Siderocytes are mature red blood cells that contain stainable iron granules (abnormal). Sideroblasts are bone marrow nucleated red cells (normoblasts) that contain small amounts of iron in the cytoplasm (normal). Ringed sideroblasts are marrow normoblasts that contain iron in the mitochondria that forms a ring around the nucleus (abnormal). Siderocytes and ringed sideroblasts are associated with iron overload problems, particularly sideroblastic anemia. Reticulocytes may contain small amounts of unused iron that is normally removed by the spleen.

36.

C. A "shift to the left" in the oxygen dissociation curve of hemoglobin means that a higher percentage of hemoglobin will retain more of its oxygen at a given pressure. Thus affinity will be greater and oxygen delivery will be reduced. A higher or more alkaline pH and a lower temperature are associated with decreased oxygen dissociation. With conditions in the lungs (increased pH, decreased 2,3-BPG, decreased temperature), hemoglobin affinity for oxygen is increased, which favors oxygen uptake. With conditions in the tissues (decreased pH, increased 2,3-BPG, increased temperature), hemoglobin affinity for oxygen is decreased, which favors release of oxygen to the tissues.

37.

D. Erythropoietin (EPO) is a hormone produced by the kidney that increases erythropoiesis in the bone marrow in response to tissue hypoxia. The CFU-E (colony-forming unit–erythroid) is a committed erythroid progenitor cell with many receptors for erythropoietin. EPO stimulation of the CFU-E produces the recognizable pronormoblast and promotes differentiation of RBC precursors. The maturation time of erythrocyte precursors (5–7 days) can be reduced in times of increased need for red cells by the action of erythropoietin.

38.

C. Increased binding of 2,3-BPG (2,3-bisphosphoglycerate) decreases the affinity of hemoglobin for oxygen, which promotes oxygen release to the tissues, a compensatory mechanism in anemic patients. Increased pH (alkalinity) enhances oxygen affinity and thus inhibits delivery to the tissues. Less oxygen is available at higher altitudes, and this affects blood saturation and delivery to tissues. An increase in erythropoietin release will affect red cell production but does not have an immediate or direct impact on oxygen delivery.

39.

A. After the nucleus is extruded, reticulocytes spend about 2 days in the bone marrow before release into the blood, where maturation continues for another day. Intense erythropoietin stimulus can cause early release of bone marrow reticulocytes. These reticulocytes are larger and contain more filamentous reticulum than a more mature reticulocyte. These "shift" or "stress" reticulocytes exhibit diffuse basophilia on the Wright's stained smear and will need more than the usual 1 day in circulation to mature (to lose RNA). A very high number of reticulocytes in the blood circulation can increase the MCV. The level of reticulocyte maturity is best assessed by the immature reticulocyte fraction (IRF), an index reported by automated cell counters that is based on RNA content.

40.

C. Heinz bodies do not stain with Wright's stain and appear as "normal" hemoglobin even though their presence causes cell rigidity and membrane damage. They can be visualized on wet preps with phase microscopy or by using supravital stains, such as crystal violet or brilliant green. Heinz bodies consist of intracellular globin or hemoglobin precipitate that results from hemoglobin denaturation (G6PD deficiency, unstable hemoglobin variants) or excess globin chains (certain thalassemic syndromes). Basophilic stippling, Cabot rings, and Pappenheimer bodies are visible with both Wright's and supravital stains.

41.

B. The presence of schistocytes (schizocytes) on the smear indicates that red cells have been subjected to some form of physical trauma that causes damage. Red cell fragmentation can be the result of impact with fibrin strands, mechanical trauma by artifical surfaces, injury by heat, partial phagocytosis, or damage by toxins and drugs. Schistocytes are characteristic of the increased red blood cell destruction that occurs in severe hemolytic anemias but are not associated with anemias that result from defective bone marrow delivery of red cells to the blood.

42.

C. Tissue hypoxia associated with low erythrocyte and hemoglobin levels causes increased renal release of erythropoietin to stimulate bone marrow erythropoiesis. Depending on severity, the bone marrow responds by increasing its activity 6–8 times normal and becomes hypercellular because of an increase in RBC precursors (erythroid hyperplasia); and the M:E ratio falls. Nucleated red cells may be released into the blood along with the outpouring of reticulocytes. The number of nucleated red cells tends to correlate with anemia severity.

43.

C. Millions of hemoglobin molecules are produced in the red cell cytoplasm during maturation. When developing erythroid cells are deprived of essential hemoglobin components, the result is the production of microcytic, hypochromic red cells. It is thought that during maturation, extra cell divisions occur until a certain hemoglobin concentration is reached. Impaired hemoglobin synthesis may be the result of heme defects (involving iron or protoporphyrin) or may be caused by globin defects. Impaired DNA synthesis is associated with macrocytic red cells, and normocytic red cells are characteristic of enzyme defects.

44.

C. The nucleated red cells seen in Color Plate 2■ would be staged as orthochromic normoblasts (metarubricytes) when in the bone marrow. This is the last stage of red cell maturation that contains a nucleus. The pyknotic, degenerated nucleus is normally extruded out of the red cell in the marrow to yield the anucleate reticulocyte. The release of nucleated red cells into the blood before reaching maturity usually indicates a high demand for red cells.

45.

B. The red cell membrane consists of a protein shell heavily coated with lipids. The membrane lipid bilayer is maintained by constant interchange with plasma lipids. Acanthocytes are the result of abnormal plasma lipids that have altered the lipid composition of the membrane, often involving increased cholesterol content. Acanthocytes (spur cells) are associated with a congenital form of acanthocytosis and with liver disease, or are seen following splenectomy.

46.

A. The mature red cell, which lacks mitochondria and Kreb's cycle activity, depends on glucose metabolism for cellular energy. The end product of the anaerobic Embden-Meyerhof pathway (EMP) is ATP, which is necessary for membrane maintenance and volume control (cation pumps). The hexose monophosphate (HMP) pathway is aerobic and reduces oxidants by providing NADPH and glutathione. The Rapoport-Luebering shunt controls the amount of 2,3-bisphosphoglycerate that regulates hemoglobin affinity for oxygen. Oxidized hemoglobin (methemoglobin) is reduced to functional hemoglobin by the methemoglobin reductase pathway.

47.

C. The polychromatophilic normoblast (rubricyte) is the last red cell stage capable of mitosis. With cellular divisions, each pronormoblast produces up to 16 erythrocytes. The polychromatophilic normoblast is also the stage in which hemoglobin is first visible. The gray-blue color of the cytoplasm when Wright's stained is due to a mixture of hemoglobin and RNA, hence the name "polychromatophilic." The reticulocyte is the last stage able to synthesize hemoglobin.

48.

A. The major adult hemoglobin, Hb A, consists of two alpha- and two beta-globin chains. The switch from gamma chains (Hb F) to beta chains occurs 3–6 months after birth, and Hb A reaches adult levels (about 97%) around 6 months of age. Most globin chains produced in a normal adult are alpha and beta types (1:1 ratio) for hemoglobin A production. Hemoglobin A_2 contains delta chains and comprises about 2% of hemoglobin in normal adults. Epsilon chains are found in early embryonic hemoglobins only.

49.

C. Impaired DNA synthesis results in nuclear maturation that lags behind cytoplasmic development (asynchrony), decreased cellular divisions, and the production of macrocytic red cells. Defective nuclear maturation (megaloblastic) is almost always caused by a deficiency of vitamin B_{12} or folic acid, which are DNA coenzymes. Macrocytic red cells that are not due to vitamin B_{12} or folic acid deficiency (non-megaloblastic) may be seen in liver disease or when reticulocytosis is pronounced. Microcytic, hypochromic red cells are the result of impaired hemoglobin synthesis.

50.

A. When iron is removed from the heme of destroyed red blood cells, it is bound to transferrin and recycled for hemoglobin production or goes to storage. The major storage form of iron is ferritin, which is a water-soluble iron complex bound in a protein shell called apoferritin. Hemosiderin is a water-insoluble complex of iron aggregates and protein that is derived from ferritin. The main site of iron stores is the liver, but storage iron is also found in the bone marrow and spleen.

51.

B. RBC indices are average values, so they have less meaning in heterogeneous RBC populations with wide size variations. Because the MCV is a mean red cell volume measurement, the presence of both microcytes and macrocytes would yield a falsely normal MCV value. One would expect the RDW (red blood cell distribution width) to be high because it is an index of variation in red cell size or anisocytosis. The RDW is low when red cells are of uniform size (a homogeneous population). The RDW is high when a heterogeneous population of red cells is present.

52.

B. Bilirubin is formed when hemoglobin degradation occurs in the reticuloendothelial system, primarily in the spleen. Unconjugated bilirubin is transported by the plasma to the liver, where it is conjugated. When excessive extravascular red cell destruction occurs, the plasma bilirubin level rises and is largely unconjugated bilirubin. When acute intravascular red cell destruction occurs, hemoglobin is released into the plasma and findings may include hemoglobinemia, hemoglobinuria, and hemosiderinuria.

53.

D. Haptoglobin forms a 1:1 complex with alpha-beta dimers of hemoglobin. The large size of this complex prevents filtration of the hemoglobin through the kidneys, where it can cause renal damage. Haptoglobin can be depleted in the plasma during major hemolytic events, such as malarial attacks, transfusion reactions, and other causes of severe intravascular red cell destruction.

54.

C. Codocytes (target cells) have an increased surface area-to-volume ratio and are associated with abnormal hemoglobin synthesis. They are found in hemoglobinopathies, especially hemoglobin S or C disorders, as well as thalassemias and iron deficiency. Target cells can also result from an increase in membrane lipids and may be seen in liver disease. Discocytes are normal biconcave red cells, and elliptocytes (ovalocytes) can be found in varying sizes. The teardrop shape of dacryocytes may occur when a red cell is stretched in the spleen and cannot regain its original shape.

55.

B. Heme synthesis begins in the mitochondria with the formation of aminolevulinic acid. Formation of the pyrrole ring structure occurs in the cytoplasm, resulting in the synthesis of coproporphyrinogen III. The final stages of porphyrin synthesis occur again in the mitochondria, culminating in the formation of heme when ferrous iron is incorporated into protoporphyrin IX in the presence of ferrochelatase.

56.

A. The red cell membrane consists of an outer bilayer of lipids with embedded, integral proteins and an underlying skeleton. Spectrin is the predominant skeletal protein that forms a cytoskeleton with other proteins, such as actin, protein 4.1, and ankyrin. The skeletal proteins are responsible for cell shape, deformability, and stability. Any defect in structure or extensive damage to the membrane cannot be repaired and may lead to premature red cell death.

57.

C. Reduced glutathione (GSH) counteracts oxidants that accumulate in the red cell. These occur as a result of normal metabolic activities and increase during infections or as a result of treatment by certain drugs. In the absence of GSH or as a result of enzyme deficiencies in the hexose monophosphate pathway (HMP), oxidant accumulation can lead to oxidation and precipitation of hemoglobin.

58.

C. Total iron-binding capacity (TIBC) represents the amount of iron that circulating transferrin could bind when fully saturated. In this test, the amount of transferrin protein in the serum is indirectly measured by adding ferric (Fe^{3+}) iron to the serum and allowing it to bind to the unsaturated sites on transferrin. Unbound iron is then removed and the sample analyzed for the remaining iron that is bound to transferrin. The serum iron level measures iron bound to transferrin. Under normal conditions, about one-third of the binding sites on transferrin are occupied with iron.

59.

B. The amount of circulating ferritin indirectly reflects the amount of storage iron in the tissues. A bone marrow exam is not essential to assess iron stores, except in complicated cases, because the serum ferritin test is considered a good indicator of iron storage status in most individuals. Because ferritin is an acute-phase reactant, it may be increased in chronic inflammatory disorders regardless of iron stores. Therefore, the serum ferritin should be interpreted with other iron tests. The percent transferrin saturation is the serum iron divided by the serum TIBC.

60.

B. Fetal hemoglobin can be distinguished from adult hemoglobin in red blood cells by the acid elution technique of Kleihauer and Betke. Only hemoglobin F remains in red blood cells after exposure to a citric acid–phosphate buffer solution at pH 3.3. Hb F has a higher oxygen affinity than Hb A (less binding of 2,3-BPG), so it carries oxygen well *in utero*. Hemoglobin F production decreases after birth, composing less than 1% of total hemoglobin in normal adults. In certain conditions (thalassemias, hemoglobinopathies), defective beta-chain production can be compensated for by increased production of gamma chains and formation of hemoglobin F (two alpha and two gamma chains).

Erythrocyte Disorders

61.

D. Deficiencies of folic acid (folate) and vitamin B_{12} result in abnormal DNA synthesis and a resultant delay in nuclear maturation in comparison to cytoplasmic development. These anemias are categorized as megaloblastic because of the giant red cell precursors observed in the bone marrow. The other anemias are characterized by defects of heme (sideroblastic anemia and iron-deficiency anemia) or globin synthesis (hemoglobin C disease).

62.

A. G6PD deficiency has a sex-linked inheritance pattern and is the most common enzyme deficiency in the hexose monophosphate (HMP) shunt. Individuals are asymptomatic unless exposed to oxidants, which compromise the ability of the glutathione reduction pathway to prevent the oxidation of hemoglobin. The oxidized hemoglobin precipitates in the form of Heinz bodies, which cause acute intravascular hemolysis. In the most common G6PD variant, the hemolytic episode is self-limiting, with old red cells that lack enzyme being destroyed and young red cells with some enzyme activity unaffected.

63.

B. Hemoglobin nomenclature indicates a number of things. The symbol α_2 or $\alpha_2{}^A$ indicates the presence of normal adult, or A, alpha chains. The designation $\beta_2{}^{26\ Glu \rightarrow Lys}$ indicates that lysine residues have replaced glutamic acid on position 26 of the beta chains. All types of E hemoglobin show a similar electrophoretic mobility and migrate closely to hemoglobins C and A_2 on cellulose acetate (alkaline pH). Hemoglobin E occurs with the greatest frequency in Southeast Asia.

64.

C. The peripheral blood as seen in Color Plate 3■ shows numerous elliptocytes (ovalocytes). If they were artifact due to smear preparation, they would be oriented in the same direction. Hereditary elliptocytosis (HE) is associated with symptomatic hemolytic anemia in only about 10–15% of the cases, but the presence of an enlarged spleen is evidence of ongoing extravascular destruction. In patients with chronic hemolysis, gallstones are a common complication because of excess bilirubin catabolism. In most persons with HE, anemia does not develop because bone marrow production of red cells compensates for the mild shortening of red cell life span.

65.

B. Cooley anemia, or beta-thalassemia major, would be the appropriate diagnosis in this case. In this condition, two beta-thalassemia genes are inherited that result in virtually no hemoglobin A production because no beta-globin chains are produced. The primary hemoglobin made is hemoglobin F. The severe microcytic anemia results from the destruction of red cell precursors in the bone marrow (ineffective erythropoiesis) and rigid red cells in the blood that contain unused alpha-globin chains. Nucleated red blood cells and target cells, as seen in Color Plate 4■, are common, as well as basophilic stippling. Infants with alpha-thalassemia major die *in utero* or shortly after birth. Hemoglobin H disease (three-gene deletion alpha-thalassemia) results from deficient, alpha-chain synthesis that leads to production of Hb H (four beta chains), an unstable hemoglobin that forms Heinz bodies and causes chronic hemolysis. No clinical manifestations are seen in patients with hereditary persistence of fetal hemoglobin (HPFH).

66.

C. As seen in Color Plate 5■, the presence of numerous target cells and SC crystals on the peripheral blood smear suggests the presence of hemoglobin SC disease. These bizarre crystals are distinguished by one or more blunt, finger-like projections that protrude from the cell membrane. Clinically, hemoglobin SC disease is not usually as severe as sickle cell disease, and electrophoresis shows equal amounts of Hb S and Hb C. Codocytes, in varying numbers, are typical of hemoglobin S, C, and E disorders.

67.

C. Pica is a clinical finding seen in some patients with iron deficiency. Pica is unusual cravings for nonfood items that may include dirt, clay, laundry starch, or, most commonly, ice. Among some cultures, pica is a custom (eating dirt) that may contribute to iron deficiency. In children, lead poisoning often results from the ingestion of dirt or lead-based paint from toys and may be related to iron deficiency. Porphyrias are a group of inherited disorders characterized by enzyme deficiencies and abnormal porphyrin metabolism. The presence of pyridoxine (pyridoxal-5'-phosphate or vitamin B_6) is important to early porphyrin synthesis.

68.

B. Folate deficiency is most commonly a result of poor dietary intake of folate alone or in combination with increased requirements as during pregnancy. Daily requirements for folate are high, and depletion of folate stores can occur within 4 months as compared to vitamin B_{12}, in which deficiency takes at least 2 years to develop (there are high stores). Thus, dietary deficiency of vitamin B_{12} is rare, but folate supplements are commonly required during pregnancy or in hemolytic anemias with excess cell turnover.

69.

C. The hemoglobin solubility test can detect the presence of hemoglobin S, which is insoluble in the dithionite reagent, whereas normal hemoglobin A is soluble. A positive screening test, however, does not distinguish between patients with hemoglobin AS trait, hemoglobin SC disease, and hemoglobin SS disease, so results must be confirmed by electrophoresis. Sickle cell trait is clinically asymptomatic with target cells only. Disorders prevalent in the malarial belt (sickle cell trait, G6PD deficiency, hereditary ovalocytosis, thalassemia minor) are thought to impart resistance to falciparum malaria. Repeated splenic infarctions by sickle cell masses in hemoglobin SS disease cause autosplenectomy by adulthood.

70.

D. The schistocytes in Color Plate 6■ are found in microangiopathic hemolytic anemia and caused by red cells shearing on fibrin strands deposited in small vessels. Widespread or localized (e.g., kidney) fibrin deposition in DIC, HUS, and TTP results in red cell fragmentation. In addition, thrombocytopenia is a usual feature of MAHA. ITP is characterized by severe thrombocytopenia that results from destruction of platelets by autoantibodies, but it is not associated with red cell damage or anemia.

71.

C. Ringed sideroblasts result from the accumulation of iron deposits in the mitochondria surrounding the nucleus of erythroid precursors. The deposits are secondary to a defect in heme synthesis and a pathological finding in sideroblastic anemia. Blocks in the protoporphyrin pathway required for heme synthesis may be hereditary (rare) or acquired and result in iron overload with increased marrow iron. Pappenheimer bodies and basophilic stippling are frequent findings on the blood smear, and increased serum iron, decreased TIBC, increased percent transferrin saturation, and increased serum ferritin are usual.

72.

D. Reticulocytosis is indicative of increased erythropoietic activity by the bone marrow. This is a normal response in conditions involving premature red cell destruction in the circulation or following blood loss due to acute hemorrhage. The reticulocyte count is consistently increased in active hemolytic disease because the marrow speeds up red cell production to supply replacement cells. Anemia develops when the rate of red cell destruction exceeds the marrow's ability to replace red cells (uncompensated hemolytic disease). The reticulocyte count is not usually elevated in pernicious anemia even though increased marrow erythropoiesis occurs. The defective cellular maturation that occurs in megaloblastic anemias results in the death of many red cells in the bone marrow (ineffective erythropoiesis).

73.

B. The major defect in hereditary stomatocytosis is altered permeability of the red cell membrane to Na^+ and K^+ ions. A net gain of sodium within the cell leads to increased water entry and the appearance of a swollen cell with a slit-like area of pallor. This is a heterogeneous group of disorders, in that a number of specific membrane defects have been postulated, and anemia varies from mild to severe. One autosomal dominant disorder is associated with Rh-null individuals.

74.

D. Polycythemia vera (PV) belongs to the group of disorders that are hematopoietic stem cell defects and commonly characterized as myeloproliferative disorders. Although the major increase in PV is in red blood cells, there is also an overproduction of granulocytes and platelets, particularly in the early stages of the disease. The increased production of red cells in PV is not due to the activity of erythropoietin. The production of erythropoietin is almost completely suppressed in this malignant condition.

75.

A. Aplastic anemia can be defined as blood pancytopenia resulting from bone marrow failure. This stem cell disorder results in a hypocellular marrow with few developing precursors and decreased production of all cell lines. The anemia is generally normocytic or slightly macrocytic with reticulocytopenia. The "defect" also affects resting hematopoietic cells in the liver and spleen, so extramedullary hematopoiesis does not occur to compensate for marrow failure.

76.

A. Because red blood cells and plasma are lost together, the hemoglobin and hematocrit will not reflect the severity of an acute hemorrhage until the lost blood volume begins to be replaced by the formation of plasma. The restoration of a normal blood volume is usually complete by 24 hours. It is then that the hemoglobin and hematocrit will reach their lowest point and will begin to rise only with the release of newly formed red cells, usually within 3–4 days.

77.

D. Elevated RBC, hemoglobin, and hematocrit values in a newborn are a carryover from intrauterine life, when a high number of red cells were needed to carry oxygen. Erythropoiesis is suppressed in response to the marked increase in oxygenation of tissues after birth, and the reticulocyte count, which is initially high, will fall along with a slow decline in the hemoglobin level. A hemoglobin value below 140 g/L (14.0 g/dL) is abnormal for a neonate. Newborn red cells are macrocytic and up to 10 nucleated red cells per differential may be seen.

78.

D. The red blood cells with single elongated projections, seen in Color Plate 7■, are dacryocytes or teardrops. Dacryocytes are often seen in disorders of marrow replacement that affect bone marrow architecture, especially myelofibrosis. Teardrops can also result from the splenic removal of inclusions and may be present in a variety of anemias. Drepanocytes or sickle cells are observed during a sickling crisis of sickle cell anemia. Acanthocytes, echinocytes, and/or codocytes can be found in liver disease (presence varies with disease severity).

79.

C. Myelophthisic anemia is an anemia of bone marrow failure. It is seen in patients who are experiencing bone marrow replacement of normal hematopoietic tissue by metastatic cancer cells, fibrosis, or leukemia. The anemia is considered a hypoproliferative anemia because there is no hemolysis involved and the cells are normocytic, normochromic. Disruption of the bone marrow by abnormal cells can result in the release of immature cells (nucleated red cells and immature neutrophils) into the blood and may involve blood cell production in extramedullary sites.

80.

A. Any idiopathic disorder is one for which there is no apparent cause. Ionizing radiation is a well-known cause of aplasia, as is chemical exposure (pesticides, benzene). Iatrogenic disorders are those that result from treatments for a different disorder; for example, aplasia can result from chloramphenicol treatment for bacterial disease. Aplastic anemia may develop as a complication from infections such as Epstein-Barr or hepatitis viruses.

81.

C. Diamond-Blackfan anemia is a congenital disorder that depresses only red blood cell production. Fanconi anemia is a congenital form of aplastic anemia that results in aplasia of all cell lines and has a high risk of developing acute myeloid leukemia or other cancers. The bone marrow distinguishes Diamond-Blackfan from the hypocellular marrow seen in aplastic anemia because there is a lack of erythroid precursors but a normal number of myeloid and megakaryocytic precursor cells.

82.

C. Dehydration is a cause of relative (pseudo) erythrocytosis due to plasma loss. High altitude adjustment, cardiac or pulmonary disease, and defective oxygen transport are all causes of absolute secondary erythrocytosis. Secondary erythrocytosis (polycythemia) is a compensatory increase in red cells, produced in an attempt to increase the amount of oxygen available to the tissues.

83.

B. In infants with homozygous alpha-thalassemia, no alpha-globin chains are produced (because of the deletion of all four alpha genes). Consequently, the infants have nearly 100% hemoglobin Bart's, which consists of four gamma-globin chains. This hemoglobin migrates farther toward the anode than Hb A. Because Hb Bart's has a very high oxygen affinity, it is useless for delivery of oxygen to the tissues, making its presence incompatible with life. Hemoglobin H, composed of four beta chains, also migrates farther than Hb A, but Hb H disease is not fatal.

84.

C. Bart's hydrops fetalis (homozygous alpha-thalassemia major) is a lethal condition in which all normal hemoglobins are absent and the presence of Bart's hemoglobin results in death due to hypoxia. Severe cases of Rh hemolytic disease of the newborn/fetus (erythroblastosis fetalis) are characterized by hemolytic anemia, high numbers of nucleated red blood cells, and hyperbilirubinemia that can cause brain damage (kernicterus). The bilirubin level is elevated but anemia is mild, if present, in ABO hemolytic disease of the newborn.

85.

B. The blood profile alone cannot distinguish folic acid and vitamin B_{12} deficiencies, because both are characterized by macrocytic ovalocytes, Howell-Jolly bodies, and hypersegmented neutrophils. Clinical severity generally differentiates the heterozygous (mild) and homozygous (severe) conditions of thalassemic and sickle cell syndromes. The anemia of acute blood loss is usually normocytic, whereas the anemia of chronic blood loss becomes microcytic due to the development of iron deficiency.

86.

C. The cause of the many spherocytes and polychromatophilic red cells seen in Color Plate 8■ would be best determined with the direct antiglobulin test (DAT). The differential diagnosis is hereditary spherocytosis (negative DAT) and warm autoimmune hemolytic anemia (positive DAT). Both of these hemolytic disorders are the result of membrane injury and would show an increased osmotic fragility result due to the spherocytes, elevated reticulocyte counts, and elevated urine urobilinogen, as well as elevated serum bilirubin levels.

87.

D. Schistocytes and spherocytes are associated with red cell destruction and would be found in clostridial septicemia (toxins), prosthetic heart valves (mechanical trauma), and thermal burns (heat). Microspherocytes can also result from the direct membrane damage caused by clostridial toxins and heat. Aplastic anemia is not a hemolytic anemia but is caused by decreased bone marrow production. Aplastic anemia is usually normocytic, with no evidence of red cell damage on the blood smear, and red cell destruction tests such as serum bilirubin would be normal.

88.

D. Erythrocytosis (polycythemia) is either absolute or relative. Absolute erythrocytosis occurs when the RBC mass increases, taking up a larger than usual proportion of the blood volume. Relative polycythemia occurs when the RBC mass stays normal but the amount of fluid volume decreases, thus increasing the proportion of the blood occupied by red cells as compared to the total blood volume which has decreased. Primary polycythemia is a condition of erythrocytosis without an underlying or contributing condition. The body produces an increased number of red cells without an increase in erythropoietin (an inappropriate response). Secondary polycythemia occurs when some underlying condition causes an increase in erythropoietin, so erythrocytosis occurs secondary to the condition (an appropriate response).

89.

A. Hemochromatosis is an excessive deposition of iron in body tissues that results in iron-laden macrophages, expansion of storage sites, and serious damage to organs (heart, liver). Iron overload can be hereditary or acquired as a complication of severe hemolytic anemias, frequent blood transfusions, or sideroblastic anemia. Hereditary hemochromatosis is caused by a mutation of the HFE gene that results in increased absorption of iron from the gastrointestinal tract and leads to iron overload. It is associated with low levels of hepcidin, an iron regulator, which causes increased iron absorption and release of iron from macrophages. The treatment for hereditary hemochromatosis is phlebotomy, and molecular testing is done for diagnosis.

90.

A. Abetalipoproteinemia, or hereditary acanthocytosis, is a rare autosomal recessive disorder of lipid metabolism. An absence of serum beta lipoprotein, a transport protein, causes abnormal plasma lipids. The numerous acanthocytes (spur cells) are the result of an alteration in the lipid content of the red cell membrane. The anemia is mild, but this disorder is associated with progressive neurologic disease.

91.

A. The majority of body iron is found in the hemoglobin of circulating erythrocytes. This means that any form of bleeding will lead to excessive iron loss. Iron balance is normally very tightly controlled through absorption rather than excretion. Iron deficiency in males is rare but, if present, it is usually the result of chronic gastrointestinal bleeding (ulcers, cancer).

92.

B. The severe transfusion-dependent anemia, which is typical of homozygous beta-thalassemia, is the result of imbalanced globin-chain synthesis and massive red cell destruction that far exceeds the rate of production. Decreased or absent beta chains lead to excess alpha chains that precipitate in red cells and subsequently are destroyed. The response is intense marrow erythroid hyperplasia, bone expansion, and erythropoiesis in extramedullary sites. A complication of continuous red cell hemolysis and repeated blood transfusions is iron overload. Patients require iron chelation therapy to prevent liver and heart failure. Splenectomy may be needed to reduce blood requirements, but it is not done before 4 years of age because of the increased risk of infection.

93.

A. Serum iron is low in both iron-deficiency anemia and the anemia of chronic disorders. The total iron-binding capacity (TIBC), which is an indirect measure of the amount of transferrin protein, is low in the anemia of chronic disease, whereas it is high in iron deficiency. Synthesis of transferrin is regulated by iron availability. Usually, when storage iron decreases, serum iron levels decrease and transferrin levels (TIBC) increase. In the anemia of chronic disorders, storage iron is normal or increased (but unavailable), and transferrin levels (TIBC) are decreased.

94.

C. Although the punctate basophilic stippling found in lead poisoning in erythrocytes is considered a classic finding, the anemia present is usually not severe unless accompanied by iron deficiency. The presence of lead inhibits several enzymes involved in the formation of heme, with a consequent increase in erythrocyte protoporphyrin and urinary aminolevulinic acid. The most significant effect of lead toxicity is the resulting neurological deficit and impairment of mental development.

95.

C. In Color Plate 9■, the dual population of red blood cells represented may also be termed dimorphic. This blood picture could be seen in a patient with microcytic, hypochromic anemia after the transfusion of normal red cells or when new normocytic, normochromic red cells are produced after successful treatment for iron deficiency. Concurrent deficiencies, such as coexisting iron and folate deficiency during pregnancy, would result in the production of both microcytic and macrocytic red cells.

96.

B. The cells visualized in Color Plate 10■ are sickle cells in the presence of target cells. The substitution of a valine for the glutamic acid normally found in the sixth position of the beta-globin chain causes red cells containing hemoglobin S to undergo the characteristic shape change that gives the sickle cell its name. A defect of both beta genes results in sickle cell disease, whereas a single gene mutation causes the sickle cell trait. Hemoglobin C results from the substitution of lysine for glutamic acid in the sixth position of the beta-globin chain.

97.

D. The number of irreversibly sickled cells (ISCs) and the proportion of S hemoglobin within the cells contribute collectively to the severity of sickle cell disorders. The classification of "trait" versus "disease" is not based on the severity of symptoms. The absence of Hb A and the presence of over 80% Hb S on electrophoresis would be classified as homozygous sickle cell disease (SS), whereas the heterozygous condition (AS) would show approximately 60% Hb A and 40% Hb S. Sickling is rare in the trait condition because of the lower concentration of Hb S. Sickle cell disease typically shows increased levels of compensatory Hb F, as does hemoglobin SC disease.

98.

B. Hyperbaric oxygen will reverse the sickling process, but it will also suppress erythropoietin, which stimulates the bone marrow to produce adequate replacement erythrocytes. Hydroxyurea reduces sickling by increasing Hb F levels and has been shown to improve the clinical course of patients plagued by painful crises. Treatment is primarily supportive and symptomatic, with efforts made to avoid those factors known to precipitate a crisis.

99.

D. In response to premature red cell destruction, the normal bone marrow can speed up red cell production. Hemolytic anemias typically have high reticulocyte counts, because the marrow can respond to the need for red cells. Generally, anemias caused by defective maturation or decreased production have inappropriately low reticulocyte counts, because the marrow fails to respond due to injury or lack of essential hematopoietic components. Low hemoglobin and hematocrit values reveal the presence of anemia but do not indicate etiology. *The reticulocyte count is particularly useful in distinguishing hemolytic anemias from other normocytic anemias that are not hemolytic.*

100.

C. Both iron deficiency and heterozygous thalassemia can present with a mild microcytic, hypochromic anemia. Target cells may be seen in both, but basophilic stippling is only found in thalassemia. The hemoglobin A_2 is normal in heterozygous alpha-thalassemia but is frequently twice the normal level in heterozygous beta-thalassemia, because these individuals compensate with increased delta-chain production due to deficient beta-globin chain synthesis. In this case, iron deficiency would likely be ruled out first with iron tests. Beta-thalassemia with hemoglobin S trait (Hb S/beta-thalassemia) produces a severe clinical picture similar to sickle cell anemia, with sickling of red cells.

101.

D. Oxidative denaturation is the primary mechanism of the hemolytic process. When glucose-6-phosphate dehydrogenase (G6PD) is deficient, the red blood cells cannot generate sufficient reduced glutathione (GSH) to detoxify hydrogen peroxide. Hemoglobin is oxidized to methemoglobin, denatures, and precipitates, forming Heinz bodies. The Heinz bodies cause the rigidity of the red cells, and hemolysis occurs as the cells try to pass through the microcirculation.

102.

B. In hereditary spherocytosis, the rigid spherocytes are being destroyed in the splenic microcirculation. Following splenectomy, the hemoglobin level should rise as the spherocytes circulate longer. Consequently, there is less need for increased red cell production by the bone marrow, and the number of reticulocytes released into the blood will fall. Approximately, 30% of platelets are normally sequestered by the spleen, so a transient increase in the platelet count occurs and red cell inclusions (Howell-Jolly and Pappenheimer bodies), normally pitted out of red cells by the spleen, will be observed.

103.

D. Unconjugated bilirubin levels will rise when either excessive intravascular or extravascular hemolysis is occurring. When hemolysis is intravascular, the free hemoglobin released into the circulation is bound by haptoglobin, and the complex is transported to the liver, where it is metabolized to bilirubin. Depletion of the haptoglobin protein will occur if use exceeds production, and then hemopexin binds hemoglobin for removal. When both haptoglobin and hemopexin are depleted, plasma hemoglobin levels will increase. The serum lactate dehydrogenase rises when red cells are broken down and intracellular LD enzymes are released.

104.

A. Hemolytic anemias can be classified by the mode of transmission (hereditary or acquired) and by the type of defect (intrinsic or extrinsic). With the exception of paroxysmal nocturnal hemoglobinuria (PNH), intrinsic defects are hereditary, and the defect that shortens survival is within the abnormal red cell. Hemolytic anemias due to extrinsic defects are acquired and caused by external agents or extracorpuscular factors that destroy the intrinsically normal red cell.

105.

C. Paroxysmal cold hemoglobinuria (PCH) is caused by an IgG biphasic antibody with P specificity known as the Donath-Landsteiner antibody. This autoantibody fixes complement to the red cells in the cold, and the complement-coated red cells lyse when warmed. PCH can be idiopathic or follow a viral infection and is characterized by acute intravascular hemolysis and hemoglobinuria after cold exposure. Cold autoantibodies usually show I specificity, whereas warm autoantibodies are often directed against Rh antigens on the red cells.

106.

B. The hypersegmented neutrophils and macrocytic ovalocytes seen in Color Plate 11■ suggest the presence of megaloblastic anemia. The two most common causes are lack of folic acid or vitamin B_{12}, which are coenzymes required for normal DNA synthesis. This patient's neurological symptoms are indicative of a vitamin B_{12} deficiency, because that vitamin is also needed for myelin synthesis (CNS).

107.

B. Pappenheimer bodies observed with Wright's stain can be confirmed with the Prussian blue stain and are composed of iron. The presence of siderotic granules in the red cells is associated with iron overload, and the serum ferritin test, which reflects the amount of storage iron, would be elevated. The test for parietal cell antibodies can be done to determine the cause of vitamin B_{12} deficiency.

108.

D. Hemoglobinopathies are a hereditary group of qualitative disorders in which genetic mutations cause the production of structurally abnormal globin chains. The three most common variant hemoglobins are Hb S, Hb C, and Hb E, all of which are due to an amino acid substitution in the beta-globin chain. Hemoglobin C is the second most common hemoglobin variant, after hemoglobin S, seen in the United States. Thalassemias are characterized by an absent or reduced rate of globin-chain synthesis.

109.

C. The structurally abnormal red cells in hereditary spherocytosis (HS) are deficient in spectrin and are abnormally permeable to sodium. The bone marrow produces red cells of normal biconcave shape, but HS cells lose membrane fragments and become more spherical as they go through the spleen and encounter stress in the blood circulation. The membrane defect is accentuated by the passage of red cells through the spleen, where they are deprived of glucose and are unable to generate sufficient ATP to pump sodium out of the cell. Ultimately, the red cells are trapped and destroyed in the spleen. The osmotic fragility is increased because of the membrane loss (reduced surface area-to-volume ratio), and the MCHC value may be increased.

110.

A. In hereditary elliptocytosis (HE), the red blood cells show increased permeability to sodium and may have one of several membrane defects linked to this heterogeneous disorder. These include deficiencies in skeletal proteins such as protein 4.1 or spectrin. The characteristic oval or elliptical shape is seen only in mature red blood cells, and it occurs in the circulation when HE red cells cannot return to a normal biconcave shape.

111.

C. The anemia of chronic disease (ACD) is very common and develops in patients with chronic infections (tuberculosis), chronic inflammatory disorders (rheumatoid arthritis, systemic lupus), and malignant disease (cancer, lymphoma). ACD has a complex etiology that includes impaired release of storage iron for erythropoiesis and a reduced response to erythropoietin. The anemia may be normocytic or microcytic, and severity depends on the underlying disorder.

112.

A. Hemoglobin C disease results from a homozygous substitution of lysine for glutamic acid at position 6 of the beta-globin chains. Numerous target cells and occasional intracellular C crystals will be found on the blood smear. Osmotic fragility is decreased (increased resistance) because of the many target cells, and electrophoresis will show an absence of Hb A and over 90% Hb C. The clinical severity of Hb CC is mild as compared to Hb SS or Hb SC diseases.

113.

D. Red cells that contain a high concentration of hemoglobin S will assume the sickle shape when deprived of oxygen (which can be reversed if reoxygenated). After repeated sickling, reversion capabilities are lost and irreversibly sickled cells (ISCs) are seen. Sickle cells are mechanically brittle, nondeformable cells that become impeded in circulation, causing blocks that restrict blood flow in vessels and leading to organs (vascular occlusive disease). They are easily trapped in the small vessels of the spleen, leading to obstructive ischemia and eventual destruction of splenic tissue.

114.

B. Aplastic anemia is bone marrow failure characterized by hypocellularity and decreased production of all cell lines. The normal M:E ratio (4:1) does not change in aplasia, because the number of both myeloid and erythroid precursors is decreased. In anemias such as sickle cell anemia, beta-thalassemia major, or megaloblastic anemia, the marrow becomes hypercellular because of an increase in erythroid precursors, and the M:E ratio falls.

115.

B. The immune hemolytic anemia indicated by the smear findings is warm autoimmune hemolytic anemia (WAIHA). Antibody-coated red cells are being partially phagocytized by macrophages (with receptors for IgG and complement), causing loss of membrane fragments. The spherocytes are ultimately destroyed, primarily in the spleen. The cause of the autoantibody production may be unknown (idiopathic), develop secondary to a disease that alters the immune response (chronic lymphocytic leukemia or lymphoma), or can be drug induced. Cold hemagglutinin disease is characterized by red cell agglutination due to a cold autoantibody.

116.

C. Intrinsic factor is a glycoprotein secreted by the parietal cells, along with HCl, that is needed to bind vitamin B_{12} for absorption. Pernicious anemia (PA), which is a megaloblastic anemia caused by the lack of intrinsic factor, is most common cause of vitamin B_{12} deficiency (cobalamin). PA is characterized by atrophy of the gastric parietal cells and achlorhydria (absence of HCl). Autoimmune factors are involved because a high percentage of patients produce autoantibodies to intrinsic factor (50%) and/or parietal cells (90%). The bone marrow erythroid precursors exhibit megaloblastic maturation, with nuclear maturation lagging behind cytoplasmic maturation (asynchrony is also seen in developing granulocytes and platelets). Many fragile red cells die in the bone marrow, and those released into the circulation have a very short survival, which causes a marked increase in lactate dehydrogenase levels.

117.

B. Hemoglobin D migrates in the same location as hemoglobin S on cellulose acetate at alkaline pH but does not cause sickling. The negative solubility test rules out the presence of hemoglobin S. Target cells are seen in large numbers in homozygous hemoglobin D disease. The quantification of 95% differentiates homozygous from heterozygous states where less than 50% hemoglobin D would be seen.

118.

B. G6PD deficiency compromises the ability of the glutathione reduction pathway to prevent the oxidation of hemoglobin. Oxidative stress may occur from infections, ingestion of mothballs, ingestion of fava beans, and certain drugs, including primaquine or sulfonamides. The oxidized hemoglobin precipitates in the form of Heinz bodies, which leads to a hemolytic crisis characterized by intravascular red cell destruction, removal of Heinz bodies by splenic macrophages, and the presence of spherocytes and fragmented red cells on the smear.

119.

B. Paroxysmal nocturnal hemoglobinuria (PNH) is an acquired defect of membrane structure in which red cells have a high affinity for complement binding. PNH is characterized by pancytopenia and chronic intravascular hemolysis with hemoglobinuria and hemosiderinuria. A stem cell mutation causes production of red cells, white cells, and platelets that are sensitive to complement lysis because of the loss of a membrane glycolipid (GPI). The sucrose hemolysis test can be used to screen for PNH red cells, but the Ham's acid serum test has been replaced by immunophenotyping for confirmation of PNH. Paroxysmal cold hemoglobinuria (PCH) is characterized by intravascular hemolysis and hemoglobinuria after cold exposure that is due to a complement-binding autoantibody. A transient finding of hemoglobinuria following forceful contact of the body with hard surfaces (as may be seen in joggers and soldiers) describes March hemoglobinuria.

120.

A. Secondary, acquired sideroblastic anemia is the result of blocks in the protoporphyrin pathway that can be identified and, therefore, are reversible if the toxin or agent is removed. Alcohol inhibits vitamin B_6 and anti-tuberculosis drugs also interfere with vitamin B_6 (pyridoxine). Lead poisoning causes multiple blocks in protoporphyrin synthesis, including inhibition of ferrochelatase, which is needed for iron incorporation into protoporphyrin to produce heme. Methotrexate is an antifolate chemotherapeutic agent that causes a drug-induced megaloblastic anemia.

121.

A. Although iron deficiency may be the most common cause of anemia in pregnancy, there is a mild form of anemia that develops during the third trimester in pregnant women with adequate iron levels. Although both erythrocytes and plasma increase during pregnancy, the plasma increases in a higher proportion, causing a relative (pseudo) anemia. This increased blood volume actually increases oxygen delivery to both the mother and the fetus.

122.

A. The anemia of chronic renal failure results from decreased production and release of erythropoietin from the diseased kidney. The drop in erythropoietin results in decreased red blood cell production by the marrow. Recombinant erythropoietin is of great value in treating anemia resulting from end-stage renal disease. Iron or folate supplements may be needed to maximize the response, especially in patients on dialysis. Uremic metabolites may cause reduced red cell survival and impairment of platelet function.

123.

D. Aplastic anemia is a stem cell defect that leads to decreased production of erythrocytes, leukocytes, and platelets (pancytopenia). The survival of red cells released into the circulation is normal. Infection is a serious problem because of the lack of neutrophils. The reduced number of platelets is responsible for the bleeding often seen. Treatment includes blood and platelet transfusions, antibiotics, growth factors, and steroids. Bone marrow transplantation may be necessary.

124.

B. The fish tapeworm competes for vitamin B_{12}, and a macrocytic (megaloblastic) anemia may develop. Hookworm infestation causes chronic blood loss and a microcytic anemia due to iron deficiency. A variety of organisms are associated with hemolysis, including malaria and clostridial infections. Viral hepatitis can cause marrow suppression and a normocytic, hypoproliferative anemia.

125.

A. A need for the increased oxygen carrying capacity provided by additional red blood cells is found in conditions such as pulmonary disease, where normal oxygenation is inhibited. A decrease in the ability of the cardiovascular system to appropriately circulate cells is another reason for increased erythrocytes. Individuals with a high level of methemoglobin, such as heavy smokers or persons with genetic disorders, cannot effectively unload oxygen. This results in a need for increasing the number of red blood cells to compensate. Renal tumors are associated with excess production of erythropoietin, leading to an inappropriate polycythemia.

126.

B. Thalassemias are a group of congenital disorders characterized by quantitative defects in globin-chain synthesis. Alpha-thalassemias result from gene deletions that cause a reduced rate of alpha-globin chain production. Beta-thalassemias result from point mutations that cause a reduced rate of beta-globin chain synthesis. Normally, equal amounts of alpha- and beta-globin chains are produced for Hb A synthesis. In alpha- or beta-thalassemias, synthesis of globin chains is imbalanced, because a decreased production rate of one type of globin chain causes an excess of the other (consequences will depend on the thalassemia type).

127.

C. Low serum iron and iron stores (represented by serum ferritin) characterize iron deficiency that is severe enough to result in anemia. The production of transferrin, the iron transport protein, increases as iron stores decrease. Transferrin saturation decreases dramatically so that transferrin is less than 15% saturated with iron.

128.

D. The switch from gamma-globin chain production for Hb F to beta-globin chain synthesis for Hb A occurs 3–6 months after birth. Clinical symptoms of a homozygous beta-globin chain defect, such as sickle cell disease or homozygous beta-thalassemia, will not be evident until about 6 months of age or shortly after. Alpha-globin chain production is normally high throughout fetal and adult life. A homozygous defect involving the alpha-globin chain will affect the infant *in utero*.

129.

A. The hemolytic crisis of malaria results from the rupture of erythrocytes containing merozoites. This event becomes synchronized to produce the fever and chill cycles that are characteristic of this infection. In severe infections, particularly those caused by *Plasmodium falciparum,* the massive intravascular hemolysis results in significant hemoglobinuria.

130.

D. The incorrect ratio of blood to anticoagulant caused the cells to shrink. This produced the crenated appearance of the red cells. This is an artifact as opposed to a significant clinical finding and can also be the result of prolonged blood anticoagulation. Spur cells (or acanthocytes) lack a central pallor area and have sharp projections, as opposed to crenated cells (or echinocytes), which have a pallor area and blunt projections.

131.

A. Pyruvate kinase (PK) is an enzyme of the Embden-Meyerhof pathway (anaerobic glycolysis). A deficiency of PK results in decreased ATP generation, which causes impairment of the cation pump and a loss of normal membrane deformability. PK-deficient cells have a shortened survival time, but clinical manifestations vary widely.

132.

B. When iron use exceeds absorption, iron stores (serum ferritin) are depleted first. At this early stage, there is no anemia (normal hemoglobin) and the transferrin level is normal. This is followed by increased transferrin synthesis (TIBC) and decreased serum iron. Finally, a microcytic, hypochromic anemia develops.

133.

A. In the anemia of chronic disease (ACD), a chronic illness causes impaired release of iron from storage. These patients have iron but are unable to use it for bone marrow erythropoiesis. Hepcidin, a hormone produced by the liver, plays a major role in the regulation of body iron by influencing intestinal absorption and release of storage iron from macrophages. Hepcidin levels increase during inflammation (positive acute-phase reactant), which causes decreased release of iron from stores. There is also impaired response of marrow red cell precursors to erythropoietin stimulation in ACD. The impaired response is thought to be related to the effects of inflammatory cytokines. Recombinant erythropoietin improves the anemia in some cases.

134.

C. In the idiopathic or primary type of sideroblastic anemia, the blocks in the protoporphyrin pathway (heme synthesis) that lead to iron overload are unknown and, therefore, are irreversible. The anemia is refractory (unresponsive) to treatment other than transfusion. Ringed sideroblasts and increased stainable iron will be found in the bone marrow when stained with Prussian blue. This primary, acquired form of sideroblastic anemia is also known as refractory anemia with ringed sideroblasts (RARS) and is classified as a myelodysplastic syndrome.

135.

B. The demand for red blood cell replacement in beta-thalassemia major during early childhood development results in a hyperproliferative marrow. Expansion of the marrow causes the bones to be thin and narrow. This may result in pathologic fractures. Facial bones have the Mongoloid appearance, with prominence of the forehead, cheekbones, and upper jaw.

136.

D. Cold autoimmune hemolytic anemia (CAIHA), or cold hemagglutinin disease, is characterized by the production of IgM cold autoantibodies that often show I specificity. The cause of the autoantibody production may be unknown (primary) or occur secondary to *Mycoplasma* pneumonia or lymphoma. Significantly high titers can result in agglutination of red cells in the extremities called Raynaud's phenomenon (acrocyanosis).

137.

A. The red blood cell distribution width (RDW) is an index of red cell size variation or anisocytosis. The RDW will be high when a heterogeneous cell population consisting of red cells with varying sizes is present (sickle cell anemia with compensation). The RDW is low when a homogeneous or single population of red cells is present that are of uniform size (thalassemia minor, anemia of chronic disease).

138.

D. Splenomegaly is a common finding in hemolytic anemias, because the spleen is the major site of extravascular red cell destruction. Patients with hereditary spherocytosis and hemoglobin SC disease often have enlarged spleens for this reason. Patients with beta-thalassemia major exhibit splenomegaly because of active splenic removal of red cells, but the spleen may also be a site of extramedullary erythropoiesis. Splenomegaly can also be due to extramedullary hematopoiesis in malignant disorders such as polycythemia vera or myelofibrosis. Splenomegaly would not be a characteristic finding in megaloblastic anemia.

Leukocytes

139.

C. The major function of leukocytes is defense, either by phagocytosis or by immune mechanisms. The phagocytic cells are the granulocytes and monocytes. The immune response is mediated by lymphocytes; however, monocytes play a role in immunity as antigen-presenting cells. Leukocytes may be classified according to granularity as granulocytes and nongranulocytes or divided based on nuclear segmentation as polymorphonuclears (PMNs) and mononuclears.

140.

D. Monocytes have a diameter up to 20 μm, making them the largest cells in the peripheral blood under normal conditions. Eosinophils and neutrophils have diameters of about 12 μm. The small lymphocyte is 8–9 μm in diameter, similar to the red blood cell, which has a diameter of 6–8 μm. Large lymphocytes range in size from 11 to 16 μm in diameter.

141.

A. After granulocytes are released from the bone marrow, they remain in the circulation one day or less. Their major function takes place in the tissues. They migrate through the vessel walls to reach areas of inflammation very soon after release. The life span of the granulocyte is short; however, eosinophils and basophils appear to survive longer in the tissues than neutrophils.

142.

C. Approximately 50% of the neutrophils in the peripheral blood are found in the circulating pool. This is the pool measured when a total WBC count is done. Another 50% are found adhering to vessel walls (marginal pool). These pools are in constant exchange. Emotional or physical stimuli can cause a shift of cells from the marginating pool to the circulating pool, causing a transient rise in the total WBC count. The total WBC count can double but returns to normal within several hours.

143.

A. Although some phagocytic activity has been attributed to the eosinophil, it is the segmented neutrophil and monocyte that have the greatest phagocytic activity. The neutrophil is the most important because of numbers and its ability to respond quickly, especially against bacterial pathogens. Monocytes arrive at the site of injury after the neutrophil to "clean up."

144.

B. The growth factor mainly responsible for regulating the production of granulocytes and monocytes is granulocyte/monocyte colony-stimulating factor (GM-CSF), which acts on the committed bipotential progenitor cell CFU-GM (colony-forming unit–GM). GM-CSF stimulation of granulocyte or monocyte production increases in response to need and can also affect the production of erythrocytic and megakaryocytic lineages. G-CSF induces granulocyte differentiation, and M-CSF supports monocyte differentiation. Erythropoietin (EPO) is a lineage-specific growth factor responsible for stimulating erythrocyte production, and thrombopoietin (TPO) is mainly responsible for regulating platelet production. Interleukins, particularly IL-3, influence multiple cell lines, including granulocytes and monocytes.

145.

A. The granulocyte mitotic pool contains the cells capable of division, which are the myeloblasts, promyelocytes, and myelocytes. The post-mitotic pool, or reserve, is the largest bone marrow pool and contains metamyelocytes, band and segmented forms. This pool is available for prompt release into the blood if needed (e.g., infection), and its early release is the cause of a "left shift." If released, the bone marrow mitotic pool can dramatically increase its activity to replenish this reserve (cytokine stimulation increases).

146.

B. A "shift to the left" means an increase in immature neutrophilic cells in the blood caused by bone marrow release of cells in response to infection or tissue damage. A redistribution of the blood pools because of emotional or physical stimuli is characterized by an increased WBC count without a left shift. A cell "hiatus" refers to a population of cells in which there is a gap in the normal maturation sequence. A cell hiatus is most often seen in acute leukemia, in which there are many blasts and a few mature cells but no intermediate stages.

147.

D. "Agranulocytosis" refers to an absence of granulocytes in both the peripheral blood and bone marrow. A deficiency of granulocytes is found in cases of aplastic anemia, in which deficiencies in red cells and platelets also occur. The early release of cells from the bone marrow will result in immature cells in the blood but is not referred to as agranulocytosis. Neutrophils that exhibit little or no granulation may be called hypogranular or agranular and are a sign of abnormal growth (dyspoiesis).

148.

B. Antibodies are synthesized by plasma cells, which are end-stage B lymphocytes that have transformed to plasma cells following stimulation by antigen. An end product of T cell activation is the production of cytokines (lymphokines) such as interleukins and colony-stimulating factors. T cells are surveillance cells that normally comprise the majority (about 80%) of lymphocytes in the blood. T cells regulate the immune response by helping (T helper or inducer cells) or suppressing (T suppressor cells) the synthesis of antibody by plasma cells.

149.

C. Absolute values for cell types are obtained by multiplying the percentage of the cell type by the total number of cells. In this case, $4000/mm^3 \times 0.65 = 2600/\mu L$ or $2.6 \times 10^9/L$. Although reference ranges vary, the normal absolute count for lymphocytes is from 1.0 to $4.0 \times 10^9/L$ and the normal percentage of lymphocytes is 20–44%. In this case, there is a relative lymphocytosis (increase in percentage), but the absolute lymphocyte value is normal. Percentages can be misleading, so the absolute number of a particular cell type should always be evaluated.

150.

B. Auer rods are seen in the cytoplasm of malignant cells, most often myeloblasts, and are composed of fused primary (nonspecific, azurophilic) granules. Hypersegmented neutrophils have five lobes or more and are associated with vitamin B_{12} or folate deficiency. Toxic granules are primary granules with altered staining characteristics that stain in late-stage neutrophils due to toxicity. Döhle bodies are agranular patches of RNA present in neutrophil cytoplasm and associated with toxic states.

151.

D. The total white blood cell count reference ranges for males and females are equivalent. WBC counts do change with age, being higher in newborns and children than in adults. Any change from basal conditions, such as exercise or emotional stress, will cause a transient leukocytosis due to a redistribution of blood pools. WBC values are lower in the morning and higher in the afternoon (diurnal variation).

152.

C. Neutropenia is associated with a risk of infection. The degree of neutropenia correlates with the infection risk from high susceptibility ($<1.0 \times 10^9/L$) to great risk ($<0.5 \times 10^9/L$). Infection increases with the degree and duration of the neutropenia. Shortness of breath and bleeding tendencies are clinical symptoms associated with severe anemia and thrombocytopenia, respectively.

153.

B. Basophils and tissue mast cells have receptors for IgE and complement components, which trigger degranulation when appropriate antigens are present and are responsible for severe hypersensitivity reactions (anaphylaxis). Basophils and tissue mast cells have morphologic similarities but represent distinct cell types. Basophils possess water-soluble granules that contain, among other substances, heparin and histamine (a vasodilator and smooth muscle contractor). Basophils have a segmented nucleus, and the granules, although often scanty, overlie the nucleus. The mast cell has a single round nucleus, contains many more granules than the basophil, and can be found in the bone marrow.

154.

C. The last stage in the granulocytic series that divides is the myelocyte. Cells before and including this stage constitute the bone marrow mitotic pool and undergo multiple cellular divisions. Nuclear chromatin progressively clumps and nucleoli are no longer present in the nondividing metamyelocyte stage that follows the myelocyte.

155.

C. The precursor cell that can first be recognized as granulocytic is the myeloblast and has no granules. Primary or nonspecific granule production begins and ends during the promyelocyte stage. The granules are distributed between daughter cells as mitotic divisions occur. Secondary or specific granule production begins with the myelocyte stage and continues during succeeding cell stages with the synthesis of products specific to the function of the particular granulocyte (neutrophil, eosinophil, or basophil).

156.

B. Eosinophils lack lysozyme, which is present in neutrophils and monocytes, and contain a distinctive peroxidase that differs biochemically from the myeloperoxidase of neutrophils and monocytes. Major basic protein is a component of the granules and is very important to the ability of eosinophils to control parasites. In addition, eosinophils play a role in modifying the allergic reactions caused by degranulation of basophils. Basophils release eosinophil chemotactic factor of anaphylaxis (ECF-A), which calls eosinophils to the site.

157.

A. Primary granules, which appear in the promyelocyte stage, may be called azurophilic or nonspecific granules. Specific or secondary granules (neutrophilic, eosinophilic, basophilic) appear in the myelocyte stage. Primary granules contain hydrolytic enzymes (e.g., myeloperoxidase, lysozyme, acid phosphatase) and are coated with a phospholipid membrane. Lactoferrin is a component of neutrophil granules. Primary granules are visible in the myelocyte stage, but in later stage cells the primary granules, although present, are less visible by light microscopy under normal conditions.

158.

A. The presence of toxic granules, Döhle bodies, and/or vacuoles in the cytoplasm of neutrophils (segmented, band, metamyelocyte, and myelocyte stages) is indicative of a neutrophilic response to inflammation. The changes observed in the "toxic" neutrophil may occur in patients with severe burns, some malignancies, exposure to toxic drugs and chemicals, and acute infection (most often bacterial). A Barr body is a "drumstick"-shaped body of nuclear material found in the neutrophils of females that represents the inactive X chromosome and is of no significance. Auer rods are seen in malignant myeloid cells, usually blasts. Hypersegmented neutrophils are associated with megaloblastic anemias but may be seen in long-term chronic infections. Pyknotic cells and vacuoles may be seen in overwhelming sepsis or in a degenerating blood specimen. Russell bodies are globular inclusions found in plasma cells that are composed of immunoglobulin.

159.

A. Diapedesis is the movement of cells (usually referring to neutrophils) from the blood stream into the tissues by squeezing through endothelial cells of the vessel wall. Chemotaxis is the movement of cells directed by chemotactic stimuli such as bacterial products, complement components, or injured tissue. Opsonization is the coating of an organism or foreign particle by IgG or complement for recognition and phagocytosis by neutrophils or monocytes. The ingestion of red cells, often coated with IgG or complement, is called erythrophagocytosis. Margination is the attachment of neutrophils to the endothelial lining of the blood vessels.

160.

D. Basophil granules contain histamine, a potent vasodilator and smooth muscle contractor, that is responsible for the systemic effects seen in immediate hypersensitivity reactions (type I), which are also termed anaphylaxis. Degranulation occurs when basophils are coated with an IgE type of antibody that recognizes a specific allergen, such as bee venom, certain plant pollens, or latex. The resulting anaphylactic shock can be life threatening.

161.

B. Morphologic criteria such as cell size, nuclear shape, and chromatin pattern or cytoplasmic granularity cannot be used to identify lymphocyte subtypes. Monoclonal antibodies (CD surface markers) to specific surface and cytoplasmic antigens can distinguish lymphocyte subpopulations and identify the development stage. For example, blood lymphocytes that are B cells express CD19 and CD20 markers, T cells express CD2 and CD3 (and either CD4 or CD8 markers), and NK cells express CD56. Natural killer cells often exhibit large granular lymphocyte morphology (LGLs).

162.

B. Plasma cells are the mature end stage of the B lymphocyte, producing immunoglobulins (antibodies) in response to activation by a specific antigen (humoral immunity). The antibody produced by a single plasma cell is of one immunoglobulin type. Natural killer (NK) cells recognize and kill tumor cells or cells infected with virus through direct contact. Virocytes are reactive lymphocytes, and thymocytes are immature T cells. T lymphocytes provide cellular (cell mediated) immunity.

163.

A. A function of the eosinophil is to modify the severe allergic reactions caused by degranulation of the basophil. Neutrophils have receptors for the opsonins IgG and complement and are the most important cell in the initial defense against acute bacterial infection. Neutrophils are nonspecific phagocytes, ingesting bacteria, fungi, dead cells, etc., and they contain hydrolytic enzymes, including muramidase (lysozyme) and alkaline phosphatase. Neutrophils die in the performance of their function and are removed by macrophages.

164.

C. The nucleus in both monocytes and reactive lymphocytes can be irregular in shape, with indentations, although a monocyte nucleus often has folds and lobulations. Reactive lymphocytes characteristically have an increased amount of dark blue cytoplasm, whereas monocyte cytoplasm is usually a blue-gray color. Lymphocytes lack the many fine granules that give monocytes a typical "ground glass" appearance of the cytoplasm, but monocytes can occasionally have larger granules. Sharp indentation of the cytoplasm by adjacent red cells and an increased number of large granules are features of reactive lymphocytes. Vacuoles, although more commonly present in monocytes, can also be seen in reactive lymphocytes.

165.

B. Indentation of the nucleus (kidney shape) is the feature that characterizes the metamyelocyte stage. Specific granules begin forming in the myelocyte and persist through later stages. Cytoplasmic color is not a reliable feature, because it is variable and may not differ significantly from the myelocyte or band stage. Nucleoli are absent in metamyelocytes and may not be visible in myelocytes (they may be indistinct).

166.

A. Young children have the highest peripheral lymphocyte concentrations, ranging from 4.0 to 10.5×10^9 cells/L at 1 year of age and declining to $2.0–8.0 \times 10^9$ cells/L by 4 years of age. Lymphocyte counts decrease with age because of a decrease in lymphocyte stimulation and processing of antigens, ranging from 1.0 to 4.0×10^9 cells/L in adults. In addition to the difference in lymphocyte number in children, the normal morphology of children's lymphocytes differs from that of adults. Patient age should be considered when deciding between normal and abnormal lymphocytes.

167.

A. Early B cell precursors would be expected to express TdT, CD10, and CD34. TdT, the enzyme marker for terminal deoxynucleotidyl transferase, and the stem cell marker CD34 are present on the earliest B or T lymphoid cells. Surface immunoglobulin (SIgM) can only be detected on B cells at later stages of development. TdT can be used to differentiate the leukemic cells of acute lymphoid leukemia from acute myeloid leukemia. CALLA (CD10 or common ALL antigen) is a marker found in precursor types of B cell ALL.

168.

C. Acid hydrolases and the number of lysosomes increase as the blood monocyte matures into a tissue macrophage. Macrophages are widely dispersed in body tissues and organs of the reticuloendothelial (RE) system (also known as the mononuclear phagocyte system). Macrophages have receptors for IgG and complement, and they serve as phagocytes by ingesting debris and dead cells (usually neutrophils) at sites of inflammation. Macrophages act in the immune response as antigen-presenting cells by ingesting and exposing antigens for recognition by lymphocytes. Monocytes/macrophages secrete complement components and cytokines, including colony stimulating factors and interleukins.

169.

B. Antigen-*independent* lymphopoiesis occurs in primary lymphoid tissue located in the thymus and bone marrow. The formation of immunocompetent T and B cells from the lymphoid progenitor cell is influenced by environment (thymus, marrow) and several interleukins. Antigen-*dependent* lymphopoiesis occurs in secondary lymphoid tissue (spleen, lymph nodes, Peyer's patches) and begins with antigenic stimulation of immunocompetent T and B cells. Lymphocytes are the only white cells that recirculate (i.e., return to the blood from the tissues).

170.

D. Myeloperoxidase is an enzyme present in the primary granules, regardless of the phagocytic activity of the cell. The products produced during the respiratory burst, many of them short lived, are generated in response to chemotactic activation and ingestion of microbes. Generation of oxygen metabolites is necessary for microbial killing.

Leukocyte Disorders

171.

C. The Epstein-Barr virus (EBV) attaches to receptors on B lymphocytes, and the virus is incorporated into the cell. The infection generates an intense immune response of T cells directed against infected B cells. It is the activated T lymphocytes that comprise the majority of reactive lymphocytes seen in the blood of patients with infectious mononucleosis. Other B cells produce nonspecific polyclonal (heterophile) antibody in response to the EBV infection.

172.

C. The malignant cells of hairy cell leukemia (HCL) stain positive with acid phosphatase in the presence of tartaric acid; that is, hairy cells contain tartrate-resistant acid phosphatase (TRAP). Normal cells stain acid phosphatase positive, but staining is inhibited by the addition of tartrate. HCL is a chronic disorder, mainly confined to the elderly. The spleen usually shows marked enlargement, but enlarged lymph nodes are very uncommon. Hairy cells are malignant B cells, and pancytopenia is usual at presentation.

173.

A. The lymphoid cells of B cell acute lymphoblastic leukemia (FAB type L3) are morphologically identical to the malignant B cells of Burkitt lymphoma (large cells with basophilic cytoplasm and cytoplasmic lipid vacuoles). Although the site of origin is the bone marrow in B cell ALL and the tissues in Burkitt lymphoma, the World Health Organization (WHO) classifies them as the same disease entity with different clinical presentations (Burkitt leukemia/lymphoma). Both chronic lymphocytic leukemia (CLL) and small lymphocytic lymphoma (SLL) are malignant proliferations of small, mature lymphocytes, and diagnosis is based on the predominant site of involvement. Mycosis fungoides and Sézary syndrome are different stages of a cutaneous T cell lymphoma in which the skin is the early site of involvement, with subsequent progression to the bone marrow and blood.

174.

D. A leukoerythroblastic blood profile, which refers to the presence of both immature neutrophils and nucleated red cells, is most commonly associated with conditions involving bone marrow infiltration by malignant cells (leukemia, cancer) or replacement by fibrotic tissue. A neutrophilic left shift is defined as the presence of increased numbers of immature neutrophils in the blood without nucleated red cells. A regenerative left shift and a neutrophilic leukemoid reaction are characterized by varying degrees of leukocytosis and a neutrophilic left shift, most often found in response to infection. In contrast, a degenerative left shift refers to leukopenia and a left shift that may occur if marrow pools are depleted in an overwhelming infection (use exceeds the bone marrow's capacity to replace).

175.

A. "True" Pelger-Huët anomaly is a benign autosomal dominant trait characterized by hyposegmentation of the granulocytes, coarse nuclear chromatin, and normal cytoplasmic granulation. The cells have no functional defect. It is of practical importance to recognize this anomaly so that it is not confused with a shift to the left due to infection. Acquired or "pseudo" Pelger-Huët is commonly associated with myeloproliferative disorders, myelodysplastic syndromes, or drug therapy. *Pelgeroid cells* are hyposegmented and the cytoplasm is frequently hypogranular.

176.

A. Eosinophils are decreased in Cushing syndrome, in which the adrenal glands secrete large amounts of adrenocorticosteroids. Eosinophils are increased in allergic disorders, various skin diseases, and certain types of parasitic infections (especially those due to intestinal and tissue-dwelling worms). Eosinophilia is also seen in chronic myelogenous leukemia and Hodgkin lymphoma.

177.

D. Chronic granulomatous disease (CGD) is a hereditary disorder in which neutrophils are incapable of killing most ingested microbes. The disease is usually fatal because of defective generation of oxidative metabolism products, such as superoxide anions and hydrogen peroxide, which are essential for killing. Chemotaxis, lysosomes, phagocytosis, and neutrophil morphology are normal. Several variants of CGD have been described, with specific enzyme defects and different modes of inheritance. The more common type of CGD has a sex-linked inheritance pattern.

178.

B. A drug-induced megaloblastic blood profile with macrocytic ovalocytes and hypersegmented neutrophils is shown in Color Plate 11■. This is a common finding in patients receiving antifolate chemotherapeutic drugs such as methotrexate. Recombinant erythropoietin is associated with a reticulocyte response and used to treat a variety of conditions, such as renal disease, anemia of chronic disease, or anemia caused by chemotherapy. Chloramphenicol is an antibiotic with a known association for aplasia due to marrow suppression.

179.

C. Primary or essential thrombocythemia (ET) is a chronic myeloproliferative disorder in which the main cell type affected is the platelet. An extremely high number of platelets are produced, but abnormal platelet function leads to both bleeding and clotting problems. The bone marrow shows megakaryocytic hyperplasia. The hemoglobin value and platelet count are increased in polycythemia vera, and CML is characterized by a high WBC count. Malignant thrombocythemia must be differentiated from a reactive thrombocytosis seen in patients with infection or following surgery. In reactive causes, the platelet count is rarely over 1 million $\times 10^9$/L, platelet function is normal, and thrombocytosis is transient.

180.

B. The bone marrow is progressively replaced by fibrotic tissue in myelofibrosis, a chronic myeloproliferative disorder. Attempts to aspirate bone marrow usually result in a "dry tap." A biopsy stain demonstrates increased fibrosis (fibroblasts are thought to be stimulated by megakaryocytes). The presence of teardrop-shaped red blood cells is an important feature of myelofibrosis. In addition, abnormal platelets, a leukoerythroblastic blood profile and myeloid metaplasia in the spleen and liver are often associated with this disease. A high LAP score (reference range 13–160) and increased RBC mass are found in polycythemia vera, but the LAP score is low in chronic myelogenous leukemia.

181.

B. A striking lymphocytosis may be seen in children with pertussis, but normal lymphocytes, rather than reactive lymphocytes, are present. A relative and/or absolute lymphocytosis with reactive lymphocytes in various stages of activation, as seen in Color Plate 12■, is characteristic of infection caused by Epstein-Barr virus (EBV), cytomegalovirus (CMV), and toxoplasmosis. A positive heterophile antibody test can help distinguish infectious mononucleosis caused by EBV from conditions with a similar blood picture. Epstein-Barr virus is also linked to Burkitt and Hodgkin lymphomas.

182.

A. B cell chronic lymphocytic leukemia (CLL) is by far the most common type found in the United States. Immune dysfunction because of hypogammaglobulinemia occurs in later stages of the disease, as does thrombocytopenia. Development of warm autoimmune hemolytic anemia is a frequent occurrence in patients with CLL. Treatment for B cell CLL is conservative and aimed at controlling symptoms. T cell CLL is a rare and is a more aggressive disease.

183.

D. Waldenström macroglobulinemia is caused by a proliferation of transitional B lymphocytes (lymphoplasmacytic or plasmacytoid lymphs) that secrete high amounts of monoclonal IgM. Because IgM is a macroglobulin, blood hyperviscosity is the cause of many of the symptoms found in this disease (bleeding and visual impairment). Plasmapheresis can reduce the IgM protein concentration. Hepatosplenomegaly is common in Waldenström disease (rather than bone lesions).

184.

C. The elevated WBC count and toxic neutrophils seen in Color Plate 13■ indicate an extreme response to severe infection (bacterial septicemia, fungal) or treatment with recombinant myeloid growth factors. GM-CSF and G-CSF are used to increase cells for peripheral stem cell transplant and reduce infection in patients after high-dose chemotherapy or during transplant. A leukemoid reaction is one that mimics the type of blood picture seen in leukemia. It is associated with extremely high leukocyte counts (often greater than 50×10^9 cells/L) and is usually found in severe infection. The most common type of leukemoid reaction is neutrophilic, but lymphocytic leukemoid reactions also occur. HIV infection is associated with leukopenia and lymphocytopenia.

185.

B. The blood shown in Color Plate 14■ is from an elderly patient with chronic lymphocytic leukemia (CLL), which is characterized by an absolute lymphocytosis and a predominance of small, mature lymphocytes with hyperclumped nuclear chromatin. Elevated leukocyte counts are usual, as are fragile, smudged lymphocytes. Acute lymphoblastic leukemia (ALL) typically occurs in children and is characterized by immature lymphoid cells. Plasmacytoid lymphocytes and red cell rouleaux may be found in the blood of individuals with Waldenström disease. Viral infections are associated with a lymphocytosis and the presence of reactive lymphocytes that are heterogeneous in morphology. Reactive lymphocytes exhibit a variety of forms with regard to size and cytoplasmic staining intensity as compared to the homogeneous cell populations present in malignant disorders such as CLL and ALL.

186.

D. Progression to acute leukemia is a very unlikely event for patients with chronic lymphocytic leukemia, even though there is no cure. Patients with chronic myelogenous leukemia typically progress to "blast crisis," most often of myeloid type, unless treated with imatinib mesylate (Gleevec®) in the chronic phase. Refractory anemia with excess blasts (RAEB) is the most likely type of myelodysplastic syndrome to develop acute myelogenous leukemia. Refractory anemia with ringed sideroblasts (RARS) is "preleukemic" but fairly stable.

187.

A. Gaucher disease is a lipid storage disorder in which there is an accumulation of glucocerebroside in the macrophages because of a genetic lack of glucocerebrosidase, an enzyme required for normal lipid metabolism. Gaucher cells are found in the liver, spleen, and bone marrow. Niemann-Pick disease is caused by a deficiency of sphingomyelinase in which "foamy" macrophages, called Niemann-Pick cells, are filled with sphingomyelin. Normal macrophages may contain iron and other cellular debris.

188.

D. The presence of Reed-Sternberg cells is the diagnostic feature of Hodgkin disease (lymphoma). The Reed-Sternberg giant cell is usually binucleated, and each lobe has a prominent nucleolus. Studies suggest that this neoplastic cell is of B cell lineage. It is not found in the blood but only in the tissues. Circulating T cells with a convoluted nucleus describe the Sézary cells seen in Sézary syndrome, the leukemic phase of mycosis fungoides. A monoclonal population of large lymphoid cells or immature B cells with nuclear clefts is most descriptive of lymphoma cells, present in certain types of peripheralized non-Hodgkin lymphoma, that have spread from the tissues to the bone marrow and blood.

189.

B. The overall reaction of the body to tissue injury or invasion by an infectious agent is known as inflammation. This response brings leukocytes to the site of infection or tissue damage and is associated with activation of inflammatory mediators, including cytokines (IL-1), molecules released by cells (histamine), and by-products of plasma enzyme systems (complement, kinins, fibrin). The plasma concentration of positive acute-phase reactants (APRs), such as C-reactive protein (CRP) and fibrinogen, can increase dramatically in response to inflammation, and levels of albumin and transferrin will fall (negative APRs). The erythrocyte sedimentation rate (ESR) will be elevated, primarily because of the rise in fibrinogen.

190.

C. The abnormal cells found in acute promyelocytic leukemia (FAB type M3) contain large numbers of azurophilic granules. These granules contain procoagulants that on release hyperactivate coagulation, resulting in disseminated intravascular coagulation. Although other acute leukemias may trigger DIC, M3 is the one most frequently associated with this life-threatening bleeding complication. If DIC is resolved, many patients with acute promyelocytic leukemia respond favorably to therapy with retinoic acid, which causes maturation of the malignant promyelocytes. The presence of t(15;17) has diagnostic and prognostic significance, and acute promyelocytic leukemia is classified with "acute myeloid leukemias with recurrent cytogenetic translocations" by the World Health Organization (WHO). Acute myeloblastic leukemia with t(8;21) is also included in this WHO category (correlates with FAB type M2).

191.

D. Although a hallmark of acute lymphoblastic leukemias (ALL), lymphadenopathy is not associated with acute myelogenous leukemias. ALL is also more likely to have central nervous system involvement, and the CNS is a potential site of relapse. Hepatomegaly and splenomegaly are associated with both types of acute leukemia, as well as with the presence of anemia, neutropenia, and thrombocytopenia. Common presenting symptoms are fatigue, infection, or bleeding. If untreated, both acute myelogenous and lymphoblastic leukemias have a rapidly fatal course.

192.

B. The blast cells shown in Color Plate 15■ are from a child with CALLA positive, precursor B acute lymphoblastic leukemia. The malignant cells would be expected to express CD10, the common ALL antigen marker; the B cell lineage marker CD19; and TdT (terminal deoxynucleotidyl transferase), a marker on early lymphoid cells. Precursor T acute lymphoblastic leukemia would express TdT and the T cell markers CD2 and CD7. CD13 and CD33 are myeloid markers, and CD14 is a marker for monocytic cells.

193.

A. The "packed" bone marrow with predominantly immature blast cells and few normal precursor cells, as seen in Color Plate 16■, is most indicative of a patient with acute leukemia. Although chronic leukemias usually have a hypercellular marrow, the malignant cells are more mature or differentiated (i.e., able to mature beyond the blast stage). Myelodysplastic syndromes are associated with a hypercellular bone marrow, but the marrow blast percent is less than 20% (using WHO criteria). Aplastic anemia is characterized by a hypocellular bone marrow with few cells.

194.

C. The secretion of large amounts of monoclonal IgG or other immunoglobulin light chains by a malignant clone of plasma cells produces a characteristic M spike on serum and urine protein electrophoresis. In some cases, only the light chains are produced in excess. Because the light chains are easily cleared by the kidneys, they may appear only in the urine (Bence-Jones proteinuria). Renal impairment in multiple myeloma is associated with the toxic effects of filtered light chains. High levels of serum beta microglobulin correlate with the myeloma tumor burden. Cryoglobulins are proteins that precipitate in the cold and may be seen in multiple myeloma and Waldenström macroglobulinemia.

195.

C. Multiple myeloma is a malignant lymphoproliferative disorder characterized by a clonal proliferation of plasma cells and multiple bone tumors. Myeloproliferative disorders are characterized by a proliferation of bone marrow cells (granulocytic, monocytic, erythrocytic, megakaryocytic), with usually one cell type primarily affected. For example, the main cell type affected in polycythemia vera is the erythrocyte, and the platelet is mainly affected in essential thrombocythemia. Transformation among the myeloproliferative disorders is frequent.

196.

B. The Philadelphia chromosome, t(9;22), is detected in almost all cases of CML (depends on detection method) and results in a mutated BCR/ABL fusion gene. The resulting fusion protein causes increased tyrosine kinase activity, which promotes cell proliferation. Imatinib mesylate (Gleevec®) is a therapeutic agent that targets the molecular defect by blocking tyrosine kinase activity and is now a first-line drug used in the chronic phase of CML. The t(15;17) that is diagnostic of promyelocytic leukemia results in a PML/RARA (retinoic acid receptor alpha) fusion gene that blocks maturation. Many PML patients respond to retinoic acid therapy, which induces promyelocyte differentiation. Nearly all cases of Burkitt lymphoma have t(8;14), which is a translocation of the MYC gene from chromosome 8 to the Ig heavy chain (IgH) region on chromosome 14. JAK2 (Janus kinase) is a point mutation in a gene regulating cell proliferation, and it is present in over 90% of polycythemia vera cases and approximately 50% of those with essential thrombocythemia and myelofibrosis. Detection of cytogenetic and molecular mutations has diagnostic and prognostic significance and is an important tool in monitoring response to treatment.

197.

B. The French-American-British (FAB) classification of acute leukemias, myeloproliferative disorders, and myelodysplastic diseases was originally based on cellular morphology and cytochemistry (immunophenotyping was later added). Using FAB criteria, acute leukemia was defined as greater than 30% bone marrow blasts. The diagnostic criteria used by the World Health Organization (WHO) includes morphologic, cytochemical, immunologic, cytogenetic, and molecular features, as well as clinical findings, to better characterize all hematologic malignancies (myeloid and lymphoid) and predict disease course. The WHO classification defines acute leukemia as the presence of 20% or more bone marrow blasts and includes diagnostic categories with recurrent cytogenetic abnormalities. According to the WHO classification, lymphoid disorders are grouped into B cell, T/NK cell, and Hodgkin lymphoma. Further division of the B and T cell neoplasms considers site of involvement and precursor cell versus mature cell conditions.

198.

A. The blood profile of both chronic myelogenous leukemia (CML) and a neutrophilic leukemoid reaction is characterized by extremely high leukocyte counts with immature neutrophils. Splenomegaly is a manifestation of the malignant disease process and associated with CML rather than a leukemoid reaction. The presence of toxic granules and Döhle bodies would be typical of a leukemoid reaction caused by a severe bacterial infection. The LAP score is low in CML and high in a neutrophilic leukemoid reaction.

199.

C. In Color Plate 17■, the malignant blast cell contains an Auer rod, composed of fused primary granules, which stains positive with both myeloperoxidase and Sudan black B. Auer rods are not seen in lymphoblasts, and their presence can be diagnostic of acute myelogenous leukemia, such as acute myeloblastic leukemia (FAB types M1 and M2) or acute myelomonocytic leukemia (FAB type M4). Multiple Auer rods may be seen in acute promyelocytic leukemia (FAB type M3). Auer rods stain negatively with LAP, which detects the enzyme alkaline phosphatase in neutrophil granules.

200.

D. At presentation, patients with chronic leukemia (e.g., CLL or CML) consistently have elevated leukocyte counts, whereas individuals with acute leukemia may present with low, normal, or high leukocyte counts. The hallmark findings of anemia, thrombocytopenia, and neutropenia are often found in patients with acute leukemia at the time of diagnosis and are due to replacement of normal marrow hematopoietic cells by blasts. Patients with chronic leukemia may have few symptoms at onset, with anemia and thrombocytopenia developing during progression of the disease.

201.

D. Leukocyte alkaline phosphatase (LAP) scores are usually low in patients with chronic myelogenous leukemia (CML). The LAP reflects alkaline phosphatase activity in neutrophils, and the score is usually elevated in conditions where neutrophils are activated and/or increased in number, such as late pregnancy, bacterial infection, and polycythemia vera. The primary use of the LAP is to distinguish between the malignant cells of CML and a severe bacterial infection (leukemoid reaction). It may also be used to distinguish between CML and other chronic myeloproliferative disorders such as polycythemia vera. The LAP may be called NAP (neutrophil alkaline phosphatase) stain.

202.

C. Acute viral hepatitis is associated with lymphocytosis. The major causes of neutrophilia are bacterial infection, neoplastic tumors, and inflammatory responses to tissue injury. "Toxic" neutrophils may be present (toxic granulation, Döhle bodies, vacuolization). Infection with organisms other than bacteria (fungi, some parasites, certain viruses) may also cause neutrophilia.

203.

D. Monocytes must be distinguished from reactive lymphocytes, which are the characteristic feature of infectious mononucleosis. Monocytosis occurring in the recovery stage of acute infections is considered a favorable sign. An increase in monocytes is associated with collagen disorders (e.g., rheumatoid arthritis), tuberculosis, and malignant conditions such as myelodysplastic syndromes and monocytic leukemias.

204.

B. The periodic acid–Schiff (PAS) stain can be used to detect intracellular glycogen deposits in the lymphoblasts of acute lymphoblastic leukemia (ALL), in which coarse clumps of PAS positive material may be observed. Myeloblasts and monoblasts usually show a faint staining reaction. The immunophenotype has a much greater diagnostic value for ALL than the cytochemical stain results. The PAS may also be used to distinguish the malignant erythroid precursors of acute erythroleukemia, which show strong PAS positivity, from normal erythrocytic cells that stain negative.

205.

D. Myelodysplastic syndromes (MDSs) are characterized by a hypercellular bone marrow and up to 20% marrow blasts that distinguish MDS from acute leukemia (using WHO criteria). The blood and bone marrow blast percentages differ, and the risk of transformation to acute leukemia varies with the types of MDSs. These disorders are characterized by one or more peripheral blood cytopenias along with features of abnormal growth (dyspoiesis) in the bone marrow. A consistent feature in all types of myelodysplasia is unexplained and refractory anemia. Abnormalities may be morphologic and/or functional. Criteria that help define the types of myelodysplastic syndromes include megaloblastoid maturation of erythroid precursors, presence of multinucleated red cells, ringed sideroblasts, hypogranular and/or hyposegmented neutrophils, monocytosis, abnormal platelet morphology, circulating micromegakaryocytes, and degree of dyspoiesis.

206.

A. Naphthol AS-D chloroacetate esterase (specific) reacts strongly in granulocytic cells, and alpha-naphthyl acetate esterase (nonspecific) stains positively in monocytic cells. The esterase stains are used to distinguish between subtypes of acute myelogenous leukemia. The cells of acute myeloblastic leukemia (FAB types M1 and M2) will stain positive with specific esterase and negative with nonspecific esterase. The cells of acute monocytic leukemia (FAB type M5) will stain positive with nonspecific esterase and negative with specific esterase. The cells of acute myelomonocytic leukemia (FAB type M4) will show positivity with both specific and nonspecific esterase. Stain results are correlated with cell morphology, immunophenotype, and karyotype for diagnosis.

207.

A. May-Hegglin anomaly is an autosomal dominant disorder in which large blue cytoplasmic structures that resemble Döhle bodies are found in the granulocytes and possibly the monocytes. Leukocytes are normal in function. Platelets are decreased in number and abnormally large. About one-third of patients have mild to severe bleeding problems because of abnormal platelet function.

208.

D. Alder-Reilly anomaly is a hereditary autosomal recessive disorder caused by a deficiency of enzymes involved in the metabolism of mucopolysaccharides. Partially degraded mucopolysaccharides accumulate in various tissues, organs, and the leukocytes that are characterized by the presence of large azurophilic granules resembling toxic granulation. The inclusions do not affect leukocyte function and are referred to as Alder-Reilly bodies. The anomaly is often associated with facial and skeletal abnormalities, such as those seen in Hunter syndrome and Hurler syndrome. Lysosomal fusion with impaired degranulation is the defect in Chédiak-Higashi syndrome and is associated with early death due to abnormal leukocyte function.

209.

C. Serum and urine protein electrophoresis detects the presence of an M spike, the first essential step in establishing the disorder as a monoclonal gammopathy such as multiple myeloma or Waldenström disease. This can be followed by immunoelectrophoresis to determine the class of immunoglobulin or chain type. Immunologic markers, cytochemical stains, and/or cytogenetics are used in conjunction with cell morphology to diagnose malignant conditions.

210.

C. The prognosis is poor for patients with stage IV Hodgkin disease, in which there is widespread disease including bone marrow involvement. Stage I or II Hodgkin disease has a very good prognosis for cure. The clinical course and treatment varies with the extent of disease and morphologic subtype (Rye classification). The peak incidence for Hodgkin lymphoma occurs in young adults (late twenties). Men have a 50% higher incidence of the disease than women. The CRP level and ESR are increased during active disease and can be used to monitor remission status.

211.

C. The acute leukemia indicated by these results is acute myelomonocytic leukemia (AMML), which has both granulocytic and monocytic features. Note the monocytic characteristics of the blast cells in Color Plate 18■. CD14 is a monocytic marker and CD33 is a marker for primitive myeloid cells. The SBB shows positive staining in both granulocytic and monocytic cells, the specific esterase stains positive in granulocytic cells, and the nonspecific esterase is positive in monocytic cells.

212.

B. Acute lymphoblastic leukemia (ALL) of children has the best prognosis. Other favorable factors include children between ages 3 and 7, mild to moderate increases in the peripheral white blood count prior to treatment, and precursor B ALL, CALLA positive type (rather than T cell ALL). Certain cytogenetic and molecular abnormalities are also associated with a better prognosis. Acute leukemia in adults is less favorable because remissions are shorter and more difficult to induce, especially in those over 70 years of age. Prognosis is poor in adults with ALL.

213.

B. The test that would be the most beneficial for the diagnosis of Hodgkin lymphoma is a lymph node biopsy. Lymphadenopathy is the major clinical presentation of Hodgkin disease, and early stages do not have bone marrow involvement. A skin biopsy would be indicated for diagnosis of mycosis fungoides, a T cell lymphoma of the skin. A bone marrow exam and spinal tap are important to the diagnosis of acute leukemias.

214.

A. A hypercellular bone marrow and high M:E ratio are most characteristic of the excessive granulocyte production that occurs in chronic myelogenous leukemia. Polycythemia vera typically has a hypercellular marrow with panhyperplasia and a normal or low M:E ratio. Beta-thalassemia major is a severe hemolytic anemia in which RBC hyperplasia of the marrow is pronounced and a low M:E ratio is usual. Aplastic anemia is associated with a hypocellular marrow with a reduction of all cell lines and normal M:E ratio.

215.

B. Refractory anemia with ringed sideroblasts (RARS) is a myelodysplastic syndrome (MDS) that may also be referred to as primary or idiopathic sideroblastic anemia. The main findings that characterize this type of MDS include refractory anemia with a heterogeneous population of red cells, a hypercellular bone marrow with $<5\%$ blasts, and the presence of $>15\%$ ringed sideroblasts in the marrow (demonstrated with Prussian blue stain). RA and RARS are the least likely MDS types to progress to acute myelogenous leukemia.

216.

C. Recent strenuous exercise or other physical and emotional stimuli cause a transient increase in the leukocyte count. This is due to a redistribution of the blood pools. Marrow injury to stem cells or marrow replacement by malignant cells causes neutropenia of varying degrees. Neutropenia may be caused by immune mechanisms (antibodies) or an overactive spleen that sequesters neutrophils. Chemotherapeutic drugs also suppress bone marrow production of neutrophils.

217.

C. Primary polycythemia (vera) is a malignant myeloproliferative disorder characterized by autonomous marrow production of erythrocytes in the presence of low erythropoietin levels. Usual findings include increased RBC mass with elevated hemoglobin values and variable degrees of leukocytosis and thrombocytosis (pancytosis). Splenomegaly, a high LAP score, thrombotic tendencies, and problems caused by blood viscosity are typical. Phlebotomy is done to reduce red cell mass.

218.

D. Basophilia (and eosinophilia) is a typical finding in patients with chronic myelogenous leukemia (CML). A progressive increase in basophil number suggests transformation of the disease to a more accelerated phase. Myeloproliferative disorders such as CML, polycythemia vera, or AML are often associated with peripheral basophilia, which is not a feature of lymphoproliferative disorders such as acute lymphoblastic leukemia, hairy cell leukemia, or plasma cell leukemia.

219.

C. More than 50% of the marrow cells are erythroid in acute erythroleukemia (FAB type M6). Giant erythroid precursors, bizarre and multinucleated red cells, and increased myeloblasts are found in the marrow and may appear in the blood. Acute erythroid leukemia is rare, and the disease typically evolves into acute myeloblastic leukemia (FAB types M1 or M2).

220.

D. Plasma cell myeloma is a clonal disease involving malignant end-stage B cells, in which overproduction of immunoglobulin is a hallmark and the presence of red cell rouleaux is a characteristic finding on the blood smear, as shown in Color Plate 20■. Excessive amounts of a monoclonal immunoglobulin result in the deposition of proteins on circulating red cells that causes red cell "coining." The erythrocyte sedimentation rate is extremely elevated because of spontaneous rouleaux formation. Multiple myeloma is characterized by bone pain and spontaneous bone fractures caused by tumors of plasma cells. Bone destruction leads to elevated calcium levels, and renal impairment can result from damage by excess light chains. Plasma cells progressively crowd out normal bone marrow precursors and may be found in the blood circulation in advanced disease. Treatment with thalidomide has improved survival.

221.

C. The production of hematopoietic cells in sites outside of the bone marrow can be referred to as myeloid metaplasia or extramedullary hematopoiesis. Hematopoiesis, with the exception of lymphopoiesis, is normally confined to the bone marrow during postnatal life. Production of erythroid, myeloid, and megakaryocytic elements can be established in the liver and spleen, similar to that which occurs during embryonic development. Myeloid metaplasia is frequently associated with myelofibrosis, a condition in which the marrow is gradually replaced by fibrotic tissue.

222.

A. Prominent lymphadenopathy is the most consistent finding in non-Hodgkin types of lymphoma at presentation, but lymphoma may also arise in the spleen, liver, or GI tract (abdominal tumor). Lymphomas begin as localized tumors involving lymphoid tissue that spread to the bone marrow and blood (depends on type). The malignant lymphoid cells are immunologically classified as B cell (most common) or T/NK cell. Clonality can also be established by demonstrating gene rearrangements via molecular analysis. Some common subtypes of non-Hodgkin lymphoma are small lymphocytic, Burkitt, follicular, and mantle cell lymphomas. Leukemias are initially systemic disorders primarily involving the bone marrow and blood at onset. Bone lesions are associated with multiple myeloma.

Methodology

223.

B. The standard assay for hemoglobin utilizes potassium ferricyanide. This solution, formerly called Drabkin's reagent, is now called cyanmethemoglobin (HiCN) reagent. The ferricyanide oxidizes hemoglobin iron from ferrous (Fe^{2+}) to ferric (Fe^{3+}), and the potassium cyanide stabilizes the pigment as cyanmethemoglobin for spectrophotometric measurement.

224.

B. The band containing hemoglobin A_2 is the slowest-migrating, staying closest to the cathode. The band containing hemoglobin A has a net negative charge at an alkaline pH, and it moves the farthest toward the anode. An adult patient without a hemoglobinopathy will have only these two bands appearing on a cellulose acetate electrophoresis.

225.

C. Hemoglobins S, D, and G all migrate to the same location on the hemoglobin electrophoresis gel at an alkaline pH. However, because hemoglobins D and G are nonsickling hemoglobins, tests based on sickle formation under decreased oxygen tension will have *negative results*. These hemoglobins can be further differentiated by their movement on agar gel at an acid pH, whereas hemoglobins D and G will migrate with hemoglobin A, not with hemoglobin S.

226.

C. When the sample is deoxygenated, reduced hemoglobin S polymerizes, resulting in a cloudy solution. A false negative can be obtained if the quantity of hemoglobin S is below the sensitivity of the method, which can be seen in newborns and anemic patients. Although this procedure is a screening test for hemoglobin S detection, it is positive in the presence of any sickling hemoglobin, such as hemoglobin C_{Harlem}.

227.

A. A slanted column increases the ESR. A clotted sample, which lacks fibrinogen, causes a falsely decreased ESR. Fibrinogen is the plasma protein that most greatly affects the ESR. The EDTA tube for ESR must be at least half-full, and the test must be set up within 4 hours of draw; failure to follow these guidelines results in poikilocytosis that will inhibit rouleaux formation.

228.

C. Some patients develop EDTA-dependent platelet agglutinins caused by an IgM or IgG platelet-specific antibody. To correct for this, the sample can be redrawn in sodium citrate and rerun. The dilution factor of blood to anticoagulant in sodium citrate is 9:1. To compensate for the 10% dilutional loss of platelets, the platelet count obtained must be multiplied by 1.1 ($300 \times 10^9/L \times 1.1 = 330 \times 10^9/L$).

229.

B. Blood smears should be made within 5 hours of collection from blood anticoagulated with EDTA. Although some of the blood cells may still be normal in blood kept longer, others (especially granulocytes) may deteriorate. Vacuolation of neutrophils can appear as an artifact in blood kept past this time. The age of the blood may also affect the visual quality when the slide is stained.

230.

A. Decreasing the angle will produce a longer, thinner smear. Increasing the angle or using a smaller drop of blood will produce a shorter, thicker smear. The angle normally used for the spreader slide when making a smear is 30–45 degrees.

231.

D. One type of Romanowsky stain is the Wright's stain. It is a polychrome stain consisting of methylene blue and eosin. This combination causes multiple colors to appear on staining. Another commonly used Romanowsky stain is the Wright's-Giemsa stain. Brilliant green and neutral red are used in a supravital stain for Heinz bodies. Crystal violet and safranin are used in Gram's stain for bacteria.

232.

C. When red blood cells are stained correctly with Wright's stain, their color is pink to orange-red. They will appear bright red in the presence of an acid buffer and stain. Staining elements such as white cells, which stain with a more basic pH, will not take up the stain adequately in this instance. Inadequate washing and an alkaline stain or buffer mixture results in a smear that is excessively blue.

233.

C. The formula for calculating a reticulocyte count in percent is

$$\frac{\text{Number of reticulocytes counted}}{\text{Total number of RBCs counted}} \times 100$$

In the case described in question 233,

$$\%\text{Reticulocytes} = \frac{60}{1000} \times 100 = 6.0\%$$

Because the error in reticulocyte counts is high, it is desirable to count a larger number of cells or use a standardized counting method such as the Miller disk.

234.

C. The formula used to calculate the absolute reticulocyte count is

$$\frac{\text{Reticulocyte percent}}{100} \times \text{RBC } (10^{12}/\text{L}) \times 1000$$

Multiplication by 1000 is done to report the results in SI units of 10^9/L.

In this case, $\frac{6.0 \times 3.00}{100} \times 1000 = 180 \times 10^9/\text{L}$

$180 \times 10^3/\mu\text{L}$ is not expressed in SI units.

235.

B. As visualized in Color Plate 19■, Sudan black B is a cytochemical stain for lipids, including steroids, phospholipids, and neutral fats. It is widely used as a tool to differentiate the blasts of acute lymphoblastic leukemia (ALL) from those of acute myelogenous leukemia (AML). Blasts in ALL are SBB negative, whereas those in AML will show some degree of positivity.

236.

C. An LAP score is determined by first multiplying the number of cells found by the degree of positivity (i.e., $20 \times 1 = 20$). These numbers are then added together to obtain a final score. In this instance, $0 + 20 + 60 + 60 + 60 = 200$.

237.

C. When stained with a mixture of potassium ferricyanide and hydrochloric acid, nonheme iron stains bright blue. This is the most common stain used for storage iron. It can be used on bone marrow to identify sideroblasts, peripheral blood to identify the presence of siderocytes, or urine to perform hemosiderin testing.

238.

C. Pappenheimer bodies are iron deposits associated with mitochondria, and they stain with both Perl's Prussian blue and Wright's stain. A cell that contains Pappenheimer bodies is called a siderocyte. Howell-Jolly bodies and basophilic stippling can be visualized with Wright's stain, whereas Heinz bodies require a supravital stain to be seen.

239.

B. Depth on a standard counting chamber is 0.10 mm. The formula to calculate volume is $V = A \times D$, where V is volume, A is area, and D is depth. When the counting chamber is used, the area may change, depending on the number of ruled squares counted, but the depth remains constant.

240.

B. The standard formula for hemacytometer counts expressed in mm^3 is

$$\frac{\text{Total number cells counted} \times \text{dilution factor}}{\text{Area counted} \times \text{depth}}$$

In this instance,

$$\frac{308 \times 20}{8\ \text{mm}^2 \times 0.10\ \text{mm}} = \frac{6160}{0.8\ \text{mm}^3} = 7700/\text{mm}^3$$

$$= 7.7 \times 10^9/\text{L}$$

241.

B. The Rule of Three states that RBC × 3 = Hgb and Hgb × 3 = Hct ±3 in error-free results. These rules apply only for normocytic, normochromic erythrocytes. One check to determine if an error has occurred is to determine the MCHC. An MCHC should be less than 37.0 g/dL in error-free results. The MCHC is calculated by dividing hemoglobin by hematocrit and multiplying by 100. In instance (B), the MCHC is 38.3 g/dL and the Rule of Three is broken. All other answers follow the Rule of Three.

242.

A. The only *true* cause of a high MCHC is the presence of spherocytes, as may be seen in hereditary spherocytosis. Because the MCHC is a calculation using the hemoglobin and hematocrit, anything causing those parameters to be wrong will affect the MCHC. The occurrence of a falsely high MCHC is much more common than the presence of spherocytes, and specimen troubleshooting procedures must be undertaken to obtain reportable results.

243.

C. Blood cells are nonconductors of electrical current; they create a resistance/impedance of current in a diluent solution that is conductive. When the suspension is forced through a small aperture, the current flow is interrupted by the presence of the cells. A pulse is generated. The number of pulses generated is proportional to the number of particles present, and the size of the pulse generated is proportional to the size of the cell.

244.

C. Using ± 2 *s*, 95% confidence limits are achieved; 95% confidence limits predict a range that values should fall within 95% of the time. For example, if a WBC count is 12.0×10^9/L with 2 *s* of ± 0.5, then a succeeding count must be less than 11.5×10^9/L or greater than 12.5×10^9/L to be considered significantly different.

245.

B. Side angle scatter of a laser beam increases with granularity of the cytoplasm. Forward angle scatter is used to determine relative size. The number of signals is proportional to the number of cells. The presence of specific antigens in the cytoplasm or on the cell surface is determined by immunofluorescence after reactions with appropriate antibodies.

246.

A. An impedance counter cannot differentiate between the nucleus of a white blood cell and the nucleus of an nRBC. Both will be counted as WBCs. The presence of 5 or more nRBCs/100 WBCs can result in a falsely elevated white blood cell count, and a correction must be made as follows:

$$\text{Corrected WBC count} = \text{Observed count} \times \frac{100}{100 + \#\,\text{nRBCs/100 WBCs}}$$

247.

D. Hemoglobin A_2 values up to 3.5% are considered normal. Values between 3.5 and 8.0% are indicative of beta-thalassemia minor. Hemoglobins C, E, and O have net electrical charges similar to hemoglobin A_2. They elute off with hemoglobin A_2 using anion exchange (column) chromatography, causing an invalid hemoglobin A_2 result. If the hemoglobin A_2 quantification using column chromatography yields a result greater than 8.0%, one of these interfering hemoglobins should be considered.

248.

A. Any condition with spherocytes can cause an increased osmotic fragility, dependent on the number of spherocytes present. Spherocytes are seen in hereditary spherocytosis, immune hemolytic anemias, and severe burns. Target cells, associated with thalassemias and hemoglobinopathies, have an increased surface area-to-volume ratio and a decreased osmotic fragility.

249.

B. The solubility test for hemoglobin S is not quantitative; it is reported as positive or negative. A clotted specimen will not affect the result. A clotted specimen will falsely decrease the other tests listed: ESR due to low fibrinogen, hematocrit due to a false low RBC count, and platelets are trapped in the clot.

250.

A. Hematology reference intervals are available in many textbooks. They are influenced by patient population, instrumentation, and reagents used. Therefore, it is ideal for each laboratory to establish its own reference intervals. The reference interval excludes the upper and lower 2.5% of the values. The remaining 95% represent the reference interval.

251.

C. Standards are commercially available to generate a hemoglobin concentration curve. The absorbance of each solution is read against a reagent blank at 540 nm on a spectrophotometer. Patient blood samples and commercial control materials can be used to assess precision and other quality control parameters.

252.

A. Anything that causes an increase in absorbance will cause a hemoglobin that is read spectrophotometrically to be falsely high. It is necessary to correct for this type of error, such as making a plasma blank in the case of lipemia or icterus. WBCs are present in the hemoglobin dilution and usually do not interfere. When the WBC count is extremely high, their presence will cause cloudiness, increasing the absorbance in the hemoglobin measuring cell and resulting in a falsely high hemoglobin concentration. Excessive anticoagulant does not affect hemoglobin readings.

253.

A. When a microhematocrit is spun at 10,000–15,000 × *g* for 5 minutes, maximum packing is achieved. Spinning a longer time has no affect on the result. A tube that is not full causes RBC shrinkage and a falsely decreased hematocrit. Hemolysis damages the RBC membrane and allows for more packing. Trapped plasma is present when optimal packing is not achieved due to inadequate speed or time of centrifugation, causing a falsely high hematocrit.

254.

C. The erythrocyte sedimentation rate (ESR) measures the rate of fall of red cells through plasma. ESR increases when cells become stacked (rouleaux, as seen in Color Plate 20■). ESR decreases when cells are not normal discocytes. Larger cells (macrocytes) and fewer cells (anemia) fall faster. Plasma containing increased proteins, such as fibrinogen and globulins, promote rouleaux formation and an elevated ESR. Hemoglobin content does not affect the ESR.

255.

A. Impedance counters measure RBCs and platelets using the same dilution. To differentiate the two, sizing thresholds are used. Particles between 2 and 20 fL are counted as platelets, and particles larger than 35 fL are counted as RBCs. Small RBCs, clumped platelets, and giant platelets fall in the overlap area between platelets and RBCs, generating a warning flag. Nucleated RBCs are larger than normal RBCs and are not mistaken for platelets.

256.

C. A platelet estimate is obtained by multiplying the average number of platelets per oil immersion field (in an erythrocyte monolayer) by 20,000. The reference range for a platelet count is 150–450 × 10^9/L. Approximately 8–20 platelets per oil immersion field will represent a normal platelet concentration of approximately 160–400 × 10^9/L. This method assumes the red blood cell count is normal. If it is not, alternate platelet estimate procedures may need to be performed.

257.

D. A living cell stain using new methylene blue is performed for reticulocyte counts. Reticulocytes should not be stained for less than 5 minutes. Howell-Jolly bodies, Pappenheimer bodies, crenated cells, and refractile artifact can be mistaken for reticulocyte inclusions. Two or more particles of reticulum constitute a reticulocyte.

258.

B. All accredited laboratories are required to perform calibration with commercially available calibrators at least once every 6 months. Calibration must be checked if any major part is replaced or if optical alignment is adjusted. A calibration procedure can be verified using commercially available controls.

259.

A. An MCHC >37.0 g/dL is most likely caused by an error in measurement. In this instance, the Rule of Three shows that the RBC × 9 matches the hematocrit, but the RBC × 3 does not match the hemoglobin. The hemoglobin does not match either the RBC or hematocrit. This indicates a hemoglobin problem, and it can be corrected with a saline replacement procedure. This specimen may be lipemic or icteric. Warming the specimen is useful in troubleshooting a high MCHC due to a cold agglutinin. A microhematocrit would be indicated if the hematocrit result was invalid.

260.

A. Hemoglobin is valid on a hemolyzed specimen, because RBC lysis is the first step in the cyanmethemoglobin method. The red blood cell count depends on the presence of intact red blood cells. Red blood cell fragments caused by hemolysis may be as small as platelets and affect instruments that use sizing criteria to differentiate the two. Therefore, samples for these procedures should be re-collected.

261.

D. Heparin is recommended for osmotic fragility and red cell enzyme studies, because it results in less lysis and less membrane stress than other anticoagulants. Heparin induces platelet clumping and is unacceptable for the platelet count. Heparin is unacceptable for coagulation test procedures because it binds with antithrombin to neutralize many enzymes, especially thrombin. This would cause very long coagulation test results. EDTA is recommended for most routine hematology procedures, especially for Wright's-stained smears. Sodium citrate or EDTA can be used for sedimentation rates.

262.

A. When counting platelets, the center square (1 mm^2) is counted on each side of the hemacytometer. Platelets appear round or oval and may have dendrites. These characteristics can help distinguish them from debris, which is irregularly shaped and often refractile. White cells are not lysed; they may be counted, using a different ruled area of the hemacytometer. Platelets will be easier to count if allowed to settle for 10 minutes, because they will have settled into one plane of focus.

263.

B. "Precision" is the term used to describe the reproducibility of a method that gives closely similar results when one sample is run multiple times. An accurate method is one that gives results that are very close to the true value. Laboratories must have procedures that are both accurate and precise.

Case Histories

264.

C. WBC counts done by an impedance cell counter must be corrected when nucleated red blood cells (nRBCs) are present (see Color Plate 4■), because such instruments do not distinguish between white and red nucleated cells. This correction is done according to the following formula:

$$\text{Corrected WBC count} = \text{Observed count} \times \frac{100}{100 + \#\ \text{nRBCs per 100 WBCs}}$$

In this instance,

$$35.0 \times \frac{100}{100 + 110} = 16.7 \times 10^9/\text{L}$$

265.

A. The appearance of red cells on a differential smear may be predicted by calculating the red cell indices.

$$\text{MCV} = \frac{\text{Hct} \times 10}{\text{RBC}} = \frac{16\% \times 10}{2.50 \times 10^{12}/\text{L}} = 64.0\ \text{fL}$$

$$\text{MCH} = \frac{\text{Hgb} \times 10}{\text{RBC}} = \frac{4.5\ \text{g/dL} \times 10}{2.50 \times 10^{12}/\text{L}} = 18.0\ \text{pg}$$

$$\text{MCHC} = \frac{\text{Hgb} \times 100}{\text{Hct}} = \frac{4.5\ \text{g/dL} \times 100}{16\%} = 28.1\ \text{g/dL}\ (281\ \text{g/L})$$

The mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), and mean corpuscular hemoglobin concentration (MCHC) are all below the reference range. This indicates a cell that is small (microcytic) with a reduced hemoglobin concentration (hypochromic). These indices refer to averages and do not necessarily reflect the actual appearance of cells in which there is great diversity in size and shape.

266.

D. Children with beta-thalassemia major, also known as Cooley anemia, do not use iron effectively to make heme. This occurs because of a genetic defect that causes a decreased rate of production of structurally normal globin chains. In addition, these children receive frequent transfusions due to the severe hemolytic anemia. The result is hemochromatosis with a high serum iron and storage iron. None of the other anemias listed would elicit the bone marrow response seen by the high number of nucleated RBCs, because they are not hemolytic.

267.

D. Beta-thalassemia major is characterized by an inability to produce beta-globin chains, resulting in a decrease or complete absence of hemoglobin A. Hemoglobin F, a compensatory hemoglobin that contains two alpha- and two gamma-globin chains, is frequently the only hemoglobin present. Hemoglobin A_2 is classically increased in heterozygous beta-thalassemia, but it is variable in homozygous beta-thalassemia.

268.

B. The predominant hemoglobin present at birth is hemoglobin F, which consists of two alpha- and two gamma-globin chains. It is not until about 6 months of age that beta-chain production is at its peak. At this point, hemoglobin A (two alpha and two beta chains) replaces hemoglobin F as the predominant hemoglobin. A deficiency in the production of these chains will not be apparent until this beta-gamma switch has occurred.

269.

D. The formula for calculation of transferrin saturation is as follows:

$$\text{Transferrin saturation \%} = \frac{\text{Serum iron } (\mu g/dL) \times 100}{\text{TIBC } (\mu g/dL)}$$

In this case,

$$\text{Transferrin saturation \%} = \frac{22 \times 100}{150} = 15\%$$

Because the reference range for saturation is 20–45%, this is a low saturation.

270.

D. In the anemia of chronic disease, patients have iron but are unable to utilize it. Hepcidin, a hormone produced by the liver, plays a role in body iron regulation. Intestinal iron absorption and release of iron from macrophages both decrease in response to increased hepcidin levels. Hepcidin is a positive acute-phase reactant, so increased levels are seen in anemia of chronic disease due to inflammation. This adversely affects iron availability.

271.

A. The most common anemia among hospitalized patients is anemia of chronic disease. Patients with chronic infections, inflammatory disorders, and neoplastic disorders develop this type of anemia. The typical presentation is a normocytic, normochromic anemia, but microcytic and hypochromic anemia can develop in long-standing cases. Chronic blood loss can cause iron deficiency and microcytic/hypochromic anemia.

272.

A. Both megaloblastic anemias and some nonmegaloblastic anemias are characterized by the presence of macrocytic, normochromic red cells. Vitamin B_{12} and folic acid are coenzymes necessary for DNA synthesis. Lack of either one causes megaloblastic anemia. Maturation asynchrony is evident in both the peripheral blood and bone marrow. The bone marrow examination is done after vitamin B_{12} and folate levels because of the test's invasive nature. Vitamin B_{12} and folic acid levels are normal or increased in nonmegaloblastic anemias. Iron studies are useful in the diagnosis of microcytic/hypochromic anemias.

273.

C. The common causes of megaloblastic anemia are pernicious anemia and folic acid deficiency. Neurological symptoms are not associated with folic acid deficiency. Folic acid is a water-soluble vitamin for which there are low body stores. A diet low in green vegetables and meat products or high in alcohol can result in folate deficiency in 2–4 months. Alcohol is a folate antagonist.

274.

A. The general classification of anemia described here is megaloblastic anemia. A deficiency of vitamin B_{12} or folic acid affects DNA production. All dividing cells will show nuclear abnormalities, resulting in megaloblastic changes. In the neutrophil, as seen in Color Plate 11■, this takes the form of hypersegmentation (five lobes or more). Enlarged, fragile cells are formed, many of which die in the bone marrow. This destruction leads to increased LD, bilirubin, and iron levels. Oval macrocytes and teardrop cells are seen. Pancytopenia and inclusions are common findings. One cause of a nonmegaloblastic macrocytic anemia, which has round cells such as target cells instead of oval cells, is liver disease.

275.

B. The pancytopenia and red blood cell morphologic findings are all consistent with megaloblastic anemia. Further investigation of serum folate and vitamin B_{12} levels is warranted. Pernicious anemia (PA) is noted for neurological complications and is seen more commonly among people of British and Scandinavian ancestry. PA is caused by a lack of intrinsic factor production in the stomach, which is necessary for the absorption of vitamin B_{12}. Because there are large body stores of vitamin B_{12}, it takes from 1 to 4 years for the deficiency to manifest itself. Aplastic anemia is also associated with pancytopenia, but not the red cell morphologic changes seen in this patient.

276.

A. Alcohol is a folic acid, not a vitamin B_{12}, antagonist. Patients with pernicious anemia (PA) are incapable of absorbing vitamin B_{12} due to a lack of intrinsic factor or antibodies to intrinsic factor or parietal cells. PA is characterized by achlorhydria and atrophy of gastric parietal cells that secrete intrinsic factor. Achlorhydria is not diagnostic for PA, because it may occur in other disorders (such as severe iron deficiency), but it is confirmatory evidence of the problem. *D. latum* competes for B_{12} in the intestines.

277.

B. Because intrinsic factor is necessary for absorption of vitamin B_{12} from the ileum, intramuscular injections of vitamin B_{12} are used to treat PA. Although oral doses of folic acid will correct the megaloblastic blood profile seen in PA, the neurological symptoms will not improve. For this reason, correct diagnosis is crucial. Methotrexate is a folic acid antagonist. Because body folic acid stores are low, a deficiency can develop quickly.

278.

A. The adult red blood cell in glucose-6-phosphate dehydrogenase (G6PD) deficiency is susceptible to destruction by oxidizing drugs. This occurs because the mechanism for providing reduced glutathione, which keeps hemoglobin in the reduced state, is defective. The anti-malarial drug primaquine is one of the best-known drugs that may precipitate a hemolytic episode. Ingestion of fava beans can also elicit a hemolytic episode in some patients.

279.

C. G6PD deficiency, a sex-linked disorder, is the most common enzyme deficiency in the hexose monophosphate shunt. Most patients are asymptomatic and go through life being unaware of the deficiency unless oxidatively challenged. Pyruvate kinase, an enzyme in the Embden-Meyerhof pathway, is necessary to generate ATP. ATP is needed for red blood cell membrane maintenance. Patients with a pyruvate kinase deficiency have a chronic mild to moderate anemia.

280.

B. Reduced glutathione levels are not maintained due to a decrease in NADPH production. Methemoglobin (Fe^{3+}) accumulates and denatures in the form of Heinz bodies. Heinz bodies cause rigidity of the RBC membrane, resulting in red cell lysis. Döhle bodies are composed of RNA; Howell-Jolly bodies are composed of DNA; Pappenheimer bodies are iron deposits.

281.

C. Iron-deficiency anemia (IDA) causes a microcytic, hypochromic anemia. It is the most common anemia found in children. IDA develops quickly in children because of rapid growth with increased dietary iron requirements. Hereditary spherocytosis results in RBCs that are normal to low-normal in size, with an MCHC possibly greater than 37.0 g/dL. Folic acid deficiency causes a macrocytic/normochromic anemia. Erythroblastosis fetalis is a hemolytic disease of the newborn caused by red blood cell destruction by antibodies from the mother; such antibodies are no longer in the circulation of a 15-month-old child.

282.

C. The development of iron deficiency occurs in stages: the iron depletion stage, the iron-deficient erythropoiesis stage, and the iron-deficiency anemia stage. Iron stores are the first to disappear, so the serum ferritin level is the earliest indicator of iron-deficiency anemia. This is followed by decreased serum iron and increased TIBC. The last abnormality seen is microcytic, hypochromic red blood cells.

283.

C. In iron-deficiency anemia, red blood cell production is restricted because of lack of iron, and the reticulocyte absolute value reflects this ineffective erythropoiesis. The formula used to calculate the absolute reticulocyte count is

$$\text{Absolute reticulocytes} = \frac{\text{Reticulocytes \%}}{100} \times \text{RBC}\ (10^{12}/\text{L}) \times 1000$$

The 1000 in the calculation is to convert to SI units (10^9/L).
In this case,

$$\frac{0.2}{100} \times 2.70\ (10^{12}/\text{L}) \times 1000 = 5 \times 10^9/\text{L}$$

The reference interval for the absolute reticulocyte count is approximately $18–158 \times 10^9$/L.

284.

A. Petechiae and ecchymoses (bruises) are primary hemostasis bleeding symptoms seen in quantitative and qualitative platelet disorders. Although estimates vary, spontaneous bleeding does not usually occur until platelet numbers are less than 50×10^9/L. The malignant disorder represented in this case is noted for thrombocytopenia.

285.

A. The bone marrow blast percent indicates the presence of an acute leukemia. The triad of symptoms seen in acute leukemia is neutropenia, anemia, and thrombocytopenia. Acute lymphoblastic leukemia is the leukemia most likely to be found in this age group. Hairy cell leukemia does not present with blasts. Myelodysplastic syndrome presents with less than 20/30 (WHO/FAB, respectively) percent marrow blasts.

286.

C. Periodic acid–Schiff stains glycogen in lymphoblasts. The myeloperoxidase stain is positive in myeloid cells. Monocytes show a positive reaction to the nonspecific esterase stain. Leukocyte alkaline phosphatase is useful in the diagnosis of chronic myelogenous leukemia.

287.

A. Terminal deoxyribonucleotidyl transferase (TdT) is a nuclear enzyme (DNA polymerase) found in stem cells and precursor B and T lymphoid cells. High levels of TdT are found in 90% of ALLs. TdT has been found in up to 10% of cases of AML (FAB M0 and M1), but in lower levels than are present in ALL. This enzyme is not found in mature lymphocytes.

288.

B. There are now more than 200 recognized human leukocyte antigens, each of which has been given a CD (cluster designation) number. CDs 2, 5, and 7 are seen on T cells. CALLA, the common acute lymphoblastic leukemia antigen, is seen in early pre–B cells. Distinct CD markers have been identified for cells of both lymphoid and myeloid stem cell lineage.

289.

B. The bone marrow blast percent is high enough to indicate an acute leukemia. Sudan black B, myeloperoxidase, and specific esterase stains are positive, indicating the presence of the myeloid cell line. The nonspecific esterase stain is negative, indicating the absence of a monocytic cell line. The bone marrow blast percent is too low for FAB M1, but it is in the range for FAB M2.

290.

A. Chromosome analysis is an important diagnostic tool in clinical medicine. Nonrandom chromosome abnormalities are recognized in many forms of cancer. t(8;21) is associated with acute myelogenous leukemia FAB M2; t(15;17) is only seen in FAB M3. The Philadelphia chromosome, t(9;22), is seen in at least 90% of patients with chronic myelogenous leukemia; t(8;14) is associated with Burkitt lymphoma.

291.

B. When a diagnosis of AML or myelodysplastic syndrome is suspected, a bone marrow examination is performed. The WHO approach to the diagnosis of acute leukemia requires the presence of >20% blasts in the bone marrow; the FAB classification requires >30%. The reference interval for bone marrow blast percent is 0–2%. Myelodysplastic syndromes have increased bone marrow blast percentages, but <20% using WHO criteria and <30% using FAB criteria.

292.

B. HTLV-I is implicated in T cell leukemia and lymphoma in Japan. The Epstein-Barr virus is associated with Burkitt lymphoma in Africa. Chronic bone marrow dysfunction can be caused by exposure to radiation, drugs, and chemicals such as benzene. Myelodysplastic syndromes and myeloproliferative disorders are "preleukemic" because they have a high incidence of terminating in acute leukemia. Paroxysmal nocturnal hemoglobinuria, aplastic anemia, multiple myeloma, and lymphoma are stem cell disorders that are particularly noted for transformation into acute leukemia. Genetic susceptibility is associated with Klinefelter and Down syndromes, both of which have chromosomal abnormalities. It is likely that more than one factor is responsible for the evolution of an acute leukemia.

293.

A. The myelodysplastic syndromes (MDSs) are pluripotential stem cell disorders characterized by one or more peripheral blood cytopenias. Bone marrow examination is necessary for diagnosis. There are prominent maturation abnormalities in all three cell lines in the bone marrow. Megaloblastoid erythrocyte maturation is present that is not responsive to B_{12} or folic acid therapies. Although many of the red blood cell inclusions noted in this case can be seen in a megaloblastic anemia such as pernicious anemia, this patient has a normal vitamin B_{12} and folate level. This patient has hyposegmentation of neutrophils, whereas megaloblastic anemias present with hypersegmentation of neutrophils. Of the disorders listed, the only one associated with dyshematopoiesis of all cell lines is myelodysplastic syndrome.

294.

D. In most cases of MDS, the bone marrow is hypercellular with erythroid hyperplasia. MDS is considered a disease of the elderly. Because normal cellularity decreases with age, interpretation of cellularity must take the age of the patient into account. WHO criteria for a diagnosis of MDS are related to bone marrow blast percent, which must be <20%.

295.

D. In RAEB at least two cell lines exhibit cytopenia, and all cell lines show evidence of dyshematopoiesis. Poor granulation and pseudo-Pelger-Huët anomaly is seen. There are less than 5% blasts in the peripheral blood, and between 5 and 19% blasts in the bone marrow. Platelets exhibit poor granulation, giant forms, and the abnormal maturation stage of micromegakaryocytes. The five FAB classifications of myelodysplastic syndrome are RA, RARS, RAEB, CMML, and RAEB-t. CML is a myeloproliferative disorder, not a myelodysplastic syndrome.

296.

C. The myelodysplastic syndromes are refractory to treatment, and patients are supported using blood products dependant on their cytopenias. The median survival rate for all types of MDSs is less than 2 years. RAEB and RAEB-t have the highest percentage of blasts and the lowest survival rates. At this time, bone marrow transplant offers the only chance for a cure, and it is the treatment of choice in patients less than 50 years old. Studies have shown that the incidence of MDS is greater than the incidence of AML in the 50–70-year-old age group. Up to 40% of the myelodysplastic syndromes transform into acute leukemia.

297.

B. Chronic myelogenous leukemia (CML) is a myeloproliferative disorder, a malignant proliferation of leukocytes not in response to infection; the leukocyte count is often greater than 100.0×10^9/L. No toxic changes are present in CML. Thrombocytosis is seen in more than half of the patients with CML. A neutrophilic leukemoid reaction represents a normal body response to a severe infection. It is a benign proliferation of WBCs with a high leukocyte count but usually less than 50.0×10^9/L. Toxic changes to the neutrophils such as toxic granulation, vacuoles, and Döhle bodies are present. These two disorders both display a "left" shift, and they can be confused with each other.

298.

D. Leukocyte alkaline phosphatase activity is increased in severe infections such as the neutrophilic leukemoid reaction and polycythemia vera, and during the last trimester of pregnancy. It is greatly reduced in chronic myelogenous leukemia, although it may increase during blast crisis of this disease. Periodic acid–Schiff and Sudan black B are used to differentiate ALL from AML. The TRAP stain is useful in the diagnosis of hairy cell leukemia.

299.

C. The Philadelphia chromosome, t(9;22), is found in the precursor cells for erythrocytes, granulocytes, and platelets in at least 90% of the cases of CML. It is an acquired chromosome abnormality that results from a reciprocal translocation between chromosomes 9 and 22, and it can be detected even when the patient is in remission. The BCR/ABL oncogene is also associated with CML.

300.

B. Patients who have the Philadelphia chromosome have a less aggressive disease and better prognosis than the rare cases that do not have the abnormality. Some cases of CML terminate in an acute lymphoblastic leukemia. However, this outcome cannot be predicted by the presence or absence of the Philadelphia chromosome.

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

Principles of Coagulation

1. The hemorrhagic problems associated with scurvy are due to a deficiency of _____, which is a cofactor required for collagen synthesis.
   A. Vitamin C
   B. Prothrombin
   C. Vitamin K
   D. Protein C

2. The number of platelets an average megakaryocyte generates is approximately
   A. 25–50
   B. 50–200
   C. 200–500
   D. 2000–4000

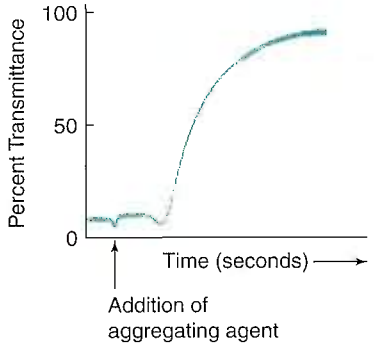
3. Which of the following is *not* a cause of thrombocytopenia?
   A. Splenomegaly
   B. Chemotherapy
   C. Increased thrombopoietin
   D. Aplastic anemia

4. Platelets interacting with and binding to other platelets is referred to as
   A. Adhesion
   B. Aggregation
   C. Release
   D. Retraction

5. In platelet aggregation studies, certain aggregating agents induce a biphasic aggregation curve. This second phase of aggregation is directly related to
   A. Formation of fibrin
   B. Changes in platelet shape
   C. Release of endogenous ADP
   D. Release of platelet factor 3

6. A platelet aggregation agent that characteristically yields a biphasic curve when used in optimal concentration is
   A. Arachidonic acid
   B. Collagen
   C. Epinephrine
   D. Ristocetin

7. The platelet aggregation pattern drawn below is characteristic of the aggregating agent
   A. ADP
   B. Collagen
   C. Ristocetin
   D. Thrombin



8. The operating principle of a platelet aggregometer is best described as
   A. Aggregation on a foreign surface: Platelet aggregation is directly proportional to the difference in platelet counts performed before and after platelet-rich plasma is passed through a column of glass beads.
   B. Change in optical density: As platelets aggregate, the optical density of the platelet-rich plasma decreases.
   C. Electrical impedance: Platelet aggregates are counted as they pass through an aperture, temporarily interrupting the flow of current between two electrodes.
   D. Pulse editing: Editing electronically generated pulses can differentiate the number of free platelets versus platelet aggregates.

9. Of the following therapeutic agents, those considered to be antiplatelet medications are
   A. Aspirin and Plavix®
   B. Coumadin® and heparin
   C. Heparin and protamine sulfate
   D. Tissue plasminogen activator and streptokinase

10. A potent inhibitor of platelet aggregation released by endothelial cells is
    A. Epinephrine
    B. Prostacyclin
    C. Ristocetin
    D. Thromboxane A_2

11. The reference value for mean platelet volume (MPV) is approximately
    A. 2–4 fL
    B. 5–7 fL
    C. 8–10 fL
    D. 11–14 fL

12. The platelet parameter PDW refers to the
    A. Average platelet volume
    B. Cell weight versus density
    C. Capacity to adhere to foreign surfaces
    D. Variation in platelet cell size

13. A normal histogram showing platelet size distribution is best described as
    A. Bimodal, nonskewed peaks
    B. Left-skewed single peak
    C. Right-skewed single peak
    D. Single peak, Gaussian distribution

14. Which of the following is *not* a normal maturation stage for platelets?
    A. Megakaryoblast
    B. Promegakaryocyte
    C. Micromegakaryocyte
    D. Megakaryocyte

15. The recommended type of microscopy for the performance of manual platelet counts is
    A. Electron
    B. Dark field
    C. Light
    D. Phase contrast

16. Twenty microliters of blood are diluted in 1.98 mL of diluent. This dilution is plated on both sides of a Neubauer-ruled counting chamber. A total of 356 cells is seen when both large center squares are counted. The platelet count *expressed in SI units* is
    A. 178×10^9/L
    B. $178 \times 10^3/\mu$L
    C. 356×10^9/L
    D. 712×10^9/L

17. The size threshold range used by electrical impedance methods to count particles as platelets is
    A. 0–10 fL
    B. 2–20 fL
    C. 15–40 fL
    D. 35–90 fL

18. In storage pool disease, platelets are primarily deficient in
    A. ADP
    B. Platelet factor 3
    C. Thrombasthenin
    D. Thromboxane A_2

19. The anticoagulant required for routine coagulation testing is
    A. Sodium heparin
    B. Sodium citrate
    C. Acid citrate dextrose
    D. Sodium fluoride

20. Which of the following is *not* synthesized in the liver?
    A. Factor VIII
    B. Plasminogen
    C. Protein C
    D. von Willebrand factor

21. When thrombin binds to thrombomodulin on the endothelial cell surface, thrombin can
    A. Activate the protein C pathway
    B. Activate factor V and factor VIII
    C. Convert fibrinogen to fibrin
    D. Stimulate platelet aggregation

22. The coagulation factors having a sex-linked recessive inheritance pattern are
    A. Factor V and factor VIII
    B. Factor VIII and factor IX
    C. Factor IX and factor X
    D. von Willebrand factor and factor VIII

23. Prekallikrein deficiency is associated with
    A. Prolonged aPTT that does not correct with a mixing study
    B. Autosomal dominant inheritance
    C. Increased risk of thrombosis
    D. Delayed bleeding at the incision site following surgery

24. Which of the following will *not* cause the thrombin time to be prolonged?
    A. Fibrin degradation products
    B. Heparin
    C. Factor I deficiency
    D. Factor II deficiency

25. The expected screening test results for a patient with a fibrin stabilizing factor deficiency are
    A. Prolonged prothrombin time
    B. Prolonged activated partial thromboplastin time
    C. Prolonged prothrombin time and activated partial thromboplastin time
    D. Normal prothrombin time and activated partial thromboplastin time

26. A patient on therapeutic warfarin will most likely have a(n)
    A. Normal PT/INR, increased aPTT, prolonged bleeding time, low platelet count
    B. Increased PT/INR, increased aPTT, normal bleeding time, normal platelet count
    C. Normal PT/INR, normal aPTT, normal bleeding time, normal platelet count
    D. Increased PT/INR, normal aPTT, prolonged bleeding time, low platelet count

27. Which of the following complexes is *not* needed for blood coagulation to occur?
    A. VIIa, tissue factor, Ca^{2+}
    B. IXa, VIII, Ca^{2+}, PF3
    C. Xa, V, Ca^{2+}, PF3
    D. XIIa, kallikrein, HMWK

28. von Willebrand factor is a
    A. Phospholipid required for multiple reactions in the coagulation sequence
    B. Plasma protein that binds platelets to exposed subendothelial collagen
    C. Plasma protein with procoagulant activity in the intrinsic coagulation system
    D. Platelet membrane glycoprotein that attaches the platelet to the injured vessel wall

29. Fibrin strands are cross-linked and the fibrin clot is stabilized by the activity of
    A. α_2-Antiplasmin
    B. Factor XIIIa
    C. Plasmin
    D. Thrombin

30. Which of the following does *not* contribute to the activation of the fibrinolytic system?
    A. XIIa
    B. XIa
    C. Kallikrein
    D. Tissue plasminogen activator

31. Which of the following enzymatically degrades the stabilized fibrin clot?
    A. Plasminogen
    B. Plasmin
    C. Prothrombin
    D. Thrombin

32. The activity of the lupus anticoagulant and anticardiolipin antibodies appears to be directed against
    A. Factor V
    B. Factor VIII
    C. Factor IX
    D. Phospholipid

33. Heparin inhibits clotting by
    A. Chelating calcium ions
    B. Preventing activation of prothrombin
    C. Causing liver synthesis of nonfunctional factors
    D. Enhancing the action of antithrombin

34. The main regulatory protein of secondary hemostasis is
    A. Antithrombin
    B. Protein C
    C. α_2-Antiplasmin
    D. Tissue plasminogen activator

35. Why is the activated partial thromboplastin time (aPTT) *not* the procedure of choice for detecting a platelet factor 3 (PF3) deficiency?
    A. Platelet-rich plasma is used for this test.
    B. The reagent contains a phospholipid substitute for PF3.
    C. PF3 is unstable in the reagent used for this test.
    D. PF3 does not function in the system being tested.

36. Measurement of the time required for fibrin formation when thrombin is added to plasma evaluates the
    A. Fibrinogen concentration
    B. Prothrombin concentration
    C. Extrinsic clotting system
    D. Intrinsic clotting system

37. A fibrinogen assay is performed on the fibrometer using the standard 1:10 dilution with Owren's buffer. The seconds obtained do not read on the standard curve. An alternate 1:20 dilution is performed and is 400 mg/dL when read off the curve. The concentration of fibrinogen to be reported in mg/dL is
    A. 160 mg/dL
    B. 200 mg/dL
    C. 400 mg/dL
    D. 800 mg/dL

38. Which of the following is *not* true of the international normalized ratio (INR)?
    A. INR is dependent on reagents and instrumentation used.
    B. INR is calculated using the PT ratio taken to the power of the ISI value.
    C. The World Health Organization recommends reporting the INR on patients on stable oral anticoagulant therapy.
    D. A therapeutic INR for a patient on Coumadin® is between 2.0 and 3.0, but may be higher depending on the cause of the patient's underlying disease state.

39. A prolonged aPTT result is obtained on a patient diagnosed with acute disseminated intravascular coagulation (DIC). The patient has not yet been treated for this disorder. The most likely cause of the prolonged aPTT is
    A. In addition to DIC, the patient is deficient in a factor required for the extrinsic pathway.
    B. DIC is characterized by synthesis of less stable coagulation factors, which deteriorate rapidly in the circulation.
    C. Systemic activation of the coagulation system depletes some factors more rapidly than the liver can synthesize them.
    D. The patient has been misdiagnosed; a prolonged aPTT indicates that the problem is deficient, not excessive, coagulation.

40. Which of the following test results is *not* characteristic of DIC?
    A. Decreased fibrinogen concentration
    B. Positive test for degradation products
    C. Decreased platelet count
    D. Increased antithrombin

41. The principle of _____ methods depends on cleavage of synthetic substrates by an active serine protease.
    A. Chromogenic
    B. Photo-optical
    C. Mechanical
    D. Immunodiffusion

42. Epsilon aminocaproic acid is the treatment of choice for
    A. von Willebrand disease
    B. Hemophilia A
    C. DIC with secondary fibrinolysis
    D. Primary fibrinogenolysis

43. A clot retraction defect is most likely due to
    A. Lack of platelet receptor glycoprotein Ib
    B. Lack of platelet receptor glycoprotein IIb/IIIa
    C. Insufficient ADP in dense bodies
    D. Absence of von Willebrand factor

44. Thrombocytosis is a characteristic of
    A. Disseminated intravascular coagulation
    B. Splenomegaly
    C. Polycythemia vera
    D. Idiopathic thrombocytopenic purpura

45. In which of the following functions are the products released by vascular endothelial cells *not* involved?
    A. Inhibition of platelet aggregation
    B. Activation of the fibrinolytic system
    C. Conversion of thrombin from a procoagulant to an anticoagulant
    D. Cross-linkage of fibrin monomers

46. If a physician suspects a qualitative platelet defect, the most useful test to order is the
    A. Platelet count
    B. Prothrombin time
    C. 5.0 M urea solubility test
    D. Bleeding time

47. The coagulation factors referred to as "vitamin K-dependent" are
    A. I, V, VIII, XIII
    B. II, V, IX, XII
    C. II, VII, IX, X
    D. XI, XII, Fletcher, Fitzgerald

48. A patient on warfarin therapy will be deficient in a functional amount of
    A. Fibrinogen and prothrombin
    B. Stable and labile factors
    C. Protein C and protein S
    D. Fletcher and Fitzgerald factors

49. A 25-year-old male presents to his physician complaining of leg pain. The physician diagnoses a deep vein thrombosis and wants to determine the cause of the thrombotic episode. Which of the following conditions would *not* be associated with such a thrombotic episode?
    A. Factor V Leiden and Prothrombin 20210 mutations
    B. Hypofibrinogenemia and hyperhomocysteinemia
    C. Lupus anticoagulant and anticardiolipin antibodies
    D. Antithrombin and protein C deficiencies

50. An 85-year-old male with slurred speech and paralysis on the right side of the body is seen in the emergency department. A stat D-dimer is ordered and is very high. The physician suspects a thromboembolic event based on the D-dimer, and needs to institute clot-dissolving therapy immediately. The most likely diagnosis and appropriate therapy for the patient is
    A. Myocardial infarction; treat with aspirin
    B. Pulmonary embolism; treat with warfarin
    C. Deep vein thrombosis; treat with heparin
    D. Stroke; treat with tissue plasminogen activator

51. Reversal of a heparin overdose can be achieved by administration of
    A. Vitamin K
    B. Protamine sulfate
    C. Antithrombin
    D. Warfarin

52. Which of the following best describes protein C?
    A. Vitamin K-dependent inhibitor to clotting
    B. Activator of factors V and VIII:C
    C. Inhibitor of fibrinolysis
    D. Synthesized by endothelial cells

53. The prothrombin time will detect deficiencies in the _____ pathway(s) when calcium and a tissue factor source such as rabbit brain are added to plasma.
    A. Extrinsic
    B. Extrinsic and common
    C. Intrinsic
    D. Intrinsic and common

54. A 65-year-old patient in the emergency department has a normal D-dimer and an elevated FDP result. These results are consistent with the presence of degradation products of
    A. Non-cross-linked fibrin
    B. Cross-linked fibrin
    C. Fibrinogen
    D. Plasmin

Specimen Acceptability

55. A specimen is received for a prothrombin time and activated partial thromboplastin time. The 5 mL tube has 2.5 mL of blood in it. Expected test results are
    A. PT and aPTT both falsely short
    B. PT and aPTT both falsely long
    C. PT and aPTT both unaffected
    D. PT unaffected, aPTT falsely short

56. A microtainer EDTA sample obtained during a fingerstick puncture is run on an automated cell counter, yielding a platelet count of 178×10^9/L. In the erythrocyte monolayer of the stained peripheral blood smear, an average of 9 platelets per field is seen under 1000× magnification. Based on these data, you should
    A. Report the results because the platelet count and platelet estimate correlate.
    B. Recollect a specimen for a repeat platelet count because the platelet count and estimate do not correlate.
    C. Examine the periphery of the blood smear for clumping because the platelet count and estimate do not correlate.
    D. Rerun the platelet count on the available specimen to confirm the results.

57. Blood for an aPTT was collected from a 5-year-old boy. During the venipuncture, he had to be restrained by several people and still managed to be a moving target. The result of the child's aPTT was 18.0 sec (reference range 22.0–38.0 sec). The aPTT controls were in range. Which of the following interpretations would apply to the aPTT result?
    A. aPTT is abnormal because of a hereditary factor deficiency.
    B. aPTT is invalid because of contamination with tissue factor.
    C. Tube is probably not full, resulting in a falsely short time.
    D. Result is within reference range for a patient of this age.

58. Laboratory tests requested on a patient scheduled for early morning surgery include a CBC with platelet count. An automated platelet count performed on the specimen is 57×10^9/L. In the monolayer area of the peripheral blood smear there are approximately 12 platelets per oil immersion field, many of which are encircling neutrophils. Controls are in range. Based on this information, the best course of action is
    A. Report all the results because the instrument is functioning properly.
    B. Alert the physician immediately so cancellation of surgery can be considered.
    C. Thoroughly mix specimen and repeat platelet count; if results remain the same, report all results and indicate that platelet count has been confirmed by repeat testing.
    D. Have the specimen redrawn using 3.2% sodium citrate as the anticoagulant.

59. Phlebotomist Forgetful Frank collected a tube of blood for an aPTT on John Smith at 10:00 A.M. The blood was collected in a sodium citrate tube. At 4:30 P.M., Frank was getting ready to leave for the day when he discovered Mr. Smith's blood specimen on his blood collection tray. So before leaving, Frank delivered the tube of blood to the laboratory for testing. Which of the following best describes the expected results?
    A. Sodium citrate is a preservative as well as an anticoagulant, so the aPTT result should be accurate.
    B. An aPTT collected in sodium citrate will give falsely long results because some factors are unstable in this anticoagulant.
    C. A falsely long aPTT is expected because some factors deteriorate rapidly at room temperature.
    D. Exposure of the plasma to erythrocytes for several hours has probably activated the factors, so the aPTT will be falsely short.

60. An aPTT and PT are requested on a patient scheduled for emergency surgery. On an optical density clot detection system, normal and abnormal controls for both tests are within range, but the patient's results exceed the upper limit of linearity. The patient's aPTT and PT have been performed in duplicate, but there still is sufficient plasma, which is grossly lipemic, to repeat the tests. What is the best course of action to follow?
    A. Report the results immediately by phone, emphasizing that the tests were run in duplicate and the controls are within range.
    B. Request a new specimen and repeat the aPTT and PT using freshly diluted controls.
    C. Repeat the aPTT and PT on an instrument that detects clot formation electromechanically.
    D. Inform the physician that accurate results are impossible.

61. A sodium citrate tube is received in the laboratory for PT and aPTT testing. Results are as follows:

Prothrombin time	>100.0 sec (control 12.0 sec)
aPTT	>200.0 sec (control 32.0 sec)

On examination, a large clot is discovered. The abnormal test results are due to deficiencies of factors
    A. I, V, VIII, IX
    B. I, II, V, VIII, XIII
    C. II, VII, IX, X
    D. VIII, IX, XI, XII

Case Studies

62. A 30-year-old female is admitted to the hospital with neurological symptoms. The following results are obtained:

Hemoglobin	60 g/L
Hematocrit	0.19 L/L
Platelet count	25×10^9/L
RBC morphology	Many schistocytes
ADAMTS-13	Markedly decreased

The most likely diagnosis for the patient is
A. Thrombotic thrombocytopenic purpura
B. Idiopathic thrombocytopenic purpura
C. Hemolytic uremic syndrome
D. von Willebrand disease

63. A 4-year-old child is seen in the emergency department with petechiae and a platelet count of 15×10^9/L. She has no previous history of bleeding problems. Three weeks earlier she had chicken pox. The physician advises the parents to keep the child off the playground to avoid injury, and says the child will recover within 2–4 weeks with no further treatment. What condition does this child most likely have?
A. Essential thrombocythemia
B. Idiopathic thrombocytopenic purpura
C. Thrombotic thrombocytopenic purpura
D. Glanzmann thrombasthenia

64. Laboratory results on a 16-year-old female with frequent nosebleeds and severe menorrhagia are as follows:

Platelet count	250×10^9/L
Bleeding time	>15 min (reference range ≤8.0 min)
Prothrombin time	13.0 sec (control 12.0 sec)
aPTT	75.0 sec (control 32.0 sec)
Platelet aggregation	Normal response to ADP, collagen, epinephrine; no response with ristocetin

These results are consistent with
A. Christmas disease
B. Hemophilia A
C. Glanzmann thrombasthenia
D. von Willebrand disease

65. Laboratory results on a 6-year-old female with petechiae and severe epistaxis are as follows:

Platelet count	145×10^9/L
Bleeding time	>15 min (reference range ≤8.0 min)
MPV	16.0 fL (reference range 8.0–10.0 fL)
Platelet aggregation	Normal response to ADP, collagen, epinephrine; no response with ristocetin
Prothrombin time	11.5 sec (control 12.0 sec)
aPTT	33.0 sec (control 32.0 sec)

These results are consistent with
A. Bernard-Soulier syndrome
B. von Willebrand disease
C. Glanzmann thrombasthenia
D. Ehlers-Danlos syndrome

66. A clot retraction defect is suspected in a newborn male experiencing severe bleeding following circumcision. The following results are obtained:

Platelet count	320×10^9/L
Bleeding time	>15 min (reference range ≤ 8.0 min)
Platelet aggregation	Normal response to ristocetin; weak response to ADP, collagen, epinephrine
Prothrombin time	12.0 sec (control 12.0 sec)
aPTT	31.0 sec (control 32.0 sec)

These results are characteristic of
A. von Willebrand disease
B. Glanzmann thrombasthenia
C. Storage pool disease
D. Christmas disease

67. Results on a 35-year-old male presenting with sudden severe hemorrhagic problems are as follows:

Platelet count	225×10^9/L
Bleeding time	6.5 min (reference range ≤ 8.0 min)
Prothrombin time	12.8 sec (control 12.0 sec)
aPTT	85.0 sec (control 32.0 sec)
aPTT 1:1 mixing study	65.0 sec

These clinical manifestations and laboratory results are consistent with
A. Lupus anticoagulant
B. von Willebrand disease
C. Hemophilia A
D. Factor VIII inhibitor

68. An 80-year-old man suffered a heart attack 1 month ago, and after the hospital stay was discharged with instructions to follow an outpatient treatment plan. He arrives at the cardiology clinic today for lab work to monitor the treatment plan. The following results are obtained:

PT	52.0 sec (control 12.0 sec)
INR	5.5 (therapeutic range 2.0–3.0)
aPTT	50.0 sec (control 32.0 sec)

This patient is most likely on a
A. Nontherapeutic dose of unfractionated heparin
B. Nontherapeutic dose of coumarin
C. Nontherapeutic dose of both unfractionated heparin and coumarin
D. Fibrinolytic agent such as tissue plasminogen activator

69. The following results are obtained on a 60-year-old male patient:

WBC	24.7×10^9/L
RBC	6.67×10^{12}/L
Hgb	200 g/L
Hct	0.61 L/L
Plt	79×10^9/L
PT	19.3 sec (control 12.0 sec)
aPTT	81.2 sec (control 32.0 sec)

The WBC, RBC, Hgb, Hct, and Plt were performed on blood collected in an evacuated tube containing EDTA. The PT and aPTT were performed on blood collected in an evacuated tube containing 3.2% sodium citrate. The standard collection procedure was followed, and all tests were performed within the appropriate time limits. Based on this information, the statement that best explains the prolonged coagulation test results is

A. Coagulation reactions require platelet factor 3; availability of this component is insufficient when the platelet count is below 100×10^9/L.
B. The ratio of anticoagulant to blood is critical; the volume of anticoagulant must be decreased when the Hct is greater than 55%.
C. The PT and aPTT evaluate the extrinsic and intrinsic pathways, respectively; prolongation of both tests indicates a deficiency of a factor common to both systems.
D. Coagulation reactions are inhibited by a product released by leukocytes; this inhibitory activity becomes significant when the leukocyte count is greater than 20.0×10^9/L.

70. The following results are obtained on a 3-year-old boy with sudden severe hemorrhagic problems:

Bleeding time	5.0 min (reference range ≤ 8.0 min)
Prothrombin time	13.0 sec (control 12.0 sec)
aPTT	95.0 sec (control 32.0 sec)
aPTT 1:1 mixing study	35.0 sec
Platelet aggregation	Normal with ristocetin, ADP, collagen, and epinephrine

These clinical manifestations and laboratory results are consistent with

A. Aspirin therapy
B. von Willebrand disease
C. Hemophilia A
D. Heparin therapy

71. Screening tests for a 46-year-old male patient admitted for minor surgery follow:

Platelet count	325×10^9/L
Bleeding time	4.5 min (reference range ≤8.0 min)
Prothrombin time	13.0 sec (control 12.0 sec)
aPTT	95.0 sec (control 32.0 sec)
aPTT 1:1 mixing study	32.0 sec

The patient has no clinical manifestations of a bleeding problem and has no personal or family history of bleeding problems, even following dental extraction. Several family members have been treated for deep vein thrombosis. Based on these laboratory results and the clinical history, the most likely cause of the prolonged aPTT is

A. Heparin present in the sample
B. Factor VIII deficiency
C. Factor XII deficiency
D. Factor XIII deficiency

72. A 25-year-old obstetrical patient at 35 weeks gestation is admitted through the emergency room. She has bleeding in the genitourinary tract, and there are visible petechiae and ecchymoses. The following laboratory results are obtained:

Platelet count	Markedly decreased
Prothrombin time	Prolonged
aPTT	Prolonged
Fibrinogen	Decreased
Thrombin time	Prolonged
D-dimer	Positive
FDP	Positive
Antithrombin	Decreased
RBC morphology	Schistocytes present

These laboratory results are consistent with

A. Primary fibrinogenolysis
B. DIC with secondary fibrinolysis
C. Factor II deficiency
D. Heparin therapy

73. A 57-year-old man with prostate cancer is admitted to the intensive care unit with severe bleeding problems. The following laboratory results are obtained:

Platelet count	Normal
Prothrombin time	Prolonged
aPTT	Prolonged
Fibrinogen	Decreased
Thrombin time	Prolonged
D-dimer	Negative
FDP	Positive
Antithrombin	Normal
RBC morphology	Schistocytes absent

These laboratory results are consistent with
A. Primary fibrinogenolysis
B. DIC with secondary fibrinolysis
C. Factor II deficiency
D. Coumadin® therapy

74. A patient in the hospital for an acute myocardial infarction is placed on standard unfractionated heparin therapy and aspirin. Laboratory results are performed before instituting therapy and then daily as shown:

	Before Therapy	Day 2	Day 3
Prothrombin Time	Normal	Normal	Normal
aPTT	Normal	Prolonged	Prolonged
Platelet Count	325×10^9/L	160×10^9/L	42×10^9/L

The most likely complication by Day 3 is
A. Disseminated intravascular coagulation
B. Primary fibrinogenolysis
C. Aspirin-induced thrombocytopenia
D. Heparin-induced thrombocytopenia

75. A 24-year-old female with painful swelling in her left leg is seen by her physician, who orders laboratory testing for PT and aPTT. The PT is normal. The aPTT is prolonged, but shortens with a 10-minute incubation of patient plasma with partial thromboplastin reagent that uses kaolin as the activator. A 1:1 aPTT mixing study corrects to normal. The most likely diagnosis is
A. Factor II deficiency
B. Factor VIII inhibitor
C. Factor XIII deficiency
D. Prekallikrein deficiency

answers & rationales

Principles of Coagulation

1.

A. Vascular integrity is influenced by vitamin C intake. In a deficiency or absence of vitamin C, collagen production is insufficient or abnormal. Vitamin C deficiency is associated with capillary fragility and the primary hemostasis bleeding symptoms of petechiae and mucosal bleeding.

2.

D. Each megakaryocyte produces approximately 2000–4000 platelets. A single megakaryocyte can generate this large number of cells because platelets are nonnucleated fragments of their cytoplasm. The number of platelets generated by a megakaryocyte depends on its cell size, which is directly related to the number of endomitotic divisions before cytoplasmic fragmentation.

3.

C. Thrombopoietin is the major humoral factor involved in platelet production. Increased thrombopoietin results in thrombocytosis; decreased thrombopoietin results in thrombocytopenia. Two-thirds of platelets, once released from the bone marrow, are in circulation. The other one-third of platelets is sequestered in the spleen. Splenomegaly is a cause of thrombocytopenia due to increased sequestration. Chemotherapeutic agents destroy both normal and malignant cells, causing thrombocytopenia, anemia, and leukopenia. Aplastic anemia occurs when the bone marrow fails to produce any of the three cell lines.

4.

B. "Adhesion" refers to platelets interacting with something other than platelets. *In vivo* platelets adhere to collagen that is exposed when vessel damage occurs. "Aggregation" refers to attachment of platelets to other platelets. Release is the process by which platelet granule contents are secreted. Retraction describes one of the final steps in coagulation in which the fibrin-platelet plug contracts, restoring normal blood flow to the vessel.

5.

C. In platelet aggregation studies, the addition of the aggregating agent may induce an initial aggregation phase followed by a secondary wave. The initial phase is due to the interaction of the aggregating agent with the platelet. The second phase is due to release of nonmetabolic ADP from platelet granules, which promotes the additional wave of aggregation.

6.

C. Epinephrine is the only aggregating agent listed that typically gives a biphasic pattern. ADP and thrombin also give biphasic patterns when used in optimal concentrations. Arachidonic acid causes a rapid monophasic platelet aggregation. Collagen and ristocetin also induce monophasic aggregatory responses.

7.

B. Collagen is the only aggregating agent that includes a single wave response preceded by a lag phase. During the lag phase collagen stimulates platelets to release their granule contents. Endogenous ADP released from the platelets then initiates irreversible platelet aggregation.

8.

B. When an aggregating agent is added to an optically dense suspension of platelet-rich plasma (PRP), the platelets normally stick to each other, forming platelet aggregates. As additional platelets aggregate, the cell suspension becomes clearer and has a few large clumps of cells. At maximum aggregation the specimen is relatively clear, allowing light transmission that is only partially obstructed by a few large platelet aggregates.

9.

A. Aspirin inhibits the enzyme cyclooxygenase in the prostaglandin pathway, preventing platelet aggregation. Plavix® (clopidogrel bisulfate) blocks the IIb/IIIa fibrinogen-binding platelet receptor, preventing platelet aggregation. Coumadin® and heparin inhibit clotting factors in secondary hemostasis. Protamine sulfate can be used to neutralize heparin. Tissue plasminogen activator and streptokinase are fibrinolytic system activators.

10.

B. Prostacyclin, also referred to as PGI_2, is the most potent inhibitor of platelet aggregation known. Injured endothelial cells release prostacyclin. Epinephrine and ristocetin are potent stimulators of platelet aggregation. Thromboxane A_2, generated by platelets via the prostaglandin pathway, also stimulates platelets to aggregate.

11.

C. The average volume of normal platelets is approximately 8–10 fL. This platelet parameter is equivalent to the erythrocyte parameter MCV (mean corpuscular volume). MPV is increased in the hereditary Bernard-Soulier syndrome and May-Hegglin anomaly; also in acquired disorders with increased need for platelet release from the bone marrow.

12.

D. PDW is an abbreviation for platelet distribution width. This parameter measures the uniformity of platelet size. It is the platelet equivalent of the red cell parameter RDW. It represents the coefficient of variation of the platelet population.

13.

C. A histogram showing platelet size distribution is made by plotting platelet size (x axis) versus number (y axis). The resulting curve is usually a single, right-skewed peak. This reflects a larger number of platelets in the lower size range with a "tail" of larger cells to the right of the majority.

14.

C. Micromegakaryocytes, also known as dwarf megakaryocytes, are thought to be megakaryocytes that have lost their ability to undergo endomitosis. They can be seen in the peripheral blood of patients with myelodysplastic syndromes or myeloproliferative disorders. They may resemble lymphocytes, but cytoplasmic blebs can help to identify them as micromegakaryocytes.

15.

D. Phase microscopy is currently recommended for manual platelet counts. This allows satisfactory discrimination between platelets and debris, a major problem in manual counts. Light microscopy may also be used; however, differentiating between platelets and debris is more difficult than with phase microscopy.

16.

A. Twenty microliters of blood (0.02 mL) added to 1.98 mL of diluting fluid gives a dilution of 1:100; the dilution factor is 100. The standard platelet counting area is the center mm^2 on both sides of the chamber. The standard formula for hemacytometer counts expressed in mm^3 is

$$\frac{\text{Total number cells counted} \times \text{Dilution factor}}{\text{Total area counted} \times \text{Depth}}$$

The correct equation for this problem is

$$\frac{356 \times 100}{2\ mm^2 \times 0.10\ mm} = 178{,}000$$

$$= 178 \times 10^3/mm^3$$

When expressed in SI units, the platelet count is 178×10^9/L. Answer B is expressed in conventional units, not SI units.

17.

B. In the electrical impedance method for counting platelets, particles between 2 and 20 fL will be classified as platelets by the analyzer's computer. The normal average platelet volume is 10 fL. One dilution is used for counting and sizing of platelets and red blood cells. In the electrical impedance method, size thresholds differentiate the two.

18.

A. Platelets in storage pool disease are deficient in dense granules. The platelets in this disorder lack nonmetabolic ADP found in dense granules and normally released when the platelets are stimulated. This accounts for a poor response to aggregating agents.

19.

B. The Clinical and Laboratory Standards Institute (CLSI) recommends 3.2% (0.109 M) sodium citrate for coagulation testing. Sodium heparin is used for many chemistry tests, but will cause times that exceed linearity if used for coagulation tests. Acid citrate dextrose (ACD) is used for HLA phenotyping, DNA analysis, and paternity tests. Sodium fluoride is used for glucose testing.

20.

D. The liver produces most of the clotting factors as well as inhibitors to clotting. A patient with liver disease has impaired synthesis of these clotting factors and inhibitors. One of the few hemostatic proteins not produced by the liver is von Willebrand factor, which is produced by endothelial cells and megakaryocytes.

21.

A. Thrombomodulin, an endothelial cell receptor, has the ability to change the specificity of thrombin from a procoagulant to an anticoagulant. Once bound to thrombomodulin, thrombin has anticoagulant properties because of its activation of protein C. Protein C, along with its cofactor protein S, then exerts negative feedback on the clotting system by inactivating factor V and factor VIII.

22.

B. Factor VIII and factor IX are the sex-linked recessive hemostatic defects. von Willebrand factor deficiency and dysfibrinogenemia are inherited as autosomal dominant disorders. Factor V, factor X, and most of the other inherited hemostatic disorders have an autosomal recessive inheritance pattern.

23.

C. Prekallikrein (Fletcher factor) deficiency is one of the many autosomal recessive disorders. The aPTT will be prolonged and will correct with a mixing study because it is a factor deficiency. Because prekallikrein is an activator of the fibrinolytic system, prekallikrein-deficient patients cannot lyse clots efficiently and are prone to thrombosis. Fibrinolytic and anticoagulant therapies are indicated in patients who develop thrombosis. Delayed post-operative bleeding at the incision site is characteristic of a factor XIII deficiency.

24.

D. The thrombin time is a test that measures fibrinogen. Thrombin reagent is added to undiluted patient plasma, and the time it takes for fibrinogen conversion to fibrin is measured. Anything that interferes with the ability of thrombin to convert fibrinogen to fibrin will prolong the test. Heparin and the degradation products X and E inhibit thrombin. Factor II cannot be measured in the thrombin time because the reagent used is its active form, thrombin (II_a).

25.

D. Thrombin converts fibrinogen to the fibrin monomer. Fibrin monomers spontaneously polymerize to form the fibrin polymer. This is the endpoint of clot-based PT and aPTT tests. This fibrin polymer is unstable. Once activated, factor XIII, also known as fibrin stabilizing factor, produces strong, covalent bonds to create a stable fibrin polymer. This occurs after the endpoint of the PT and aPTT has been reached. A factor XIII deficiency is suspected when delayed post-operative bleeding occurs at the incision site, and the deficiency can be confirmed with the 5 M urea clot solubility test.

26.

B. Warfarin is a vitamin K antagonist and affects liver synthesis of the prothrombin group factors II, VII, IX, and X. The factors are produced but are nonfunctional. Warfarin therapy is monitored with the prothrombin time (PT), which will detect nonfunctional II, VII, and X. The aPTT, though used to monitor heparin therapy, can detect nonfunctional II, IX, and X caused by warfarin therapy. Warfarin does not affect platelet function or quantity. Aspirin and other antiplatelet medications such as clopidogrel bisulfate (Plavix®) will affect platelet function and prolong the bleeding time.

27.

D. The intrinsic, extrinsic, and common pathways each have a complex that must form for blood coagulation to occur. The intrinsic complex of IXa, VIII, and Ca^{2+} forms on the platelet surface (PF3) and activates factor X. The extrinsic complex of VIIa, tissue factor, and Ca^{2+} activates factor X as well as factor IX and is the dominant pathway *in vivo*. Factor IX activation by the extrinsic complex provides a link between the intrinsic and extrinsic systems and minimizes the importance of the contact factors *in vivo*. The prothrombin-converting complex of Xa, V, Ca^{2+}, and PF3 is responsible for converting prothrombin to thrombin.

28.

B. von Willebrand factor (vWF) is a portion of the plasma protein known as the factor VIII/von Willebrand factor complex. Its function is to bind to platelet membrane glycoprotein Ib and form a bridge between the platelet and exposed subendothelial collagen. vWF is a carrier protein for factor VIII:C, but vWF does not have coagulant activity in secondary hemostasis as factor VIII:C does.

29.

B. Activated factor XIII is a transglutaminase that cross-links fibrin monomers between glutamine and lysine residues. Fibrin monomers that are not cross-linked lack the stability to maintain the hemostatic plug, as evidenced by the bleeding problems experienced by individuals deficient in factor XIII. Thrombin contributes to the formation of the fibrin clot, which is degraded by plasmin. Once the fibrin clot has been lysed and plasmin is free in circulation, α_2-antiplasmin quickly neutralizes plasmin.

30.

B. Once activated, three of the four contact factors activate both the intrinsic clotting system and provide intrinsic activation of the fibrinolytic system. The only contact factor that does not activate the fibrinolytic system is factor XI. Extrinsic activation of the fibrinolytic system is achieved by the release of tissue plasminogen activator by damaged endothelial cells.

31.

B. Plasmin, the active form of plasminogen, is the enzyme responsible for degrading fibrin into several different fragments. The D-dimer test is abnormal when there is excessive fibrinolytic activity. Prothrombin is the inactive precursor of thrombin that cleaves fibrinogen to form fibrin, which is stabilized by the activity of factor XIII.

32.

D. The lupus anticoagulant was first discovered in patients with systemic lupus erythematosus. It is actually seen in more patients without SLE, but the original name remains. Lupus anticoagulant and anticardiolipin antibodies belong to the antiphospholipid antibody family. Their activity appears to be directed against the phospholipid portion of the prothrombinase complex (Xa-V-phospholipid-calcium). The antibodies are usually IgG, but can also be IgM. They are found in autoimmune disorders, neoplasms, and some infections. They can also be medication related and can be found in apparently normal individuals. Their presence is suspected when the patient is experiencing thrombosis and the aPTT is prolonged with no correction of the mixing study.

33.

D. Heparin forms a complex with antithrombin to inhibit coagulation. The heparin-antithrombin complex rapidly inhibits thrombin and other serine proteases. Coagulation may be inhibited by coumarin and related vitamin K antagonists that cause liver synthesis of nonfunctional prothrombin group factors II, VII, IX, and X. Several anticoagulants, such as sodium citrate and EDTA, prevent fibrin formation by chelating calcium ions, which serve as cofactors in several reactions in the coagulation cascade.

34.

A. Antithrombin is the most important naturally occurring inhibitor to clotting and accounts for 80% of negative feedback in the coagulation cascade by inhibiting serine proteases. Protein C and its cofactor, protein S, inhibit cofactors V and VIII. α_2-Antiplasmin is responsible for neutralizing plasmin once the clot has been lysed. Tissue plasminogen activator activates the fibrinolytic system in response to clot formation.

35.

B. The reagent used in the aPTT procedure is a phospholipid extract that substitutes for platelet factor 3 in coagulation reactions. Platelet-poor plasma is used, so thrombocytopenia does not affect the aPTT. Platelet factor 3 functions in the intrinsic and prothrombin-converting complexes that must form for blood coagulation to occur.

36.

A. When thrombin is added to patient plasma, fibrinogen is converted to fibrin. No factors above fibrinogen in the cascade are measured, including prothrombin. Both the thrombin time and fibrinogen test use thrombin reagent; both tests measure only one factor, fibrinogen.

37.

D. A 1:20 dilution is used when the time obtained on a patient sample is less than the shortest time used in preparation of the standard curve. A 1:20 dilution is diluted by a factor of 2 when compared to the usual 1:10 dilution. The value read off the curve must be multiplied by 2 to take into account the alternate dilution used.

$$400 \times 2 = 800 \text{ mg/dL}$$

38.

A. The PT seconds are dependent on the reagent or instrument used. Because of this, the World Health Organization recommends using the INR to monitor patients on stabilized Coumadin® therapy because it is *independent* of the reagent or instrument used. The INR is a means of standardizing the reporting of prothrombin times (PTs) worldwide. The INR is calculated by many instruments and laboratory information systems but can be calculated manually as follows:

$$\text{INR} = \left[\frac{\text{Patient PT (in seconds)}}{\text{Control PT (in seconds)}}\right]^{\text{ISI}}$$

where

INR = the International Normalized Ratio

ISI = the International Sensitivity Index of the thromboplastin source. This value is determined by the manufacturer for each lot number of thromboplastin reagent. The closer to 1.00 the ISI, the more sensitive the thromboplastin reagent is in detecting factor deficiencies.

Patient PT = the prothrombin time in seconds for the patient.

Control PT is the geometric mean of the reference interval.

39.

C. As coagulation occurs *in vivo,* some factors are consumed just as they are when blood is allowed to clot in a test tube *in vitro.* The factors consumed during coagulation are I, II, V, VIII, and XIII. Results of laboratory procedures relying on one or more of these factors will be affected. All these factors except factor XIII contribute to the reactions evaluated in the aPTT procedure.

40.

D. The PT and aPTT are prolonged in DIC because of consumption of factors I, II, V, and VIII. Platelets are trapped in forming clots and are removed from circulation. The fibrinolytic system is activated by systemic intravascular coagulation; fibrin and fibrinogen degradation products are elevated. FDP and D-dimer tests will both be positive. The regulatory proteins antithrombin, protein C, and protein S are depleted trying to turn off systemic clotting.

41.

A. The proteolytic activity of antithrombin after activation to a serine protease can be assayed via methods that employ synthetic substrates. The cleavage of the synthetic substrate by an active serine protease will yield a chromogenic compound. Chromogenic methods can also be used to assay plasminogen, protein C, and heparin.

42.

D. Epsilon aminocaproic acid (EACA) is a specific inhibitor of plasmin and is used to turn off inappropriate lysing that occurs in primary fibrinogenolysis. Fibrinolysis seen in DIC is an appropriate body response to systemic clotting. If EACA is administered to a patient in DIC, clots that form will not be lysed, and this could be quickly fatal to the patient.

43.

B. Glanzmann thrombasthenia is a disorder characterized by absent or defective GP IIb/IIIa platelet receptors for fibrinogen binding and subsequent platelet aggregation. Clot retraction in these patients is abnormal due to the lack of the contractile protein actomyosin/thrombasthenin. Neither insufficient ADP in dense bodies, absence of von Willebrand factor, nor absence of the platelet receptor glycoprotein Ib affects clot retraction. Lack of glycoprotein Ib, the von Willebrand factor receptor site, causes the platelet adhesion defect seen in Bernard-Soulier disease.

44.

C. Polycythemia vera, a hemopoietic stem cell disorder characterized by excessive production of erythrocytic, granulocytic, and megakaryocytic cells in the bone marrow, is usually accompanied by thrombocytosis. ITP, DIC, and splenomegaly are all characterized by thrombocytopenia. In ITP, platelet destruction is mediated by immune mechanisms. Platelets are consumed in DIC and sequestered in an individual with an enlarged spleen.

45.

D. Endothelial cells release prostacyclin, which is a potent inhibitor of platelet aggregation. They also release tissue plasminogen activators, which initiate the fibrinolytic system. When thrombin binds with thrombomodulin on the endothelial cell surface, the specificity of thrombin changes. Rather than converting fibrinogen to fibrin, thrombin complexed with thrombomodulin activates protein C. Protein C along with protein S inhibits clotting by "turning off" factors V and VIII. Factor XIII is responsible for cross-linking fibrin monomers but it is not a component of endothelial cells.

46.

D. The bleeding time test is a primary hemostasis screening test for platelet and vascular function. The platelet count is a quantitative test that cannot be used to determine platelet function. The prothrombin time is a secondary hemostasis screening test and does not evaluate platelet function because platelet-poor plasma is used for testing. The 5 M urea solubility test will detect a factor XIII deficiency.

47.

C. The prothrombin group factors II, VII, IX, and X are called vitamin K–dependent factors. Vitamin K is needed by the liver to synthesize functional circulating forms of these factors. In the absence of vitamin K, the liver synthesizes the prothrombin group factors, but they are nonfunctional because they lack the carboxyl (COOH) groups needed for binding to Ca^{2+} on phospholipid membranes. The oral anticoagulant warfarin is a vitamin K antagonist and causes liver synthesis of these nonfunctional factors.

48.

C. Vitamin K is required for liver synthesis of regulatory proteins C and S and functional clotting factors II, VII, IX, and X. A deficiency of vitamin K decreases the concentrations of these proteins and subsequently affects test results that measure one or more of them. Fibrinogen, Labile, Fletcher, and Fitzgerald factors do not require vitamin K for their synthesis.

49.

B. All of the conditions listed are associated with thrombosis except hypofibrinogenemia. Although hereditary dysfibrinogenemia frequently causes thrombosis, hypofibrinogenemia causes bleeding tendencies. The most common hereditary thrombotic disorder, factor V Leiden, is caused by synthesis of an abnormal factor V molecule that is resistant to the inhibitory affects of protein C. The Prothrombin 20210 mutation, in which an abnormal factor II molecule is synthesized, is the second most common hereditary thrombotic disorder.

50.

D. A high D-dimer level indicates the presence of a thrombus (deep vein thrombosis, pulmonary embolism) but is not useful in determining the location. If a thrombus breaks away and travels to the brain, a stroke occurs, causing the symptoms described in this patient. Tissue plasminogen activator will activate the fibrinolytic system to lyse the clot. It should be administered within hours of onset of symptoms to prevent irreversible brain damage. Because of the small window of treatment time, a D-dimer performed on a possible stroke patient should be done STAT. Aspirin, warfarin, and heparin can be administered to prevent the formation of new clots but will not lyse existing clots.

51.

B. A heparin overdose can result in hemorrhage. If bleeding becomes life threatening, protamine sulfate can be given. Heparin will dissociate from antithrombin if protamine sulfate is administered, because heparin has a higher affinity for protamine sulfate. Vitamin K can be administered in the management of bleeding for patients who overdose with warfarin, which is a synonym for Coumadin®.

52.

A. Protein C, a glycoprotein produced in the liver, is a potent inhibitor of coagulation. The activation of protein C, by the thrombin/thrombomodulin complex, will cause the inactivation of factors V and VIII:C. Protein C and its cofactor, protein S, are vitamin K–dependent proteins.

53.

B. The prothrombin time test measures the coagulant activity of the extrinsic and common pathway factors of I, II, V, VII, and X. The reagent used for the prothrombin time test contains calcium and tissue thromboplastin. Thromboplastin is an extract of tissue such as brain or placenta. The activated partial thromboplastin time test measures all coagulation factors present in the intrinsic and common pathways except factor XIII. Calcium, a phospholipid source, and an activating agent, such as kaolin, silica, or celite, are present in the reagents used for the activated partial thromboplastin time.

54.

C. The D-dimer is a specific marker of fibrinolysis. A normal D-dimer test can be used to rule out the formation of a clot. The D-dimer test will be elevated when a clot has formed, factor XIII has cross-linked fibrin, and the fibrinolytic system is lysing the clot. The FDP detection test will be abnormal whether fibrin degradation products or fibrinogen degradation products are present. Because the D-dimer is normal in this patient, fibrin degradation products have not formed, but fibrinogen degradation products are present resulting in the elevated FDP result.

Specimen Acceptability

55.

B. A 9:1 ratio of blood to anticoagulant is needed for sodium citrate to bind all available calcium in the blood sample and prevent coagulation. When the 9:1 ratio is not maintained due to the tube not being full, excess sodium citrate present will bind reagent calcium in the test system. This will cause falsely prolonged PT and aPTT results.

56.

A. The results should be reported. A platelet estimate is obtained by multiplying the average number of platelets per oil immersion field (in an erythrocyte monolayer) by 20,000. This number is based on a normal erythrocyte count, which must be considered when comparing the platelet count and estimate. The estimate in this example is 180×10^9/L. This agrees with the platelet count.

57.

B. A clean venipuncture is required for coagulation testing. The description of the traumatic venipuncture indicates that the result might be invalid because of exposure to tissue thromboplastin/tissue factor, resulting in a falsely short test result. A tube that is not full would cause a falsely long time. A factor deficiency causes long clotting times. This 5-year-old would have the same reference range as adults.

58.

D. Platelets encircling neutrophils is a phenomenon referred to as platelet satellitosis. This "pseudothrombocytopenia" occurs when the blood of some individuals is anticoagulated with EDTA. Recollecting the specimen using sodium citrate often corrects this problem. If sodium citrate is used, the platelet count obtained must be multiplied by 1.1 for reporting purposes. Multiplying by 1.1 adds back the 10% loss of platelets seen when sodium citrate, with a 9:1 ratio of blood to anticoagulant, is used.

59.

C. Factors V and VIII are labile and deteriorate rapidly at room temperature. Blood for aPTT testing should be tested within 4 hours of draw. Sodium citrate is the appropriate anticoagulant for coagulation procedures.

60.

C. The lipemic plasma may interfere with the detection of fibrin clot formation by instruments measuring a change in optical density. The change in optical density may be insufficient for detection. The most accurate results in this situation can be obtained by performing the procedure on an electromechanical fibrin clot detection instrument, such as a fibrometer. Requesting a new specimen will be of no use because the redraw will most likely also be lipemic.

61.

B. A synonym for the fibrinogen group of factors is the consumed factors, because they are totally used up in clot formation. In the conversion of fibrinogen to fibrin, all of prothrombin is converted to thrombin. Thus, factors I, V, VIII, XIII, and II are all missing in serum.

Case Studies

62.

A. With the severe anemia and many schistocytes, a microangiopathic hemolytic anemia should be considered (TTP, HUS). HUS is seen in children after a gastrointestinal infection, frequently caused by *E. coli 0157:H7,* and results in renal damage. TTP is seen in young adults and is more common in women than men. TTP causes neurological damage. Patients with TTP have unusually large multimers of von Willebrand factor because they have a deficiency of a metalloprotease, ADAMTS-13, responsible for breaking down the multimers. These large multimers of vWF bind strongly to platelets, causing platelet aggregation and thrombotic complications in multiple organs.

63.

B. Acute idiopathic thrombocytopenic purpura is mainly seen in young children. A viral infection often precedes the onset of symptoms by several weeks. In 90% of patients with acute ITP, there is an increase in IgG immunoglobulin attached to the surface of the platelets. Spontaneous remission occurs in most patients within 2–6 weeks of the onset of the illness. A chronic form of ITP, believed to be a different disease, is seen in adults.

64.

D. The patient's platelet count is within the reference range, but the bleeding time is prolonged. This indicates a platelet function problem. The coagulation tests indicate a problem in the intrinsic clotting system (factors XII, XI, IX, VIII, Fitzgerald, and Fletcher). The one disorder in which both platelet function and the coagulant property of factor VIII:C are affected is von Willebrand disease. A synonym for von Willebrand factor is the ristocetin cofactor. In its absence, platelets will not aggregate with ristocetin. The platelet aggregation pattern confirms this diagnosis.

65.

A. Bernard-Soulier syndrome is a platelet adhesion defect that can be mistaken for von Willebrand disease. Platelets in this syndrome lack the glycoprotein Ib receptor, which is necessary for von Willebrand factor to attach to the platelet. Both disorders give identical platelet aggregation patterns. Bernard-Soulier syndrome is noted for giant platelets (note the increased MPV) and varying degrees of thrombocytopenia. Because von Willebrand factor is present in Bernard-Soulier syndrome, the aPTT is normal.

66.

B. Because the platelet count is within the reference range, the prolonged bleeding time is due to a qualitative platelet disorder. Poor clot retraction is characteristic of Glanzmann thrombasthenia. Clot retraction is normal in storage pool disease and von Willebrand disease. Christmas disease is caused by factor IX deficiency and does not affect platelet function. Glanzmann thrombasthenia can be further differentiated from von Willebrand disease by the platelet aggregation study results. The PT and aPTT results rule out a secondary hemostasis defect.

67.

D. Failure of normal plasma to correct the aPTT indicates the presence of a circulating inhibitor. A factor VIII inhibitor is associated with hemorrhagic problems and would result in noncorrection of the mixing study. Hemophilia A and von Willebrand disease are both caused by factor deficiencies that would result in correction when a 1:1 mixing study is performed. The lupus anticoagulant is a circulating inhibitor that prolongs the aPTT with little or no correction of the mixing study, but it is associated with thrombosis, not severe hemorrhagic problems.

68.

B. Coumarin interferes with the function of vitamin K in the synthesis of prothrombin group factors II, VII, IX, and X. Tests that measure one or more of these factors will be prolonged. These factors are synthesized but are nonfunctional. Of the factors affected by coumarin, IX, X, and II are measured in the aPTT. VII, X, and II are measured in the PT. The patient is on a nontherapeutic dose of coumarin, and the INR demonstrates this. Unfractionated heparin (administered intravenously) and fibrinolytic activators are not used on an outpatient basis.

69.

B. The required blood-to-anticoagulant ratio for coagulation testing is 9:1. If a volume of blood contains an elevated number of RBCs, generally considered to be a hematocrit greater than 55%, this ratio will be affected. Excess sodium citrate in the patient plasma, which acts as an anticoagulant by binding calcium ions, will bind the reagent calcium added back to the test plasma during the procedure. Falsely prolonged results are obtained. The specimen needs to be redrawn using less sodium citrate.

70.

C. Hemophilia A is inherited as a sex-linked recessive disorder of factor VIII:C. Mothers are carriers who pass the disease on to male offspring. This disorder is strictly a secondary hemostasis defect, so tests for primary hemostasis such as the bleeding time and platelet aggregation studies are normal. Factor deficiencies correct when a 1:1 mixing study is performed; presence of heparin in the sample would result in little or no correction of the mixing study. Aspirin affects platelets in primary hemostasis.

71.

C. An abnormal aPTT that corrects with a 1:1 mixing study, and a normal PT indicate a deficiency in the intrinsic coagulation pathway. Because factor XII is an activator of the fibrinolytic system, a deficiency can result in thrombosis. Presence of heparin in the sample would cause noncorrection of the mixing study. Factor VIII deficiency causes bleeding into muscles and joints, not thrombosis. Factor XIII can be eliminated because it is not measured in the aPTT or PT, and a deficiency would cause a bleeding disorder.

72.

B. DIC is a consumption coagulopathy in which the fibrinogen group of factors I, V, VIII, XIII, as well as II in the prothrombin group, is consumed in systemic clotting faster than the liver can synthesize them. When the clotting system is activated, the fibrinolytic system is simultaneously activated. Degradation products form causing the FDP and D-dimer tests for degradation products to be positive. The thrombin time measures fibrinogen and is prolonged due to low fibrinogen and the presence of degradation products. Antithrombin is quickly depleted in an attempt to stop systemic clotting. Platelets are consumed in clotting, and schistocytes form as red blood cells fragment as they encounter fibrin strands in circulation.

73.

A. Primary fibrinogenolysis is an unusual disorder in which the fibrinolytic system is activated in the absence of clot formation. Plasmin degrades factors V, VIII, and fibrinogen. The D-dimer test is positive if fibrin degradation products are present; they are absent in this disorder. The FDP test is positive in the presence of either fibrin *or* fibrinogen degradation products. Tests that are abnormal in DIC due to the systemic clotting are normal in primary fibrinogenolysis.

74.

D. Up to 5% of patients receiving unfractionated heparin therapy for more than 5 days develop an IgG antibody that can cause platelet activation, leading to thrombosis in the microvasculature. If this occurs, the platelet count drops quickly. Patients receiving heparin therapy should be monitored with daily platelet counts. Direct thrombin inhibitors such as lepirudin, bivalirudin, and argatroban can be used in place of heparin. Enoxaparin sodium, a low-molecular-weight heparin, is contraindicated as a treatment for heparin-induced thrombocytopenia. Aspirin therapy causes a qualitative, not quantitative, platelet defect.

75.

D. The patient's symptoms suggest a possible deep vein thrombosis. An increased incubation time of patient plasma with aPTT reagent using kaolin is considered to be a screening test for prekallikrein deficiency. The additional incubation time allows for increased activation of the contact factors, thus shortening the aPTT. Because prekallikrein is an activator of fibrinolysis, a deficiency causes an increased thrombotic risk in the patient. A factor II deficiency would not give a normal PT result. A factor XIII deficiency can only be detected by performing a 5.0 M urea solubility test. A factor VIII inhibitor would cause noncorrection of the mixing study. All conditions listed except prekallikrein deficiency would cause bleeding symptoms.

review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

1. Color Plate 21■ depicts a monomeric immunoglobulin molecule. The portion of the molecule indicated by the dotted red circle and the red arrow is called the
   A. Fab fragment
   B. Fc fragment
   C. Heavy chain
   D. Hinge region

2. A hapten is
   A. Half of an immunoglobulin molecule
   B. A carrier molecule for an antigen that is not antigenic alone
   C. An immunoglobulin functional only in the presence of complement
   D. A determinant capable of stimulating an immune response only when bound to a carrier

3. Which of the following is characteristic of B cells?
   A. Phagocytic
   B. Participate in antibody-dependent cellular cytotoxicity (ADCC) reactions
   C. Contain surface immunoglobulins
   D. Secrete the C5 component of complement

4. A lymphokine is
   A. A soluble mediator produced by granulocytes and affecting lymphocytes
   B. A soluble mediator produced by lymphocytes
   C. A soluble mediator produced by plasma cells
   D. An antibody that reacts with lymphocytes

5. Monocytes and macrophages play a major role in the mononuclear phagocytic system. For an antibody-coated antigen to be phagocytized, what part of the antibody molecule fits into a receptor on the phagocytic cell?
   A. Fc region
   B. Fab region
   C. Hinge region
   D. Variable region

6. Cell-mediated immunity is primarily mediated by
   A. B cells
   B. T helper cells
   C. Plasma cells
   D. Dendritic cells

7. The HLA complex is located primarily on
   A. Chromosome 3
   B. Chromosome 6
   C. Chromosome 9
   D. Chromosome 17

8. HLA antigens are found on
   A. All nucleated cells
   B. Red blood cells only
   C. Solid tissue only
   D. White blood cells only

9. Which of the following is more likely to be diagnostic of an acute infection?
   A. A total acute antibody titer of 2 followed by a convalescent titer of 16
   B. A total acute antibody titer of 80 followed by a convalescent titer of 40
   C. A total antibody titer of 80
   D. An IgG antibody titer of 80

10. A young woman shows increased susceptibility to pyogenic infections. Upon assay, she shows a low level of C3. Which of the following statements is probably *true?*
    A. She has an autoimmune disease with continual antigen-antibody activity causing consumption of C3.
    B. She has DiGeorge syndrome.
    C. She has decreased production of C3.
    D. She may produce an inactive form of C2, a precursor of C3.

11. What is the predominant type of antibody found in the serum of neonates born after full-term gestation?
    A. Infant IgA
    B. Infant IgG
    C. Infant IgM
    D. Maternal IgG

12. An important part of the nonspecific immune response is(are)
    A. B cells
    B. Basophils
    C. Complement cascade
    D. Cytotoxic T lymphocytes

13. The major class of immunoglobulin found in adult human serum is
    A. IgA
    B. IgE
    C. IgG
    D. IgM

14. Which class of immunoglobulin possesses delta heavy chains?
    A. IgA
    B. IgD
    C. IgE
    D. IgG

15. Which class of immunoglobulin possesses 10 antigenic binding sites?
    A. IgA
    B. IgD
    C. IgG
    D. IgM

16. Color Plate 22■ represents a dimeric IgA molecule. The structure printed in red and indicated by the red arrow is called the
    A. J-piece
    B. Hinge region
    C. Heavy chain
    D. Light chain

17. Which class of immunoglobulin binds to basophils and mast cells to mediate immediate hypersensitivity reactions?
    A. IgA
    B. IgD
    C. IgE
    D. IgG

18. Type I hypersensitivity is
    A. Associated with complement-mediated cell lysis
    B. Due to immune complex deposition
    C. Mediated by activated macrophages
    D. An immediate allergic reaction

19. When performing the enzyme-multiplied immunoassay technique (EMIT), how is the ligand in the patient's serum detected?
    A. Agglutinates by binding to antibody-coated latex beads
    B. Binds to enzyme-labeled antibody
    C. Competes with enzyme-labeled antigen for binding to a specific antibody
    D. Forms antibody-antigen complex and precipitates

20. Severe combined immunodeficiency (SCID) is an
    A. Immunodeficiency with decreased B cells and neutrophils
    B. Immunodeficiency with lymphocytopenia and eosinophilia
    C. Immunodeficiency with decreased or dysfunctional T and B cells
    D. Immunodeficiency with decreased lymphocytes and decreased complement concentration

21. An example of immune injury due to the deposition of antigen-antibody complexes is
    A. Acute glomerulonephritis
    B. Bee-sting allergy
    C. Contact dermatitis
    D. Penicillin allergy

22. The serologically detectable antibody produced in rheumatoid arthritis (RA) is primarily of the class
    A. IgA
    B. IgE
    C. IgG
    D. IgM

23. In bone marrow transplantation, immunocompetent cells in the donor marrow may recognize antigens in the recipient and respond to those antigens. This phenomenon is an example of
    A. Acute rejection
    B. Chronic rejection
    C. Graft versus host disease
    D. Hyperacute rejection

24. Multiple myeloma is a
    A. Lymphoproliferative disease of T cells
    B. Cancer of plasma cells characterized by increased antibody concentration
    C. Lymphoproliferative disease resulting in a decrease in antibody production
    D. Cancer of monocytes characterized by increased kappa and lambda chain synthesis

25. Which one of the following describes a direct immunofluorescence assay?
    A. Conjugated reagent antigen reacts with antibodies to form antigen-antibody complexes
    B. Antigens react with unlabeled antibody forming antigen-antibody complexes that attach to labeled antibodies
    C. A dye is attached to a molecule and it reacts with an immune complex to produce a color
    D. Conjugated reagent antibody reacts with antigen to form antigen-antibody complexes

26. In individuals allergic to pollen, hyposensitization protocols may be initiated. These individuals receive injections of
    A. Allergen
    B. Pooled human antisera
    C. Monoclonal antibody directed against human T cells
    D. Monoclonal antibody directed against human B cells

27. After exposure to antigen, the first antibodies that can be detected belong to the class
    A. IgA
    B. IgE
    C. IgG
    D. IgM

28. Corneal tissue may be transplanted successfully from one patient to another because
    A. The cornea is nonantigenic
    B. Corneal antigens do not activate T cells
    C. Anticorneal antibodies are easily suppressed
    D. The cornea occupies a privileged site not usually seen by the immune system

29. A kidney transplant from one identical twin to another is an example of a(n)
    A. Allograft
    B. Autograft
    C. Isograft
    D. Xenograft

30. In Bruton disease, measurement of serum immunoglobulins would show
    A. Elevated levels of IgE
    B. Elevated levels of IgG
    C. Normal levels of IgG and IgM but reduced levels of IgA
    D. The absence of all immunoglobulins

31. Diagnosis of group A streptococci (*Streptococcus pyogenes*) infection is indicated by the presence of
    A. Anti-protein A
    B. Anti-DNase B
    C. Anti-beta-toxin
    D. C-reactive protein

32. A molecule found in human serum sometimes used as a tumor marker is
    A. α-Fetoprotein
    B. HBsAg
    C. Biotin
    D. CD1

33. Which cell is the principal source of interleukin 2?
    A. B cell
    B. T cell
    C. Monocyte
    D. Plasma cell

34. Diagnostic reagents useful for detecting antigen by the coagglutination reaction may be prepared by binding antibody to killed staphylococcal cells via the Fc receptor of staphylococcal protein A. The class of antibody bound by this protein is
    A. IgA
    B. IgD
    C. IgG
    D. IgM

35. A major advantage of passive immunization compared to active immunization is that
    A. Antibody is available more quickly
    B. Antibody persists for the life of the recipient
    C. IgM is the predominant antibody class provided
    D. Oral administration can be used

36. The strength with which a multivalent antibody binds a multivalent antigen is termed the
    A. Affinity
    B. Avidity
    C. Reactivity
    D. Valence

37. How does the secondary humoral immune response differ from the primary response?
    A. The lag phase (the time between exposure to immunogen and production of antibody) is longer in the secondary immune response.
    B. IgM is the predominant antibody class produced in the secondary immune response.
    C. The antibody levels produced are higher in the secondary immune response.
    D. Cytotoxic T lymphocytes play an important role in the secondary response.

38. After activation of the complement system, leukocytes and macrophages are attracted to the site of complement activation by
    A. C1
    B. C5a
    C. C8
    D. IgM

39. The type of immunity that follows the injection of an immunogen is termed
    A. Artificial active
    B. Natural active
    C. Artificial passive
    D. Innate

40. The type of immunity that follows the injection of antibodies synthesized by another individual or animal is termed
    A. Artificial active
    B. Natural adaptive
    C. Artificial passive
    D. Natural passive

41. Innate immunity includes
    A. Anamnestic response
    B. Antibody production
    C. Cytotoxic T cell activity
    D. Phagocytosis by polymorphonuclear cells

42. The agglutination pattern shown in Color Plate 23■ was observed while performing an antibody titration. This agglutination pattern is an example of
    A. A prezone reaction
    B. A prozone reaction
    C. A postzone reaction
    D. Incomplete complement inactivation

43. The antibody most frequently present in systemic lupus erythematosus is directed against
    A. Surface antigens of bone marrow stem cells
    B. Surface antigens of renal cells
    C. Nuclear antigen
    D. Myelin

44. The rapid plasma reagin assay for syphilis does not need to be read microscopically because the antigen is
    A. Cardiolipin
    B. Complexed with latex
    C. Complexed with charcoal
    D. Inactivated bacterial cells

45. The Venereal Disease Research Laboratory (VDRL) test for syphilis is classified as a(n)
    A. Agglutination reaction
    B. Flocculation reaction
    C. Hemagglutination reaction
    D. Precipitation reaction

46. One cause of a false-positive VDRL test is
    A. Brucellosis
    B. *Treponema pallidum* infection
    C. Rocky Mountain spotted fever
    D. Systemic lupus erythematosus

47. The portion of an antigen that binds to an antibody or T cell receptor is called a(n)
    A. Allergin
    B. Avidin
    C. Epitope
    D. Valence

48. Identical antibodies produced from a single clone of plasma cells describes
    A. Reagin
    B. Cold agglutinins
    C. Heterophile antibodies
    D. Monoclonal antibodies

49. IgM antibodies react well in complement fixation (CF) tests. Because of this, CF tests for antibodies should
    A. Be positive early in the course of the disease
    B. Be useful in identifying antibodies responsible for a delayed hypersensitivity reaction
    C. Be useful in identifying antibodies responsible for anaphylactic reactions
    D. Detect transplacental antibodies

50. Which of the following serologic tests is commonly performed by an immunofluorescence method?
    A. Anti-HBs
    B. Antinuclear antibody (ANA)
    C. Antistreptolysin O (ASO)
    D. C-reactive protein (CRP)

51. The Fab portion of an antibody
    A. Binds T cell receptor
    B. Consists of two light chains only
    C. Consists of two heavy chains only
    D. Contains the hypervariable reagion

52. In the enzyme-linked immunosorbent assay (ELISA), the visible reaction is due to a reaction between
    A. Enzyme and antibody
    B. Enzyme and substrate
    C. Fluorescent dye and antigen
    D. Latex particles and antibody

53. Elevated IgE levels are typically found in
    A. Type I hypersensitivity reactions
    B. Type II hypersensitivity reactions
    C. Type III hypersensitivity reactions
    D. Type IV hypersensitivity reactions

54. Loss of self-tolerance results in
    A. Autoimmune disease
    B. Graft-versus-host disease
    C. Immunodeficiency
    D. Tumors

55. A human cell with CD8 on its surface is most likely a
    A. B cell
    B. Monocyte
    C. T helper cell
    D. Cytotoxic T cell

56. Which of the following statements about immunoglobulin light chains is *true?*
    A. Each immunoglobulin monomer has either one kappa or one lambda chain.
    B. There are two types: kappa and lambda.
    C. They consist of constant regions only.
    D. They form part of the Fc fragment.

57. Which of the following statements applies to the Fc fragment of an immunoglobulin molecule?
    A. It consists of the entire heavy chain.
    B. It contains the variable region of the heavy chain.
    C. It contains the antigen binding sites of the molecule.
    D. It is the region of the molecule that binds to receptors on various white blood cells.

58. Monoclonal antibodies are produced by
    A. Cultured T cells
    B. Human plasma cells
    C. Mouse plasma cells
    D. Hybridomas

59. Antibodies that bind to the same epitope are of the same
    A. Allotype
    B. Autotype
    C. Idiotype
    D. Isotype

60. Skin testing is a useful diagnostic tool in a number of disorders, such as tuberculosis. Which of the following statements about skin testing is *true?*
    A. A positive test depends on preformed antibody.
    B. Reactivity to a particular antigen may be transferred from one individual to another by sensitized lymphocytes.
    C. The intensity of the response correlates directly with the clinical activity of the disease.
    D. The maximum response will occur immediately.

61. The activity of natural killer (NK) cells
    A. Does not require previous exposure to an antigen
    B. Involves phagocytosis and killing of bacteria
    C. Requires interaction with cytotoxic T cells
    D. Requires interaction with B cells

62. Interaction between B and T helper cells involves
    A. MHC II molecule on B cell binding to MHC I molecule on the T cell
    B. MHC II molecule on B cell binding to CD3 on the T cell
    C. Foreign antigen on B cell binding to T cell receptor
    D. CD3 molecule on B cell binding to T cell receptor

63. Which of the following is a characteristic of T cells?
    A. Synthesize antibody
    B. Mature in the thymus
    C. Able to bind unprocessed antigen
    D. Primarily protect against extracellular parasites

64. The primary mechanism responsible for pathology in systemic lupus erythematosus is
    A. Allergic reaction to foreign molecules
    B. Antibodies directed against self antigens
    C. Polyclonal activation of cytotoxic T cells
    D. Lack of intracellular killing after neutrophil phagocytosis of bacteria

65. Which complement protein is present in the greatest concentration in human serum?
    A. C1
    B. C2
    C. C3
    D. C4

66. An autoimmune disease causing destruction of pancreatic cells can result in
    A. Hashimoto disease
    B. Multiple sclerosis
    C. Myasthenia gravis
    D. Type 1 diabetes

67. An Ouchterlony gel diffusion plate is depicted in Color Plate 24■. The center well contains antibody, and the peripheral wells contain antigens labeled 1 through 4. What is the relationship between the antigens in wells 2 and 3?
    A. 2 is part of 3.
    B. 3 is part of 2.
    C. They are identical.
    D. They are unrelated.

68. An Ouchterlony gel diffusion plate is depicted in Color Plate 24■. The center well contains antibody, and the peripheral wells contain antigens labeled 1 through 4. What is the relationship between the antigens in wells 2 and 4?
    A. Cannot be determined.
    B. They are identical.
    C. They are unrelated.
    D. They react incompletely with the antibody.

69. Which of the following complement proteins is part of the membrane attack complex?
    A. C1
    B. C3
    C. C4
    D. C5

70. Which of the following is characteristic of contact hypersensitivity reactions?
    A. Caused by preformed IgE antibody
    B. Characterized by infiltration of neutrophils into the area of reaction
    C. The primary symptoms often occur in the respiratory tract.
    D. Usually due to a hapten

71. Which of the following statements about the test for C-reactive protein (CRP) is *true?*
    A. It correlates with neutrophil phagocytic function.
    B. It is an indicator of ongoing inflammation.
    C. It is diagnostic for rheumatic fever.
    D. Levels decrease during heart disease.

72. In the classical pathway of complement activation,
    A. C3 is activated by binding C-reactive protein
    B. The sequence of activation is C1, C2, C3, C4
    C. C1q is activated by the presence of a single Fab region
    D. Activation by antibody requires one IgM or two IgG molecules

73. The alternative complement pathway
    A. Can be activated by bacterial capsule polysaccharides
    B. Uses C5b as a C3 convertase
    C. Bypasses steps C3 through C5
    D. Is inactivated by properdin

74. A cut on a person's finger becomes contaminated with the bacterium *Staphylococcus aureus.* The first response by the immune system consists of activity of
    A. B cells
    B. Monocytes
    C. Neutrophils
    D. T cells

75. Incompatible blood transfusions are examples of
    A. Type I hypersensitivity reactions
    B. Type II hypersensitivity reactions
    C. Type III hypersensitivity reactions
    D. Type IV hypersensitivity reactions

76. A soluble antigen and soluble antibody reacting to form an insoluble product describes
    A. Agglutination reactions
    B. Heterophile reactions
    C. Labeled reactions
    D. Precipitation reactions

77. Which of the following is an example of a treponemal antigen test used for the diagnosis of syphilis?
    A. CRP
    B. RPR
    C. VDRL
    D. FTA-ABS

78. A serum sample is positive for HBsAg. This result indicates that the person from whom the serum was taken
    A. Had a hepatitis B infection in the past but overcame the infection
    B. Has either active or chronic hepatitis B infection
    C. Was immunized recently against the hepatitis B virus
    D. Is not infectious for the hepatitis B virus

79. What is the indicator system used in the complement fixation test?
    A. Sensitized sheep red blood cells
    B. Fluorescent-labeled antihuman globulin
    C. Enzyme-labeled antihuman globulin
    D. Guinea pig complement

80. The isotype of an immunoglobulin antibody
    A. Is defined by the heavy chain
    B. Is defined as different alleles of the same antibody type (e.g., IgG)
    C. Is constant for all immunoglobulins of an individual
    D. Is the variation within the variable region

81. A patient report states the presence of serum antibodies to OspC. What disease does the patient most likely have?
    A. Syphilis
    B. Strep throat
    C. Lyme disease
    D. Rubella

82. Patient serum is mixed with a suspension of guinea pig antigen. When the sample is then mixed with horse red blood cells, agglutination occurs. This is suggestive of an infection caused by
    A. *Borrelia burgdorferi*
    B. Hepatitis B virus
    C. Hepatitis C virus
    D. Epstein-Barr virus

83. Hashimoto disease is an autoimmune disease primarily involving the
    A. Kidneys
    B. Liver
    C. Lungs
    D. Thyroid gland

84. Rheumatic fever sometimes occurs after group A streptococcal infections. In this condition, an autoimmune response attacks the tissue of the heart valves. This phenomenon is an example of
    A. Epitope spreading
    B. Molecular mimicry
    C. Polyclonal B cell activation
    D. Preferential activation of T helper cells

85. "Superantigens" are toxins produced by some strains of *Staphylococcus aureus* and group A streptococci and cause damage by
    A. Molecular mimicry
    B. Polyclonal T cell activation
    C. Lysing white blood cells and platelets
    D. Lysing red blood cells

86. The first serologic marker to appear in patients with acute hepatitis B virus infection is
    A. Anti-HB
    B. Anti-HBc
    C. Anti-HBe
    D. HBsAg

87. A living donor is being sought for a child who requires a kidney transplant. The best odds of finding an MHC-compatible donor occur between the child and
    A. A sibling (brother or sister)
    B. An unrelated individual
    C. The child's father
    D. The child's mother

88. Cells that can act as antigen-presenting cells for exogenous antigens include
    A. All nucleated cells
    B. Endothelial cells
    C. B lymphocytes
    D. T lymphocytes

89. In patients with human immunodeficiency virus infection, immune status can be monitored by measuring the ratio of
    A. CD3+ cells to CD8+ cells
    B. CD4+ cells to CD8+ cells
    C. Lymphocytes to monocytes
    D. T cells to B cells

90. Why does vaccination against hepatitis B virus (HBV) also prevent hepatitis D virus (HDV) infections?
    A. An immunogen from HBV in the vaccine is also associated with HDV.
    B. The HBV vaccine induces formation of heterophile antibodies that cross react with HDV.
    C. The HBV vaccine stimulates liver cells to produce antiviral molecules active against all hepatitis viruses.
    D. HDV requires the host to be concurrently infected with HBV.

91. B lymphocytes and T lymphocytes are derived from
    A. Hematopoietic stem cells
    B. Macrophages or monocytes
    C. Mucosa-associated lymphoid tissue
    D. The fetal liver

92. Contact dermatitis is mediated by
    A. B lymphocytes
    B. T lymphocytes
    C. Macrophages
    D. Polymorphonuclear cells

93. In a competitive radioimmunosorbent test (RIST), what does a high signal suggest?
    A. The patient sample has a low concentration of IgE.
    B. The patient sample has a low concentration of IgM.
    C. The patient sample has a high concentration of IgE.
    D. The patient sample has a high concentration of total antibody.

94. An antibody titration is depicted in Color Plate 25■. In this titration, a 0.2 mL aliquot of a patient's serum sample was added to 0.8 mL of saline, and this mixture was placed into tube #1. A 0.5 mL sample was removed from tube #1 and placed into tube #2, containing 0.5 mL of saline. This procedure was repeated through tube #10. The dilutions were assayed for antibody to an infectious agent. How should the antibody titer be reported?
    A. 256
    B. 512
    C. 640
    D. 1280

95. In a chemiluminescent immunologic assay, what is the signal detected?
    A. Light
    B. An electric signal
    C. A purple-colored compound
    D. A yellow-colored compound

96. A 28-year-old female complains to her family physician of abdominal pain, loss of appetite, and low-grade fever. Physical examination reveals abdominal tenderness and a low-grade fever. Her physician orders a hepatitis profile and obtains the results below.

Anti-HAV	Nonreactive
Anti-HBc	Reactive
Anti-HBs	Nonreactive
HBsAg	Reactive
HBeAg	Reactive
Anti-HCV	Nonreactive

Which of the following is the most likely conclusion?

A. Acute HAV infection
B. Acute HBV infection
C. Chronic HBV infection
D. Immunity to HBV due to past infection

97. An 11-year-old female presents with fever, sore throat, lethargy, and tender cervical lymphadenopathy. Relevant findings include splenomegaly and lymphocytosis, with many large reactive (atypical) lymphocytes. A heterophile antibody test was negative. Further laboratory results were as follows:

	IgG Titer	IgM Titer
Cytomegalovirus (CMV)	20	0
Epstein-Barr virus (EBV) VCA	0	80
Mono spot	0	0

What conclusion can be made concerning the diagnosis?

A. Acute CMV infection
B. Acute EBV infection
C. Chronic CMV infection
D. Chronic EBV infection

98. A male infant had been well until about 5 months of age, at which time he was diagnosed as having otitis media and bronchitis caused by *Haemophilus influenzae*. Over the next several months he presented with streptococcal pneumonia several times. At 10 months of age a serum protein electrophoresis showed a virtual lack of gamma globulins. Quantitative serum levels were as follows: 75 mg/dL IgG and undetectable levels of IgM, IgA, and IgE. There were a normal number of T cells, and they exhibited normal mitogen stimulation. What disease does this child most likely suffer from?

A. Combined immunodeficiency
B. DiGeorge syndrome
C. Iatrogenic immunodeficiency
D. X-linked agammaglobulinemia

99. A 25-year-old male presents to his family physician complaining of fatigue, diarrhea, and weight loss of a few months duration. On physical examination the patient is found to have a fever and abdominal discomfort. Laboratory results indicate a white blood cell count of 14.3×10^9/L (reference range $4.8–10.8 \times 10^9$/L). Assays for HBSAg and anti-HCV are negative. An ELISA test for antibodies to the human immunodeficiency virus (HIV) performed on the patient's serum is found to be reactive. What step should be taken next?
    A. Call the physician with the HIV result.
    B. Repeat the HIV ELISA test on the sample.
    C. Test the patient's serum for anti-HBs.
    D. Contact the patient to collect a second sample.

100. A 38-year-old woman visited her physician because of fatigue, fever, and joint pain (proximal interphalangeal, wrist, and knee joints). She also noticed sensitivity to the sun and reported having a rash following recent exposure. The physician noted a rash over her nose and cheeks. Laboratory results included white blood cell count 5.5×10^9/L (reference range $4.8–10.8 \times 10^9$/L) and red blood cell count 4.5×10^{12}/L (reference range $4.0–5.4 \times 10^{12}$/L). Urinalysis results were within reference ranges, except for 4+ protein and 1+ RBCs, 0–3 hyaline casts/lpf and 0–1 RBC cast/lpf on microscopic examination. Which of the following tests would be most helpful in diagnosing this patient's condition?
    A. Anti-nuclear antibody
    B. α-Fetoprotein
    C. Anti-streptolysin O
    D. Hepatitis profile

answers & rationales

1.

A. The basic structure of all immunoglobulins is two light chains joined to two heavy chains by disulfide bonds. The amino terminus of both the heavy and light chains, together, constitutes the Fab fragment (fragment of antigen binding). The carboxy-terminus of the heavy chains constitutes the Fc fragment. The hinge region is the area at the center of the "Y," near the carboxy-terminus of the light chains.

2.

D. Haptens are substances that are not immunogenic by themselves. These molecules are not large or complex enough to stimulate the immune system. When bound to a carrier, they are capable of stimulating a specific immune response.

3.

C. B cells carry surface immunoglobulins that react to a specific antigen. The antigen can then be internalized processed and presented to an appropriate T helper cell. B cells are not phagocytic, nor do they participate in antibody-dependant cellular cytotoxicity (ADCC) reactions. Complement proteins are secreted by hepatocytes.

4.

B. Lymphokines are soluble mediators of immune reactions. They are produced most often by T lymphocytes. Antibodies are produced by plasma cells.

5.

A. The Fc region of an IgG molecule fits into an Fc receptor (FcR) on macrophages and monocytes. The Fc receptor binds to specific amino acid residues in the Fc region of the immunoglobulin. The variable region of immunoglobulin binds to the antigen.

6.

B. T helper cells are the primary mediators of cell-mediated immunity (CMI). They secrete several different lymphokines that stimulate a number of other cells, such as cytotoxic T lymphocytes and monocytes. B cells differentiate into plasma cells during a humoral-mediated immune response. Dendritic cells are important antigen presenting cells, but they are not the primary mediators of a CMI response.

7.

B. The HLA system is part of a larger region known as the major histocompatibility complex. It is located on chromosome 6. The region is located on the short arm of the chromosome. Chromosome 15 contains one HLA gene, B2M.

8.

A. Human leukocyte antigens (HLAs) are a group of antigens originally described on human white cells. It is now known that they are found on all nucleated cells of the body, including solid tissue cells. HLAs are not found on red blood cells.

9.

A. The most significant indicator of acute or recent infection is the presence of a rising antibody titer. A fourfold or greater rise in titer, from 2 to 16, is significant. Even relatively high antibody titers of IgG may indicate past infection. IgM is produced first following infections, so a high IgM titer is also suggestive of an acute infection.

10.

C. C3 may be decreased due to a genetic defect that causes deficient production. In certain autoimmune disorders, such as systemic lupus erythematosus, continual complement activation leads to low levels; however, susceptibility to pyogenic infections is not a feature of autoimmune diseases. DiGeorge syndrome is a deficiency in T cells, and complement protein C2 is not a precursor of C3.

11.

D. Antibody production is immunogen induced. Because the fetus develops in a sequestered site, it makes very little immunoglobulin. Maternal IgG crosses the placenta and is the primary antibody found in infant's circulation.

12.

C. Important parts of an animal's nonspecific immune response include phagocytosis, inflammation, and complement activation. In a nonspecific immune response, the animal responds in much the same way to all invaders. B cells and cytotoxic T lymphocytes responded to specific antigens and are, therefore, involved in the specific immune response. Basophils are involved in type I hypersensitivity reactions.

13.

C. Immunoglobulin G is the predominant class of immunoglobulin found in serum. It accounts for approximately 80% of the total serum immunoglobulin. The normal range is 800–1600 mg/dL.

14.

B. The heavy chains divide human immunoglobulin molecules into separate classes and subclasses. The delta (Δ) heavy chain corresponds to IgD. The remaining classes IgA, IgE, IgG, and IgM correspond to α, ε, γ, and μ, respectively.

15.

D. The IgM molecule is a pentamer that contains 10 binding sites. However, the actual valence falls to 5 with larger antigen molecules, probably because of steric restrictions. IgA, IgG, IgD, and IgE monomers each have two antigenic binding sites.

16.

A. IgA is found in mucous secretions as a dimer stabilized by the J-piece. IgA is synthesized locally by plasma cells and dimerized intracellularly. IgM is also held together by a J-piece, but it exists as a pentamer.

17.

C. Mast cells and basophils have surface receptors (FcεRI) for the Fc portion of IgE. When IgE molecules, attached to the surface of mast cells and basophils, bind the allergen they are specific for, this triggers the cells to degranulate, producing the symptoms of immediate type I hypersensitivity. The main function of IgE appears to be the ability to trigger an immune response, thereby recruiting plasma factors and effector cells to areas of trauma or parasite infection.

18.

D. Type I hypersensitivity reactions occur immediately after second exposure to an allergen. On the first, or primary, exposure, IgE specific to the allergen is produced. The IgE binds to Fc receptors on the surface of basophils and mast cells. Immune complexes and complement are not involved in the response.

19.

C. In the EMIT, a ligand (antigen) in a sample competes with an enzyme-labeled ligand for binding to a specific antibody. The labeled ligand is designed so that following antibody binding, the enzyme is inactive. As the ligand concentration in the test sample increases, more enzyme-labeled ligand remains unbound, resulting in greater enzyme activity.

20.

C. SCID is defined as a condition in which adaptive immune responses (i.e., cell-mediated and humoral-mediated immune responses) do not occur because of a lack of T and B cell activity. A number of genetic defects can lead to this condition. Children born with SCID need to live in a *sterile environment*, and they have a short life expectancy.

21.

A. Acute glomerulonephritis is caused by the presence of a soluble circulating antigen (Ag) that provokes and combines with antibody (Ab). As these Ag-Ab complexes reach a critical size, they are deposited in the glomerular membranes of the kidney. Upon deposition, an acute inflammatory reaction occurs because of complement activation. Bee-sting and penicillin allergies are examples of IgE-mediated anaphylactic reactions. Contact dermatitis is mediated by T cells, not antibody.

22.

D. Rheumatoid factor (RF) is an immunoglobulin that reacts with antigenic determinants on an IgG molecule. Although they may be of several types, the one that is easily serologically detectable is IgM. This is because of the agglutination activity of the molecule. RF tests are commonly used in the diagnosis of rheumatoid arthritis.

23.

C. Bone marrow transplants by their nature contain immunologically competent cells: B cells and T cells in particular. Unless the transplanted marrow is HLA-matched perfectly to the donor, the immunocompetent cells in the transplant will recognize and react against the nonself HLAs of the recipient's tissues. This phenomenon is known as graft-versus-host disease, because the graft attempts to reject its host. Acute rejection, chronic rejection, and hyperacute rejection are examples of mechanisms a recipient's immune system uses to reject a graft.

24.

B. Plasma cells are normally end-stage cells; they live a few days and die. During multiple myeloma, plasma cells become cancerous and continue to secrete antibody. The cells also secrete excess light chains that can be found in the urine; these proteins are called Bence Jones proteins.

25.

D. In a direct immunofluorescence assay, a fluorescent molecule is linked to an antibody. This complex is often called a conjugate. Clinical material is fixed onto a microscope slide, and the conjugate is added. After a wash step, the slide is examined with a microscope using UV light. If antigen specific to antibody was present in the clinical specimen, fluorescence will be seen.

26.

A. Hyponsensitization, allergy injections, involves the administration of gradually increasing concentrations of an allergen. The goal is for the patient to become tolerant of the allergen and no longer exhibit an allergic response to the allergen. It is hypothesized that patients will ultimately develop high concentrations of IgG to the allergen, blocking IgE from binding and thereby preventing the allergic reaction.

27.

D. The first B cells to respond to antigen differentiate into plasma cells that produce IgM antibody. Later in the immune response, stimulated B cells undergo a phenomenon called "class switching" and begin to produce antibodies of the IgG, IgA, and IgE classes. High concentration of IgM in patient serum is indicative of a recent infection.

28.

D. Corneas are readily transplanted from one individual to another. This is because the cornea is nonvascularized and is a sequestered site. Thus the immune system of the host does not "see" the cornea and recognize it as foreign.

29.

C. Identical twins have the same genetic makeup. Grafts between them would be isografts or syngeneic grafts. Autografts are transplantations from one site to another in the same individual. Xenograft refers to transplantation between different species. Transplantation between two nonidentical individuals of the same species is called an allograft.

30.

D. Bruton disease is a congenital form of agammaglobulinemia. It is a sex-linked phenomenon that affects males. Because B cells are not produced, affected males have levels of IgA, IgD, IgE, and IgM undetectable by routine assays. IgG may be absent or present at very low levels.

31.

B. The serological diagnosis of group A streptococcal infection can be made by demonstrating anti-DNase B. The antistreptolysin O (ASO) assay can also be used; however, ASO response is poor in skin infections. C-reactive protein is an acute-phase protein indicating inflammation.

32.

A. α-Fetoprotein (AFP) and carcinoembryonic antigen (CEA) are oncofetal antigens that become expressed after malignant transformation. Approximately 70% of patients with primary hepatoma have elevated levels of AFP. However, the major use of determining AFP levels is in monitoring patients undergoing cancer treatment.

33.

B. Interleukin 2 (IL-2) is a lymphokine produced by activated T helper cells. IL-2 principally affects T cells, including the cell that released IL-2, acting on its target cells via the IL-2 receptor. This receptor is not present on resting cells.

34.

C. Staphylococcal protein A binds only the IgG class (subclasses IgG1, IgG2, and IgG4) of immunoglobulin. Binding occurs via the Fc portion of the antibody molecule, leaving the Fab portion available to bind antigen in an immunologic assay. Binding of the Fab portion to test antigen causes agglutination of the staphylococcal cells (coagglutination).

35.

A. In passive immunization, preformed antibody is delivered to the recipient, making the antibody available immediately. In active immunization, a period of days is required before antibody production occurs. Passive immunity is short-lived, in contrast to the possibly lifelong persistence of actively induced antibody. Because passive immunization involves the transfer of antibodies, the oral route cannot be used—antibodies are digested in the gastrointestinal tract. The antibodies administered by passive immunization consist largely of the IgG class.

36.

B. "Avidity" is used to describe the strength of binding between a multivalent antibody and multivalent antigen. "Affinity" describes the bond between a single antigenic determinant and an individual combining site. "Valence" refers to the number of antigenic determinants on an antigen.

37.

C. The secondary immune response is characterized by the predominance of IgG over IgM. In addition, because of the formation of memory cells following the primary response, the secondary response occurs much more quickly and strongly. This is the basis for immunization as a protection against various infectious diseases. Cytotoxic T lymphocytes are not involved in humoral immunity.

38.

B. The complement-activation product C5a is chemotactic for neutrophils and macrophages. Neither C1 nor C8 (which occur in the plasma before complement activation) possesses such chemotactic properties. IgM antibody, although capable of activating complement by the classical pathway, is not a chemotactic factor for phagocytic cells.

39.

A. Active immunity follows exposure to an antigen that stimulates the recipient to develop his or her own immune response. Vaccines are an example of artificial immunity in that the animal was exposed to the immunogen by the actions of a healthcare provider (unnatural). Surviving infections can result in natural active immunity. Protection is due to the formation of memory cells.

40.

C. Artificial passive immunity results following the injection of antibody synthesized by another individual or animal. This type of immunity is only temporary but may be very important in providing "instant" protection from an infectious agent before the recipient would have time to actively synthesize antibody. The injected antibodies are treated as foreign proteins and are eventually cleared from the body.

41.

D. Innate, or nonspecific, immunity refers to host defenses that are in general present at birth and do not require immunogen stimulation. Phagocytosis of bacteria by polymorphonuclear cells is an example. Cytotoxic T cell activity is part of the adaptive cell-mediated immune response, and antibody production is the mechanism of protection in the adaptive humoral-mediated immune response.

42.

B. Prozone occurs when an extremely high titer of antibody is present. In the first tubes of the titration, not enough antigen is present to allow for cross-linking and lattice formation. The antibody effectively blocks all the antigen sites present, so agglutination does not occur. Complement is not involved in antibody titration.

43.

C. Antinuclear antibody (ANA) is the most consistent feature of systemic lupus erythematosus (SLE). Although renal or nerve pathology may occur, that pathology is secondary to deposition of antigen-antibody complexes and subsequent activation of complement proteins. Bone marrow stems cells are not involved in the pathology of SLE.

44.

C. The rapid plasma reagin (RPR) and Venereal Disease Research Laboratory (VDRL) tests use a cardiolipin antigen. However, in the RPR test, charcoal particles are included with the antigen. When antibody in the patient sample combines with the antigen, the charcoal is trapped in the immune complex, allowing the reaction to be read macroscopically.

45.

B. The cardiolipin antigen is particulate, not soluble, in the VDRL test. However, the particles are too small to make macroscopic agglutinates when combined with antibody. This type of reaction is called a flocculation reaction and needs to be read with low-power microscopy.

46.

D. Patients with connective tissue disorders such as systemic lupus erythematosus may show a false positive reaction in the VDRL test. Other causes of false positives include rheumatic fever, infectious mononucleosis, malaria, and pregnancy. *Treponema pallidum* subsp. *pallidum* is the causative agent of syphilis.

47.

C. Antigens can have multiple epitopes. Each epitope can be unique, binding an antibody with a different idiotype. "Valence" refers to the number of epitopes on an antigen.

48.

D. Monoclonal antibodies are derived from a single clone of plasma cells. Plasma cells are fused with a cancerous myeloma cell. Reagin has two meanings: it can refer to the antibody produced during syphilis or it can refer to IgE. Cold agglutinins are antibodies that agglutinate in cold temperatures (e.g., 4°C). Heterophile antibodies are antibodies produced following exposure to an immunogen that are able to bind a similar but different molecule.

49.

A. In most infections, IgM antibodies will develop first followed by IgG, which develop higher titers and are longer lasting. Anaphylactic reactions are caused by IgE antibody. Delayed hypersensitivity reactions are caused by T cells. Transplacental antibodies belong to the IgG class. IgG antibodies, although they can be detected by complement fixation (CF), do not fix complement efficiently.

50.

B. Testing for antinuclear antibodies (ANAs) is commonly performed by the immunofluorescence method—using fluorescein-conjugated antihuman antibody to detect patient antibody bound to nuclear components of test cells. Antistreptolysin O tests are performed with red cells or, more recently, by latex agglutination. Anti-HBs assays are generally performed by ELISA, and C-reactive protein assays are generally performed by latex agglutination, turbidimetry, or nephelometry.

51.

D. The Fab portion of an antibody contains the hypervariable region. This portion of the molecule has a variable sequence of amino acids that affects the three-dimensional structure of the molecule and, therefore, determines the specificity (idiotype) of the antibody. This region contains the amino terminal portion of the two light chains and the two heavy chains.

52.

B. The indicator system in an ELISA test consists of an enzyme and its substrate. If the enzyme-labeled antibody has complexed with the immobilized antigen, the addition of substrate will produce a colored end product. Alkaline phosphatase is an enzyme frequently used in ELISA tests. Latex particles, fluorescent dyes, and red blood cells are not used in ELISA tests but in other test methodologies.

53.

A. Elevated IgE levels are found in type I hypersensitivity reactions. The antibody binds via the Fc portion of the molecule to Fc receptors on mast cells and basophils. When the attached antibody binds its specific allergen, the cell degranulates.

54.

A. The immune system recognizes host cells as self and is tolerant to antigens on those cells. The loss of tolerance will result in an autoimmune disease in which the immune system mounts an immune response against self cells. Graft-versus-host disease occurs when a bone marrow graft is incompatible with the host tissue and attacks the host.

55.

D. The CD8 molecule is found primarily on cytotoxic T lymphocytes. T helper cells possess CD4 on their surface, as do several other cell types. CD3 is a marker found on most T cells.

56.

B. Light chains are of two distinct types: kappa and lambda. Either type may combine with any of the heavy chains, but in any one molecule, only one type is found. Each immunoglobulin monomer contains two light chains, either kappa or lambda. They extend into the Fab, or antigen-binding, site. This half of the chain is highly variable, whereas the carboxy-terminal portion of the molecule is a constant region.

57.

D. The Fc (crystalline) fragment of an immunoglobulin is produced by papain digestion of an immunoglobulin monomer. The Fc portion of antibodies binds to specific Fc receptors on the surface of some white blood cells. Only part of the heavy chain is found in the Fc fragment. The Fab fragment contains the antigen-combining sites of both the heavy chains and the light chains.

58.

D. A monoclonal antibody is produced by a single cell or clone. Plasma cells obtained from an immunized animal and subsequently fused with myeloma cells result in a hybrid myeloma or hybridoma that will indefinitely secrete a specific antibody. Hybridomas have been prepared from mouse and human plasma cells fused with myeloma cells. T cells do not produce antibodies.

59.

C. Idiotype of an antibody refers to the antigen specificity of the molecule. The isotype is the different classes and subclasses of antibodies (e.g., IgG, IgM, etc.). "Allotype" refers to different alleles of the same isotype. Genetically different individuals will produce antibodies of the same isotype, but they would have a different allotype.

60.

B. Skin testing is based upon the presence of T cells sensitized to antigen. Their activation produces a delayed hypersensitivity reaction, which reaches its peak in about 48 hours. There is no correlation of the amount of the reaction with clinical disease. If the sensitized T cells are transferred from one individual to another, the recipient individual will manifest the same delayed hypersensitivity as the donor.

61.

A. The natural killer (NK) cells destroy target cells through an extracellular nonphagocytic mechanism. NK cells are part of the host's innate resistance and, therefore, do not need previous exposure to an antigen to be active. They also do not need interaction with B or cytotoxic T cells.

62.

C. B cells have the ability to present antigen (immunogen) to T helper cells. This interaction involves several surface molecules. The antigen is complexed with MHC II on the surface of the B cell. CD4 on the T cell interacts with MHC II, whereas the T cell receptor binds the antigen.

63.

B. T cells are produced in the bone marrow and mature in the thymus. Plasma cells, not T cells, produce antibody, and T cells can only react to antigen processed by an antigen-presenting cell. The cell-mediated immune response, which requires the activity of T cells, is primarily helpful in fighting against intracellular parasites.

64.

B. Antibodies directed against self antigens form immune complexes and activate complement. Circulating immune complexes, composed of nuclear antigen and antinuclear antibody, deposit in various organ systems, activate complement, and produce organ pathology. T cells are not directly involved in this process. Allergens, phagocytosis, and killing of ingested bacteria by neutrophils do not play a role in the pathogenic process.

65.

C. Complement protein C3 has a serum concentration of about 1300 μg/mL, which makes it the complement protein present in the greatest concentration. The second highest concentration of complement protein is C4 (600 μg/mL). C3 is cleaved into fragments: C3a and C3b.

66.

D. Destruction of the beta cells in the pancreas results in type 1 diabetes. An autoimmune response destroys the insulin-producing cells. The immune response is probably due to molecular mimicry. Cytotoxic T cells and antibodies directed against an infectious agent cross react to the beta cells.

67.

D. The two antigens are not related. There are two different antibodies that are able to react with the two antigens, forming precipitin lines that cross. If the antigens were identical, a smooth curve precipitation line would have formed.

68.

B. When two antigens are identical, a smooth curved line of precipitation is formed between them. In the diagram, the antigen in well 2 is identical to one of two antigens in well 1. The same antigen in well 1 is identical to antigen in well 4. Therefore, it follows that antigens 2 and 4 are identical.

69.

D. The membrane attack complex forms following the binding of C5 to a biologic membrane. The complex is formed by the sequential addition of C6, C7, C8, and C9. When C5–C8 complex with C9, a tubule is formed that bridges the cell membrane.

70.

D. Contact dermatitis is a cell-mediated hypersensitivity reaction. The offending substance is typically a hapten that combines with a carrier molecule on the skin surface. The hapten-carrier complex is recognized by T cells. IgE mediates immediate hypersensitivity reactions such as hay fever and some forms of asthma.

71.

B. CRP is an acute-phase reactant. Although it is elevated in inflammation, its presence is not diagnostic for any one disease, such as rheumatic fever. It does not correlate with antibody levels or with neutrophil phagocytic function. CRP levels are sometimes elevated during heart disease.

72.

D. Complement attaches to the Fc portion of the antibody molecule. At least two Fc binding sites are required for C1q to attach. Therefore, activation requires two IgG molecules or a single molecule of IgM, which is a pentamer. The C proteins were named in order of discovery. The correct reaction sequence is C1, C4, C2, C3. As the last step of this reaction sequence, C3 is split into C3a and C3b.

73.

A. The alternative pathway for complement activation is a more nonspecific defense mechanism, in that it does not require the presence of antibody for activation. It can be activated by a variety of substances, including complex polysaccharides found in bacterial capsules and cell walls. These materials activate C3 directly. Properdin protein stabilizes some of the active complement proteins, and C4b2a is a C3 convertase.

74.

C. The first response to invading bacteria is mounted by the innate immune system. The innate immune system, although it lacks the specificity of the adaptive immune system, is nonetheless effective at handling many invading bacteria. The first response by the innate immune system consists of an influx of neutrophils into the tissue invaded by bacteria. Monocytes and macrophages, although they are phagocytic cells and part of the innate immune system, play only a minor role in the initial response to bacterial invasion.

75.

B. Incompatible blood transfusions are examples of a type II hypersensitivity reaction. These reactions are characterized as the antigen being a part of a cell. Antibody binds to the antigen, complement is activated, and the red blood cells are lysed.

76.

D. Precipitation reactions involve both soluble antigens and antibodies. These reactions are typically detected in agarose gels. With agglutination reactions, one of the reactants is soluble and the other is insoluble. A reactant is made insoluble by combining with a carrier particle such as latex beads.

77.

D. The fluorescent treponemal antibody absorbance (FTA-ABS) test is often used as a confirmatory test for syphilis. *Treponema pallidum* subsp. *pallidum,* the causative agent of syphilis, is the source of the antigen. The rapid plasma reagin (RPR) and Venereal Disease Research Laboratory (VDRL) are diagnostic tests for syphilis that use nontreponemal antigen. C-reactive protein (CRP) is not involved in syphilis testing.

78.

B. Hepatitis B surface antigen (HBsAg) is a marker for active or chronic infection by the hepatitis B virus; it indicates ongoing viral replication. A person positive for this marker is infectious. If the person had overcome a past infection, he or she would have antibody to the surface antigen (anti-HBs) but not the surface antigen. Immunization causes formation of anti-HBs antibody, and the surface antigen would not be present in serum.

79.

A. The first step in the complement fixation test, the test system, involves the reaction of antibody in the patient's serum to the corresponding antigen in the presence of guinea pig complement. If antibody-antigen binding occurs, complement will bind to the immune complexes. The second step is the addition of sensitized sheep red blood cells (the indicator system). If complement bound to the immune complexes in the first step, it is not available to lyse the sensitized red blood cells. If antibody was not present in the patient sample, complement will not bind to the immune complexes, and it will be free to lyse the sensitized cells.

80.

A. The isotype of an antibody is determined by which heavy chain is present. The term "idiotype" refers to the variable region of an immunoglobulin molecule. The variable region is the portion of immunoglobulin that binds antigen. Every immunoglobulin with a given antigenic specificity has a unique idiotype.

81.

C. IgM antibody to OspC is an important early marker in the diagnosis of Lyme disease. This antibody, along with several others, is often detected by Western blot. Antibodies to p35, p39, and the flagellin subunits p37 and p41 are also useful in diagnosing this disease.

82.

D. The Davidsohn differential test can be used to detect heterophile antibodies produced during infectious mononucleosis. These antibodies are not adsorbed by guinea pig antigens. Therefore, the antibodies are available to agglutinate horse red blood cells. Forssman heterophile antibodies are absorbed by guinea pig antigens and would not agglutinate the horse red blood cells.

83.

D. Hashimoto disease is a type of thyroiditis due to an autoimmune disease. Patients produce autoantibodies and T cells that respond to thyroid antigens. This results in inflammation and swelling of the thyroid gland (goiter). The autoantibody blocks the uptake of iodine, which results in a decrease in the production of thyroid hormones (hypothyroidism).

84.

B. Group A streptococci contain antigenic determinants that are similar to antigenic determinants found on heart valve tissue in some individuals. The immune response occurring during the course of a group A streptococcal infection may be extensive enough to include an immune-mediated attack on the heart valves—rheumatic heart disease. "Molecular mimicry" is the term given to this phenomenon, whereby an immune response directed against one antigen may be extended to include activity against closely related antigens.

85.

B. Some strains of *Staphylococcus aureus* and group A streptococci produce toxins that have the properties of "superantigens." Superantigens react with T cells directly without processing by an antigen presenting cell. These toxins can stimulate many T cells, rather than only those T cells bearing T cell receptors specific for the bacterial toxins. The result is a massive T cell response, leading to the release of cytokines and resulting in disease entities known as toxic shock syndrome (in the case of *S. aureus* infection) and toxic shock–like syndrome in the case of group A streptococci.

86.

D. HBsAg is the first serologic marker occurring in patients with hepatitis B virus infection. The antigen appears about 3–5 weeks before symptoms appear. About 2–4 weeks later, anti-HBc, primarily of the IgM class, begins to appear.

87.

A. Because the human leukocyte antigen (HLA) system is extremely polymorphic, the odds are greatly against finding an HLA-compatible donor in unrelated individuals. The genes coding for HLA antigens are inherited from one's parents and are expressed co-dominantly. Between an offspring and either parent, there is, statistically, a 25% chance of an HLA match. Between siblings, there is a 50% chance of an HLA match.

88.

C. Exogenous antigens are nonself antigens derived from infectious agents or immunizing preparations. Exogenous antigens are processed for presentation to specific T cells by specialized cells collectively referred to as antigen-presenting cells (APCs). APCs for exogenous antigens include B cells, macrophages, monocytes, and dendritic cells.

89.

B. Human immunodeficiency virus preferentially infects T helper cells, which are positive for the surface marker CD4. As the infection progresses, the number of CD4+ cells in the peripheral bloodstream decreases. CD8 is a marker found on another subset of T cells, cytotoxic T cells. The reference ratio of CD4:CD8 cells is 2:1. A decrease in the ratio indicates a decline in immune function.

90.

D. HDV requires HBsAg produced by HBV-infected cells. HDV, therefore, requires the host to be concurrently infected with HBV. The HBV vaccine prevents HBV infection and also HDV infection.

91.

A. The stem cells of the bone marrow give rise to both T and B cells, as well as other cells in the bloodstream. Macrophages and monocytes also arise from hematopoietic stem cells, but they do not differentiate into lymphocytes. Mucosa-associated lymphoid tissue contains mature lymphocytes, particularly B cells, but is not the source of lymphocytes. The fetal liver is a maturation site for B lymphocytes during fetal life but is not the source of those lymphocytes.

92.

B. Contact dermatitis is a delayed-type hypersensitivity reaction mediated by T cells. Antibody is not involved in this type of hypersensitivity, so B cells play no role in it. Neither macrophages nor neutrophils are involved in this type of hypersensitivity.

93.

A. The competitive RIST assay is used to determine the concentration of total IgE. Patient sample containing IgE is mixed with labeled IgE. Both labeled and unlabeled IgE are captured by antihuman IgE. After a wash step, the signal from the label is detected. A high signal indicates a low concentration of unlabeled IgE from the patient sample.

94.

C. The titer of this assay is the reciprocal of the highest dilution demonstrating the desired result, in this case tube #8. The dilution is determined as shown below.

Dilution for tube #1: 0.2 mL serum in a total volume of 1.0 mL = 1:5 dilution.

Dilutions in succeeding tubes: 0.5 mL diluted serum in a total volume of 1.0 mL = 1:2 dilution.

The dilutions in the series of tubes are as follows:

Tube #1, 1:5; tube #2, 1:10; tube #3, 1:20; tube #4, 1:40; tube #5, 1:80; tube #6, 1:160; tube #7, 1:320; tube #8, 1:640; tube #9, 1:1280; tube #10, 1:2560

The reciprocal of the dilution in tube #8 (1:640) is 640.

95.

A. In chemiluminiscent assays, light is the end product. These assays require special instruments to measure the light emitted in the reaction. Chemicals used to generate light include luminol and luciferase.

96.

C. The presence of HBsAg indicates viral replication and that the patient is infectious; this marker can be seen in both acute and chronic infections. The lack of anti-HBs indicates that the patient is not immune to the infection. The presence of anti-HBc and HBeAg with HBsAg indicates a chronic infection.

97.

B. The symptoms of fever of unknown origin, lymphocytosis, and lymphadenopathy suggest EBV or CMV infection and lymphoma or leukemia. Heterophile antibodies become positive later than antibodies to viral core antigen (VCA) of EBV. In addition, only about 50% of children less than 12 years of age form heterophile antibodies following EBV infection. The IgM titer of 80 for EBV is consistent with acute EBV infection.

98.

D. The case history is typical of a child with X-linked agammaglobulinemia. He presented with chronic and recurrent infections beginning at 5 months of age, when transplacentally acquired IgG had declined. Normal IgG serum level is about 800–1200 mg/dL. The infant had normal T cell function, which rules out combined immunodeficiency and DiGeorge syndrome. "Iatrogenic immunodeficiency" refers to an immunodeficiency following therapy prescribed by a physician.

99.

B. When a standard screening test for human immunodeficiency virus infection, such as an ELISA, is positive, it is recommended that the sample be repeated in duplicate. If one or both of the repeated tests is reactive, the sample is considered to be repeatedly reactive and needs to be confirmed by a confirmatory test (e.g., Western blot or immunofluorescent antibody). If this test is positive, the sample can be reported as positive. If the confirmatory test is negative, an additional confirmatory test should be performed if the patient has risk factors for HIV infection. If the repeated ELISA tests are both negative, the sample is reported as negative.

100.

A. The presence of arthritis is suggestive of a number of autoimmune diseases. Protein, RBCs, and casts in the urine are indicative of kidney inflammation. These signs and symptoms along with the rash on the face are characteristic of systemic lupus erythematosus (SLE). A commonly used sensitive screening test for SLE is the antinuclear antibody (ANA) test. The ANA, however, is not specific for SLE. If the ANA were positive, additional autoantibody tests specific for SLE (e.g., anti-Smith) should be performed.

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

Blood Collection, Preservation, Processing, Component Preparation, and Quality Control

1. A woman wants to donate blood. Her physical examination reveals the following: weight—110 lb, pulse—73 bpm, blood pressure—125/75 mm Hg, hematocrit—35%. Which of the following exclusions applies to the prospective donor?
   A. Pulse too high
   B. Weight too low
   C. Hematocrit too low
   D. Blood pressure too low

2. A potential donor has no exclusions, but she weighs only 95 pounds. What is the allowable amount of blood (including samples) that can be drawn?
   A. 367 mL
   B. 378 mL
   C. 454 mL
   D. 473 mL

3. Donors who have received blood or blood products within 12 months of when they desire to donate are deferred to protect the recipient because the
   A. Blood could have transmitted hepatitis (HBV or HCV) or HIV
   B. Blood may have two cell populations
   C. Donor may not be able to tolerate the blood loss
   D. Donor red cell hemoglobin level may be too low

4. Which of the following conditions would contraindicate autologous presurgical donation?
   A. Weight of 100 pounds
   B. Age of 14 years
   C. Hemoglobin of 12 g/dL
   D. Mild bacteremia

5. Which of the following donors would be deferred indefinitely?
   A. History of syphilis
   B. History of gonorrhea
   C. Accutane® treatment
   D. Recipient of human growth hormone

6. Which of the following viruses resides exclusively in leukocytes?
   A. CMV
   B. HIV
   C. HBV
   D. HCV

7. A donor indicates that he has taken two aspirin tablets per day for the last 36 hours. The unit of blood
   A. May not be used for pooled platelet concentrate preparation
   B. Should not be drawn until 36 hours after cessation of aspirin ingestion
   C. May be used for pooled platelet concentrate preparation
   D. May be used for red blood cells and fresh-frozen plasma production, but the platelets should be discarded

8. Which of the following best describes what must be done with a unit of blood drawn from a donor who is found to be at high risk of contracting acquired immune deficiency syndrome (AIDS)?
   A. Hold unit in quarantine until donor diagnosis is clarified.
   B. Use the blood for research dealing with AIDS.
   C. Properly dispose of unit by autoclaving or incineration.
   D. Use the plasma and destroy the red blood cells.

9. Which of the following is *least* likely to transmit hepatitis?
   A. Cryoprecipitate
   B. RBC
   C. Plasma protein fraction (PPF)
   D. Platelets

10. A pooled sera product from 16 donors has a repeatedly positive nucleic acid test (NAT) for HCV. The next action that should be taken is to
    A. Permanently exclude all the donors in the pool
    B. Test each donor in the pool for HCV
    C. Label all the donors as HCV positive
    D. Confirm the positive using recombinant immunoblot assay (RIBA)

11. Although cryoprecipitate has primarily been used for treatment of hypofibrinogenemia and hemophilia A, it contains other blood proteins useful in the treatment of coagulopathies. Which of the following is *not* found in cryoprecipitate?
    A. Fibronectin
    B. Factor XIII
    C. Factor VIII:vW
    D. Antithrombin III

12. Even though it is properly collected and stored, which of the following will fresh-frozen plasma (FFP) *not* provide?
    A. Factor V
    B. Factor VIII
    C. Factor IX
    D. Platelets

13. Blood needs to be prepared for intrauterine transfusion of a fetus with severe HDN. The red blood cell unit selected is compatible with the mother's serum and has been leuko-depleted. An additional step that must be taken before transfusion is to
    A. Add pooled platelets and fresh-frozen plasma
    B. Check that the RBC group is consistent with the father's
    C. Irradiate the RBCs before infusion
    D. Test the RBC unit with the neonate's eluate

14. The addition of adenine in an anticoagulant-preservative formulation aids in
    A. Maintaining ATP levels for red cell viability
    B. Maintaining platelet function in stored blood
    C. Reducing the plasma K^+ levels during storage
    D. Maintaining 2,3-BPG levels for oxygen release to the tissues

15. The pilot tubes for donor unit #3276 break in the centrifuge. You should
    A. Label the blood using the donor's previous records
    B. Discard the unit because processing procedures cannot be performed
    C. Discard the red cells and salvage the plasma for fractionation
    D. Remove sufficient segments to complete donor processing procedures

Use the following information to answer questions 16 and 17.

A satellite bag containing 250 mL of fresh plasma is selected for quality control of cryoprecipitate production. Cryoprecipitate is prepared according to standard operating procedures. The final product has a total volume of 10 mL. The factor VIII assays are 1 IU/mL before and 9 IU/mL after preparation.

16. What is the percent yield of factor VIII in the final cryoprecipitate?
    A. 11%
    B. 25%
    C. 36%
    D. 80%

17. Does this product meet AABB *Standards* for cryoprecipitate production?
    A. Yes
    B. No; the percent recovery is too low.
    C. No; the final factor VIII level is too low.
    D. Data are insufficient to calculate.

Use the following information to answer questions 18–21.

A centrifuge used for platelet preparation has been returned after major repair. A unit of whole blood (450 mL; platelet count 200,000/μL) is selected for calibration of platelet production. The platelet-rich plasma (PRP) contains 250 mL with a platelet count of 300,000/μL. The final platelet concentrate prepared from the PRP contains 50 mL with a platelet count of 900,000/μL.

18. What is the percent yield of platelets in the PRP from this unit?
    A. 33%
    B. 45%
    C. 66%
    D. 83%

19. What is the percent yield of platelets in the final product from the PRP?
    A. 30%
    B. 45%
    C. 50%
    D. 60%

20. Does this product meet AABB *Standards* for platelet concentrate production?
    A. Yes
    B. No; the count on the final product is too low.
    C. No; the percentage recovery in the PRP is too low.
    D. Data are insufficient to calculate.

21. The final product was prepared with a PRP spin time of 2 minutes at 2500 rpm. To increase the percent platelet yield in the final product, one would
    A. Increase the time and/or rpm for the first spin
    B. Increase the time and/or rpm for the second spin
    C. Decrease the time and/or rpm for the first spin
    D. Decrease the time and/or rpm for the second spin

22. When 2,3-BPG levels drop in stored blood, which of the following occurs as a result?
    A. Red blood cell K^+ increases.
    B. Red blood cell ability to release O_2 decreases.
    C. Plasma hemoglobin is stabilized.
    D. ATP synthesis increases.

23. The last unit of autologous blood for an elective surgery patient should be collected no later than _____ hours before surgery.
    A. 24
    B. 36
    C. 48
    D. 72

24. For which of the following patients would autologous donation *not* be advisable?
    A. Patients with an antibody against a high-incidence antigen
    B. Patients with uncompensated anemia
    C. Open heart surgery patients
    D. Patients with multiple antibodies

25. It is generally asymptomatic but has a very high carrier rate (70–80% have chronic infections). About 10% of the carriers develop cirrhosis or hepatocellular carcinoma. These statements are most typical of which of the following transfusion-transmitted infections?
    A. HAV
    B. HBV
    C. HCV
    D. HEV

26. Biochemical changes occur during the shelf life of stored blood. Which of the following is a result of this "storage lesion"?
    A. Increase in pH
    B. Increase in plasma K^+
    C. Increase in plasma Na^+
    D. Decrease in plasma hemoglobin

27. It has been determined that a patient has posttransfusion hepatitis and received blood from eight donors. There is nothing to indicate that these donors may have been likely to transmit hepatitis. What action must be taken initially?
    A. Defer all donors indefinitely from further donations.
    B. Repeat all hepatitis testing on a fresh sample from each donor.
    C. Notify the donor center that collected the blood.
    D. Interview all implicated donors.

28. The temperature range for maintaining red blood cells and whole blood during shipping is
    A. 0–4°C
    B. 1–6°C
    C. 1–10°C
    D. 5–15°C

29. Platelets play an important role in maintaining hemostasis. One unit of donor platelets derived from whole blood should yield _____ platelets.
    A. 5.5×10^6
    B. 5×10^8
    C. 5.5×10^{10}
    D. 5×10^{10}

30. The pH of four platelet concentrates is measured on the day of expiration. The pH and plasma volumes of the four units are as follows: pH 6.0, 45 mL; pH 5.5, 38 mL; pH 5.8, 40 mL; pH 5.7, 41 mL. What corrective action is needed in product preparation to meet AABB *Standards* for platelet production?
    A. No corrective action is necessary.
    B. Recalibrate pH meter.
    C. Increase final plasma volume of platelet concentrates.
    D. Decrease final plasma volume of platelet concentrates.

31. During preparation of platelet concentrate, the hermetic seal of the primary bag is broken. The red blood cells
    A. Must be discarded
    B. May be labeled with a 21-day expiration date if collected in CPD
    C. Must be labeled with a 24-hour expiration date
    D. May be glycerolized within 6 days and stored frozen

32. The blood bank procedures manual must be
    A. Revised annually
    B. Revised after publication of each new edition of AABB *Standards*
    C. Reviewed prior to a scheduled inspection
    D. Reviewed annually by an authorized individual

33. Previous records of patients' ABO and Rh types must be immediately available for comparison with current test results
    A. For 6 months
    B. For 12 months
    C. For 10 years
    D. Indefinitely

34. Which of the following weak D donor units should be labeled Rh-positive?
    A. Weak D due to transmissible genes
    B. Weak D as position effect
    C. Weak partial D
    D. All the above

35. In order to meet the current AABB *Standards* for leukocyte reduction to prevent HLA alloimmunization or CMV transmission, the donor unit must retain at least _____ of the original red cells and leukocytes must be reduced to less than _____.
    A. 85%, 5×10^8
    B. 80%, 5×10^6
    C. 75%, 5×10^5
    D. 70%, 5×10^4

36. Which of the following tests is/are *not* performed during donor processing?
    A. ABO and Rh grouping
    B. HB_sAg
    C. HIV-1-Ag
    D. HB_sAb

37. A 70-kg man has a platelet count of 15,000/μL, and there are no complicating factors such as fever or HLA sensitization. If he is given a platelet pool of 6 units, what would you expect his posttransfusion count to be?
    A. 21,000–27,000/μL
    B. 25,000–35,000/μL
    C. 45,000–75,000/μL
    D. 75,000–125,000/μL

38. Which of the following tests on donor red blood cells must be repeated by the transfusing facility when the blood was collected and processed by a different facility?
    A. Confirmation of ABO group and Rh type of blood labeled D-negative
    B. Confirmation of ABO group and Rh type
    C. Weak D on D-negatives
    D. Antibody screening

INSTRUCTIONS: Each numbered group of incomplete statements (questions 39–63) is followed by four suggested responses. Select the *best* answer or completion statement in each case. Lettered responses may be used once, more than once, or not at all.

For the following components prepared from whole blood (questions 39–43), indicate the required storage temperature.

---

39. Red blood cells (RBCs), liquid
40. Red blood cells, frozen
41. Fresh-frozen plasma
42. Cryoprecipitate
43. Platelet concentrate

    A. 1–6°C
    B. 20–24°C
    C. –18°C or colder
    D. –65°C or colder

For the following components prepared from whole blood (questions 44–48), indicate the shelf life.

44. Red blood cells in CPDA-1
45. Fresh-frozen plasma
46. Cryoprecipitate
47. Fresh-frozen plasma, thawed
48. Platelet concentrate in PL-732 (with agitation)

A. 24 hours
B. 5 days
C. 35 days
D. 1 year

Using the specified anticoagulant/preservative (questions 49–52), indicate the allowable shelf life for blood for transfusion therapy.

49. CPD (citrate phosphate dextrose)
50. CPDA-1 (citrate phosphate dextrose adenine)
51. AS-1 (Adsol®)
52. EDTA

A. 21 days
B. 35 days
C. 42 days
D. Not an approved anticoagulant

For the following situations (questions 53–59), indicate whether the individual volunteering to donate blood for allogeneic transfusion should be accepted or deferred. Assume results of the physical examination to be acceptable unless noted.

53. A 65-year-old man whose birthday is tomorrow
54. A 45-year-old woman who donated a unit during a holiday appeal 54 days ago
55. A 50-year-old man who had sex with another man in 1980
56. A 25-year-old man who says he had yellow jaundice right after he was born
57. An 18-year-old with poison ivy on his hands and face
58. A woman who had a baby 2 months ago
59. A 35-year-old runner (pulse 46 bpm)

A. Defer temporarily
B. Defer for 12 months
C. Defer indefinitely
D. Accept

For the following patients (questions 60–63), indicate the component of choice for transfusion therapy.

60. Patients with warm autoimmune hemolytic anemia (AIHA) due to α-methyldopa (Aldomet®) with hemoglobins of 8.5 g/dL or above
61. Patients requiring transfusion with RBC that will not transmit cytomegalovirus (CMV)
62. Patients with normovolemic anemia
63. Patients who are thrombocytopenic secondary to the treatment of acute leukemia

A. Platelet concentrate
B. RBC
C. Leukocyte-reduced RBC
D. Transfusion *not* indicated

Blood Groups, Genetics, Serology

64. Most blood group antibodies are of what immunoglobulin classes?
    A. IgA and IgD
    B. IgA and IgM
    C. IgE and IgD
    D. IgG and IgM

65. The following family study is performed:

Mother	Father	Child 1	Child 2
K + k +	K − k +	K + k −	K − k +

All other indications are that these children are both the products of this mating. Possible explanations for these results would include
    A. A dominant inhibitor gene has been passed to child 1
    B. Father has one *k* gene and one K^0 gene
    C. Father has the McLeod phenotype
    D. Mother has a *cis-Kk* gene

66. Which of the following blood groups reacts *least* strongly with an anti-H produced in an A_1B individual?
    A. Group O
    B. Group A_2B
    C. Group A_2
    D. Group A_1

67. How many genes encode the following Rh antigens: D, C, E, c, e?
    A. One
    B. Two
    C. Three
    D. Four

Use the following information to answer questions 68 and 69.

A patient arrives in the OB clinic 3 months pregnant. This is her first pregnancy, and she has never been transfused. Her prenatal screen includes the following results:

Cell Typing Results		Serum Typing Results	
Anti-A	Anti-B	A_1 cells	B cells
0	0	4+	4+

	Screening Cell I	Screening Cell II	Auto-control
IS	4+	4+	0
37°C LISS	4+	4+	0
AHG	4+	4+	0
Check Cells			√

68. The test results could be due to
    A. Cold autoantibody
    B. Inheritance of *sese* genes
    C. Inheritance of *hh* genes
    D. Rouleaux

69. If the patient's RBCs were tested against anti-H lectin and did not react, this person would be identified as a(an)
    A. Acquired B
    B. O_h phenotype
    C. Secretor
    D. Subgroup of A

70. If a person has the genetic makeup *Hh, AO, LeLe, sese,* what substance will be found in the secretions?
    A. A substance
    B. H substance
    C. Le^a substance
    D. Le^b substance

71. The following results were obtained when typing a patient's blood sample.

Cell Typing Results		Serum Typing Results	
Anti-A	**Anti-B**	**A_1 cells**	**B cells**
4+	2+	0	4+

    The tech suspects that this is a case of an acquired B antigen. Which of the following would support this suspicion?
    A. A positive autocontrol test
    B. Secretor studies show that the patient is a nonsecretor.
    C. A patient diagnosis of leukemia
    D. The patient's red cells give a negative result, with a monoclonal anti-B reagent lacking the ES-4 clone.

72. Lectins are useful in determining the cause of abnormal reactions in blood bank serology. These lectins are frequently labeled as anti-H, anti-A_1, etc. The nature of these lectins is explained by which of the following?
    A. An early form of monoclonal antibody produced in nonvertebrates
    B. A plant substance that chemically reacts with certain RBC antigens
    C. Naturally occurring antibodies in certain plants
    D. The ability of plants to respond to RBC antigens by antibody production

73. Which of the following sugars must be present on a precursor substance for A and B antigenic activity to be expressed?
    A. D-Galactose
    B. *N*-Acetylgalactosamine
    C. Glucose
    D. L-Fucose

74. An antigen-antibody reaction alone does not cause hemolysis. Which of the following is required for red blood cell lysis?
    A. Albumin
    B. Complement
    C. Glucose-6-phosphate dehydrogenase (G6PD)
    D. Antihuman globulin (AHG)

75. A white female's red blood cells gave the following reactions upon phenotyping: D+ C+ E– c+ e+. Which of the following is the most probable Rh genotype?
    A. *DCe/Dce*
    B. *DCe/dce*
    C. *DCe/DcE*
    D. *Dce/dCe*

76. A black patient has the following Rh phenotype: D+ C+ E+ c+ e+. Which of the following genotypes is the *least* probable?
    A. *DCE/dce*
    B. *DCe/DcE*
    C. *DCe/dcE*
    D. *DcE/dCe*

77. An individual of the *dce/dce* genotype given *dCe/dce* blood has an antibody response that appears to be anti-C plus anti-D. What is the most likely explanation for this?
    A. The antibody is anti-G.
    B. The antibody is anti-partial D.
    C. The antibody is anti-Cw.
    D. The reactions were read incorrectly.

78. If a patient has the Rh genotype *DCe/DCe* and receives a unit of red blood cells from a *DCe/dce* individual, what Rh antibody might the patient develop?
    A. Anti-C
    B. Anti-c
    C. Anti-d
    D. Anti-E

Use the following information to answer questions 79–81.

The following results were obtained when testing the individuals below:

	Anti-					Test for
	D	C	E	c	e	Weak D
Husband	0	+	0	+	+	+
Wife	0	0	0	+	+	0
Infant	+	0	0	+	+	N/A

79. What percentage of this couple's offspring can be expected to be D-negative?
    A. 0%
    B. 25%
    C. 50%
    D. 75%

80. Which of the following conclusions regarding the family typing is most likely?
    A. The husband is not the infant's father.
    B. The husband is proved to be the infant's father.
    C. The husband cannot be excluded from being the infant's father.
    D. The D typing on the infant is a false positive.

81. Which, if any, of these three individuals can make anti-D?
    A. Husband
    B. Husband and wife
    C. Wife
    D. None

82. If a D-positive person makes an anti-D, this person is probably
    A. Partial D
    B. D-negative
    C. Weak D as position effect
    D. Weak D due to transmissible genes

83. A serum containing anti-k is not frequently encountered. This is because
    A. People who lack the k antigen are rare
    B. People who possess the k antigen are rare
    C. The k antigen is not a good immunogen
    D. $Kell_{null}$ people are rare

84. A victim of an auto accident arrives in the emergency department (ED) as a transfer from a hospital in a rural area. The patient has been in that facility for several weeks and has received several units of red blood cells during that time. The ED resident orders 2 units of RBCs for transfusion. The sample sent to the blood bank is centrifuged and the cell-serum interface is not discernable. A subsequent sample produces the same appearance. You would suspect that the patient has
    A. Autoimmune hemolytic anemia
    B. Anti-Fy^a
    C. Anti-Jk^a
    D. Paroxysmal nocturnal hemoglobinuria

85. Which of the following is a characteristic of the Xg^a blood group system?
    A. The Xg^a antigen has a higher frequency in women than in men.
    B. The Xg^a antigen has a higher frequency in men than in women.
    C. The Xg^a antigen is enhanced by enzymes.
    D. Anti-Xg^a is usually a saline-reacting antibody.

86. Testing needs to be done with an antiserum that is rarely used. The appropriate steps to take in using this antiserum include following the manufacturer's procedure and
    A. Performing a cell panel to be sure that the antiserum is performing correctly
    B. Performing the testing on screen cells
    C. Testing in duplicate to ensure the repeatability of the results
    D. Testing a cell that is negative for the antigen and one that is heterozygous for the antigen

87. Which of the following is a characteristic of Kidd system antibodies?
    A. Usually IgM antibodies
    B. Corresponding antigens are destroyed by enzymes.
    C. Usually strong and stable during storage
    D. Often implicated in delayed hemolytic transfusion reactions

88. Which of the following statements is *not* true of anti-Fy^a and anti-Fy^b?
    A. Are clinically significant
    B. React well with enzyme-treated panel cells
    C. Cause hemolytic transfusion reactions
    D. Cause a generally mild hemolytic disease of the newborn

89. Which of the following antibodies can be neutralized with pooled human plasma?
    A. Anti-Hy and anti-Ge:1
    B. Anti-Ch^a and anti-Rg^a
    C. Anti-Co^a and anti-Co^b
    D. Anti-Do^a and anti-Js^b

90. Which of the following statements is *not* true about anti-U?
    A. Is clinically significant
    B. Is only found in black individuals
    C. Only occurs in S–s– individuals
    D. Only occurs in Fy(a–b–) individuals

91. A patient had an anti-E identified in his serum 5 years ago. His antibody screening test is now negative. To obtain suitable blood for transfusion, what is the best procedure to use?
    A. Type the patient for the E antigen as an added part to the crossmatch procedure.
    B. Type the donor units for the E antigen and crossmatch the E-negative units.
    C. Crossmatch donors with the patient's serum and release the compatible units for transfusion.
    D. Perform the crossmatch with enzyme-treated donor cells, because enzyme-treated red cells react better with Rh antibodies.

92. A patient's red blood cells are being typed for the Fy^a antigen. Which of the following is the proper cell type of choice for a positive control of the anti-Fy^a reagent?
    A. Fy(a+b–)
    B. Fy(a+b+)
    C. Fy(a–b+)
    D. Fy(a–b–)

93. Which of the following antibodies has been clearly implicated in transfusion reactions and hemolytic disease of the newborn?
    A. Anti-I
    B. Anti-K
    C. Anti-Le^a
    D. Anti-N

Refer to red cell panel chart 1 to answer questions 94–96.

RED CELL PANEL CHART 1

	Rh					MNSs				P	LEWIS		LUTHERAN		KELL					DUFFY		KIDD						
Cell #	D	C	E	c	e	M	N	S	s	P_1	Le^a	Le^b	Lu^a	Lu^b	K	k	Kp^a	Js^a	Fy^a	Fy^b	Jk^a	Jk^b	Xg^a	IS	37	AHG	CkC
1	0	+	0	+	+	+	+	+	0	0	+	0	0	+	0	+	0	0	+	+	+	+	+	0	0	0	√
2	+	+	0	0	+	0	+	0	+	+	0	+	+	+	0	+	0	0	+	+	+	+	+	0	0	0	√
3	+	+	0	0	+	0	+	0	+	+	0	+	0	0	0	+	0	0	0	+	+	0	+	0	0	0	√
4	+	0	+	+	0	+	0	+	0	+	+	0	0	+	+	+	0	0	0	+	+	0	+	0	0	4+	
5	0	0	+	+	+	0	+	0	+	+	+	0	0	+	0	+	0	0	+	0	0	+	0	0	0	3+	
6	0	0	0	+	+	+	0	0	+	+	0	+	0	+	0	+	0	0	+	0	+	+	0	0	0	0	√
7	0	0	0	+	+	+	+	0	+	+	0	0	0	+	+	0	0	0	+	+	+	+	+	0	0	2+	
8	0	0	0	+	+	+	+	0	+	+	0	+	0	+	0	+	0	+	0	0	+	+	0	0	0	0	√
9	0	0	+	+	+	+	0	+	0	0	0	+	0	+	0	+	0	0	0	+	0	+	0	0	0	3+	
10	0	0	0	+	+	+	0	+	+	0	0	+	+	+	0	+	0	0	+	0	+	0	0	0	0	0	√
11	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	/	0	0	0	√
A/C																								0	0	0	√

94. Which of the following antibodies would require additional testing in order to be ruled out?
    A. Anti-E, -K, -Kp^a, -Js^a, -Jk^b
    B. Anti-E, -S, -Le^b, -K, -Kp^a, -Fy^a
    C. Anti-E, -S, -Le^a, -K, -Kp^a, -Js^a, -Fy^a, -Jk^a
    D. Anti-E, -Le^a, -K, -Kp^a, -Js^a, -Fy^b, -Jk^a, -Jk^b

95. The most likely antibody(ies) in the patient's serum is(are)
    A. Anti-S and anti-E
    B. Anti-E and anti-K
    C. Anti-Fy^b showing dosage
    D. Anti-K, anti-Js^a, and anti-Le^a

96. From the cells in red cell panel chart 2, choose a selected cell panel to help identify the antibody(ies) in the patient described in question 95.
    A. 1, 2, 5, 9, 10
    B. 2, 6, 7, 10
    C. 1, 4, 7
    D. 2, 3, 4, 6, 9

RED CELL PANEL CHART 2

	Rh					MNSs				P	LEWIS		LUTHERAN		KELL				DUFFY		KIDD		
Cell #	D	C	E	c	e	M	N	S	s	P_1	Le^a	Le^b	Lu^a	Lu^b	K	k	Kp^a	Js^a	Fy^a	Fy^b	Jk^a	Jk^b	Xg^a
1	0	+	0	+	+	+	+	+	0	0	+	0	0	+	0	+	0	0	0	+	0	+	+
2	+	+	0	0	+	0	+	0	+	+	0	+	+	+	0	+	0	0	+	+	+	+	+
3	+	+	0	0	+	0	+	0	+	+	0	+	0	0	0	+	0	0	0	+	+	0	+
4	+	0	+	+	0	+	0	0	+	+	0	+	0	+	0	+	0	0	+	0	+	0	+
5	0	0	+	+	+	0	+	0	+	+	+	0	0	+	0	+	0	0	+	0	0	+	0
6	0	0	0	+	+	+	0	0	+	+	0	+	0	+	0	+	0	0	+	0	+	+	0
7	0	0	0	+	+	+	+	0	+	+	0	0	0	+	+	0	0	0	+	0	+	0	+
8	0	0	0	+	+	+	+	+	0	+	0	+	0	+	0	+	0	0	0	0	0	+	0
9	0	0	+	+	+	+	0	+	0	0	0	+	0	+	0	+	0	0	0	+	0	+	0
10	0	0	0	+	+	+	0	+	+	0	0	+	+	+	0	+	0	0	+	0	+	0	0

97. Often when trying to identify a mixture of antibodies, it is useful to neutralize one of the known antibodies. Which one of the following antibodies is neutralizable?
    A. Anti-D
    B. Anti-Jk^a
    C. Anti-Le^a
    D. Anti-M

98. Which of the following antibodies does *not* match the others in terms of optimal reactive temperature?
    A. Anti-Fy^a
    B. Anti-Jk^b
    C. Anti-N
    D. Anti-U

99. A recently transfused patient's serum has a positive antibody screen. The panel performed at IS, in LISS at 37°C, and at AHG shows a strong anti-Fy^a and a weak possible anti-C. To confirm the anti-C, you would perform an
    A. Elution
    B. Absorption
    C. Antigen typing
    D. Enzyme panel

100. The antiglobulin test does not require washing or the addition of IgG-coated cells in which of the following antibody detection methods?
    A. Solid-phase red cell adherence assays
    B. Gel test
    C. Affinity column technology
    D. Polyethylene glycol (PEG) technique

101. Which set of antibodies could you possibly find in a patient with no history of transfusion or pregnancy?
    A. Anti-I, anti-s, anti-P_1
    B. Anti-Le^b, anti-A_1, anti-D
    C. Anti-M, anti-c, anti-B
    D. Anti-P_1, anti-Le^a, anti-I

102. Lymphocytotoxicity testing can be used to detect the presence of antibodies to
    A. Wr^a and Wr^b
    B. HLA antigens
    C. Bg^a, Bg^b, and Bg^c
    D. JMH antigen

103. In which of the following instances may mixed-field (mf) agglutination be observed?
    A. Direct antiglobulin test (DAT) result of patient undergoing delayed hemolytic transfusion reaction
    B. Indirect antiglobulin test (IAT) result of patient who has anti-Le^a
    C. DAT result of patient on high doses of α-methyldopa
    D. Typing result with anti-A of patient who is A_2 subgroup

104. The antibody produced during the secondary response to a foreign antigen is usually
    A. IgM
    B. A product of T lymphocytes
    C. Produced a month or more after the second stimulus
    D. Present at a higher titer than after a primary response

105. In which situation(s) may the ABO serum grouping *not* be valid?
    A. The patient has hypogammaglobulinemia.
    B. IgM alloantibodies are present.
    C. Cold autoantibodies are present.
    D. All the above

106. A group A, D-negative obstetric patient with anti-D (titer 256) is carrying a fetus who needs an intrauterine transfusion. Which of the following units should be chosen?
A. Group A, D-negative RBC
B. Group A, D-negative whole blood
C. Group O, D-negative RBC
D. Group O, D-negative whole blood

107. Which of the following is generally detected at the antiglobulin phase of testing?
A. Anti-Jk^a
B. Anti-M
C. Anti-P_1
D. Anti-I

108. Which of the blood group systems is associated with antibodies that are generally IgM?
A. Rh
B. Duffy
C. Kell
D. Lewis

109. Some antigens that are primarily found on white blood cells can occur on erythrocytes. Which of the following are the red blood cell equivalents of human leukocyte antigens (HLAs)?
A. Le^a, Le^b
B. Bg^a, Bg^b, Bg^c
C. Kp^a, Kp^b, Kp^c
D. Do^a, Do^b

110. The following phenotypes resulted from blood typing a mother, 6-month-old baby, and alleged father in a case of paternity testing.

	ABO	Rh	HLA
Mother	A	ce	A2, A29, B12, B17
Baby	O	ce	A2, A3, B12, B15
Alleged Father	A	DCce	A3, A9, B5, B27

Which of the following statements is true? The alleged father
A. Is excluded by the ABO system
B. Is excluded by the Rh system
C. Is excluded by the HLA system
D. Cannot be ruled out

INSTRUCTIONS: Each numbered group of incomplete statements (questions 111–123) is followed by four suggested responses. Select the *best* answer in each case. Lettered responses may be used once, more than once, or not at all.

Eight blood samples are received in the laboratory for ABO grouping. For each patient (questions 111–118), indicate the most likely cell and serum reactions selected from the lettered reaction matrix.

---

111. A patient with an acquired antigen due to infection with gram-negative bacteria

112. A patient with multiple myeloma

113. A newborn

114. An A_2 individual making an anti-A_1

115. A patient with antibodies to acriflavin (a yellow dye)

116. A patient who is immunodeficient

117. A patient with an unexpected IgM antibody in his serum

118. A patient with cold hemagglutinin disease (CHD)

	Cell Typing Results		Serum Typing Results		
	Anti-A	Anti-B	A_1 cells	B cells	O cells
A.	+	+	+	+	+
B.	+	0	+	+	0
C.	+	+	0	+	0
D.	0	+	0	0	0

For the following items (questions 119–123), select the answer that most closely corresponds to the description.

119. Found predominantly in whites

120. Associated with weak Kell system antigenic expression

121. Associated with the presence of chronic granulomatous disease

122. Linked with *MN*

123. A rare allele of *M* and *N*

A. McLeod phenotype
B. M^g
C. Kp^a
D. *Ss*

Antibody Identification, Transfusion Therapy, Transfusion Reactions

For questions 124–132, refer to red cell panel chart 3.

124. The racial origin of the donor of Cell #3 is most likely
A. Black
B. Eskimo
C. Oriental
D. White

125. The donor of Cell #5 is homozygous for which combination of the following genes?
A. *Ce*, P_1, *M*, *s*, *k*, Jk^a, Fy^a, Le^b
B. *Ce*, P_1, *s*, *k*, Jk^a, Fy^a, Le^b
C. *Ce*, *s*, *k*, Jk^a, Fy^a, Le^b, P_1
D. *Ce*, *s*, *k*, Jk^a, Fy^a

126. After testing a patient's serum with the panel, one observes there are no reactions at IS or 37°C with Cells #1–8. There is a 1+ AHG reaction with Cells #1 and #6 and a 3+ AHG reaction with Cells #4 and #5. All other Cells, #2, #3, #7, and #8, are negative at AHG. Which of the following statements is *true?*
A. Anti-Fy^a appears to be present.
B. Anti-Fy^a is present as well as an antibody that is reacting with an undetermined antigen on Cells #4 and #5.
C. Ficin will enhance the reactions of the antibody(ies) present.
D. Anti-Fy^a is present but can be ignored because most people are Fy(a–b–).

127. The serum of a patient tested with the reagent red cell panel using a low-ionic-strength-saline (LISS) additive demonstrates 3+ reactivity with Cells #1–8 at the antiglobulin phase. The autocontrol is negative. This pattern of reactivity is most likely due to
   A. Rouleaux formation
   B. Warm autoantibody
   C. Alloantibody directed against a high-frequency antigen
   D. Antibody directed against a preservative present in LISS

128. A patient's serum reacts with all the panel cells except Cell #7 at the antiglobulin phase only. Which of the following techniques would be most helpful at this point?
   A. Treat the panel cells with a proteolytic enzyme and repeat the panel with untreated serum.
   B. Treat the panel cells with dithiothreitol (DTT) and repeat the panel with untreated serum.
   C. Treat the patient's serum with dithiothreitol (DTT) and repeat the panel with treated serum.
   D. Treat the patient's serum with a proteolytic enzyme and repeat the panel with treated serum.

RED CELL PANEL CHART 3

Cell #	D	C	E	c	e	M	N	P_1	S	s	K	k	Jk^a	Jk^b	Fy^a	Fy^b	Lu^a	Le^a	Le^b	Xg^a
1	+	+	0	+	+	+	+	+	0	+	+	+	0	+	+	+	0	0	+	0
2	+	+	0	0	+	0	+	0	+	+	0	+	0	+	0	+	0	0	+	+
3	+	+	0	+	+	+	0	+	+	0	0	+	+	+	0	0	0	+	0	+
4	+	0	+	+	0	+	+	0	+	0	0	+	+	0	+	0	0	0	+	0
5	0	+	0	0	+	+	+	+	0	+	0	+	+	0	+	0	0	0	+	0
6	0	0	+	+	0	+	0	+	+	+	0	+	0	+	+	+	0	0	0	0
7	0	0	0	+	+	0	+	+	+	0	+	0	+	+	0	+	+	0	+	+
8	0	0	0	+	+	+	+	0	+	+	0	+	+	0	0	+	0	+	0	+

In addition to red cell panel chart 3, use the following information to answer questions 129–132.

The patient is group A, D-negative and has not been recently transfused. Cells #5, #6, and #7 are negative in all phases with this patient's serum. The autocontrol is negative. Other cell results are as follows:

Cell #	IS	37°C LISS	AHG
1	0	1+	4+
2	0	1+	4+
3	2+	1+	4+
4	0	1+	4+
8	2+	0	0

129. From the reactions given, it appears that there is(are)
   A. One antibody reacting
   B. One antibody reacting that shows dosage
   C. "Cold" and "warm" antibodies reacting
   D. Two "warm" antibodies reacting

130. The antibody that reacts at immediate spin is most likely
   A. Anti-D
   B. Anti-P_1
   C. Anti-Le^a
   D. Anti-Le^b

131. The antibody that reacts at 37°C and with AHG is
   A. Anti-C
   B. Anti-D
   C. Anti-CD
   D. Anti-K

132. What should you do to increase the probability that an antibody identification is correct?
   A. Make an eluate.
   B. Do saliva testing.
   C. Run an additional panel.
   D. Type the patient's cells for the corresponding antigens.

133. The following results were obtained upon testing a specimen of a patient, being admitted after a car accident, who had no recent history of transfusion or medical problems.
   ABO group: A
   Rh type: D-positive
   Antibody screening test: Positive, one screening cell only
   Direct antiglobulin test: Negative
   Antibody identification: Anti-K identified; 3 K+ cells that reacted with the patient serum and 3 K– cells that did not react with the patient serum were on the panel.
   Patient's cell phenotyping: K+

   What is the most likely cause of the discrepant results?
   A. Failure to read panel at antiglobulin phase
   B. Failure to use positive and negative controls with anti-K
   C. Panel cell reactions interpreted incorrectly
   D. Patient has circulating donor cells that are K+

134. False negative results at the antiglobulin phase of an antibody screening test are most likely due to
   A. Excessive washing of the red cells
   B. Inadequate washing of the red cells
   C. Warm autoantibody present in the patient's serum
   D. Failure to allow the blood to clot properly

135. What is the process of removing an antibody from the red blood cell membrane called?
A. Absorption
B. Adsorption
C. Elution
D. Immunization

136. At the end of an antiglobulin test, IgG-coated control cells are added to the negative tests and centrifuged. What does it mean if no agglutination occurs?
A. Test is valid.
B. Antiglobulin reagent was working properly.
C. Cells were not washed thoroughly.
D. Control cells are contaminated.

137. The crossmatch is performed using
A. Donor's serum and recipient's red cells
B. Donor's red cells and recipient's serum
C. Donor's serum and reagent red cells
D. Recipient's serum and reagent red cells

138. A male trauma victim whose blood type is group AB, D-negative has a negative antibody screening test. He has been transfused with both of the group AB, D-negative units in inventory within the last hour. He is now in surgery and expected to need large amounts of blood. Of the following available units in inventory, which type should be given next?
A. 30 units of group O, D-positive
B. 26 units of group A, D-positive
C. 10 units of group O, D-negative
D. 5 units of group A, D-negative

139. Which of the following will the crossmatch do?
A. Prevent immunization
B. Prevent delayed transfusion reactions
C. Guarantee normal survival of the red blood cells
D. Frequently verify donor ABO compatibility

140. Given that a patient's antibody screening test is negative, which of the following may cause a false positive result in a compatibility test?
A. Incorrect ABO typing of the donor or patient
B. An alloantibody against a low-frequency antigen on the donor cells
C. Prior coating of IgG antibody on the donor cells
D. All the above

141. Which of the following will be incompatible in the crossmatch?

	Donor	Recipient
A.	Group A, D-negative	Group A, D-positive
B.	Group O, D-positive	Group A, D-positive
C.	Group AB, D-positive	Group A, D-positive
D.	Group A, D-positive	Group A, D-negative

142. A resident physician hand-delivers a blood sample, drawn by the attending physician, for pretransfusion testing from a patient who is difficult to draw. The sample is unlabeled. One should
   A. Discard the sample and request that the resident obtain a new sample, adhering to proper guidelines for labeling
   B. Label the specimen with the information the resident provides
   C. Label the specimen with information from the accompanying transfusion request form
   D. Request the sample be returned to the nursing station to be labeled

143. A specimen of blood is received in the blood bank with request slips initialed by the phlebotomist. The tube has the patient's first and last name and medical records identification number on the label. What else must be on the tube label as required by AABB *Standards?*
   A. Patient's room number
   B. Date of phlebotomy
   C. Initials of phlebotomist
   D. Attending physician's name

144. A physician calls the blood bank and wants an additional unit of RBC crossmatched for a patient. Several specimens from that patient are identified that have been drawn over the past month. Which of the following available samples is the oldest acceptable specimen that may be used for crossmatching?
   A. 1 day old
   B. 4 days old
   C. 1 week old
   D. 1 month old

145. A patient has a hemoglobin value of 8.1 g/dL. The surgeon wants to raise the hemoglobin to 10 g/dL before surgery. How many units of RBC need to be administered to this patient to raise the hemoglobin to the required level?
   A. 1
   B. 2
   C. 3
   D. 4

146. A patient with an anti-K and an anti-Jk^a in her plasma needs 2 units of RBC for surgery. How many group-specific units would need to be screened to find 2 units of RBC? The frequency of Jk(a+) is 77%; the K+ frequency is 10%.
   A. 6
   B. 10
   C. 20
   D. 36

Use the following information to answer questions 147 and 148.

A postpartum female is bleeding because of disseminated intravascular coagulation (DIC). The attending physician orders cryoprecipitate for fibrinogen replacement. The freezer inventory contains the following cryoprecipitate: 6 bags Group A, 8 bags Group O, 6 bags Group AB, 12 bags Group B.

147. How many bags (units) should be thawed and pooled to provide 2 g of fibrinogen?
   A. 2
   B. 4
   C. 8
   D. 10

148. The patient is group A. Which cryoprecipitate units would most appropriately be used to treat this patient?
   A. Group A only
   B. Group AB only
   C. Group A and Group O
   D. Group A and Group AB

149. If 98% of the red blood cells are viable in a unit of RBC at the time of transfusion, what percentage of red cells will remain viable 28 days posttransfusion?
   A. 10%
   B. 30%
   C. 50%
   D. 70%

150. What is the component of choice for someone who needs a RBC transfusion when there is a history of febrile transfusion reactions?
   A. RBCs less than 5 days old
   B. Leukocyte-reduced RBCs
   C. RBCs 30 to 35 days old
   D. Frozen RBCs that have been thawed and deglycerolized

151. Which of the following is the component of choice when a physician is concerned about restoring or maintaining oxygen-carrying capacity?
   A. Albumin
   B. Cryoprecipitate
   C. Whole blood
   D. Red blood cells

152. The serum of a patient contains an antibody that reacts with all random donor cells and panel cells that have been tested. The best possibility to find compatible blood would be to test
   A. Grandparents
   B. Parents
   C. Siblings
   D. Spouse

Use the following information to answer questions 153 and 154.

A resident physician on the trauma team runs a pretransfusion blood sample from a male trauma victim to the blood bank and wants 6 units of blood to be issued immediately. He indicates that he is willing to sign for uncrossmatched blood. He also indicates that he wants 6 units ready at all times. The patient has been admitted to this institution previously for GI bleed.

153. The resident says the victim has a donor card in his wallet indicating a group B, D-positive blood type. What should be done immediately?
   A. Issue 6 units of uncrossmatched group B, D-positive whole blood.
   B. Check patient and donor records to confirm the blood type, then issue 6 units of uncrossmatched group B, D-positive blood.
   C. Withhold blood until ABO and compatibility testing are completed.
   D. Issue 6 units of uncrossmatched group O RBCs.

154. What should be the next step in the work-up of this emergency department patient?
   A. Prepare 6 units uncrossmatched group B, D-positive whole blood.
   B. Check blood bank records for any previous patient information.
   C. Type and screen the patient sample.
   D. Prepare 6 more units of uncrossmatched group O blood.

155. Four units of fresh-frozen plasma have been ordered to correct factor V deficiency in a group O patient. One should thaw and issue _____ plasma.
   A. Group O only
   B. Group O and/or group A
   C. Group O and/or group AB
   D. Any blood group available

156. Which of the following is acceptable to be given intravenously with a blood transfusion?
   A. 5% dextrose in water
   B. Physiologic saline
   C. Ringer's solution
   D. Potassium chloride in saline

157. Hemolytic transfusion reactions are the most serious type of reactions to blood transfusion. The majority of hemolytic transfusion reactions are caused by _____ errors.
   A. Blood typing
   B. Antibody identification
   C. Clerical
   D. Crossmatching

158. What type of transfusion reaction is often diagnosed by a positive DAT and a gradual drop in the patient's hemoglobin level?
   A. Anaphylactic
   B. Febrile
   C. Delayed hemolytic
   D. Acute hemolytic

159. What antibody, labile both in stored serum and the patient's plasma, is a frequent cause of delayed hemolytic transfusion reactions?
   A. Anti-A
   B. Anti-D
   C. Anti-Jk^a
   D. Anti-K

160. Occasionally, patients have an anaphylactic reaction to a specific immunoglobulin class during a transfusion. Which immunoglobulin class is most often implicated?
   A. IgA
   B. IgD
   C. IgE
   D. IgG

161. A transfusion of which of the following is *least* likely to transmit HIV, HCV, or HBV?
   A. Pooled plasma, solvent/detergent treated
   B. Cryoprecipitate
   C. Leukocyte-reduced RBCs
   D. Platelets

Use the following information to answer questions 162–164.

A transfusion reaction is reported by the nursing unit on patient A. D. The nurse reports that the patient had chills, fever, and back pain within a few minutes of starting the unit. The nurse asks what s/he should do.

162. You tell the nurse to immediately
   A. Collect posttransfusion blood samples
   B. Monitor the pulse and blood pressure
   C. Discontinue the unit, keep the line open
   D. Page the patient's physician for instructions

163. Which of the following directives would *not* be included in the additional activities you would request the nurse to do?
A. Return the unit to the blood bank.
B. Obtain a posttransfusion blood sample in EDTA.
C. Obtain a posttransfusion urine sample.
D. Obtain a fresh unit from the blood bank for immediate infusion.

164. All paperwork checks on this transfusion reaction are OK. The pretransfusion sample has straw-colored plasma. The posttransfusion sample has red-tinged plasma. This is indicative of a(an)
A. Uncomplicated transfusion
B. Intravascular transfusion reaction
C. Error in which drugs have been infused with the blood
D. Febrile transfusion reaction

165. Although use of autologous blood transfusions generally has several advantages, which of the following is *not* avoidable?
A. Transmission of disease
B. Clerical error
C. Allergic reactions
D. Graft-versus-host disease

166. Before blood is issued for transfusion, a patient's previous blood bank records must be reviewed. Which of the following is *not* included in this review process?
A. ABO group and Rh type
B. Clinically significant antibodies
C. Serious adverse reactions
D. Hepatitis testing

167. Which of the following would *not* cause a positive hemagglutination reaction in the crossmatch?
A. Incorrect ABO grouping of the donor
B. Unexpected antibodies in the recipient serum
C. A positive DAT on the recipient red cells
D. A positive DAT on the donor red cells

168. Which of the following blood types necessitates that a separate Rh control tube be set up when using a monoclonal anti-D reagent?
A. Group O, D-positive
B. Group A, D-positive
C. Group B, D-positive
D. Group AB, D-positive

169. Six units of blood were ordered stat for a young female patient who has the following tube typing results (the tube typing procedure uses a washed red cell suspension with monoclonal reagents). The physician has just called requesting emergency release of 2 units of RBCs.

Cell Typing Results				Serum Typing Results	
Anti-A	Anti-B	Anti-D	Rh control	A_1 cells	B cells
2+	4+	3+	2+	4+	4+

Which of the following should be done first?
A. Perform a DAT on the patient's red cells.
B. Tell the physician that no blood can be released until a full work-up has been done.
C. Begin the antibody screening test.
D. Select 2 units of group O, D-negative RBCs for emergency release.

170. Referring to the tube typing results in question 169, the most probable cause of the patient's positive Rh control test is that the patient has
   A. A positive DAT result with anti-IgG
   B. A cold autoantibody
   C. Leukemia
   D. Multiple myeloma

171. A patient experiences severe rigors and goes into shock after receiving part of a unit of RBC. The patient's temperature, which was 37.5°C pretransfusion, is now 40.0°C. Which of the following is the most likely type of reaction?
   A. Hemolytic
   B. Anaphylactic
   C. Septic
   D. Embolic

172. Referring to the reaction described in question 171, the incidence of this type of reaction is highest with which of the following components?
   A. RBC
   B. FFP
   C. Cryoprecipitate
   D. Platelets

173. The serum of a patient transfused 2 weeks ago reacts 3+ on immediate spin and 1+ at the antiglobulin phase of testing with all reagent red cells except for the *ii* cell. The autocontrol reacts similarly to the panel cells. In order to crossmatch this patient, one should
   A. Use autoadsorbed serum
   B. Use the prewarmed technique
   C. Identify the antibody and obtain blood from the rare donor file
   D. Use a LISS additive

INSTRUCTIONS: Each numbered set of test results or conditions (questions 174–184) is followed by four or five lettered responses. Select the *best* answer in each case. Lettered responses may be used once, more than once, or not at all.

Six units of blood from volunteer donors are tested for ABO group, Rh type, and unexpected antibodies. For each set of test results (questions 174–179), indicate the final disposition of the donated unit. Assume additional FDA required testing is nonreactive, unless noted.

	Cell Typing Results			Serum Typing Results		
	Anti-A	Anti-B	Anti-D	A_1 cells	A_2 cells	B cells
174.	0	0	3+	4+	3+	4+
175.	0	0	0	4+	4+	4+
	Weak D test = 3+, DAT = 0					
176.	0	0	0	4+	4+	4+
	Weak D test = 1+, DAT = 1+					
177.	0	0	3+	2+	0	4+
178.	0	0	3+	0	0	4+
179.	0	0	3+	4+	4+	4+
	Antibody screen = positive Antibody identification = anti-Fy^a					

   A. Label group O, D-positive
   B. Label group O, D-negative
   C. Label the RBC group O, D-positive; do not use the plasma
   D. Perform additional testing
   E. Discard the unit

For the following conditions (questions 180–184), select the blood component of choice for treatment.

180. von Willebrand disease

181. Hypofibrinogenemia

182. Factor V deficiency

183. Liver disease

184. Hemorrhagic episode during intensive chemotherapy

A. Platelet concentrate
B. RBC
C. Cryoprecipitate
D. Fresh-frozen plasma

Hemolytic Disease (Hemolytic Disease of the Newborn, Immune Hemolytic Anemia)

185. Which of the following would *not* be included when routine testing is performed early in a pregnancy?

A. ABO and Rh testing
B. Antibody screening
C. Amniocentesis
D. Weak D testing on apparent Rh-negative patients

186. In which of the following blood group systems may the red blood cell typing change during pregnancy?

A. P
B. MNS
C. Lewis
D. Duffy

187. Which of the following is *not* considered a useful predictor of hemolytic disease of the newborn (HDN) during the gestational period?

A. Anti-A
B. Anti-D
C. Anti-Fy^a
D. Anti-U

188. Which is the class of immunoglobulin uniquely associated with hemolytic disease of the newborn (HDN)?

A. IgA
B. IgD
C. IgE
D. IgG

189. A neonate with a positive direct antiglobulin test (DAT) indicates that there was an incompatibility between a mother and her fetus. The system that is most commonly associated with an incompatibility is

A. ABO
B. Rh
C. Kell
D. Kidd

190. The cord blood of an infant of a D-negative mother with anti-D, titer 2048, is submitted to the laboratory along with a sample of maternal blood with a request to select blood for possible exchange transfusion. The neonate appears to be D-negative. The weak D status cannot be determined because the DAT result is positive (4+). What is the most likely explanation for this?

A. Wharton's jelly contaminated the sample.
B. The baby has ABO HDN.
C. The baby has a "blocked D" antigen.
D. A different antibody is causing the positive DAT.

191. A newborn is group O, D-positive and has a 3+ DAT. The mother's antibody screening test is negative. Assuming the antibody detection test is valid, one should consider HDN due to an antibody directed against

A. Fy^b antigen
B. K antigen
C. Low-incidence antigen
D. A or B antigen

192. The most conclusive way to demonstrate the antibody that is causing a positive DAT in a newborn is to perform an antibody
A. Titration using the mother's serum
B. Panel using the mother's serum
C. Panel using an eluate from the mother's red cells
D. Panel using an eluate from the baby's red cells

193. Which two of the following conditions are the most serious immediate consequences of HDN?
A. Anemia and a positive DAT
B. Hyperbilirubinemia and anemia
C. Hyperbilirubinemia and jaundice
D. Hyperbilirubinemia and kernicterus

194. A premature infant with *hydrops fetalis* and a bilirubin of 20 mg/dL is referred to an intensive care unit. The neonatologist wants to perform an exchange transfusion to correct anemia and prevent kernicterus. No blood specimen from the mother is available. The infant's serum has a positive antibody screen. The DAT is 4+. What would be the best approach in this situation?
A. Identify the antibody in the serum and crossmatch blood negative for the offending antigen, using the serum in a crossmatch.
B. Issue group O, D-negative blood for the exchange.
C. Refuse to issue blood for exchange until a sample can be obtained from the mother.
D. Identify the antibody in the serum and eluate and crossmatch blood negative for the offending antigen, using both the serum and eluate in a crossmatch.

195. Which of the following is *not* true of an exchange transfusion when an infant is suffering from HDN?
A. Removes unconjugated bilirubin
B. Reduces the amount of incompatible antibody in the baby's circulation
C. Removes antibody-coated red blood cells
D. Provides red blood cells of the baby's type

196. A massive fetomaternal hemorrhage in a D-negative woman who had a D-positive infant should be suspected if the
A. Infant is premature
B. Infant has a positive acid elution slide test
C. Mother requires a transfusion following childbirth
D. Weak D test on the maternal blood shows a mixed-field reaction microscopically

197. A D-negative woman who received antepartum RhIG gave birth to a D-positive infant and received one vial of RhIG the same day. Because of postpartum hemorrhage, her physician ordered two units of RBCs for her 2 days later. The antibody screening test was positive, but the crossmatches were both compatible. The most likely cause for the positive antibody screening test was the presence of a(an)
A. Clinically significant anti-K
B. Actively acquired anti-D
C. Passively acquired anti-D
D. Rh antibody other than anti-D

198. What is the principle of the Kleihauer-Betke stain?
   A. Fetal hemoglobin is more resistant to alkaline buffer than adult hemoglobin.
   B. Adult hemoglobin is more resistant to alkaline buffer than fetal hemoglobin.
   C. Fetal hemoglobin is more resistant to erythrosin and hematoxylin staining than adult hemoglobin.
   D. Adult hemoglobin is more soluble in acid buffer than fetal hemoglobin.

199. Which of the following antibodies present in a multitransfused obstetric patient would be most likely to cause HDN in her infant?
   A. Anti-Le^a
   B. Anti-c
   C. Anti-P_1
   D. Anti-K

Use the following information to answer questions 200 and 201.

A Kleihauer-Betke acid elution stain for postpartum fetomaternal hemorrhage (FMH) is reported to be 1.3%.

200. What is the total volume of FMH?
   A. 6.5 mL
   B. 13 mL
   C. 26 mL
   D. 65 mL

201. With this amount of FMH, how many vials of a standard dose of RhIG should be administered to the mother within 72 hours of childbirth? (Presume the infant to be D-positive.)
   A. 1
   B. 2
   C. 3
   D. 4

Use the following information to answer questions 202–205.

A 64-year-old female is seen in the emergency department with a hemoglobin value of 8.9 g/dL. The resident sends down a request for 2 units of packed red cells. She types as group O, D-positive using monoclonal antisera. Her ABO group and Rh type match previous records. She has not been transfused in the past 5 years. However, her antibody screen produces the following results:

	Screen Cell I	Screen Cell II	Screen Cell III	Auto-control
IS	0	0	0	0
37°C LISS	0	0	0	0
AHG	4+	4+	4+	4+
CC	NT*	NT	NT	NT

*NT = not tested

202. What is the most likely cause for these results?
   A. Polyagglutination
   B. Rouleaux
   C. Transfusion reaction
   D. Warm autoantibody

203. To demonstrate whether the antibody(ies) has/have become attached this patient's red blood cells *in vivo*, which of the following tests would be most useful?
   A. Direct antiglobulin test
   B. Complement fixation test
   C. Elution procedure
   D. Indirect antiglobulin test

204. How would you identify the antibody(ies) on this woman's cells?
   A. Autoabsorption followed by a panel on the absorbed serum
   B. Elution followed by a panel on the eluate
   C. Enzyme-treated panel on her serum
   D. Perform a panel on the serum.

205. What is the best treatment for this woman's anemia?
   A. Transfusion with packed red cells
   B. Infusion of fresh-frozen plasma
   C. Steroid administration
   D. Plasma exchange

206. The specificity of the antibody in warm autoimmune hemolytic anemia (WAIHA) is most often associated with which of the following blood group systems?
   A. ABO
   B. Kell
   C. Kidd
   D. Rh

207. What is the most important consideration in patients suffering from life-threatening anemia and whose serum contains warm autoantibodies?
   A. Determine the specificity of the autoantibody.
   B. Determine the immunoglobulin class of the autoantibody.
   C. Exclude the presence of alloantibody(ies).
   D. Avoid transfusion.

208. The serum and eluate from a male patient with a 3+ DAT on α-methyldopa therapy demonstrates anti-e specificity. The patient denies knowledge of having received blood transfusions. To determine whether the anti-e is an auto- or alloantibody, one should
   A. Type the patient's red cells with a low-protein anti-e reagent
   B. Adsorb the serum with the patient's red cells
   C. Adsorb the eluate with R_2R_2 red cells
   D. Adsorb the eluate with rr red cells

209. A patient has a 2+mf DAT with anti-IgG. He was transfused 1 week ago with 2 units of RBCs during surgery. His eluate would most likely contain
   A. No antibody
   B. Autoantibody
   C. Alloantibody
   D. Drug-related antibody

Use the following information to answer questions 210 and 211.

An antibody screen was performed on a 25-year-old male referred to the hospital for elective surgery. Refer to the gel reactions below seen in an antibody screen test. (SC = screening cell, AC = autocontrol)

210. Which of the following is the correct interpretation of the reaction in SC I?
   A. 0
   B. 1+
   C. 2+
   D. 3+

211. What has happened to the matrix in SC I that caused a difference in its appearance from that of SC II?
   A. Antibody-coated cells have been trapped in the gel matrix of SC I.
   B. A too heavy red cell suspension was used in SC I; correct suspensions were used in SC II and AC.
   C. Hemolysis is occurring in SC I, but not in SC II or AC.
   D. SC II shows a mixed-field reaction.

212. How is cold hemagglutinin disease (CHD) different from paroxysmal cold hemoglobinuria (PCH)?
   A. PCH is a common form of cold autoimmune anemia whereas CHD is rare.
   B. PCH is a warm autoimmune hemolytic anemia.
   C. The offending antibody in PCH is an IgG antibody unlike the IgM antibody in CHD.
   D. The offending antibody in PCH is an IgM antibody while an IgG antibody is common in CHD.

213. If during a Donath-Landsteiner test there is hemolysis in both the test and control tubes at the conclusion of the test, this indicates that the test is
   A. Positive
   B. Negative
   C. Invalid
   D. False negative

214. A patient has a positive DAT due to cephalosporin therapy and a negative antibody screening test result. Two units of RBCs have been ordered. In order to crossmatch this patient, one should crossmatch with
   A. The eluate from the patient's red cells and donor cells
   B. Autoadsorbed patient's serum and untreated donor cells
   C. Untreated patient's serum and untreated donor cells
   D. Cephalosporin-treated donor cells and untreated patient's serum

215. A patient being treated with α-methyldopa has a 4+ DAT result. You would expect an eluate from his red cells to react most likely with
   A. All the untreated panel cells tested
   B. Just the untreated D-positive cells tested
   C. All panel cells treated with α-methyldopa
   D. All panel cells when α-methyldopa is added to the eluate

216. A patient with drug-induced hemolytic anemia has the following DAT results:
   Polyspecific AHG = 3+
   Anti-IgG = 3+
   Anti-C3d = 0
   Which of the following drugs is most likely to be the cause?
   A. Phenacetin
   B. Quinidine
   C. Penicillin
   D. Tolmetin

217. A patient with cold hemagglutinin disease (CHD) has a positive DAT when tested with a polyspecific AHG. Which of the following would most likely be detected on her red cells?
   A. IgM
   B. IgG
   C. IgA
   D. C3

218. If a patient's red blood cells are DAT+ due to penicillin antibody, the
   A. Serum will react if penicillin is added to the test system
   B. Serum will react with all red cells
   C. Eluate will react with penicillin-coated red cells
   D. Eluate will react with all red cells

219. A patient's preoperative antibody screening test is negative, but the autocontrol is positive. A DAT performed on his red cells is 2+ with anti-IgG. His last transfusion was 9 months ago, and he has a negative drug history. Which of the following would most likely be present in his eluate?
   A. No antibody
   B. Alloantibody
   C. Alloantibody and autoantibody
   D. Autoantibody

220. A patient with WAIHA has a history of an anti-Jk^a in her autoabsorbed serum, and an anti-e in her eluate. Her autoabsorbed serum today is not showing anti-Jk^a on prewarmed panel, but the eluate is still showing anti-e. What blood would be selected for cross-matching packed red cells today?
   A. e-Negative
   B. Jk(a–)
   C. Jk(a–) and e-negative
   D. No screening is necessary because all transfused cells will be destroyed anyway.

INSTRUCTIONS: The numbered group of incomplete statements (questions 221–228) is followed by four suggested responses. Select the *best* answer in each case. Lettered responses may be used once, more than once, or not at all.

For the following situations (questions 221–228), indicate whether the women are candidates for Rh immune globulin (RhIG) prophylaxis. Assume that D-negative mothers have a negative test for weak D and a nonreactive antibody screening test unless noted.

221. Mother D-negative; infant weak D

222. Mother weak D (strong); infant D-positive

223. Mother D-negative; twin #1 D-negative, twin #2 D-positive

224. Mother D-negative with anti-Fy^a; infant D-positive

225. Mother group O, D-negative; infant group A, DAT = 2+, monoclonal anti-D-negative at immediate spin, weak D test not performed

226. Mother D-negative, with anti-D, titer 2, history of RhIG injection postamniocentesis procedure at 30 weeks; infant D-positive

227. Female D-negative; miscarriage at 11 weeks

228. Mother D-negative; infant D-positive; rosette test = 1−2 rosettes per field

A. Yes, 50-μg dose
B. Yes, 300-μg dose
C. Yes, additional testing necessary to determine dose.
D. RhIG is not indicated.

answers & rationales

Blood Collection, Preservation, Processing, Component Preparation, and Quality Control

1.

C. Some potential donors are rejected to protect the recipient, and others are rejected to protect themselves. In this case, the woman meets the criteria except that her hematocrit is too low, and the loss of a unit of blood may have a detrimental effect on her. The minimum acceptable hematocrit is 38%.

2.

C. Donors are allowed to donate no more than 10.5 mL/kg of their body weight. This amount includes the samples used for testing drawn at the time of collection. The calculation for a 95-lb donor is

95 lb ÷ 2.2 lb/kg = 43.2 kg

43.2 kg × 10.5 mL/kg = 453.6 mL

If less than 300 mL is to be collected, the anticoagulant must be reduced proportionately.

3.

A. Hepatitis viruses and HIV have extended incubation periods in which exposure has occurred but neither serological nor clinical manifestations of the disease are evident. The current screening tests, although quite sensitive, are unable to detect the viruses if testing is performed during this incubation period. To safeguard against the possibility that the donor received blood or blood products collected during the incubation period, a 12-month deferral is incurred to allow for fulmination of the disease.

4.

D. During autologous presurgical donation, a different set of criteria is used for donor acceptability. All the conditions listed are acceptable with the exception of the bacteremia. The bacteria may proliferate in the stored blood and be reinfused into the donor (patient) during or after the surgery. Even treatment for a suspected bacteremia is a contraindication for autologous donation.

5.

D. Recipients of human growth hormone are deferred indefinitely because of the risk of transmission of Creutzfeldt-Jakob disease. Recipients of recombinant growth hormone incur no deferral. A history of either syphilis or gonorrhea causes a deferral of 12 months from completion of treatment. Accutane®, a drug used to treat acne, may be a teratogen and requires a 1-month deferral after receipt of the last dose.

6.

A. Of the viruses listed, CMV is the only one that resides exclusively in leukocytes. Although CMV transmission is not a problem for most patients, it can cause serious disease in low-birth-weight neonates of CMV-negative mothers and immunocompromised patients. These patients should be transfused with CMV seronegative or leukocyte-reduced cellular components.

7.

C. Donors who have ingested aspirin within 36 hours of donation need not be excluded for whole blood donation. The platelets prepared from such donors should be labeled and may be used in a multiple pool prepared for adult transfusion. Because aspirin affects platelet function, a single unit of platelet concentrate from this donor should not be used for platelet therapy for infants and neonates. This donor should not be the sole source of platelets and, therefore, would be temporarily deferred as a plateletpheresis donor.

8.

C. Under no circumstances should any blood component from a high-risk donor be released from the donor center to a transfusion unit. Donors in high-risk groups for AIDS must be deferred from donating. If high-risk activity becomes known retrospective to blood donation (such as in the self-exclusion process), the blood components from the donation must be retrieved and destroyed.

9.

C. Plasma protein fraction (PPF) and albumin preparations (5% and 20%) provide colloid replacement and volume expansion with virtually no risk of viral transmission. These are pooled products and are pasteurized by heating to 60°C for 10 hours. Other products, such as clotting factor concentrates, are usually treated by solvent-detergent method to inactivate viruses with lipid envelopes such as HBV, HCV, HIV, and HTLV-I.

10.

B. It is acceptable according to FDA and AABB *Standards* to screen donors for infectious diseases in pools of 16 to 24 donor sera. If a donor pool is positive for HCV, all individual donors making up the pool are tested individually using the same nucleic acid test (NAT) to find the positive donor. When that donor is identified, s/he is excluded from donating henceforth and all components from that donation are retrieved and destroyed.

11.

D. Cryoprecipitate provides the only known concentrated source of fibronectin, useful in the phagocytic removal of bacteria and aggregates by the reticuloendothelial system. It also contains factors VIII:C, VIII:vW, and XIII. Antithrombin III (AT III), necessary to prevent a thromboembolic disorder, is depleted in DIC and liver disease. Transfusion sources of AT III are fresh-frozen plasma (FFP) and commercial concentrates, but AT III is not present in cryoprecipitate.

12.

D. FFP contains all the plasma clotting factors. FFP's primary use is for patients with clotting factor deficiencies for which no concentrate is available and patients who present multiple factor deficiencies such as in liver disease. Platelets are cellular elements, not a plasma clotting factor, and they must be maintained at 20–24°C with continuous gentle agitation to maintain their viability.

13.

C. The AABB *Standards* require that when a patient is likely at risk for graft-versus-host disease (GVHD), all cellular blood components must be irradiated before transfusion. This includes components for patients who are immunodeficient or immunoincompetent, such as a patient on immunosuppressive therapy and a fetus who receives intrauterine transfusion. Irradiation of RBCs for exchange transfusion is not required by AABB *Standards,* although many hospital transfusion services do so. Immunocompetent individuals require irradiated components if they are to receive cellular components from someone who may be homozygous for a shared HLA haplotype, such as a blood relative or an HLA-matched donor. Gamma irradiation of cellular components is the *only* way to prevent transfusion-associated GVHD that occurs when immunocompetent donor T cells survive in the patient's circulation and mount an immune response against the host cells. A minimum of 25 Gy delivered to the midplane of the container and at least 15 Gy to all other areas will prevent GVHD.

14.

A. The limiting criterion for *in vitro* storage of blood is the survival in the recipient of at least 75% of the transfused red cells for at least 24 hours after transfusion. Additional adenine in an anticoagulant-preservative formulation provides a substrate for the continued generation of ATP *in vitro*. The overall effect is improved viability.

15.

D. Donor blood may not be labeled according to test results obtained from previous donations. Several segments removed from the donor unit will provide sufficient sample for all required testing but will limit the number of segments available for crossmatching. After centrifugation, the plasma may be removed from the segments and clotted with calcium chloride or a similar commercial product for use in test procedures requiring serum. Alternatively, institutions with sterile connecting devices may attach a small bag and remove an aliquot sufficient for testing.

16–17.

(16:C, 17:A) *In vitro* recovery of factor VIII must be assayed monthly to ensure proper control of conditions during cryoprecipitate production. A minimum of 80 international units (IU) per container must be present in the final product. One international unit is defined as the clotting activity of 1 mL of fresh plasma. The total number of factor VIII units is calculated from the formula:

$$\text{Factor VIII (IU/mL)} \times \text{volume (mL)} = \text{Total IU Factor VIII}$$

In this case, 9 IU/mL × 10 mL = 90 IU per container in the final product. This exceeds the required 80 IU/container and so meets AABB *Standards*.

Although there is no existing standard for percent recovery of factor VIII during production, this information may be helpful in monitoring various stages of production when the monthly quality control assays fall below the acceptable standard. Recovery can be calculated by the formula:

$$\frac{\text{Post (F VIII IU/mL} \times \text{volume mL)}}{\text{Pre (F VIII IU/mL} \times \text{volume mL)}} \times 100 = \%\ \text{Factor VIII recovery}$$

In this instance,

$$\frac{9\ \text{IU/mL} \times 10\ \text{mL}}{1\ \text{IU/mL} \times 250\ \text{mL}} \times 100 = 36\%\ \text{Factor VIII recovery}$$

18–21.

(18:D, 19:D, 20:B, 21:B) Each centrifuge used for platelet production must be calibrated upon receipt or after major repair. Calculating percent recovery during various stages of platelet production is a valuable troubleshooting tool when monthly quality control assays consistently fall below the minimum standard of 5.5×10^{10} platelets per platelet concentrate. In this exercise, the first spin gives an 83% platelet yield, and the second spin produces a 60% yield from the PRP. Increasing the time or rpm of the PRP will result in a greater percent yield for the platelet concentrate. Yield is calculated from the following formulas:

$$\text{Platelet count}/\mu\text{L} \times 1000\ \mu\text{L/mL} \times \text{Volume (mL)} = \text{Total count in component}$$

$$\frac{\text{Total count in PRP}}{\text{Total count in whole blood}} \times 100 = \%\ \text{yield from whole blood}$$

$$\frac{\text{Total count in platelet concentrate}}{\text{Total count in PRP}} \times 100 = \%\ \text{yield from PRP}$$

In the example above, the total count in PRP would be

$$300{,}000/\mu\text{L} \times 1000\ \mu\text{L/mL} \times 250\ \text{mL} = 7.5 \times 10^{10}\ \text{platelets}$$

The total count in whole blood is

$$200{,}000/\mu\text{L} \times 1000\ \mu\text{L/mL} \times 450\ \text{mL} = 9 \times 10^{10}\ \text{platelets}$$

The percent yield from whole blood is

$$\frac{7.5 \times 10^{10}}{9 \times 10^{10}} \times 100 = 83\%$$

The total count in the platelet concentrate is

$$900{,}000/\mu\text{L} \times 1000\ \mu\text{L/mL} \times 50\ \text{mL} = 4.5 \times 10^{10}\ \text{platelets}$$

The percent yield from PRP is

$$\frac{4.5 \times 10^{10}}{7.5 \times 10^{10}} \times 100 = 60\%$$

At least 75% of prepared platelet packs must contain a minimum of 5.5×10^{10} cells. Because there are only 4.5×10^{10} cells in this platelet concentrate, the platelet yield would be considered too low.

22.

B. A low red blood cell concentration of 2,3-BPG increases red cell affinity for O_2, causing less O_2 to be released to the tissues. As blood is stored, 2,3-BPG levels fall. Once the blood is transfused, red cells regenerate 2,3-BPG and ATP, which are fully restored in about 24 hours. Other metabolic changes that occur as blood is stored are an increase in plasma K^+ as red cells leak K^+, an increase in plasma hemoglobin, and a decrease in ATP.

23.

D. Autologous blood should not be drawn later than 72 hours prior to surgery. The reason is to allow time for adequate volume repletion. However, the medical director may decrease this time if the patient's condition warrants it.

24.

B. Preoperative autologous donation is commonly done for orthopedic surgery, radical prostatectomy, and open heart surgery. Patients with uncompensated anemia and hemoglobin levels below 11.0 g/dL are unable to donate because they do not have sufficient red blood cells to maintain oxygen-carrying capacity after the donation. Because it is difficult to find donors for patients with multiple antibodies or an antibody to a high-incidence antigen, these individuals, if anemic, may be given supplemental iron and allowed to donate once their hemoglobin levels are above 11.0 g/dL. Their cells can also be frozen for later use.

25.

C. Both Hepatitis B virus (HBV) and hepatitis C virus (HCV) are transfusion transmitted. However, only HCV is associated with hepatocellular carcinoma and cirrhosis of the liver in many of the chronically infected. Because nucleic acid testing for HCV is done on all donor samples, there is only a small risk (<1 in 2,000,000) of transmission from tested donors. The acute phase of the disease is frequently asymptomatic, but most of these patients become chronic carriers, with 70–80% having persistent infections. About 10% of those chronically infected eventually develop cirrhosis and/or hepatic carcinoma.

26.

B. Red blood cells continue to metabolize, albeit at a slower rate, during storage at 1–6°C. Decreased ATP levels result in loss of RBC viability. Plasma hemoglobin, ammonia, and K^+ levels increase, whereas plasma Na^+ and pH and 2,3-BPG levels decrease. These biochemical changes are collectively referred to as the "storage lesion" of blood.

27.

C. Because the patient received eight units of blood and none of the donors has been implicated in other cases of hepatitis, none of these donors would be deferred. The donor center should be immediately notified so it can enter in each donor's record that s/he has been implicated in a case of transfusion-transmitted hepatitis. After a second implication, the donor would be indefinitely deferred. If only one donor had been implicated, s/he would have been indefinitely deferred.

28.

C. Red blood cells and whole blood must be stored between 1 and 6°C in a monitored refrigerator with a recording thermometer and audible alarm system. During transportation between collection and transfusion facilities, blood must be packed in well-insulated containers designed to maintain a temperature range of 1–10°C. Wet ice in a leak-proof plastic bag is placed on top of the blood. The amount of ice to be used is dictated by the transportation time, the number of units packed, and the ambient outside temperature.

29.

C. In addition to the minimum number of platelets that should be present, 5.5×10^{10}, the pH of the unit must be 6.2 or higher in at least 75% of the units. The units should be assayed at the end of the allowable storage period. A donor who has taken aspirin should not be the sole donor of platelets for a patient. Aspirin has an adverse effect on platelet aggregation.

30.

C. Platelets must be stored in sufficient plasma volume to prevent the pH from falling below 6.2 at the time of expiration. Lactic acid is a by-product of anaerobic glycolysis during platelet storage, causing a drop in plasma pH and a loss of discoid shape, and hence viability. Second generation platelet bags allow better gas exchange, permitting platelets to be stored for longer periods of time at a favorable pH.

31.

C. Red blood cells expire 24 hours from the time the hermetic seal is broken, provided they are maintained at 1–6°C during the storage period. The new expiration date and time must be placed on the label and in the appropriate records. An open system exposes the blood to possible bacterial contamination. Blood may be frozen for up to 6 days after collection when maintained at 1–6°C in a closed system. If the seal is inadvertently broken on a rare unit during component preparation, the red cells may be salvaged by glycerolization and freezing, providing this is accomplished within the 24-hour restriction.

32.

D. The AABB *Standards* require that an authorized individual (such as a supervisor or medical director) review the standard operating procedures (SOPs), policies, and process annually and document the review. The SOPs should be reviewed and revised as needed to reflect the techniques used by the laboratory. It is prudent to conduct a review before a scheduled inspection and following publication of each new edition of AABB *Standards* to ensure conformance with new requirements.

33.

B. Previous ABO and Rh records of patients must be retained for 10 years and be immediately available for 12 months as a check to confirm the identity of the current pretransfusion sample. Records of unexpected antibodies identified in the serum of intended recipients and of serious adverse reactions to blood components must be retained indefinitely. Consulting records may prevent a delayed hemolytic transfusion reaction when the antibody is no longer demonstrable.

34.

D. In the United States, the weak D test is performed routinely when a donor appears to be Rh-negative, and all weak D donor units are labeled Rh-positive. Weak D units are much less immunogenic than normal D units. In many countries, neither donors nor recipients are tested for weak D.

35.

B. In order to meet the current AABB *Standards* for leukocyte reduction to prevent HLA alloimmunization or CMV transmission, the donor unit must retain at least 80% of the original red cells and the leukocytes must be reduced to less than 5×10^6. Leukocyte reduction may also prevent febrile reactions in two ways: (1) By reducing the number of leukocytes in the component to a low enough level, one can prevent febrile reactions when patients have leukocyte antibodies. (2) Cytokines are also known to cause febrile reactions. If prestorage leukocyte reduction is done, cytokine generation should be prevented.

36.

D. ABO grouping must be determined by doing both cell and serum grouping. The Rh type must be determined by direct agglutination with anti-D; if negative, the test is incubated and converted to the antiglobulin test to detect weak D phenotypes. Performing an antibody screening test on the serum or plasma of a donor is required when the donor has a history of transfusion or pregnancy. For practical purposes, most donor centers screen all donors for clinically significant antibodies. The absence of hepatitis B surface antigen (HB_sAg) and HIV must be confirmed using a method currently licensed by the FDA. The test for hepatitis B surface antibody (HB_sAb) is not required.

37.

C. In the average-size adult (70 kg), a unit of platelet concentrate should raise the platelet count by 5000–10,000/μL if there are no other complicating factors to cause decreased survival. Complicating factors include fever, sepsis, disseminated intravascular coagulation (DIC), and HLA sensitization. One apheresis platelet unit is equivalent to 6–8 units of pooled platelet concentrate and has the advantage of decreased donor exposure.

38.

A. The ABO group on all units and the Rh type on all D-negative units must be repeated by the transfusing facility for units of RBCs or whole blood collected and processed at another facility. This is generally accomplished by repeating the cell grouping only. To save time and reagent cost, it is convenient to test units labeled group O with anti-A,B only. Confirmatory testing for weak D is not required. The Rh type of units labeled D-positive need not be confirmed. Repeat antibody screening and viral testing are not required.

39–43.

(39:A, 40:D, 41:C, 42:C, 43:B) The storage temperature for whole blood, modified whole blood, RBCs, including leukocyte-reduced and deglycerolized products, is between 1 and 6°C. This range may be extended to 10°C during brief periods of transport. RBCs are frozen in a glycerol solution. These units must be stored at –65°C or lower. Fresh-frozen plasma (FFP) and cryoprecipitate are stored at –18°C or colder with a 1-year expiration. Although this temperature meets AABB *Standards,* optimal storage temperature is –30°C or below. In fact, FFP expiration may be extended to 7 years if kept at –65°C or lower. Frozen storage at low temperatures maintains optimum levels of the labile coagulation factors V and VIII in FFP and VIII in cryoprecipitate. Plasma should be frozen within 8 hours of collection when collected in CPD or CPDA-1. Platelet concentrates are stored at room temperature (20–24°C). They need to be agitated during storage.

44–48.

(44:C, 45:D, 46:D, 47:A, 48:B) Whole blood and RBCs may be stored up to 35 days when collected in CPDA-1, as long as the hermetic seal remains unbroken. Adenine added to the anticoagulant increases the viability of the cells. Cells stored only in CPD have a shorter allowable storage of 21 days. Addition of adsol solution extends expiration of red blood cells to 42 days. Fresh-frozen plasma (FFP) and cryoprecipitate expire 12 months from the date of collection if stored at –18°C or colder. The expiration time for these components is based on the deterioration of the labile factor VIII. Units stored beyond 12 months may have reduced levels of factor VIII unless stored at much lower temperatures. FFP has been approved for 7-year storage if kept at –65°C or lower. Once thawed, FFP expires in 24 hours when stored at 1–6°C. The type of plastic used in the manufacture of the bag affects the allowable storage time for platelets. The older type of bag (polyvinylchloride) does not allow as effective a gas exchange as the newer types of plastic. Platelet concentrates, prepared in PL–732 bags and stored at 20–24°C with agitation, expire 5 days from the date and time of collection.

49–52.

(49:A, 50:B, 51:C, 52:D) Blood cells continue to metabolize *in vitro.* Plasma glucose and ATP are depleted. Intermediary metabolites are generated. These may interfere with the production of energy via glycolysis. This results in a gradual loss of red blood cell viability. Storage at lowered temperatures (1–6°C) slows metabolism. ACD and CPD solutions contain sufficient glucose to support RBC viability for 21 days. CPDA-1 also contains adenine, which allows extension of the shelf life to 35 days. Adenine maintains viability by ATP regeneration. Red blood cells prepared with additive solutions such as AS-1 have a shelf life of 42 days. EDTA is not an approved solution for the storage of blood for transfusion.

53–59.

(53:D, 54:A, 55:C, 56:D, 57:D, 58:D, 59:D) Donors may be accepted after age 17, provided all results of the physical examination are normal. There is no upper age limit. Elderly donors may participate in a blood program at the discretion of the local blood bank physician. Many senior citizens obtain written permission from their personal physicians and present approval at the time of donation. The interval between donation of blood for allogeneic transfusion is 8 weeks, or 56 days. This time period is designed to protect the health of the donor. Exceptions at the discretion of the blood bank and personal physician may be made if the blood is intended for autologous use. A man who had a history of sex with another man after 1977 must be indefinitely deferred because of the possibility of transmitting the HIV virus. A history of jaundice in the first days of life is indicative of hemolytic disease of the newborn and is not a cause for deferral. A mild skin rash caused by acne, poison ivy, psoriasis, or other allergies is not a cause for donor deferral, as long as the disorder does not extend into the antecubital area at the venipuncture site. Final acceptance or deferral may be made at the phlebotomist's discretion, dependent upon whether the arm can be properly prepared to maintain sterility of the product without undue discomfort to the donor. A woman who has been pregnant is deferred until 6 weeks following conclusion of the pregnancy unless her blood is needed for her infant and the donation is physician approved.

The acceptable limits of the physical examination include:

Temperature: 37.5°C (99.5°F) or less
Pulse: 50–100 bpm
Blood pressure: systolic ≤180 mm Hg, diastolic ≤100 mm Hg

Runners or other athletes may be accepted when the pulse rate is less than 50 bpm, as long as no irregularity in beats is detected. These parameters are incorporated in the AABB *Standards* for the safety of the donor and are in general use by all blood-collecting facilities. For donor suitability, the FDA and AABB require only that the hemoglobin level be no less than 12.5 g/dL (with no sex differentiation) and that the temperature and blood pressure be within normal limits as determined by a qualified physician or by persons under his or her supervision.

60–63.

(60:D, 61:C, 62:B, 63:A) Patients with warm autoimmune hemolytic anemia (AIHA) secondary to α-methyldopa respond rapidly following cessation of the drug. They can usually be managed without transfusion. The DAT (direct antiglobulin test) may not revert to negative for up to 6 months or even longer. Leukocyte-reduced blood components ($\leq 5 \times 10^6$) are indicated in order to avoid repeated febrile episodes, CMV transmission, and alloimmunization to leukocytes. Leukocytes can be removed by filtration, centrifugation, or washing. Currently, the preferred and most efficient method is filtration with commercially available adsorption filters capable of reducing leukocytes to the required level. Patients with normovolemic anemia should be transfused with RBCs, which provide the red blood cells needed to correct the anemia in the smallest volume. These patients may not be able to tolerate whole blood because of the volume increase. It is not necessary to use leukocyte-reduced RBC for patients with normovolemic anemia. Thrombocytopenia means there is a lack of platelets. Often platelet counts drop in acute leukemia and during the subsequent treatment. Platelet counts below 20,000/μL are not uncommon under the circumstances, and the patient is considered to have severe thrombocytopenia. Leukocyte-reduced platelets will lower the chance of alloimmunization and are routinely given prophylactically to leukemia patients.

Blood Groups, Genetics, Serology

64.

D. The body makes five different immunoglobulins: IgA, IgD, IgE, IgG, and IgM. IgG makes up about 80% of the total serum immunoglobulin. Although IgA is more abundant than IgM (13% versus 6%), IgM is more common as a blood group antibody.

65.

B. From his phenotype, the father appears to be homozygous *kk* genetically. However, he is actually K^0k and has passed the K^0 gene to child 1. The K^0 gene at the *Kell* locus does not appear to result in the formation of any Kell system antigens. Child 1 has the genotype K^0K, having received the *K* gene from the mother, and has a phenotype expressing on the K antigen. Child 2 can have either the genotype *kk* or K^0k, with the mother contributing a *k* gene and the father either K^0 or *k*. The McLeod phenotype would result in weakened expression of K or k antigens. There has been no *cis-Kk* gene discovered, nor any dominant inhibitor gene that represses the expression of Kell system genes.

66.

D. All red blood cells contain some amount of H substance. The only exception is the very rare O_h (Bombay) individual because these persons lack the *H* gene that codes for H substance. Group O cells contain the most H substance, and A_1B cells contain the least amount of H substance. The order of decreasing reactivity with anti-H is: O > A_2 > A_2B > B > A_1 > A_1B.

67.

B. Two genes control Rh antigen activity. *RHD* controls the expression of D antigen, and *RHCE* determines the C, E, c, and e antigens. *RHD* is absent or inactive in D-negative individuals. Alleles of *RHCE* are *RHCe, RHcE,* and *RHce*. The RH is often dropped (for example *CE, Ce, cE, ce*).

68.

C. This individual does not have a cold autoantibody, as demonstrated by the negative autocontrol at all phases of testing. Being a nonsecretor does not affect the ABO or Rh typing, nor will it cause the appearance of unexpected antibodies in the patient's plasma. Rouleaux is ruled out because reactions are still seen at AHG after all of the patient serum or plasma has been washed away. Of the choices given, the most likely is that the patient is a Bombay phenotype individual, having inherited one *h* gene from each parent. The only other possibility is that the patient is a group O with a strong unexpected antibody or antibodies in his/her serum; however, that was not one of the choices given as an answer.

69.

B. Bombay (O_h) individuals' red blood cells not only lack A and B substances, but they also lack H substance. Bombays are genetically *hh* and, therefore, are unable to produce the precursor H substance upon which the A and B transferases act to produce A and B substances. In their serum, they will have anti-A, anti-B, anti-A,B, and an equally strong anti-H, which will react with normal group O cells. Neither O nor O_h red blood cells react with anti-A,B or anti-A_1 lectin. However, O_h red blood cells give a negative reaction with anti-H lectin, whereas O cells are positive, allowing differentiation of the two.

70.

C. The *Le* gene codes for a transferase enzyme, L-fucosyl transferase, which attaches fucose to the subterminal sugar on the Type 1 precursor substance producing Le^a substance. This occurs independently of the ABH secretor status. For Le^b as well as ABH substances to be present in the secretions, both the *Se* gene and the *Le* gene must be present. The *Se* gene produces a transferase that attaches a fucose to the terminal sugar on precursor substance, forming H substance in the secretions. Type 1H and Type 2H are the precursors for A and B substance. The *Le* gene can act upon Type 1H as well to form Le^b substance; therefore, a nonsecretor who has a *Le* gene will only secrete Le^a, whereas a secretor will secrete a little Le^a and a lot of Le^b substance.

71.

D. Monoclonal reagents containing the ES-4 clone react well with acquired B cells, and those lacking that clone do not react. Most human anti-B will react, but not the individual's own anti-B. Acquired B antigens are often associated with carcinoma of the colon, gram-negative infection, and intestinal obstructions. Also, B substance will not be found in the saliva of a person with an acquired B antigen if the patient is an ABH secretor. Acquired B occurs in group A people when microbial enzymes deacetylate the A determinant sugar (*N*-acetylgalactosamine) so that it resembles the B sugar (D-galactose).

72.

B. Lectins are proteins present in plants, often derived from the seeds of plants. Lectins can also be found in lower forms of animal life. The specificity of lectins is for carbohydrate moieties and, used undiluted, they will often react with all human red blood cells. The lectins used in blood banking are most often derived from the seeds of a plant, then diluted to achieve the desired specificity.

73.

D. The sugar L-fucose is attached to the terminal sugar of precursor substance by a fucosyl transferase. The fucosyl transferase coded by the *H* gene adds the fucose to the precursor substance on the red cells. The fucosyl transferase encoded by the *Se* gene adds a fucose to the precursor substance in the same configuration in the secretions. In both cases, the resulting configuration is called H substance. Without H substance present, the sugars giving A or B antigenic activity cannot attach.

74.

B. The complement cascade has many functions in the body associated with immunity and inflammation. The last stages of the complement cascade ultimately lead to RBC lysis. IgM is the immunoglobulin that most readily activates complement. The IgG immunoglobulins can activate complement to a lesser extent. IgG3 activates complement more efficiently than the other IgG subclasses. Although glucose-6-phosphate dehydrogenase deficiency can result in the lysis of RBCs in the presence of fava beans and certain drugs, the enzyme itself does not lyse RBCs. Albumin and antihuman globulin serum can be used in blood bank testing and do not harm RBCs.

75.

B. The answer is based upon the frequencies of genes. The genes that code for the haplotypes *DCe* and *dce* are high in the white population. A *DCe/dce* genotype has a frequency of approximately 31.1% in the general white population. The other two possible choices among the answers that would fit the typing results are *DCe/Dce* and *Dce/dCe* and have frequencies of approximately 3.4% and 0.2%, respectively. *DCe/DcE* is incorrect because the typing does not indicate that the E antigen is present.

76.

C. All these genotypes have a low frequency in the black population. *DCe/dcE* is the rarest, with a frequency of <0.1%. *DCe/DcE* is the most frequent, with an occurrence of 3.7%.

77.

A. Red blood cells that have either the C or D antigen also have the G antigen. When anti-G is made, it is capable of reacting with the G antigen on both C-positive and D-positive red blood cells, therefore appearing to be anti-C plus anti-D. In the stated case, the immunizing red blood cells were D-negative and C-positive. Therefore, what appears to be a combination of anti-D and anti-C is anti-G or a combination of anti-C and anti-G.

78.

B. The unit from the *DCe/dce* donor has the c antigen that the patient lacks. This antigen is a good immunogen. Although this patient can form the anti-E antibody, the donor cells lack the E antigen. Thus, the donor cells cannot stimulate the production of anti-E. Remember, "d" simply implies the absence of D and is not an antigen.

79.

C. Fifty percent of the children can be expected to be D-positive (*DCe/dce*) and 50% can be expected to be D-negative (*dce/dce*). The following chart clearly illustrates how the percentages were determined. The mother can pass on only *dce* haplotype, whereas the father can pass on *DCe* or *dce*.

	Father	
Mother	***DCe***	***dce***
dce	*DCe/dce*	*dce/dce*
dce	*DCe/dce*	*dce/dce*

80.

C. The husband's genotype is most likely *Dce/dCe*. He is weak D because of position effect and has a normal *D* gene. The *C* in trans position to a normal *D* gene often causes a weakened expression of D antigen. The infant has inherited the father's normal *D* gene but does not have *C* in trans position, and therefore has a normal D antigen expression. Thus, the husband is not excluded and is probably the father.

81.

C. The wife is dce/dce and, lacking the D antigen, can make anti-D. The child is D+ and therefore cannot make anti-D. The husband, although being weak D, has passed a normal D antigen to the child. This indicates that the husband's "weak D" antigen is not the result of a partial D, and therefore he cannot make anti-D.

82.

A. Occasionally, D-positive people make an apparent anti-D. The D antigen is made up of several epitopes or antigenic determinants. Those individuals missing one or more of these epitopes are called partial D. When individuals lack one or more of these epitopes, they can make an antibody, after appropriate stimulation, against the epitope or epitopes that they lack. All of these antibodies to D epitopes will react with "normal" D-positive cells that have all of the epitopes on the D antigen. Therefore, a D-positive or weak-D person appears to make an anti-D.

83.

A. The k antigen is a high-frequency or "public" antigen present in greater than 99% of the random population. The probability of encountering an individual who is k-negative and capable of producing the corresponding antibody after red blood cell stimulation is very low. Kell system antigens are good immunogens, second only to those of the Rh system. Although it is true that $Kell_{null}$ individuals are very rare, they do not make a separable anti-k.

84.

C. Anti-Jk^a often declines in the serum to below detectable levels. Therefore, when a patient has been transfused and makes anti-Jk^a and then is not transfused again for a long time, the subsequent antibody screen may not reveal the presence of the anti-Jk^a. An intravascular delayed transfusion reaction is characteristic of the Kidd antibodies because, after a second stimulation, there is a slow rise in the antibody titer and they activate complement very well. In the case of the patient in question, the antibody has been missed and he received Jk(a+) cells at some point during his stay in the other hospital. This caused a severe delayed hemolytic transfusion reaction with intravascular hemolysis.

85.

A. The Xg^a antigen is produced by a gene on the X chromosome. Because women inherit two X chromosomes, there is a higher incidence of the antigen in females. The antibody is usually detected by an antiglobulin test, and the antigenic activity is depressed by enzymes.

86.

D. To ensure that an antiserum is reacting properly, positive and negative controls must be tested. The antiserum must be tested against a cell that is negative for the corresponding antigen to ensure that no interfering substances are present that will cause false positives. It must also be tested against a cell positive for the corresponding antigen. A heterozygous cell is used to determine whether or not the antiserum will be reactive with the smaller number of antigen sites on the RBCs seen in heterozygotes. For example, when using anti-K, you would test a *kk* (K–k+) cell and a *Kk* (K+k+) cell.

87.

D. Kidd antibodies are often weak and deteriorate during storage. They are usually IgG, antiglobulin reactive only, and complement dependent. Reactions are enhanced when enzyme-treated panel cells are used. Kidd antibodies also show dosage, and the titer may drop to undetectable levels after the primary response. For this reason, they are often implicated in delayed hemolytic transfusion reactions when there is no previous record of the presence of the antibody. A Kidd antibody rarely occurs singly in a patient's serum but is often seen accompanying other antibodies.

88.

B. Enzymes denature the Fy^a and Fy^b antigens and render panel cells Fy(a–b–). Therefore, anti-Fy^a and anti-Fy^b will not react with enzyme-treated red cells. These Duffy antibodies are clinically significant. They can cause hemolytic transfusion reactions and mild hemolytic disease of the newborn. They are usually IgG antibodies and are best detected by the antiglobulin technique.

89.

B. Chido (Ch^a) and Rodgers (Rg^a) antigens are actually "pieces" of complement component 4 (C4) and are present on some RBCs, although not an RBC antigen. Anti-Ch^a and anti-Rg^a have a relatively high titer with a low avidity (HTLA) for the corresponding antigen, often not stronger than 1+ agglutination. Clinically, these antibodies are usually not significant. They are IgG and are detected by the indirect antiglobulin test. They can be neutralized with pooled normal human plasma, because complement components are always present in some amount in human plasma. Pooled plasma is used because the amount of C4 in the plasma varies between individuals. All of the other antibodies mentioned in the answers are RBC antigens and cannot be neutralized in this way.

90.

D. Anti-U is a clinically significant IgG antibody causing hemolytic transfusion reactions and hemolytic disease of the newborn. All white people appear to be U+, because no U negatives have been found. However, about 99% of blacks are U+ and 1% are U–. Those people who are U– are also S–s– and lack the entire Ss sialoglycoprotein (glycophorin B) except for very rare genetic mutations.

91.

B. Although the patient's antibody screening is negative at this time, previous records show that the patient had an anti-E. Anti-E is a significant IgG antibody; only blood negative for the E antigen should be transfused to the patient. Failure to give E-negative blood could result in a serious delayed transfusion reaction due to an anamnestic response.

92.

B. For control purposes the cell should have the weakest expression of the antigen in question; that would be an Fy(a+b+) cell. A weaker cell from a heterozygote is used because a weak antiserum might detect an antigen from a homozygote but not from a heterozygote (dosage effect). If this should happen, then red blood cells might be mistyped as Fy(a–) when in fact the cells are Fy(a+).

93.

B. Most antibodies in the Kell system are red blood cell stimulated. They are generally IgG antibodies and usually detected in the antiglobulin phase of testing. Because of their nature, they have been implicated in both transfusion reactions and hemolytic disease of the newborn. The other choices are usually IgM antibodies that cannot cross the placenta and are rarely involved in transfusion reactions.

94.

A. Antibodies can be ruled out using only one cell that is homozygously antigen positive and nonreactive at all phases of testing with the patient serum. Preferably, two or three cells that are positive for the antigen and nonreactive with the patient serum increase confidence. Antibodies in the Duffy, Kidd, and MNSs systems often show dosage. Because there are more antigenic sites on homozygous cells than on heterozygous cells, these antibodies react more strongly with homozygous cells [cells with 2 "doses" of the antigen, such as Jk(a+b−)] than with cells that carry a single "dose" of the antigen [such as Jk(a+b+)]. These antibodies can be ruled out only using homozygous cells that are positive for the antigen and do not react with the patient serum. Of the answers available, **"A"** is the best choice. Additional antibodies may not be ruled out when a standard of three cells that are positive for the antigen and nonreactive with the patient serum is used.

95.

B. The most likely combination of antibodies is anti-E plus anti-K. More than one antibody is likely because the reactions seen are of varying strengths. The panel cells that did not react with the patient serum are all lacking both the E and K antigens. The E antigen is present on Cells #4, #5, and #9. The K antigen is present on Cells #4 and #7. Note that Cell #4 has both the E and K antigens and reacts 4+ at AHG. Cells #5 and #9 have the E antigen, but not the K antigen, and both cells react 3+ at AHG. Cell #7 has the K antigen but not the E antigen and reacts 2+ at AHG. Therefore, the anti-E is stronger than the anti-K, and when both antigens are present the reaction is stronger than a reaction with one antibody alone.

96.

C. Cells #1, #4, and #7 are the cells from this panel that will be helpful in confirming the antibodies, anti-E and anti-K, and ruling out the other possible antibodies. Cell #1 is both E- and K-negative and should not react with the patient's serum. However, Cell #1 is also S+s–, Le(a+), Jk(a–b+), and Fy(a–b+) and can help to rule out all of the other possible antibodies. Cell #4 is E+, K–, S–, s+, Le(a–), Jk(b–), and Fy(b–). This cell can help confirm the presence of anti-E. Cell #7 is E–, S–, Le(a–), K+, Jk(b–), and Fy(b–). This cell can help confirm the presence of anti-K.

97.

C. Anti-Le^a, -Le^b, and -P_1 may all be neutralized by commercially available soluble substances. Le^a and Le^b are not RBC antigens but are plasma substances that are absorbed onto RBCs in the circulation. Soluble antigens are more available to the antibodies and can attach to soluble antibodies more readily than particulate antigens. Thus, the plasma (soluble) Le^a and Le^b can be used to bind to the soluble antibodies, leaving no antibody to react with the particulate antigen on the RBCs. Soluble P_1 can be obtained from several sources and can be used in the same way to preferentially bind the anti-P_1, leaving no anti-P_1 to react with the particulate P_1 antigen on the RBCs.

98.

C. Anti-N is the only antibody listed that is generally a room temperature saline agglutinin. The remaining choices, anti-Fy^a, anti-Jk^b, and anti-U, are best detected at the antiglobulin phase of testing. Remember, this is where these antibodies are optimally reactive; it does not mean they will never react at other phases of testing. Some antibodies just don't read the books!

99.

D. The Fy^a antigen is destroyed by enzyme treatment. Therefore, the anti-Fy^a seen in the initial panel will not react in an enzyme-treated panel. Enzyme-treated cells react extremely well with antibodies of the Rh system. Because the second antibody is suspected of being a weak anti-C, the antibody will react more strongly with an enzyme-treated panel than it did in the initial LISS panel. Although one cannot rule out antibodies to Duffy system, MNSs system, or Xg^a using an enzyme-treated panel, all other major blood group system antibodies present in the patient's serum should react in this medium. An elution removes antibody from the RBCs; in our scenario, the antibody is seen in the serum. Absorption of the anti-Fy^a from the patient serum could be useful; however, finding cells that lack all of the antigens lacked by the patient to avoid missing an alloantibody is a difficult and time-consuming task. Additionally, absorption will unavoidably dilute the patient's serum slightly and may dilute the weak second antibody to a point where it cannot be identified. Antigen typing the patient's cells is useful in determining whether or not the patient can form anti-C and anti-Fy^a, but it will not exclude any other antibodies nor confirm the presence of any specific antibody.

100.

B. Solid-phase red cell adherence assays, the gel test, and affinity column technology are all third-generation antibody detection methods. They have equal or greater sensitivity for clinically significant antibodies than first- and second-generation techniques. In general, they have the following advantages: less hands-on time, smaller sample size, improved safety, and stable endpoints, and they can be automated. In the gel test, the antiglobulin test does not require washing or the addition of IgG coated cells, because unbound globulins are trapped in the viscous barrier at the top of the gel column. Upon centrifugation, the anti-IgG in the column traps red cells that have been coated with IgG during the incubation period. In affinity column technology, the viscous barrier traps unbound IgG, but *Staphylococcus aureus* derived protein A and protein G are in the column instead of anti-IgG and react with the Fc portion of IgG-coated red cells. The other two techniques, solid-phase red cell adherence and polyethylene glycol, require a washing step.

101.

D. The three antibodies, anti-P_1, anti-Le^a, and anti-I, are most often non-red-cell-stimulated IgM antibodies. All of these antibodies can also, however, be stimulated by exposure to red blood cells carrying the corresponding antigen. Each of the other answers has at least one antibody that will be formed only due to exposure, either through transfusion or pregnancy, to the corresponding red cell antigen.

102.

B. Lymphocytotoxicity testing is performed by adding patient mononuclear leukocytes to wells containing sera that have antibodies to various HLA antigens and then adding guinea pig complement and indicator dye. The cells that take up the dye have had their cell membranes perforated by the action of the antigen-antibody reaction and complement, indicating that they carry the HLA antigen corresponding to the antibody in the well. Although the antibodies to Bg system antigens are antibodies to HLA antigens, the Bg terminology is only used for the remnant HLA antigens found on RBCs. The Wright system and JMH antigens are RBC antigens and are not related to the HLA system.

103.

A. Mixed-field agglutination refers to an agglutination pattern where there are two distinct cell populations, one agglutinated and one not. The appearance is clumps of cells among many unagglutinated cells. In a delayed hemolytic transfusion reaction, surviving donor cells will be coated with patient antibody and the patient's own cells will not, yielding a mixed-field DAT result. Other examples of mixed-field agglutination are seen in patients who have been transfused with blood of another ABO group, in patients with Lutheran antibodies, and in D-negative mothers with D-positive infants where there was a large fetomaternal bleed. Also, A_3 subgroup RBCs may demonstrate a mixed-field reaction with anti-A.

104.

D. The first antibody to become detectable in a primary immune response to a foreign blood group antigen is IgM followed by IgG, usually detectable from less than a week to several months after immunization. After secondary exposure to the same antigen, the antibody titer usually increases rapidly within several days. Antibodies are produced by plasma cells. The antibody produced by the plasma cells in the secondary response is IgG. Plasma cells are the terminal differentiation of the B lymphocytes. Differentiation occurs in the presence of certain cytokines when B lymphocytes are stimulated by "seeing" the antigen that corresponds to the binding site of the immunoglobulin on the B cell surface.

105.

D. All the conditions listed affect the agglutination of A and B cells in serum grouping. The gamma-globulin fraction of the serum contains the immunoglobulins. When it is reduced, there will be fewer molecules of blood group antibodies, leading to weakened or negative reactions. Both cold autoagglutinins and cold reactive IgM alloantibodies, which will react at room temperature (such as anti-M), may agglutinate the cells used because of the presence of the corresponding antigen on the group A and/or B red blood cells. Cold auto- and alloantibodies are the most common causes of ABO discrepancies.

106.

C. Blood for intrauterine transfusion should be group O, D-negative (because the fetus's blood group is unknown) and negative for the antigen corresponding to any other IgG antibody in the maternal serum. It should be recently drawn and administered as RBC (Hct 75–85%) to minimize the chance of volume overload. It should be irradiated, CMV safe, and known to lack hemoglobin S.

107.

A. Anti-Jk^a is an IgG antibody and is nearly always detected in the antiglobulin phase. Rarely, it can be detected at the 37°C phase of testing. Anti-M, anti-P_1, and anti-I are generally IgM antibodies and react at room temperature and below by direct agglutination.

108.

D. Lewis system antibodies are generally IgM. Antibodies in the Rh, Duffy, and Kell systems are generally IgG. There may be rare IgM exceptions.

109.

B. Bg antibodies react with the red blood cell equivalents of HLA antigens. Bg^a corresponds with HLA-B7, Bg^b with HLA-B17, and Bg^c with HLA-A28. These antibodies can be frustrating in that few panel cells will react and the Bg type of panel cells is often not listed.

110.

C. There is no ABO exclusion. Although the alleged father and mother are group A, they could both be heterozygous (*AO*) with the baby inheriting the *O* gene from each parent. The child appears to be of the Rh genotype *dce/dce*. One of these haplotypes is inherited from the mother. It is feasible for the alleged father to be *DCe/dce*. He could then contribute the second haplotype. The baby can inherit the *A2B12* haplotype from the mother. Although *A3* can come from the alleged father, *B15* cannot and therefore there is a direct HLA exclusion.

111–118.

(111:C, 112:A, 113:D, 114:B, 115:C, 116:D, 117:A, 118:A)

Discrepancies in ABO blood grouping may occur for numerous reasons. Any discrepancy between cell and serum grouping must be resolved before blood is identified as belonging to a particular ABO group. The presence of an acquired B antigen on cells that are normally group A can be found in some disorders, where gram-negative bacteria have entered the circulation. The serum will contain an anti-B, which will not agglutinate the patient's own cells that have the acquired B antigen. The red cell reaction with anti-B reagent may be weaker than usual. Protein abnormalities of the serum such as are present in multiple myeloma may cause the presence of what appear to be additional antibodies. The rouleaux of the red blood cells caused by the excess globulin may appear to be agglutination. Saline replacement of the serum and resuspension of the cells will usually resolve the problem in the serum grouping. Washed red blood cells should be used for the cell grouping. Infants do not begin making antibodies until they are 3 to 6 months of age. Newborns therefore will not demonstrate the expected antibody(ies) on reverse grouping. The antibody that is present is probably IgG from the mother that has crossed the placenta. An A_2 individual has the ability to make an antibody that agglutinates A_1 red cells. This anti-A_1 will cause a serum grouping discrepancy, but the antibody is almost always naturally occurring and clinically insignificant. A patient's serum may have antibodies to the yellow dye used to color anti-B reagents. If serum or plasma suspended red cells are used in the cell grouping, a false positive reaction may occur. Using washed cells will eliminate the problem. Patients who are immunodeficient may have such depressed immunoglobulins that their serum does not react with the expected A and B reagent red cells. An unexpected IgM antibody in the serum will react at room temperature and may interfere with ABO typing. Reverse grouping cells carry all of the normal RBC antigens. Therefore, they can react at room temperature with anti-M, anti-N, anti-P_1, etc. O cells may also react if they carry the antigen corresponding to the antibody in the patient's serum. Thus a patient with anti-M in his serum could react with both reverse grouping cells and the O cells if all were positive for the M antigen. A patient with cold hemagglutinin disease (CHD) may have a discrepancy affecting both cell and serum groupings. The red blood cells should be washed with warm saline before typing; the serum and reagent A and B cells should be prewarmed before mixing and testing and converted to the antiglobulin test if necessary.

119–123.

(119:C, 120:A, 121:A, 122:D, 123:B) The Kell system has a number of antigens, among which is Kp^a (Penney). This antigen has not been reported in blacks. The corresponding antibody is very rare because so few individuals have the antigen that stimulates its production. When it is present, it is not a serious problem because Kp(a–) blood is easily found. The McLeod phenotype is one in which all the Kell-associated antigens are expressed only weakly. McLeod cells are missing a precursor substance called Kx. Kx is coded for by a gene present on the X chromosome. Some of the male children afflicted with chronic granulomatous disease are of the McLeod phenotype, but exactly how the two are associated is not clear. The *Ss* locus is closely linked with the *MN* locus, and they are considered part of the same blood group system. M^g (Gilfeather) is a rare allele in the MN system. When the M^g antigen is present it can cause typing difficulties because it will not react with either anti-M or anti-N. Because the MN antigens are well developed at birth, they were often used in paternity testing. The presence of an M^gM or M^gN combination can look like a homozygous *M* or *N*, leading to a second-order (indirect) exclusion unless the red cells are tested with anti-M^g. The presence of the M^g antigen on the red blood cells of the alleged father and child practically proves paternity. Currently, most paternity testing is done by DNA analysis, not by red cell antigen testing.

Antibody Identification, Transfusion Therapy, Transfusion Reactions

124.

A. The racial origin of this donor is probably black. This origin can be determined by looking at the Duffy (Fy) phenotype. About 70% of American blacks are Fy(a–b–). This phenotype is extremely rare in whites.

125.

D. Donor 5 is homozygous for the following genes: *Ce, s, k,* Jk^a, Fy^a, because the corresponding antigen is produced and the antithetical antigen is not being produced by an allele (for example, C+c– implies homozygosity: *CC*). The donor cannot be homozygous for *M*, because its allele is producing N antigen. There is no way to tell whether P_1 is homozygous, because it lacks a co-dominant allele and P_1 does not show dosage. There is no Le^b gene. The antigen is produced by the action of the *Le* gene on Type 1 H. The Lewis genes are *Le* and the amorph *le*, and dosage is not observed.

126.

A. Anti-Fy^a can be identified by eliminating specificities where the corresponding antigens appear on the panel cells that do not react. The differences in the strength of reactivity can be explained by the fact that the Duffy antigens show dosage (react stronger with cells from homozygotes). Cells #1 and #6 are from Fy^a heterozygotes [Fy(a+b+)]. Cells #4 and #5 are from Fy^a homozygotes [Fy(a+b−)]. When eliminating an antibody specificity known to show dosage, it is best to have a negative reaction with a panel cell from a donor who is homozygous for the corresponding gene. Fy^a and Fy^b antigens are destroyed by enzymes. Although the Fy(a–b–) type is common in blacks, the frequency of Fy^a in whites is about 66%. Anti-E and anti-s should be ruled out with Fy(a–) cells from individuals who are homozygous for *E* and *s* (in other words, E+e– and S–s+).

127.

C. Serum must be present to cause rouleaux formation; it should not occur at the antiglobulin phase of testing when the rouleaux-producing properties have been removed by washing. Warm and cold autoantibodies result in a positive autocontrol, usually equal in strength to reactivity observed with reagent red cells. Antibodies directed against preservatives in potentiating media should also react in the autocontrol. When the autocontrol is nonreactive and all panel cells are uniformly positive, one should suspect the presence of an alloantibody directed against a "public," or high-frequency, antigen. A selected panel of red cells, each lacking a different high-frequency antigen, should be tested until a compatible cell is found. The patient's red cells may be typed for a variety of high-frequency antigens. If such an antigen is found to be missing on the red cells, the corresponding serum antibody is likely that specificity.

128.

B. Cell #7 is negative for the high-frequency antigen k (cellano). Many other specificities cannot be ruled out because there is only one negative reaction. Treating the panel cells with dithiothreitol (DTT) destroys Kell system antigens. If no reactions are seen when the panel is repeated with DTT-treated cells, then many other clinically significant antibodies can be ruled out and the presence of anti-k would be supported. If the patient has not recently been transfused, his cells should be typed with anti-k and would be expected to be k-negative. Proteolytic enzymes neither destroy Kell system antigens nor enhance their reactions with Kell system antibodies. Treating serum with DTT will destroy IgM antibodies by cleaving disulfide bonds of the pentamer and would not be helpful because anti-k is generally IgG.

129.

C. From the presence of positive reactions taking place at two different temperatures, it appears that there are two different antibodies reacting. There is a cold antibody reacting with Cells #3 and #8 at immediate spin and a warm antibody reacting with Cells #1, #2, #3, and #4. It is unlikely that the cold antibody is carrying over to a warmer phase, because there is no 37°C reaction with Cell #8.

130.

C. Anti-Le^a, -Le^b, and -P_1 are antibodies that react at immediate spin (room temperature or below). Of these, P_1 and Le^b antigens are present on Cell #7, which shows negative reactivity. This makes these specificities unlikely to be present in the patient's serum. Le^a antigen is present on Cells #3 and #8, both of which show a positive immediate-spin reaction. Anti-D is usually IgG and reacts best at 37°C and AHG phases of testing.

131.

B. All the antibodies listed react at warm temperatures. The K antigen is present only on Cells #1 and #7 and is absent from Cells #2, #3, and #4 that reacted at 37°C and AHG phases of testing. Also, Anti-K and anti-k do not usually react without the addition of AHG. Anti-C and -D may react at 37°C without AHG, but usually only if albumin or enzymes are used as potentiators. Anti-C and -D are often found together. In this instance, however, there would be a positive reaction with Cell #5 if anti-C were present as well as anti-D.

132.

D. A patient's red blood cells should be negative for the antigen corresponding to the antibody identified as long as the autocontrol is also negative. In this case, one already knows that the patient is group A, D-negative (does not have D antigen). A standard approach has been to require three antigen-positive cells that react and three antigen-negative cells that do not react for each antibody identified to establish probability that the antibody(ies) has (have) been correctly identified. There are only two Le(a+) donor cells on this panel. The anti-Le^a reacts only at immediate spin and the anti-D does not. Presumably the screening cells have an additional Le(a+) cell. Because this antibody appears to be clinically insignificant, many would simply ignore it by eliminating the IS. At any rate, it would certainly not be necessary to run another panel.

133.

B. The patient's positive antibody screening test is consistent with an anti-K, and this is what was identified in the antibody identification. Three K antigen-positive and three K antigen-negative cells were tested and reacted appropriately. The antibody identification could have been misinterpreted, but it seems unlikely. The panel must have been read at the AHG phase of testing, because most examples of anti-K do not react at any other phase of testing. Positive and negative control cells (K+k+ and K–k+) should be tested with the anti-K at the same time as the patient's cells to be certain of the specificity of the anti-K antiserum. There is no indication that this has been done, and the patient's phenotype should not be K+. If the patient had circulating K+ donor cells, the K typing would have shown a mixed-field reaction, which has not been indicated.

134.

B. Although there are many potential sources for error in performing an indirect antiglobulin test, the most common error leading to a false negative reaction is the failure to wash the red blood cells adequately before the addition of AHG reagent. Traces of free human globulin can neutralize the AHG reagent. Red cells known to be coated with IgG antibody (Coombs' control cells, check cells) are added to all negative tests. Agglutination of these control cells confirms that AHG was present in the system and that proper washing procedures were performed.

135.

C. Elution is a process in which bound antibody is released from red blood cells. The eluate produced can then be further tested to identify the specificity of the antibody. Some elution methods use temperature, chemicals, or manipulation of the pH to dissociate antibodies from red cells.

136.

C. If the antiglobulin test was performed properly and the antiglobulin reagent is working properly, the IgG-coated control red blood cells should be agglutinated; thus, this test is invalid. Unagglutinated cells after the addition of the control cells might mean that the cells were not washed well and that the antiglobulin reagent has been neutralized or that the antiglobulin reagent may have been omitted. The test must be repeated if this happens.

137.

B. The crossmatch is performed by testing the serum of the recipient with a suspension of the donor's red blood cells. The serum and red cells are usually tested at the immediate-spin (IS) phase to detect ABO mismatches, if the patient has no history of having unexpected antibodies in his/her serum and the current antibody screen is negative. Additional testing is done at 37°C and antiglobulin phases to detect the presence of clinically significant antibodies, if the patient has a positive antibody screen or a history of ever having an unexpected antibody in his/her serum. Because clinically significant antibodies (other than anti-A and anti-B) are almost always detected during the antibody screening test, AABB *Standards* sanctions performing only the immediate-spin crossmatch (for ABO compatibility) when the patient has a negative antibody screening test. An antiglobulin crossmatch *must be performed* when a patient has a positive antibody screening test because of a clinically significant antibody, or if the patient has a history of a clinically significant antibody. Compatible units must also be phenotyped for the corresponding antigen and shown to be negative. When an antiglobulin crossmatch is performed, potentiating media such as albumin, polyethylene glycol (PEG), or LISS may be added to the test system to enhance sensitivity and/or decrease incubation time.

138.

B. A group AB individual can receive red blood cells from donors of all ABO groups. Because the patient does not have anti-D, it would be best to next select group A, D-positive units because the need for large amounts of blood is anticipated. These units should be given as RBCs, because the plasma has anti-B. If necessary, the patient may be later switched to group O, D-positive RBCs. It would not be wise to deplete the D-negative supply, because D-negative women of child-bearing age may need blood and should not be exposed to the D antigen. The decision to transfuse D-positive blood to a D-negative patient must be approved by the physician in charge of the transfusion service.

139.

D. The crossmatch, which is the recipient's serum with the donor's cells, will reveal only if the patient has a detectable antibody against some antigen on the donor cells. In the presence of a negative antibody screening test, an incompatible crossmatch at the immediate-spin phase will most likely be due to an ABO mismatch between the recipient's serum and the donor's cells. For this reason, AABB *Standards* mandate performing only the immediate-spin crossmatch when the patient has a negative antibody screening test and no history of clinically significant antibodies. The crossmatch will not guarantee *in vivo* response to the transfused red blood cells. Also, it will not detect all ABO typing errors, and it will not detect most Rh typing errors.

140.

C. A false positive crossmatch could occur if the donor has a positive direct antiglobulin test (DAT). A DAT should be done on the donor cells and, if positive, the unit should be removed from inventory. Another possible cause of a false positive crossmatch could be contaminants in dirty glassware causing clumping of red cells. The other responses are true positives. If a strong incompatibility is immediately present, one should check the ABO type of the patient and the donor. If the antibody screening test was negative, one might suspect an antibody against a low-incidence antigen on the donor's cells.

141.

C. The crossmatch consists of testing donor cells with recipient serum. A group A individual will have anti-B in his/her serum, which will agglutinate AB cells. D-positive cells given to a D-negative person may cause antibody stimulation, but there will not be a visible reaction without a preformed antibody.

142.

A. The most critical step to ensuring safe transfusion is obtaining a properly labeled blood sample from the correct patient. Transfusion accidents due to ABO mismatches are usually the result of a patient receiving the wrong blood. The identity of the patient must be verified, both verbally and by comparison of the wristband with the transfusion request form. Tubes must be labeled properly at the bedside with the full name, another acceptable identifier such as the medical record number, and the date.

143.

B. Sufficient information for unique identification of the patient (including two independent identifiers) and the date of sample collection must be on the label. The phlebotomist's signature or initials must appear on either the tube of blood or on the request slips. It is not necessary for both to be signed. The physician's name, the patient's room number, and the time of the phlebotomy may be helpful but are not required by AABB *Standards*.

144.

A. According to AABB *Standards,* specimens used for antibody screening and crossmatching must be less than 3 days old if the patient has been transfused or pregnant within the past 3 months. Either serum or plasma may be used. The specimen must be labeled properly at the bedside at the time of collection. Specimens are required to be retained for only 7 days posttransfusion.

145.

B. In general, one unit of red blood cells should raise a patient's hemoglobin by 1 g/dL. In this instance, a 2-g/dL rise is required, so two units would need to be given. This rule is true for patients of average size. A very large or heavy individual with an expanded blood volume may require additional units to attain the same level. Conversely, a pediatric patient may require less.

146.

B. The percentage of compatible blood is obtained by multiplying the frequencies of antigen negative. In this instance, one wants to find Jk(a–), K– blood. The incidence of Jk(a+) blood is 77%; therefore, the incidence of Jk(a–) blood is 23%. Likewise, K+ incidence is 10%; K– would be 90%. Multiply these two frequencies together to get the frequency for Jk(a–), K– units: $0.23 \times 0.90 = 0.21$, or an incidence of 21 units in 100. Divide 2 by this figure because 2 units are needed: $2 \div 0.21 = 9.5$, or 10 units must be screened to find 2 compatible units.

147.

C. Cryoprecipitate provides a source of fibrinogen and fibronectin in addition to factors VIII and XIII. This component is indicated for use in bleeding disorders associated with hypofibrinogenemia, such as DIC, when excessive fibrinogen consumption is occurring. Each unit contains an average of 250 mg of fibrinogen or 0.25 g. The AABB *Standards* require a minimum of 150 mg per individual collection. The amount of pooled product to administer is calculated by the formula:

$$\frac{\text{Total grams desired}}{0.25\ \text{g/unit}} = \text{Total number of cryoprecipitate units to administer}$$

For example,

$$\frac{2\ \text{g}}{0.25\ \text{g/unit}} = 8\ \text{Units}$$

148.

D. Plasma compatible with the recipient's ABO group is preferred when large volumes are transfused. Both group O and group B plasma contain anti-A that can cause a positive direct antiglobulin test (DAT) when infused into either group A or AB recipients. Compatibility testing is not required before cryoprecipitate administration. Plasma compatibility is not as important with cryoprecipitate as with platelet concentrates. Approximately 10 mL of plasma is in a cryo unit and 50 mL in a single platelet concentrate.

149.

D. Approximately 1% of transfused red cells are cleared daily from the circulation of a recipient. The clearance rate may be increased in patients with autoimmune hemolytic anemia, pernicious anemia, aplastic anemia, hemorrhage, splenomegaly, and fever. Transfused cells survive normally in patients with anemia because of intrinsic red cell enzyme defects, spherocytosis, and paroxysmal nocturnal hemoglobinuria.

150.

B. Febrile reactions are brought about by the interaction of antibodies in the recipient directed against antigens on donor leukocytes or by cytokines secreted by leukocytes. The antigens involved are both the HLA and granulocyte-specific antigens. Leukocyte-reduced RBCs are the component of choice for a patient with repeated febrile transfusion reactions. Although frozen RBCs that have been thawed and deglycerolized are considered leukocyte reduced, the cost and time involved in preparation make them an unpractical choice.

151.

D. Red blood cells are the component of choice to maintain or restore oxygen-carrying capacity. This component has the least effect on blood volume and the maximum effect on the oxygen-carrying capacity of all the products available for transfusion. In some patients, increasing the total blood volume more than what is absolutely necessary could have a detrimental effect. Examples are patients with chronic anemia or congestive heart failure.

152.

C. Children inherit half their genetic characteristics from each parent. Because the parents are not identical in antigen composition (a situation only found in identical twins), the child cannot be totally compatible with either parent. Siblings, however, have access to the same genetic material from each parent and so may have identical genes and antigens. A spouse genetically would be equivalent to a random donor.

153.

D. Because time is of the essence when a trauma victim is severely hemorrhaging, blood bank personnel must respond promptly. Group-specific blood may not be issued on the basis of previous patient or donor records. If the situation is so urgent as to preclude performing an ABO and Rh typing, or when a blood sample cannot be obtained, group O RBCs may be issued. The decision as to whether group O, D-positive or O, D-negative RBCs should be used will depend upon inventory and the age and sex of the trauma victim. Blood banks located in a trauma center should have a written procedures manual with well-defined criteria. All staff must be familiar with these guidelines.

154.

B. The first step in any pretransfusion work-up is to check the blood bank records for previous information on the patient. This information must match the ABO group and Rh type on the sample obtained currently. This helps to ensure that patients are not given the wrong blood type. A mismatch may indicate that the wrong patient was drawn or the label was incorrectly applied. After the initial release of 6 units of uncrossmatched group O blood, there should be ample time to at least check records and obtain an ABO group and Rh type on the patient sample before more blood is required.

155.

D. Large volumes of transfused plasma should be ABO compatible with the recipient's red blood cells. Isoagglutinins present in the plasma will attach to the corresponding antigen on the patient's red cells *in vivo* and cause a positive DAT and perhaps hemolysis. Plasma of any blood group can be given to a group O patient, because his/her red cells will not be agglutinated by anti-A or anti-B in donor plasma.

156.

B. Physiologic saline is the only generally acceptable solution that is allowed to be added to blood or blood components. Ringer's solution causes small clots to develop in anticoagulated blood, and 5% dextrose causes red cell lysis. Other solutions and medication should not be added to blood unless they have been proved safe and are sanctioned by the FDA.

157.

C. The majority of deaths due to hemolytic transfusion reactions are caused by clerical errors, not laboratory errors. Patients, blood samples, and lab records, if misidentified, may lead to the wrong ABO type blood being administered to the patient. These deaths most often occur in areas of high stress, such as in emergency departments and surgical suites.

158.

C. A delayed hemolytic transfusion reaction is generally the result of a patient's second exposure to an antigen present on donor red blood cells. The patient at some time previously had been exposed to the antigen, and this is his/her anamnestic response. This reaction usually occurs from 3 to 14 days after transfusion and is accompanied by extravascular red blood cell destruction. Often the patient is asymptomatic. The DAT is usually positive in a delayed hemolytic reaction, because the reaction is extravascular and coated cells are present in the peripheral circulation. An acute hemolytic transfusion reaction is usually intravascular, and the coated cells are destroyed by complement, leaving the DAT negative or at most weakly positive. Anaphylactic and febrile reactions do not involve red blood cell antibodies and do not cause a positive DAT.

159.

C. Kidd antibodies are generally IgG, complement dependent, and warm reacting. However, they are usually weak and labile. Because of this, they may go undetected in pretransfusion testing and the patient may inadvertently be transfused with antigen-positive blood, leading to a delayed transfusion reaction.

160.

A. Some people are genetically deficient in IgA. If these individuals have anti-IgA in their plasma, they may suffer a severe anaphylactic reaction when subsequently exposed to IgA in donor plasma. Once these people are identified, they must receive IgA-deficient components such as multiple-washed or frozen-thawed RBC or components drawn from IgA-deficient donors.

161.

A. Viral inactivation methods such as the use of a solvent/detergent combination have eliminated the risk of transmission of viruses with a lipid envelope in clotting factor concentrates. This method has been applied to group-specific frozen plasma. Pooled plasma, solvent/detergent-treated is much safer than the other components listed from the standpoint of HIV, HBV, and HCV, because the process destroys lipid-enveloped viruses. It does not destroy non-lipid-enveloped viruses such as parvovirus B19. However, it has been withdrawn from the market in the United States and is not currently being used. Another approach to safety is "FFP-Donor Retested," which means that the FFP (fresh-frozen plasma) has been held for 90 days or more and released only after the donor has been retested negative for infectious disease markers. It is not a pooled product. The retesting should show that the donor was not in an infectious window period when the plasma was drawn.

162.

C. The transfusion of the unit should be stopped, and the transfusionist should keep the patient's intravenous (IV) line open with physiologic saline in case medications must be given quickly to counteract the transfusion reaction. The unit must then be returned to the blood bank along with all of the transfusion set and any attached IVs, such as any physiologic saline that was being infused along with the unit. The patient's physician should be notified, but after the transfusion is discontinued and the new IV of physiologic saline has been hung. Monitoring pulse and blood pressure is a good idea but is not an immediate necessity.

163.

D. Returning the unit to the blood bank and obtaining posttransfusion blood and urine samples is required by AABB *Standards*. No more blood or blood products should be transfused into the patient until the reaction has been investigated. If the patient has an antibody that caused the transfusion reaction, any red blood cell units transfused must be negative for the corresponding antigen.

164.

B. Red-tinged plasma is indicative of hemolysis. When this is seen in the posttransfusion sample but not in the pretransfusion sample, it is evidence that an intravascular hemolytic transfusion reaction has occurred. The antibody attached to antigen on the patient's RBCs, and the resulting antigen-antibody complex activated the complement cascade. This resulted in hemolysis of the coated RBCs in the patient's vasculature. Other likely laboratory findings in this situation include hemoglobinuria and decreased plasma haptoglobin. Although intravascular hemolysis can be seen in situations where a drug has been administered with the blood, there are several possible causes of an intravascular hemolytic transfusion reaction. In the scenario given, there is no evidence that would suggest any particular cause.

165.

B. Autologous transfusions are the safest form of transfusion available, although they are not always the most practical. When administered properly, they eliminate disease transmission, immunization to foreign antigens, allergic reactions, and graft-versus-host reactions. Clerical error is still a significant risk. An AABB survey revealed that 1.2% of respondents reported an erroneous autologous transfusion. Although preoperative autologous collection *is feasible for* elective surgery, this form of autologous transfusion is not possible in cases of unexpected or massive blood loss. Intraoperative blood collection is another form of autologous transfusion used during operations where the estimated blood loss is great. Another advantage of autologous transfusions is that allogeneic donor blood is available for other patients.

166.

D. Reviewing the previous records of a patient may help to confirm the identity of the current pretransfusion sample. Records should be checked for ABO group and Rh type, clinically significant antibodies that were present but may no longer be detectable, and adverse reactions to previous units transfused. ABO records from the past 12 months must be immediately available and retained for 5 years; antibody and adverse reaction records must be available indefinitely for review before issuing blood for transfusion. It is not necessary to check hepatitis records of the patient because the hepatitis status of the patient does not affect transfusion of blood to him as a recipient.

167.

C. The crossmatch is performed using the recipient's serum and the donor's red blood cells. Therefore, a positive DAT on the recipient's cells will not affect the crossmatch results. A positive reaction may be obtained when the recipient has an antibody directed against a corresponding antigen on the donor's red cells. If this is a low-frequency antigen, the crossmatch may be incompatible and the antibody screening result negative. A positive reaction may also indicate that the donor's red cells are coated with human globulin. This can be confirmed by performing a direct antiglobulin test (DAT) on the donor's red cells. Units of blood demonstrating a positive DAT should be returned to the collecting facility.

168.

D. Monoclonal anti-D reagents are low-protein reagents, therefore, a negative reaction with anti-A and/or anti-B (also low-protein) serves as a control. When the patient appears to be group AB, D-positive, it is necessary to set up a separate control. A drop of the patient's cell suspension with his/her own serum (autocontrol) or with 6–8% albumin makes a suitable control.

169.

D. Both the ABO grouping and Rh typing are in question. Because the transfusion need is urgent, group O, D-negative donor units should be selected initially for this young woman of childbearing age. They should be transfused, if necessary, before the problem has been resolved or crossmatching performed. In some cases, the risk of withholding transfusion is far greater than the risk of a transfusion reaction in a patient with an unresolved antibody problem. The physician must sign an emergency release form indicating that the clinical situation was such to warrant the release of blood.

170.

B. The patient most likely has a potent cold autoagglutinin. The antibody screening test and crossmatches with group O, D-negative donor units should be set up as soon as possible by prewarmed technique. In the past, when Rh typing was primarily done with high-protein reagents, an Rh control, containing all the potentiating ingredients found in the Rh reagent except for the anti-D, was tested in parallel. The most likely cause of a positive Rh control with a high-protein reagent is a strongly positive DAT result. This would not be the cause in this case because monoclonal anti-D is a low-protein reagent. The usual cause of false positive reactions with low-protein reagents is a potent cold autoagglutinin. A single wash may not remove all the antibody from the patient's red cells. The cells should be washed with warm saline; and if they are still autoagglutinated, antibody can be removed by 45°C heat elution or treatment with a sulfhydryl reagent such as dithiothreitol (DTT), which destroys IgM antibodies. Because washed red cells were used when typing the patient's red cells, multiple myeloma could not be the cause of the false positive, because the abnormal protein causing the pseudo-agglutination (rouleaux) would have been washed away.

171.

C. Although rigors and shock may be caused by hemolytic or anaphylactic reactions, bacterial sepsis is the most likely cause in this case. The sudden rise of the patient's temperature from normal to 40°C or above is typical of such an infection. Bacterial sepsis is an important cause of transfusion reactions, with about one-fourth of these reactions resulting in death.

172.

D. The incidence of bacterial sepsis is highest with platelet components. It is higher with pooled platelets than platelets collected by apheresis. Pooled platelets usually involve 6 or more donations from different donors, multiplying the chance of contamination. Most bacteria grow better at room temperature (the normal storage temperature for platelets) than refrigerator temperature. Sepsis from RBCs is usually due to *Yersinia enterocolitica,* which grows well at refrigerator temperature.

173.

B. The reactions are most likely all caused by the cold autoagglutinin anti-I. The I antigen is not present on *ii* cells. Autoadsorption of the patient's serum with his/her own cells should not be performed following recent transfusion. Alloantibody may be adsorbed onto circulating donor red cells, resulting in false negative reactions with repeat testing of the autoadsorbed serum and reagent red cells. The weak reactions at the AHG phase of testing are most likely due to complement being bound at room temperature by the cold autoantibody reacting with the anti-C3d in polyspecific AHG reagent. A prewarmed technique, in which the donor's cells and patient's serum are warmed separately to 37°C before combining, is commonly used to eliminate interference from cold agglutinins. Many transfusion services use an anti-IgG monospecific AHG reagent, instead of a polyspecific reagent that contains anti-IgG and anti-C3d, in order to avoid such problems, but the prewarmed crossmatch should eliminate complement from being bound. Because the patient was recently transfused, there is a slight possibility that the reactions at AHG could be caused by a high-incidence alloantibody causing delayed hemolysis. Such an antibody would still react by prewarmed technique.

174–179.

(174:A, 175:A, 176:E, 177:D, 178:D, 179:C) The ABO group and Rh type must be determined by the blood-collecting facility with every donation. The unit must be labeled using the interpretation of current testing, not with previous donor records from repeat donors. When the immediate-spin (IS) reaction of the donor red cells is positive with anti-D (with a negative Rh control), the unit may be labeled D-positive. If the red cells fail to agglutinate anti-D directly, the test must be incubated and converted to the antiglobulin test to detect weak D phenotype. All units tested with anti-D that are IS negative but are found to be weak D-positive must be labeled D-positive to avoid sensitizing an intended D-negative recipient to the D antigen. A direct antiglobulin test (DAT) should be performed as a control along with the weak D test. For the test to be valid, the DAT must be negative. If the donor is DAT-positive, the weak D status cannot be interpreted because the donor's red cells are coated with antibody before the incubation with anti-D. DAT-positive units of blood should be discarded. Two different test methods, a cell grouping and a serum grouping, must be used for ABO grouping; the results of these methods must be in agreement before a label is applied to the unit. Although testing the red cells with anti-A,B and testing the serum with A_2 red cells is not required, many collecting facilities incorporate these additional reagents to detect discrepancies due to subgroups of A or B. When the cell and serum groupings are not in agreement, additional testing to resolve the discrepancy is required. Weak or missing red-cell reactions with anti-A,B or anti-A, or both, accompanied by serum reactions with A_1 cells, but not A_2 cells, are an indication that the donor may be a subgroup of A with anti-A_1. Extended incubation of the cell grouping, testing with additional A_1 cells, A_2 cells, and anti-A_1 lectin, and adsorption/elution/titration/secretor studies are techniques used to resolve discrepancies due to subgroups. Donor units found to contain unexpected antibodies should be processed into RBCs with small amounts of plasma. They should be labeled to indicate the antibody specificity. It is helpful to attach a tie tag with this information to the RBCs. Transfusing large amounts of antibody containing plasma (such as anti-Fy^a) into a Fy(a+) recipient may cause decreased red cell survival and, therefore, is not used for individual transfusion to patients. Plasma from units with antibodies may be salvaged for reagent use or source plasma.

180–184.

(180:C, 181:C, 182:D, 183:D, 184:A) There are three parts to the factor VIII molecule: F VIII:C, F VIII:Ag, and F VIII:vW. Individuals manifesting the X-linked (gene carried on the X chromosome) disorder known as hemophilia A are deficient in F VIII:C. The clinical severity, resulting in hemorrhage either spontaneously or following trauma, depends upon the level of F VIII:C present. Deficiency in F VIII:vW is known as von Willebrand disease. It is not X-linked and is the most common inherited coagulopathy. Deficiency in F VIII:vW results in impaired platelet adhesion and aggregation, leading to prolonged bleeding. Cryoprecipitate contains both F VIII:C and F VIII:vW and may be used for treatment of these disorders, although it is not the preferred treatment. F VIII concentrates have become safer with improved viral inactivation processes, and some now have therapeutic amounts of F VIII:vW as well. Cryoprecipitate also contains an average of 250 mg of fibrinogen per unit, as well as Factor XIII and fibronectin, and currently it is primarily used to treat hypofibrinogenemia. Although Factor V deficiency is rare, it can present severe manifestations leading to hemarthrosis. Treatment of choice is fresh-frozen plasma (FFP) because F V is a labile factor not found in cryoprecipitate. FFP can be used to correct the factor deficiencies found in liver disease (factors II, VII, IX, and X). Because all these are stable factors, the plasma need not be fresh even though FFP is commonly used. Platelet concentrates are used to correct thrombocytopenia following chemotherapy. Fresh whole blood is seldom available. Specific components are instead provided to give the patient exactly what is needed and conserve blood resources.

Hemolytic Disease (Hemolytic Disease of the Newborn, Immune Hemolytic Anemia)

185.

C. ABO testing, Rh testing (for weak D when applicable), and antibody screening should all be performed early in a pregnancy. Amniocentesis should be done only when clinically indicated. Furthermore, amniocentesis generally is not done before the third trimester, although in recent years the procedure has been done as early as 14 weeks.

186.

C. The Lewis typings of a pregnant woman may appear to be Le(a–b–), even though the original typing may have been Le(a–b+). When women are pregnant, they have an increased plasma volume and increased amount of lipoprotein in relation to red blood cell mass. Because Lewis antigens are adsorbed onto red cells and lipoprotein from plasma, the dilutional effect and greater lipoprotein mass would lead to less adsorption of Le^b onto red cells. After the pregnancy, the woman will return to her original type.

187.

A. Prenatal testing for all pregnant women should include ABO, Rh, and antibody screening to exclude the presence of unexpected antibodies with the potential for causing hemolytic disease of the newborn (HDN). The presence of an unexpected antibody does not indicate that the infant will be affected. Testing the red blood cells of the father, to determine whether the corresponding antigen is expressed and, if so, whether he is a homozygote or heterozygote, should indicate the probability for the presence of the antigen on infant cells. ABO-HDN is not predictable until postpartum, when the blood type of the infant is determined.

188.

D. IgG is the only immunoglobulin that is transported across the placenta. It does not cross the placenta because of low molecular weight or simple diffusion, as evidenced by higher concentrations of antibody present in cord than in maternal serum. IgG molecules are actively transported via the Fc portion beginning in the second trimester. Therefore, potentially any IgG blood group antibody produced by the mother could cause HDN, if the fetus possesses a well-developed corresponding antigen. The disease varies widely in severity, being dependent on multiple factors.

189.

A. Although the ABO system is most often implicated in fetomaternal incompatibilities, it very rarely causes clinical symptoms. ABO-HDN generally occurs when group O mothers have group A or B children. Although Rh-HDN can be prevented, there is no prevention for ABO-HDN, and generally there is none needed because exchange transfusion is rarely necessary.

190.

C. When D+ red blood cells are sufficiently coated with antibody, leaving no or few remaining sites to react with D antiserum, the cells are referred to as having a "blocked D" and may react weakly or not at all with a low-protein anti-D reagent. One may suspect this phenomenon, confirmed by elution of anti-D from the red cells, when the DAT is strongly positive. Enough antibody may be removed either with a gentle heat elution (45°C) or using chloroquine diphosphate to permit accurate D typing of the coated red cells.

191.

C. Given that all prenatal and neonatal testing is valid, one should consider an antibody against a low-incidence antigen. The low-incidence antigen was of paternal origin, and it stimulated the mother to form an IgG antibody. To prove this theory, an eluate from the baby's cells should be tested with the father's cells. Also, the mother's serum and the baby's eluate could be tested with a panel of cells positive for various low-incidence antigens to identify the specificity of the antibody.

192.

D. An antibody panel performed on an eluate made from the baby's red blood cells is the most conclusive way to identify positively the antibody causing the positive DAT. This would be especially helpful in a case where the mother has several antibodies that could cause hemolytic disease of the newborn. However, RBCs for transfusion in the neonatal period should be negative for any antigen corresponding to any IgG antibody that crossed the placenta.

193.

B. During the first few hours of life, the primary risk to a baby with hemolytic disease of the newborn is heart failure caused by severe anemia. After the first 24 hours, in which the anemia can be compensated, the highest risk to the infant comes from hyperbilirubinemia. Kernicterus, which is brought on by hyperbilirubinemia (generally >18 mg/dL of unconjugated bilirubin) in a full-term infant in the first days of life, can cause irreversible brain damage. Depending on the severity of the hyperbilirubinemia, one or more exchange transfusions may be needed.

194.

D. A positive DAT on the infant's red blood cells indicates that IgG antibody has crossed the placenta and coated the neonate's red cells. Identification of the antibody in the maternal serum and elution of the same antibody from the infant's red cells confirms the specificity of the offending antibody. In lieu of a maternal blood sample, the identity of the antibody may be confirmed by testing the infant's eluate and serum. The eluate contains the antibody(ies) responsible for the clinical HDN; the serum may contain additional maternal antibody(ies) directed against antigens absent on the infant's cells but present on the donor's cells. Blood compatible with both serum and eluate should be prepared for exchange transfusion.

195.

D. Providing blood of the baby's type is exactly what one does not want to do. This would defeat the purpose of the exchange transfusion. For example, if a D+ infant was suffering from HDN because of an anti-D, the transfusion of D-positive cells would allow transfused cells to be coated with anti-D. The transfused cells would then be removed from the circulation by the RES and would have decreased survival.

196.

D. A mixed-field weak D test on maternal blood indicates the presence of D-positive baby cells circulating with the mother's D-negative cells, suggestive of a large fetomaternal hemorrhage. If a mother does demonstrate a positive weak D test when previously it was negative, a Kleihauer-Betke acid elution test should be done on the mother's red blood cells. This test is used to quantify the amount of the fetal blood that has entered the mother's circulation. The results of the test will determine how many vials of Rh immune globulin should be administered to the patient. One vial will protect against approximately 30 mL of fetal blood (or 15 mL of red cells) that have entered the mother's circulation. A more sensitive method to identify fetomaternal hemorrhage (FMH) than the test for weak D is the rosette test. A maternal red cell suspension is incubated with an anti-D of human source, allowing antibody to coat D+ fetal cells. D+ indicator cells are added that bind to the coated D+ cells, forming rosettes. This is a qualitative test and must also be followed by a quantitative test, such as the Kleihauer-Betke acid elution test.

197.

C. The most likely cause for the positive antibody screening test is the presence of a passively acquired anti-D. Because the mother received antepartum Rh immune globulin (RhIG), anti-D from that injection may still be present at childbirth. Depending on how the RhIG was injected postpartum, the anti-D could already be present in the patient's serum. Because antepartum RhIG was given, it is unlikely that active immunization has occurred. Passively acquired anti-D rarely has an antiglobulin titer above 4 and should be entirely IgG. When in doubt about whether anti-D is passive or represents active immunization, it is always better to administer RhIG at the appropriate time. The crossmatches are compatible because D-negative RBCs would have been chosen for transfusion.

198.

D. The Kleihauer-Betke acid elution stain is used to quantify the amount of fetal cells present in the maternal circulation postpartum to calculate the correct dose of RhIG to administer. Adult hemoglobin is soluble in acid buffer, whereas fetal hemoglobin is resistant to acid elution. A thin blood smear is subjected to acid elution, pH 3.2, and then is stained with erythrosin B and Harris hematoxylin. Normal adult cells appear as pale ghosts microscopically; fetal cells are bright pink. The number of fetal cells in 2000 maternal cells is calculated. The volume of fetal hemorrhage is calculated as follows:

$$\frac{\text{Number of fetal cells}}{\text{Number of maternal cells}} \times \text{material blood volume (estimate 5000 mL)} = \text{fetal bleed}$$

This is equivalent to: Fetal cells expressed as a percentage of maternal cells × 50 = mL of fetal whole blood. One vial of a standard 300-μg dose protects the D-negative mother against sensitization to the D antigen for a 30-mL bleed. Therefore, the fetal hemorrhage volume is divided by 30 to determine the number of vials.

199.

B. Neither anti-P_1 nor anti-Le^a is likely to cause HDN. They are almost exclusively IgM antibodies (cannot cross the placenta), and the corresponding antigens are not well developed on neonatal red blood cells. Both anti-K and anti-c are almost exclusively IgG antibodies and are capable of causing serious HDN. However, the K antigen has a much lower frequency (<10%) in the population than the c antigen (>80%), so the infant is much more likely to be c+.

200.

D. The Kleihauer-Betke acid elution stain is used to estimate the amount of fetal red blood cells present in the circulation of a D-negative mother postpartum. Failure to quantify the FMH may result in the administration of insufficient Rh immune globulin. Sensitization to the Rh antigen may occur, leading to HDN in subsequent pregnancies. The fetal bleed is calculated using the formula: KB% × 50 = milliliters fetal blood present or 1.3 × 50 = 65 mL.

201.

C. One standard dose of Rh immune globulin (300 μg) protects the mother from a 30-mL bleed. Because the precision of a Kleihauer-Betke stain is poor, a margin of safety is employed to prevent RhIG prophylaxis failure. The total bleed in milliliters is divided by the level of protection in one dose (30 mL). For decimals less than five, round down and add one dose (e.g., 2.3 rounds down to 2 + 1 dose = 3 vials total dose); for decimals five or greater, round up and add one dose (e.g., 2.6 rounds up to 3 + 1 dose = 4 vials total dose).

Example:

$$\frac{65 \text{ mL bleed}}{30 \text{ mL}} = 2.2; \text{ give 3 vials RhIG}$$

202.

D. A positive autocontrol reacting at the antiglobulin phase of testing indicates that the patient has a positive DAT. All screening cells react at the same strength at the same phase of testing as the autocontrol. This indicates the presence of a warm (IgG) autoantibody, both on the patient's cells and in her serum. This patient's serum would be expected to react 4+ at AHG with all cells tested. Polyagglutinable cells will not react in the autocontrol, because the patient will lack the antibody corresponding to the antigen causing the polyagglutination. Rouleaux will not be present in the antiglobulin phase because all of the serum proteins are washed away before the AHG is added. A transfusion reaction may demonstrate a positive autocontrol and/or antibody screen; however, the reaction will be mixed field, because the patient's cells will not react with her alloantibody but the donor cells will react with the patient's antibody.

203.

A. The direct antiglobulin test (DAT) is the easiest and the quickest way to detect *in vivo* sensitization of red blood cells. The indirect antiglobulin test is used to detect *in vitro* sensitization of red blood cells and is most commonly used in antibody detection tests and crossmatching. Uses of the direct antiglobulin test include investigation of hemolytic disease of the newborn, autoimmune hemolytic anemia, transfusion reactions, and drug-induced sensitization of red blood cells.

204.

B. The only method that can be used to identify an antibody coating red blood cells is to perform an elution on those cells and then test the eluate against a panel. Autoabsorption of the patient's serum and then a panel on the autoabsorbed serum would be the method of choice for the identification of alloantibodies in her serum because she has not been recently transfused. Any panel on her serum would simply react with all panel cells tested because of the presence of the autoantibody.

205.

C. Most patients having warm autoimmune hemolytic anemia (WAIHA) respond well to steroids, which decrease the autoantibody production. Transfusing these individuals usually will cause the autoantibody to be produced in greater amounts. Transfusion in these cases is often counterproductive because the donor cells are destroyed as quickly as the patient's own cells and the expected increment in hemoglobin from the transfusion does not occur. Because this woman needs more oxygen-carrying capacity, transfusion with FFP will not be of help to her. Plasma exchange may remove antibody from her circulation but will not provide any oxygen-carrying capacity.

206.

D. Most antibodies in WAIHA have specificity that appears to be directed toward the Rh blood group system antigens. Sometimes the antibody has a simple specificity; anti-e is the most common warm autoantibody of simple specificity. Usually the antibody has a broad specificity, reacting with blood cells of most Rh phenotypes except for rare phenotypes such as Rh_{null}. In addition, there are mimicking or relative specificities—for example, an apparent auto anti-e that can be adsorbed onto e-negative red cells.

207.

C. Although it is best to avoid transfusing patients with warm autoantibodies, life-threatening anemia may develop, which necessitates them to receive blood. The primary concern in patients demonstrating serum warm autoantibody is to detect and identify the presence of underlying alloantibodies that may be masked by the reactions of the autoantibody. One must be certain to differentiate between auto- and alloantibodies. If the autoantibody appears to have a simple specificity such as anti-e, alloantibody identification can be accomplished by testing reagent red cells that are e-negative and antigen positive for the alloantibodies to be excluded. Alternatively, when sufficient e-negative panel cells are not available to do a thorough rule out, warm autoadsorption of the serum with the untransfused patient's red cells will remove autoantibody but not alloantibody. Alloantibodies will cause the transfused RBCs carrying the corresponding antigen to be removed from the recipient's circulation even more rapidly than the patient's own RBCs and will not provide the oxygen-carrying capacity the patient requires.

208.

A. Autoantibodies are directed against antigens present on the patient's own red blood cells. The easiest way to determine whether the anti-e is auto or allo when the patient has not been recently transfused is to type the patient's red cells with a low-protein anti-e reagent. A high-protein reagent should not be used, since a false positive reaction is likely to occur. The preparation of an eluate when patients have no history of recent transfusion is not necessary; it is presumed that the eluate contains only autoantibody.

209.

C. When a patient has been recently transfused, a mixed-field positive DAT may indicate the presence of clinically significant alloantibody attached to circulating donor's cells. A positive DAT may also reflect autoantibody attached to patient's or donor's cells, either because of drug sensitivity or secondary to clinical disease, which is less likely in this case. It is important to obtain information, including diagnosis, clinical condition, medication history, and laboratory data such as hematocrit, bilirubin, haptoglobin, and reticulocyte count, both pre- and posttransfusion to determine whether a delayed hemolytic transfusion reaction (DHTR) has occurred. A DHTR commonly demonstrates a mixed-field DAT result.

210.

C. The reaction in screening cell I is 2+. This reaction is characterized by the presence of some agglutinates in the bottom of the gel microtube along with antigen-antibody complexes that are distributed throughout the gel. The 2+ reaction represents agglutinates of many different sizes. This is different than a 1+ reaction in which all of the agglutinates are confined to the lower half of the gel microtube. Larger agglutinates are trapped at the top of the gel, and the smaller the agglutinates, the further they fall through the matrix.

211.

A. When centrifuged, unagglutinated cells, like those in SC II, fall completely through the gel matrix and form a pellet at the bottom of the microtube. The agglutinated cells in SC I have been trapped in the gel according to the size of the agglutinates. The gel matrix is porous and will suspend antigen-antibody agglutinates in the gel when they are too large to fall further through the matrix.

212.

C. Paroxysmal cold hemoglobinuria (PCH) is the rarest form of cold autoimmune hemolytic anemia. It is characterized by an IgG antibody that usually exhibits anti-P specificity. The anti-P attaches to the RBC during exposure to cold and then activates complement when the patient rewarms, causing hemolysis. A diagnostic test used to aid in the identification of PCH is the Donath-Landsteiner test. CHD is usually caused by anti-I, which is IgM.

213.

C. The Donath-Landsteiner (D-L) test is a diagnostic test for PCH. A characteristic of the antibody involved is that it will bind to red blood cells at cold temperatures; at 37°C it will cause red blood cell lysis. In the D-L test, a tube with the patient's serum and test cells is incubated at 4°C whereas a control tube is incubated at 37°C. After the 4°C incubation, the tube is placed in the 37°C incubator. After incubation, the tubes are centrifuged and examined for hemolysis. If there is hemolysis noted in the test and none in the control, the test is positive. If no hemolysis is noted in either tube, the test is negative, and if hemolysis is present in both tubes, the test is invalid.

214.

C. Cephalosporin antibodies will generally *not* react with red blood cells unless the RBCs have been treated with the drug. The negative antibody screening test indicates that crossmatches will be similarly compatible. An eluate from the patient's red cells would be expected to be nonreactive with untreated RBCs as well. An eluate would not be prepared unless the patient had been recently transfused and a DHTR was suspected.

215.

A. The drug α-methyldopa (Aldomet®), a blood pressure medication that is no longer frequently used, induces autoantibody formation that is indistinguishable from antibody produced in WAIHA. The antibody produced usually reacts with all panel cells tested. Some of the other drugs that induce autoantibody formation are L-dopa, procainamide, and some nonsteroidal anti-inflammatory drugs.

216.

C. Of the drugs listed, only penicillin has the observed DAT profile. IgG alone is also generally seen when α-methyldopa and cephalosporins are the implicated drugs. When phenacetin, quinidine, and tolmetin are the drugs in question, only C3 (complement) is usually detected on the patient's red cells. Antibody in the patient's serum and/or eluate will not react with reagent red cells unless the RBCs are first treated with penicillin. This occurs because the antibody produced by the patient reacts with epitopes formed by the penicillin coating the RBCs.

217.

D. Although the cold autoagglutinin is generally IgM, complement is usually the only protein detected on the red blood cells. The IgM antibody binds complement at lower temperatures in the extremities (such as fingers exposed to cold temperatures) and then dissociates when the red cells circulate. The polyspecific AHG is not required to have anti-IgM but must have anti-IgG and anti-C3d.

218.

C. Antibody directed against drugs may be suspected if patients present a positive DAT with a nonproductive eluate. The eluate and serum antibody may demonstrate reactivity only with reagent red cells coated with the appropriate drug and not with uncoated red cells. A current medication history and demonstration of eluate reactivity against the appropriate drug-coated red cells confirm the cause of the positive DAT. Drug-related antibodies are seldom clinically significant, and elaborate testing usually is not performed unless the patient shows symptoms of active hemolysis.

219.

D. Autoantibody(ies) directed against self-antigens may attach to a patient's own red blood cells *in vivo,* resulting in a positive DAT. A variety of elution techniques can remove this antibody from the red cells for testing. When a patient has not been transfused recently (within 3 to 4 months), it may be presumed that only autoantibody is present in the eluate. If recently transfused, donor's cells may be present in the patient's circulation. Alloantibody may be attached to donor cells, and autoantibody may be attached to patient's or donor's cells. When this is the case, alloantibody and autoantibody both could be recovered in the eluate.

220.

B. A patient with a history of having had anti-Jk^a in her serum must always be given Jk(a–) blood. Anti-Jk^a in the plasma often drops to below detectable levels of antibody in the absence of stimulus. However, if Jk(a+) cells were to be given to the patient, the antibody would quickly rebound to high levels and cause a delayed hemolytic transfusion reaction. It is difficult to procure e negative blood because ~80% of the population is e+. Add that to the need for Jk(a–) blood and the likelihood of finding suitable units drops. It is much more important to avoid transfusing against the alloantibody, anti-Jk^a, than the autoantibody. The autoantibody will destroy transfused cells at the same rate it is destroying the patient's RBCs; however, the alloantibody will destroy the transfused cells more rapidly.

221–228.

(221:B, 222:D, 223:B, 224:B, 225:B, 226:B, 227:A, 228:C)

Mothers who are not candidates to receive Rh immune globulin (RhIG) are (1) D-positive, (2) D-negative giving birth to a D-negative infant, or (3) D-negative who have produced alloanti-D. For purposes of determining RhIG candidacy, the weak D phenotype is considered to be D-positive. If the mother is D-negative and has not made anti-D, she must receive RhIG when giving birth to a D-positive or weak D phenotype infant. When only one of the infants in a multiple birth is D-positive and the other(s) is (are) D-negative, the mother is a candidate for RhIG. The presence of antibodies other than anti-D in the pre- or postpartum serum does not preclude a D-negative woman from receiving RhIG. When the infant's red cells are DAT+, it may be difficult to determine the correct Rh status. Because RhIG is a low-risk product, experts recommend a fail-safe approach and give RhIG when in doubt. Most obstetricians routinely inject RhIG at 28 weeks antenatally to prevent sensitization during a third-trimester fetal bleed. This injected anti-D is sometimes detectable postpartum and does not preclude additional administration of RhIG at childbirth. Whenever doubt exists as to the weak D status of the infant or the origin of the anti-D, postpartum RhIG should be administered. The preferred method for determining fetomaternal hemorrhage (FMH) is the rosette test, which is more sensitive than a microscopic weak D test. The rosette test will detect FMH of >10 mL. A positive test would show one or more rosettes per field when an enhancing medium is used. Additional testing must be performed to quantify the amount of FMH to prevent sensitization to the D antigen. The Kleihauer-Betke acid elution stain is the standard technique available to determine whether a bleed of more than 15 mL red cells (30 mL whole blood) occurred, necessitating additional 300-μg doses beyond the standard one-vial dose. Other methods include the enzyme-linked antiglobulin test (ELAT) and flow cytometry. The 50-μg dose of RhIG is used only when a female of 12-weeks gestation or less suffers a miscarriage. It is not intended for administration postpartum or for injection antenatally, either at 28 weeks or after amniocentesis.

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

Aerobic Gram-Positive Bacteria

1. A test for the hydrolysis of esculin in the presence of bile is especially useful in identifying species of the genus
   A. *Abiotrophia*
   B. *Corynebacterium*
   C. *Enterococcus*
   D. *Staphylococcus*

2. The organism associated with a disease characterized by the presence of a pseudomembrane in the throat and the production of an exotoxin that is absorbed into the bloodstream with a lethal effect is
   A. *Arcanobacterium haemolyticum*
   B. *Staphylococcus aureus*
   C. *Streptococcus pyogenes*
   D. *Corynebacterium diphtheriae*

3. Enterotoxin produced by *Staphylococcus aureus* is responsible for causing
   A. Carbuncles
   B. Enterocolitis
   C. Impetigo
   D. Scalded skin syndrome

4. *Abiotrophia,* formerly known as nutritionally variant streptococci, will not grow on routine blood or chocolate agars because they are deficient in
   A. Hemin
   B. Pyridoxal
   C. Vitamin B_{12}
   D. Thiophene-2-carboxylic hydrazide

5. Exfoliatin produced by *Staphylococcus aureus* is responsible for causing
   A. Enterocolitis
   B. Toxic shock syndrome
   C. Scalded skin syndrome
   D. Staphylococcal pneumonia

6. *Streptococcus pyogenes* can be presumptively identified using a(an)
   A. PYR disk
   B. ONPG disk
   C. SPS disk
   D. Optochin disk

7. A gram-positive coccus that is catalase positive, nonmotile, lysostaphin resistant, and modified oxidase positive is best identified as a member of the genus
   A. *Micrococcus*
   B. *Lactococcus*
   C. *Pediococcus*
   D. *Staphylococcus*

8. *Nocardia asteroides* infections in humans characteristically produce
   A. Carbuncles
   B. Draining cutaneous sinuses
   C. Septic shock
   D. Serous effusions

9. *Erysipelothrix* infections in humans characteristically produce
   A. Pathology at the point of entrance of the organism
   B. Central nervous system pathology
   C. Pathology in the lower respiratory tract
   D. The formation of abscesses in visceral organs

10. In the CAMP test, a single streak of a beta-hemolytic *Streptococcus* is placed perpendicular to a streak of beta-lysin-producing *Staphylococcus aureus*. After incubation, a zone of increased lysis in the shape of an arrowhead is noted; this indicates the presumptive identification of
    A. *S. agalactiae*
    B. *S. bovis*
    C. *S. equinus*
    D. *S. pyogenes*

11. *Staphylococcus saprophyticus,* a recognized pathogen, is a cause of
    A. Furuncles
    B. Impetigo
    C. Otitis media
    D. Urinary tract infections

12. Color Plate 26■ shows the Gram stain of a blood culture on a 23-year-old pregnant woman who presented with fever and flulike symptoms in her ninth month. The isolate on blood agar produced small, translucent beta-hemolytic colonies. Which of the following is the most likely etiologic agent in this case?
    A. *Listeria monocytogenes*
    B. *Propionibacterium acnes*
    C. *Streptococcus agalactiae*
    D. *Streptococcus pyogenes*

13. The etiologic agent most commonly associated with septicemia and meningitis of newborns is
    A. *Streptococcus agalactiae*
    B. *Streptococcus bovis* group
    C. *Streptococcus pneumoniae*
    D. *Streptococcus pyogenes*

14. Which of the following is the most commonly isolated species of *Bacillus* in opportunistic infections such as bacteremia, post-traumatic infections of the eye, and endocarditis?
    A. *B. circulans*
    B. *B. cereus*
    C. *B. licheniformis*
    D. *B. subtilis*

15. Loeffler's serum medium is recommended for the cultivation of
    A. *Abiotrophia* sp.
    B. *Corynebacterium diphtheriae*
    C. *Leuconostoc* sp.
    D. *Streptococcus agalactiae*

16. On Tinsdale agar, colonies of *Corynebacterium diphtheriae* are characterized by the observance of
    A. Liquefaction of the agar surrounding the colonies on the medium
    B. Opalescent colonies with a white precipitate in the surrounding agar
    C. Black colonies on the culture medium surrounded by brown halos
    D. Pitting of the agar medium surrounding the colonies

17. Precipitates of diphtheria toxin and antitoxin formed in agar gels are an *in vitro* means for detecting toxigenic strains of *Corynebacterium diphtheriae.* The name of this test procedure is the
    A. D-test
    B. Elek test
    C. Hodge test
    D. Nagler test

18. The etiologic agent of the disease erysipelas is
    A. *Staphyloccus aureus*
    B. *Streptobacillus moniliformis*
    C. *Streptococcus agalactiae*
    D. *Streptococcus pyogenes*

19. *Staphylococcus aureus,* when present, could most likely be recovered from a stool sample if the primary plating medium included
    A. Bismuth sulfite
    B. Phenylethyl alcohol
    C. Thiosulfate citrate bile salts sucrose
    D. Xylose-lysine-desoxycholate

20. A common member of the normal flora of the upper respiratory tract is
    A. *Corynebacterium jeikeium*
    B. *Lactobacillus*
    C. *Staphylococcus epidermidis*
    D. Viridans streptococcus

21. Streptococci obtain all their energy from the fermentation of sugars to
    A. Formic acid
    B. Lactic acid
    C. Succinic acid
    D. Valeric acid

22. Streptococci are unable to synthesize the enzyme
    A. Catalase
    B. Kinase
    C. Hyaluronidase
    D. Lipase

23. The beta-hemolysis produced by group A *Streptococcus* seen on the surface of a sheep blood agar plate is primarily the result of streptolysin
    A. H
    B. M
    C. O
    D. S

24. When an infection due to *Streptomyces* is suspected, isolates can be separated from most other bacteria by
    A. Heat shocking the culture
    B. Incubating the culture at 25°C
    C. Incubating the culture at greater than 37°C
    D. Drying the specimen before inoculating the culture media

25. The production of H_2S is one characteristic used to differentiate which of the aerobic gram-positive bacilli?
    A. *Corynebacterium*
    B. *Erysipelothrix*
    C. *Lactobacillus*
    D. *Nocardia*

26. Growth in a 48-hour semisolid agar stab culture at room temperature reveals lateral filamentous growth away from the stab near the top of the medium. This observation is most characteristic of which organism?
    A. *Rhodococcus* sp.
    B. *Corynebacterium urealyticum*
    C. *Enterococcus faecalis*
    D. *Listeria monocytogenes*

27. A former species of *Corynebacterium* pathogenic for swine, horses, and cattle is also known to cause disease in compromised hosts. This organism when grown on culture media produces pale pink colonies that help to presumptively identify it as
    A. *Arcanobacterium hemolyticum*
    B. *Actinomyces naeslundii*
    C. *Gardnerella vaginalis*
    D. *Rhodococcus equi*

28. Which one of the following is *not* appropriate when describing *Streptococcus pneumoniae?*
    A. Bile-resistant
    B. Alpha-hemolytic
    C. Lancet-shaped, gram-positive diplococcus
    D. Virulent strains are encapsulated.

29. *Nocardia* can be differentiated from *Actinomyces* based on
    A. *Nocardia* being an obligate anaerobe
    B. The partial-acid fast staining reaction of *Actinomyces*
    C. The production of sulfur granules in cases of nocardiosis
    D. *Nocardia* being catalase positive

30. *Enterococcus faecium* is characteristically
    A. Inhibited by the presence of bile in culture media
    B. Able to grow in the presence of high concentrations of salt
    C. PYR negative
    D. Beta-hemolytic

31. A negative PYR (L-pyrolidonyl-α-naphthylamide) test is demonstrated by
    A. *Enterococcus faecalis*
    B. *Enterococcus faecium*
    C. *Streptococcus pyogenes*
    D. Viridans streptococci

32. A Gram stain of a sputum specimen from a patient with a suspected case of lobar pneumonia reveals many white blood cells and many gram-positive cocci, which are primarily diplococci. Which of the following statements would be appropriate, given these findings?
    A. A PYR test should be performed on the culture isolate.
    B. An Elek test should be performed on the culture isolate.
    C. An optochin test should be performed on the culture isolate.
    D. A hippurate hydrolysis test should be performed on the culture isolate.

33. A child presented in August at the pediatric clinic with a superficial skin infection of the neck. The large, itchy lesions were cultured, and the diagnosis of impetigo was made. One of the etiologic agents of this clinical condition is
    A. *Erysipelothrix rhusiopathiae*
    B. *Corynebacterium diphtheriae*
    C. *Staphylococcus saprophyticus*
    D. *Streptococcus pyogenes*

34. An identifying characteristic of *Staphylococcus aureus* is
    A. DNase negative
    B. Coagulase negative
    C. Mannitol fermentation postive
    D. Growth inhibition in presence of increased salt

35. Which of the following organisms is able to hydrolyze sodium hippurate to benzoic acid and glycine?
    A. *Streptococcus agalactiae*
    B. *Streptococcus pneumoniae*
    C. *Listeria monocytogenes*
    D. *Enterococcus faecalis*

36. Which of the following is *not* characteristic of *Listeria monocytogenes?*
    A. CAMP test postive
    B. Catalase negative
    C. Esculin hydrolysis positive
    D. Motile

37. Which of the following is *not* associated with *Staphylococcus aureus?*
    A. Endotoxin production
    B. Clumping factor production
    C. Deoxyribonuclease production
    D. Hemolysin production

38. Which of the following is a characteristic of staphylococci that would help in their isolation from clinical specimens?
    A. Bile resistance
    B. Growth at 55°C
    C. High salt tolerance
    D. Resistance to novobiocin

39. Which of the following species of *Bacillus* is nonmotile?
    A. *B. cereus*
    B. *B. subtilis*
    C. *B. anthracis*
    D. *B. thuringiensis*

40. Which one of the following diseases involves erythrogenic toxin?
    A. Cutaneous anthrax
    B. Diphtheria
    C. Impetigo
    D. Scarlet fever

41. Cultures of the posterior pharynx are most commonly submitted to the clinical laboratory for the detection of
    A. *Corynebacterium diphtheriae*
    B. *Staphylococcus aureus*
    C. *Streptococcus pneumoniae*
    D. *Streptococcus pyogenes*

42. *Streptococcus sanguis,* a viridans streptococcus, is most commonly associated with which of the following clinical conditions?
    A. Otitis media
    B. Pharyngitis
    C. Relapsing fever
    D. Subacute bacterial endocarditis

43. Rust-colored sputum in cases of lobar pneumonia is characteristic of which of the following possible etiologic agents?
    A. *Corynebacterium jeikeium*
    B. *Staphylococcus aureus*
    C. *Streptococcus pneumoniae*
    D. *Streptococcus pyogenes*

44. A urine culture from a 23-year-old female grew a catalase-positive gram-positive coccus (>100,000 cfu/mL), which would most likely be
    A. *Staphylococcus saprophyticus*
    B. *Enterococcus faecalis*
    C. *Streptococcus bovis* group
    D. *Streptococcus* viridans

45. Cystine-tellurite blood agar plates are recommended for the isolation of
    A. *Corynebacterium diphtheriae*
    B. *Streptococcus agalaciae*
    C. *Streptococcus pyogenes*
    D. Group D streptococci

46. The pulmonary form of anthrax is known as
    A. Valley fever
    B. Walking pneumonia
    C. Farmers' lung
    D. Woolsorters disease

47. Pleomorphic gram-positive bacilli in a Gram stain best describes
    A. *Bacillus anthracis*
    B. *Bacillus subtilis*
    C. *Listeria monocytogenes*
    D. *Corynebacterium pseudodiphtheriticum*

48. An aerobic gram-positive rod known to cause bacteremia in hospitalized immunocompromised patients is
    A. *Bacillus anthracis*
    B. *Corynebacterium jeikeium*
    C. *Corynebacterium ulcerans*
    D. *Corynebacterium urealyticum*

49. A bone marrow transplant patient on immunosuppressive therapy developed a pulmonary abscess with symptoms of neurologic involvement. A brain abscess was detected by MRI, and aspirated material grew an aerobic, filamentous, branching gram-positive organism, which stained weakly acid-fast. The most likely etiologic agent in this case would be
    A. *Actinomyces israelii*
    B. *Nocardia asteroides*
    C. *Mycobacterium tuberculosis*
    D. *Propionibacterium acnes*

50. Which of the following is catalase negative?
    A. *Bacillus*
    B. *Corynebacterium*
    C. *Leuconostoc*
    D. *Listeria*

51. Colonies of *Listeria monocytogenes* on a sheep blood agar plate most closely resemble colonies of
    A. *Corynebacterium diphtheriae*
    B. *Streptococcus agalactiae*
    C. *Streptococcus bovis* group
    D. *Rhodococcus equi*

52. The most common etiologic agent of infections associated with the surgical insertion of prosthetic devices such as artificial heart valves and cerebrospinal fluid shunts is
    A. *Corynebacterium urealyticum*
    B. *Staphylococcus capitis*
    C. *Staphylococcus epidermidis*
    D. *Streptococcus mutans*

53. The description of "Medusa head" colonies on solid agar is most characteristic of
    A. *Bacillus anthracis*
    B. *Enterococcus faecalis*
    C. *Staphylococcus saprophyticus*
    D. *Streptococcus agalactiae*

54. Which of the following is most likely to be isolated in cultures from the anterior nares of healthcare workers?
    A. *Bacillus cereus*
    B. *Streptococcus pneumoniae*
    C. *Staphylococcus aureus*
    D. *Staphylococcus saprophyticus*

55. Ethylhydrocupreine HCl susceptibility is a presumptive test for the identification of
    A. Viridans streptococci
    B. *Streptococcus pyogenes*
    C. *Streptococcus agalactiae*
    D. *Streptococcus pneumoniae*

56. Solubility in the presence of sodium desoxycholate is characteristic of
    A. *Enterococcus faecalis*
    B. *Streptococcus agalactiae*
    C. *Streptococcus mutans*
    D. *Streptococcus pneumoniae*

57. Family members attending a picnic became ill about 2 hours after eating. The illness was characterized by rapid onset of violent vomiting. The most likely bacterial cause of such symptoms would be food poisoning caused by
    A. *Enterococcus faecium*
    B. *Bacillus subtilis*
    C. *Listeria monocytogenes*
    D. *Staphylococcus aureus*

58. The novobiocin susceptibility test is used for the identification of
    A. *Corynebacterium diphtheriae*
    B. *Streptococcus pyogenes*
    C. *Streptococcus pneumoniae*
    D. *Staphylococcus saprophyticus*

59. Tellurite reduction is used for the presumptive identification of
    A. *Bacillus anthracis*
    B. *Corynebacterium diphtheriae*
    C. *Erysipelothrix rhusiopathiae*
    D. *Staphylococcus saprophyticus*

60. The etiologic agent of the majority of adult joint infections is
    A. *Abiotrophia* sp.
    B. *Leuconostoc* sp.
    C. *Staphylococcus aureus*
    D. *Streptococcus pneumoniae*

61. Which of the following is associated with infections in humans often linked to deli meats and improperly pasteurized dairy products?
    A. *Bacillus subtilis*
    B. *Listeria monocytogenes*
    C. *Leuconostoc*
    D. *Streptococcus agalactiae*

62. *Bacillus cereus* has been implicated as the etiologic agent in cases of
    A. Food poisoning
    B. Impetigo
    C. Pelvic inflammatory disease
    D. Toxic shock syndrome

63. The causative agent of "malignant pustule" is
    A. *Bacillus anthracis*
    B. *Corynebacterium ulcerans*
    C. *Erysipelothrix rhusiopathiae*
    D. *Listeria monocytogenes*

64. An infant was hospitalized with a severe, tender erythema. The child's epidermis was loose, and large areas of skin could be peeled off. The condition described is most consistent with a clinical syndrome associated with
    A. *Streptococcus pyogenes*
    B. *Staphylococcus aureus*
    C. *Bacillus anthracis*
    D. *Erysipelothrix rhusiopathiae*

65. A catalase-negative gram-positive coccus is isolated from a urine sample of a hospitalized patient. The bacterium produced a black pigment on bile-esculin agar and formed acid from glucose in the presence of 6.5% NaCl. What is the most likely identification of this bacterium?
    A. *Abiotrophia* sp.
    B. *Enterococcus faecalis*
    C. Group B streptococci
    D. Group D streptococci

Aerobic Gram-Negative Bacteria

66. In suspected cases of brucellosis, the optimal specimen to be collected for the isolation of the etiologic agent is
    A. Blood
    B. Urine
    C. Cerebrospinal fluid
    D. Nasopharyngeal exudates

67. The majority of clinical isolates of *Klebsiella* are
    A. *K. ozaenae*
    B. *K. pneumoniae*
    C. *K. aerogenes*
    D. *K. oxytoca*

68. The enterotoxins of both *Vibrio cholerae* O1 and noninvasive (toxigenic) strains of *Escherichia coli* produce serious diarrhea by what mechanism?
    A. Stimulation of adenylate cyclase, which gives rise to excessive fluid secretion by the cells of the small intestine
    B. Penetration of the bowel mucosa
    C. Stimulation of colicin production
    D. Elaboration of a dermonecrotizing toxin

69. Colonies of *Neisseria* sp. turn color when a redox reagent is applied. The color change is indicative of the activity of the bacterial enzyme
    A. Beta-galactosidase
    B. Urease
    C. Cytochrome oxidase
    D. Phenylalanine deaminase

70. Which of the following is *not* true of *Shigella sonnei?*
    A. Large numbers of organisms must be ingested to produce disease.
    B. The organism produces an inflammatory condition in the large intestine with bloody diarrhea.
    C. The organism produces disease most commonly in the pediatric population.
    D. The organism is a delayed lactose fermenter.

71. An environmental sampling study of respiratory therapy equipment produced cultures of a yellow, nonfermentative (at 48 hours), gram-negative bacillus from several of the nebulizers, which would most likely be species of
    A. *Chryseobacterium*
    B. *Pseudomonas*
    C. *Alcaligenes*
    D. *Moraxella*

72. The characteristics of being lactose negative, citrate negative, urease negative, lysine decarboxylase negative, and nonmotile best describe which organism?
    A. *Proteus vulgaris*
    B. *Yersinia pestis*
    C. *Salmonella enterica*
    D. *Shigella dysenteriae*

73. A fermentative gram-negative bacillus that is oxidase positive, motile, and grows well on MacConkey agar is
    A. *Aeromonas hydrophila*
    B. *Pseudomonas aeruginosa*
    C. *Stenotrophomonas maltophilia*
    D. *Yersinia enterocolitica*

74. Fecal cultures are inoculated on thiosulfate-citrate-bile salts-sucrose agar specifically for the isolation of
A. *Shigella*
B. *Vibrio*
C. *Campylobacter*
D. *Salmonella*

75. The K antigen of the family *Enterobacteriaceae* is
A. Heat labile
B. The somatic antigen
C. Located on the flagellum
D. The antigen used to group *Shigella*

76. The causative agent of melioidosis is
A. *Burkholderia cepacia*
B. *Burkholderia pseudomallei*
C. *Moraxella catarrhalis*
D. *Stenotrophomonas maltophilia*

77. Which microorganism will grow only on culture media supplemented with either cysteine or cystine?
A. *Actinobacillus lignieresii*
B. *Bartonella bacilliformis*
C. *Francisella tularensis*
D. *Kingella kingae*

78. A culture of a decubitus ulcer grew a gram-negative facultative bacillus. On TSI it produced an acid slant, acid butt, and gas. Test reactions in other media were as follows:

Citrate	negative
Indole	positive
Urease	negative
ONPG	positive
Voges-Proskauer	negative

The organism was identified as
A. *Enterobacter cloacae*
B. *Escherichia coli*
C. *Citrobacter (diversus) koseri*
D. *Providencia stuartii*

79. An example of an oxidase-positive, glucose nonfermenting organism is
A. *Aeromonas hydrophila*
B. *Escherichia coli*
C. *Klebsiella pneumoniae*
D. *Pseudomonas aeruginosa*

80. A fastidious gram-negative bacillus was isolated from a case of periodontal disease, which upon darkfield examination was noted to have gliding motility. The most likely identification of this etiologic agent would be
A. *Capnocytophaga*
B. *Chromobacterium*
C. *Kingella*
D. *Plesiomonas*

81. The species of *Vibrio* closely associated with rapidly progressing wound infections seen in patients with underlying liver disease is
    A. *V. alginolyticus*
    B. *V. cholerae*
    C. *V. vulnificus*
    D. *V. parahaemolyticus*

82. Severe disseminated intravascular coagulation often complicates cases of septicemia caused by
    A. *Acinetobacter* sp.
    B. *Moraxella* sp.
    C. *Neisseria gonorrhoeae*
    D. *Neisseria meningitidis*

83. The *Haemophilus influenzae* vaccine protects against which serotype?
    A. Serotype a
    B. Serotype b
    C. Serotype c
    D. Serotype d

84. *Salmonella* Typhi exhibits a characteristic biochemical pattern, which differentiates it from the other salmonellae. Which of the following is *not* characteristic of *S.* Typhi?
    A. Large amounts of H_2S are produced in TSI agar.
    B. Agglutination in Vi grouping serum
    C. Lysine decarboxylase positive
    D. Citrate negative

85. The sexually acquired disease characterized by genital ulcers and tender inguinal lymphadenopathy, which is caused by a small, gram-negative bacillus, is known as
    A. Chancroid
    B. Bacterial vaginosis
    C. Syphilis
    D. Trachoma

86. Which of the following diseases is most likely to be acquired from a hot tub or whirlpool?
    A. Q fever
    B. Erysipelas
    C. *Acinetobacter* cellulitis
    D. *Pseudomonas* dermatitis

87. *Campylobacter* spp. are associated most frequently with cases of
    A. Osteomyelitis
    B. Gastroenteritis
    C. Endocarditis
    D. Appendicitis

88. An organism occasionally misidentified as an enteric pathogen because it produces a large amount of H_2S is
    A. *Burkholderia cepacia*
    B. *Burkholderia pseudomallei*
    C. *Pseudomonas putida*
    D. *Shewanella putrefaciens*

89. The etiologic agent of whooping cough is
    A. *Bordetella pertussis*
    B. *Brucella suis*
    C. *Francisella tularensis*
    D. *Haemophilus ducreyi*

90. An important characteristic of *Neisseria gonorrhoeae* or the infection it produces is
    A. A Gram stain of the organism reveals gram-negative bacilli.
    B. Asymptomatic infections are common in females.
    C. Produces disease in humans and domestic animals
    D. The bacteria survive long periods outside the host's body.

91. Which of the following organisms would most likely produce the biochemical reactions shown in Color Plate 27■?
    A. *Citrobacter freundii*
    B. *Proteus mirabilis*
    C. *Providencia rettgeri*
    D. *Salmonella*

92. A gram-negative, "kidney bean" cellular morphology is a distinguishing characteristic of
    A. *Neisseria meningitidis*
    B. *Yersinia pestis*
    C. *Bartonella* spp.
    D. *Actinobacter* spp.

93. Which of the following nonfermenters is rarely isolated in the U.S.?
    A. *Pseudomonas aeruginosa*
    B. *Stenotrophomonas maltophilia*
    C. *Burkholderia mallei*
    D. *Burkholderia cepacia*

94. Erythromycin eye drops are routinely administered to infants to prevent infections by
    A. *E. coli*
    B. *Haemophilus influenzae*
    C. *Pseudomonas aeruginosa*
    D. *Neisseria gonorrhoeae*

95. *Neisseria lactamica* closely resembles *Neisseria meningitidis* but can be differentiated from it by its ability to metabolize
    A. Maltose
    B. Lactose
    C. Glucose
    D. Sucrose

96. A causative agent of the form of conjunctivitis known as pinkeye is
    A. *Haemophilus aegyptius*
    B. *Moraxella lacunata*
    C. *Chlamydia trachomatis*
    D. *Klebsiella ozaenae*

97. The single species in the genus *Hafnia* is
    A. *alvei*
    B. *gergoviae*
    C. *ruckeri*
    D. *tarda*

98. *Acinetobacter baumannii*
    A. Requires cysteine
    B. Is oxidase negative
    C. Ferments glucose
    D. Does not grow on MacConkey agar

99. *Legionella pneumophila* is the etiologic agent of both Legionnaires disease and
    A. Swine fever
    B. Pontiac fever
    C. Rift Valley fever
    D. San Joaquin Valley fever

100. In suspected cases of brucellosis, what is the most sensitive specimen to submit for culture?
    A. Bone marrow
    B. Nasopharyngeal swab
    C. Sputum
    D. Stool

101. Hemolytic uremic syndrome is a complication after infection with
    A. *E. coli* O157:H7
    B. *Salmonella* Typhi
    C. *Vibrio cholerae* O1
    D. *Yersinia enterocolitica*

102. Identify the fermentative agent that may infect reptiles and fish as well as humans when they are exposed to contaminated soil or water.
    A. *Aeromonas*
    B. *Chromobacterium*
    C. *Chryseobacterium*
    D. *Enterobacter*

103. *Campylobacter jejuni* is
    A. Nonmotile
    B. Oxidase negative
    C. Hippurate hydrolysis positive
    D. A straight gram-negative bacillus

104. Which of the following has a negative oxidase test?
A. *Aeromonas*
B. *Chryseobacterium*
C. *Hafnia*
D. *Vibrio*

105. Which of the following is *true* of *Neisseria gonorrhoeae?*
A. Adversly affected by fatty acids in clinical specimens
B. Rapid growth on sheep blood agar
C. Ferments glucose and maltose
D. ONPG positive

106. The method of serogrouping *Shigella* used in the clinical laboratory is based on
A. Bacteriocins
B. H antigens
C. K antigens
D. O antigens

107. The symptom of diffuse, watery diarrhea that produces a relatively clear stool containing mucus flecks is suggestive of an infection caused by
A. Enterohemorrhagic *Escherichia coli*
B. *Shigella dysenteriae*
C. *Vibrio cholerae*
D. *Yersinia enterocolitica*

108. An example of a halophilic microorganism is
A. *Morganella morganii*
B. *Plesiomonas shigelloides*
C. *Vibrio parahaemolyticus*
D. *Yersinia pestis*

109. Which of the following statements is *not* true of *Brucella?*
A. Infection may occur via abrasions of the oral mucosa, conjunctiva, and genitals.
B. They are fastidious and require supplemented media for isolation.
C. The risk of accidental laboratory infection is no greater than with any other organism.
D. Phage and dye sensitivity tests are used for identification to the species level.

110. Infection of the gastric mucosa leading to gastritis or peptic ulcers is most commonly associated with
A. *Campylobacter jejuni*
B. *Helicobacter pylori*
C. *Salmonella* Typhi
D. *Shigella sonnei*

111. Which of the the following does *not* describe *Acinetobacter* sp.?
A. Commonly susceptible to most antimicrobials
B. Generally coccobacillary in morphology
C. Oxidase negative
D. Infections associated with use of medical devices

112. Explosive watery diarrhea with severe abdominal pain after eating raw shellfish is most characteristic of infection caused by
A. *Campylobacter jejuni*
B. *Helicobacter pylori*
C. *Shigella dysenteriae*
D. *Vibrio parahemolyticus*

113. An unheated suspension of *Salmonella* Typhi typically produces agglutination of Vi antisera. After heating the same suspension, agglutination will occur in which grouping sera?
A. A
B. B
C. C_1
D. D

114. The species of *Campylobacter* noted to produce septicemia, septic arthritis, meningitis, jaundice with hepatomegaly, and thrombophlebitis in debilitated patients is
A. *C. coli*
B. *C. fetus*
C. *C. laris*
D. *C. sputorum*

115. *Shigella sonnei* is differentiated from other *Shigella* sp. by
A. A positive ONPG
B. Its positive phenylalanine deaminase reaction
C. Its negative oxidase reaction
D. Its ability to demonstrate motility at 22°C

116. A positive Voges-Proskauer reaction is characteristic of
A. *Enterobacter aerogenes*
B. *Escherichia coli*
C. *Proteus vulgaris*
D. *Providencia rettgeri*

117. Which of the following is *not* true regarding virulent strains of *Vibrio cholerae?*
A. Adherent to enterocytes
B. Mucinase production
C. Nonmotile
D. Toxigenic

118. The classic toxigenic strains of which serogroup are implicated in epidemic infections of *Vibrio cholerae?*
A. O1
B. O2
C. O3
D. O4

119. *Brucella* spp. are
A. The etiologic agents of relapsing fever
B. Small spiral organisms
C. Primarily a cause of endogenous human infections
D. Intracellular pathogens

120. Which of the following reactions is typical for *Escherichia coli?*
A. Beta-hemolytic on sheep blood agar
B. Colorless colonies on MacConkey agar
C. Colorless colonies on xylose-lysine-desoxycholate agar
D. Green colonies with black centers on Hektoen enteric agar

121. *Yersinia pestis* is characteristically
A. Urease negative
B. Hydrogen sulfide positive
C. Motile at 20–25°C
D. Oxidase positive

122. Lack of motility is characteristic of
A. *Enterobacter cloacae*
B. *Klebsiella oxytoca*
C. *Morganella morganii*
D. *Providencia stuartii*

123. In cases of legionellosis
A. Person-to-person transmission is common
B. Farm animals are important animal reseroirs
C. Patients can present with a self-limited nonpneumonic febrile illness
D. Specimens may be cold enriched to enhance recovery of the organism

124. Which of the following is *not* a characteristic of *Klebsiella* (*Calymmatobacterium*) *granulomatis?*
A. Is often sexually transmitted
B. Is isolated on chocolate agar
C. Is the causative agent of granuloma inguinale
D. Can be diagnosed by detection of Donovan bodies in clinical specimens

125. A positive DNase would be seen with
A. *Escherichia coli*
B. *Klebsiella oxytoca*
C. *Proteus mirabilis*
D. *Serratia marcescens*

126. A negative citrate reaction is characteristic of
A. *Citrobacter freundii*
B. *Enterobacter aerogenes*
C. *Serratia marcescens*
D. *Shigella boydii*

127. Isolation of *Neisseria gonorrhoeae*
A. Is enhanced by cold enrichment
B. Requires incubation under increased CO_2
C. From contaminated sites is made easier by the use of CIN agar
D. Is not affected if clinical specimen is refrigerated before culturing

128. A positive indole reaction is characteristic of
A. *Escherichia coli*
B. *Proteus mirabilis*
C. *Salmonella* Choleraesuis
D. *Serratia marcescens*

129. Which one of the following organisms would produce a yellow slant and a yellow butt on TSI agar after incubating 18 hours?
A. *Escherichia coli*
B. *Proteus mirabilis*
C. *Salmonella* Typhimurium
D. *Shigella sonnei*

130. Pyocyanin is characteristically produced by
A. *Pseudomonas aeruginosa*
B. *Pseudomonas fluorescens*
C. *Shewanella putrefaciens*
D. *Stenotrophomonas maltophillia*

131. Color Plate 28■ shows the Gram stain of cerebrospinal fluid from a 1-year-old girl suspected of having meningitis. After 24 hours of growth, small tan colonies were isolated on chocolate agar incubated in CO_2. Sheep blood agar also incubated in CO_2 had no growth. Which of the following organisms should be suspected?
A. *Brucella canis*
B. *Bordetella parapertussis*
C. *Haemophilus influenzae*
D. *Neisseria meningitidis*

132. Which of the following is *true* concerning *Campylobacter jejuni?*
A. Catalase negative
B. Isolated best at 24°C
C. Hydrogen sulfide positive
D. A leading cause of bacterial diarrhea worldwide

133. Of the following microorganisms, which one will turn a dark purple when tetramethyl-*p*-phenylenediamine hydrochloride is applied?
A. *Acinetobacter baumannii*
B. *Stenotrophomonas maltophilia*
C. *Moraxella catarrhalis*
D. *Yersinia enterocolitica*

134. *Cardiobacterium hominis,* an inhabitant of the upper respiratory tract of humans, has been recovered as the etiologic agent from cases of endocarditis. An identifying characteristic of the organism is
A. Positive oxidase
B. Positive catalase
C. Indole negative
D. Inability to grow on sheep blood agar

135. *Vibrio vulnificus* is a well-established human pathogen that is known to cause
A. Gastroenteritis
B. Pneumonia
C. Pyelonephritis
D. Wound infections

136. Which of the following organisms produce a positive phenylalanine deaminase reaction?
A. *Citrobacter freundii*
B. *Klebsiella pneumoniae*
C. *Providendia stuartii*
D. *Yersinia enterocolitica*

137. Which of the following non-lactose-fermenting organisms does *not* produce fluorescein?
A. *Pseudomonas alcaligenes*
B. *Pseudomonas aeruginosa*
C. *Pseudomonas fluorescens*
D. *Pseudomonas putida*

138. Which of the following organisms is unable to grow on MacConkey agar?
A. *Bordetella bronchiseptica*
B. *Burkholderia cepacia*
C. *Kingella denitrificans*
D. *Plesiomonas shigelloides*

139. Which of the the following is *not* true of *Haemophilus* spp.?
A. Obligate parasites
B. Grow well on sheep blood agar
C. Small, pleomorphic, gram-negative coccobacilli
D. Many are found as normal flora in the human respiratory tract.

140. *Legionella pneumophila*
A. Infections are most often acquired from environmental sources
B. Metabolizes a number of carbohydrates
C. Stains easily on the routine Gram stain
D. Does not autofluoresce

141. Which of the following is the optimal clinical specimen for the recovery of *Legionella pneumophila?*
A. Stool
B. Blood
C. Bronchial washings
D. Nasopharyngeal swab

142. Which of the following is *not* an appropriate medium for primary isolation of *Bordetella pertussis?*
A. Regan-Lowe
B. Bordet-Gengou
C. Modified Skirrow's medium
D. Buffered charcoal yeast extract

143. Production of a yellow pigment is characteristic of which of the following *Enterobacter* sp.?
A. *E. aerogenes*
B. *E. cloacae*
C. *E. sakazakii*
D. *E. taylorae*

144. A gram-negative bacillus was recovered from the urine of a child with a history of recurrent urinary tract infections. The organism was oxidase negative, lactose negative, urease positive, and motile. The most likely identification of this agent would be
A. *Escherichia coli*
B. *Klebsiella pneumoniae*
C. *Proteus mirabilis*
D. *Pseudomonas aeruginosa*

145. *Acinetobacter baumannii* is characteristically
A. Motile
B. Oxidase positive
C. Sensitive to penicillin
D. Able to grow on MacConkey agar

146. Which of the following is *not* characteristic of *Eikenella corrodens?*
A. It is a thin, gram-negative bacillus.
B. Colonies sometimes pit the surface of the agar medium.
C. It is found in the mouth and upper respiratory tract of humans.
D. It is often found in pure culture when recovered from infections.

147. Which of the following species of *Pasteurella* is associated with human infections following cat bites?
A. *P. avium*
B. *P. canis*
C. *P. multocida*
D. *P. stomatis*

148. Which of the following *Legionella* spp. is positive for hippurate hydrolysis?
A. *L. micdadei*
B. *L. longbeachae*
C. *L. gormanii*
D. *L. pneumophila*

149. Which of the following is *not* characteristic of *Haemophilus influenzae* biogroup aegyptius?
A. Indole negative
B. Requires V factor
C. Requires X factor
D. Aminolevulinic acid positive

150. *Kingella denitrificans* can be differentiated from *Neisseria gonorrhoeae* because it is
A. Able to grow on Thayer-Martin agar
B. Able to reduce nitrates
C. Oxidase positive
D. Glucose positive

151. *Plesiomonas* was recently moved to which family?
A. *Brucelleaceae*
B. *Enterobacteriaceae*
C. *Legionellaceae*
D. *Vibrionaceae*

152. In the past, povidone iodine, tincture of iodine prep pads and swabs, and other disinfectants have been recalled because of microbial contamination. The most likely organism to be isolated in these cases is
A. *Bordetella bronchiseptica*
B. *Klebsiella pneumoniae*
C. *Pseudomonas aeruginosa*
D. *Serratia marcescens*

153. Foodborne outbreaks of brucellosis are most commonly associated with eating
A. Raw shellfish
B. Imported cheese
C. Contaminated potato salad
D. Improperly cooked hamburger

154. *Serratia* spp. are unique in the family *Enterobacteriaceae* because of their ability to produce extracellular hydrolytic enzymes. Which of the following is *not* produced by *Serratia* species?
A. DNase
B. Gelatinase
C. Lipase
D. NADase

155. *Haemophilus ducreyi* is the causative agent of
A. Chancroid
B. Lymphogranuloma venereum
C. Trachoma
D. Whooping cough

156. The optimal specimen for the recovery of *Bordetella pertussis* is
A. Anterior nares swab
B. Blood
C. Expectorated sputum
D. Nasopharyngeal swab

157. Which of the following is the most common indicator of bacterial vaginosis?
A. *Eikenella*
B. *Capnocytophaga*
C. *Gardnerella*
D. *Listeria*

158. Blood cultures are recommended for the recovery of which of the following microorganisms?
A. *Acinetobacter baumannii*
B. *Brucella canis*
C. *Chlamydia trachomatis*
D. *Yersinia enterocolitica*

159. *Yersinia pseudotuberculosis* is known to manifest commonly as which of the following clinical conditions?
A. Epiglottitis
B. Hepatitis
C. Mesenteric lymphadenitis
D. Pseudomembranous colitis

160. Cefsulodin-irgasan-novobiocin (CIN) agar is recommended for the recovery of
A. *Brucella suis*
B. *Cardiobacterium hominis*
C. *Rhodococcus equi*
D. *Yersinia enterocolitica*

161. Why are cultures for *Gardnerella vaginalis* not recommended?
A. The bacteria grow so slowly that results take too long to be relevant.
B. Isolation of the bacteria may not be clinically significant.
C. It is unsafe to grow this bacterium.
D. Artificial media are not available.

162. Buffered charcoal yeast extract agar is the recommended medium for the recovery of
A. *Hafnia alvei*
B. *Legionella pneumophila*
C. *Neisseria meningitidis*
D. *Vibrio cholerae*

163. Swimmer's ear, a form of external otitis is commonly caused by
A. *Acinetobacter baumannii*
B. *Bordetella bronchiseptica*
C. *Haemophilus influenzae*
D. *Pseudomonas aeruginosa*

164. A number of vacationers who have traveled outside the U.S. have had their vacations interrupted by a case of "traveler's diarrhea," which is commonly associated with which etiologic agent?
A. *Aeromonas hydrophila*
B. *Escherichia coli*
C. *Proteus mirabilis*
D. *Vibrio parahemolyticus*

165. On Gram stain, a morphology that resembles "seagull wings" is most characteristic of
A. *Campylobacter jejuni*
B. *Neisseria gonorrhoeae*
C. *Plesiomonas shigelloides*
D. *Yersinia pseudotuberculosis*

166. The flattened adjacent sides of the cellular appearance of which microorganism are said to resemble kidney beans?
A. *Aeromonas hydrophila*
B. *Campylobacter jejuni*
C. *Neisseria gonorrhoeae*
D. *Pasteurella multocida*

167. Which of the following is *not* urease positive?
A. *Salmonella*
B. *Helicobacter pylori*
C. *Proteus mirabilis*
D. *Yersinia enterocolitica*

168. Which of the *Neisseria* spp. produces acid from glucose but not from maltose, lactose, or sucrose?
A. *Neisseria gonorrhoeae*
B. *Neisseria lactamica*
C. *Neisseria meningitidis*
D. *Neisseria sicca*

169. Which of the *Neisseria* spp. produces acid from glucose and maltose but not sucrose or lactose?
   A. *Neisseria gonorrhoeae*
   B. *Neisseria lactamica*
   C. *Neisseria meningitidis*
   D. *Neisseria sicca*

170. The causative agent of the septicemic, hemolytic disease known as Oroya fever is
   A. *Bartonella bacilliformis*
   B. *Burkholderia mallei*
   C. *Haemophilus aegyptius*
   D. *Yersinia pestis*

171. A positive gelatin reaction is characteristic of
   A. *Morganella morganii*
   B. *Proteus vulgaris*
   C. *Salmonella* Typhimurium
   D. *Serratia liquefaciens*

172. A positive phenylalanine deaminase reaction is characteristic of
   A. *Hafnia alvei*
   B. *Moraxella catarrhalis*
   C. *Proteus vulgaris*
   D. *Salmonella* Enteritidis

173. Lack of motility is a characteristic of
   A. *Enterobacter*
   B. *Klebsiella*
   C. *Salmonella*
   D. *Serratia*

174. Violet-colored colonies are typically produced by
   A. *Chromobacterium violaceum*
   B. *Chryseobacterium meningosepticum*
   C. *Pseudomonas aeruginosa*
   D. *Serratia marcescens*

175. Which of the following is *not* true about *Pasteurella multocida?*
   A. Most common human infections occur in soft tissues, bones, and joints.
   B. Humans harbor the organism as part of their normal flora.
   C. It is the most virulent of the species in the genus.
   D. It grows on sheep blood agar.

176. Pus was aspirated from an empyema. A Gram stain of the aspirated material showed many white blood cells and numerous gram-negative bacilli. The culture grew many colonies producing a soluble green pigment. The most likely etiologic agent in this case would be
   A. *Chromobacterium violaceum*
   B. *Legionella pneumophila*
   C. *Pseudomonas aeruginosa*
   D. *Serratia marcescens*

177. Some strains of *Serratia marcescens* produce a red-colored pigment. Pigment production is enhanced by
   A. Incubation at 22°C
   B. Incubation at 42°C
   C. Growth on sheep blood agar
   D. Growth on typtic soy agar without blood

178. Which of the following is *not* a correct description regarding *Aeromonas hydrophila?*
   A. Beta-hemolytic
   B. Catalase positive
   C. ONPG negative
   D. Oxidase positive

179. *Edwardsiella tarda* is occasionally isolated in stool specimens and can biochemically be confused with
   A. *Salmonella*
   B. Enterohemorraghic *E. coli*
   C. *Vibrio cholerae*
   D. *Yersinia enterocolitica*

180. The porphyrin test determines an organism's requirement for
A. Cystiene
B. Hemin
C. NAD
D. Thiol

181. Which of the following is an ocasional cause of respiratory tract infections and is rapidly urea positive?
A. *Bordetella bronchiseptica*
B. *Brucella abortus*
C. *Campylobacter fetus*
D. *Escherichia coli*

182. One of the most common etiologic agents of community-acquired uncomplicated cases of cystitis is
A. *Enterobacter aerogenes*
B. *Escherichia coli*
C. *Klebsiella pneumoniae*
D. *Proteus vulgaris*

183. The characteristic growth pattern known as "satelliting" is associated with
A. *Burkholderia pseudomallei*
B. *Campylobacter jejuni*
C. *Haemophilus influenzae*
D. *Yersinia pestis*

184. Colonies that are said to resemble "droplets of mercury" are characteristic of
A. *Bordetella pertussis*
B. *Burkholderia cepacia*
C. *Campylobacter jejuni*
D. *Yersinia pestis*

185. When an epidemiologic survey for the detection of upper respiratory tract carriers of *Neisseria meningitidis* or *Bordetella pertussis* is being conducted, the optimal type of specimen to be obtained for culture is
A. Anterior nares
B. Buccal cavity
C. Nasopharyngeal
D. Throat

186. Chronic carriers, persons who remain infected with an organism for long periods, are typically associated with the dissemination of
A. *Bordetella pertussis*
B. *Campylobacter jejuni*
C. *Salmonella* Typhi
D. *Yersinia pestis*

187. Milk has classically been the primary food associated with the transmission of some diseases, especially for those diseases of cattle transmissible to humans, such as
A. Brucellosis
B. Glanders
C. Meliodosis
D. Pontiac fever

188. Association with faucet aerators and humidifiers used with ventilators in intensive care units is commonly a factor in outbreaks of infections with which of the following microorganisms?
A. *Klebsiella pneumoniae*
B. *Pseudomonas aeruginosa*
C. *Salmonella* spp.
D. *Serratia marcescens*

189. Which of the following is *not* appropriate for the cultivation of *Neisseria gonorrheae?*
A. Chocolate agar
B. Cefsulodin-irgasan-novobiocin agar
C. Martin-Lewis agar
D. Modified Thayer-Martin agar

190. The selective nature of Hektoen enteric agar is due to the inclusion of which one of the following?
A. Bile salts
B. *Bis*-sodium metasulfate
C. Bromthymol blue
D. NaCl

191. For the selective isolation of *Vibrio* spp. the recommended agar is
   A. Thiosulfate-citrate-bile salt-sucrose agar
   B. Charcoal yeast extract agar
   C. Mannitol salt agar
   D. Tinsdale agar

192. When performing the oxidase test, which of the following would *not* be appropriate?
   A. The reagent used is *o*-nitrophenyl-β-D-galactopyranoside.
   B. A nichrome wire loop should be used to acquire inoculum for testing.
   C. Colonies from sheep blood agar can be used.
   D. A positive colony turns dark purple within 10 seconds after application of the reagent.

193. The porphyrin test is most useful for the identification of which of the following?
   A. *Campylobacter*
   B. *Haemophilus*
   C. *Moraxella*
   D. *Neisseria*

194. Which of the following is *not* true about the laboratory diagnosis of pertussis?
   A. Regan-Lowe medium is the recommended medium.
   B. DFA test results are definitive and do not need to be confirmed by culture.
   C. Calcium alginate or Dacron swabs are recommended over cotton-tipped swabs for specimen collection.
   D. Material collected from the nasopharynx is optimal for detection of the causative agent.

195. Reptiles kept as pets are sometimes associated with the transmission of
   A. *Campylobacter*
   B. *Helicobacter*
   C. *Salmonella*
   D. *Vibrio*

196. A young man developed keratitis associated with the use of contact lenses that had been immersed in a contaminated cleaning solution. The most common bacterial etiologic agent in such cases is
   A. *Chryseobacterium meningosepticum*
   B. *Pseudomonas aeruginosa*
   C. *Francisella tularensis*
   D. *Escherichia coli*

197. The organism that is linked to peptic ulcers and is the most frequent cause of gastritis is
   A. *Campylobacter jejuni*
   B. *Helicobacter pylori*
   C. *Salmonella* Typhimurium
   D. *Yersinia enterocolitica*

198. New York City agar was developed for the isolation of
   A. *Bordetella pertussis*
   B. *Campylobacter fetus*
   C. *Haemophilus influenzae*
   D. *Neisseria gonorrhoeae*

199. Besides *Pseudomonas aeruginosa,* which of the following is an important cause of lower respiratory tract infections in patients with cystic fibrosis?
   A. *Actinobacillus actinomycetemcomitans*
   B. *Burkholdia cepacia*
   C. *Chryseobacterium meningosepticum*
   D. *Serratia marcescens*

200. *Eikenella corrodens* is an opportunistic pathogen, but it is most noted for causing
   A. Abscesses of the oral cavity
   B. Pneumonia
   C. Postsurgical wound infections
   D. Urinary tract infections

Mycobacteria

201. Mycobacteria have a large amount of a component in their cell wall that other bacteria lack. That component is
A. Lipids
B. Murein
C. Sterols
D. Teichoic acid

202. The species of *Mycobacterium* that would be most commonly associated with contamination of the hot water system in large institutions such as hospitals is
A. *M. haemophilum*
B. *M. marinum*
C. *M. ulcerans*
D. *M. xenopi*

203. *Mycobacterium fortuitum,* a rapidly growing *Mycobacterium,* grows on MacConkey agar in 5 days. Which other species of *Mycobacterium* is able to demonstrate growth within the same time period on MacConkey agar?
A. *M. bovis*
B. *M. chelonei*
C. *M. kansasii*
D. *M. tuberculosis*

204. A slowly growing, orange-pigmented, acid-fast bacillus was isolated from a cervical lymph node of a child with symptoms of cervical adenitis. The most likely etiologic agent in this case would be
A. *Mycobacterium avium-intracellulare* complex
B. *Mycobacterium chelonei*
C. *Mycobacterium fortuitum*
D. *Mycobacterium scrofulaceum*

205. When clinical specimens are processed for the recovery of *Mycobacterium tuberculosis,* the generally recommended method for digestion and decontamination of the sample is
A. 6% NaOH
B. HCl
C. NALC-NaOH
D. Trisodium phosphate

206. The etiologic agent of Hansen disease is
A. *Mycobacterium bovis*
B. *Mycobacterium fortuitum*
C. *Mycobacterium leprae*
D. *Mycobacterium tuberculosis*

207. The finding of five to six acid-fast bacilli per field (× 800 to × 1000) in a fuchsin smear of expectorated sputum should be reported as
A. 1+
B. 2+
C. 3+
D. 4+

208. Mycobacteria can be examined by using the
A. Dieterle stain
B. Gimenez stain
C. Kinyoun stain
D. Wright's stain

209. Which of the following is *not* characteristic of *Mycobacterium kansasii?*
A. Slow growing
B. Obligate aerobe
C. Nonchromogenic
D. Infections usually acquired by respiratory route

210. Which of the following mycobacteria has an optimal growth temperature 30–32°C?
A. *M. avium* complex
B. *M. bovis*
C. *M. ulcerans*
D. *M. xenopi*

211. Which one of the following drugs is *not* considered as primary antimycobacterial therapy?
A. Isoniazid
B. Kanamycin
C. Rifampin
D. Pyrazinamide

212. In the decontamination and digestion procedure for the isolation of mycobacteria from sputa samples, what is the role of NALC?
A. Bactericidal for normal flora
B. Liquefies mucus
C. Growth stimulant
D. Neutralizes pH to prevent damage to mycobacteria

213. Which one of the following tests would be appropriate in the diagnosis of a mycobacterial infection?
A. Elek test
B. CAMP test
C. Nagler test
D. PPD test

214. Which of the following is associated with livestock and causes a zoonosis?
A. *M. bovis*
B. *M. fortuitum*
C. *M. gordonae*
D. *M. xenopi*

215. Which of the following *Mycobacterium* is most noted for being associated with patients with acquired immunodeficiency syndrome?
A. *M. avium-intracellularae* complex
B. *M. marinum*
C. *M. kansasii*
D. *M. bovis*

216. Which of the following *Mycobacterium* produces an orange pigment and is most commonly recovered from water?
A. *M. intracellulare*
B. *M. gordonae*
C. *M. asiaticum*
D. *M. kansasii*

217. Which of the following *Mycobacterium* appears as buff-colored colonies after exposure to light and is niacin positive?
A. *M. bovis*
B. *M. scrofulaceum*
C. *M. tuberculosis*
D. *M. ulcerans*

218. The *Mycobacterium* that is the etiologic agent of "swimming pool granuloma" is
A. *M. fortuitum*
B. *M. kansasii*
C. *M. marinum*
D. *M. xenopi*

219. Susceptibility to thiophene-2-carboxylic acid hydrazide (T2H) is characteristic of which of the following mycobacteria?
A. *M. avium-intracellularae* complex
B. *M. bovis*
C. *M. kansasii*
D. *M. tuberculosis*

220. Which of the following is a fluorescent stain for mycobacteria?
A. Auromine-rhodamine
B. Calcofluor white
C. Fluorescein isothiocyanate
D. Ziehl-Neelsen

221. The most common photochromogenic *Mycobacterium* isolated in the U.S. is
A. *M. bovis*
B. *M. kansasii*
C. *M. tuberculosis*
D. *M. xenopi*

222. Skin cultures for the recovery of *Mycobacterium* spp. should be incubated at
A. 22°C
B. 30°C
C. 35°C
D. 42°C

223. Which of the following media would *not* be routinely used to culture *Mycobacterium* spp.?
A. Lowenstein-Jensen-Gruft
B. Lowenstein-Jensen
C. Middlebrook 7H11
D. Chocolate agar

224. Which of the following specimens is routinely decontaminated when trying to recover *Mycobacterium* spp.?
A. Sputum
B. Pleural fluid
C. Lung biopsy
D. Cerebrospinal fluid

225. Which of the following is *not* true of *Mycobacterium leprae?*
A. Causes Hansen disease
B. Difficult to grow *in vitro*
C. Easily transmitted from person to person
D. Usually grows in peripheral limbs of infected patients

Anaerobic Bacteria

226. The potentially lethal intoxication type of food poisoning often associated with improperly canned food is caused by
A. *Bacteroides fragilis*
B. *Clostridium botulinum*
C. *Clostridium perfringens*
D. *Clostridium septicum*

227. Which of the following is *not* considered a zoonotic disease?
A. Anthrax
B. Botulism
C. Brucellosis
D. Leptospirosis

228. An anaerobically incubated blood agar plate shows colonies surrounded by an inner zone of complete red cell lysis and an outer zone of incomplete cell lysis (double zone of hemolysis). The most likely presumptive identification of this isolate would be
A. *Clostridium perfringens*
B. *Clostridium tetani*
C. *Fusobacterium nucleatum*
D. *Prevotella melaninogenica*

229. A cervical mucosal abscess specimen was sent to the laboratory for bacteriologic examination. The culture of this sample grew an anaerobic gram-negative bacillus that was inhibited by bile, produced a black pigment, and was negative for indole production and positive for glucose, sucrose, and lactose fermentation. This isolate would most likely be
A. *Bacteroides fragilis* group
B. *Bacteroides ureolyticus*
C. *Porphyromonas gingivalis*
D. *Prevotella melaninogenica*

230. Which one of the following is *not* true of *Clostridium tetani?*
A. It produces rapid tissue necrosis.
B. It is a gram-positive, spore-forming bacillus.
C. Microorganisms in soil contaminate puncture wounds.
D. Disease is caused by an exotoxin acting on the central nervous system.

231. The characteristic colony morphology of *Actinomyces israelii* on solid agar resembles
A. "Medusa head"
B. A molar tooth
C. A fried egg
D. Ground glass

232. What is the predominant indigenous flora of the colon?
   A. Anaerobic, gram-negative, non-spore-forming bacteria
   B. Anaerobic, gram-positive, non-spore-forming bacteria
   C. Aerobic, gram-negative, non-spore-forming bacteria
   D. Aerobic, gram-positive, spore-forming bacteria

233. Obligately anaerobic, gram-negative bacilli, recovered from an abdominal wound, were found to be resistant to penicillin. Growth of this organism was not inhibited by bile. What is the most likely identification of this isolate?
   A. *Bacteroides fragilis* group
   B. *Clostridium septicum*
   C. *Eubacterium lentum*
   D. *Fusobacterium nucleatum*

234. Which of the following is described as obligately anaerobic gram-positive cocci?
   A. *Capnocytophaga*
   B. *Peptostreptococcus*
   C. *Propionibacterium*
   D. *Veillonella*

235. Color Plate 29 ■ shows the filamentous gram-positive rod recovered from an aspirate of a closed chest abscess. It grew only under anaerobic conditions and was not acid-fast. What is the most likely presumptive identification of the isolate seen?
   A. *Actinomyces israelii*
   B. *Bacteroides fragilis*
   C. *Clostridium septicum*
   D. *Propionibacterium acnes*

236. Growth on kanamycin-vancomycin laked blood agar incubated anaerobically is primarily used for
   A. *Bacteroides fragilis*
   B. *Bifidobacterium dentium*
   C. *Clostridium perfringens*
   D. *Peptostreptococcus anaerobius*

237. The diagnosis of pseudomembranous colitis (*Clostridium difficile* associated disease) is often made by
   A. Serology
   B. Culturing blood specimens
   C. Assays to detect toxin in stool
   D. Acid-fast stain of fecal material

238. The anaerobic, gram-negative, curved, motile bacilli associated with bacterial vaginosis belong to the genus
   A. *Actinomyces*
   B. *Bifidobacterium*
   C. *Lactobacillus*
   D. *Mobiluncus*

239. An infant was seen in the emergency department with symptoms of neuromuscular weakness and constipation. The diagnosis of infant botulism was confirmed by the demonstration of toxin in the child's stool. The child most likely contracted this disease by
   A. A scratch wound caused by a cat
   B. Ingestion of spores that germinated in the intestine
   C. A puncture wound with a contaminated household item
   D. Ingestion of preformed toxin found in a contaminated jar of pureed vegetables

240. The majority of the gram-positive, non-spore-forming, anaerobic bacilli isolated from clinical material will be
A. *Bifidobacterium dentium*
B. *Capnocytophagia ochracea*
C. *Eubacterium limosum*
D. *Propionibacterium acnes*

241. Which of the following clostridia has a terminal spore that causes the cell to swell?
A. *C. botulinum*
B. *C. difficile*
C. *C. perfringens*
D. *C. tetani*

242. The gram-negative, non-spore-forming, anaerobic bacillus frequently implicated in serious clinical infections such as brain and lung abscesses is
A. *Bacteroides urealyticus*
B. *Eubacterium lentum*
C. *Fusobacterium nucleatum*
D. *Peptostreptococcus anaerobius*

243. Which one of the following is a non-spore-forming, gram-positive, anaerobic bacillus?
A. *Clostridium*
B. *Fusobacterium*
C. *Propionibacterium*
D. *Veillonella*

244. Which of the following statements is *not* characteristic of *Clostridium botulinum?*
A. Infant botulism is the most common clinical form.
B. Pathogenicity is related to a potent neurotoxin.
C. Oval spores are located terminally.
D. Of the seven toxogenic types, types A, B, E, and F are associated with human botulism.

245. A tube of semisolid medium that contains resazurin appears pink. What does this indicate?
A. Acid environment
B. Alkaline environment
C. Motility
D. Presence of oxygen

246. Identify the *Fusobacterium* sp. considered to be the most frequent isolate recovered from clinical infections.
A. *F. varium*
B. *F. nucleatum*
C. *F. mortiferum*
D. *F. necrophorum*

247. Septicemia caused by which of the following is generally associated with an underlying malignancy?
A. *Bifidobacterium dentium*
B. *Clostridium septicum*
C. *Eubacterium lentum*
D. *Lactobacillus catenaforme*

248. Which of the following is the most potent bacterial exotoxin known?
A. Botulinum toxin
B. Erythrogenic toxin
C. *C. difficile* toxin B
D. *C. perfringens* alpha-toxin

249. Which *Clostridium* sp. is most commonly recovered from cases of gas gangrene?
A. *C. bifermentans*
B. *C. perfringens*
C. *C. sordellii*
D. *C. difficile*

250. Which of the following organisms is *not* gram-positive?
A. *Eubacterium lentum*
B. *Bifidobacterium dentium*
C. *Propionibacterium acnes*
D. *Suttonella wadsworthensis*

251. Which of the following statements is *not* true of clostridia?
   A. Pseudomembranous colitis is due to a toxin produced by *C. difficile*.
   B. Clinically significant clostridia are found in the normal flora of the colon and in the soil.
   C. Botulism is caused by ingesting preformed toxin and can be prevented by boiling food prior to eating.
   D. *C. tetani* spores will form in the presence of oxygen; therefore, anaerobiosis in a wound is not required to cause tetanus.

252. Gram-positive bacilli with central spores are seen in the direct Gram stain of a tissue biopsy. After 24 hours of incubation, no growth is seen on the sheep blood agar plate incubated aerobically and the chocolate agar plate incubated in increased CO_2. Which of the following is the likely cause of the infection?
   A. *Bacillus*
   B. *Clostrdium*
   C. *Lactobacillus*
   D. *Prevotella*

253. Small alpha-hemolytic colonies are seen on a sheep blood agar plate after 48 hours of incubation on a vaginal culture. This describes which of the following?
   A. *Bacteroides*
   B. *Clostridium*
   C. *Lactobacillus*
   D. *Porphyromonas*

254. Which of the following statements is true regarding *Clostridium perfringens?*
   A. There are five serologic types.
   B. Spores are terminally located.
   C. Alpha-toxin is produced by all strains.
   D. Spores are readily seen in laboratory media.

255. Which is a correct statement regarding *Clostridium tetani?*
   A. It is proteolytic
   B. It is lecithinase positive
   C. It is characteristically nonmotile
   D. It produces terminal spores

256. Which anaerobic, gram-negative rod can be presumptively identified by its Gram stain morphology, and inhibition by bile and a 1-μg kanamycin disk?
   A. *Bacteroides fragilis* group
   B. *Eubacterium lentum*
   C. *Fusobacterium nucleatum*
   D. *Porphyromonas gingivalis*

257. Which of the following tests is most appropriate for the presumptive identification of *Prevotella melaninogenica?*
   A. SPS sensitivity test
   B. Nagler test
   C. Cytotoxin assay
   D. Fluorescence test

258. Which of the following tests is most appropriate for the identification of *Clostridium difficile?*
   A. SPS sensitivity test
   B. Nagler test
   C. Cytotoxin assay
   D. Fluorescence test

259. Which of the following tests is most appropriate for the presumptive identification of *Clostridium perfringens?*
   A. SPS sensitivity test
   B. Reverse CAMP test
   C. Cytotoxin assay
   D. Esculin hydrolysis

260. Which of the following tests is most appropriate for the presumptive identification of *Peptostreptococcus anaerobius?*
A. SPS disk
B. Colistin disk
C. Kanamycin disk
D. Vancomycin disk

261. A curved appearance on Gram stain is characteristic of which of the following?
A. *Actinomyces israelii*
B. *Clostridium septicum*
C. *Fusobacterium nucleatum*
D. *Propionibacterium acnes*

262. Purulent material from a cerebral abscess was submitted to the laboratory for smear and culture. On direct Gram stain, gram-positive cocci in chains and gram-negative bacilli with pointed ends were seen. Plates incubated aerobically exhibited no growth at 24 hours. On the basis of the organisms seen on the smear, what is the most likely presumptive identification of the etiologic anaerobic agents?
A. *Veillonella* sp. and *Clostridium* sp.
B. *Eubacterium* sp. and *Veillonella* sp.
C. *Peptostreptococcus* sp. and *Nocardia* sp.
D. *Fusobacterium* sp. and *Peptostreptococcus* sp.

263. Which of the following is an important virulence factor of *Bacteroides fragilis?*
A. Endotoxin
B. Exotoxins
C. Polysaccharide capsule
D. Protease

264. Which of the following is *not* true of *Bacteroides fragilis?*
A. Lipase and lecithinase negative
B. Anaerobic gram-negative bacillus
C. Commonly associated with intra-abdominal infections
D. Among the most antimicrobial-sensitive anaerobic bacteria

265. Aspiration of vomitus can lead to pneumonia. Which of the following would *not* be a likely causative agent in aspiration pneumonia?
A. *Bacteroides gracilis*
B. *Mobiluncus* sp.
C. *Porphyromonas* sp.
D. *Prevotella melaninogenica*

266. To ensure that anaerobic conditions have been achieved in anaerobic jars or chambers, an oxygen-sensitive indicator is employed, such as
A. Bromcreosol purple
B. Methylene blue
C. Methyl red
D. Phenol red

267. Egg yolk agar showing a precipitate in the medium surrounding the colony is positive for
A. Lecithinase production
B. Lipase production
C. Protease activity
D. Starch hydrolysis

268. After 72 hours of anaerobic incubation, small olive-green to black colonies are seen. A Gram stain reveals gram-postive cocci. What is the most likely identification of this organism?
A. *Fingoldia magna*
B. *Peptococcus niger*
C. *Peptostreptococcus anaerobius*
D. *Veillonella parvula*

269. Which of the following is an important cause of food poisoning?
A. *Bacteroides fragilis*
B. *Bacteroides ureolyticus*
C. *Clostridium perfringens*
D. *Clostridium histolyticum*

270. Which bacterium is part of the normal vaginal flora that helps resist the onset of bacterial vaginosis?
   A. *Peptostreptococcus* sp.
   B. *Peptococcus* sp.
   C. *Lactobacillus* sp.
   D. *Mobiluncus* sp.

Chlamydia, Rickettsia, and *Mycoplasma*

271. *Chlamydia trachomatis* causes which of the following?
   A. Rat-bite fever
   B. Inclusion conjunctivitis
   C. A skin disease found predominantly in tropical areas
   D. Zoonosis in birds and parrot fever in humans

272. Which one of the following microorganisms *cannot* be cultivated on artificial cell-free media?
   A. *Chlamydia trachomatis*
   B. *Mycoplasma hominis*
   C. *Mycoplasma pneumoniae*
   D. *Ureaplasma urealyticum*

273. The etiologic agent of primary atypical pneumonia is
   A. *Chlamydia trachomatis*
   B. *Chlamydiophila psittaci*
   C. *Mycoplasma pneumoniae*
   D. *Ureaplasma urealyticum*

274. The recommended medium for the recovery of *Mycoplasma pneumoniae* from clinical specimens is
   A. Charcoal yeast extract medium
   B. Fletcher semisolid medium
   C. Middlebrook
   D. SP4 agar

275. *Chlamydophila (Chlamydia) psittaci* infections in humans most commonly result after exposure to infected
   A. Amphibians
   B. Arthropods
   C. Avians
   D. Mammalians

276. Which of the following is *not* true of *Coxiella burnetii?*
   A. It is an obligate intracellular parasite.
   B. It is transmitted from animals to humans by inhalation.
   C. A rash appears first on the extremities and then on the trunk.
   D. Is the etiologic agent of Q fever, which may be acute or chronic.

277. Which of the following is *true* about mycoplasmas?
   A. Resistant to penicillin
   B. Not able to survive extracellularly
   C. Easily stained using the Gram stain
   D. Grow on routine nonselective culture media

278. Corneal scrapings are collected and examined microscopically using a direct fluorescent test to detect inclusion bodies for the diagnosis of infection caused by
   A. *Chlamydia trachomatis*
   B. *Ehrlichia chaffeensis*
   C. *Mycoplasma hominis*
   D. *Rickettsia prowazekii*

279. Which of the following *Mycoplasmataceae* has *not* been connected with human genital infections?
   A. *Mycoplasma genitalium*
   B. *Mycoplasma hominis*
   C. *Mycoplasma pneumoniae*
   D. *Ureaplasma urealyticum*

280. Which of the following is *not* true about *Chlamydophila (Chlamydia) pneumoniae?*
   A. Common agent of lower respiratory tract infection
   B. Humans become infected from animal reservoirs.
   C. Tetracycline and erythromycin are effective treatments.
   D. Research has found an association with artherosclerosis.

281. What is the reference method for detection of *Chlamydia trachomatis* in cases of sexually transmitted disease?
   A. Tissue culture
   B. Nonculture EIA methods
   C. DNA-amplification techniques
   D. Culture on modified Thayer-Martin agar

282. Colonies said to have the appearance of a "fried egg" are characteristic of
   A. *Ehrlichia chaffeensis*
   B. *Mycoplasma genitalium*
   C. *Mycoplasma hominis*
   D. *Ureaplasma urealyticum*

283. Human infection with the causative agent of Q fever is acquired by
   A. Inhalation of infectious material
   B. The bite of a mite (chigger)
   C. The bite of a body louse
   D. The bite of the arthropod *Phlebotomus*

284. For nonspecific staining of *Rickettsia* the recommended stain is
   A. Gimenez stain
   B. Gomori silver stain
   C. Gram stain
   D. Kinyoun stain

285. Rocky Mountain spotted fever is transmitted by the bite of a tick infected with
   A. *Rickettsia akari*
   B. *Rickettsia conorii*
   C. *Rickettsia prowazekii*
   D. *Rickettsia rickettsii*

286. Transmission of the sylvatic form of typhus infection caused by *Rickettsia prowazekii* is associated with
   A. Bats
   B. Rabbits
   C. Raccoons
   D. Squirrels

287. The mild type of typhus fever that is caused by recrudescence of an initial attack of epidemic typhus is known as
   A. Brill-Zinsser disease
   B. Q fever
   C. São Paulo typhus
   D. Tsutsugamushi disease

288. The causative agent of endemic or murine typhus is
   A. *Rickettsia akari*
   B. *Rickettsia conorii*
   C. *Rickettsia prowazekii*
   D. *Rickettsia typhi*

289. A screening test for the identification of *Mycoplasma pneumoniae* isolates is the
   A. Acid-fast stain
   B. Gram stain
   C. Lysis of red blood cells
   D. Catalase test

290. A genital specimen is inoculated into 10 B broth. After overnight incubation, an alkaline reaction is noted without turbidity. What is the most likely explanation?
   A. pH change due to molecules in the clinical specimen
   B. Presence of *Mycoplasma genitalium*
   C. Presence of *Mycoplasma hominis*
   D. Presence of *Ureaplasma urealyticum*

Spirochetes

291. Detection of antibody against cardiolipin is useful for the diagnosis of which of the following diseases?
   A. Leptospirosis
   B. Lyme disease
   C. Relapsing fever
   D. Syphilis

292. During the first week of leptospirosis, the most reliable way to detect the presence of the causative agent is by the direct
   A. Culturing of blood
   B. Culturing of urine
   C. Examination of blood
   D. Examination of cerebrospinal fluid

293. Serious congenital infections are associated with
   A. *Borrelia burgdorferi*
   B. *Borrelia recurrentis*
   C. *Treponema pallidum* subsp. *pallidum*
   D. *Treponema pallidum* subsp. *pertenue*

294. A helicoidal, flexible organism was demonstrated in a blood smear. This motile organism was approximately 12 μm long, approximately 0.1 μm wide, and had semicircular hooked ends. The description of this organism corresponds most closely to the morphology of
   A. *Borrelia*
   B. *Leptonema*
   C. *Leptospira*
   D. *Treponema*

295. The etiologic agent of epidemic relapsing fever is *Borrelia recurrentis,* which is commonly transmitted by
   A. Fleas
   B. Lice
   C. Mosquitoes
   D. Ticks

296. Which of the following is *not* true of the VDRL test?
   A. False-positive tests are more frequent than with the FTA-ABS test.
   B. The antibody titer will decline if the patient is adequately treated.
   C. Inactivated *Treponema pallidum* serves as the antigen.
   D. The test is usually positive in secondary syphilis.

297. *Borrelia burgdorferi,* a spirochete transmitted by *Ixodes dammini* in the northeastern U.S., is the etiologic agent of
   A. Lyme disease
   B. Rat-bite fever
   C. Relapsing fever
   D. Q fever

298. The axial fibrils of spirochetes most closely resemble which bacterial structure?
   A. Cytoplasmic membrance
   B. Flagellum
   C. Pilus
   D. Sporangium

299. Which of the following is *not* correct regarding spirochetes?
   A. Motility is via axial filaments.
   B. Spirochetes are gram positive.
   C. They are visualized best using dark field or phase optics.
   D. Those associated with human disease are 0.1–0.5 μm in diameter and 5–30 μm in length

300. A positive VDRL test for syphilis was reported on a young woman known to have hepatitis. When questioned by her physician, she denied sexual contact with any partner symptomatic for a sexually transmitted disease. Which of the following would be the appropriate next step for her physician?
A. Treat her with penicillin
B. Identify her sexual contacts for serologic testing
C. Test her serum using a fluorescent treponemal antibody-absorbed assay
D. Reassure her that it was a biologic false-positive caused by her liver disease

Antimicrobial Agents and Antimicrobial Susceptibility Testing

301. A suspension of the test organism for use in broth dilution and disk diffusion testing is adjusted to match the turbidity of a
A. #0.5 McFarland standard
B. #1.0 McFarland standard
C. #2.0 McFarland standard
D. #3.0 McFarland standard

302. When testing the antimicrobial susceptibility of *Haemophilus influenzae* strains by disk-agar diffusion, the recommended medium is
A. Chocolate agar
B. Charcoal yeast extract agar
C. Mueller-Hinton base supplemented with 5% sheep blood
D. Mueller-Hinton base supplemented with hematin, NAD, and yeast extract

303. The chemotherapeutic agents structurally similar to the vitamin *p*-aminobenzoic acid that act by inhibiting bacteria via inhibition of folic acid synthesis are
A. Aminoglycosides
B. Penicillins
C. Macrolides
D. Sulfonamides

304. Which of the following organisms would *not* routinely undergo antimicrobial susceptibiity testing?
A. *Escherichia coli*
B. *Proteus mirabilis*
C. *Staphylococcus aureus*
D. *Streptococcus pyogenes*

305. Penicillin is active against bacteria by
A. Inhibition of protein synthesis at the 30S ribosomal subunit
B. Reduction of dihydrofolic acid
C. Inhibition or peptidoglycan synthesis
D. Inhibition of nucleic acid function

306. The minimum bactericidal concentration (MBC) of an antimicrobial agent is defined as the lowest concentration of that antimicrobial agent that kills at least _____ of the original inoculum.
A. 95.5%
B. 97%
C. 99.9%
D. 100%

307. Resistance to clindamycin can be induced *in vitro* by
A. Ampicillin
B. Erythromycin
C. Gentamicin
D. Penicillin

308. The term that denotes a situation in which the effect of two antimicrobial agents together is greater than the sum of the effects of either drug alone is
A. Additivism
B. Antagonism
C. Sensitivity
D. Synergism

309. Beta-lactamase-producing strains of *Haemophilus influenzae* are resistant to
A. Chloramphenicol
B. Erythromycin
C. Penicillin
D. Trimethoprim sulfamethoxazole

310. The agar recommended by the Clinical and Laboratory Standards Institute for routine susceptibility testing of nonfastidious bacteria is
A. MacConkey agar
B. Middlebrook 7H10 agar
C. Mueller-Hinton agar
D. Trypticase soy agar

311. The pH of the agar used for the Kirby-Bauer test should be
A. 7.0–7.2
B. 7.2–7.4
C. 7.4–7.6
D. 7.6–7.8

312. Which drug known to be active against parasitic infections has importance as a therapeutic agent in cases of disease caused by anaerobic bacteria?
A. Isoniazid
B. Metronidazole
C. Rifampin
D. Trimethoprim

313. An example of a bactericidal antibiotic is
A. Chloramphenicol
B. Erythromycin
C. Tetracycline
D. Tobramycin

314. The extended-spectrum beta-lactamases confer resistance to
A. Amoxicillin
B. Ceftriaxone
C. Erythromycin
D. Rifampin

315. Which of the following media should be used for *in vitro* susceptibility testing of *S. pneumoniae?*
A. Chocolate agar
B. Charcoal yeast extract agar
C. Mueller-Hinton base supplemented with 5% lysed horse blood
D. Mueller-Hinton base supplemented with 1% hemoglobin and 1% IsoVitaleX

316. Rapid testing for beta-lactamase production is recommended, before initiation of antimicrobial therapy, for isolates of
A. *Serratia marcescens*
B. *Haemophilus influenzae*
C. *Staphylococcus epidermidis*
D. *Streptococcus pyogenes*

317. The phenomenon of bacterial resistance to the bactericidal activity of penicillins and cephalosporins, with only inhibition of the organism's growth, is known as
A. High-level resistance
B. Intrinsic resistance
C. Inducible resistance
D. Tolerance

318. The supervisor of a microbiology laboratory has been asked to begin performing *in vitro* antimicrobial susceptibility testing of *Mycobacterium tuberculosis* because of an increase in the reported resistance in his community. Which of the following methods would be appropriate for this testing?
    A. Broth microdilution method using Mueller-Hinton broth
    B. Kirby-Bauer method
    C. BACTEC method
    D. Schlichter method

319. Clavulanic acid is classified as a
    A. Beta-lactam
    B. Beta-lactamase inhibitor
    C. Macrolide
    D. Aminoglycoside

320. Which of the following antimicrobial agents acts by inhibiting cell wall synthesis?
    A. Clindamycin
    B. Gentamicin
    C. Naladixic acid
    D. Vancomycin

321. Metronidazole is most commonly recommended for treatment of infections caused by
    A. Aerobic microorganisms
    B. Microaerophilic microorganisms
    C. Obligate anaerobic microorganisms
    D. Obligate intracellular microorganisms

322. Which of the following antimicrobial agents acts by inhibiting protein synthesis?
    A. Gentamicin
    B. Methicillin
    C. Rifampin
    D. Ampicillin

323. Chloramphenicol is an important antimicrobial agent for the treatment of meningitis as well as several other serious infections. Unfortunately, chloramphenicol exhibits significant complications that limit its clinical usefulness. These effects include
    A. Allergic reactions and anaphylaxis
    B. Bone marrow suppression and aplastic anemia
    C. Significant gastrointestinal manifestations
    D. Photosensitivity

324. Which of the following is *not* one of the standard control organisms used for the weekly testing of antimicrobial disks?
    A. *Staphylococcus epidermidis* (ATCC 25833)
    B. *Pseudomonas aeruginosa* (ATCC 27853)
    C. *Escherichia coli* (ATCC 25922)
    D. *Enterococcus faecalis* (ATCC 29212)

325. When using the rapid chromogenic cephalosporin method for the detection of beta-lactamase production by an organism, a positive test is indicated by the color
    A. Yellow
    B. Green
    C. Red
    D. Blue

Procedures and Biochemical Identification of Bacteria

326. Which of the following body sites is *not* normally colonized by large numbers of normal flora organisms?
    A. Colon
    B. Skin
    C. Lungs
    D. Vagina

327. During childbearing years, the normal flora of the vagina is predominantly
A. *Enterococcus*
B. *Lactobacillus*
C. *Propionibacterium*
D. Coagulase-negative *Staphylococcus*

328. The MIDI identification system is based on
A. Antibiograms
B. Colony pigment on ChromAgar
C. Fatty acid analysis
D. Multiple biochemical tests

329. Which of the following is *not* a correct statement regarding blood cultures?
A. Collection of 10–20 mL per culture for adults is recommended
B. Two or three blood cultures are recommended as optimum
C. Volume of blood cultured is more criticial than timing of culture
D. Blood drawn for culture may be allowed to clot

330. In capnophilic incubators, carbon dioxide concentrations should be maintained between
A. 1% and 5%
B. 5% and 10%
C. 10% and 15%
D. 15% and 20%

331. The recommended anticoagulant for use when a body fluid or joint fluid that may clot is sent for microbiologic examination is
A. Heparin
B. Sodium polyethanolsulfonate (SPS)
C. Sodium EDTA
D. Sodium citrate

332. Suggested quality control organisms for Simmons citrate agar slants are
A. *Shigella sonnei; Escherichia coli*
B. *Klebsiella pneumoniae; Escherichia coli*
C. *Escherichia coli; Edwardsiella tarda*
D. *Morganella morganii; Edwardsiella tarda*

333. Laboratory professionals are at risk for disease transmission. The majority of cases of laboratory-related infections are associated with
A. Infectious aerosols
B. Contamination of abraded skin
C. Puncture wounds
D. Person-to-person transmission

334. The ability of a microorganism to deaminate phenylalanine can be assessed by inoculating a phenylalanine agar slant with the test organism. Following incubation, if the organism is positive, a green color develops with the addition of
A. 10% ferric chloride
B. 2% sulfanilamide
C. 2 N sodium carbonate
D. 5% alpha-naphthol

335. Which of the following media is both selective and differential?
A. Sheep blood agar
B. Chocolate agar
C. Mannitol salt agar
D. Mueller-Hinton agar

336. The paper strip test for the demonstration of hydrogen sulfide production is impregnated with a solution of
A. Sodium desoxycholate
B. Lead acetate
C. Potassium tellurite
D. Sodium thiosulfate

337. In the nitrate test, reduction is demonstrated by the development of a red color, following the addition of
   A. Alpha-naphthol and potassium hydroxide
   B. *p*-Dimethylaminobenzaldehyde and amyl alcohol
   C. Ninhydrin and acetone
   D. Alpha-naphthylamine and sulfanilic acid

338. The Moeller test for the detection of decarboxylase activity is dependent upon
   A. An alkaline pH shift in the medium
   B. The oxidation of gluconate
   C. An acid pH basal medium
   D. A deamination of tryptophan

339. The medium used to determine whether an organism is oxidative or fermentative with respect to its metabolic activities is
   A. CTA medium
   B. OF medium
   C. HE medium
   D. XLD medium

340. Methyl red-Voges Proskauer (MRVP) broth is inoculated and incubated for 48 hours. What two reagents must be added to determine if the bacterium is VP positive?
   A. Creatine and 1 N HCl
   B. 10% $FeCl_3$ and alpha-naphthol
   C. Kovac's reagent and zinc dust
   D. Alpha-naphthol and 40% KOH

341. Which of the following is *not* true of blood cultures for the recovery of bacteria?
   A. No more than three cultures should be drawn in 1 day.
   B. Cultures should be drawn before the expected fever spike.
   C. Cultures should be incubated aerobically and anaerobically.
   D. Collect 5 mL of blood for optimal recovery of pathogen.

342. Continuous blood culture systems that detect changes in headspace pressure in bottles are refered to as
   A. Fluorescence methods
   B. ^{14}C detection methods
   C. Colorimetry methods
   D. Manometric methods

343. Which of the following statements does *not* apply to the acridine orange stain?
   A. Binds to the teichoic acid of the cell wall
   B. Requires the use of a fluorescence microscope
   C. Is more sensitive than the Gram stain
   D. Is recommended for fluid and exudates with low bacterial concentrations

344. In the catalase test, the formation of bubbles is due to
   A. Production of oxygen (O_2)
   B. Production of hydrogen gas (H_2)
   C. Breakdown of water
   D. Oxidation of hydrogen peroxide

345. Which of the following would *not* be appropriate for a positive control in the PYR test?
   A. *Staphylococcus aureus*
   B. *Staphylococcus lugdunensis*
   C. *Streptococcus pyogenes*
   D. *Enterococcus faecalis*

Case Studies

346. A purulent aspirate of joint fluid from a 28-year-old female with joint pain was sent for microbiologic examination. The Gram stain of this sample revealed many polymorphonuclear cells with intracellular and extracellular gram-negative diplococci. Given the specimen type and microscopic findings, the appropriate selective medium for primary isolation would be
    A. Mannitol salt agar
    B. Potassium tellurite agar
    C. Modified Thayer-Martin agar
    D. Cefsulodin-irgasan-novobiocin

347. A 21-year-old sexually active woman came to the university student health service with a 2-day history of urinary frequency with urgency, dysuria, and hematuria. She had no history of prior urinary tract infection. Laboratory test showed a white blood cell count of 10×10^9/L. The urine sediment contained innumerable white cells. Cultures yielded more than 10^5 colony-forming units/mL of a lactose-fermenting gram-negative rod. The most likely etiologic agent in this case is
    A. *Escherichia coli*
    B. *Klebsiella pneumoniae*
    C. *Morganella morganii*
    D. *Proteus mirabilis*

348. A 36-year-old man was seen in the emergency department. He complained of fever and headache. He had returned 1 week previously from a 6-week visit to a village in India. Among the differential diagnoses was typhoid fever. What is the most critical laboratory test necessary to establish or eliminate the diagnosis?
    A. Blood cultures
    B. Sputum cultures
    C. Stool cultures
    D. Urine cultures

349. A 45-year-old man was seen in the emergency department with fever, chills, nausea, and myalgia. He reported that 2 days earlier he had eaten raw oysters at a popular seafood restaurant. On admission he was febrile and had hemorrhagic, fluid-filled bullous lesions on his left leg. The patient had a history of diabetes mellitus, chronic hepatitis B, and heavy alcohol consumption. The patient, who had a temperature of 102.2°F, was admitted to the intensive care unit for presumed sepsis, and treatment was begun. A curved gram-negative rod was isolated from blood cultures drawn on admission and fluid from the bullous leg wound. On the third day, disseminated intravascular coagulation developed, and he died. The source of the oysters eaten by the deceased patient was the Gulf of Mexico. The most likely etiologic agent in this case would be
    A. *Aeromonas hydrophila*
    B. *Plesiomonas shigelloides*
    C. *Vibrio vulnificus*
    D. *Yersinia enterocolitica*

350. A 3-year-old was brought to the emergency department by her parents. She had been febrile with a loss of appetite for the past 24 hours. Most recently the parents noted that it was difficult to arouse her. She attended a daycare center, and her childhood immunizations were current. On examination she demonstrated a positive Brudzinski sign indicative of meningeal irritation. Cultures of blood and cerebrospinal fluid (CSF) were sent to the laboratory. Her CSF was cloudy, and the Gram stain showed many polymorphonuclear cells containing gram-negative diplococci. The white blood cell count was 25×10^9/L, with 88% polymorphonuclear cells. The CSF protein was 100 mg/dL, and the glucose was 15 mg/dL. Cultures of the blood and CSF grew the same organism. The most likely etiologic agent in this case is
A. *Haemophilus influenzae*
B. *Listeria monocytogenes*
C. *Moraxella catarrhalis*
D. *Neisseria meningitidis*

351. Gram-positive rods were recovered from the chest fluid drawn from a teenager with right lower lobe pneumonia who lived on a dairy farm. At 24 hours, pinpoint colonies grew on sheep blood agar that showed faint zones of beta-hemoloysis. The isolate was catalase negative and demonstrated a positive CAMP test. Which of the following is the most likely etiologic agent in this case?
A. *Listeria monocytogenes*
B. *Streptococcus agalactiae*
C. *Arcanobacterium pyogenes*
D. *Streptobacillus moniliformis*

352. A young woman complaining of symptoms of sudden onset of fever, vomiting, diarrhea, and rash was seen by her gynecologist. She was admitted to the hospital, where a culture of vaginal discharge grew many coagulase-positive staphylococci. The most likely diagnosis in this case would be
A. Kawasaki disease
B. Pelvic inflammatory disease
C. Scalded skin syndrome
D. Toxic shock syndrome

353. A 32-year-old male was seen in the emergency department with symptoms of lower right quadrant abdominal pain and diarrhea. A complete blood count showed a leukocytosis with an increased number of neutrophils. He was admitted, and a stool culture was obtained. The culture showed many gram-negative bacilli, which were oxidase negative, citrate negative, and indole negative. The triple sugar iron reaction was acid over acid, but there was no evidence of gas or H_2S production. The organism was positive for urease and ONPG and negative for phenylalanine. The characteristic symptomatology and the biochemical reactions confirmed that the etiologic agent was
A. *Salmonella* Paratyphi
B. *Shigella dysenteriae*
C. *Vibrio parahaemolyticus*
D. *Yersinia enterocolitica*

354. In August, a patient presented at a community hospital in New England with symptoms of a skin rash, headache, stiff neck, muscle aches, and swollen lymph nodes. A silver-stained biopsy of a skin lesion showed spirochetes. On the basis of the clinical syndrome and laboratory detection of a causative agent, the patient was diagnosed as having
A. Lyme disease
B. Plague
C. Rabbit fever
D. Relapsing fever

355. Several international participants in an Eco-Challenge adventure race in Borneo became ill with symptoms of chills, diarrhea, headaches, and eye infections. The racers hiked in the mountains and jungles, swam in rivers, and slogged through flooded streams for 2 weeks. Contact with contaminated water and soil during the race was highly associated with illness. What is the most likely etiologic agent in this case?
A. *Borellia recurrentis*
B. *Brucella canis*
C. *Franciscella tularensis*
D. *Leptospira interrogans*

356. An anemic patient was transfused with packed red blood cells. Approximately 1 hour after the transfusion began, the patient developed fever and hypotension consistent with endotoxic shock. The red blood cells had been stored at 4°C for approximately 30 days before their use. The organism most likely to be involved in this case is
A. *Campylobacter fetus*
B. *Neisseria meningitidis*
C. *Pseudomonas aeruginosa*
D. *Yersinia enterocolitica*

357. A college student got a summer job working at a marina. While repairing the outboard motor on a rental boat, he received several lacerations on his right forearm. No medical treatment was sought at the time of the injury, but after several weeks he noted that the lesions were not healing and he sought the opinion of his physician. A biopsy of one of the lesions revealed a cutaneous granulomatous condition. Given the history, which of the following microorganisms would most likely be the etiologic agent in this case?
A. *Mycobacterium marinum*
B. *Nocardia asteroides*
C. *Pseudomonas aeruginosa*
D. *Vibrio vulnificus*

358. A woman, who had recently returned from a vacation in Mexico, was admitted to the hospital. She was febrile and complained of flulike symptoms. Her case history revealed that she had eaten cheese that had been made from unpasteurized milk while on vacation. The most likely etiologic agent in this case would be
A. *Bordetella pertussis*
B. *Listeria monocytogenes*
C. *Staphylococcus aureus*
D. *Yersinia enterocolitica*

359. A 7-year-old female became ill with an intestinal illness after visiting a petting zoo featuring farm animals such as calves, lambs, and chickens. She had bloody diarrhea and went on to develop hemolytic uremic syndrome. The most likely etiologic agent in this case is
A. *Eschericia coli* 0157:H7
B. *Shigella dysenteriae*
C. *Vibrio cholerae* 01
D. *Vibrio cholerae* non01

360. A middle-aged man with a history of smoking and drinking for over 40 years developed shortness of breath, fever, frontal headache, diarrhea, and cough. He worked in the produce section of a supermarket, which routinely misted the fresh greens. His medical history included a kidney transplant several years ago for which he remains on antirejection therapy. His sputum Gram stain showed numerous polymorphonuclear cells but rare microorganisms. An X-ray of his chest showed an infiltrate in the left lower lobe, and a diagnosis of atypical pneumonia was made. Which of the following is the most likely etiologic agent in this case?
A. *Bordetella pertussis*
B. *Klebsiella pneumoniae*
C. *Legionella pneumophila*
D. *Moraxella catarrhalis*

answers & rationales

Aerobic Gram-Positive Bacteria

1.

C. *Enterococcus* and other group D streptococci can be presumptively identified based on their ability to hydrolyze esculin in the presence of 1–4% bile salts. The medium is made selective for enterococci by the addition of either sodium azide or 4% bile salts. Organisms able to grow on this medium and hydrolyze esculin produce esculetin, which reacts with an iron salt to form a black color in the agar.

2.

D. Strains of *Corynebacterium diphtheriae* infected by a lysogenic bacteriophage produce an extremely potent exotoxin. Absorption of the toxin may cause a rapidly fatal hypertoxic disease characterized by myocarditis and neuritis. This disease most commonly affects children aged 1 to 10 years. Transmission is by contact with a human carrier or with contaminated fomites.

3.

B. Staphylococcal enterocolitis food poisoning cases result from the ingestion of contaminated foods containing preformed thermostable enterotoxin. This form of intoxication causes a perfuse and watery diarrhea due to the loss of electrolytes and fluids into the lumen. In many cases, the causative agent may never be recovered from patient specimens.

4.

B. Nutritionally variant streptococci (NVS) are now termed *Abiotrophia*. These clinically significant microorganisms, which account for 5–6% of the cases of endocarditis, are frequently not able to be recovered because of insufficient quantities of vitamin B_6 in the culture medium. The routine use of a pyridoxal disk, a streak of *Staphylococcus,* or vitamin B_6–supplemented culture media is required for isolation.

5.

C. Scalded skin syndrome is a form of dermatitis produced by strains of *Staphylococcus aureus* that elaborate exfoliative toxin. Two types of this toxin have been identified: exfoliation A and exfoliation B. This potent toxin acts by disturbing the adhesive forces between cells of the stratum granulosum, which causes the appearance of the clear, large, flaccid bullae and the skin to peel off. Infants and children are most commonly affected with this form of dermatitis, beginning about the face and trunk and subsequently spreading to the extremities.

6.

A. Presumptive identification of group A streptococci can be achieved through the PYR (L-pyrrolidonyl-β-naphthlylamide) disk test. The use of a 0.04-unit bacitracin disk is no longer recommended because groups C and G streptococci are also susceptible to this agent. A positive test result is interpreted as a bright red color change within 5 minutes.

7.

A. Staphylococci and micrococci are both catalase positive gram-positive cocci. Staphylococci are more clinically significant, so it is important to differentiate *Micrococcus* from *Staphylococcus*. Micrococci are modified oxidase positive, whereas staphylococci are negative.

8.

B. Nocardiosis is characterized by mycetoma or chronic suppurative infection. Draining sinus tracts in the subcutaneous tissue are a common manifestation of the disease. *Nocardia* spp. are soil saprophytes that may produce disease in humans either by the inhalation of contaminated material or through skin abrasions. Microscopic examination of pus from suspected cases will demonstrate partially acid-fast, gram-positive, branching filamentous or coccoid organism.

9.

A. Infection caused by *Erysipelothrix rhusiopathiae* in humans is primarily erysipeloid. Erysipeloid is usually the result of contact with an infected animal or contaminated animal product. The characteristic presentation is cutaneous spreading lesions of the fingers or hand that are raised and erythematous. Although generally confined to the skin, *E. rhusiopathiae* has been implicated in rare cases of endocarditis.

10.

A. *Streptococcus agalactiae* isolates can be presumptively identified by the demonstration of a positive CAMP reaction. CAMP is an acronym for the scientists (Christie, Atkins, Munch-Petersen) who discovered this phenomenom. Group B streptococci elaborate the CAMP factor, which acts to enhance the zone of hemolysis produced by beta-lysin-producing strains of *Staphylococcus aureus*. Incubation of test plates should be carried out in ambient air, because increased CO_2 and anaerobic incubation increase the rate of false-positive CAMP reactions by group A streptococci.

11.

D. The recovery rate of coagulase-negative *Staphylococcus saprophyticus* from urinary tract infections in young females is second only to that of *Escherichia coli*. The organism has a predilection for the epithelial cells of the urogenital tract and is often seen in large numbers adhering to these cells on Gram stain. Key to the identification of this coagulase-negative *Staphylococcus* is its resistance to novobiocin.

12.

A. The organism seen in Color Plate 26■ is *Listeria monocytogenes*. *Listeria* is an important animal and human pathogen that is known to cause abortion, meningitis, and septicemia in humans. This gram-positive rod is actively motile at room temperature (but not at 35°C), hydrolyzes esculin, produces catalase, and is oxidase negative. When recovered on sheep blood agar plates from clinical samples, it is often initially confused with group A or group B streptococci because of its beta-hemolysis.

13.

A. *Streptococcus agalactiae* (group B *Streptococcus*) is a principal cause of bacterial meningitis and septicemia in neonates. The organism, which is a part of the indigenous microbial flora of the vagina, is transmitted by the mother before birth, usually as the baby passes through the birth canal. Neonatal infection with group B streptococci may occur either as an early-onset disease (at birth) or as a delayed-onset syndrome that manifests itself weeks after birth.

14.

B. The vegetative cells and spores of *Bacillus cereus* are widely distributed in the environment. The virulence mechanisms of *B. cereus* are an enterotoxin and a pyogenic toxin. Accidents in nature resulting in cuts or abrasions contaminated with soil or vegetation, intravenous drug abuse, ingestion of contaminated foods, and traumatic introduction into a normally sterile site through the use of contaminated medical equipment are associated with infection.

15.

B. The formation of the characteristic *Corynebacterium diphtheriae* granules and cellular morphology seen in methylene blue stains is enhanced when the organism is grown on Loeffler's serum medium. Although this medium is primarily designed for the recovery of *C. diphtheriae* from clinical samples, it is not a differential medium. The agar slant, when inoculated, may demonstrate growth of corynebacteria within 8 to 24 hours.

16.

C. Tinsdale medium, for the primary isolation of *Corynebacterium diphtheriae,* not only inhibits indigenous respiratory flora but differentiates colonies of *C. diphtheriae*. The potassium tellurite in the medium is taken up by colonies of *Corynebacterium,* causing them to appear black. Colonies of *C. diphtheriae* are presumptively identified when black colonies surrounded by a brown halo are seen on this agar medium. However, other corynebacteria and some staphylococci will produce a similar reaction.

17.

B. The Elek immunodiffusion test is recommended for detecting toxigenic strains of *Corynebacterium diphtheriae*. In the test, diphtheria antitoxin is impregnated on a sterile filter paper strip, which is pressed onto the surface of an Elek agar plate. Test and control strains are then inoculated perpendicular to the strip on both sides and without touching the strip. A positive reaction by toxigenic strains produces a precipitin line at a 45-degree angle to the inoculum streak.

18.

D. Erysipelas results from person-to-person transmission of group A streptococci. Symptoms occur when nasopharyngeal infection spreads to the face. The rare complication of an upper respiratory infection with *Streptococcus pyogenes* is characterized by sensations of burning and tightness at the site of invasion. Erythema associated with this superficial cellulitis rapidly spreads with an advancing elevated margin. *Erysipelothrix rhusiopathiae* causes a similar disease referred to as erysipeloid.

19.

B. Phenylethyl alcohol agar (PEA) is a selective medium for the isolation of gram-positive cocci. Blood agar medium is supplemented with 0.15% phenylethyl alcohol, which is inhibitory to most gram-negative aerobic bacilli. This medium is particularly helpful when a specimen containing gram-positive cocci is contaminated with a *Proteus* spp. due to the inhibition of swarming by PEA.

20.

D. Viridans streptococci are the most common normal flora in upper respiratory cultures. They are opportunistic pathogens with low virulence. Subacute endocarditis is seen in patients with previously damaged heart valves.

21.

B. Whether growing aerobically or anaerobically, streptococci obtain all their energy from the fermentation of sugars to lactic acid. Streptococci are all catalase negative and grow on coventional media such as sheep blood agar. Most are part of the normal flora of human skin, throat, and intestine but produce a wide variety of infections when introduced in tissues or blood.

22.

A. Organisms that synthesize the enzyme catalase are able to protect themselves from the killing effects of H_2O_2 by converting it to H_2O and O_2. Streptococci are unable to synthesize the heme prosthetic group for this enzyme and are catalase negative. Therefore, they grow better on blood-containing media because of the catalase-like activity of hemoglobin.

23.

D. Streptolysin S is primarily responsible for the beta-hemolysis seen on the surface of a sheep blood agar plate inoculated with a group A streptococcus. Of the two hemolysins secreted by beta-hemolytic group A streptococcus, streptolysin S is stable in the presence of atmospheric oxygen. Streptolysin O is inactivated in the presence of oxygen, and it is best demonstrated when the agar has been stabbed and subsurface hemolysis is revealed.

24.

B. *Streptomyces* are weak pathogens rarely associated with disease. The bacteria normally inhabit the soil. The most common human infection is myectoma, which is most frequently caused by *S. somaliensis*. Some strains of *Streptomyces* grow better at 25°C than at 35°C.

25.

B. *Erysipelothrix* is a nonmotile, catalase-negative, gram-positive bacillus that often appears as long filaments. Unlike other aerobic gram-positive bacilli, this organism produces H_2S, which can be demonstrated in triple sugar iron agar. Erysipeloid, a skin disease of the hands usually associated with the handling of infected animals, is the human infection produced most commonly by this agent.

26.

D. *Listeria monocytogenes* is motile at room temperature. When inoculated into a semisolid medium, growth away from the stab is characteristic of motility. Motility is generally enhanced just below the agar surface, giving the growth pattern an "umbrella" appearance. *L. monocytogenes* is nonmotile at 35°C.

27.

D. *Rhodococcus equi* is found in soil and commonly produces disease among livestock. These gram-positive bacilli can demonstrate primary mycelia and were formerly in the genus *Nocardia*. This species is characterized by its pink pigmentation on culture media and its inability to ferment carbohydrates.

28.

A. *Streptococcus pneumoniae,* a primary etiologic agent of lobar pneumonia, is an encapsulated, gram-positive, lanceolate diplococcus. Fastidious in its growth requirements, the organism on sheep blood agar produces characteristic alpha-hemolytic colonies, which are convex and often mucoid in appearance and bile soluble. Upon aging, colonies of *S. pneumoniae* undergo autolytic changes. There are approximately 80 types of pneumococci based on specific capsular antigens.

29.

D. Species of the genus *Nocardia* are ubiquitous in the soil and thus characteristically produce exogenous forms of infection as a result of inhalation of contaminated fomites or a traumatic incident with soil contamination. A diagnostic characteristic, depending on the species, is the acid fastness of the filamentous bacilli or coccoid forms. Unlike *Actinomyces* spp., which are catalase-negative, gram-positive, non-spore-forming anaerobic bacilli, *Nocardia* spp. are catalase-positive aerobic organisms. "Sulfur granules" are characteristic of actinomycotic pus and upon examination would reveal non-acid-fast branching filaments.

30.

B. *Enterococcus faecium* is an important agent of human infection. Their differentiation from other enterococcal strains is of importance because of their resistance to most clinically useful antimicrobial agents, including vancomycin. The ability to tolerate a high concentration of salt is characteristic of the clinically significant species of *Enterococcus. E. faecium* is PYR positive and is usually nonhemolytic.

31.

D. Viridans streptococci do not produce the enzyme pyroglutamyl aminopeptidase and, therefore, in the PYR test do not produce a positive or red color. The PYR test is used predominantly for the presumptive identification of group A streptococci and *Enterococcus. Micrococcus* and *Lactococcus* are known to produce a positive reaction as well, although the reaction may be delayed.

32.

C. *Streptococcus pneumoniae* is a leading cause of lobar pneumonia as well as other serious bacterial infections. The Gram stain smear of clinical specimens can provide a rapid presumptive diagnosis when the characteristic morphology and Gram reaction is observed. The optochin disk test can be performed to presumptively identify this organism. Optochin lyses pneumococci, producing a zone of inhibition around the disk.

33.

D. Bacteriologic cultures of a typical impetigo lesion may yield either a pure culture of *Streptococcus pyogenes* or a mixed culture of *S. pyogenes* and *Staphylococcus aureus*. The thick crust form of impetigo, which is most commonly seen, is primarily caused by *S. pyogenes*. It is the bullous form of impetigo for which *S. aureus* is the etiologic agent. The route of infection is direct inoculation of the causative agents into abraded or otherwise compromised areas of the skin.

34.

C. Identifying characteristics of *Staphylococcus aureus* include the production of the extracellular enzymes coagulase and DNase and its ability to grow in the presence of high salt concentrations. Differential and selective media, such as mannitol salt agar, have been developed for the recovery of this organism. Selective media and rapid identification tests are important for this widely recognized opportunistic pathogen.

35.

A. Group B streptococci (*Streptococcus agalactiae*), unlike other streptococci, can hydrolyze sodium hippurate to benzoic acid and glycine. If glycine is produced, the addition of ninhydrin to the medium will reduce the glycine to produce a purple color. The use of ninhydrin to detect glycine is a sensitive and rapid test of hippurate hydrolysis.

36.

B. *Listeria monocytogenes* and *Streptococcus agalactiae* produce an extracellular factor known as the CAMP factor. The test is performed by making a streak of the test isolate perpendicular to a streak of *Staphylococcus aureus*. A positive CAMP reaction is indicated by a zone of enhanced beta-hemolysis (arrowhead shape) at the point where the zone of hemolysis produced by *S. aureus* joins with that produced by the beta-hemolytic test isolate. Unlike *S. agalactiae, L. monocytogenes* is catalase positive.

37.

A. The production of hemolysins and the enzymes coagulase and DNase is associated with the virulence of staphylococci. The coagulase-producing staphylococci are most commonly producers of staphylolysins, which produce beta-hemolysis when the isolate is grown on sheep blood agar. Many factors contribute to staphylococcal virulence by overcoming the host's natural defenses. Endotoxin is found in the cell wall of gram-negative bacteria.

38.

C. The physiology of staphylococci enables them to remain infectious in the environment longer than many other pathogenic bacteria. Staphylococci are somewhat heat resistant and can survive dry conditions. Their high salt tolerance enables strains to grow in salt-preserved foods and causes cases of food poisoning. Staphylococci, however, cannot resist temperatures as high as 55°C for long periods, and they are not bile resistant. Most species are sensitive to novobiocin.

39.

C. Motility is a key test for the differentiation of *Bacillus anthracis* from other species of *Bacillus*. Suspect *Bacillus* colonies are inoculated in a broth medium and allowed to grow to a visible turbidity. A sample of this actively growing culture should be examined using the hanging-drop technique for motility. *B. anthracis* is nonmotile and can therefore be easily differentiated from commonly encountered motile species.

40.

D. The rash of scarlet fever is a result of the action of an erythrogenic toxin produced by group A streptococci. Because of the rapid diagnosis and treatment of group A streptococci infections, scarlet fever is rare in most developed countries. The other diseases listed do not involve an erythrogenic toxin.

41.

D. Cultures of the tonsillar fossae and posterior pharynx are most commonly obtained in suspected cases of streptococcal pharyngitis. *Streptococcus pyogenes* is most often associated with cases of pharyngitis but is also the agent of scarlet fever and erysipelas in addition to wound infections (e.g., necrotizing fasciitis). Rapid identification of this organism and prompt antimicrobial therapy are required to prevent sequelae (i.e., rheumatic fever and acute glomerulonephritis).

42.

D. Subacute bacterial endocarditis is an inflammation of the lining membrane of the heart, which most often is caused by a member of the viridans group of streptococci. *Streptococcus sanguis* is one of several species that may lodge in an abnormal heart or on valves damaged by previous infection. Viridans streptococci are normal inhabitants of the human upper respiratory tract.

43.

C. *Streptococcus pneumoniae* is most commonly associated with cases of lobar pneumonia. Patients characteristically produce blood-tinged, rust-colored sputum in which the characteristic gram-positive lanceolate diplococci can be found. *S. pneumoniae* forms alpha-hemolytic colonies when grown on sheep blood agar.

44.

A. *Staphylococcus saprophyticus* is recognized as an etiologic agent of uncomplicated cystitis cases in young females. These nonhemolytic, coagulase-negative staphylococci closely resemble *S. epidermidis* on sheep blood agar. Identification of *S. saprophyticus* is facilitated by demonstrating its resistance to novobiocin.

45.

A. Clinical material sent to the laboratory for the recovery of *Corynebacterium diphtheriae* should be inoculated on cystine-tellurite agar plates or Tinsdale medium. On tellurite-containing media, colonies of this pathogen will appear dark-brown to black, which aids in their differentiation. Suspicious colonies should be further tested for their biochemical activity and toxin production.

46.

D. *Bacillus anthracis* is the causative agent of woolsorters disease or the pulmonary form of anthrax. The mode of infection is the inhalation of spores by the patient, usually during the performance of his/her occupation (sheep shearing or processing of animal hair). Prompt diagnosis and treatment of this disease is needed because it is known to progress rapidly to a fatal form of septicemia.

47.

D. *Corynebacterium pseudodiphtheriticum* is morphologically similar to all other members of the genus *Corynebacterium*. They are all gram-positive, non-spore-forming bacilli that characteristically resemble Chinese characters or palisades. These bacteria often stain irregularly and have a pleomorphic club-shaped appearance.

48.

B. *Corynebacterium jeikeium* is a low virulence organism resistant to multiple antimicrobials. Its multiple drug resistance allows it to remain in hospital environments, and it is often cultured from the skin of hospitalized patients. In compromised patients it has been implicated in cases of septicemia, wound infections, and endocarditis in association with intravenous catheter use.

49.

B. In immunocompromised patients, *Nocardia asteroides* can cause invasive pulmonary infection and can often spread hematogenously throughout the body. Lesions in the brain are commonly associated with dissemination and have a poor prognosis. The organism is ubiquitous in nature, and infection is acquired by traumatic inoculation or inhalation.

50.

C. Of the genera listed, only *Leuconostoc* is catalase negative. *Leuconostoc* is vancomycin resistant and associated with infections in hospitalized patients. It has also been linked to septicemias in neonates.

51.

B. *Listeria monocytogenes* is a small, gram-positive bacillus that is actively motile at room temperature. When grown on sheep blood agar, this organism produces small, translucent beta-hemolytic colonies, which may be visually mistaken for beta-hemolytic streptococci. Biochemically *L. monocytogenes* differs from streptococci because it possesses the enzyme catalase.

52.

C. *Staphylococcus epidermidis* is a saprophytic microorganism found on the skin and mucous membranes of humans. This coagulase-negative *Staphylococcus* is seen frequently as a contaminant in blood cultures when proper venipuncture technique has not been used. *S epidermidis* has been implicated in serious human infections associated with the surgical insertion of prosthetic devices.

53.

A. *Bacillus anthracis* is the etiologic agent of human anthrax that occurs in any of three forms: cutaneous, pulmonary, and gastrointestinal. On Gram stain this organism appears as a large, spore-forming, gram-positive bacillus that characteristically grows in long chains. Colonies on agar plates are large and opaque with fingerlike projections referred to as "Medusa head" forms.

54.

C. Staphylococci colonize various skin and mucosal surfaces in humans. *S. aureus* is carried as transient flora in the anterior nares. *S. saprophyticus* is less likely found as normal flora and is associated with urinary tract infections. Hospital personnel may harbor resistant strains of *S. aureus,* and person-to-person contact is a substantial infection control concern. Cultures of the anterior nares are recommended when screening for carriers in the hospital environment.

55.

D. The susceptibility of alpha-hemolytic streptococcal isolates to optochin, or ethylhydrocupreine HCl, is a presumptive test for the differentiation of *Streptococcus pneumoniae* from viridans streptococci. Viridans streptococci are typically resistant to this agent and show no zone of inhibition or a zone of less than 10 mm with a 6-mm disk. *S. pneumoniae* characteristically is susceptible and produces a zone of inhibition greater than 14 mm.

56.

D. Solubility of *Streptococcus pneumoniae* colonies by surface-active agents, such as sodium desoxycholate, is a widely used presumptive identification procedure. When a 10% solution of this reagent is applied to test colonies, *S. pneumoniae* will be totally dissolved. Colonies of viridans streptococci typically remain intact when bile is applied.

57.

D. The ingestion of food contaminated with enterotoxin produced by *Staphylococcus aureus* is the most likely cause of the disease in the case described. *S. aureus* multiplies rapidly in improperly stored food. Within a few hours, levels of 10^5 organisms per gram of food can be found. Enterotoxin is elaborated when the organism reaches stationary growth phase. Ingestion of small amounts of toxin results in a rapid onset (1–6 hours) of vomiting and diarrhea as a result of a neural response.

58.

D. Most strains of *Staphylococcus saprophyticus* are resistant to novobiocin. This organism is frequently found in urine culture of young women and may be misidentified as *S. epidermidis*. A 5-μg disk is used in the test, and a zone of 16 mm or less determines resistance.

59.

B. On serum-cystine-sodium thiosulfate-tellurite medium (Tinsdale medium), *Corynebacterium diphtheriae* is differentiated from other cornybacteria and other bacteria of the respiratory tract by its ability to produce black colonies surrounded by a brown-black halo after 48 hours of incubation. Growth factors needed by *C. diphtheriae* are provided by the addition of the serum. Potassium tellurite is inhibitory to many gram-positive and gram-negative bacteria, but corynebacteria are resistant.

60.

C. *Staphylococcus aureus* is the predominant pathogen involved in joint infections of adults. Bacterial arthritis can occur following infection in other parts of the body or bacteremia. *Streptococcus pyogenes* and *Neisseria gonorrhoeae* each account for a significant number of adult infections, whereas *Streptococcus pneumoniae* and *Haemophilus influenzae* predominate in childhood infections.

61.

B. *Listeria monocytogenes* is a cause of human and bovine abortion. In humans, the mother's symptoms are usually mild, resembling the flu and causing a low-grade fever. The organism can be isolated from aborted fetuses as well as from the maternal placenta. When infection with this etiologic agent is detected early, appropriate therapy can be initiated, which may prevent the death of the fetus.

62.

A. *Bacillus* spp. are gram-positive, spore-forming bacilli widely found in the environment. *Bacillus cereus* is of particular interest as an etiologic agent of human cases of food poisoning. This enterotoxin-producing microorganism is most commonly associated with cases of food poisoning following ingestion of reheated rice served at Asian restaurants.

63.

A. *Bacillus anthracis* infects humans by three routes: respiratory, gastrointestinal, and cutaneous. Malignant pustule is the name given to lesions seen in cutaneous anthrax in humans. The lesion is, however, neither malignant nor a pustule. The disease produces a localized abscess on the skin, which forms a characteristic black eschar surrounded by a red raised ring.

64.

B. Scalded skin syndrome is the dermatitis associated with the effects of the exfoliative toxin produced by strains of *Staphylococcus aureus*. Exfoliatin acts in humans to disrupt the adhesive forces between cells of the stratum granulosum, creating large flaccid bullae. This syndrome occurs primarily in infants and children; the primary infection is usually unrelated to the areas where lesions appear.

65.

B. *Enterococcus faecalis* and *E. faecium* grow in the presence of bile, hydrolyze esculin, and produce acid from glucose in the presence of high salt concentration. These bacteria also express streptococcal group D antigen. The ability to tolerate high salt concentrations differentiates the enterococci from the group D streptococci like the *S. bovis* group.

Aerobic Gram-Negative Bacteria

66.

A. The etiologic agents of brucellosis are the brucellae, which are small, nonmotile, gram-negative coccobacilli that are facultative intracellular parasites. Isolation of these organisms is difficult. In suspected cases, which are generally job related, multiple blood cultures are recommended for optimal recovery of the agent. Bone marrow cultures have been found to be positive when cultures of blood failed to recover the organism.

67.

B. *Klebsiella pneumoniae* is the species most frequently recovered from the vast majority of clinical cases. Members of the genus *Klebsiella* have a capsule and appear mucoid on cultures. This highly encapsulated organism can cause severe pneumonia, nosocomial infections of several types, infantile enteritis, and other extraintestinal infections.

68.

A. Enterotoxins are produced in the intestinal tract and primarily cause diarrhea. The heat-labile enterotoxin of *Escherichia coli,* which resembles cholera toxin, acts to stimulate the enzyme adenylate cyclase. The stimulation of the enzyme adenylate cyclase by the toxin increases the production of cyclic AMP, causing rapid gastrointestinal fluid loss. Diarrhea results following stimulation of the secretion of chloride ions by the cells lining the small intestine.

69.

C. The genus *Neisseria* contains organisms that possess cytochrome oxidase activity. Colonies can be identified by the development of a dark purple color following the application of tetramethyl-*p*-phenylenediamine dihydrochloride. The reaction relies on the property of the molecule to substitute for oxygen as an electron acceptor. In the presence of the enzyme and atmospheric oxygen, the molecule is oxidized to form indophenol blue.

70.

A. *Shigella* has a low infecting dose and has been reported to cause outbreaks in daycare centers and can be spread to family members. These organisms are found in humans only at the time of infection; they are not part of the normal flora. Transmission is by the fecal-oral route, typically by ingestion of contaminated foods or water.

71.

A. *Chryseobacterium* spp. are ubiquitous in the environment and are especially associated with moist soil and water. *Chryseobacterium* (formerly *Flavobacterium*) *meningosepticum,* a known nosocomial pathogen, has been implicated in outbreaks of meningitis in hospitals and is associated with the use of contaminated respiratory therapy equipment. Adult human infections are rare; these opportunistic microorganisms occur primarily in immunocompromised patients.

72.

D. *Shigella dysenteriae,* the type species of the genus, is a causative agent of bacillary dysentery. Differential and selective media for the recovery of enteric pathogens from stool samples would demonstrate *Shigella* species as H_2S negative, non-lactose-fermenting, gram-negative bacilli. Further biochemical testing would generally show these organisms to be unable to use citrate as their sole carbon source, unable to decarboxylate the amino acid lysine, and urease negative.

73.

A. *Aeromonas hydrophila* is typically found in fresh water and has been implicated in human infections. Growth on MacConkey agar and a positive oxidase reaction are characterisic of this organism. A positive oxidase reaction differentiates this organism from all of the *Enterobacteriaceae,* except the recently added *Plesiomonas shigelloides*. On sheep blood agar, many strains of *Aeromonas* produce beta-hemolysis.

74.

B. A highly selective medium, thiosulfate-citrate-bile salt-sucrose (TCBS) is used for the isolation of *Vibrio* spp. Species able to ferment sucrose, such as *V. cholerae,* produce yellow colonies. Non-sucrose-fermenting organisms produce green colonies.

75.

A. The K (capsule) antigen surrounds the bacterial cell and masks the somatic antigens of the cell wall, which are used to group members of the *Enterobacteriaceae*. These heat-labile antigens can be removed by heating a suspension of the culture at 100°C for 10–30 minutes. Antisera that contain K antibody can be used to demonstrate the presence of the capsular antigens.

76.

B. *Burkholderia pseudomallei* is the causative agent of melioidosis. The bacterium is found in soil and water in subtropical areas of Southeast Asia and Australia. Melioidosis exhibits several forms, from skin abscesses to abscess formation in internal organs.

77.

C. *Francisella tularensis* requires cysteine or cystine for growth. Glucose-cysteine with thiamine and cystine heart media are commercially available for suspected cases of tularemia. They both require the addition of 5% sheep or rabbit blood. Buffered charcoal yeast extract also supports the growth of *F. tularensis,* a medium generally used by clinical laboratories for the cultivation of *Legionella* spp.

78.

B. Decubitus ulcers frequently contain normal intestinal flora. The biochemical results are characteristic of *Escherichia coli*. *E. coli* is associated with a variety of diseases; it is the predominant organism associated with cases of neonatal meningitis, specticemia, cystitis, appendicitis, and endocarditis.

79.

D. *Pseudomonasa aeruignosa* is the most commonly encountered gram-negative species that is not a member of the family *Enterobacteriaceae.* It is ubiquitous in nature and is found in homes and hospitals. It is an opportunistic pathogen responsible for nosocomial infections.

80.

A. *Capnocytophaga* spp. are fermentative gram-negative bacteria that inhabit the oral cavity of humans. These organisms have been identified as a cause of disease in the oral cavity, and in compromised hosts they have been implicated in systemic disease isolated from cerebrospinal fluid, pleural fluid, and pulmonary secretions. The gliding motility is best observed during the log phase of growth and can be demonstrated by darkfield microscopy and on sheep blood agar by the production of concentrically spreading growth around primary colonies.

81.

C. *Vibrio vulnificus* is implicated in wound infections and septicemia. The organism is found in brackish or salt water. Ingestion of contaminated water or seafood is the typical mode of transmission. Wound infections are associated with contamination at the site with organisms in water.

82.

D. The Waterhouse-Friderichsen syndrome of disseminated intravascular coagulation occurs in cases of fulminant meningococcemia—*Neisseria meningitidis* septicemia. Invasion of the circulatory system by *N. meningitidis* may produce only a transient bacteremia or meningitis or may go on to cause a rapidly fatal infection. In cases of meningococcemia with intravascular coagulation, acute adrenal insufficiency due to hemorrhage into the adrenal gland may result.

83.

B. Before the development of an effective vaccine, the strain of *Haemophilus influenzae* found to be implicated in the majority of cases of bacterial meningitis in children 1–6 years of age was serotype B. This serotype is surrounded by a weakly immunogenic polyribitol phosphate capsule. The widespread use of *Haemophilus influenzae* type b (Hib) vaccine beginning in 1985 has significantly reduced the incidence of invasive *H. influenzae* type b disease.

84.

A. Unlike other salmonellae, *Salmonella* Typhi produces only a small amount of hydrogen sulfide, produces no gas from glucose, is citrate negative, and possesses a capsular antigen (Vi). Identification of *Salmonella* Typhi, the etiologic agent of typhoid fever, may be delayed if laboratory professionals do not have a good appreciation of its atypical characteristics. It is also important to note that the bacilli appear in the patient's circulatory system several days before a stool culture will be positive.

85.

A. Chancroid or soft chancre is caused by *Haemophilus ducreyi,* a small, gram-negative coccobacillus. Painful genital lesions and painful swelling of the inguinal lymph nodes characterize the disease. The incubation period following contact with an infected person ranges from 1 to 5 days, after which the patient notes the painful, round, nonindurated primary lesion on the external genitalia. Signs of regional lymphadenitis appear in about one-half of the cases a few days after the appearance of the primary lesion.

86.

D. Pseudomonads are ubiquitous microorganisms generally associated with moist environments. Cases have been increasing as the popularity of health spas increases. In some cases the pattern of dermatitis caused by these organisms matches the areas covered by the individual's swimsuit. When not properly maintained, whirlpools create a favorable environment for the growth of these organisms.

87.

B. *Campylobacter jejuni* rivals *Salmonella* as the most common bacterial cause of diarrheal disease in humans. *Campylobacter* enterocolitis is characterized by fever, bloody diarrhea, and abdominal pain. Special selective culture media and incubation under a microaerophilic atmosphere at 42°C are required for the recovery of this organism from clinical samples.

88.

D. Screening procedures for the recovery of the enteric pathogen *Salmonella* rely heavily on differential media, which indicate lactose fermentation and the production of H_2S. Isolates of *Shewanella putrefaciens* recovered from stool samples on a medium such as Hektoen enteric agar would resemble *Salmonella* in that the organism is not able to ferment lactose and does produce a significant amount of H_2S. However, unlike *Salmonella,* some strains of *S. putrefaciens* will produce acid from sucrose, forming yellow colonies on Hektoen enteric agar.

89.

A. Whooping cough, or pertussis, is caused by *Bordetella pertussis,* a minute, encapsulated, nonmotile, gram-negative, pleomorphic bacillus. The best identification method is the polymerase chain reaction. Regan-Lowe medium is recommended for the isolation of this agent.

90.

B. *Neisseria gonorrhoeae* is a primary pathogen of the urogenital tract. It is an important cause of sexually transmitted diseases. Surface structures such as pili aid in attachment to mucosal epithelial cells and invasion of submucosa to produce infection.

91.

D. Most serotypes of *Salmonella* produce hydrogen sulfide in triple sugar iron agar. As seen in Color Plate 27■, they demonstrate a positive lysine and a negative urease reaction, which differentiates them from *Proteus* spp., which also produce H_2S. It is important to be able to quickly differentiate those organisms resembling *Salmonella* from other H_2S-producing organisms such as *Citrobacter freundii* and *Edwardsiella tarda.*

92.

A. *Neisseria meningitidis* and *N. gonorrhoeae* are most commonly described as having a "kidney bean" cellular morphology. Occasionally some *Moraxella* spp. will exhibit this morphology. These gram-negative coccal organisms appear characteristically as diplococci, with the paired cells having adjacent walls that are flattened. *Neisseria* are important human pathogens.

93.

C. *B. cepacia* is the most common *Burkolderia* spp. in clinical specimens. *P. aeruginosa* is the most common gram-negative bacillus that is not in the family *Enterobacteriaceae* and *Stenotrophomonas maltophilia* the second most common. *B. mallei* has not been isolated recently in the U.S.

94.

D. Ophthalmia neonatorum, a form of conjunctivitis, is associated with *Neisseria gonorrhoeae.* The infection is transmitted to the newborn by the mother as it passes through the birth canal. The use of an ophthalmic solution of erythromycin is recommended for the prevention of this form of conjunctivitis.

95.

B. *Neisseria lactamica* is part of the normal nasopharyngeal flora of humans. In the laboratory this agent may be mistakenly identified as *Neisseria meningitidis,* an organism of significant pathogenicity. Differentiation of these two species is easily accomplished by demonstrating the fermentation of lactose or an ONPG (*o*-nitrophenyl-beta-galactopyranoside) positive test.

96.

A. *Haemophilus aegyptius* is the causative agent of "pinkeye." This form of conjunctivitis is highly contagious and is frequently seen in children attending daycare centers. The agent is an aerobic gram-negative bacillus that is nonmotile and requires both hemin (X factor) and nicotine adenine dinucleotide (NAD, V factor) for growth.

97.

A. *Hafnia* is a member of the family *Enterobacteriaceae*. *Hafnia alvei* is the only species in the genus. The characteristics of this organism are positive motility and lysine, ornithine, ONPG, and KCN reactions.

98.

B. *Acinetobacter* spp. are opportunistic pathogens for humans and are important causes of nosocomial infections. They are oxidase negative and will grow on most laboratory media, including MacConkey agar. *Acinetobacter* spp. are nonfermenters, but *A. baumannii* can form pink to purple colonies on MaConkey agar that can be mistaken for lactose fermentation. Many strains of *A. baumannii* will oxidize glucose.

99.

B. Pontiac fever is caused by *Legionella pneumophila,* as is Legionnaires disease, but it is not as serious an infection. This febrile illness is characteristically self-limited and does not demonstrate significant pulmonary symptoms. The incubation period, unlike that for Legionnaires disease, is short, followed by symptoms of malaise, muscle aches, chills, fever, and headache.

100.

A. *Brucella* spp. are fastidious, gram-negative, coccobacillary organisms. They are predominantly animal pathogens, but occasionally produce disease in humans. The usual specimens for recovery of *Brucella* are blood and bone marrow, with the latter considered the more sensitive.

101.

A. *E. coli* O157:H7 produces a toxin similar to Shiga toxin produced by *Shigella dysenteriae*. It is most commonly transmitted by ingestion of undercooked ground beef or raw milk. Hemorrhagic colitis is characteristic of infection, but infection can also lead to hemolytic uremic syndrome resulting from toxin-mediated kidney damage.

102.

A. *Aeromonas* spp. are found in bodies of fresh water and salt water that can be flowing or stagnant and contaminated with sewage. These organisms are known as one of the animal pathogens that cause "red leg disease" in frogs. The largest number of human cases occurs between May and November and seems to be highly associated with exposure to water or soil.

103.

C. *Campylobacter jejuni* are small, curved, motile gram-negative rods that are hippurate hydrolysis positive. They are found in the gastrointestinal tract of a variety of animals. Campy agar is used for isolation from stool and is incubated at 42°C under microaerophilic conditions (10% CO_2, 5% O_2 with balance N_2) for 72 hours.

104.

C. *Hafnia alvei* is a member of the family *Enterobacteriaceae* and is oxidase negative. It has been isolated from a variety of clinical specimens but is generally considered clinically insignificant. *Aeromonas, Chryseobacterium,* and *Vibrio* are all oxidase positive.

105.

A. *Neisseria gonorrhoeae* is the causative agent of gonorrhoea and is very sensitive to drying, temperature variations, and fatty acids in clinical material. *N. gonorrhoeae* will grow on chocolate agar but not sheep blood agar. Incubation under CO_2 is required for recovery, and selective media like Thayer-Martin are recommended.

106.

D. The most commonly used method for serogrouping *Shigella* is based on the somatic oligosaccharide or O antigens. The O antigens are also used to serogroup *Escherichia coli* and *Salmonella*. Because *Shigella* spp. are nonmotile, the H or flagella antigens cannot be used; however, H antigens are used for *E. coli* and *Salmonella*. The capsule (K) antigens are used to serogroup *Klebsiella pneumoniae*.

107.

C. *Vibrio cholerae* produces an exotoxin that causes infected individuals to lose massive amounts of fluids. Severe dehydration is usually the cause of death in untreated patients. Proper therapy begins with intravenous fluids to restore the patient's water volume and electrolyte balance. The microorganism does not invade the intestinal mucousa but is attached to the surface of enterocytes. The other bacteria listed produce toxins or invasive diseases that often produce inflammation resulting in bloody diarrhea.

108.

C. *Vibrio parahaemolyticus* is classified as a halophilic *Vibrio* sp. requiring increased osmotic pressure, in the form of salt, for growth. This makes routine biochemical test media less than optimal because of their low NaCl content. Growth in the presence of 1% NaCl but no growth in media without the added Na^+ is the test for the differentiation of halophilic organisms. Marine water is the normal habitat of most *Vibrio* spp.

109.

C. *Brucella* spp. are harzardous, especially in aerosol-generating procedures. It is important for the laboratory to be notified whenever brucellosis is suspected. Most laboratories send isolates to a reference laboratory for confirmation or definitive identification because they lack specialized media and containment facilities.

110.

B. *Helicobacter pylori* is found in the human gastric mucosa colonizing the mucous layer of the antrum and fundus but does not invade the epithelium. Approximately 50% of adults over the age of 60 years are infected, with the incidence of gastritis increasing with age. *H. pylori* has been cultured from feces and dental plaque, supporting the theory of a fecal-oral or oral-oral route of transmission.

111.

A. *Acinetobacter* is widely distributed in nature and commonly colonizes hospitalized patients. Infection occurs mainly in compromised hosts. Its resistance to many of the commonly used antimicrobial agents limits the selection of therapeutic agents.

112.

D. *Vibrio parahemolyticus* is found in brackish or salt water. The mode of transmission is the ingestion of contaminated water or seafood. *V. parahemolyticus* is halophilic.

113.

D. Organisms biochemically resembling *Salmonella* are typically tested using a polyvalent antiserum composed of antibodies against the commonly isolated strains, including antisera against the Vi antigen. The Vi antigen is a heat-labile capsular antigen associated with *Salmonella* Typhi. After heating a *Salmonella* Typhi suspension, the Vi antigen has been removed, and the organism can now react with the somatic grouping antisera. *Salmonella* Typhi demonstrates a positive agglutination reaction in D-grouping sera.

114.

B. *Campylobacter fetus* subsp. *fetus* is occasionally implicated in human disease. This organism, unlike *C. jejuni,* is characterized as producing extraintestinal symptoms. Those persons most at risk of infection are those with preexisting disease who are in a debilitated condition.

115.

A. *Shigella sonnei* is a group D *Shigella* and is characterized by its ability to ferment lactose. *S. sonnei* is ONPG positive but is a delayed lactose fermenter. This is the most commonly isolated species of *Shigella* in the U.S. The genus *Shigella* is characterized biochemically by being negative for citrate, urease, motility, and lysine decarboxylation.

116.

A. The Voges-Proskauer (VP) test is a broth test that detects the presence of acetoin from the metabolism of glucose in the medium. A red color indicates a positive reaction. The most common clinical isolates from the genera *Providencia, Escherichia, Salmonella,* and *Proteus* are generally VP negative, whereas most members of the *Klebsiella, Enterobacter,* and *Serratia* are positive. *Enterobacteriaceae* that are VP positive are typically methyl red negative.

117.

C. Pathogenic mechanisms of *V. cholerae* include adherence to enterocytes via pili, motility, enzymes such as protease and mucinase, and the production of an enterotoxin. Epidemic strains colonize the small intestine where they elaborate choleragen, an enterotoxin. Disease is produced when the enterotoxin stimulates the secretion of large volumes of fluids into the intestinal lumen.

118.

A. Classic epidemic strains of *Vibrio cholerae* are included in the antigenic O group 1. The Ogawa and Inaba strains are considered the predominant epidemic strains. In the last few years the strain O139 has also been associated with outbreaks of cholera.

119.

D. *Brucella* spp. are small, gram-negative intracellular parasites implicated in zoonotic infection of humans. Brucellosis presents as an undulant febrile illness. In the U.S., disease caused by *Brucella* sp. is mainly job related or involves food or animal associations, such as in hunters or those who drink raw milk.

120.

A. *Escherichia coli* is a lactose-fermenting member of the family *Enterobacteriaceae*. Various selective and differential agars are available for the differentiation of lactose fermenters from those that do not degrade lactose. In some media H_2S production may be demonstrated. Isolates of *E. coli* would produce yellow colonies at 24 hours on xylose-lysine-desoxycholate (XLD) agar. Non–lactose fermenters such as *Shigella* would produce red colonies on XLD agar. On MacConkey agar, lactose fermenters produce pink colonies; on Hektoen enteric agar, colonies would be orange. Most strains of *E. coli* are beta-hemolytic on sheep blood agar.

121.

A. *Yersinia pestis* is the causative agent of plague. The organism is endemic in rodents and is transmitted to humans by the rat flea. This oxidase-negative organism, unlike other *Yersinia* spp., is nonmotile at 20–25°C. It is also negative for H_2S and urease.

122.

B. Motility can be important in the identification of microorganisms. Of the *Enterobacteriaceae,* the genera *Klebsiella* and *Shigella* are characteristically nonmotile, as is *Tatumella.* Motility of the *Enterobacteriaceae* can normally be detected by the use of a semisolid motility medium, which is grossly observed for the determination of motility. The hanging-drop method is perhaps the most accurate means of detecting motility of nonfermentative microorganisms.

123.

C. Pneumonic legionellosis and the nonpneumonic illness known as Pontiac fever are the two clinical forms of disease caused by *Legionella pneumophila.* The optimal temperature for cultivation is 35°C, and cold enrichment is not appropriate. Direct fluorescent antibody is often used diagnostically, and erythromycin is the drug of choice for therapy.

124.

B. *Klebsiella granulomatis,* formerly *Calymmatobacterium granulomatis,* is the etiologic agent of the sexually transmitted disease granuloma inguinale. It is a pleomorphic, gram-negative, encapsulated bacillus, although it does not Gram stain well. First seen as inclusions (Donovan bodies) in mononuclear cells from genital ulcers stained with the Giemsa or Wright stain, these organisms are extremely difficult to recover.

125.

D. When streaked on DNase test medium, colonies of *Staphylococcus aureus* and *Serratia marcescens* will demonstrate a positive reaction for DNase activity. Inoculated plates are incubated 18 to 24 hours, after which the plates are flooded with a 0.1% solution of HCl. DNase-producing organisms are differentiated by the development of a clear zone in the agar around the colonies.

126.

D. Citrate utilization is a common test used for the differentiation of members of the family *Enterobacteriaceae.* Both *Escherichia coli* and *Shigella boydii* are incapable of using citrate as the sole source of carbon as an energy source. Organisms such as *Citrobacter freundii, Enterobacter aerogenes,* and *Serratia marcescens* are able to grow on citrate agar and produce a color change from green to blue in the medium. Many *Enterobacteriaceae* that are Voges-Proskauer positive are also citrate positive.

127.

B. *Neisseria gonorrhoeae* is a fastidious organism requiring the addition of serum or blood to the culture media in order to grow. A selective medium such as modified Thayer-Martin or GC-Lect should be used for primary isolation, especially from sites that may be contaminated with normal flora. Collection and processing of specimens must be done under optimal conditions, because this organism is sensitive to drying and low temperatures.

128.

A. The indole reaction is a widely used method for differentiating lactose-positive *Escherichia coli* from other members of the family *Enterobacteriaceae*. Organisms such as *E. coli,* which possess the enzyme tryptophanase, are able to metabolize the amino acid tryptophan with the production of indole, pyruvic acid, and ammonia. Indole represents the "I" in the IMViC reactions, a battery of tests used for the identification of the *Enterobacteriaceae*.

129.

A. *E. coli* produces an acid over acid (A/A) reaction on TSI agar that indicates that glucose and either lactose or sucrose or both have been fermented. Bacteria that ferment lactose or sucrose produce large amounts of acid in the medium. The enteric pathogens *Salmonella* and *Shigella* can be ruled out when such a reaction is observed, because they are generally not able to use either lactose or sucrose within 18 hours.

130.

A. Pyocyanin is the nonfluorescent, blue-green, diffusable pigment produced by *Pseudomonas aeruginosa.* It is the only bacterium able to produce this pigment. Pyocyanin mixes with the yellow pigment fluorescein to turn culture media green. Most *P. aeruginosa* strains can be identified presumptively by their characteristic grape-like odor, colony morphology, and blue-green pigment.

131.

C. Color Plate 28■ is a Gram stain of a cerebrospinal fluid specimen revealing many white blood cells. All of the bacteria listed are fastidious; however, *Haemophilus influenzae* would be expected to grow on chocolate agar but not sheep blood agar (SBA). *Neisseria meningitidis,* also an important cause of meningitis, would be expected to grow on SBA incubated in CO_2. *Bordetella parapertussis* and *Brucella* sp. would likely grow on both SBA and chocolate agar, and both are uncommon isolates.

132.

D. *Campylobacter jejuni* is an important human pathogen most commonly associated with cases of bloody diarrhea, fever, and abdominal pain in humans. Special handling of cultures suspected to contain this organism is required for optimal recovery. Cultures should be incubated at 42°C in a microaerophilic atmosphere and examined at 24 and 48 hours for spreading nonhemolytic colonies, which may be slightly pigmented. Wet mounts demonstrate the typical "darting" motility of this isolate.

133.

C. *Moraxella catarrhalis* possesses the enzyme indophenol oxidase. When a 1% solution of tetramethyl-*p*-phenylenediamine (oxidase reagent) is applied to colonies of these organisms, the colonies turn a purple color, which rapidly darkens. The other species listed are oxidase negative.

134.

C. *Cardiobacterium hominis* is a rare pathogen that is recovered predominantly from cases of endocarditis. It is characterized as a fermentative, gram-negative bacillus that is nonmotile, catalase negative, oxidase positive, and weakly indole positive. *C. hominis* will grow on sheep blood agar, but growth is enhanced by the addition of yeast extract to media.

135.

D. *Vibrio vulnificus* is a halophilic lactose-fermenting organism. The isolate is associated with two distinct clinical conditions: primary septicemia and wound infection. Septicemia with this organism appears to be correlated in most cases with preexisting hepatic disease. Septicemia due to *V. vulnificus* characteristically produces a fulminant disease with a high mortality rate. Wound infection with this organism is usually associated with trauma and contact with a marine environment.

136.

C. Members of the tribe *Proteae* are characteristically positive for phenylalanine deaminase (PDA). This includes *Proteus vulgaris, Providencia stuartii,* and *Morganella morganii. Tatuella ptyseos* belongs to the family *Enterobacteriaceae* and is also PDA positive.

137.

A. The water-diffusible yellow pigment fluorescein (pyoverdin) is produced by members of the *Pseudomonas* fluorescent group, which includes *Pseudomonas aeruginosa, P. fluorescens,* and *P. putida.* The production of pyoverdin can be detected when a culture of the organism is exposed to a short-wavelength, ultraviolet light source. The production of fluorescent pigments is dependent upon nutritional factors; therefore, special media, such as cetrimide, should be used when trying to detect the pigment. Cationic salts such as magnesium sulfate intensify luminescence.

138.

C. *Kingella* spp. are gram-negative bacilli or coccobacilli that may appear in short chains. *K. denitrificans* can be isolated from the human upper respiratory tract, will grow on modified Thayer-Martin agar, and is oxidase-positive. The growth of this organism is inhibited by MacConkey agar, and growth is poor on triple sugar iron agar.

139.

B. Chocolate agar is the preferred culture medium for *Haemophilus.* Unlike 5% sheep blood agar, it provides both hemin (X factor) and NAD (V factor) required for growth. *H. ducreyi* grows best in a special medium, Mueller-Hinton-based chocolate agar, supplemented with 1% IsoVitaleX and 3 μg/mL of vancomycin. *Haemophilus* spp. are obligate parasites of animals and are found primarily in the upper respiratory tract and oral cavity.

140.

A. *Legionella pneumophila* requires the use of special laboratory media for cultivation and does not stain well by the conventional Gram stain. Most *Legionella* spp. are motile, are biochemically inert, and autofluoresce.The primary mode of transmission is by the airborne route, usually in association with an environmental source of bacteria.

141.

C. Tissue samples from the lower respiratory tract (lung biopsy) have the greatest yield of positive cultres for *Legionella pneumophila.* However, these specimens require invasive procedures and are not commonly performed. Cultures of lower respiratory tract specimens, such as bronchial wash and expectorated sputum, are appropriate for the isolation of *L. pneumophila.* The bacteria are seldom recovered from blood specimens.

142.

C. The diagnosis of pertussis, or whooping cough, is confirmed by culture. Regan-Lowe, a charcoal-based medium, provides the best results. Cephalexin is often added to make the medium selective. Isolation of the etiologic agent is best done within the first week of the illness. Modified Skirrow's medium is a primary plating media for *Campylobacter* spp.

143.

C. *Enterobacter sakazakii* produces a yellow pigment that aids in its presumptive identification. Some strains of *E. cowanii* are also pigmented. *E. sakazakii* is an occasional clinical isolate that has been linked to respiratory tract infections and wounds.

144.

C. *Proteus mirabilis* is commonly associated with urinary tract infections as well as infections in other parts of the body. It is a motile organism that characteristically swarms across the surface of sheep blood agar plates. Members of the genus *Proteus* are characteristically rapidly urea positive, lactose negative, and phenylalanine deaminase positive.

145.

D. *Acinetobacter baumannii* is not able to reduce nitrate. This species will oxidize but not ferment glucose. *Acinetobacter* spp. are able to grow on MacConkey agar, and they are oxidase negative, nonmotile, and characteristically resistant to penicillin.

146.

D. *Eikenella corrodens* is a facultatively anaerobic gram-negative bacillus that requires hemin in the culture medium to grow aerobically. This organism, which is a part of the normal indigenous flora of humans, is seldom found in pure culture. It is commonly associated with polymicrobial infections following bite or clenched-fist wounds. Infections of the face and neck may also involve this organism, which produces pitting of the agar on which it is isolated.

147.

C. In the genus *Pasteurella, P. multocida* is the species commonly recovered in clinical specimens. This gram-negative coccobacillus is a normal inhabitant of the oral cavity of domestic animals. Humans most often become infected from a bite or scratch of a cat or dog, which produces a rapidly progressing, painful, suppurative wound infection. Penicillin is an effective drug for the treatment of *Pasteurella* infections.

148.

D. *Legionella pneumophila* is able to hydrolyze hippurate. *L. pneumophila* will also autofluoresce. Although most of the studies done on legionellosis are based on this species, *L. pneumophila* is not the only one associated with human disease.

149.

D. *Haemophilus influenzae* biogroup aegyptius causes a conjunctivitis followed by invasive disease known as Brazilian purpuric fever. The bacterium requires both X and V factors and is therefore negative for delta-aminolevulinic acid (ALA). It resembles *H. influenzae* biotype III in that it is indole negative, urease positive, and ornithine decarboxylase negative.

150.

B. *Kingella denitrificans* is most often associated with endocarditis. It is morphologically similar to *Neisseria gonorrhoeae* both on Gram stain and colonies on culture media. Confusion is further compounded by its ability to grow on modified Thayer-Martin medium and its positive oxidase and glucose reaction. The ability of *K. denitrificans* to reduce nitrates is a key test for its differentiation from *N. gonorrhoeae.*

151.

B. *Plesiomonas* was previously in the family *Vibrionaceae*. Based on nucleic acid and antigenic studies, it was recently moved to the family *Enterobacteriaceae*. *P. shigelloides* is the only species in this genus. Infection in humans has manifested mainly as diarrheal diseases, and a waterborne mode of transmission is often the source.

152.

C. The recalls described illustrate the ubiquitous nature of *Pseudomonas aeruginosa* in the environment and its resistance to many disinfectants. In addition, the bacterium has minimal nutritional requirements and the ability to tolerate a wide range of temperatures (4–42°C). *P. aeruginosa* is an opportunistic pathogen commonly associated with hospital-acquired infections.

153.

B. Ingestion of contaminated unpasteurized (raw) milk or cheese is one of the primary routes of infection. Brucellosis is found worldwide, and symptoms vary from asymptomatic to a debilitating systemic infection. Only four of the six species are typically pathogenic for humans: *B. abortus, B. melitensis, B. suis,* and *B. canis*.

154.

D. The production of DNase, lipase, and gelatinase differentiates the genus *Serratia* from other *Enterobacteriaceae*. *Serratia* spp., especially *S. marcescens,* have a close association with nosocomial infections. *Serratia* can produce severe infections such as septicemia and meningitis and are frequently difficult to eradicate because of the characteristic antimicrobial-resistant strains found in the hospital environment.

155.

A. *Haemophilus ducreyi* is the causative agent of chancroid, a serious sexually transmitted disease. The disease is more prevalent in the tropics than in temperate parts of the world. The bacteria produce buboes in the groin and can cause a septicemia.

156.

D. Posterior nasopharyngeal cultures are recommended for the recovery of *Bordetella pertussis* in suspected cases of pertussis (whooping cough). Swabs of the nasopharynx are inoculated on the selective agar Regan-Lowe. Cephalexin is added to the culture medium to inhibit the growth of contaminating indigenous flora.

157.

C. *Gardnerella vaginalis* is associated with cases of bacterial vaginosis (BV) formerly called "nonspecific vaginitis." Although *G. vaginalis* is probably not involved in the pathogenesis of BV, its presence in high numbers is considered a presumptive diagnosis. These small, gram-negative bacilli are frequently seen in great numbers on the surface of epithelial cells ("clue cells") taken from the vagina.

158.

B. Cultures of blood and bone marrow are the recommended specimens for the isolation of *Brucella* spp. Inoculation of a blood culture bottle for a continuous monitoring system is the most senstitive recovery method. The lysis-centrifugation method (Isolator®, Wampole Laboratories) is more sensitive than a biphasic culture bottle.

159.

C. Mesenteric lymphadenitis is one of the common manifestations of human *Yersinia pseudotuberculosis* infections. Symptoms produced by this agent closely resemble those of acute appendicitis. This gram-negative coccobacillus grows well on routine culture media and has an optimal growth temperature of 25–30°C.

160.

D. Cefsulodin-irgasan-novobiocin (CIN) agar is recommended for the primary isolation of *Yersinia* and *Aeromonas. Y. enterocolitic* produces "bull's-eye" colonies at 48 hours; colonies show a dark red center surrounded by a translucent border. This is a selective and differential agar that supresses the growth of normal fecal flora and differentiates colonies of *Y. enterocolitica.*

161.

B. *Gardnerella vaginalis* is associated with bacterial vaginosis, but cultures are not recommended for diagnosis. Many women carry *G. vaginalis* as normal vaginal flora; therefore, the isolation of the organism may not be clinically significant. The disease can be diagnosed by detecting "clue" cells, vaginal epithelial cells with gram-variable bacilli attached to their surface.

162.

B. *Legionella pneumophila,* the causative agent of Legionnaires disease, can be recovered from respiratory tract secretions. The bacterium is fastidious and, like *Francisella tularensis,* requires cystiene or cystine for growth. The culture medium most commonly recommended is buffered charcoal yeast extract (BCYE) agar, which is incubated in a moist chamber at 35°C. Growth on this medium may not be visible for 3 to 4 days, after which further identification procedures may be carried out.

163.

D. Swimmer's ear is a form of external otitis common to persons who swim and fail to completely dry their ear canals when they get out of the water. The organism most commonly associated with this condition is *Pseudomonas aeruginosa.* It is an organism known to be an opportunistic pathogen and one that favors a watery environment.

164.

B. Traveler's diarrhea is caused by strains of toxin-producing invasive or enteropathogenic *Escherichia coli.* Enterotoxigenic *E. coli* can produce one or two exotoxins: one is heat stable and one is heat labile. Contaminated food products and water in foreign countries seem to be the major vehicle for human infection with these agents.

165.

A. Fresh isolates of *Campylobacter jejuni* on Gram stain characteristically reveal a "gull-wing" appearance. These gram-negative bacilli are motile with a typical darting pattern on wet mounts. They stain poorly using the Gram stain method, and it is recommended that carbolfuchsin or basic fuchsin be substituted for the counterstain safranin.

166.

C. *Neisseria gonorrhoeae* is said to resemble a kidney bean on Gram stain because of its characteristic gram-negative diplococcal morphology in which the adjacent sides are flattened. Typically these organisms are found intracellularly when direct smears of clinical material are examined. Smears from the female genital tract must be interpreted with caution, however, because other normal flora microorganisms are morphologically similar.

167.

A. *Salmonella* is urease negative and H_2S positive. The negative urease reaction would help to differentiate it from an H_2S-positive *Proteus* spp. *Yersinia enterocolitica* and *Y. pseudotuberculosis* are both urease positive. A rapid positive urease is a key test for the identification of *H. pylori*.

168.

A. *Neisseria gonorrhoeae* is identified in the clinical laboratory by its ability to ferment only glucose. The diagnosis of the sexually transmitted disease caused by this agent can be definitively made only by the isolation and identification of *N. gonorrhoeae* in the clinical laboratory. Morphologically, all members of the genus are alike, and all are oxidase positive, which makes definitive identification procedures necessary. Nucleic acid amplification tests are also used frequently to diagnose gonorrhea.

169.

C. *Neisseria meningitidis* is a human pathogen most commonly associated with meningitis. These oxidase-positive, gram-negative diplococci are identified either by fermentation tests or serologic methods that use specific antisera. *N. meningitidis* ferments both glucose and maltose.

170.

A. *Bartonella bacilliformis* is the causative agent of Oroya fever and verruga peruana. It is a pleomorphic, gram-negative rod that is an intracellular parasite of red blood cells and can be cultured from blood in the acute stage of the disease. The disease is rare and occurs primarily in South America.

171.

D. *Serratia liquefaciens,* as its name implies, is able to liquefy gelatin. This bacterium is not thought to be a primary pathogen and is only rarely isolated from clinical specimens. The positive arabinose reaction is one way of differentiating this isolate from *Serratia marcescens*.

172.

C. *Proteus vulgaris* is able to deaminate phenylalanine. The test is performed by inoculating the isolate on a slant of medium containing phenylalanine and, after incubation, adding a 10% solution of ferric chloride. A dark green color on the slant after addition of the reagent is a positive result.

173.

B. *Klebsiella* spp. are all nonmotile, which aids in their identification. *Klebsiella* spp. produce a capsule resulting in mucoid colonies. *Shigella,* another genus in the family *Enterobacteriaceae,* is also nonmotile.

174.

A. *Chromobacterium violaceum* is a motile, gram-negative bacillus found in soil and water that can be pathogenic for humans. The production of a non-water-soluble violet pigment by these organisms aids in their identification. *Chromobacterium* is catalase and oxidase positive and generally attacks carbohydrates fermentatively.

175.

B. *Pasteurella multocida* is the species in the genus most often encountered in the clinical laboratory. It is normal oral flora in animals, not humans, and it is an opportunistic pathogen. The mode of transmission generally involves traumatic inoculation of the organism through the skin. *P. multocida* grows on sheep blood agar but not on MacConkey agar.

176.

C. *Pseudomonas aeruginosa* has not only a characteristic grapelike odor but also a blue-green color. These oxidative, motile organisms are oxidase positive and are able to grow at 42°C. In humans these opportunistic organisms cause many types of infections, but they are primarily associated with burn wound infections.

177.

A. *Serratia marcescens* is a chromogenic member of the family *Enterobacteriaceae. S. marcescens* is the most clinically significant of the genus and is frequently involved in nosocomial infection. The red pigment produced is not water soluble and is demonstrated more readily by incubation at room temperature than at 35°C.

178.

C. *Aeromonas* can be differentiated from many other fermentative gram-negative bacilli, such as the *Enterobacteriaceae,* in that they are oxidase positive. Isolates are ONPG and catalase positive. On sheep blood agar medium, colonies are beta-hemolytic. *A. hydrophila* is found in soil and water and has been isolated from a variety of human infections.

179.

A. *Edwardsiella tarda* is a motile member of the family *Enterobacteriaceae* and as such is characteristically peritrichously flagellated. These organisms are infrequently isolated in the clinical laboratory. Biochemically they may initially resemble *Salmonella* in many ways, such as hydrogen sulfide production and the inability to ferment lactose.

180.

B. The porphyrin test is commonly used to test for the X factor (hemin) requirement of *Haemophilus* spp. A positive test result indicates that the organism possesses the enzymes to convert aminolevulinic acid (ALA) into porphyrins and, therefore, would not require hemin. If porphyrins are produced, this rapid test will show red fluorescence under UV light after a 4-hour incubation period.

181.

A. *Bordetella bronchiseptica* in humans produces either a respiratory illness or wound infections. The organism is a part of the normal respiratory flora of laboratory animals such as rabbits and guinea pigs. *B. bronchiseptica* may cause problems for researchers because it can cause outbreaks of bronchopneumonia in experimental animals. It also causes kennel cough in cannines.

182.

B. *Escherichia coli* is frequently the etiologic agent of community-acquired cystitis. This agent can be easily recognized by its fermentation of lactose, negative citrate reaction, and positive indole test. On eosin methylene blue agar, *Escherichia coli* produces characteristic dark colonies with a metallic sheen.

183.

C. "Satellitism" is the name given to the appearance of colonies of *Haemophilus influenzae* on sheep blood agar medium around colonies of organisms that provide an essential growth factor. *H. influenzae* requires both hemin and NAD. Colonies of some organisms, such as *Staphylococcus* and *Neisseria,* produce NAD, which diffuses into the surrounding agar and enables *H. influenzae* to grow.

184.

A. *Bordetella pertussis* is the etiologic agent of pertussis, or whooping cough. On Bordet-Gengou or Regan-Lowe agars, the organism forms small, round colonies that resemble mercury droplets. A nasopharyngeal swab is recommended as the optimal specimen for the recovery of this agent.

185.

C. *Neisseria meningitidis* is the etiologic agent of one form of inflammation of the meninges, known as epidemic cerebrospinal meningitis. Infection with *Bordetella pertussis* produces the highly contagious upper respiratory infection pertussis. Both diseases are spread by droplet infection or fomites contaminated with respiratory secretions. The microorganisms are present in greatest numbers in the upper respiratory tract, and specimens for isolation and identification should be collected on nasopharyngeal swabs.

186.

C. *Salmonella* Typhi is commonly spread by chronic carriers. Without treatment, this enteric bacillus can be carried throughout a person's lifetime and is sequestered most often in the gallbladder. Carriers are usually asymptomatic, and the presence of the organism can be confirmed only by isolation and identification in the clinical laboratory.

187.

A. *Brucella* infects cattle and may be transmitted to humans by the ingestion of contaminated milk or other dairy products. Milk is able to support the growth of many clinically significant microorganisms, which may often be ingested in unpasteurized dairy products. Meliodosis and glanders are caused by *Burkholderia pseudomallei* and *B. mallei,* respectively. Pontiac fever is caused by *Legionella pneumoniae*. None of these is transmitted by milk.

188.

B. *Pseudomonas aeruginosa* is a major cause of hospital-acquired infections. These opportunistic organisms are able to survive in moist environments for prolonged periods and may be transferred to immunocompromised patients. *Pseudomonas* infections in recent years have accounted for as much as 10% of nosocomial infections.

189.

B. A variety of media has been developed to aid in the isolation of *Neisseria gonorrhoeae* from specimens containing mixed flora. Examples include Martin-Lewis, modified Thayer-Martin, GC-Lect, and New York City media. The most commonly used nonselective medium for the isolation of *N. gonorrhoeae* is chocolate agar. Cefsulodin-irgasan-novobiocin (CIN) is a selective and differential medium for the isolation of *Yersinia enterocolitica* and *Aeromonas*.

190.

A. Hektoen enteric agar was developed to improve the isolation of *Shigella* and *Salmonella* from stool specimens. The selective nature of this agar is due to bile salts. The medium also contains three carbohydrates—lactose, sucrose, and salicin—along with a pH indicator to detect carbohydrate fermentation. Fermentative organisms turn the medium yellow. Ferric ammonium citrate and sodium thiosulfate are included in the medium to detect H_2S production. H_2S-producing organisms appear as black-centered colonies.

191.

A. Thiosulfate-citrate-bile salt-sucrose (TCBS) agar is recommended for use in the selective isolation of *Vibrio* spp. associated with cholera, diarrhea, or food poisoning. The selective agent in this medium to inhibit gram-positive organisms is oxgall, a naturally occurring substance containing bile salts and sodium cholate. Sucrose is the carbohydrate in the medium. *V. cholerae* and *V. alginolyticus* ferment sucrose and appear as large yellow colonies. *V. parahemolyticus* is unable to ferment sucrose and exhibits colonies with blue to green centers.

192.

A. The oxidase test detects those organisms that produce the enzyme cytochrome oxidase. A 1% solution of dimethyl- or tetra-methyl-*p*-phenylenediamine dihydrochloride is applied to filter paper, and the test organism is then rubbed into the impregnated area. Because Nichrome wire may cause a false-positive result, a platinum or plastic loop or wooden applicator stick should be used to pick the colony. The rapid development of a dark purple color in the area where the organism was inoculated is a positive oxidase test.

193.

B. Strains of *Haemophilus* able to synthesize heme are identified by the porphyrin test. Species such as *H. influenzae,* which require heme, would give a negative test result, whereas *H. parainfluenzae* would be positive. A red color is indicative of a positive reaction in this test.

194.

B. Direct fluorescent antibody (DFA) test results for *Bordetella pertussis* are rapid but presumptive. Both positive and negative test results must be confirmed by culture. The quality of the test result depends greatly on the experience of the microscopist, the quality of the antibody, and the microscope. Material collected from the nasopharnyx with calcium alginate swabs for DFA and culture is recommended.

195.

C. Exotic pets such as iguanas, snakes, and turtles are known to carry *Salmonella*. Young children who do not practice good handwashing after touching family pets are particularly at risk for infection. Natural medicinal products made from snakes or other animals known to carry *Salmonella* have been implicated in cases of salmonellosis.

196.

B. Keratitis is a serious clinical condition that is characterized by inflammation of the cornea, which, if not appropriately treated, may lead to loss of vision. *Pseudomonas aeruginosa* is the most common agent of bacterial keratitis associated with lens-cleaning solution. Pseudomonads are opportunistic pathogens that are commonly associated with contaminated fluids.

197.

B. *Helicobacter pylori* is implicated as an etiologic agent of gastritis and peptic ulcer disease. This organism can be demonstrated in gastric biopsy specimens. *H. pylori* produces a strong positive urease test result.

198.

D. New York City (NYC) medium was developed by the New York City Public Health Laboratory for the isolation of *Neisseria gonorrhoeae*. It is a horse serum–based medium that is selective by the addition of colistin, vancomycin, and amphotericin B. Modified Thayer-Martin, another commonly used selective medium for *N. gonorrhoeae,* is chocolate based.

199.

B. Like *Pseudomonas aeruginosa, Burkholderia cepacia* is a ubiquitous opportunistic organism. Although *P. aeruginosa* is by far the most important cause of lower respiratory tract infections in patients with cystic fibrosis, *B. cepacia* is also a significant cause of morbidity. Both of these bacteria are oxidase positive and will grow on MacConkey agar. *P. aeruginosa* typically produces a green discoloration of the medium it is grown on.

200.

A. *Eikenella corrodens* can be normal flora of the oral cavity of humans. It is a weak pathogen that is associated with polymicrobial abscesses of the oral cavity. *E. corrodens* will grow on sheep blood and chocolate agars. Some strains will produce pitting of the agar.

Mycobacteria

201.

A. Mycobacteria characteristically possess a high lipid content, unlike gram-positive cocci and gram-negative bacteria. The high lipid content acts to protect these organisms from dehydration and the lethal effects of alkali, various germicides, alcohol, and acids. Thus, these bacteria do not stain well with the Gram stain, and an acid-fast staining technique must be used.

202.

D. The optimal growth temperature of *Mycobacterium xenopi* is 42°C, which enables its survival and replication as an environmental contaminant in hot water systems. Human infections caused by *M. xenopi* are rare. The majority of clinically significant *Mycobacterium* spp., those not known to cause cutaneous infections, have an optimal growth temperature of 37°C.

203.

B. Growth on MacConkey agar is a test used for differentiation of rapidly growing mycobacteria. The MacConkey agar used for mycobacteria identification is a different formulation than that used for enterics, in that crystal violet is omitted. A MacConkey agar plate is inoculated with a 7-day broth culture of the test organism. The inoculated plate is then incubated at 37°C. Plates are checked for growth at 5 days, and if no growth is detected, they are checked daily until day 11, at which time they are discarded as negative. *M. fortuitum* and *M. chelonei* are the only mycobacteria able to grow on MacConkey agar in 5 days.

204.

D. *Mycobacterium scrofulaceum* is defined as a scotochromogen because of its characteristic of producing pigmentation in the dark. This slowly growing *Mycobacterium* is a cause of cervical adenitis and other types of infections predominantly in children. Therapy may require susceptibility studies that include the secondary drugs, because the organism is known in some cases to be resistant to isoniazid and streptomycin.

205.

C. The *N*-acetyl-L-cysteine-sodium hydroxide (NALC-NaOH) method is recommended because the addition of NALC allows the concentration of NaOH to be reduced to 2%. The NALC is a mucolytic agent that frees trapped organisms in the sample, and the NaOH acts as a decontaminant. The optimal treatment reduces the numbers of indigenous microorganisms present in the sample without significantly reducing the number of tubercle bacilli.

206.

C. Hansen disease (leprosy) is caused by *Mycobacterium leprae.* Chronic skin lesions and sensory loss characterize this disease. Skin or biopsy specimens taken from within the margin of a lesion will demonstrate the causative agent. Cultures of this agent on artificial media, unlike other mycobacteria, have not been successful. Cultivation can be accomplished by injecting bacilli into the foot pads of mice or systemically into armadillos.

207.

C. The Centers for Disease Control and Prevention has adopted the diagnostic standards recommended by the American Thoracic Society as published in 1981. This is a method of reporting the number of acid-fast bacilli observed in fuchsin-stained smears of clinical material. Up to nine acid-fast bacilli per field should be reported as a positive, at 3+.

208.

C. Acid-fast bacilli can be demonstrated in stained smears of clinical material using the Ziehl-Neelsen or Kinyoun acid-fast stains. The Kinyoun carbol-fuchsin method uses a higher concentration of phenol in the primary stain to accelerate the staining process. Therefore, unlike the Ziehl-Neelsen stain, the Kinyoun stain does not need to be heated.

209.

C. Members of the genus *Mycobacterium* are characterized as obligate aerobic bacilli that, because of the high lipid content of their cell wall, exhibit acid fastness when stained. Most species pathogenic for humans are slowly growing. *Mycobacterium kansasii* is a slowly growing photochromogen.

210.

C. *Mycobacterium ulcerans* and *Mycobacterium marinum* have both been implicated in skin infections. Their predilection for surface areas of the body is related to their optimal growth temperature range of 30–32°C. At body temperature (37°C) or higher, these organisms grow poorly, if at all.

211.

B. Rapid development of drug resistance is a concern in the treatment of tuberculosis. Patients are treated generally with a combination of at least two of the primary drugs, such as isoniazid, rifampin, ethambutol, and pyrazinamide. Because of the slowly growing nature of the bacteria, they are innately resistant to a number of agents.

212.

B. NALC (*N*-acetyl-L-cysteine) is a mucolytic agent used in decontamination and digestion procedures for the recovery of mycobacteria. NALC liquefies mucus, releasing trapped bacteria. NaOH, between 2 and 4%, is frequently used as a bactericidal agent to prevent the overgrowth of normal flora in clinical specimens.

213.

D. A positive tuberculin skin test reaction is an example of a hypersensitivity reaction. Tuberculin preparations are prepared from culture filtrates, which are precipitated with trichloroacetic acid and are known as purified protein derivative (PPD). A positive test demonstrates an area of induration following an intradermal injection of PPD.

214.

A. *Mycobacterium bovis* causes tuberculosis in cattle. This agent is an etiologic agent of tuberculosis in humans as well, and it must be differentiated from *M. tuberculosis* when recovered from clinical material. Unlike *M. tuberculosis, M. bovis* is negative for niacin production and nitrate reduction.

215.

A. The *M. avium* complex is sometimes referred to as *Mycobacterium avium-intracellulare* complex. These slowly growing bacilli are uncommon in immunocompetent individuals. These bacteria cause disseminated infections in patients with acquired immunodeficiency syndrome and are important causes of morbidity and mortality in these patients.

216.

B. *Mycobacterium gordonae* has been recovered from water stills, faucets, and bodies of water in nature, which is why it has been called the "tap water scotochromogen." These organisms are not considered to be pathogenic for humans, but because they may be recovered as contaminants, their identification is recommended. Members of Runyon group II, they are slow growing and form yellow-orange colonies that do not depend on exposure to light.

217.

C. The human tubercle bacillus is *Mycobacterium tuberculosis*. Growth of this well-known human pathogen appears in 2 to 3 weeks when incubated at 35°C. These niacin-positive mycobacteria form dry heaping colonies that are buff colored.

218.

C. *Mycobacterium marinum* is the causative agent of "swimming pool granuloma." Typically, patients with abraded skin come in contact with water containing this agent and develop granulomatous skin lesions. Lesions generally occur on the extremities, because the skin temperature is close to the organism's optimal growth temperature of 25–32°C.

219.

B. *Mycobacterium bovis* is susceptible to 5 µg/mL of thiophene-2-carboxylic acid hydrazide (T2H). This *Mycobacterium* is associated with cattle and is rarely isolated from humans in the U.S. Growth occurs only at 35°C and is differentiated from other mycobacteria by its susceptibility to T2H.

220.

A. Auromine-rhodamine is a fluorescent stain used to visualize the mycobacteria. The bacteria retain the stain and will appear bright yellow against a black background. Because it is easier to see the bacilli, this stain is more sensitive than a fuschin-based stain (e.g., Ziehl-Neelsen). The calcofluor white stain is a fluorescent stain used to visualize fungi.

221.

B. *M. kansasii* is the most commonly isolated photochromogen in the U.S. It is the second most commonly isolated nontuberculosis *Mycobacterium* sp. behind *M. avium* complex. *M. kansasii* produces chronic lung disease resembling classic tuberculosis.

222.

B. Skin cultures for the recovery of *Mycobacterium* spp. should be incubated at 30°C. The mycobacteria associated with these type of infections include *M. ulcerans, M. marinum,* and *M. haemophilium.* The optimal temperature for these slow growers is 30°C.

223.

D. Lowenstein-Jensen, Lowenstein-Jensen-Gruft, and Middlebrook media are commonly used for the isolation of the mycobacteria. Chocolate agar will support the growth of *Mycobacterium haemophilium;* however, chocolate agar is not routinely used for cultures of mycobacteria. This species requires ferric ammonium citrate or hemin for growth and also has an optimal temperature of 30°C.

224.

A. The mycobacteria are only slightly more resistant to the decontamination procedures than other bacteria. Therefore, it is only appropriate to decontaminate specimens for mycobacteria that are contaminated with normal flora. Because sputum passes through the oral cavity, it contains a large amount of normal oral flora. The other specimens listed are typically sterile and lack normal flora.

225.

C. *Mycobacterium leprae* is the causative agent of Hansen disease (leprosy). This bacterium cannot be grown on artifical media and requires laboratory animals for cultivation. The optimal temperature for *M. leprae* is lower than the core body temperature of 37°C; therefore, infections generally occur in the skin in the extremeties. The bacteria are likely spread from nasal secretions and not the lesions; they are not highly contagious, as most people believe.

Anaerobic Bacteria

226.

B. Improperly home-canned foods, especially low-acid-content vegetables, cause the majority of the cases of food-borne botulism. The ubiquitous nature of *Clostridium botulinum* enables the spores to contaminate a variety of foods. Contamination and subsequent germination under anaerobic conditions stimulate toxin formation. The patient becomes ill following the ingestion of food that contains nanograms of preformed toxin.

227.

B. Zoonotic diseases are diseases of animals that are transmissible to humans. Leptospirosis is primarily a disease of small animals such as rabbits. It is contracted by humans through contact with infected carcasses or contaminated water. *Bacillus anthracis* is found in the environment. Anthrax is transmitted to humans by exposure to contaminated animal products such as cattle hides, goat hair, or wool. Brucellosis is associated with a variety of animals.

228.

A. Isolates of the anaerobic, spore-forming bacillus *Clostridium perfringens* characteristically produce a pattern of double zone hemolysis on sheep blood agar plates. A Gram stain of such colonies should demonstrate a medium-sized gram-positive bacillus that does not contain spores. For further identification the isolate should be inoculated on an egg yolk agar plate to detect lecithinase production.

229.

D. *Prevotella melaninogenica* was isolated from this cervical abscess. This anaerobic organism is part of the indigenous microflora of the respiratory, gastrointestinal, and genitourinary tracts and is considered a significant human pathogen. The black pigment appears after several days when growing on laked blood agar plates. Prior to pigmentation, this isolate can be presumptively identified by its brick-red fluorescence under UV light. Pigmented *Porphyromonas* spp. are asaccharolytic.

230.

A. *Clostridium tetani* is an obligate anaerobe. Spores are widespread in nature and cause disease by contaminating puncture wounds. The exotoxin, tetanospasmin, produced by this organism is one of the most powerful bacterial toxins known.

231.

B. The gram-positive, non-spore-forming, anaerobic bacillus *Actinomyces israelii* is a slowly growing organism that is considered to be an opportunistic pathogen. Colonies may not be visible before 5 to 7 days or longer. When colonies are seen, they appear white, opaque, lobate, irregular, and shiny and are described as resembling a molar tooth. *A. israelii* is part of the indigenous flora of the human mouth, and a few *Actinomyces* spp. have been found to inhabit the vagina. Pathogenesis generally involves trauma to tissues of a mucous membrane and the introduction of this endogenous organism.

232.

A. The predominant indigenous flora of the human intestinal tract is anaerobic, gram-negative, non-spore-forming bacilli. The *Bacteroides fragilis* group, in particular, predominates in the fecal flora. Trauma involving the intestinal area or bowel surgery predisposes patients to an endogenous anaerobic infection. Although these organisms are present in large numbers, their routine identification in fecal cultures is of no diagnostic value.

233.

A. Members of the *Bacteroides fragilis* group, the most commonly isolated anaerobes and a predominant part of the indigenous fecal flora in humans, are not inhibited by the presence of bile. Bile-esculin agar plates are used for the selection and presumptive identification of the *B. fragilis* group. Although not used as a component of selection media for the *Bacteroides* group, it is important to note that, in general, gram-negative, non-spore-forming, anaerobic bacilli are susceptible to penicillin. The *B. fragilis* group is an exception in that it is known to be resistant to penicillin.

234.

B. The second most commonly encountered group of anaerobes in human infections is the anaerobic, gram-positive cocci. They may account for one-fourth of all anaerobes isolated in clinical laboratories. Estimating their clinical significance, however, is often difficult. Important isolates include *Fingoldia magna* (formerly *Peptostreptococcus magnus*) and *Peptostreptococcus anaerobius*.

235.

A. The closed chest abscess described is characteristic of human actinomycosis, which is caused by *Actinomyces israelii,* an anaerobic, gram-positive, non-spore-forming bacillus. The organism is not acid-fast, which helps to differentiate it from *Nocardia* spp. Actinomycotic pus characteristically shows "sulfur granules" or solid yellow particles made up of masses of the filamentous bacilli seen on the Gram stain in Color Plate 29■.

236.

A. Kanamycin-vancomycin laked blood (KVLB) agar is selective for the *Prevotella* and *Bacteroides* spp. Presumptive identification of *B. fragilis* group can be accomplished utilizing its antimicrobial resistance pattern. *Bacteroides* spp. are resistant to vancomycin and kanamycin, unlike *Fusobacterium* spp., which are resistant to vancomycin but susceptible to kanamycin. A KVLB agar plate should be part of the primary plating media for anaerobic cultures.

237.

C. *C. difficile* is an important cause of a hospital-acquired infection commonly called pseudomembranous colitis. Hospitalized patients treated with broad-spectrum antimicrobial agents become colonized when their normal intestinal flora is diminished. The most rapid and accurate diagnostic method is detecting toxins A and/or B in stool specimens. Cycloserine-cefoxitin-fructose agar (CCFA) is the recommended selective medium for *C. difficile,* although few laboratories attempt isolation.

238.

D. Although it has a gram-positive-like cell wall, *Mobiluncus* stains gram-variable to gram-negative. This curved and motile bacillus seems to contribute to the pathology of bacterial vaginosis (BV). A Gram stain of the discharge that is produced in this condition can be used for the detection of these distinctively curved organisms. The presence of "clue cells," gram-variable pleomorphic bacilli on vaginal epithelial cells, is diagnositic of BV.

239.

B. Infant botulism or "floppy infant" syndrome is seen in children up to 6 months of age. This infectious process begins with the ingestion of food contaminated with spores of *Clostridium botulinum*. Following ingestion, viable spores are carried to the lower bowel, where they germinate and elaborate the powerful neurotoxin that produces the characteristic flaccid paralysis.

240.

D. *Propionibacterium acnes* is the most frequently isolated of all the gram-positive, non-spore-forming, anaerobic bacilli. It is a part of the normal human bacterial flora and predominates on the surface of the body, but may also be recovered from the upper respiratory tract, intestines, and urogenital tract. This organism is a common contaminant of blood cultures because of its presence on the skin. Care in the preparation of the skin before venipuncture helps to eliminate confusion caused by the recovery of this anaerobic isolate.

241.

D. The spore of *Clostridium tetani* is located terminally and is larger than the sporangium. Characteristically, when seen on Gram stain, the cells of *Clostridium tetani* resemble a drumstick or tennis racket. Spores can be readily seen in late growth phase cultures incubated at 37°C.

242.

C. *Fusobacterium nucleatum,* a gram-negative, anaerobic bacillus, is part of the indigenous microbial flora of the respiratory, gastrointestinal, and genitourinary tracts. It is frequently implicated as the causative agent in metastatic suppurative infections such as brain abscesses. These pale-staining bacilli characteristically appear as long, thin bacilli with pointed ends.

243.

C. *Propionibacterium* spp. are non-spore-forming, anaerobic, gram-positive bacilli. *Clostridium* spp. typically form spores, although it is difficult to induce some species to form spores *in vitro*. *Veillonella* is a gram-negative coccus, and *Fusobacterium* is a gram-negative bacillus.

244.

C. *Clostridium botulinum* is the causative agent of botulism, a disease produced by an exotoxin that acts on the central nervous system. Types A, B, E, and F are causes of human botulism; types C and D and less commonly types A and B are associated with disease in animals and birds. Type G has not been associated with disease in humans or animals. This anaerobic organism produces oval, central, or subterminal, spores that germinate in food products or less commonly in wounds.

245.

D. Resazurin is an E_h indicator used in anaerobic culture media. When the oxygen concentration is reduced, the resazurin indicator is colorless. A pink color in the medium indicates aeration and an unsuitable environment for the preservation of obligate anaerobic organisms.

246.

B. *Fusobacterium nucleatum* is the most frequent clinical isolate within the genus *Fusobacterium*. These anaerobes are part of the indigenous flora of human mucous membranes, oral cavity, intestine, and urogenital tract. *F. necrophorum* is, however, much more virulent.

247.

B. *Clostridium septicum* is isolated in the clinical laboratory in cases of serious or often fatal infections. Bacteremia is seen in association with an underlying maglignancy. The most common types of cancer are colon or cecum, breast, and leukemia or lymphoma.

248.

A. Botulinal toxin is the most potent exotoxin known. When absorbed, this exotoxin produces the paralyzing disease botulism. Toxin acts in the body by blocking the release of acetylcholine in the neuromuscular junction of the peripheral nervous system, causing muscle paralysis.

249.

B. *Clostridium perfringens* is the species most commonly associated with clostridial myonecrosis or gas gangrene. These soil and water saprophytes most frequently gain entrance to the human body through traumatic wounds. Once they have been introduced into injured tissue, the characteristic syndrome of myonecrosis due to the elaboration of exotoxins may occur. Other species involved with myonecrosis are *C. septicum, C. novyi, C. sordellii,* and *C. histolyticum.*

250.

D. *Eubacterium, Bifidobacterium,* and *Propionibacterium* are all anaerobic, gram-positive, non-spore-forming bacilli. This group of anaerobic microorganisms is difficult to identify in the clinical laboratory and often requires the use of gas chromatography. These organisms are rarely isolated. *Suttonella wadsworthensis* is an anaerobic, gram-negative bacillus.

251.

D. *Clostridium tetani* is an obligate anaerobe. Spores are widespread in the soil. When introduced into a puncture wound, the spores require the reduced oxygen environment produced by the necrotic tissue and poor blood supply in the wound. Cleaning and debridement of the wound is important, as is the administration of a tetanus toxoid booster. Many clostridia require anaerobic conditions for spore formation.

252.

B. Most *Clostridium* spp. are gram-positve, and they generally form spores. Because they are obligate anaerobes, they will not grow on sheep blood or chocolate agars incubated aerobically. *Bacillus* spp. also form spores, but they are facultative anaerobes and would therefore grow on media incubated aerobically. *Lactobacillus* is a non-spore-forming, gram-positive bacillus, and *Prevotella* is a gram-negative bacillus.

253.

C. *Lactobacillus* spp. are normal flora of the vagina and digestive tract and are rarely pathogenic. They are aerotolerant anaerobes and will produce alpha-hemolysis on sheep blood agar plates incubated aerobically. These organisms can also produce a green discoloration on chocolate agar.

254.

C. *Clostridium perfringens* produces spores that are oval and central in location but that are rarely seen in foods or on laboratory cultures. This organism is divided into five types, A to E, based on the quantities and types of exotoxins produced. Type A is responsible for human cases of myonecrosis and food poisoning. Alpha-toxin or lecithinase is produced by all strains of *C. perfringens*.

255.

D. *Clostridium tetani* is a strict anaerobe that is motile and produces terminal round spores. Biochemically it does not utilize carbohydrates, with the rare exception of glucose. *C. tetani* is gelatinase and indole positive but is nonproteolytic and H_2S negative. The clinical manifestations of tetanus are the result of the release of a neurotoxic exotoxin.

256.

C. *Fusobacterium nucleatum* characteristically appears on Gram stain as a gram-negative rod with pointed ends. Its growth is inhibited by a 1-μg kanamycin disk and the presence of bile. The *Bacteroides fragilis* group and the pigmented species *Prevotella* and *Porphyromonas* are not inhibited by kanamycin.

257.

D. *Prevotella melaninogenica* can be rapidly presumptively identified on media containing laked blood with the use of an ultraviolet light source. This important anaerobic pathogen can be differentiated after 5 to 7 days' incubation by its black pigmentation. The use of ultraviolet light enables a more rapid differentiation because of the appearance of a brick red fluorescence before the pigment is demonstrated.

258.

C. The symptoms of *Clostridium difficile* infection are toxin mediated. This organism is known to cause pseudomembranous colitis associated with the use of antimicrobial therapy. Cytotoxins can be directly detected in stools by enzyme immunoassays. These tests have generally replaced the use of cell culture monolayers, which were examined after incubation for evidence of cytotoxicity (i.e., cytopathic effect).

259.

B. A reverse CAMP test aids in the identification of *Clostridium perfringens*. In this test, a single straight streak of *Streptococcus agalactiae* is made down the center of the plate. Suspected *C. perfringens* isolates are inoculated at right angles to the *S. agalactiae* inoculum. After anaerobic incubation, *C. perfringens* will exhibit enhanced hemolysis at the intersection where the two species meet.

260.

A. The identification of *Peptostreptococcus anaerobius* is made easier by the use of the sodium polyethanol sulfonate (SPS) disk. The test is performed by growing the organism in the presence of a disk impregnated with SPS. A zone of inhibition of 12–18 mm around the disk is considered sensitive and a presumptive identification of this organism.

261.

C. *Fusobacterium nucleatum* is a thin gram-negative rod with pointed ends and a slightly curved appearance in fresh isolates. As the bacteria are subcultured, they may lose their curved appearance and appear as thin rods. *F. nucleatum* is found in human specimens and is considered clinically significant.

262.

D. Anaerobes are a major cause of brain abscess. *Peptostreptococcus* spp. are associated with human disease, usually in polymicrobial infections, and can be seen on a Gram stain of clinical material. The characteristic Gram stain morphology of *Fusobacterium* would enable a physician to make a presumptive identification of the presence of anaerobic flora in this clinical case.

263.

C. *Bacteroides fragilis* stimulates abscess formation. The capsule is a contributing factor to the pathology produced by this anaerobe. *B. fragilis* is the most common anaerobic gram-negative bacillus isolated in the clinical laboratory.

264.

D. The *Bacteroides fragilis* group is among the most antimicrobial-resistant anaerobes. Beta-lactamase production is responsible for their resistance to the penicillins. These anaerobes are also resistant to first-generation cephalosporins and aminoglycosides.

265.

B. The common agents in cases of aspiration pneumonia are oral anaerobes, such as the black-pigmented *Prevotella* and *Porphyromonas,* and *Bacteroides,* fusobacteria, and anaerobic streptococci. These endogenous organisms, when in an abnormal site, possess virulence factors that enable them to produce disease. Often these are polymicrobic infections mixing anaerobes with aerobic or facultative organisms such as *Enterobactericeae* or *Staphylococcus aureus. Mobiluncus* is not associated with aspiration pneumonia.

266.

B. Methylene blue strips are the most commonly used oxidation-reduction (E_h) indicators. When anaerobic conditions are achieved, the methylene blue indicator will turn from blue (oxidized) to white, indicating reduction. Resazurin, another E_h indicator, is used in anaerobic transport systems and anaerobic culture media such as the prereduced anaerobically sterilized (PRAS) system. Resazurin when oxidized is pink; when reduced, the color fades to white, indicating anaerobiosis.

267.

A. An area of precipitate in the agar around the colonies indicates that the organism produced lecithinase. Lecithinase (alpha-toxin) cleaves lecithin in the medium, producing an insoluble product. *Clostridium perfringens* is positive for lecithinase.

268.

B. *Peptostreptococcus niger* produces a pigment that begins olive-green and gradually becomes black. This is the only species in the genus. It is a weak pathogen sometimes found in polymicrobial infections.

269.

C. *Clostridium perfringens* is one of the most important causes of food-borne diseases in the U.S. The bacterial spores can survive cooking (typically found in meats and gravies), and upon cooling they germinate into vegetative cells. When the bacteria are ingested, they sporulate in the intestinal tract. The enterotoxin is a spore coat protein made in excess and released by the bacteria.

270.

C. *Lactobacillus* spp. are found as normal flora in the gastrointestinal and female genital tract. The bacteria produce acids from the metabolism of carbohydrates, resulting in an acid environment in the vagina. If the population of lactobacilli decreases, the vaginal pH will rise toward neutrality. This favors the growth of other bacteria, such as *Mobiluncus,* that can result in bacterial vaginosis.

Chlamydia, Rickettsia, and *Mycoplasma*

271.

B. *Chlamydia trachomatis* is the causative agent of inclusion conjunctivitis, trachoma, and genital tract infections, including lymphogranuloma venereum. Trachoma is a primary cause of blindness worldwide. The disease is preventable, but when it is not treated, the organism produces hypertrophy of the lymphoid follicles on the inner surface of the upper eyelid. This process causes the upper eyelid to evert (entropion), which ultimately leads to blindness.

272.

A. The *Chlamydia* and *Chlamydiophila* are obligate intracellular parasites. They require ATP from their host cell. As such, these bacteria cannot be grown on artificial media. They can be cultivated in cell cultures.

273.

C. *Mycoplasma pneumoniae* causes primary atypical pneumonia. The pneumonia is atypical in that it is milder than the pneumonia caused by *Streptococcus pneumoniae.* Chest X-rays of patients with atypical pneumonia may show bilateral infiltrates, although physical examination reveals few chest findings.

274.

D. Provided that arginine is added for *Mycoplasma hominis,* SP4 agar or broth can be used for the growth of *M. pneumoniae* and *M. hominis. M. pneumoniae* is a slow-grower, so most infections are diagnosed by serologic assays. M. *pneumoniae* is an important respiratory tract pathogen of humans. It is found only in humans and is typically spread person to person.

275.

C. Human infections with *Chlamydiophila psittaci* (psittacosis) occur after exposure to infected birds and their droppings. A true zoonosis, psittacosis is a disease of birds that may be contracted by humans. The disease produced by this organism may be mild or fulminant, the latter of which has a high mortality rate. Clinical manifestations of the disease include severe headache, weakness, and mild pulmonary symptoms.

276.

C. Unlike rickettsial diseases, no rash occurs in *Coxiella burnetti* infections. The organism is an obligate intracellular parasite that is able to survive for long periods in the environment. It causes a zoonosis and is transmitted to humans by inhalation and contact with fomites. Infections can also be acquired by ingestion of unpasteurized milk.

277.

A. Mycoplasmas are small, pleomorphic organisms that lack a cell wall and are best visualized by darkfield or phase microscopy. Penicillin is not an effective treatment because of their lack of a cell wall, and isolation requires media supplemented with peptone, yeast extract, and serum. Species of the genus *Mycoplasma* are well-known human pathogens that cause a variety of disease processes.

278.

A. *Chlamydia trachomatis,* a leading cause of blindness, can be detected in corneal scrapings of suspected cases of trachoma and inclusion conjunctivitis. Clinical material can be examined directly using fluorescent antibody techniques or can be cultured on McCoy cells. Trachoma is a chronic inflammatory process of the conjunctiva that results in corneal involvement.

279.

C. *Mycoplasma hominis, M. genitalium,* and *Ureaplasma urealyticum* have been linked to human genital infections. These species can also be isolated from asymptomatic sexually active adults. *M. pneumoniae* is primarily a respiratory tract pathogen.

280.

B. *Chlamydiophlia pneumoniae* is an important cause of sporadic and epidemic lower respiratory tract disease characterized as atypical pneumonia. The organism is a human pathogen spread person to person. Most infections are diagnosed serologically. Tetracycline and erythromycin are effective treatments. The organism has been associated epidemiologically to coronary heart disease.

281.

C. Polymerase chain reaction DNA amplification has been shown to be more sensitive than cell culture and nearly 100% specific for the detection of *Chlamydia trachomatis*. Suitable specimens for detection are cervical secretions and urine. When confirmation of *C. trachomatis* is needed, tissue culture remains the method of choice.

282.

C. *Mycoplasmas* are implicated in a variety of human infections. *M. pneumoniae,* in particular, is a clinically important respiratory tract pathogen. When grown on culture media, colonies, most notably *M. hominis,* are said to have a "fried egg" appearance because the central portion of the colony has grown into the agar and thus appears more dense and is slightly raised.

283.

A. Q fever is caused by infection with *Coxiella burnetii,* which has unique characteristics. Unlike other rickettsiae, this organism is able to resist heat and drying for long periods and does not rely on an arthropod vector for transmission. Infectious fomites such as dust from contaminated cattle hides and fluids released during birth are considered the primary modes of infection.

284.

A. Direct microscopic examination for *Rickettsia* organisms is possible using such stains as Giemsa, Machiavello, or Gimenez. The recommended procedure is the nonspecific Gimenez stain, which colors the organisms a brilliant red against a green background. The staining technique calls for flooding a thin smear, which has been air dried, with a solution of carbol-fuchsin for 1–2 minutes. After washing with tap water, malachite green is added for 6–9 seconds before the final washing with tap water.

285.

D. Transovarian passage from generation to generation in ticks perpetuates *Rickettsia rickettsii* for several generations outside an animal host. A blood meal serves to reactivate the rickettsiae carried by the arthropod vector. Rodents and small mammals are the natural reservoirs for the rickettsiae that cause this form of spotted fever.

286.

D. Flying squirrels, *Glaucomys volans,* are associated with cases of the sylvatic form of typhus in the U.S. The squirrel louse transmits the organism among the squirrel population. Humans contract the disease through association with infected squirrels. The disease is more common in the winter months, when squirrels seeking shelter enter dwellings.

287.

A. Humans who have had the classic form of typhus may remain infected with the causative agent *Rickettsia prowazekii.* Relapses or recrudescence of disease may occur in these persons years or decades after the initial attack. The latent form of infection is known as Brill-Zinsser disease and may serve as an interepidemic reservoir for epidemic typhus.

288.

D. Murine typhus is transmitted to humans by fleas infected with *Rickettsia typhi.* Prevalent in the southern U.S., it is primarily a disease of rodents and is sometimes transmitted to humans. Control of disease outbreaks is related to rodent (rat) control and the related rat flea population. The symptoms of murine or endemic typhus are similar to those of the classic epidemic form seen in Europe.

289.

C. *Mycoplasma pneumoniae* produces hydrogen peroxide, which causes lysis of red blood cells. Suspected *M. pneumoniae* growing on SP4 agar can be overlayed with a molten agar containing guinea pig red blood cells. After incubation, a small zone of beta-hemolysis will be seen around the colonies.

290.

D. 10 B broth is used with genital specimens to isolate *Ureaplasma urealyticum.* The bacterium requires urea and produces a strong alkaline pH because of the activity of urease. The bacteria are slow growers and form tiny colonies. The broth will typically not appear turbid.

Spirochetes

291.

D. Cardiolipin is a tissue lipid produced as a byproduct of treponemal infection. Nontreponemal tests for syphilis take advantage of antibodies made to cardiolipin. The most commonly used tests are the rapid plasma reagin (RPR) for serum and the Venereal Disease Research Laboratory (VDRL) for cerebrospinal fluid.

292.

A. *Leptospira* spp. are most reliably detected during the first week of illness by the direct culturing of a blood sample. The media of choice are Fletcher semisolid and Stuart liquid medium, both of which are supplemented with rabbit serum. One or two drops of the patient's blood are added to 5 mL of culture medium, which is incubated in the dark at 30°C or room temperature for up to 6 weeks. After the first week of disease and lasting for several months, the urine becomes the specimen of choice for isolation of the organism. Direct microscopic examination is not reliable for detection because of the low numbers of organisms normally present in body fluids.

293.

C. Syphilis is caused by *Treponema pallidum* subsp. *pallidum.* Congenital syphilis occurs when a pregnant woman has a septicemia, and the spirochetes cross the placenta and infect the fetus. Infection can affect fetal development and cause premature birth or fetal death, or the pregnancy may go to term. Following *in utero* infection, the infant is most often born with lesions characteristic of secondary syphilis; perinatal death is not an uncommon consequence of infection.

294.

C. The description given is characteristic of members of the genus *Leptospira.* Blood and other fluids, such as cerebrospinal fluid and urine, are examined by direct darkfield microscopy and stained preparations for the presence of these organisms in suspected cases of leptospirosis. The number of organisms present in clinical samples is low, and detection is difficult even when concentration methods are used. Cultural and serologic tests are available for the diagnosis of disease produced by these organisms.

295.

B. The human body louse, *Pediculus humanus,* is the vector for *Borrelia recurrentis*. Pathogenic species not only have specific vectors but also well-defined geographical distributions. Epidemic relapsing fever is found in Ethiopia, Sudan, and parts of South America.

296.

C. The antigen in the VDRL test is cardiolipin. In this flocculation test, reagin, an antibody-like protein, is produced by infected patients. Reagin binds to cardiolipin-lecithin-coated cholesterol particles, causing the particles to flocculate and indicating a positive test result. The test must be read microscopically.

297.

A. Lyme disease was first described in 1975 following an outbreak in Lyme, Connecticut. The etiologic agent, *Borrelia burgdorferi,* is transmitted to humans by the tick vector *Ixodes dammini.* Clinically the disease peaks in the summer and produces an epidemic inflammatory condition characterized by skin lesions, erythema, headache, myalgia, malaise, and lymphadenitis. Rat-bite fever is caused by *Spirillum minus.* Relapsing fever is caused by *Borrelia,* and Q fever is caused by *Coxiella burnetti.*

298.

B. The basic structure of spirochetes is an outer membrane, cytoplasmic membrane-peptidoglycan complex, cytoplasm, and axial fibrils. The fibrils are attached to the cytoplasmic membrane close to the ends of the cell, extending along the body under the outer membrane. The axial fibrils most closely resemble bacterial flagella and are associated with motility of the organism.

299.

B. Spirochetes are gram-negative, but most do not stain with the Gram stain. Silver impregnation can be used to visualize them in smears. The direct observation using darkfield or phase microscopy is recommended to view these delicate, coiled cells in body fluids or tissue sections.

300.

C. Infections other than syphilis can cause a positive VDRL result. The VDRL test detects an antibody that is not directed against *T. pallidum* antigens. It is a good screening test for syphilis, but it is not highly specific. Confirmation with a specific treponemal test, such as the fluorescent treponemal antibody-absorbed (FTA-ABS) assay, is required.

Antimicrobial Agents and Antimicrobial Susceptibility Testing

301.

A. Standardization of the susceptibility testing procedure is essential for determining the susceptibility of an organism to antimicrobial agents. A #0.5 McFarland standard is used when adjusting the turbidity of the suspension of test organism. A #0.5 McFarland standard has a turbidity consistent with approximately 1.5×10^8 organisms/mL of broth or saline.

302.

D. Hemophilus Test Medium is recommended for use in the disk-agar diffusion susceptibility testing procedure of *Haemophilus*. The testing of *Haemophilus* spp. requires supplemented media to support the growth of these fastidious organisms. *In vitro* growth of *H. influenzae* requires the presence of accessory growth factors: X factor (hemin) and V factor (NAD).

303.

D. Sulfonamides act to interfere with the ability of bacteria to use *p*-aminobenzoic acid, which is a part of the folic acid molecule, by competitive inhibition. These chemotherapeutic agents are bacteriostatic and not bactericidal. The drug sulfisoxazole is a member of this group and is used in the treatment of urinary tract infections, especially those caused by *Escherichia coli,* which must synthesize folic acid for growth.

304.

D. Antimicrobial susceptibility testing is not routinely performed on all bacteria. Certain organisms are predictably susceptible to a variety of antimicrobial agents. Therefore, testing is not usually performed even when these organisms are the etiologic agents of infection. Bacteria for which susceptibility tests are usually not performed include *Streptococcus pyogenes* (group A *Streptococcus*), *Streptococcus agalactiae* (group B *Streptococcus*), and *Neisseria meningitidis.*

305.

C. Inhibitors of peptidoglycan synthesis such as penicillin act to inhibit cell wall development. Bacteria unable to produce peptidoglycan for their cell walls are subject to the effects of varying osmotic pressures. The peptidoglycan component of the cell wall protects the bacterium from lysis.

306.

C. The requirement of 99.9% killing defines the minimum bactericidal concentration (MBC) of an antimicrobial agent. The MBC test is an additional quantitative assessment of the killing effect of a drug on a specific patient isolate. This test, done to evaluate a drug's activity, is sometimes requested in cases of life-threatening infections.

307.

B. Even though clindamycin and erythromycin are in different classes, the mechanisms of resistance are similar. The presence of erythromycin can induce clindamycin resistance. The D-zone test is used to detect the presence of this inducible resistance.

308.

D. The therapeutic effect of antimicrobial therapy is often increased by the use of a combination of drugs. A combination of antimicrobials is said to be synergistic when the sum of their effects is greater than that derived from either drug when tested independently. A tenfold decrease in the number of viable cells from that obtained by the most effective drug in the combination is the definition of synergism. Synergistic combinations of antimicrobials are used primarily in the treatment of tuberculosis, enterococcal endocarditis, and certain gram-negative bacillus infections.

309.

C. Beta-lactamase production by strains of *Haemophilus influenzae* renders them resistant to the antibacterial effect of penicillin and ampicillin. It is recommended that rapid beta-lactamase testing be performed on isolates in life-threatening clinical infections such as meningitis. The rapid tests all rely on this enzyme's ability to act on a beta-lactamase ring and in turn produce a color change, which denotes a positive result due to the production of penicilloic acid.

310.

C. The recommended plating medium for use in both the disk diffusion and tube dilution susceptibility test procedures is Mueller-Hinton. Low in tetracycline and sulfonamide inhibitors, this medium has been found to show only slight batch-to-batch variability. For the susceptibility testing of fastidious organisms (e.g., *Streptococcus pneumoniae*), 5% lysed sheep blood may be added.

311.

B. The Kirby-Bauer or disk-agar diffusion susceptibility test requires that the pH of the agar be tested at room temperature to ensure an optimal range of 7.2–7.4 before use in the procedure. A sample of the Mueller-Hinton medium can be tested by macerating it in distilled water and testing with a pH meter electrode; a surface electrode is acceptable for direct testing. Another acceptable method is to allow the agar to solidify around the electrode of a pH meter and then obtain a reading.

312.

B. Metronidazole, a drug recommended for the treatment of amebic dysentery and trichomoniasis, is a synthetic compound that acts by inhibiting DNA synthesis. The use of this drug for treating anaerobic infections has gained emphasis in light of resistance patterns of many of the commonly recovered anaerobes. Metronidazole is consistently active against all gram-negative, anaerobic bacilli; is able to cross the blood-brain barrier; and is the only agent consistently bactericidal against susceptible isolates.

313.

D. Tobramycin, an aminoglycoside, is the only antibiotic, of those listed, that is bactericidal. Bactericidal antibiotics actually destroy the bacteria, whereas bacteriostatic drugs only arrest the growth of the microorganism. All aminoglycosides, with the exception of spectinomycin, are bactericidal in their activity.

314.

B. The extended spectrum beta-lactamases (ESBLs) confer resistance to the extended spectrum cephalosporins such as ceftriaxone and cefotaxime. ESBLs cleave the antibiotic, inactivating it. So far, ESBLs have only been found in gram-negative bacteria.

315.

C. Most fastidious bacteria do not grow satisfactorily in standard *in vitro* susceptibility test systems that use unsupplemented media. For certain species, such as *Haemophilus influenzae, Neisseria gonorrheae, Streptococcus pneumoniae,* and other *Streptococcus* species, modifications have been made to the standard Clinical and Laboratory Standards Institute (CLSI) methods. In the case of *S. pneumoniae,* current CLSI broth dilution test conditions include cation-supplemented Mueller-Hinton broth with 5% lysed horse blood.

316.

B. *Haemophilus influenzae* should be tested for beta-lactamase production. The test can be performed directly, and the methods are rapid and reliable for the detection of penicillin and ampicillin resistance. Rapid test methods, in general, rely on a color change to detect the presence of this enzyme. A pH indicator may be used to detect the penicilloic acid produced when the beta-lactam ring of penicillin is cleaved, or a color change can be observed when the beta-lactam ring of a chromogenic cephalosporin is hydrolyzed by the enzyme.

317.

D. Tolerance is described as the ability of certain strains of organisms to resist lethal concentrations of antimicrobial agents like penicillin. The growth of these organisms is only inhibited by these cidal drugs. This mechanism of bacterial resistance is attributed to a deficiency of cell wall autolysins.

318.

C. The rise in antimicrobial-resistant isolates of *Mycobacteria tuberculosis* has been an important public health crisis. The accepted methods for determining the *in vitro* antimicrobial susceptibility of mycobacteria are based on the growth of the microorganisms on solid or in liquid media containing a specified concentration of a single drug. Two such methods that have been described and are in common use in the U.S. are the agar proportion method and the BACTEC 460TB radiometric method.

319.

B. Clavulanic acid is a beta-lactamase inhibitor. It can be administered with amoxicillin or ticarcillin and is effective in treating infections caused by beta-lactamase-producing bacteria such as staphylococci, *Klebsiella,* and *Haemophilus influenzae*. Sulbactam and tazobactam are also beta-lactamase inhibitors.

320.

D. Vancomycin, which acts to inhibit cell wall synthesis of susceptible bacteria, is produced by an actinomycete. The main activity of this drug is to inhibit peptidoglycan synthesis, but it also has an effect on other aspects of bacterial metabolism. Vancomycin is a bactericidal antibiotic.

321.

C. Metronidazole, a nitroimidazole derivative, is active against most of the clinically significant anaerobes. Only some of the non-spore-forming, gram-positive anaerobic bacilli and gram-positive anaerobic cocci are resistant to this agent. This drug acts to disrupt bacterial DNA through the production of cytotoxic intermediates.

322.

A. Gentamicin is a member of the aminoglycoside group of antibiotics. These drugs act on the 30S ribosomal subunit to inhibit protein synthesis. Gentamicin is particularly effective against a wide variety of gram-negative bacilli.

323.

B. Bone marrow toxicity is the major complication of chloramphenicol. Reversible bone marrow suppression with anemia, leukopenia, and thrombocytopenia occurs as a direct result of the agent on hematopoiesis. The second form of bone marrow toxicity is a rare but usually fatal aplastic anemia. The mechanism of this response is not known.

324.

A. Standard quality control strains maintained by the American Type Culture Collection (ATCC) should be tested routinely as recommended by the Clinical and Laboratory Standards Institute. Guidelines developed for the quality assurance of the disk-diffusion antimicrobial susceptibility test procedure recommended that the following organisms be used for this purpose: *Pseudomonas aeruginosa* (ATCC 27853), *Staphylococcus aureus* (ATCC 25923), *Escherichia coli* (ATCC 25922), and *Enterococcus fecalis* (ATCC 29212). Cultures of these organisms should be frozen or lyophilized to maintain their antimicrobial susceptibility pattern. Testing should not be done from stored cultures but rather from freshly grown 18- to 24-hour cultures.

325.

C. The chromogenic cephalosporin method is the most sensitive test for detecting the production of beta-lactamase enzymes. This yellow compound will become red if the organism produces the enzyme that breaks the beta-lactam ring. Nitrocefin, the commonly used compound, has a high affinity for most bacterial beta-lactamases.

Procedures and Biochemical Identification of Bacteria

326.

C. Normal flora can offer the host protection against infections by providing competition to pathogenic bacteria. The lungs are not typically colonized with bacterial flora. When diagnosing lower respiratory track infections, procedures such as bronchoscopy or percutaneous transtracheal aspitate are used to obtain a specimen that is not contaminated by upper respiratory tract flora.

327.

B. The flora of the female genital tract changes with age and the associated effects of pH and estrogen concentration in the mucosa. *Lactobacillus* spp. are the predominant flora during childbearing years. Earlier and later in life, staphylococci and corynebacteria predominate.

328.

C. The MIDI system is based on the analysis of fatty acids in the cell wall of microorganisms. The bacteria are grown under standardized conditions, and the fatty acids are extracted. The *Mycobacterium* fatty acids are analyzed by high-performance liquid chromatography. Gas liquid chromatography is used for other bacteria and yeasts. Results are compared to a computerized database.

329.

D. Most commercially available blood culture media contain the anticoagulant sodium polyanetholsulfonate (SPS). Anticoagulation is important because certain bacteria do not survive well within clotted blood. Within the clot, neutrophils and macrophages remain active and phagocytosis can occur.

330.

B. Incubation of inoculated bacteriologic culture media requires that attention be given to optimal temperature ranges, adequate moisture, and proper atmospheric conditions for growth. The optimal atmosphere for many clinically significant isolates is one that contains 5–10% carbon dioxide. Capnophilic environments may be obtained by using incubators equipped with a tank of carbon dioxide and a regulator. Candle jars produce only about a 3% concentration of carbon dioxide. The portable Fyrite carbon dioxide gas analyzer may be used for the daily monitoring of capnophilic incubators.

331.

B. Microbiologic examination of body fluids is less effective when bacteria become trapped in clotted specimens. The most effective anticoagulant for use in the microbiology laboratory is sodium polyanetholsulfonate (SPS) in a concentration of 0.025–0.05%. Fluids known to clot on standing should be transported to the laboratory in a sterile tube containing SPS. This polyanionic anticoagulent is also anticomplementary and antiphagocytic.

332.

B. One of the metabolites in the tricarboxylic acid cycle, citrate can serve as an energy source for some bacteria. The assessment of the ability of an organism to use citrate as its sole carbon source aids in the identification of the family *Enterobacteriaceae*. *Klebsiella pneumoniae* is able to use citrate with the production of alkaline by-products. A blue color and/or growth of the isolate on the streak line or both are indicative of a positive reaction. *Escherichia coli* is citrate negative.

333.

A. Infectious aerosols put laboratory professionals at risk for acquiring many diseases. The handling of clinical specimens that require pipetting, centrifugation, or decanting may produce infectious aerosols. Bacteria frequently are present in greater numbers in aerosol droplets than in the liquid medium.

334.

A. The deamination of the amino acid phenylalanine results in the formation of phenylpyruvic acid. Detection of the activity of this deaminase enzyme is accomplished by adding a 10% solution of ferric chloride to the growth on an overnight agar culture. Formation of a green color in the liquid on the agar slant indicates the presence of phenylpyruvic acid.

335.

C. Mannitol salt agar is highly selective and differential. It is used for the isolation and identification of staphylococcal species. The 7.5% concentration of sodium chloride results in inhibition of most bacteria other than staphylococci. Mannitol fermentation, as indicated by a change in the phenol red indicator, aids in the differentiation of staphylococcal species because most *S. aureus* isolates ferment mannitol (changing the color of the medium to yellow) and most coagulase-negative staphylococci are unable to ferment mannitol.

336.

B. When inoculated on a sulfur-containing medium, organisms that produce hydrogen sulfide will demonstrate a partial blackening of a strip impregnated with a 5% solution of lead acetate. The strip is inserted above the medium in the tube and is secured under the closure, which seals the tube. The presence of liberated, dissolved sulfide gas reacts with the lead on the strip, and a black (lead sulfide) color develops. This test is useful for the detection of weak hydrogen sulfide-producing organisms, because the triple sugar iron agar is not a sensitive indicator.

337.

D. Nitrate reduction is a general characteristic of members of the family *Enterobacteriaceae*. An organism with this ability reduces nitrates, and nitrites are produced in the medium. After incubation, α-naphthylamine and sulfanilic acid are added to the medium. The presence of nitrites is indicated by the production of a red color within 30 seconds. If the reduction has gone on to nitrogen gas, the color change will not occur. The addition of zinc dust with no resulting color change confirms the reduction of nitrate to nitrogen gas.

338.

A. The decarboxylase activities of members of the family *Enterobacteriaceae* are important tests for their identification. When a decarboxylase broth is inoculated with a test organism, the organism first ferments the glucose present, which produces a color change from purple to yellow. The yellow color indicates acid production. An organism that possesses decarboxylase activity will then be able to attack the amino acid present, producing alkaline amines. The lowered pH (acid) activates the decarboxylase enzyme. The amines in turn raise the pH, and a color change from yellow to dark purple results.

339.

B. Oxidative-fermentative (OF) medium was first devised by Hugh and Leifson in an attempt to detect weak acid production from nonfermentative bacilli. By decreasing the amount of peptone (0.2%) used in conventional media, the formation of oxidative products from amino acids, which may neutralize the weak acids produced by the organism, is reduced, and the metabolic reaction can be demonstrated. Demonstration is further facilitated by an increase in the concentration of carbohydrate (1.0%) in the medium, along with a semisolid consistency. Bromthymol blue is the pH indicator.

340.

D. Glucose metabolism by certain organisms produces acetyl methyl carbinol (acetoin) as the chief end product. To demonstrate this reaction, 40% potassium hydroxide is added, which oxidizes acetoin to diacetyl. The prior addition of α-naphthol, which acts as a catalyst, produces a red color complex if acetoin is oxidized. Members of the genera *Klebsiella, Enterobacter, Hafnia,* and *Serratia* produce a positive reaction in this test.

341.

D. Blood cultures are one of the most important specimen types sent for bacteriologic examination. Knowledge of the various clinical conditions that produce bacteremia is essential for optimal recovery of the causative agent. Not all conditions produce continuous bacteremia, and organisms may be present in low numbers. If possible, cultures should be drawn before antimicrobial therapy is initiated and optimally before a fever spike. There is a period of 1–2 hours from the time of the release of bacteria into the bloodstream and the subsequent physiologic chill response. Between 10 and 20 mL of blood is recommended. Blood cultures should be held in the laboratory at least 1 week before being reported as negative.

342.

D. Bacterial metabolism of carbohydrates in the culture media produces the by-product carbon dioxide, which is captured as head gas in sealed culture vials. Manometric systems measure the head space pressure. It is possible to detect bacterial metabolism in these systems within only a few hours of inoculation.

343.

A. The acridine orange stain is used to detect low numbers of microorganisms in fluid and exudate samples. The application of this fluorescent dye enables the microbiologist to screen samples at low-power microscopic magnification. This technique is recommended for the routine screening of blood cultures and cerebrospinal fluid sediment smears because of its superior sensitivity as compared to the Gram stain.

344.

A. In the calalase test, hydrogen peroxide is reduced to water and oxygen. The formation of oxygen produces the bubbles seen in a positive test. The catalase test is used to differentiate the staphylococci (positive) from the streptococci (negative).

345.

B. Pyrrolidonyl-α-naphthylamide (PYR) is the substrate. The test detects the presence of the enzyme L-pyrrolidonyl arylamidase. The PYR test helps differentiate *Staphylococcus aureus* (positive) from *S. lugdunensis* (negative).

Case Studies

346.

C. Disseminated gonococcal infection produces symptoms of arthritis, especially in the major joints of the body. Samples of joint fluid from these patients should be inoculated to a selective medium for the isolation of *Neisseria gonorrhoeae* in addition to nonselective media. Thayer-Martin agar has a chocolate agar base formulated to support the growth of fastidious species of *Neisseria* while suppressing the growth of normal or indigenous flora by the addition of antimicrobial agents. Only about 50% of patients with gonococcal arthritis will have positive synovial fluid cultures.

347.

A. The anatomy of the female urethra allows bacteria from the perirectal region to reach the bladder easily. *E. coli* is the most common pathogen in uncomplicated community-acquired urinary tract infections. Other organisms are more prevalent in nosocomial or recurrent infections.

348.

A. *Salmonella* Typhi, the causative agent of typhoid fever, is commonly associated with invasion of the bloodstream. The presence of organisms is the result of an extravascular site of infection. The extravascular sites in the case of typhoid fever are the small intestine, the regional lymph nodes of the intestine, and the reticuloendothelial system. The bacteremic phase is seen before the organism can be recovered in stool.

349.

C. *Vibrio vulnificus* is responsible for septicemia after consumption of contaminated raw oysters. Infections are most severe in patients with hepatic disease, hematopoietic disease, or chronic renal failure and those receiving immunosuppressive drugs. Mortality in patients with septicemia can be as high as 50% unless antimicrobial therapy is started rapidly.

350.

D. *Neisseria meningitidis* is a leading cause of bacterial meningitis. Disease is transmitted by respiratory droplets among people in prolonged close contact, such as in daycare centers. Chemoprophylaxis with rifampin is appropriate for those in close contact with the patient: household members, daycare staff, and classmates.

351.

C. *Arcanobacterium pyogenes* has been reclassified several times. It was formerly a member of the genera *Corynebacterium* and *Actinomyces*. *A. pyogenes* is a well-known animal pathogen causing soft tissue infections in a wide variety of farm animals. Mode of transmission to humans is unknown, but most cases occur in a rural environment and include a history of abrasion or undetected wounds with animal exposure. *Listeria monocytogenes* is also a gram-postive bacillus that is CAMP positive; however, it is catalase positive.

352.

D. *Staphylococcus aureus* has been isolated from a majority of the reported cases of the clinical syndrome described—toxic shock syndrome. First reported in the late 1970s, the disease was linked to the use of a specific brand of tampons. Symptoms are associated with the production of a pyrogenic exotoxin (toxic shock syndrome toxin-1; TSST-1) by the coagulase-positive *Staphylococcus aureus*.

353.

D. The etiologic agent in this case is *Yersinia enterocolitica.* Disease caused by this organism frequently mimics the symptoms of appendicitis, although it has been implicated in a variety of clinical illnesses such as bacteremia, cholecystitis, and mesenteric lymphadenitis. *Y. enterocolitica* grows slowly at 35°C and, unless in large numbers or pure culture, may be overlooked in the laboratory. A key finding for *Y. enterocolitica* is a positive urease.

354.

A. Lyme disease is an inflammatory disease seen predominantly in the northeast and mid-Atlantic U.S. during the summer months. The initial symptoms of this disease may be followed months later by more serious complications, such as meningoencephalitis, myocarditis, and arthritis of the large joints. The etiologic agent of this tick-borne disease is *Borrelia burgdorferi.* The spirochetes causing Lyme disease have not been demonstrated in peripheral blood smears. An indirect immunofluorescence test and an ELISA test are available for the detection of specific antibody in the patient's serum. The western blot assay is often used for serologic confirmation.

355.

D. Human infections caused by *Leptospira* characteristically produce the clinical symptoms of fever, anemia, and jaundice. Weil disease is another name for leptospirosis. Infections result from contact with the urine or tissue of infected animals like rats and mice or from water contaminated with urine of these animals. Most infections resolve in about a week, but they can go on for much longer and can cause fatal kidney and liver damage.

356.

D. *Yersinia enterocolitica* causes a variety of infections. This organism is able to grow at refrigerator temperatures (4°C). Contamination of stored blood units is not visually detected because the organism is able to reproduce in red blood cells without causing lysis or a color change.

357.

A. *Mycobacterium marinum* produces lesions on the skin or the extremities of humans. This species of *Mycobacterium* is a free-living organism found in salt or brackish water. Human infection characteristically follows trauma to the body in or around water.

358.

B. *Listeria monocytogenes* has been associated with human disease following the ingestion of unpasteurized dairy products. The organism is capable of replicating at refrigerator temperatures and is commonly found in low numbers in animal products. Listeriosis associated with contaminated food, in uncompromised patients, usually produces a self-limiting, nonspecific febrile illness.

359.

A. *Escherichia coli* 0157:H7 is associated with hemolytic uremic syndrome. These strains produce verotoxin and are associated with outbreaks of diarrheal disease following ingestion of undercooked hamburger at fast-food restaurants and contact with calves at petting zoos. Cattle infected with this strain serve as the reservoir, and humans become infected by eating products made from their meat or contaminated with their excretions.

360.

C. The clinical presentation suggests the etiologic agent is *Legionella pneumophila.* The Gram stain is not helpful in making the diagnosis because of the poor staining quality of this microorganism. Examination of the sputum using fluorescent antibody to *L. pneumophila* could provide a rapid positive identification.

review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

1. A bulldozer operator became ill while working on a new highway in the San Joaquin Valley. He developed chest pain, anorexia, headache and general malaise, and myalgia with fever. Chest X-ray showed pneumonic infiltrate and a single, well-defined nodule in the left lower lobe. His leukocyte count and sedimentation rate were slightly elevated. Although no fungus was seen in direct examination of a sputum specimen, processing included a culture on Sabouraud dextrose agar with chloramphenicol and cycloheximide. Within 3 days at 30°C, this culture produced moist, grayish growth, and white aerial mycelia began to develop (see Color Plate 30■). A lactophenol cotton blue wet mount of this organism is seen in Color Plate 31■. What is the most likely identification of this fungus?
   A. *Asperigillus fumigatus*
   B. *Blastomyces dermatitidis*
   C. *Coccidioides immitis*
   D. *Histoplasma capsulatum*

2. A 38-year-old male from Ohio presented to his physician with a mild influenzalike illness that included headache and malaise. His chest X-ray showed no infiltrates. His past medical history was unremarkable. He had no history of travel but reported recently cleaning the bell tower at his church, which was littered with bird excrement. The most likely agent causing his disease is
   A. *Aspergillus fumigatus*
   B. *Coccidioides immitis*
   C. *Candida albicans*
   D. *Histoplasma capsulatum*

3. A 44-year-old gardener pricked herself with a rose thorn. A subcutaneous fungal infection characterized by the development of necrotic ulcers followed this direct inoculation of fungal spores into the skin. The causative fungus was cultured as a small yeast form at 35°C (see Color Plate 32■) and as a mould at room temperature with delicate hyphae and conidia. This disease is
   A. Blastomycosis
   B. Chromomycosis
   C. Mycetoma
   D. Sporotrichosis

4. A yeastlike fungus was isolated from a sputum sample. No hyphae were produced on cornmeal agar with Tween 80. The isolate was negative for nitrate assimilation and positive for inositol assimilation and produced urease at 37°C. These findings are typical of
   A. *Candida krusei*
   B. *Cryptococcus terreus*
   C. *Cryptococcus neoformans*
   D. *Trichosporon beigelii*

5. A 24-year-old Vietnamese refugee was seen at a clinic in Houston. His chief complaints were weight loss and fever. A complete blood count confirmed he was suffering from anemia as well. Multiple skin lesions were present on his arms, some of them draining pus. Gram stain of the pus revealed what appeared to be yeastlike cells. A culture of the pus grew a green mould at 22°C, which produced a red soluble pigment (see Color Plate 33■). A lactophenol cotton blue wet mount of this organism is seen in Color Plate 34■. The causative agent in this case is
   A. *Aspergillus fumigatus*
   B. *Fusarium* sp.
   C. *Trichoderma* sp.
   D. *Penicillium marneffei*

6. A section of a lymph node stained with the Gomori silver and hematoxylin and eosin stains is shown in Color Plate 35■. A lactophenol cotton blue wet mount of a mould that grew from this specimen is shown in Color Plate 36■. Large, one-celled, smooth to tuberculate macroconidia and smooth or echinulate microconidia are typical of mycelial phase growth of
   A. *Blastomyces dermatitidis*
   B. *Coccidioides immitis*
   C. *Histoplasma capsulatum*
   D. *Paracoccidioides brasiliensis*

7. Which of the following types of *Candida albicans* infection is commonly acquired from an exogenous source?
   A. Diaper rash
   B. Neonatal thrush
   C. Perianal infection
   D. Urinary tract infection

8. In a direct examination of a KOH wet mount of a nail specimen, *Epidermophyton floccosum* could be detected as
   A. Arthroconidia
   B. Blastoconidia
   C. Macroconidia
   D. Microconidia

9. The mould phase of the systemic fungus *Blastomyces dermatitidis* can be confused with
   A. *Scedosporium apiospermum*
   B. *Sporothrix schenckii*
   C. *Aspergillus* sp.
   D. *Penicillium notatum*

10. It is usually difficult or impossible to identify a fungal culture before it is mature. However, hyaline, septate hyphae, and a young conidiophore with a foot cell (see Color Plate 37■) and a swollen vesicle are excellent clues to the identification of
    A. *Acremonium*
    B. *Aspergillus*
    C. *Paecilomyces*
    D. *Penicillium*

11. Zygomycetes are rapidly growing, airborne saprobes. In clinical specimens they
    A. Are common as normal, human microflora
    B. Are found only as contaminants
    C. May be seen in a dimorphic tissue phase
    D. May be found as a cause of rapidly fatal infection

12. *Trichophyton rubrum* and *T. mentagrophytes* may be differentiated by the
    A. Consistently different appearance of their colonies
    B. Endothrix hair infection produced by *T. rubrum*
    C. Fluorescence of hairs infected with *T. rubrum*
    D. *In vitro* hair penetration by *T. mentagrophytes*

13. Broad, coenocytic hyphae found in tissue would be most typical of infection with
    A. *Aspergillus*
    B. *Blastomyces*
    C. *Microsporum*
    D. *Rhizopus*

14. A fungus infecting only skin and nails typically produces in culture
    A. Spindle-shaped, hyaline, echinulate macroconidia and microconidia
    B. Cylindrical or club-shaped, smooth, thin-walled macroconidia and microconidia
    C. Many microconidia in clusters or along the hyphae
    D. Large, thin-walled, club-shaped macroconidia without microconidia

15. The most useful finding for prompt, presumptive identification of *C. albicans* is its
    A. Failure to assimilate sucrose
    B. "Feathering" on EMB
    C. Production of chlamydospores
    D. Production of germ tubes

16. Identify the dimorphic fungus that typically has a tissue phase in which the large mother cells have one to a dozen narrow-necked buds and a slowly growing mycelial form with intercalary chlamydoconidia and coiled hyphae.
    A. *Blastomyces dermatitidis*
    B. *Coccidioides immitis*
    C. *Histoplasma capsulatum*
    D. *Paracoccidioides brasiliensis*

17. Which of the following stains greatly enhances the visibility of fungi by binding to the cell walls, causing the fungi to fluoresce blue-white or apple green?
    A. Rhodamine-auramine
    B. Warthin-Starry
    C. Calcofluor white
    D. Periodic acid-Schiff

18. The formation of arthroconidia is *not* an important characteristic in the identification of
    A. *Coccidioides*
    B. *Geotrichum*
    C. *Trichosporon*
    D. *Sporothrix*

19. A black pigment produced by colonies growing on bird seed agar is due to
    A. Urease
    B. Phenol oxidase
    C. Sucrose assimilation
    D. Arthroconidia production

20. Which of the following fungi is *not* considered an opportunistic pathogen?
    A. *Absidia*
    B. *Aspergillus*
    C. *Coccidioides*
    D. *Fusarium*

21. Observation of hyaline or dematiaceous hyphae is an early clue in the identification of common, airborne fungi. Which of the following genera contains species found as dematiaceous contaminants?
    A. *Alternaria*
    B. *Aspergillus*
    C. *Fusarium*
    D. *Penicillium*

22. Which of the following fungi is most likely to be found as a common saprobe and as an agent of keratitis?
    A. *Exophiala*
    B. *Phialophora*
    C. *Fusarium*
    D. *Wamgiella*

23. The microscopic identification of *Pneumocystis jirovecii* is based on the detection of
    A. Arthroconidia in subcutaneous tissue biopsies
    B. Cysts and trophozoites in respiratory specimens
    C. Yeasts in respiratory specimens
    D. Tuberculate macroconidia in lung biopsies

24. Fungi that undergo asexual reproduction are termed
    A. Imperfect
    B. Perfect
    C. Aseptate
    D. Septate

25. Hyaline septate hyphae, branched or unbranched conidiophores, and multicelled banana-shaped conidia are characteristic of which of the following?
    A. *Fusarium*
    B. *Curvularia*
    C. *Acremonium*
    D. *Trichophyton*

26. Which of the following does *not* correctly describe the yeast *Rhodotorula rubra?*
    A. It has been isolated from dairy products, soil, and water.
    B. It is the most common fungal cause of diaper rash.
    C. It has been identified as a nosocomial pathogen.
    D. It has been found as a contaminant or commensal in specimens of urine, sputum, and feces.

27. A 21-year-old male member of a university track team presents to student health services with a light brown circular lesion on his upper back. The agent most likely responsible for this condition is
    A. *Candida albicans*
    B. *Fusarium* spp.
    C. *Geotrichum candidum*
    D. *Malassezia furfur*

28. Which of the following is likely to be found in clinical specimens as normal microflora and as clinically significant isolates?
    A. *Aspergillus niger*
    B. *Paracoccidioides brasiliensis*
    C. *Penicillium marneffei*
    D. *Candida albicans*

29. A 4-year-old child's hair is falling out in patches. The hair fluoresces when subjected to the UV light from a Wood's lamp. When the hair is cultured, a white cottony mould grows at 25°C on potato dextrose agar. Microscopically, rare microconidia, septate hyphae, and terminal chlamydospores are seen. Macroconidia are absent. The mould fails to grow on polished rice grains. The causative agent is
    A. *Microsporum audouinii*
    B. *Microsporum gypseum*
    C. *Trichophyton mentagrophytes*
    D. *Trichophyton rubrum*

30. In tissues infected with *Histoplasma capsulatum*
    A. The hyphae usually invade blood vessels
    B. Encapsulated yeast cells are typical
    C. Tuberculate macroconidia are typical
    D. The fungus is usually intracellular

For each numbered mycosis below, choose the letter of the environment most commonly associated with an increased incidence of that infection.

31. Blastomycosis
32. Coccidioidomycosis
33. Cryptococcosis

    A. Lower Sonoran Life Zone
    B. Mississippi and Ohio River basins
    C. Pigeon roosts
    D. Bat roosts

For each numbered mycosis below, choose the letter of the environment most commonly associated with an increased incidence of that infection.

34. Histoplasmosis
35. Sporotrichosis

    A. Sphagnum moss
    B. Starling roosts
    C. Stagnant fresh water
    D. Colorado River Valley

For each numbered incomplete statement, select the letter of the most appropriate species.

36. The cause of white piedra
37. The cause of black piedra
38. The cause of tinea nigra

    A. *Hortaea werneckii*
    B. *Trichosporon* sp.
    C. *Piedraia hortae*
    D. *Fonsecaea compacta*

For each numbered incomplete statement, select the letter of the most appropriate species.

39. The cause of tinea versicolor
40. A keratinophilic saprophyte
41. A cause of otomycosis

A. *Aspergillus niger*
B. *Malassesia furfur*
C. *Microsporum gypseum*
D. *Geotrichum candida*

The incomplete statements below describe the appearance of growth of yeast or yeastlike fungi in morphology agar, such as rice agar or cornmeal agar with Tween 80, a finding helpful in the presumptive identification of these organisms. For each numbered description, select the letter of the most appropriate species.

42. True hyphae and arthroconidia only
43. True hyphae, arthroconidia, and blastoconidia
44. Pseudohyphae, blastoconidia, and chlamydospores

A. *Candida albicans*
B. *Geotrichum*
C. *Trichosporon*
D. *Aspergillus fumigatus*

The incomplete statements below describe the appearance of growth of yeast or yeastlike fungi on morphology agar, such as rice agar or cornmeal agar with Tween 80, a finding helpful in the presumptive identification of these organisms. For each numbered description, select the letter of the most appropriate species.

45. Pseudohyphae and blastospores only
46. Blastospores only, without hyphae or pseudohyphae

A. *Mucor* sp.
B. *Candida tropicalis*
C. *Cryptococcus neoformans*
D. *Candida albicans*

Select the letter of the most appropriate specimen source for isolation of each numbered species description.

47. *Cryptococcus neoformans*
48. *Histoplasma capsulatum*
49. *Pseudallescheria boydii*
50. *Trichophyton mentagrophytes*

A. Bone marrow
B. Cerebrospinal fluid
C. Chronic draining sinus tract of foot
D. Chronic interdigital lesion of foot

answers & rationales

1.

C. Areas of the San Joaquin Valley are highly endemic for *Coccidioides immitis,* and infectious arthroconidia of this fungus can be distributed in dust aerosols produced by construction and other disturbances. Symptomatic pulmonary disease patterns vary, but the signs and symptoms given are found in many cases. The fungus grows more rapidly than do other systemic fungal pathogens, and the aerial mycelium will typically produce the characteristic barrel-shaped arthrospores.

2.

D. The distribution of *Histoplasma capsulatum* is probably worldwide, but most clinical disease occurs in the western hemisphere. Most cases in the U.S. occur in the Ohio and Mississippi River valleys. This organism is found in areas contaminated by large amounts of bird excrement, such as starling and blackbird roosts. Inhalation of the spores results in a respiratory illness usually with clinical symptoms within 2 weeks of exposure. Disease ranges from a mild influenzalike illness to acute fulminant lung infection resembling tuberculosis.

3.

D. *Sporothrix schenckii* is the agent of sporotrichosis. It usually enters the skin by traumatic implantation. This fungus grows *in vitro* as small yeasts at 35°C and as a mould at room temperature (22–30°C) with delicate hyphae and conidia.

4.

C. All species listed may be urease positive, but *C. terreus* does not grow at 35°C and may assimilate nitrate. *C. krusei* is inositol negative, and these species of *Candida* and *Trichosporon* produce hyphae on morphology agar. *C. neoformans* typically does not produce hyphae and is nitrate negative, is inositol and urease positive, and grows at 37°C.

5.

D. Infections due to *Penicillium marneffei* seem to originate in eastern and southeastern Asia. This fungus was first isolated in 1959 from a hepatic lesion from a bamboo rat, a rodent found throughout Southeast Asia. Clinical disease includes fever, weight loss, anemia, and death if untreated. Skin lesions may be present and may drain pus. Diagnosis is made via culture or histopathologic exam of lesions of skin, bone, or liver. The yeastlike cells of *P. marneffei* are oval (3–8 μm) and scattered throughout tissue. Elongated, sausage-shaped cells often contain cross-walls. At 22–30°C, structures typical of the genus *Penicillium* develop. At 35–37°C, round or oval yeastlike cells are seen.

6.

C. Diagnostic features of *H. capsulatum* include large, 8- to 14-μm macroconidia with tuberculate projections. Tuberculate and smooth macroconidia may be seen in the same colony. Microconidia are also produced.

7.

B. Neonatal thrush is the oral candidiasis most commonly associated with mothers having vaginal *Candida,* and the newborn acquires the organism from the mother. Diaper rash due to *C. albicans* usually follows oral and perianal candidiasis of the infant. The other three infections are associated with physiologic changes in the host that permit proliferation of *C. albicans* already present in the host's microflora.

8.

A. KOH wet mounts should be used routinely for direct examination of nails, skin, or hair for fungal elements. KOH digests the keratinous tissue and facilitates observation of any fungi present. *Epidermophyton floccosum* and *Trichophyton* spp. invade nails, and the former typically is found as chains of arthroconidia in nail tissue.

9.

A. At 25–30°C, *Blastomyces dermatitidis* forms septate hyphae with delicate conidiophores of various lengths that bear round or oval conidia. It is important not to confuse the mould phase of *B. dermatitidis* with either *Scedosporium apiospermum* or *Chrysosporium* sp. *S. apiospermum* appears as septate hyphae with simple conidiophores of various lengths that bear oval conidia singly or in groups. *S. apiospermum* is the causative agent of mycetoma and can infect brain, bones, eyes, lungs, etc. *Chrysosporium* sp. appears as septate hyphae with simple to branched conidiophores that bear oval conidia. *Chrysosporium* sp. is commonly considered a contaminant.

10.

B. Conidiophores of *Aspergillus* arise from a foot cell and terminate in a vesicle. The vesicle produces phialides; the phialides then produce the conidia. Before the culture is mature, the presence of a young conidiophore with a foot cell and vesicle is a good clue to the identity of the fungus.

11.

D. Although generally found as laboratory contaminants, the zygomycetes can be clinically significant. Zygomycosis (mucormycosis) is an acute disease that often results in death within a few days in acidotic patients. Fungal agents of mucormycosis include *Rhizopus, Mucor,* and *Absidia,* which are common fungi found in the environment.

12.

D. When speciation of *T. mentagrophytes* or *T. rubrum* is not certain on morphology alone, the *in vitro* hair perforation test is useful; *T. mentagrophytes* is positive and *T. rubrum* is negative. Urease production by *T. mentagrophytes* is less reliable. Neither species produces endothrix infection, and *T. rubrum* rarely infects hair.

13.

D. *Rhizopus* and other fungal agents of mucormycosis are characterized by having coenocytic (nonseptate) hyphae. The finding of broad, nonseptate hyphal elements in sterile body fluids or tissue can provide rapid confirmation of a clinical diagnosis of mucormycosis. The other moulds listed have septate hyphae.

14.

D. *Epidermophyton floccosum* infects skin and nails. This dermatophyte produces thin-walled macroconidia, usually in clusters, but no microconidia. *Microsporum* spp. produce infections in hair and skin. *Trichophyton* spp. may produce infection of the nails, hair, and skin.

15.

D. Essentially all strains of *Candida albicans* produce germ tubes within 2 hours of incubation at 37°C in serum. Chlamydospores are produced by most strains of *C. albicans* after 24–48 hours at 22–26°C on cornmeal Tween 80 agar or a similar substrate. Use of eosin methylene blue medium to screen for *C. albicans* may require 24–48 hours of incubation.

16.

D. The dimorphic pathogenic fungi include the species listed. The parasitic or tissue phase of *P. brasiliensis* produces large, multiple-budding yeasts, 20–60 μm long. The saprophytic or mycelial phase colonies resemble *B. dermatitidis,* but all cultures produce intercalary chlamydoconidia and coiled hyphae, and conidia development is delayed or absent. Clinical types of paracoccidioidomycosis include relatively benign primary pulmonary infection; progressive pulmonary disease; disseminated disease; or an acute, fulminant, juvenile infection. The disease is endemic in certain areas of Central and South America.

17.

C. The calcofluor white stain requires the use of a fluorescence microscope. It is a rapidly staining method, requiring only one minute to complete. Stain binds to chitin in the cell wall of fungi.

18.

D. Barrel-shaped arthroconidia, alternating with empty cells, are typical of the mature mycelial phase of *Coccidioides immitis*. Species of *Geotrichum* produce chains of hyaline arthroconidia, and *Trichosporon* is characterized by production of hyaline arthroconidia, blastoconidia, hyphae, and pseudohyphae. *Aureobasidium* produces dematiaceous arthroconidia. Sporothrix is the sole member of the list that does not produce arthroconidia.

19.

B. Phenol oxidase breaks down the substrate found in niger seeds producing melanin. This result is characteristic of *C. neoformans*. *C. neoformans* is urease positive, but that reaction is not detected on this medium.

20.

C. *Absidia* and *Mucor* can cause the uncommon disease mucormycosis in debilitated patients. Rhinocerebral syndrome is the form of this infection most often seen in the U.S. Species of *Aspergillus* are ubiquitous and opportunistic and cause a variety of human infections. *Fusarium* is one of the saprobic fungi most often found in external mycotic keratitis following corneal trauma. *Coccidioides* is considered a true pathogen that can infect healthy people.

21.

A. Observation of dark pigmented hyphae in a culture is evidence that the fungus is in one of the dematiaceous genera. Typically, the reverse of a plate will be black. *Alternaria* is a common dematiaceous contaminant.

22.

C. Mycotic keratitis due to *Fusarium* has been reported following injury or cortisone treatment. An ulcerative lesion develops on the cornea. Corneal scrapings may be received for direct exam and culture.

23.

B. *Pneumocystis jirovecii* produces cysts and trophozoites that can be found in respiratory tract specimens. The fungus primarily infects the lungs, so specimens from the lower respiratory tract are most productive (e.g., brochoalveolar lavage). Specimens can be stained with a silver stain or Giemsa stain.

24.

A. Fungi with only an asexual stage of reproduction are referred to as the imperfect fungi. Fungi able to reproduce sexually are called the perfect fungi. "Septate" and "aseptate" refer to the presence or absence (respectively) of crosswalls in hyphae.

25.

A. Diagnostic features of *Fusarium* spp. include hyaline septate hyphae and sickle- or banana-shaped macroconidia. Macroconidia are multiseptate with long or short branched or unbranched conidiophores. Microconidia (one or two celled) are also produced.

26.

B. *Rhodotorula rubra* has been isolated from soil, water, and a number of food sources, especially dairy products and as a contaminant of skin, lung, urine, or feces. *Rhodotorula* fungemia has been caused by contaminated catheters, intravenous solutions, and dialysis machines. *C. albicans* is a more common cause of diaper rash.

27.

D. *Malassezia furfur* is the causative agent of tinea or pityriasis versicolor—a superficial skin infection that occurs commonly on the upper back, chest, shoulders, upper arms, and abdomen. Initially lesions are discrete but in time may coalesce. Lesions may be hyper- or hypopigmented. *M. furfur* is part of the normal skin flora of over 90% of adults. There may be an association between the disease and excessive sweating. The disease is more common in tropical and subtropical areas.

28.

D. *C. albicans* is an endogenous species causing a variety of opportunistic infections. Infection is usually secondary to a predisposing debility. *Aspergillus* spp. are common saprophytic contaminants. *Paracoccidioides brasiliensis* and *Penicillium marneffei* are dimorphic fungi that cause systemic mycoses.

29.

A. *Microsporum audouinii* most commonly affects children. Only rarely are adults infected. Colonies are flat, downy to silky, and gray to white in color. Colony reverse is salmon to brown with a reddish-brown center. Microscopic examination reveals septate hyphae, terminal chlamydoconidia, and occasional microconidia (borne singly). Macroconidia are very rare or absent. Infected hair fluoresces. Growth on polished rice grains aids in differentiating *M. audouinii* from other *Microsporum* species that grow well on rice grains.

30.

D. *Histoplasma capsulatum* is found primarily within histiocytes and in macrophages or monocytes in specimens from bone marrow aspirates, biopsies, or the buffy coat of centrifuged blood. Unstained cell wall of the tissue (yeast) form of *H. capsulatum* may be mistaken for a capsular halo in stained preparations. Only the mould phase would exhibit hyphae and macroconidia.

31.

B. *Blastomyces dermatitidis* is rarely found in the environment, and there is no reliable skin test for screening for past or subclinical blastomycosis. Outbreaks occur most frequently following exposures to moist environments like streams and rivers. The incidence of clinical cases in the U.S. is highest in the Mississippi and Ohio River basins and part of the Missouri River drainage.

32.

A. The most highly endemic regions of coccidioidomycosis are semiarid, with dry, hot seasons and wetter, cooler seasons above freezing. The areas of the southwestern U.S. and northern Mexico with this typical Lower Sonoran Life Zone climate have the highest incidence of coccidioidomycosis. The peak endemic period is fall, when the fungus becomes airborne from the desert surface.

33.

C. Although *Cryptococcus neoformans* does not appear to infect pigeons, it apparently passes unharmed through their gut. It has been found in large numbers, even as the predominant microorganism, from the debris of old pigeon roosts. Viable, virulent, desiccated cells, small enough to be inhaled into the alveoli, can be present in the dust of these roosts.

34.

B. The most highly endemic areas of histoplasmosis (Missouri, Kentucky, southern Illinois, Indiana, and Ohio) also have the most starlings, whose flocks produce large accumulations of guano. *Histoplasma capsulatum* has been found growing in almost pure culture in accumulated starling guano. Exposure to aerosols containing many spores of this fungus has been associated with a number of "common source" outbreaks of histoplasmosis.

35.

A. In temperate countries, including the U.S., sporotrichosis is an occupational hazard of gardeners and nursery workers and is frequently associated with contact with sphagnum moss. In Mexico, it has been associated with working with grass, and a well-known epidemic in South Africa involved gold mine workers in contact with untreated mine poles. *Sporothrix schenckii* produces subcutaneous infections that begin at the site of traumatic implantation.

36.

B. White piedra is frequently caused by *Trichosporon ovoides* and *T. inkin. T. beigelii* was the name formerly used for the species infecting humans. The disease is characterized by soft, white to light brown nodules around and in the hair shaft. The nodules are composed of hyphae, yeastlike arthroconidia, and sometimes blastoconidia. The beard and body hair are more often affected than scalp hair.

37.

C. Black piedra is caused by *Piedra hortae,* which produces brown to black, gritty nodules on the outside and under the cuticle of the hair shaft. Scalp hair is the site most often involved. Direct microscopic examination of portions of these nodules in KOH wet mounts can show septate dematiaceous hyphae *and* ascospores.

38.

A. Tinea nigra is a superficial skin infection caused by *Hortaea werneckii.* The pigmented, painless lesion, which usually occurs on the palms or fingers, may be mistaken for melanoma. Accurate laboratory findings in a KOH preparation of a skin scraping are important in preventing surgical mutilation of the patient. Microscopic examination of skin scrapings from tinea nigra shows dematiaceous, septate hyphae and budding cells.

39.

B. Tinea versicolor is a chronic, mild, superficial skin infection caused by *Malassesia furfur,* which may also be found on normal skin. Despite the name "tinea versicolor," the causative fungus is not a dermatophyte. Skin scrapings from the lesions demonstrate characteristic yeastlike cells and hyphae.

40.

C. *Microsporum gypseum* is a keratinophilic fungus (dermatophyte). It has a geophilic species that has been isolated from human infections. It is a moderately rapid grower, producing numerous thick-walled rough macroconidia.

41.

A. *Aspergillus niger* causes approximately 90% of the otomycoses and external ear infections due to fungi. *Aspergillus fumigatus* also causes otomycosis. Other fungi, far less often involved, include *Scopulariopsis, Penicillium, Rhizomucor, Candida,* and other species of *Aspergillus.*

42.

B. *Geotrichum* spp. typically produce numerous hyphae and arthroconidia. Germinating arthroconidia of *Geotrichum,* however, may be mistaken for blastoconidia production. This may cause confusion between *Geotrichum* and *Trichosporon.*

43.

C. *Trichosporon* spp. produce hyphae and arthroconidia. They may also produce blastoconidia, although these may be rare. If present, blastoconidia can differentiate *Trichosporon* from *Geotrichum.*

44.

A. *Candida albicans* and *C. dubliniensis* both produce pseudohyphae and are germ tube positive. Both are capable of producing chlamydospores and blastoconidia. These two species are difficult to differentiate.

45.

B. *Candida tropicalis* typically produces long-branched pseudohyphae. Blastoconidia are produced singly or in short chains. This species does not produce chlamydospores. The carbon assimilation pattern of *C. tropicalis* resembles that of *C. albicans,* and some strains of *C. tropicalis* may produce a positive germ tube test if incubated more than 3 hours.

46.

C. *Cryptococcus neoformans* produces only blastoconidia when growing on morphology agar (e.g., cornmeal agar with Tween 80). This species is usually identified by its encapsulated cells, production of urease, failure to assimilate nitrate, and production of brown pigment on bird seed agar. Cryptococcosis can lead to systemic infections in immunocompromised patients.

47.

B. The most frequently diagnosed form of cryptococcosis is central nervous system infection. Few or many organisms may be in the cerebrospinal fluid, but a clinical diagnosis of meningitis can often be confirmed by the cryptococcal antigen test. In the past, the use of a microscopic examination of a spun specimen with India ink has been used. The cryptococcal antigen test is much more sensitive and is the recommended test.

48.

A. *Histoplasma capsulatum* is a parasite of the reticuloendothelial system and is seldom extracellular. Specimens such as sternal bone marrow, lymph node, liver and spleen biopsies, or buffy coat of blood should be stained with Giemsa or Wright's stain and examined for small, intracellular yeast cells.

49.

C. *Pseudoallescheria boydii* is the most common cause of eumycotic mycetoma in the U.S. Mycetoma is a clinical syndrome of localized abscesses, granulomas, and draining sinuses that develops over months or years. It usually occurs on the foot or hand after traumatic implantation of soil organisms.

50.

D. *Trichophyton mentagrophytes* is a common cause of intertriginous tinea pedis or athlete's foot. This is a chronic dermatitis most often affecting the areas between the fourth and fifth and third and fourth toes. The acute inflammation often subsides, but recurrences are common.

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

1. *Enterobius vermicularis* infection is usually diagnosed by finding
   A. Eggs in perianal specimens
   B. Larvae in perianal specimens
   C. Larvae in feces
   D. Eggs in the feces

2. The best direct diagnosis of *Echinococcus granulosus* infection in humans is made by identification of
   A. Adult worms in the intestine
   B. Adult worms in tissues
   C. Eggs in feces
   D. Hydatid cysts in tissues

3. Which statement is correct for specimen collection and processing?
   A. Stool samples can contain urine.
   B. Stools can be frozen without affecting parasitic structure.
   C. Liquid stools are best for detecting ameba and flagellate trophozoites.
   D. Unpreserved stools can remain at room temperature for up to 72 hours.

4. Cysts are the infective stage of this intestinal flagellate.
   A. *Balantidium coli*
   B. *Dientamoeba fragilis*
   C. *Entamoeba coli*
   D. *Giardia lamblia*

5. Eggs or larvae recovered in the stool are *not* routinely used to diagnose infections caused by which one of the following helminths?
   A. *Trichinella spiralis*
   B. *Strongyloides stercoralis*
   C. *Necator americanus*
   D. *Ascaris lumbricoides*

6. Many parasites have different stages of growth within different hosts. The host where the sexual reproductive stage of the parasites exists is called the
   A. Commensal
   B. Definitive host
   C. Intermediate host
   D. Vector

7. Species identification of an immature amebic cyst can be very difficult. The presence of a large glycogen mass is sometimes seen in
   A. *Dientamoeba fragilis*
   B. *Endolimax nana*
   C. *Entamoeba coli*
   D. *Entamoeba histolytica*

8. Which of the following is typical in cysts of *Iodamoeba bütschlii?*
   A. A glycogen mass
   B. Blunt chromatoidal bars
   C. Four nuclei with large karyosomes
   D. Many ingested bacteria

9. Which of the following is the most important feature in differentiating cysts of *Entamoeba histolytica* from *E. dispar?*
   A. Number of nuclei
   B. Size of the cyst
   C. Shape of the karyosome
   D. Distinguishing surface antigens by immunologic assays

10. Which of the following findings in a peripheral blood smear is especially associated with tissue-invading helminths but may also be found in a variety of allergic conditions and other diseases?
    A. Eosinophilia
    B. Leukopenia
    C. Lymphocytosis
    D. Neutropenia

11. A 48-year-old man from Texas developed fever and weakness 16 days after a hunting trip in northwest Tanzania. Several days after the onset of fever, he noticed a raised, tender, erythematous nodule (6–8 cm in diameter) on the posterior aspect of his right arm. He was hospitalized in Africa and treated for 5 days with a cephalosporin for presumed cellulitis. After little improvement, he returned to Texas. On arrival, the patient had a temperature of 38.9°C (102°F), a morbilliform rash of the trunk, and right-sided, anterior cervical lymphadenopathy. Cerebrospinal fluid contained 12 red cells and 18 mononuclear cells/μL and a normal protein level (32 mg/dL). Laboratory tests of peripheral blood revealed a hemoglobin level of 107 g/L, a white cell count of 2.4×10^9/L, and a platelet count of 75×10^9/L. The diagnosis was made by finding the extracellular flagellate parasite in a peripheral blood smear. Which of the following is the most probable etiologic agent of this infection?
    A. *Leishmania donovani*
    B. *Trypanosoma brucei*
    C. *Trypanosoma cruzi*
    D. *Toxoplasma gondii*

12. Which species of malaria parasite usually has ameboid trophozoites and produces small reddish dots in the red blood cell cytoplasm?
    A. *Plasmodium knowlesi*
    B. *Plasmodium falciparum*
    C. *Plasmodium malariae*
    D. *Plasmodium vivax*

13. With a fecal specimen, which one of the following is the most dependable procedure for the accurate, specific diagnosis of an intestinal amebic infection?
    A. Direct saline wet mount
    B. Direct iodine wet mount
    C. Permanently stained smear
    D. Formalin-ethyl acetate sedimentation technique

14. In an examination of stained blood films, *Babesia* spp. are likely to resemble
    A. *Leishmania donovani*
    B. *Plasmodium falciparum*
    C. *Toxoplasma gondii*
    D. *Trypanosoma cruzi*

15. Which of the following is a mercury-containing fixative used to preserve parasites in stool specimens?
    A. Formalin
    B. Sodium acetate
    C. Buffered glycerol
    D. Polyvinyl alcohol

16. Examination of a fecal smear following acid-fast stain reveals round acid-fast positive structures 8–10 μm in diameter. You should suspect
    A. *Cryptosporidium*
    B. *Cyclospora*
    C. *Isospora*
    D. Microsporidia

17. A 22-year-old male presents to his family physician complaining of fatigue, muscle pain, periorbital edema, and fever. He denies travel outside the U.S. The physician suspects infectious mononucleosis; however, serologic tests for infectious mononucleosis are negative. The complete blood count revealed a slightly elevated white blood count, and there were 10% eosinophils on the differential. Which of the following should be considered part of the differential diagnosis?
    A. *Ascaris lumbricoides*
    B. *Taenia solium*
    C. *Trichinella spiralis*
    D. *Trypanosoma cruzi*

18. Which *Schistosoma* species has a large terminal spine?
    A. *S. haematobium*
    B. *S. japonicum*
    C. *S. mansoni*
    D. *S. mekongi*

19. Elephantiasis is a complication associated with which of the following?
    A. Cysticercosis
    B. Guinea worm
    C. Hydatid cyst disease
    D. Filariae

20. A patient with history of human immunodeficiency virus infection presents with a 5-day history of diarrhea and weight loss. A series of stool specimens is collected and examined for the presence of ova and parasites. An acid-fast stain on direct smear reveals pink-stained round structures approximately 4 μm in diameter. The most likely pathogen is
    A. *Blastocystis hominis*
    B. *Cryptosporidium* sp.
    C. *Isospora* sp.
    D. Microsporidium

21. A 55-year-old female presents to her physician complaining of a fever that "comes and goes" and fatigue. A complete blood count reveals decreased red blood cell count and hemoglobin. History reveals the patient recently traveled through Europe and Africa. You should suspect
    A. Cutaneous larval migrans
    B. Filariasis
    C. Malaria
    D. Trichinella

22. The disease most commonly associated with *Acanthamoeba* sp. is
    A. Diarrhea
    B. Keratitis
    C. Liver abscess
    D. Meningoencephalitis

23. A modified trichrome stain of a fecal smear can be used to detect microsporidia. Which of the following would describe the appearance of this parasite in this stain?
    A. Purple circles, 10–15 μm in diameter
    B. Pink ovals, 1–3 μm in diameter
    C. Blue ovals, 4–6 μm in diameter
    D. Fluorescent circles, 8–12 μm in diameter

24. Hydatid cysts in humans are due to ingestion of a tapeworm stage normally found in canines. This stage is the
    A. Cercaria
    B. Cercocystis
    C. Cysticercus
    D. Embryonated egg

25. Oocysts, the infective stage of *Toxoplasma gondii,* is found in
    A. Cat feces
    B. Human feces
    C. Undercooked pork
    D. Undercooked beef

26. A 15-μm pear-shaped flagellate with a visible parabasal body and "falling leaf" motility in a direct saline mount of a diarrheal stool specimen is most probably
    A. *Balantidium coli*
    B. *Chilomastix mesnili*
    C. *Giardia lamblia*
    D. *Trichomonas hominis*

27. This parasitic infection may result in vitamin B_{12} deficiency, and individuals with pernicious anemia are predisposed to more severe symptoms.
    A. *Diphyllobothrium latum*
    B. *Echinococcus granulosus*
    C. *Hymenolepis diminuta*
    D. *Taenia saginata*

28. Knowledge of nocturnal periodicity is especially important in the diagnosis of certain infections caused by
    A. *Babesia*
    B. *Plasmodium*
    C. Microfilariae
    D. *Trypanosoma*

29. For which of the following diseases is close contact with an infected human host the most important mechanism of transmission?
    A. Schistosomiasis
    B. Toxoplasmosis
    C. Trichinosis
    D. Trichomoniasis

30. Which of the following helminths produces an elongate, barrel-shaped egg (50 × 22 μm) with a colorless polar plug at each end?
    A. *Ascaris lumbricoides*
    B. *Hymenolepis nana*
    C. *Necator americanus*
    D. *Trichuris trichiura*

31. Which species of *Plasmodium* may readily be identified when crescent-shaped gametocytes are found in stained blood films?
    A. *P. falciparum*
    B. *P. malariae*
    C. *P. ovale*
    D. *P. vivax*

32. Cysts of *Giardia lamblia*
    A. Contain four nuclei
    B. Are motile by flagella
    C. Have an undulating membrane
    D. Are rarely found in stool specimens

33. Migration of larva through the skin can sometimes produce allergic reactions called larva migrans; this is associated with
    A. *Strongyloides stercoralis*
    B. *Dracunculus medinensis*
    C. *Onchocerca volvulus*
    D. *Loa loa*

34. A free-living ameba that causes primary amebic meningoencephalitis is
    A. *Dientamoeba fragilis*
    B. *Entamoeba coli*
    C. *Entamoeba histolytica*
    D. *Naegleria fowleri*

35. Decontamination of drinking water, fruits, and vegetables before consumption is necessary in countries without well-developed public sanitation. Which of the following diseases would probably be least affected by that kind of precaution?
    A. Amebiasis
    B. Ascariasis
    C. Filariasis
    D. Giardiasis

36. Which stage of *Taenia saginata* is usually infective for humans?
    A. Cysticercus larva
    B. Embryonated egg
    C. Filariform larva
    D. Rhabditiform larva

37. This amebic cyst has an average size of 6–8 μm and is usually spherical. When mature, it has four nuclei, but immature cysts with one or two nuclei are often seen. The nuclei have fine uniform granules of peripheral chromatin and small, discrete, usually central karyosomes. Chromatoidal bars with bluntly rounded ends are sometimes present. Name the species.
    A. *Endolimax nana*
    B. *Entamoeba coli*
    C. *Entamoeba hartmanni*
    D. *Entamoeba histolytica*

38. Which stage of *Trichuris trichiura* is infective for humans?
    A. Proglottid
    B. Filariform larva
    C. Rhabditiform larva
    D. Embryonated ovum

39. An intestinal parasite is seen in an iodine-stained fecal wet mount that is described as being 25 μm in diameter with a homogenously stained central body surrounded by a thin ring of cytoplasm containing a number of nuclei. This best describes
    A. *Blastocystis hominis*
    B. *Endolimax nana*
    C. *Entamoeba dispar*
    D. *Iodamoeba bütschlii*

40. Sanitary disposal of human feces is the most important factor in decreasing the incidence of most infections caused by intestinal parasites. Which of the following diseases would *not* be affected by that kind of sanitation?
    A. Ascariasis
    B. Taeniasis
    C. Trichinosis
    D. Hookworm infection

41. Which species of *Plasmodium* is characterized by the presence of Schüffner's dots in the infected erythrocytes?
    A. *Plasmodium falciparum*
    B. *Plasmodium knowlesi*
    C. *Plasmodium malariae*
    D. *Plasmodium ovale*

42. Which of the following is the largest intestinal protozoa infecting humans?
    A. *Balantidium coli*
    B. *Dientamoeba fragilis*
    C. *Entamoeba histolytica*
    D. *Giardia lamblia*

43. The rhabditiform larvae of *Strongyloides stercoralis*
    A. Mate and produce ova
    B. Are infective for humans
    C. Are the diagnostic form found in feces
    D. Are found in the blood of infected humans

44. Which species of *Plasmodium* can have exoerythrocytic stages capable of causing relapses months or years after initial infection?
    A. *P. falciparum*
    B. *P. ovale*
    C. *P. malariae*
    D. *P. cynomolgi*

45. A Giemsa-stained thick blood film showed many ring forms with no older stages, and a number of the rings had double chromatin dots. These findings are characteristic of
    A. *Plasmodium falciparum*
    B. *Plasmodium vivax*
    C. *Plasmodium malariae*
    D. *Plasmodium ovale*

46. Which of the following nematode parasites is acquired from eating inadequately cooked, infected pork?
    A. *Strongyloides stercoralis*
    B. *Taenia saginata*
    C. *Taenia solium*
    D. *Trichinella spiralis*

47. Which of the following pairs of helminths *cannot* be reliably differentiated by the appearance of their eggs?
    A. *Ascaris lumbricoides* and *Necator americanus*
    B. *Hymenolepis nana* and *H. diminuta*
    C. *Necator americanus* and *Ancylostoma duodenale*
    D. *Diphyllobothrium latum* and *Fasciola hepatica*

48. Which of the following forms of *Toxoplasma gondii* are produced in infected humans?
    A. Bradyzoites
    B. Macrogametes
    C. Sporoblasts
    D. Oocysts

49. Hematuria is a typical sign of human infection caused by
    A. *Trypanosoma cruzi*
    B. *Trichinella spiralis*
    C. *Trichomonas vaginalis*
    D. *Schistosoma haematobium*

50. Which of the following is the vector for *Babesia?*
    A. Fleas
    B. Lice
    C. Ticks
    D. Mosquitoes

51. Chagas disease (American trypanosomiasis) is caused by
    A. *Trypanosoma brucei*
    B. *Trypanosoma cruzi*
    C. *Leishmania braziliensis*
    D. *Dracunculus medinensis*

52. Which of the following is the preferred anticoagulant for preparing blood smears for diagnosing malaria?
    A. EDTA
    B. Heparin
    C. Sodium citrate
    D. Sodium fluoride

53. Refer to Color Plate 38■. This is a photomicrograph of a trichrome stain of a fecal smear, magnification 400×. The parasite measures 65 × 45 μm. What is the identification of this parasite?
    A. *Balantidium coli*
    B. *Diphyllobothrium latum*
    C. *Giardia lamblia*
    D. *Schistosoma japonicum*

54. Refer to Color Plate 39■. This is a photomicrograph of an iodine wet prep made from a stool sample; magnification is 400×. The ovum is about 70 × 50 μm. What is the identification of the parasite?
    A. Hookworm
    B. *Enterobius vermicularis*
    C. *Trichuris trichiura*
    D. *Ascaris lumbricoides*

55. Refer to Color Plate 40■. This is a photomicrograph of an iron hematoxylin stain from a fecal smear. The magnification is 1000×. The parasite is approximately 20 μm long and 15 μm wide. What is the identification of this parasite?
    A. *Chilomastix mesnili* trophozoite
    B. *Giardia lamblia* trophozoite
    C. *Trichomonas hominis* trophozoite
    D. *Trichomonas tenax* trophozoite

56. Refer to Color Plate 41■. This is a photomicrograph of an iron hematoxylin stain from a fecal smear. The magnification is 1000×. The parasite is approximately 12 μm in diameter. What is the identification of this parasite?
    A. *Entamoeba histolytica* trophozoite
    B. *Entamoeba hartmanni* trophozoite
    C. *Dientamoeba fragilis* trophozoite
    D. *Entamoeba coli* trophozoite

57. Refer to Color Plate 42■. This is a photomicrograph of an iodine wet-mount from a fecal sample. The magnification is 1000×. The parasite is approximately 25 μm in diameter. What is the identification of this parasite?
    A. *Entamoeba histolytica* cyst
    B. *Entamoeba histolytica* trophozoite
    C. *Entamoeba coli* cyst
    D. *Entamoeba coli* trophozoite

58. Refer to Color Plate 43■. This is a photomicrograph of an iodine wet mount from a fecal sample. The magnification is 400×. The ovum is approximately 70 × 38 μm. What is the identification of this parasite?
    A. Hookworm
    B. *Ascaris lumbricoides*
    C. *Diphyllobothrium latum*
    D. *Taenia solium*

59. Refer to Color Plate 44■. This is a photomicrograph of a trichrome stain from a fecal smear. The magnification is 1000×. The parasite is approximately 15 μm in diameter. What is the identification of this parasite?
    A. *Entamoeba hartmanni*
    B. *Dientamoeba fragilis*
    C. *Iodamoeba bütschlii*
    D. *Blastocystis hominis*

60. Refer to Color Plate 45■. This is a photomicrograph of a blood smear stained with Wright's stain. Identify the parasite.
    A. *Babesia* sp.
    B. *Plasmodium malariae*
    C. *Plasmodium falciparum*
    D. *Trypanosoma cruzi*

answers & rationales

1.

A. Because the eggs of *E. vermicularis* are usually deposited on the perianal area, cellulose tape slides are recommended for collecting the eggs. Recovery is best if specimens are collected late in the evening or before bathing or defecating in the morning. The gravid female worms usually migrate at night to the perianal region to deposit eggs. Because their migration is sporadic, several consecutive collections may be necessary to detect the infection.

2.

D. When *E. granulosus* eggs are ingested by an intermediate herbivorous host, including humans, they usually develop into hydatid cysts in which invaginated larval scolices are produced. These cysts are most often in the liver or lung. Although clinical findings can provide a presumptive diagnosis, this is best confirmed by the finding, at surgery, of encysted larval scolices. Each scolex is capable of developing into an adult worm after ingestion by a dog or related animal, the definitive host.

3.

C. Fresh liquid stools are more likely to contain motile protozoan trophozoites that can be detected in saline wet mounts. Urine in stool specimens can damage parasite morphology, as does freezing. Unpreserved stool specimens should only be left at room temperature up to a couple of hours before examining or placing into a preservative.

4.

D. Cysts are the infective stage of most intestinal parasites. *D. fragilis* is currently classified as a flagellate, even though it does not produce a flagellum. However, *D. fragilis* does not have a cyst stage. *G. lamblia* is the only flagellate in the list of protozoa with a cyst stage. *B. coli* is a ciliate and *E. coli* is an ameba.

5.

A. Although *T. spiralis* adults live in the intestinal mucosa, they are rarely seen. The female deposits living larvae into the mucosa or lymphatic vessels, from which they normally enter the bloodstream and are disseminated throughout the body. They then burrow into muscle fibers. Although larvae may occasionally be liberated into the intestinal lumen, the definitive diagnostic procedure is the demonstration of larvae in skeletal muscle, not in feces.

6.

B. In parasites with a sexual and asexual stage of development, the definitive host is the host in which the sexual stage of the parasite occurs. The intermediate host is the host in which the asexual stage of the parasite is found. Vectors are arthropods, like mosquitoes and ticks that transmit infectious agents. A commensal is an organism that benefits from an existence with a host but does not damage the host.

7.

C. Young cysts of *E. coli* can have a large glycogen mass that pushes two nuclei to the outer edge of the cell. No cyst stage is known for *D. fragilis*. The cysts of *E. nana* and *E. histolytica* do not contain glycogen.

8.

A. Mature cysts of *I. bütschlii* are usually ovoid, with a single nucleus with a large eccentric karyosome. The cytoplasm contains a compact mass of glycogen, which appears as a clear area in unstained or permanently stained preparations but stains dark brown with iodine. Chromatoid bodies are not present.

9.

D. *E. histolytica* and *E. dispar* cannot be morphologically differentiated. The cyst stage of both organisms has four nuclei with a centrally located karyosome. *E. histolytica* is a well-recognized intestinal parasite, whereas *E. dispar* is considered nonpathogenic. Immunologic assays to detect antigens or molecular biology assays are necessary to differentiate these two species.

10.

A. Although the condition may vary from patient to patient, eosinophilia is often found in association with infections with tissue-invading nematodes. Eosinophilia of 40–80% is not unusual in trichinosis and in visceral larva migrans. It may also be present in strongyloidiasis, early in *Ascaris* and hookworm infections, and in filariasis, which may also cause pulmonary eosinophilia.

11.

B. The symptoms and history for this patient are compatible with trypanosomiasis (African sleeping sickness) caused by *T. brucei*. The trypomastigote form of the parasite was found in peripheral blood smears from this patient. Another key clinical sign is the presence of swollen lymph nodes at the posterior base of the neck; this is called Winterbottom's sign.

12.

D. The trophozoites of *P. vivax* often develop fine pseudopodia and large vacuoles and are described as ameboid; infected red blood cells (RBCs) contain clumps of malarial pigment called Shüffner's dots. *P. malariae* cytoplasm is much more compact, and infected RBCs lack Shüffner's dots. *P. ovale* resembles *P. vivax*. Shüffner's dots are generally found in *P. vivax* and *P. ovale*–infected RBCs; however, *P. ovale*–infected RBCs have fimbriated edges. Growing trophozoites of *P. falciparum* seen in the peripheral blood remain in the ring form, and infected RBCs lack malarial pigment. *P. knowlesi* is rarely a human pathogen.

13.

C. The permanently stained smear is especially recommended for identification of trophozoites, for confirmation of species, and for keeping a permanent record of the organisms found. Species identification of amebic trophozoites can rarely be made from a single feature; permanent stains enable one to observe the cytoplasm and cytoplasmic inclusions and the nuclear morphologic features of many cells. Iron hematoxylin and trichrome are commonly used stains.

14.

B. *Babesia* spp. are sporozoan parasites of RBCs that have been recognized as causing febrile illness in humans. *B. microti* has caused a number of tick-borne infections in the U.S. The parasites often appear as small rings within infected RBCs, resembling *P. falciparum* trophozoites. The pathognomic form of *Babesia* is the "Maltese cross," four ring forms inside a single RBC.

15.

D. Polyvinyl alcohol (PVA) is a commonly used fixative for stool specimens. This preservative contains mercury and is used to fix fecal samples for making permanently stained smears. Formalin is commonly used to preserve stool samples in preparation for concentration procedures. Sodium acetate is used with formalin to preserve fecal specimens, but it does not contain mercury and is, therefore, less toxic. Buffered glycerol is sometimes used as a transport medium for stool samples when performing a bacterial culture.

16.

B. Although all the organisms listed have some degree of acid-fast positivity, only *Cyclospora* forms oocysts in the size range of 8–10 μm. The oocysts of *Cryptosporidium* are generally 4–6 μm in diameter and are generally strongly acid-fast positive. Oocysts of *Isospora* are much larger, approximately 25 × 18 μm. Microsporidia are acid-fast variable, and this stain is not recommended for detecting microsporidia. The spores of microsporidia are generally 1–3 μm in diameter.

17.

C. The early symptoms of trichinosis can resemble infectious mononucleosis. The presence of periorbital edema (swelling around the eyes) in this patient and hemorrhaging in the nail beds is suggestive of trichinosis. Eosinophilia also indicates a parasitic infection.

18.

A. The ova of *Schistosoma* contain a spine. *S. haematobium* ova have a large prominent spine on one end. *S. mansoni* has a prominent lateral spine, whereas the spine of *S. japonicum* is small and inconspicuous. *S. mekongi* is a rare human pathogen.

19.

D. Adult filarial helminths typically inhabit the lymph vessels. They produce inflammation and swelling of the lymph vessels, often in the legs and sometimes the scrotum. This condition is referred to as elephantiasis. The larvae are highly motile and migrate through the body.

20.

B. Although all these organisms are potential pathogens of immunocompromised patients, only *Cryptosporidium* produces acid-fast positive oocysts about 4–6 μm in diameter. The oocysts of *Isospora* measure approximately 25 × 18 μm. The spores of microsporidia are generally 1–3 μm in diameter. *B. hominis* is generally considered to be pathogenic in high numbers. The diagnostic form of this intestinal parasite measures 6–40 μm in diameter and is not acid-fast positive.

21.

C. One of the classic signs of malaria is a fever that occurs in cycles. As the infection is developing, all the parasites are in approximately the same stage of development. The fever spikes correspond to the release of the merozoites from infected RBCs. *Plasmodium* is an obligate intracellular parasite of RBCs; therefore, infections can result in decreased RBC counts and hemoglobin.

22.

B. *Acanthamoeba* is a free-living ameba rarely causing human infections. This organism has been associated with granulomatous infections of the skin and lung, as well as meningoencephalitis. However, the most common presentation is keratitis, infection of the cornea. Most keratitis cases have been associated with contact lenses.

23.

B. The small size and variable staining of the microsporidia make their detection difficult. Tissue examination by electron microscopy is the most specific diagnostic method. In the modified trichrome stain, one of the stains (chromotrope 2R) is used at 10 times the normal concentration. In addition, the staining time is increased to 90 minutes. Alternatively, 15 minutes in heated stain can be used. Under these staining conditions, the spores of microsporidia stain as pinkish ovals, 1–3 μm.

24.

D. *Echinococcus granulosus* is a tapeworm that lives as an adult in the small intestines of carnivores, primarily dogs, wolves, and other canines. When the embryonated egg from the feces of a carnivore is ingested by the intermediate host, usually an herbivore but sometimes a human, the liberated embryo can develop into a hydatid cyst. These cysts are most often hepatic or pulmonary, and the resulting symptoms are comparable to those of a slowly growing tumor.

25.

A. *T. gondii* is a protozoan parasite of humans and a variety of lower animals. Human infections can be congenital or can result from ingestion of material containing oocysts from cat feces or from eating undercooked beef, lamb, or pork containing toxoplasma cysts. The life cycle of this parasite includes asexual multiplication in a number of hosts and sexual multiplication only in domestic cats and some closely related species, which then excrete potentially infectious oocysts. Pregnant women should take precautions to avoid infection.

26.

C. All the flagellates listed are pear shaped, but only *C. mesnili* and *G. lamblia* are usually as large as 15 μm. *B. coli* is a ciliate. The typical motion of *G. lamblia* is described as "falling leaf"; *C. mesnili* has a stiff rotary motion. *G. lamblia* is known to cause diarrheal disease and malabsorption, and the trophozoites may be found in diarrheal feces.

27.

A. *D. latum* is a tapeworm that has been linked to vitamin B_{12} deficiencies in individuals of Scandinavian descent. *T. saginata* and *H. diminuta* are tapeworms that infect the gastrointestinal tract of humans but have not been linked to vitamin B_{12} deficiencies. *E. granulosus* is a dog tapeworm that causes hydatid disease in the tissue of humans.

28.

C. Transmission of filariasis depends on the presence of microfilariae in the bloodstream at the time the vector bites, and the periodicity of microfilariae in the peripheral blood varies with the species and sometimes with the geographic area. Nocturnal periodicity is marked in *W. bancrofti* in Africa, Asia, and the western hemisphere, and thick blood films for detection of these microfilariae should be made between 10 P.M. and midnight. The other choices do not exhibit nocturnal periodicity.

29.

D. Sexual intercourse with infected men is thought to be the most important mode of transmission of *Trichomonas vaginalis* to women. Other routes of infection are direct contact with infected females or contact with infected toilet articles or toilet seats; these are considered rare modes of transmission. Infants may become infected while passing through the birth canal. Toxoplasmosis occurs as a congenital infection, but it is more commonly acquired by the ingestion of infected, undercooked meat or by swallowing oocysts excreted by infected cats. Schistosomiasis and trichinosis are not passed from person to person.

30.

D. Typical eggs of *T. trichiura* are yellow to brown, with colorless polar plugs. They are shaped like a football or a barrel, and they are in the cell, or unsegmented stage, when passed in the feces. The usual egg range is 49–65 × 35–45 μm.

31.

A. The gametocytes of *P. vivax, P. malariae,* and *P. ovale* are round and somewhat similar in appearance. Those of *P. falciparum* have a typical crescent shape. The gametocytes of *P. falciparum* may remain in the peripheral blood a month or more and are often found with the ring stages.

32.

A. The cysts of *G. lamblia* contain four nuclei and are passed in the stool of infected animals. The trophozoites are motile by flagella. *G. lamblia* does not possess an undulating membrane.

33.

A. The life cycle of a number of human nematodes includes migration through the skin and peripheral bloodstream. In individuals who have suffered several infections, allergic reactions can produce inflammation in the skin. *S. stercoralis* produces symptoms in the skin sometimes called larva migrans.

34.

D. *Naegleria fowleri* is found in freshwater ponds and lakes, especially those with disturbed or suspended soil. It has caused a number of cases of meningoencephalitis in people who have swum in these bodies of water. Essentially, all these infections have been fulminating and fatal, and they are often not diagnosed until autopsy.

35.

C. Infectious cysts of amebae and *Giardia lamblia* and eggs of *Ascaris lumbricoides* may all be ingested in fecally contaminated water and/or on fecally contaminated plants. These infections are most prevalent in areas lacking good public sanitation; that is, sanitary disposal of human waste and adequately treated and protected drinking water. Filiariasis is transmitted by blood-feeding insects (vectors).

36.

A. Humans are infected with *T. saginata* by eating beef containing live cysticerci, the infectious larval stage of this parasite. Cattle become infected by ingesting viable eggs from human feces. Unlike *Taenia solium,* if humans ingest *T. saginata* ova, infection does not develop.

37.

C. Cysts of *E. hartmanni* are differentiated from cysts of *E. histolytica* by their small size; they are otherwise morphologically identical. *E. hartmanni,* which was formerly called "small race ameba," is considered to be nonpathogenic. The size range for *E. hartmanni* cysts is 5–10 μm, and for *E. histolytica,* the range is 10–20 μm.

38.

D. The fertilized ova of *T. trichiura* are unsegmented when released, and embryonic development occurs outside of the host. In moist, warm, shaded soil, the first-stage larva develops within the egg in about 2 weeks. This fully embryonated egg is infective when ingested by a susceptible host, and it hatches in the small intestine. During development from larva to adult, the worm usually passes to the cecum, where it embeds its slender anterior portion in the intestinal mucosa.

39.

A. The most common form of *B. hominis* seen in human feces is called the "classic form." This form contains a central body that was previously thought to be a vacuole. The central body can take up to 90% of the volume of the cell, displacing the nuclei to the outer edge of the cell.

40.

C. Excretion in human feces of the eggs of the hookworms, *Taenia solium, T. saginata,* and *Ascaris lumbricoides,* is an essential or important factor in perpetuating the cycle of infection with these parasites. Trichinosis is caused by ingestion of the live larvae of *Trichinella spiralis* encysted in the muscles of a flesh-eating host. The adults live in the host's intestine, and the viviparous females, after fertilization, produce larvae that migrate into the host's muscle tissue.

41.

D. Typically, RBCs infected with *P. ovale* are larger than uninfected cells, pale and often misshapen, and frequently contain Schüffner's dots or stippling in any stage from young ring forms onward. RBCs infected with *P. vivax* are also larger than normal, oval, and contain Schüffner's dots. Ovale malaria, however, is a comparatively rare disease.

42.

A. *Balantidium coli* is the largest intestinal protozoan infecting humans. Cysts range in size from 43 to 65 μm, and the trophozoites are typically 50–100 μm in length and 40–70 μm in width. Pigs seem to be the most important reservoir for *B. coli.*

43.

C. The rhabditiform larvae of *S. stercoralis* are the diagnostic stage typically passed in the feces of infected persons. The larvae measure up to 380 μm long × 20 μm wide. They have a short buccal cavity and a prominent, ovoid, genital primordium midway along the ventral wall of the body. The infective stage is the filariform larva, which differs from the hookworm filariform larva by having a notched tail tip and a long esophagus.

44.

B. A malaria relapse is parasitemia developing from exoerythrocytic stages in the liver. These persistent stages are found in *P. ovale* and *P. vivax* infections, and they may cause relapses up to 4 or 5 years after the primary infection. For infections caused by these species, treatment with primaquine is used to prevent relapses (recurrences) after clinical cure with chloroquine or an alternate drug.

45.

A. *P. falciparum* infections tend to produce a large number of rings that frequently have double chromatin, which is only occasionally found in other species. *P. falciparum* differs from other plasmodia of humans in that only early trophozoites (ring forms) and gametocytes are found in peripheral blood except in severe cases. Sex differentiation of the gametes, when present, is difficult.

46.

D. *T. spiralis* is a nematode parasite whose infectious larvae may be found encysted in the muscles of flesh-eating mammals. Humans are infected most often by eating infected, undercooked pork. *Taenia* spp. are cestodes (tapeworms). *S. stercoralis* and *N. americanus* are roundworms whose infectious larvae usually develop in the environment and infect by penetration of human skin, although internal autoinfection may also occur.

47.

C. *N. americanus* and *A. duodenale* are two species of hookworms infecting humans. Their eggs are so similar when found in stool specimens that they are reported as "hookworm ova." The two hookworms can be differentiated by the morphologic characteristics of the adult worms, which are intestinal parasites.

48.

A. The life cycle of *T. gondii* includes five forms or stages, but only bradyzoites and tachyzoites appear in the tissue phase during human infections. The crescent-shaped tachyzoites are characteristic of acute infection. The slowly multiplying bradyzoites develop within cysts and are typical of chronic infections. Oocysts, merozoites, and gametes have been found only in the cat, where the sexual cycle of *T. gondii* occurs.

49.

D. A common sign of *S. haematobium* infection is the presence of blood in the urine. This is due to the damage caused when the eggs break out of the blood vessels of the vesicular plexus into the bladder. Falciparum malaria may also cause severe hematuria or "blackwater fever."

50.

C. *B. microti* is a sporozoan parasite commonly found in voles and field mice. The vector is the tick *Ixodes,* normally a parasite of deer. Humans are accidental hosts when bitten by an infected tick. Many *B. microti* infections within the U.S. occur in the Northeast. It is important to differentiate this parasite from *Plasmodium* in a stained blood film. Antimalarial drugs are not effective in babesiosis.

51.

B. Chagas disease is found throughout the American continents. The infectious agent, *T. cruzi,* is transmitted to humans by reduviid bugs, primarily the triatomids. Chagas disease can be acute or chronic.

52.

A. Collection of blood by finger stick is preferred for preparing blood smears for the detection of malaria. When a venipuncture is performed, the preferred anticoagulant for malarial blood smears is EDTA (ethylenediaminetetraacetic acid). Heparin can be used, but it may cause distortion of some parasite forms.

53.

A. *B. coli* is the only ciliate that is pathogenic for humans. It is relatively easy to detect in stool samples because of its large size. The trophozoite, seen in Color Plate 38■, is generally oval and measures 50–100 × 40–70 μm. A cytosome is present on the anterior end.

54.

D. Color Plate 39■ demonstrates a fertilized egg of *A. lumbricoides*. Eggs measure 45−75 × 35−50 μm. Unfertilized *Ascaris* eggs typically do not float in the zinc sulfate concentration technique.

55.

B. Color Plate 40■ demonstrates a *G. lamblia* trophozoite; notice the two prominent nuclei. Trophozoites of *C. mesnili* are approximately 6–24 μm in length but have a single nucleus, whereas *G. lamblia* trophozoites have two nuclei. Trophozoites of *Trichomonas* are about the same size as *G. lamblia,* but they are more round than the pear-shaped trophozoites of *G. lamblia* and *C. mesnili*. *Trichomonas* spp. have a single nucleus, and *T. tenax* is found in the oral cavity.

56.

C. Color Plate 41■ demonstrates a *D. fragilis* trophozoite. Although this organism lacks a flagellum and morphologically resembles the ameba, based on its ultrastructure and molecular biology studies, it is classified as a flagellate. Like the trichomonads, *D. fragilis* does not have a cyst stage. Most trophozoites of *D. fragilis* have two nuclei, like the one in this image.

57.

C. Color Plate 42■ demonstrates an *E. coli* cyst. These cysts most closely resemble *E. histolytica* and *E. dispar*. The key distinguishing feature is that *E. coli* cysts contain up to eight nuclei, whereas *E. histolytica* and *E. dispar* have up to four nuclei. It is often necessary to use the fine adjustment to see all the nuclei. In this image, six nuclei can be seen. Trophozoites of all three species only contain one nucleus.

58.

A. Color Plate 43■ demonstrates a hookworm ovum. In addition to size, key characteristics are the thin ovum shell and nearly symmetrical shape. *Enterobius vermicularis,* pinworm, ova appear similar, except that they have a flattened side and thicker shell. In addition, the ova are slightly smaller. Ova of *D. latum* are unembryonated, operculated, and slightly larger than hookworm ova.

59.

D. Color Plate 44■ demonstrates the "classic form" of *B. hominis*. After years of taxonomic uncertainty, this organism is currently classified as an ameba; however, rRNA studies indicate it is related to algae and water moulds. The classic form usually seen in human feces varies in size from 6 to 40 μm in diameter. It contains a large central body, resembling a vacuole that pushes several nuclei to the periphery of the cell.

60.

B. Color Plate 45■ demonstrates a *P. malariae* trophozoite. A trophozoite stretching across the infected RBC, called a band form, is a key characteristic of *P. malariae*. Other important characteristics include a lack of malarial pigment, and infected RBCs are about the same size as uninfected RBCs. During *Babesia* infections, only ring forms are seen.

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

1. The retrovirus responsible for causing acquired immune deficiency syndrome is a member of the family
   A. *Orthomyxoviridae*
   B. *Paramyxoviridae*
   C. *Retroviridae*
   D. *Flaviviridae*

2. The appearance of Koplik spots in the oral mucosa of patients is characteristic of infection with what viral agent?
   A. Hepatitis
   B. Measles
   C. Rabies
   D. Smallpox

3. Characteristics of this DNA hepatitis virus include infections spread by contaminated body fluids, 50- to 180-day incubation period, and chronic infections.
   A. HAV
   B. HBV
   C. HCV
   D. HEV

4. Which of the following has been declared eradicated by the World Health Organization?
   A. Smallpox
   B. Human T cell lymphotropic virus
   C. Hepatitis G virus
   D. Eastern equine encephalitis

5. Rotavirus is the most common etiologic agent of
   A. Acute nonbacterial encephalitis in children
   B. Acute nonbacterial gastroenteritis in infants and young children
   C. Chronic nonbacterial pharyngitis in children and young adults
   D. Chronic nonbacterial retinitis in children

6. Kaposi sarcoma is associated with infection by
   A. Adenovirus
   B. Cytomegalovirus
   C. Hepatitis E virus
   D. Human herpes virus 8

7. The molecular receptor of the virus causing acquired immune deficiency syndrome is
   A. CD 4
   B. CD 8
   C. Fc receptor
   D. Complement receptor

8. The type of cell culture that best supports the growth of cytomegalovirus is
   A. HeLa cells
   B. HEp-2 cells
   C. Human fibroblast cells
   D. Primary monkey kidney (PMK) cells

9. Which of the following viruses is predominantly associated with respiratory disease and epidemics of keratoconjunctivitis?
   A. Adenovirus
   B. Molluscum contagiosum virus
   C. Norwalk virus
   D. Rotavirus

10. A 25-year-old patient presented with multiple vesicles around the mouth. Material from the lesions was obtained by needle aspiration and inoculated to MRC-5 cells. After 1 day, the cytopathic effect included foci of "ballooned" and lysed cells. These observations suggest infection with
    A. Adenovirus
    B. Cytomegalovirus
    C. Epstein-Barr virus
    D. Herpes simplex virus

11. The Sabin polio vaccine uses which of the following?
    A. Formalin-inactivated viruses
    B. Attenuated viruses
    C. Recombinant viral antigens
    D. DNA

12. Which of the following is caused by a herpes virus?
    A. Cold sores
    B. Hemorrhagic fever
    C. Polio
    D. Rabies

13. Which of the following is *not* a general characteristic of a virus?
    A. Obligate intracellular parasite
    B. Does not produce ATP
    C. Genome is surrounded by a protein coat
    D. Can self-replicate in the appropriate host cell

14. The viral disease shingles, which causes extreme tenderness along the dorsal nerve roots and a vesicular eruption, has the same etiologic agent as
    A. Rubeola
    B. Vaccinia
    C. Varicella
    D. Variola

15. The etiologic agents of many common colds are RNA viruses that grow better at 33°C than at 37°C. These viruses are
    A. Adenoviruses
    B. Orthomyxoviruses
    C. Paramyxoviruses
    D. Rhinoviruses

16. Influenza A virus undergoes recombination events that produce new strains; this is referred to as
    A. Antigenic drift
    B. Antigenic shift
    C. Reactivation
    D. Viral latency

17. Negri bodies may be found in brain tissue of humans or animals infected with
    A. Adenovirus
    B. Filovirus
    C. Measles virus
    D. Rabies virus

18. Molluscum contagiosum virus is a member of the
    A. Adenoviruses
    B. Herpesviruses
    C. Papovaviruses
    D. Poxviruses

19. A clinical specimen is received in viral transport medium for viral isolation. The specimen cannot be processed for 72 hours. At what temperature should it be stored?
    A. –80°C
    B. –20°C
    C. 4°C
    D. 22°C

20. Arboviruses
    A. Only infect humans
    B. Often cause hepatitis
    C. Typically infect lymphocytes
    D. Are transmitted by arthropods

21. Mumps is characterized by an infection of the
    A. Central nervous system
    B. Parotid glands
    C. Pancreas
    D. Thymus

22. Which of the following hepatitis viruses is typically transmitted by the fecal-oral pathway?
    A. HAV
    B. HBV
    C. HCV
    D. HGV

23. Enteroviruses are most often associated with
    A. Acute nonspecific febrile syndrome
    B. Bronchitis and pneumonia
    C. Lower respiratory tract infections
    D. Upper respiratory tract infections

24. The "core window" refers to the time
    A. During hepatitis B virus infection when anti-HBc IgM is the only serologic marker
    B. During hepatitis B virus infection when HBc is the only serologic marker
    C. During hepatitis A virus infection when HAc is the only serologic marker
    D. During hepatitis C virus infection when the virus is latent

25. Human herpesviruses 6 and 7 are associated with a childhood disease called
    A. Chickenpox
    B. Measles
    C. Roseola
    D. Zoster

26. A baby was admitted to the hospital in February for dehydration due to severe diarrhea. Cultures for bacterial pathogens revealed normal fecal flora at 24 hours. Which of the following additional tests would be most appropriate given the case history?
    A. Heterophile antibody test
    B. Rotavirus antigen assay of stool specimen
    C. McCoy cell inoculation for cytomegalovirus (CMV)
    D. Urine microscopic analysis for presence of CMV cellular inclusion bodies

27. Rhabdovirus is most noted for causing infections of the
    A. Central nervous system
    B. Gastrointestinal tract
    C. Lower respiratory tract
    D. Upper respiratory tract

28. Jaundice is a common clinical symptom of which of the following viral diseases?
    A. Hepatitis A
    B. Infectious mononucleosis
    C. Rabies
    D. Varicella

29. An 18-year-old male presents to his family physician complaining of sore throat and fatigue. The patient is found to have a fever and swollen cervical lymph nodes. A complete blood count and differential reveal lymphocytosis and many reactive (atypical) lymphocytes. The physician should suspect an infection caused by
    A. Adenoviruses
    B. Epstein-Barr virus
    C. Parainfluenza virus
    D. Varicella-zoster virus

30. The poliovirus, an RNA virus, is a(n)
    A. Adenovirus
    B. Coxsackie virus
    C. Enterovirus
    D. Rhinovirus

31. The virus that causes hepatitis B is characterized as a
    A. Defective DNA virus requiring delta virus to complete its replication cycle
    B. DNA virus utilizing reverse transcriptase
    C. Nonenveloped DNA virus
    D. Single-stranded RNA virus

32. Hepatitis C virus infections
    A. Are commonly diagnosed during the acute stage
    B. Are uncommon in the U.S.
    C. Are most often acquired by contact with blood
    D. Seldom results in chronic infection

33. Characteristic cytopathic effect associated with respiratory syncytial virus is
    A. Giant multinucleated cells
    B. Basophilic intranuclear inclusions
    C. Eosinophilic cytoplasmic inclusions
    D. Shrunken cells with multilobed nuclei

34. The virus associated with warts is
    A. Flavivirus
    B. Morbillivirus
    C. Mumps virus
    D. Papillomavirus

35. RNA-dependent DNA polymerase is also called
    A. Gyrase
    B. Neuraminidase
    C. Reverse transcriptase
    D. Transaminase

36. Coxsackie viruses are associated with
    A. Gastrointestinal disease
    B. Hepatitis
    C. Myocarditis
    D. The common cold

37. The tubular cells of the human kidney shed which of the following viruses for prolonged periods?
    A. Adenovirus
    B. Cytomegalovirus
    C. Epstein-Barr virus
    D. Rubella virus

38. The togavirus known to produce fetal defects is
    A. Influenza
    B. Rotavirus
    C. Rubella
    D. Varicella

39. An 8-week-old infant was admitted to the hospital with symptoms of low birth weight, jaundice, and neurologic defects. Intranuclear inclusions were found in epithelial cells from the urine. The most likely diagnosis in this case would be infection by
    A. Cytomegalovirus
    B. Epstein-Barr virus
    C. Herpes simplex virus
    D. Rubella virus

40. The most common cause of cervical cancer is
    A. Cytomegalovirus
    B. Enterovirus
    C. Molluscum contagiosum
    D. Papillomavirus

41. Select the statement that is correct concerning the influenza A viruses.
    A. Humans are the only animal hosts for influenza A viruses.
    B. Pandemics are characteristically produced by influenza A.
    C. The incidence of infection peaks in the summer months.
    D. They are DNA viruses.

42. An example of a virus associated with latent infections is
    A. Influenza
    B. Rotavirus
    C. Rubella
    D. Varicella-zoster

43. The use of cell cultures has enabled virologists to isolate and identify many clinically important viruses. However, because some viruses cannot be grown in cell cultures, these agents are best diagnosed by serologic testing. Such an agent is
    A. Cytomegalovirus
    B. Hepatitis C virus
    C. Herpes simplex virus 2
    D. Respiratory syncytial virus

44. Which of the following is associated with the rubella virus?
    A. It is a DNA virus.
    B. It is a member of the same taxonomic family as measles virus.
    C. It is known to produce defects in fetuses during the early stages of pregnancy.
    D. It is transmitted by an arthropod vector.

45. Which of the following is *not* a step involved in virus replication?
    A. Attachment
    B. Mitosis
    C. Penetration
    D. Release

46. Which of the following opportunistic diseases is *not* closely associated with acquired immune deficiency syndrome?
    A. Cryptococcosis
    B. Cryptosporidiosis
    C. Malaria
    D. Mycobacteriosis

47. Although there have been no natural cases of this serious disease in about 30 years, which of the following is considered a potential bioterrorism disease
    A. Dengue
    B. Ebola hemorrhagic fever
    C. Shingles
    D. Smallpox

48. Poliovirus is a member of the family
    A. *Flaviviridae*
    B. *Paramyxoviridae*
    C. *Picornaviridae*
    D. *Reoviridae*

49. Which of the following has *not* been successfully used to detect viruses in clinical specimens?
    A. Cytopathic effect
    B. Enzyme-linked immunosorbent assay
    C. Growth on selective agar media
    D. Immunofluorescence

50. Which of the following diseases is *not* associated with herpes simplex virus?
    A. Cold sores
    B. Encephalitis
    C. Genital herpes
    D. Thrush

answers & rationales

1.

C. Retroviruses are RNA viruses that replicate by means of DNA intermediates produced by the viral enzyme reverse transcriptase. The viruses associated with acquired immune deficiency syndrome are human immunodeficiency viruses (HIVs). These viruses belong to the family *Retroviridae*.

2.

B. Measles (rubeola) is a highly infectious childhood disease. Infection with this virus is followed by a prodromal syndrome characterized by cough, coryza, conjunctivitis, and fever. The most characteristic lesions, Koplik spots, are seen on the buccal mucosa. Koplik spots are diagnostic for measles infection and represent necrotic vesicles with a white center surrounded by erythema.

3.

B. Hepatits B virus is an enveloped DNA virus transmitted by contact with blood or via sexual contact. Most hepatitis viruses have a long incubation period of several weeks to months. Hepatitis A and hepatitis E viruses are RNA viruses primarily spread via the fecal-oral route. Hepatitis C virus is an RNA virus transmitted by contact with blood.

4.

A. The last natural case of smallpox was in 1977, and the World Health Organization declared the world smallpox free in 1979. Elimination of the virus was due to a worldwide vaccination program. Because of the highly contagious nature of variola virus, the cause of smallpox, the ability of the virus to produce severe infections, and the termination of routine vaccinations, the virus is considered a potential bioterrorism agent.

5.

B. One of the major viral agents associated with cases of acute gastroenteritis in children is rotavirus. In particular, this agent is the cause of epidemic nonbacterial gastroenteritis in infants and young children that occurs most commonly during the winter months. Rotavirus belongs to the family of RNA viruses known as *Reoviridae*. Rotavirus has a fecal-oral route of transmission and has been documented as a nosocomial pathogen in pediatric areas of hospitals.

6.

D. During acquired immune deficiency syndrome (AIDS), as the immune system becomes weakened, the patient presents with chronic and recurrent infections and various neoplasms. Kaposi sarcoma, a relatively common cancer in patients with AIDS, has been linked to human herpes virus 8. Viral genome has been found in the cancerous growths in these patients.

7.

A. The human immunodeficiency viruses cause AIDS. The major target of the virus is the T helper cell, which would normally function to control disease. The virus initially binds to CD 4 found on the surface of T helper cells. Other coreceptors are also important for attachment.

8.

C. Commercially available cell cultures of human fibroblasts are optimal for the cultivation of cytomegalovirus (CMV). CMV will not replicate in other cell cultures such as HeLa or HEp-2. CMV can be identified with a high level of confidence solely on the basis of its characteristic cytopathology. Infected cells in the monolayer appear enlarged, rounded, and refractile.

9.

A. Adenoviruses are well known as respiratory pathogens and have been the cause of acute respiratory disease among military recruit populations. Also associated with adenoviral infection is the severe ocular disease keratoconjunctivitis, which typically occurs in epidemic form. Adenoviruses may remain in tissues, lymphoid structures, and adenoids and become reactivated.

10.

D. Cell cultures recommended for the isolation of herpes simplex virus (HSV) are human embryonic fibroblasts. The usual period needed to detect HSV destruction of the cell monolayer is 1 to 2 days. The more common of the two recognized types of HSV cytopathic effect begins with a granulation of the cytoplasm followed by cell enlargement and a ballooned appearance. Monoclonal antibodies and immunofluorescence are commonly used to differentiate between HSV-1 and HSV-2.

11.

B. The Salk vaccine utilizes a formalin-inactivated poliovirus. The Sabin polio vaccine uses an attenuated virus; therefore, the virus is still able to infect cells and cause an asymptomatic infection. The Sabin vaccine provides a stronger immune response than the Salk vaccine.

12.

A. Members of the herpes virus group are responsible for a number of diseases, including cold sores. Hemorrhagic fevers are caused by a number of tropical viruses, such as Ebola and dengue. Polio is caused by a picornavirus, and rabies is caused by the rhabdovirus.

13.

D. Viruses are obligate intracellular parasites that cannot self-replicate. They cannot produce ATP, and their genome is surrounded by a protein capsid. Most viruses contain either DNA or RNA; however, some large DNA viruses do contain viral mRNA and microRNAs.

14.

C. Zoster or shingles occurs predominantly in adults, whereas varicella occurs more commonly in children. The varicella-zoster virus, following the primary infection known as chicken pox, remains latent in the sensory ganglia. Reactivation of this virus, which occurs years later, is usually associated with a slightly immunocompromised state.

15.

D. Rhinoviruses, members of the picornavirus group, are a common cause of the respiratory disease known as the common cold. Hand transmission, not aerosols, appears to be the primary means of transmission. In contrast to other picornaviruses, the optimum temperature for rhinoviruses is 33°C.

16.

B. Influenza A virus undergoes recombination events that produce significant changes in the RNA genome of the virus. These changes lead to alteration of surface antigens. This process is referred to as antigenic shift. Antigenic drift is a slight change in a gene, usually a point mutation. Influenza A, B, and C viruses can undergo antigenic drift.

17.

D. Rabies is a neurotropic virus that causes extensive destruction in the brain. Negri bodies are seen in the cytoplasm of large ganglion cells and are demonstrated by Seller's stain. Rabies in humans or lower animals can be diagnosed by demonstration of these characteristic inclusions. However, the more sensitive direct fluorescent antibody test is more commonly used.

18.

D. Molluscum contagiosum is an infectious disease with a worldwide distribution caused by a poxvirus. Nodules develop in the epidermis of the face, arms, back, and buttocks, which undergo necrosis. Examination of epithelioid cells from affected areas will show characteristic eosinophilic cytoplasmic inclusions (molluscum bodies).

19.

A. Prolonged storage of clinical specimens for viral isolation requires –80°C. Specimens can be stored at 4°C for approximately 48 hours without appreciable loss of viability. Specimens should not be stored at –20°C for any length of time.

20.

D. Arbovirus is short for arthropod-borne virus. These genetically diverse viruses share a common feature: They are transmitted by arthropods (e.g., mosquitoes and ticks). Arboviruses include West Nile virus and western equine encephalitis virus.

21.

B. The mumps virus infects the parotid glands. Infection results in swelling of the neck. Mumps is primarily a childhood infection, and swelling of the parotid glands is diagnostic.

22.

A. Hepatitis A virus is typically transmitted by the fecal-oral pathway. Hepatitis B, C, and G viruses are generally transmitted by blood contact. Hepatitis E virus is also transmitted by the fecal-oral pathway.

23.

A. Most enterovirus infections are probably asymptomatic. Despite the name, these viruses are rarely associated with infections of the gastrointestinal tract. When symptomatic, they are most noted for producing acute nonspecific febrile syndrome.

24.

A. During the course of acute hepatitis B virus infection, hepatitis B surface antigen (HBsAg) is the first marker detected. The host will ultimately begin to produce antibody (anti-HBs) to the antigen. As the antibody titer increases, there is a corresponding decrease in the antigen. However, there is a time period when neither of these markers is detectable. During this time period the only serologic marker is antibody to the hepatitis B core antigen (anti-HBc). This period is called the core window.

25.

C. Human herpes viruses (HHVs) 6 and 7 cause the childhood disease roseola, also called sixth disease. The disease is characterized by fever, rash, and sore throat. More cases are caused by HHV 6 than HHV 7.

26.

B. Rotavirus is the cause of diarrheal disease in at least half of all infants and young children admitted to the hospital with dehydration requiring fluid replacement therapy. Because rotaviruses are difficult to propagate in cell culture, the method of choice for the detection of rotavirus infection is the direct examination of stool for the presence of viral antigen. Commonly used rotavirus antigen assay tests include latex agglutination and enzyme immunoassay.

27.

A. Rhabdovirus causes rabies, an infection of the central nervous system. The virus is transmitted in the saliva of an infected animal during a bite. At the bite site, the virus initially infects muscle tissue, but will move to the peripheral nerves. The virus then migrates along the peripheral nerves to the central nervous system.

28.

A. Hepatitis A is one of several infectious diseases characterized by liver damage and icterus (jaundice). The appearance of jaundice, in the icteric phase, is correlated by liver biopsy with extensive parenchymal destruction. Convalescence is usually accompanied by subsequent complete regeneration of the diseased organ.

29.

B. The Epstein-Barr virus, which is associated with Burkitt lymphoma and nasopharyngeal carcinoma, is the etiologic agent of infectious mononucleosis. Infectious mononucleosis is an acute disease most commonly affecting children and young adults. The virus is thought to be transmitted by intimate contact and has been called the "kissing disease." The patient's blood demonstrates a leukocytosis with a marked increase in T lymphocytes, and serologically the disease is characterized by a positive heterophile antibody and antibodies to various viral antigens.

30.

C. Poliovirus, an enterovirus, is shed by both respiratory and fecal routes. Laboratory identification relies on isolation (especially from feces) and subsequent virus neutralization in tissue culture. Spread of the disease is associated with poor sanitary conditions and crowding.

31.

B. The hepatitis B virus is an enveloped, partially double-stranded DNA virus. During viral replication, full-length RNA transcripts of the viral genome are inserted into maturing virus particles. The viral enzyme reverse transcriptase then transcribes these RNA transcripts to a full-length DNA strand but only partially completes synthesis of the complementary DNA strand—hence a partially double-stranded DNA genome.

32.

C. Hepatitis C virus infections, unlike hepatitis A or hepatitis B infections, do not commonly produce jaundice. There are tens of thousands of individuals in the United States chronically infected with hepatitis C; chronic infection appears to be the rule rather than the exception. Transmission of the virus at present occurs mainly through needle sharing. Cases also occur among healthcare workers who contact infected blood.

33.

A. Respiratory syncytial virus causes fusion of adjacent cells. This produces giant multinucleated cells called syncytia. Basophilic intranuclear and eosinophilic cytoplasmic inclusions are characteristic of cytomegalovirus infection.

34.

D. The etiologic agents for the numerous benign cutaneous and mucosal lesions known as warts are the human papillomaviruses (HPVs). The diagnosis of lesions caused by these agents is based on clinical appearance and histopathology, because there are no *in vitro* systems available for isolation. Some HPV types are strongly associated with squamous cell carcinoma of the cervix and anus.

35.

C. RNA-dependent DNA polymerase is also known as reverse transcriptase. The enzyme uses an RNA template to synthesize the complementary DNA strand. The retroviruses require this enzyme for replication.

36.

C. The coxsackieviruses are enteroviruses named after the town of Coxsackie, New York, where they were first isolated. The viruses are divided into groups A and B on the basis of viral and antigenic differences. The group B coxsackieviruses are strongly associated with myocarditis that may cause sufficient damage to require heart transplantation. The group A coxsackieviruses are associated with various diseases, characterized by vesicular lesions, such as herpangina. Neither group of coxsackieviruses is associated with gastrointestinal disease.

37.

B. Cytomegalovirus infections may be asymptomatic for normal healthy hosts. Infections tend to be more severe in patients who are immunosuppressed or in neonates infected perinatally. Cytomegalovirus is readily isolated from urine because it is shed by the tubular cells of infected hosts.

38.

C. The rubella virus is an RNA virus and a member of the family *Togaviridae*. In adults and children, rubella infections are generally a mild contagious rash disease. When a pregnant woman becomes infected, the consequences become more serious. If the fetus is infected during the first trimester of pregnancy, a variety of congenital defects may result. Anatomic abnormalities produced by this agent include cataracts, deafness, and cardiac problems.

39.

A. Infants usually acquire cytomegalovirus infections before birth or at the time of childbirth. These infections may lead to death during the first month of life or may result in residual neurologic impairment. The virus can be isolated from several different body fluids, with urine being the most commonly examined.

40.

D. Papillomaviruses are responsible for warts, including genital warts. Some serotypes of papillomavirus are associated with cervical cancer. A vaccine providing protection against these serotypes is available for women.

41.

B. Influenza viruses are RNA viruses able to infect humans and other animals, such as birds and pigs. Influenza viruses are associated with epidemic and pandemic disease. There are two main types of influenza viruses (A and B), which differ antigenically and in epidemic periodicity. All recorded pandemics have been caused by influenza A viruses. The incidence of respiratory disease caused by these agents peaks during the winter months.

42.

D. Herpes simplex viruses, cytomegalovirus, and varicella-zoster viruses (VZV) produce latent infections. The genomes of these viruses can remain dormant in host cells for decades. Shingles (zoster) represents reactivation of latent VZV.

43.

B. Hepatitis C virus has never been grown in culture. All knowledge of the virus and diagnostic reagents has been attained through molecular techniques applied to the RNA genome. Hepatitis C virus infection can be diagnosed by detecting antibody to the virus or by amplifying viral RNA from plasma. Cytomegalovirus, herpes simplex virus, and respiratory syncytial virus are readily grown in culture.

44.

C. The rubella virus causes an exanthematous disease resembling a milder form of measles in children. This single-stranded RNA virus, transmitted from person to person, is of medical importance to females of childbearing years because of the teratogenic effects it has on the fetus. Congenital rubella, resulting from an intrauterine fetal infection, is most severe when contracted during the first trimester of gestation.

45.

B. Lacking essential components for the synthesis of macromolecules, viruses are not able to reproduce by binary fission. Host cells are required to provide the synthesis of viral components. The replicative cycle has four stages: absorption, penetration (uncoating), eclipse (biosynthesis), and release (maturation). Viral replication in the host cell may result in cell death, chronic infection with no observable changes, or transformation of the infected cell into a cancerous cell.

46.

C. The immunologic abnormalities demonstrated by patients with AIDS predispose them to a variety of opportunistic pathogens. The absence of a cellular and humoral immune response enables opportunistic organisms to cause extensive infection. Malaria is not an opportunistic infection.

47.

D. Although smallpox has not caused a natural infection since 1977, it is regarded as a potential bioterrorism agent. Routine vaccination against smallpox is no longer in effect, so the majority of the world's population is again susceptible. The U.S. has stockpiles of vaccine available to vaccinate everyone in the country in case of an attack.

48.

C. Poliovirus is a member of the family *Picornaviridae*. These are small RNA viruses lacking an envelope. Other members of the family include coxsackieviruses, echoviruses, enteroviruses, and rhinoviruses.

49.

C. Because of their nature as obligate intracellular parasites, successful cultivation of viruses requires living cells. Cell cultures provide host cell systems, which are easily handled, stable for long periods, and not susceptible to host factors such as stress or physiologic changes. Viruses are not like bacteria; they will not grow on any cell-free medium.

50.

D. Herpes simplex viruses (HSVs) are noted for causing a number of different diseases, notably cold sores (primarily HSV-1) and genital infections (primarily HSV-2). In neonates and immunocompromised individuals, they can produce encephalitis. They are also linked to severe eye infections. Thrush is an infection caused by the yeast *Candida*.

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

1. If 20% of the nucleotides in an organism are adenine, predict the percentage of nucleotides that are guanine.
   A. 20%
   B. 30%
   C. 40%
   D. 60%

2. Which of the following is *not* required for DNA replication by PCR?
   A. Oligonucleotide primers
   B. DNA polymerase
   C. DNA ligase
   D. Deoxynucleotides

3. In naming restriction endonucleases, the first letter of the name comes from the
   A. Bacterial genus
   B. Bacterial species
   C. Scientist who discovered it
   D. Geographic location of its discovery

4. A restriction enzyme recognizes the sequence, 5' CT^ATAG 3', and cuts as indicated. Predict the ends that would result on the *complementary* DNA strand.
   A. 3' G 5' 3' ATATC 5'
   B. 3' GA 5' 3' TATC 5'
   C. 3' GATA 5' 3' TC 5'
   D. 3' GATAT 5' 3' C 5'

5. The absorbance of a 1:100 dilution of isolated dsDNA solution, measured at 260 nm, is 0.062. What is a reasonable estimate for the dsDNA concentration of the sample, expressed in μg/mL?
   A. 3.1
   B. 6.2
   C. 310
   D. 5000

6. In the isolation of RNA, diethylpyrocarbonate (DEPC) is used to
   A. Inhibit RNase
   B. Lyse the cells
   C. Precipitate the DNA
   D. Remove buffer salts

7. Purification resins used to isolate DNA take advantage of the fact that DNA is
   A. Double stranded
   B. Negatively charged
   C. Higher in concentration than RNA
   D. Higher molecular weight than RNA

8. After performance of DNA electrophoresis, the isolated bands in the kilobase size range appear too close together. Which of the following can be done with the next run to improve the appearance/separation of the bands in the samples?
   A. Increase the percent agarose concentration of the matrix
   B. Increase the running time of the electrophoresis assay
   C. Increase the sample volume applied to the gel
   D. Decrease the sample volume applied to the gel

9. Which of the following is commonly used as a label in molecular tests?
   A. Biotin
   B. DNase
   C. RNase
   D. ^{125}I

10. Which of the following is *not* an example of target amplification?
    A. Reverse transcription-PCR (RT-PCR)
    B. Transcription mediated amplification (TMA)
    C. Branched chain DNA amplification (bDNA)
    D. Polymerase chain reaction (PCR)

11. In forensic testing, DNA fingerprinting can identify individuals with high accuracy because
    A. Human genes are highly conserved
    B. Only a small amount of sample is needed
    C. Human gene loci are polymorphic
    D. DNA is stable and not easily contaminated

12. The technique that makes ssDNA from an RNA template is called
    A. Strand displacement amplification
    B. Polymerase chain reaction
    C. Ligase chain reaction
    D. Reverse transcription

13. A 5850-base plasmid possesses *Eco*RI restriction enzyme cleavage sites at the following base pair locations: 36, 1652, and 2702. Following plasmid digestion, the sample is electrophoresed in a 2% agarose gel. A DNA ladder marker, labeled M in Color Plate 56■, is included in the first lane, with base pair sizes indicated in lanes A through D. Which lane represents the sample pattern that is most likely the digested plasmid?
    A. A
    B. B
    C. C
    D. D

14. Which of the following is characteristic of DNA chips (i.e., DNA microarrays)?
    A. Allow detection and discrimination of multiple genetic sequences at the same time.
    B. Thousands of oligonucleotide probes are labeled and placed on glass or silicon surfaces.
    C. Unlabeled target sequences within the patient sample are detected by hybridization to labeled probes.
    D. All the above

15. The most useful feature of the molecules streptavidin and biotin is that they bind
    A. Specifically to nucleic acids
    B. Only in neutral pH conditions
    C. To each other with very high affinity
    D. Directly to DNA immobilized on nitrocellulose

16. What is the theoretic estimation of the number of DNA target sequences present (per original double-stranded DNA in solution) following 15 cycles of PCR?
    A. 30
    B. 2^{10} (i.e., 1024)
    C. 2^{15} (i.e., 32,768)
    D. 2^{20} (i.e., 1,048,576)

17. "Star activity" for a restriction enzyme refers to
    A. An ability to cleave DNA at sequences different from their defined recognition sites
    B. The enzyme's specificity for sites of methylation within the nucleotide sequence
    C. The temperature and pH conditions at which the enzyme will function optimally
    D. The percent increased accuracy of the enzyme when placed in ideal conditions of pH

18. What enzyme recognizes and cuts overlapping DNA sequences formed between mutant or normal probes and target sequences within samples?
    A. Restriction endonuclease
    B. DNA ligase
    C. Cleavase
    D. RNase H

19. If a DNA probe is added to nitrocellulose after the transfer step but before the blocking step, which of the following will occur?
    A. The probe will nonspecifically bind to its DNA target.
    B. Unoccupied spaces on the nitrocellulose will bind the probe.
    C. The DNA target on the nitrocellulose will be unable to bind the probe.
    D. Bound probe will be washed away in the next wash step.

20. Which of the following items is *not* used in the preparation of a DNA probe for Southern blotting using random hexamer primers?
    A. Template DNA
    B. Three unlabeled deoxynucleotides
    C. Dideoxynucleotides, with one of them labeled
    D. DNA polymerase

21. Which of the following is considered a "high stringency" condition for DNA probe protocols?
    A. Using wash buffer with highly acidic pH
    B. Washing the matrix with high-salt buffer
    C. Radiolabeling the probe with ^{35}S rather than ^{32}P
    D. Washing the transfer membrane (e.g., nitrocellulose or nylon) at high temperature

22. When compared to Southern blot hybridization testing, PCR
    A. Is less sensitive to DNA degradation than Southern blot
    B. Includes transfer of DNA onto a nylon membrane
    C. Requires no specialized equipment
    D. Is more labor intensive

23. Which of the following specimen types is *not* used routinely as source material for molecular genetic tests?
    A. Whole blood
    B. Buccal scrapings
    C. Amniocytes
    D. Rectal swabs

24. In the presence of salt, DNA is precipitated from solution by
    A. 10 mM Tris, 1 mM EDTA
    B. 0.1% sodium dodecyl sulfate (SDS)
    C. Alkaline buffers, such as 0.2 N NaOH
    D. Alcohols, such as 95% ethanol or isopropanol

25. TaqMan probes used to increase specificity of real-time PCR assays generate a fluorescent signal
    A. At the beginning of each cycle during the denaturation step
    B. When the probes bind to the template (i.e., during annealing)
    C. When the probe is digested by 5' → 3' exonuclease activity during extension of primers (i.e., DNA synthesis)
    D. When the reporter fluorophor on the probe is separated from the quencher molecule by a restriction enzyme

26. For the purpose of diagnosing genetic diseases, what component of whole blood is used for the extraction of DNA?
    A. Leukocytes
    B. Plasma
    C. Platelets
    D. Red blood cells

27. Which of the following statements best describes characteristics of RNase?
    A. It degrades mRNA but not rRNA.
    B. It is found in large concentrations on hands.
    C. Its activity can be eliminated by autoclaving.
    D. Its activity occurs in a limited temperature range between 25 and 65°C.

28. Which of the following is the *least* likely inhibitor of PCR?
    A. Heme
    B. Sodium heparin
    C. DEPC (diethylpyrocarbonate)
    D. EDTA (ethylenediaminetetraacetic acid)

29. Frequently, DNA probes are used to detect target sequences in Northern or Southern blots. Hybridization occurs between DNA probe and RNA or DNA on the blot, respectively. To ensure that only exactly matched complementary sequences have bound together, the blot is washed under stringent conditions. Stringency of the wash steps to remove unbound and mismatched probe can be increased by
    A. High temperature, high NaCl concentration, and high detergent (i.e., SDS) solution
    B. High temperature, low NaCl concentration, and high detergent (i.e., SDS) solution
    C. High temperature, high NaCl concentration, and low detergent (i.e., SDS) solution
    D. Low temperature, high NaCl concentration, and high detergent (i.e., SDS) solution

30. In RNA, which nucleotide base replaces thymine of DNA?
    A. Adenine
    B. Cytosine
    C. Guanine
    D. Uracil

31. The component parts of a dNTP include a purine or pyrimidine base, a
    A. Ribose sugar, and one phosphate group
    B. Deoxyribose sugar, and three phosphate groups
    C. Ribose sugar, and two phosphate groups
    D. Deoxyribose sugar, and two phosphate groups

32. When comparing two dsDNA sequences of equal length, the strand that has a higher
    A. G + C content has a higher melting temperature (T_m)
    B. A + T content has a higher T_m
    C. A + T content has more purines than pyrimidines along its length
    D. G + C content has more purines than pyrimidines along its length

33. Molecular typing of bacterial strains is based on restriction fragment length polymorphisms (RFLPs) produced by digesting bacterial chromosomal DNA with restriction endonucleases. Which of the following techniques is used to separate the large DNA fragments generated?
    A. Ribotyping
    B. DNA sequencing
    C. Pulsed field gel electrophoresis
    D. Reverse transcription–polymerase chain reaction

34. Which of the following amplification methods does *not* employ isothermal conditions?
    A. Nucleic acid sequence–based amplification (NASBA)
    B. Polymerase chain reaction (PCR)
    C. Strand displacement amplification (SDA)
    D. Transcription mediated amplification (TMA)

35. The coding region of a human gene is called
    A. Exon
    B. Intron
    C. SNP
    D. VNTR

36. The central dogma is that DNA is used to make RNA, which is then used to make protein. In this scheme the two processes that are involved (i.e., DNA to RNA and RNA to protein) are termed
    A. Replication and transcription
    B. Synthesis and encryption
    C. Transcription and translation
    D. Initiation and elongation

37. How many chromosomes are contained in a normal human somatic cell?
    A. 22
    B. 23
    C. 44
    D. 46

38. An ordered sequence of events makes up the cell cycle. Which of the following describes the correct sequence of events starting at G1?
    A. G1, G2, S, M
    B. G1, S, G2, M
    C. G1, M, G2, S
    D. G1, S, M, G2

39. Purified DNA remains stable indefinitely when stored as
    A. Small aliquots at 4°C
    B. Large aliquots at 25°C
    C. Small aliquots at –70°C
    D. Large aliquots at –20°C

40. In Color Plate 57■ the procedure of Southern blotting is diagrammed. In the upper panel, restricted genomic DNA fragments have been separated by electrophoresis in an agarose gel. In lane 1 is a molecular weight marker, in lanes 2–4 are three patient samples, and in lane 5 is a positive control DNA sequence for the probe used. After electrophoresis, DNA was transferred from the gel onto a nylon membrane and then hybridized with a radiolabeled probe that recognizes CGG trinucleotide repeat. Fragile X syndrome is the most frequently inherited form of mental retardation in males (1:1000–1:1500 individuals). In affected individuals, expansions of the trinucleotide repeat within the fragile X gene increase to greater than 200 repeats. The bottom panel shows the resultant autoradiogram after a series of high-stringency washes. The three patient samples (lanes 2–4) are DNA from individuals of a single family, one of them suffering from fragile X syndrome. In which lane is the mentally handicapped patient's sample?
    A. Lane 2
    B. Lane 3
    C. Lane 4
    D. Cannot be determined by the results given

41. An advantage of amplification technologies for clinical laboratories is that
    A. They require inexpensive test reagents
    B. They lend themselves to automated methods
    C. Each target molecule sought requires a unique set of primers
    D. Contamination is not a concern when performing these assays

42. The assay method that detects the expression of a gene rather than the mere presence or structure of a gene is termed
    A. RT-PCR
    B. TMA
    C. Multiplex PCR
    D. Ribotyping

43. Which of the following assays *cannot* be accomplished using PCR methods employing only *Taq* polymerase?
    A. Diagnosis of *Chlamydia trachomatis* and *Neisseria gonorrhoeae* infection
    B. Detection of single base pair gene mutations, such as in cystic fibrosis
    C. Detection of HLA-A, B and DR genotypes
    D. Determination of viral load for HCV

44. One method to prevent "false-positive" PCR results includes the use of dUTP in the reaction mix, resulting in amplicons containing U in place of T. The enzyme used to decrease contamination is
    A. Uracil-*N*-glycosylase
    B. *Taq* polymerase
    C. S1 nuclease
    D. DNase

For questions 45–47, refer to Color Plates 58a■ and b■.

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45. What temperature is best for use in Step 1?
    A. 35°C
    B. 55°C
    C. 75°C
    D. 95°C

46. What temperature range is most appropriate for Step 2?
    A. 25–35°C
    B. 55–65°C
    C. 70–80°C
    C. 90–100°C

47. What substance within the PCR mix influences the accuracy of cDNA?
    A. Oligonucleotide primers
    B. Monovalent cation K^+
    C. Divalent cation Mg^{2+}
    D. Deoxyribonucleotide triphosphate molecules

48. The following question refers to Color Plate 59■. Factor V Leiden mutation causes increased risk of thrombosis. It is caused by a single base mutation in which guanine (G) is substituted for adenine (A) with a subsequent loss of a restriction site for the enzyme *Mnl*I. Primers used in this example generate a 223 bp PCR product from patient DNA. After resulting PCR products are digested with *Mnl*I, normal patients produce the following DNA fragments: 104 bp, 82 bp, 37 bp. In Color Plate 59■, the 37 bp fragment is not seen in all lanes because it is sometimes below detectable levels. Lane identities are as follows: M (molecular weight marker), 1–5 (patient 1 to patient 5, respectively), + (positive control showing 104, 82, and 37 bp fragments), Neg (sterile water used in place of sample DNA). Which patient is heterozygous for the factor V Leiden mutation?
    A. Patient 1
    B. Patient 2
    C. Patient 3
    D. Patient 4

49. The translocation resulting in the Philadelphia chromosome is detected by
    A. Southern blot analysis only
    B. Cytogenetic analysis (e.g., karyotyping) only
    C. PCR, Southern blot, and cytogenetic analysis
    D. RT-PCR, Southern blot, and cytogenetic analysis

50. Which sample in Color Plate 60■ contains the largest amount of cytomegalovirus?
    A. Sample 4
    B. Sample 5
    C. Sample 11
    D. Only qualitative results can be determined in this assay.

answers & rationales

1.

B. Because of the base pairing property within DNA, the presence of 20% adenine (A) means there must also be 20% thymine (T) in the organism. This means 40% of the DNA is A or T, leaving 60% of the DNA to be cytosine (C) or guanine (G). Because there must be an equal amount of each base type within the base pair, 60% divided by 2 gives 30% each of cytosine and guanine.

2.

C. DNA ligase is an enzyme that catalyzes the reaction between the 5'-phosphate end of one DNA fragment and the 3'-hydroxyl end of the next. This "nick sealing" requires energy from ATP hydrolysis, thus remaking the broken phosphodiester bond between the adjacent nucleotides. Ligase is a very important enzyme in DNA repair, but it is not used in a polymerase chain reaction (PCR). PCR does require a DNA template, two primers to anneal to nucleotide sequences flanking the desired amplification sequence, deoxynucleotide triphosphates (dNTPs) to be used as building blocks for the growing DNA chain, DNA polymerase, and magnesium chloride as an essential cofactor for DNA polymerase activity.

3.

A. The first letter of a restriction endonuclease's name comes from the bacterial genus from which it originated. The second and third letters derive from the bacterial species. The last letter indicates the subspecies or strain from which the enzyme was obtained. The last Roman numeral represents the numerical place the enzyme has among those which have been isolated from that bacterial genus/species/strain. For example, *Eco*RI is the first restriction endonuclease isolated from the bacterium *Escherichia coli,* strain R, whereas *Eco*RV is the fifth such enzyme to be discovered.

4.

C. The complementary strand for this DNA sequence would be, read left to right, 3' GATATC 5'. Restriction endonucleases require dsDNA, because they use as their substrate palindromic molecules, meaning a molecule that will "read" the same left to right or right to left. In this instance, the complementary strand, read 5' to 3' (right to left), reads the same as the sense strand, read 5' to 3'. If the enzyme cuts the sense strand as indicated, between the thymine and adenine, it will cut the complementary strand identically. This will leave, on the sense strand, the two sequences 5' CT 3' and 5' ATAG 3'. The complementary strand will show 3' GATA 5' and 3' TC 5'. Check again to see that these new sequences actually are identical, read 5' to 3', on both strands.

5.

C. The concentration of dsDNA can be estimated by taking its absorbance reading at 260 nm and multiplying that absorbance by a factor of 50, because one absorbance unit at 260 nm equals approximately 50 μg/mL. To solve this problem: 100 (dilution factor) × 0.062 (sample Abs. at 260 nm) × 50 μg/mL (conversion factor for dsDNA) = 310 μg/mL.

6.

A. In the isolation of RNA, it is very important to remove all RNase activity. Such enzymes are considered ubiquitous, so precautions must always be taken. Diethylpyrocarbonate (DEPC), or diethyl oxydiformate, will inactivate RNase, thus protecting RNA from degradation. It is used in solution at 0.1–0.2% (w/v) concentration.

7.

B. Resin-based purification of DNA takes advantage of the fact that DNA, at alkaline pH, possesses a net negative charge. Cells are incubated with detergent, which causes their lysis. RNA is digested with RNase, and the solution is neutralized with potassium acetate. This salt solution will precipitate the detergent and all proteins. Lysate is added to the exchange resin, and both DNA and residual RNA will bind. RNA and ssDNA are removed with a wash buffer, and the desired dsDNA will be eluted from the resin with either water or pH 8.0–8.5 buffer.

8.

B. The rate of electrophoretic separation when using polyacrylamide or agarose gels is affected by time, current, and the percent matrix used. Sample volume will not affect rate of separation but only makes the resulting bands more visible when stained. Achieving increased separation can be accomplished by increasing the time or current used. It can also be achieved by decreasing the percent matrix, because the "pores" present in a 1% agarose gel will be larger than those in a 5% gel. This larger size pore will allow easier molecular passage of DNA molecules during electrophoresis. Conversely, achieving a tighter band pattern (i.e., higher resolution of smaller DNA molecules) can be accomplished by decreasing time or current, or increasing percent matrix used.

9.

A. There are three essential parts to any molecular test performed: (1) a target, (2) a probe, and (3) a signal that can be detected. There are many ways a probe can be labeled in order for a signal to be produced and an analyte measured. Radioactive isotopes, such as ^{32}P, ^{33}P, ^{35}S, and ^{125}I, have traditionally been used to label probes. Positive signals are measured using X-ray exposure or scintillation counting. However, because of environmental factors, costs, and safety concerns, radioactive labels are being used with decreasing frequency. Nonradioactive probes are often labeled with haptens (e.g., digoxigenin), biotin, fluorescein, rhodamine, or a chemical such as acridinium esters. Detection of the hybridization (i.e., a positive test) is dependent on the type of label used, but it is generally colorimetric, fluorescent, or chemiluminescent. Hapten-labeled and biotin-labeled probes are detected by enzyme-conjugated antihapten antibodies and enzyme-conjugated streptavidin. Enzyme conjugates used are horseradish peroxidase and alkaline phosphatase.

10.

C. Target amplification refers to a process that increases the number of copies of the target DNA or RNA nucleotide sequence. Examples include the polymerase chain reaction (PCR), reverse transcription-PCR (RT-PCR), transcription-mediated amplification (TMA), and nucleic acid sequence–based amplification (NASBA). Signal amplification will cause increased signal strength without increasing the number of target molecules. One example of signal amplification is the branched chain DNA reaction. Probe amplification will increase the number of copies of the probe that is complementary to the target. One example of probe amplification is the ligase chain reaction.

11.

C. Although human genes are highly conserved in gene coding regions, human gene loci are polymorphic, which means many forms of the gene can exist at a given locus, making each person "unique." Only identical twins are not "unique." Short tandem repeats (STRs) account for the many polymorphisms used in DNA fingerprinting. STRs are short, repetitive sequences of 3–7 base pairs and are abundant in the human genome. There are STR kits commercially available from several manufacturers. The common loci used in forensics to obtain DNA fingerprints are the CTT triplex and the FFv triplex. These contain the following loci: CSF1PO (proto-oncogene *CSF-1*), *TPOX* (thyroid peroxidase gene), *TH01* (tyrosine hydroxylase gene), *F13A01* (coagulation factor XIII gene), *FES/FPS* (proto-oncogene *c-fes/fps*), and *v-WA* (von Willebrand gene). This testing does not require large quantities or high-quality DNA for successful results. It uses PCR, which is highly sensitive; however, this characteristic also makes the PCR method prone to contamination.

12.

D. The process whereby a strand of RNA is synthesized from template DNA is called transcription. The enzyme involved is RNA polymerase. It is possible, however, as retroviruses have shown, to produce DNA using template RNA. This reversal of the central nucleic acid dogma is called "reverse transcription," and the enzyme that performs this is called reverse transcriptase. After synthesizing a single-stranded DNA molecule from RNA, a different enzyme (DNA polymerase) then synthesizes a complementary strand to produce a DNA double helix. The other three answers describe amplification methods designed to increase the sensitivity of molecular diagnostic tests. They accomplish this by making copies of either the target nucleic acids (e.g., PCR), or the probe molecules (e.g., ligase chain reaction and strand displacement amplification), or the signal produced (e.g., branched chain DNA reaction).

13.

C. To solve this problem, it is necessary to recognize that plasmid DNA exists as a closed circle. This means that base pair #1 is adjacent to base pair #5850. If the enzyme cleaves the plasmid at positions 36, 1652, and 2702, this will result in three pieces of DNA. One piece will contain base pairs (bp) 37 through 1652 (with a size of 1616 bp), a second will contain bp 1653 through 2702 (with a size of 1050 bp), and the third will span the sequence from bp 2703 through 5850 and from 1 to 36 (with a size of 3184 bp). Note that to determine the size of each piece, subtract the numbers corresponding to each adjacent cut site (e.g., 1652 − 36 = 1616 and 2702 − 1652 = 1050). For the third piece, subtract the highest numbered cut site (i.e., 2702) from the total size of the plasmid (i.e., 5850), and add the size of the piece beginning at bp #1 through bp #36. Use the DNA ladder marker (lane M) in Color Plate 56■ to predict the placement of these pieces (3184 bp, 1616 bp, and 1050 bp) of DNA on the final electrophoresis pattern.

14.

A. DNA chips (i.e., DNA microarrays) allow detection and discrimination of multiple genetic sequences at the same time. DNA chips have thousands of oligonucleotide probes arranged on glass or silicon surfaces in an ordered manner. Target sequences within the patient sample are fluorescently labeled in solution. The labeled sequences in solution are then incubated with the DNA chips containing the oligonucleotide probes attached to the silicon or glass surface. Hybridization will occur between labeled complementary sequences within the patient samples and their corresponding probe on the chip. The DNA chip is placed in an instrument that scans the surface with a laser beam. The intensity of the signal and its location are analyzed by computer and provide a quantitative description of the genes present. Because placement of the oligonucleotides is known, identification of the gene or organism may be determined.

15.

C. Biotin is a vitamin involved physiologically in single carbon transfers. Streptavidin is a protein derived from *Streptomyces avidinii,* consisting of four subunits, each of which can bind one biotin molecule. This bond formation is rapid and essentially irreversible. The interaction between streptavidin and biotin is the strongest known noncovalent biologic interaction between a protein and its ligand. *In vitro* assays take advantage of this strong and specific binding by covalently attaching streptavidin to a reporter molecule (e.g., a primary antibody) and then incubating this with a secondary fluorescent-labeled antibody conjugated to biotin. Each streptavidin molecule will bind four biotin-conjugated molecules, thereby increasing fourfold the signal generated.

16.

C. Assuming 100% efficiency, each cycle of the polymerase chain reaction doubles the number of DNA molecules present in the solution. Starting with one DNA template molecule, there would be $2^2 = 4$ DNA molecules present after two cycles. After 5 cycles, this would result in $2^5 = 32$. Based on a starting single molecule of double-stranded DNA, after 15 cycles there would theoretically be 2^{15} molecules (32,768). Actual yield is somewhat less than theoretical yield because PCR products created in the first two PCR cycles are slightly longer than the target amplicon. Thus, yield may be better calculated as $2^{(n-2)}$. Actual yield may be decreased by a plateau effect that may occur in later PCR cycles when some components of the PCR become reaction limiting.

17.

A. Restriction enzymes will show specificity for a target nucleotide sequence when used under optimal conditions of temperature and glycerol, salt, and substrate concentrations. If these conditions are not optimal, some enzymes will lose their specificity and begin to cleave more randomly. This undesirable, nonoptimal digestion is called "star activity." Such activity is evident when the following parameters are altered in the reaction environment: more than 100 units of enzyme per microgram of DNA, more than 5% glycerol content, less than 25 mM salt concentrations, pH >8.0, presence of dimethyl sulfoxide (DMSO), ethanol, or other organic solvents.

18.

C. Cleavase is an enzyme isolated from bacteria that is likely important in DNA repair *in vivo*. The enzyme recognizes overlapping sequences of DNA and cleaves in the overlapping sequence. Third Wave Technologies has exploited the use of this enzyme in their Invader® system. Target nucleic acid is mixed with Invader and signal probes. When the Invader and signal probes bind the target, the 5' end of the signal overlaps with the Invader probe, and cleavase cleaves the signal probe. In the next step, the cleaved signal probe binds a fluorescent-labeled reporter probe containing complementary sequences and a quencher molecule, thus forming an overlapping structure. This molecule is subsequently cut by cleavase, which removes the reporter molecule from the quencher. The signal generated is directly related to the amount of target sequences in the original sample. Restriction endonucleases are also bacterial enzymes that recognize specific sequences within DNA and cut DNA near or within the recognized sequence. DNA ligase catalyzes the formation of a phosphodiester bond between adjacent 3' hydroxyl and 5' phosphate groups of adjacent nucleotides. RNaseH hydrolyzes RNA strands of a RNA:DNA hybrid molecule.

19.

B. After DNA is transferred to a nitrocellulose or nylon membrane, many sites on the membrane will not be occupied. Adding a probe at this point will not only allow for specific binding of the probe to the target DNA sequence, but also the nonspecific binding of the probe to the available binding sites on the membrane. This will cause nonspecific signal generation throughout the matrix. To prevent this, the membrane must first be treated with blocking agents. Denhardt solution and denatured nonhomologous DNA (e.g., salmon sperm DNA) are often used to bind up all the available sites on the matrix and allow for specific binding of the probe in the next step.

20.

C. Preparation of a DNA probe using random hexamer primers requires a DNA template containing the desired target sequence; four deoxynucleotides, at least one of which must be labeled (e.g., radionuclide, fluorescent, biotin, etc.); and DNA polymerase. The double-stranded DNA template (25–50 ng) is denatured, and a mixture of random oligonucleotides of six bases in length anneal to the template DNA. The primers are extended by the action of a DNA polymerase (e.g., T7 DNA polymerase) in the presence of one or two labeled deoxynucleotide triphosphates. Random primed probes are generally 500 nucleotides in length. Solutions containing the labeled probes are incubated with the blot. Hybridization of the labeled probe will occur if the gene being sought is present in the DNA on the blot to give a positive signal. Dideoxynucleotides are used in preparing samples for DNA sequencing by the Sanger method and cause DNA polymerization to cease.

21.

D. In Southern blots, hybrids can form between molecules with similar but not necessarily identical sequences. The washing conditions used after adding the labeled probe can be varied so that hybrids with differing mismatch frequencies are controlled. The higher the wash temperature or the lower the salt concentration in the wash buffer, the higher the stringency. Increasing the stringency will decrease the number of mismatches that form between the probe and the target DNA.

22.

A. Standard Southern blot techniques recommend the use of 10 μg of high-quality genomic DNA when studying single-copy genes. In a subsequent step the genomic DNA is restricted (i.e., cut into small fragments of predictable size). The resulting fragments of the gene of interest generally range in size from 1.0 to 10.0 kilobases. In contrast, the gene sequence of interest to be amplified in a routine PCR targets a smaller portion of the gene (generally 150–500 bases in length). Because the target is a smaller size, partially degraded DNA (i.e., genomic DNA samples of lesser quality) can be amplified successfully. Long-range PCR methods are available that extend the range of PCR products synthesized from 5 to 35 kilobases. Because PCR targets are usually a few hundred bases in length, high-molecular-weight DNA is not necessary for successful PCR. It requires a thermocycler to take the reaction through the cycles of three temperatures needed for denaturation, hybridization, and elongation steps. Turn-around time is also an advantage of PCR reactions because results can be completed in less than 4 hours, whereas Southern blotting takes up to 1 week to complete because of multiple steps required for this procedure.

23.

D. Many frequently used protocols in molecular biology involve PCR. Several substances can inhibit this reaction. For example, because of the nature of fecal material, it is not routinely used, and materials in swabs have also been reported to inhibit PCR. Therefore, a more appropriate specimen that could be used for PCR would be a stool filtrate. Nucleated cells are necessary for isolation of DNA. Whole blood is an acceptable specimen. White blood cells are the source of DNA in this type of specimen and must be separated from red blood cells as soon as possible because hemoglobin will inhibit PCR. For diagnosis of blood parasites, such as *Babesia* and *Plasmodium,* a hemolyzed and washed red blood cell sample is preferred for recovery of the DNA from the parasites. Amniocytes are used for molecular cytogenetic testing to prenatally screen for genetic diseases. Noninvasive collection of cells for genetic and forensic testing can be obtained from the buccal (oral) mucosa.

24.

D. Alcohol precipitation of nucleic acids is a standard method in molecular biology. Sterile water, 10 mM Tris, 1 mM EDTA, or 0.1% SDS can be used to rehydrate DNA; 1 mM EDTA and 0.1% SDS are included in these mixes to inhibit DNases. Alkaline solutions, such as 0.2 N NaOH, are used to denature nucleic acids.

25.

C. Real-time PCR or quantitative PCR (qPCR) is a modification of PCR that allows quantification of input target sequences without addition of competitor templates or multiple internal controls. qPCR is used to measure copy numbers of diseased human genes and viral and tumor load and to monitor treatment effectiveness. The accumulation of double-stranded PCR products during PCR as they are generated can be measured by adding fluorescent dyes that are dsDNA-specific to the reaction mix, such as SYBR green. However, misprimed products or primer dimers will also generate fluorescence and give falsely high readings. Thus, more specific systems utilizing probes to generate signal, such as the TaqMan probes, Molecular Beacons, and Scorpion-type primers, have been developed. In the TaqMan probe-based system, specific primers are present to prime the DNA synthesis reaction catalyzed by *Taq* polymerase, thus forming the cDNA product. The TaqMan probe binds to a smaller region within the target sequence. The TaqMan probe has a 5' reporter fluorophor and 3' quencher molecule. During extension of the primers by *Taq* polymerase to form cDNA product (i.e., DNA synthesis), the 5' → 3' exonuclease activity of *Taq* polymerase digests the TaqMan probe separating the reporter molecule from the quencher to generate a fluorescent signal. Molecular Beacon probes form hairpin structures due to short inverted repeat sequences at each end. The probe has a reporter dye at its 5' end and a quencher dye at its 3' end. In the unbound state, fluorescence is suppressed because reporter and quencher dyes are bound closely together by the short inverted repeat sequences. In qPCR assays, fluorescence occurs when molecular beacon probe binds the denatured template during the annealing step because reporter dye is separated from the quencher molecule. Scorpion primers, which contain a fluorophore and a quencher, are covalently linked to the probe. In the absence of the target, the quencher absorbs fluorescence emitted by the fluorophore. During the PCR reaction, in the presence of the target, the fluorophore and the quencher of the Scorpion primers separate, resulting in an increase in the fluorescence emitted. All of these systems require excess concentrations of the labeled probe/primer, so fluorescence emitted is directly proportional to the amount of template available for binding.

26.

A. Leukocytes are routinely used for extraction of DNA from human blood. Mature red blood cells and platelets have no nuclei. Plasma or serum can be used for detection of viremia, but it is not used for analysis of genetic diseases.

27.

B. The highest concentration of RNase is found on hands; thus, it is imperative that gloves be worn when working with RNA. RNases are ubiquitous and can act at temperatures below freezing (–20°C) and above boiling. For long-term storage, purified RNA is best stored at −70°C or below. RNases plague experiments in which RNA is used. Simple autoclaving does not eliminate RNase activity. To remove RNases, glassware must be pretreated with an RNase inhibitor, such as DEPC, followed by autoclaving; alternatively, baking glassware in a >250°C oven for 4 hours will destroy RNase. To prevent RNA degradation, isolation of RNA should be done using chaotropic agents (e.g., guanidine isothiocyanate) that inhibit RNase activity. When analyzing RNA in a gel, formaldehyde or other agents that denature RNases must be included in the gel. High-quality (i.e., undegraded) RNA will appear as a long smear with two or three distinct areas that correspond to the ribosomal RNA subunits: 28S (~4800 bases), 18S (~1800 bases), and 5.8S (~160 bases), whereas degraded RNA will appear as a smear at the bottom of the gel.

28.

D. EDTA and ACD (acid citrate dextrose) are the preferred anticoagulants for specimens that will undergo PCR. These reactions can be inhibited by a variety of substances. PCR inhibitors are concentration dependent; inhibition can often be overcome by simply diluting the DNA sample. Heme and sodium heparin can inhibit PCR. However, laboratory methods can be used to remove these inhibitors, if necessary. Diethylpyrocarbonate (DEPC) is a substance used to inhibit RNases; it can also inhibit PCR.

29.

B. Stringency of hybridization is accomplished at two steps in the blotting technique. The first step is hybridization conditions of the labeled probe in solution with the transferred RNA or DNA targets on the membrane. The second step occurs when the membrane is washed to remove unbound probe. In the hybridization reaction, formamide and temperature can be used to increase stringency. During wash steps, increasing temperature and increasing detergent concentration (e.g., 1% SDS) will increase stringency; whereas lowering NaCl concentration also increases stringency. At the end of the highest stringency wash, only specific hybrids of interest should remain on the blot.

30.

D. The four nucleotide bases found in RNA are adenine (A), guanine (G), cytosine (C), and uracil (U). The purines A and G are the same as in DNA. C is present in both DNA and RNA; however, in RNA, the DNA nucleotide base thymine (T) is replaced by uracil (U). RNA is usually single stranded, although double-stranded areas can occur. A pairs with U, and C pairs with G.

31.

B. dNTP stands for deoxyribonucleotide triphosphate. Nucleotides are the building blocks of nucleic acids. They are composed of phosphate groups, a 5-sided sugar molecule, and a nitrogenous base. Nitrogenous bases are either purines (A, G) or pyrimidines (C, T, or U, an RNA-specific base). The sugar molecules are either ribose (in RNA) or deoxyribose (in DNA), with the only difference in structure being the lack of a hydroxyl group at position 2' in the deoxyribose molecule. When the sugar is bound to a base without the phosphate group, the molecule is called a nucleoside. A nucleotide can have 1, 2, or 3 phosphate groups, which are termed monophosphate, diphosphate, and triphosphate, respectively.

32.

A. DNA is composed of two strands of polynucleotides coiled in a double helix. The outside backbone is composed of sugar-phosphate moieties, whereas the purine and pyrimidine bases are stacked inside the helix. The size and stability of the DNA molecule is such that only specific bases can hydrogen bond to each other to hold the two strands together (A-T, C-G, and vice versa). This is referred to as complementary base pairing. An A-T base pair is less stable than a C-G base pair, because three hydrogen bonds form between C-G and only two hydrogen bonds form between A-T. The increased stability between C-G causes the melting temperature (T_m) to be greater in a double-stranded DNA (ds DNA) segment with more C-G pairs than a segment with more A-T pairs. In all dsDNA molecules, the number of purines (A + G) equals the number of pyrimidines (C + T).

33.

C. Pulsed field gel electrophoresis (PFGE) is used to separate extremely large DNA molecules by placing them in an electric field that is charged periodically in alternating directions, forcing the molecule to reorient before moving through the gel. Larger molecules take more time to reorient; thus they move more slowly. Bacterial DNA is digested by restriction enzymes in agarose plugs. The PFGE of the digested fragments provides a distinctive pattern of 5 to 20 bands ranging from 10 to 800 kilobases. DNA sequencing determines the exact nucleotide sequence base by base of any organism; however, it is too laborious for epidemiologic purposes. Ribotyping is a Southern blot type of analysis using rRNA probes to detect ribosomal operons (i.e., sequences coding for 16S rRNA, 25S rRNA, and one or more tRNAs) of individual bacterial species. Its discriminatory power is less than PFGE. Reverse transcription–polymerase chain reaction (RT-PCR) is a method that determines whether a gene is being expressed. The starting material for RT-PCR is ssRNA.

34.

B. PCR requires a thermocycler because cycling at three different temperatures is the basis for this technique. First, template DNA (i.e., which may contain the target sequence) is denatured at 94°C. Next, the temperature is lowered to allow specific primers to anneal to the single-stranded target, generally at temperatures near 55°C. In the third portion of the cycle, primers are extended using deoxynucleotide triphosphate molecules to form a complementary copy of DNA under the direction of a thermostable DNA polymerase enzyme, such as *Taq* polymerase. The optimal temperature at which *Taq* polymerase acts to extend the primers is 72°C. Thus, at the end of one cycle, one molecule of dsDNA has now become two molecules of dsDNA. Cycles are generally repeated about 30 times to theoretically yield 2^{30} DNA molecules. The three steps of each cycle are termed denaturation (94°C), annealing of primers (~55°C), and extension of primers (72°C). The other methods listed, nucleic acid sequence–based amplification, strand displacement amplification, and transcription mediated amplification, are also amplification methods; however, they have been modified so all reactions take place at a single temperature (isothermal).

35.

A. The coding regions of eucaryote genes are called exons. The noncoding intervening regions are called introns. In eucaryotes, the introns and exons are transcribed into mRNA; however, before mRNA is translated, the introns are removed and the exons are spliced together. SNP is an abbreviation for single nucleotide polymorphism, and VNTR refers to variable number tandem repeats.

36.

C. Central dogma describes the flow of genetic information from DNA to RNA to protein. Individual DNA molecules serve as templates for either complementary DNA strands during replication or complementary RNA molecules during transcription. In turn, RNA molecules serve as templates for ordering of amino acids by ribosomes to form polypeptides during protein synthesis, also known as translation.

37.

D. DNA in human somatic cells is compartmentalized into 22 pairs of chromosomes, referred to as autosomes. They are numbered 1 through 22. In addition, humans have two sex chromosomes, both an X and Y (in males) or two X chromosomes (in females). Thus, the total number of chromosomes is 46 in a normal diploid cell. The genetic information of one set of chromosomes comes from the mother of the individual and the other set from the father. Gametes (i.e., eggs and sperm) are haploid and contain only one set of chromosomes (23 chromosomes in human gametes), so that upon fertilization, a diploid zygote is formed.

38.

B. Most of the lifetime of a cell is spent in G1 phase, during which the cells can produce their specialized proteins and accomplish their essential functions. However, when the signal is received for cell division, the cell enters S phase. In S phase the DNA in all chromosomes is duplicated. At the end of S phase, the duplicated chromosomes remain attached at the centromere. A time delay, G2, separates events of the actual separation of individual chromosomes from their duplicated pairs. Next, the M phase or mitosis is when the two members of each pair of chromosomes go to opposite ends of the original cell. This separates 46 chromosomes into two sets of 23 in each cell. Finally, a cleavage furrow is formed and separates the original cell into two daughter cells. Each cell contains a copy of all the genetic information from each parent.

39.

C. Purified DNA is relatively stable provided it is reconstituted in buffer that does not contain DNases. Therefore, high-quality reagents and type I sterile water should be used in preparing buffers used for this purpose. Experiments have shown that purified DNA is stable for as long as 3 years at refrigerated temperature (4°C). However, long-term storage of purified DNA is best accomplished at –20 to –70°C in a freezer that is not frost free to avoid freeze-thaw cycles that may damage DNA and by dividing the original DNA sample into multiple small aliquots for storage.

40.

C. Refer to Color Plate 57■. Given that the probe used will recognize the trinucleotide repeat found in the fragile X gene, *FMR-1,* the location of positive signals will give information about the size of the repeat sequence within each person's DNA. The normal allele for *FMR-1* has 6–50 trinucleotide repeats (found in normal individuals), the premutation for *FMR-1* contains 50–200 trinucleotide repeats (found in unaffected individuals), and the disease allele (found in affected individuals) has >200 repeats. Because electrophoresis separates DNA by size such that the larger fragments travel shorter distances than smaller fragments, then the larger fragment in the affected individual caused by the expansion of the trinucleotide repeat would be represented in Color Plate 57■ by lane 4 of the diagram.

41.

B. Amplification methods can be automated and standardized, which is proven by the variety of test systems presently on the market. Amplification methods are very sensitive and theoretically can detect one target DNA molecule in a sample. However, increased sensitivity raises the likelihood of false positive results due to contamination of testing areas with PCR amplicons. In addition, most amplification methods can be completed within 4–6 hours and can detect microorganisms that do not grow readily by standard culture techniques. At this time, test reagents are still quite expensive, although if decreased turn-around time would translate into shorter hospital stays, then resultant healthcare costs could be reduced by use of these methods in the clinical laboratory. A disadvantage of amplification technologies is that they require a unique set of primers for each target DNA being sought. Thus, amplification techniques may be replaced by use of DNA microarrays because thousands of genes can be assessed at one time, rather than a limited number of molecules of interest being assayed.

42.

A. Reverse transcription–polymerase chain reaction (RT-PCR) is used to detect gene expression; genes are expressed by transcription into mRNA. The starting material for RT-PCR is mRNA. The only method listed whose target sequence is found in mRNA is RT-PCR. Transcription mediated amplification targets are usually ribosomal RNA. In ribotyping, rRNA probes detect ribosomal RNA genes present in total bacterial DNA; bacteria can be grouped on the basis of banding patterns that result. Multiplex PCR describes a method in which DNA is the target or template, and several different primer sets are included in the reaction mix. An example of multiplex PCR would include methods that detect *Chlamydia trachomatis* and *Neisseria gonorrhoeae* in one reaction mix.

43.

D. Hepatitis C virus (HCV) has an RNA genome, and thus a reverse transcription step is needed to convert RNA into complementary DNA for use in the subsequent PCR that makes multiple copies of the target sequence. RT-PCR is both highly specific and sensitive. Viral load testing also requires that the methodology be quantitative. Quantification can be accomplished by qPCR techniques or by inclusion of a known amount of a synthetic nucleic acid, a quantification standard (QS), in the sample. The QS binds the same primers as the viral target, and so the kinetics of amplification for both may be assumed to be approximately equal. The viral target and QS are coamplified in the same reaction, and the raw data are manipulated mathematically to determine the viral load present in the specimen. To detect genetic sequences specific for the human leukocyte antigen (HLA) loci, bacteria, and gene mutations, the starting material is usually DNA; therefore, PCR methods, rather than RT-PCR, would be employed.

44.

A. The sensitivity of amplification techniques can be viewed as a double-edged sword. On one hand, the techniques have allowed detection of genetic sequences that are found in limited numbers within a sample. However, because the method creates large amounts of target sequence, the areas within the laboratory can become contaminated with amplicons. Amplicon contamination produces false positive results. The use of dUTP in the reaction mix results in PCR products (i.e., amplicons) containing uracil in place of thymidine. The enzyme used to decrease contamination of previously generated dU-containing amplicons is uracil-*N*-glycosylase (UNG). Samples are pretreated with this enzyme before their use in subsequent PCR reactions to remove contaminating dU-containing amplicons if present. Pretreatment with UNG has no effect on sample DNA containing thymidine residues. Other procedures necessary to avoid contamination include dedicated areas for reagent preparation, impeccable technique, amplification and post-amplification analysis, and use of aerosol-barrier pipette tips. Treatment of work surfaces, equipment, and pipettors with UV light can also be used to prevent contamination.

45–47.

(45:D, 46:B, 47:C)

Questions 45–47 are associated with Color Plates 58a■ and b■. The reaction depicted in Color Plates 58a■ and b■ is the polymerase chain reaction (PCR). It is the prototype of target amplification methods. Other target amplification methods include transcription-mediated amplification (TMA) and nucleic acid sequence–based analysis (NASBA). Traditional PCR is the *in vitro* equivalent of DNA replication *in vivo*. The components of PCR are a DNA template (containing target sequence of interest), a set of oligonucleotide primers (that flank the region of interest), building blocks of deoxynucleotide triphosphates (dATP, dCTP, dGTP, TTP, collectively referred to as dNTPs), and a heat-stable DNA polymerase (e.g., *Taq* polymerase) and buffer providing optimal conditions for primer annealing and DNA synthesis. Optimal salt concentrations (e.g., KCl) are needed for annealing of primers. In addition, optimal concentrations of the divalent cation Mg^{2+}, a cofactor for *Taq* polymerase that determines the fidelity of DNA replication *in vitro,* are essential. PCR consists of cycles of three steps: (1) denaturation of ds template, (2) annealing of ss oligonucleotide primers to complementary sequence in denatured template, and (3) DNA synthesis catalyzed by *Taq* polymerase. During synthesis, primers are extended in the 5' → 3' direction, adding adenine, guanine, cytosine, and thymidine nucleotide bases into the growing chain according to the complementary sequence to which it is bound. The steps generally occur at the following temperatures and duration: denaturation ≥ 94°C (20–60 seconds); annealing (dependent upon base sequences of primers) between 55–65°C (20–90 seconds); and DNA synthesis at 72°C, the optimal temperature for *Taq* polymerase (10–60 seconds). The resulting cDNA products are called amplicons. Starting with one dsDNA molecule (1A:1B) containing a target sequence, denaturation separates the dsDNA template into two ssDNA strands (1A and 1B). Next, the temperature is lowered to a point where specific hybridization of each primer to its complementary sequence in the denatured strands occurs. Then, the temperature is increased to 72°C, and extension of the primers ensues. At the end of cycle #1, two dsDNA parent (1) daughter (2) hybrid molecules result (i.e., 1A:2B and 1B:2A). At the end of cycle #2, four dsDNA products are produced. Intermediate-length products, larger than the target sequence of interest (4A and 4B length) but smaller than the original DNA template (1A:1B), are present. In subsequent cycles, the precise length target sequence is amplified most efficiently and becomes the preferred target for amplification. At each cycle, the number of copies doubles such that after N doublings, 2^{N-2}, copies of target are produced. For example, after 30 cycles, 2^{28} amplicons = 2.68×10^8 (over a billion) copies are made.

In transcription-mediated amplification methods, such as TMA and NASBA, starting material and primary products are both RNA. Enzymes used in these methods are currently avian myeloblastosis virus (AMV) reverse transcriptase with inherent RNase H activity and RNA polymerase (e.g., T7 RNA polymerase). Advantages of TMA and NASBA are that (1) they are isothermal (i.e., do not require a thermocycler); and (2) they target RNA for direct detection of RNA viruses, such as hepatitis C virus (HCV) and human immunodeficiency virus (HIV), or rRNA of bacteria, resulting in increased sensitivity. In addition to target amplification, there are other amplification methods in which either probe or signals are amplified. Ligase chain reaction, strand displacement amplification, and Q-beta replicase are examples of probe amplification systems. Signal amplification includes methods such as a hybrid capture assay for human papillomavirus; branched chain DNA methods used for viral load assays for HCV and HIV-1; cleavage-based amplification primarily for factor V Leiden mutation detection; and the cycling probe method of amplification, which has been used to detect genes associated with antimicrobial resistance in bacteria, such as *mec*A in *Staphylococcus aureus* and *van*A and *van*B in *Enterococcus*.

48.

C. Refer to Color Plate 59■. Factor V Leiden mutation (A506G) causes activated protein C resistance that results in increased risk of hypercoagulability. The mutation destroys a *MnlI* restriction enzyme site in an amplified 223 bp PCR product from patient DNA. From the electrophoretic pattern, wild-type or normal factor V will show three bands after *MnlI* digestion (104 bp, 82 bp, 37 bp), as in patients 1, 4, and 5. The pattern seen with patient 2 is that of a homozygous mutant with two bands (141 bp and 82 bp). In the heterozygous patient 3, one allele is normal and the other is mutant. Thus, the banding pattern results in four bands (141 bp, 104 bp, 82 bp, and 37 bp). Sometimes the 37 bp fragment band is not seen because it is below detectable levels.

49.

D. The translocation resulting in the Philadelphia chromosome can be detected by reverse transcription–polymerase chain reaction (RT-PCR), Southern blot, and cytogenetic analysis. The presence of a Philadelphia (Ph) chromosome confirms the diagnosis of chronic myelogenous leukemia (CML). The Ph chromosome is a shortened chromosome 22 that arises from a reciprocal translocation involving the long arms of chromosomes 9 and 22. This translocation involves the proto-oncogene c-*ABL,* normally present on chromosome 9q34, and the *BCR* gene on chromosome 22q11. The juxtaposition of *ABL* with *BCR* results in the formation of a *BCR-ABL* fusion gene, which is subsequently transcribed into a chimeric BCR-ABL mRNA that is ultimately translated into a chimeric BCR-ABL protein product. Traditionally, this rearrangement can be seen cytogenetically by visualization of the patient's karyotype (i.e., metaphase spread of patient's chromosomes). Recent techniques have been developed in which fluorescent-labeled probes for this gene rearrangement can be used to probe the patient's metaphase or interphase spread, called fluorescence *in situ* hybridization (FISH). Molecular methods to check for this gene rearrangement include Southern blotting and RT-PCR. PCR cannot be used for this particular gene rearrangement because BCR/*abl* breakpoints span large segments of DNA, which prevents direct PCR testing. Instead, RT-PCR is used. The BCR/*abl* chimeric mRNA is used as a template because primer annealing sites in the breakpoint region of the mRNA are a smaller size, suitable for amplification.

50.

B. Color Plate 60■ is graphic display of a real-time PCR (i.e., qPCR) run for cytomegalovirus (CMV). Real-time PCR assays can measure the amount of starting target sequence (i.e., template in sample) accurately. Rather than measuring PCR product generated at the stationary or end-point of the PCR assay, qPCR analysis is done as PCR products are formed (i.e., during the exponential phase) where accumulation of fluorescence is inversely proportional to the amount of starting template (i.e., the shorter the time to accumulate signal, the more starting material). Optimal threshold level is based on the background or baseline fluorescence and the peak fluorescence in the reaction and is automatically determined by the instrument. Using 10-fold dilutions of known positive standards, a standard curve can be made. The qPCR cycle at which sample fluorescence crosses the threshold is the threshold cycle (C_T). Using the standard curve, the starting amount of target sequence in each sample can be determined by its C_T. Fluorescence versus C_T is an inverse relationship. The more starting material, the fewer cycles it takes to reach the fluorescence threshold (i.e., large amounts of fluorescence accumulate in a short time). The C_T for sample 5 is 21, sample 4 is 25, and sample 11 is 38; therefore, sample 5 has more CMV copies than sample 4, which has more CMV copies than all the other samples with C_T values indicated, including sample 11 with the least CMV. The samples below the threshold fluorescence of 30 are negative for CMV.

review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

1. Why is the first-voided morning urine specimen the most desirable specimen for routine urinalysis?
   A. Most dilute specimen of the day and therefore any chemical compounds present will not exceed the detectability limits of the reagent strips
   B. Least likely to be contaminated with microorganisms because the bladder is a sterile environment
   C. Most likely to contain protein because the patient has been in the orthostatic position during the night
   D. Most concentrated specimen of the day and therefore it is more likely that abnormalities will be detected

2. The physical characteristic of color is assessed when a routine urinalysis is performed. What substance is normally found in urine that is principally responsible for its yellow coloration?
   A. Bilirubin
   B. Melanin
   C. Carotene
   D. Urochrome

3. In certain malignant disorders, what substance is found in the urine that turns the urine dark brown or black on exposure of the urine to air?
   A. Urobilinogen
   B. Indican
   C. Melanin
   D. Porphyrin

4. What is the expected pH range of a freshly voided urine specimen?
   A. 3.5–8.0
   B. 3.5–9.0
   C. 4.0–8.5
   D. 4.5–8.0

5. Urine specimens should be analyzed as soon as possible after collection. If urine specimens are allowed to stand at room temperature for an excessive amount of time, the urine pH will become alkaline because of bacterial decomposition of
   A. Protein
   B. Urea
   C. Creatinine
   D. Ketones

6. Which term is defined as a urine volume in excess of 2000 mL excreted over a 24-hour period?
   A. Anuria
   B. Oliguria
   C. Polyuria
   D. Hypersthenuria

7. The reagent test strips used for the detection of protein in urine are most reactive to
   A. Albumin
   B. Hemoglobin
   C. Alpha-globulins
   D. Beta-globulins

8. A urine specimen that exhibits yellow foam on being shaken should be suspected of having an increased concentration of
   A. Protein
   B. Hemoglobin
   C. Bilirubin
   D. Nitrite

9. How should controls be run to ensure the precision and accuracy of the reagent test strips used for the chemical analysis of urine?
   A. Positive controls should be run on a daily basis and negative controls when opening a new bottle of test strips.
   B. Positive and negative controls should be run when the test strips' expiration date is passed.
   C. Positive and negative controls should be run on a daily basis.
   D. Positive controls should be run on a daily basis and negative controls on a weekly basis.

10. The colorimetric reagent strip test for protein is able to detect as little as 5–20 mg of protein per deciliter. What may cause a false-positive urine protein reading?
    A. Uric acid concentration is greater than 0.5 g/day.
    B. Vitamin C concentration is greater than 0.5 g/day.
    C. Glucose concentration is greater than 130 mg/day.
    D. pH is greater than 8.0.

11. "Isosthenuria" is a term applied to a series of urine specimens from the same patient that exhibit a
    A. Specific gravity of exactly 1.000
    B. Specific gravity less than 1.007
    C. Specific gravity greater than 1.020
    D. Fixed specific gravity of approximately 1.010

12. A urine specimen is tested by a reagent strip test and the sulfosalicylic acid test to determine whether protein is present. The former yields a negative protein, whereas the latter results in a reading of 2+ protein. Which of the following statements best explains this difference?
    A. The urine contained an excessive amount of amorphous urates or phosphates that caused the turbidity seen with the sulfosalicylic acid test.
    B. The urine pH was greater than 8, exceeding the buffering capacity of the reagent strip, thus causing a false-negative reaction.
    C. A protein other than albumin must be present in the urine.
    D. The reading time of the reagent strip test was exceeded (the reading being taken at 2 minutes), causing a false-negative reaction to be detected.

13. Which of the following is the major organic substance found in urine?
    A. Sodium
    B. Potassium
    C. Glucose
    D. Urea

14. Each of the following is included in the quality assurance program for a urinalysis laboratory. Which one represents a preanalytical component of testing?
    A. Setting collection guidelines for 24-hour urines
    B. Setting a maintenance schedule for microscopes
    C. Reporting units to be used for crystals
    D. Requiring acceptable results for control specimens before any patient results are reported out

15. The presence of ketone bodies in urine specimens may be detected by use of a reagent strip impregnated with sodium nitroprusside. This strip test is sensitive to the presence of
    A. Acetoacetic acid and beta-hydroxybutyric acid
    B. Acetoacetic acid and acetone
    C. Diacetic acid and beta-hydroxybutyric acid
    D. Beta-hydroxybutyric acid and acetone

16. A routine urinalysis is performed on a young child suffering from diarrhea. The reagent test strip is negative for glucose but positive for ketones. These results may be explained by which of the following statements?
    A. The child has Type 1 diabetes mellitus.
    B. The child is suffering from lactic acidosis, and the lactic acid has falsely reacted with the impregnated reagent area for ketones.
    C. The child is suffering from increased catabolism of fat because of decreased intestinal absorption.
    D. The reagent area for ketones was read after the maximum reading time allowed.

17. The principle of the colorimetric reagent strip test for hemoglobin is based on the peroxidase activity of hemoglobin in catalyzing the oxidation of a dye with peroxide to form a colored compound. This method may yield false-positive results for the presence of hemoglobin when the urine specimen contains
    A. Ascorbic acid
    B. Tetracycline
    C. Myoglobin
    D. Nitrite

18. A reagent test strip impregnated with a diazonium salt such as diazotized 2,4-dichloroaniline may be used to determine which analyte?
    A. Glucose
    B. Ketone
    C. Hemoglobin
    D. Bilirubin

19. Which of the following will contribute to a specimen's specific gravity if it is present in a person's urine?
    A. 50–100 RBC/hpf
    B. 85 mg/dL glucose
    C. 3+ amorphous phosphates
    D. Moderate bacteria

20. With infections of the urinary system, white blood cells are frequently seen in the urine sediment. What type of white blood cell is seen the most frequently in urine sediment?
    A. Eosinophil
    B. Lymphocyte
    C. Monocyte
    D. Neutrophil

21. A random urine is collected from a patient and the results obtained are as follows: urine albumin = 16 mg/dL and urine creatinine = 140 mg/dL. These findings are consistent with
    A. Microalbuminuria
    B. Macroalbuminuria
    C. Nephrotic syndrome
    D. Obstructive jaundice

22. To detect more easily the presence of casts in urine sediments, which microscopic method can be used?
    A. Fluorescent microscopy
    B. Phase-contrast microscopy
    C. Polarized microscopy
    D. Brightfield microscopy

23. Which substance found in urinary sediment is more easily distinguished by use of polarized microscopy?
    A. Lipids
    B. Casts
    C. Red blood cells
    D. Ketone bodies

24. "Glitter cell" is a term used to describe a specific type of
    A. Ketone body
    B. Oval fat body
    C. Fatty droplet
    D. Neutrophil

25. The final phase of degeneration that granular casts undergo is represented by which of the following casts?
    A. Fine
    B. Coarse
    C. Cellular
    D. Waxy

26. A 40-year-old female patient with a history of kidney infection is seen by her physician because she has felt lethargic for a few weeks. She has decreased frequency of urination and a bloated feeling. Physical examination shows periorbital swelling and general edema, including a swollen abdomen. Significant urinalysis results show the following: color = yellow; appearance = cloudy/frothy; specific gravity = 1.022; pH = 7.0; protein = 4+; 0–3 WBC/hpf; 0–1 RBC/hpf; 0–2 renal epithelial cells/hpf; 10–20 hyaline casts/lpf; 0–1 granular casts/lpf; 0–1 fatty casts/lpf; occasional oval fat bodies. Her serum chemistries show significantly decreased albumin, increased urea nitrogen, and increased creatinine. These findings suggest which condition?
    A. Multiple myeloma
    B. Glomerulonephritis
    C. Nephrotic syndrome
    D. Chronic renal failure

27. A 47-year-old female patient with controlled type 2 diabetes mellitus complains of urinary frequency and burning. She provides a first-morning, clean-catch specimen. Results show color = yellow; appearance = cloudy; pH = 6.5; a representative microscopic high-power field is shown in Color Plate 46■. Which of the following is *true* for this patient?
    A. The number of bacteria seen would result in a positive nitrite.
    B. The major formed elements are white blood cells and yeast.
    C. The type and number of epithelial cells suggest incorrect sample collection.
    D. The red blood cells would be sufficient to give a positive blood result on the reagent strip.

28. Alkaptonuria, a rare hereditary disease, is characterized by the urinary excretion of
    A. Alkaptone
    B. Phenylalanine
    C. 5-Hydroxyindole acetic acid
    D. Homogentisic acid

29. A 22-year-old female clinical laboratory student performs a urinalysis on her own urine as part of a lab class. Significant results include: color = yellow; appearance = cloudy; pH = 7.5; nitrite = positive; leukocyte esterase = 2+; 25–40 WBC/hpf; 0–3 RBC/hpf; 2–5 squamous epithelial cells/hpf; moderate bacteria. All other chemistries and microscopic results were normal. These findings suggest
    A. Glomerulonephritis
    B. Upper urinary tract infection
    C. Lower urinary tract infection
    D. Nephrolithiasis

30. Metastatic carcinoid tumors arising from the enterochromaffin cells of the gastrointestinal tract are characterized by increased excretion of urinary
    A. Serotonin
    B. 5-Hydroxytryptophan
    C. Homogentisic acid
    D. 5-Hydroxyindole acetic acid

31. Some clinical conditions are characterized by unique urinalysis result patterns. Which of the following shows such a relationship?
    A. Nephrotic syndrome: positive protein on reagent strip, negative protein with sulfosalicylic acid
    B. Intensive dieting: increased ketones, negative glucose
    C. Multiple myeloma: positive protein by both reagent strip and sulfosalicylic acid
    D. Cystitis: positive nitrite and protein

32. Nitrite in a urine specimen suggests the presence of
    A. White blood cells
    B. Red blood cells
    C. Bacteria
    D. Yeasts

33. If a fasting plasma glucose level of 100 mg/dL is obtained on an individual, what is the expected fasting cerebrospinal fluid (CSF) glucose level in mg/dL?
    A. 25
    B. 50
    C. 65
    D. 100

34. A 35-year-old man has just experienced severe crush injuries sustained in a car accident. He has a broken pelvis and right femur and has numerous abrasions and contusions. A random urinalysis specimen shows a brown color and clear appearance. pH is 6.0, protein is 1+, and blood is 3+. There is, however, only 0–1 RBC/hpf, along with 0–3 WBC/hpf. Casts found include hyaline (0–2/lpf) and granular (0–1/lpf). Other urine results are normal. Which of the following is *true* about this patient?
    A. The positive blood result is from a hemolytic anemia.
    B. The bilirubin result should have also been positive for this patient.
    C. Rhabdomyolysis may be a cause for the discrepant chemical/microscopic blood findings.
    D. The bone crushing led to the increased protein result.

35. A 67-year-old male has routine testing done and shows an estimated glomerular filtration rate (eGFR) of 42 mL/min/1.73 m^2. Which of the following is *true* for this patient?
    A. This test requires a 24-hour urine collection.
    B. The patient does not have chronic kidney damage, based on these results.
    C. Similar results would be obtained using the Cockgroft-Gault formula.
    D. The patient is in Stage 3 chronic kidney damage.

36. Which is *true* about the formed element shown in Color Plate 47■?
    A. May be found in normal alkaline urine
    B. Associated with renal pathology
    C. Characteristic of glomerulonephritis
    D. Associated with lung pathology

37. The major formed element in the high-power field shown in Color Plate 48■ is most likely a
    A. Granular cast
    B. Hyaline cast
    C. Waxy cast
    D. Fiber artifact

38. Which of the following is *true* about the final concentrating of urine in the kidney?
    A. The distal convoluted tubule, through active transport, reabsorbs water.
    B. Water is reabsorbed under the direct influence of angiotensin II.
    C. Vasopressin controls the collecting duct reabsorption of water.
    D. Water reabsorption is influenced by urine filtrate levels of potassium.

39. If a urine specimen is left standing at room temperature for several hours, which of the following changes may occur?
    A. Multiplication of bacteria
    B. An increase in the glucose concentration
    C. Production of an acid urine
    D. Deterioration of any albumin present

40. The formed element shown in Color Plate 49■ would usually be found in the patient's urine along with which soluble biochemicals?
    A. Phenylalanine and tyrosine
    B. Ornithine and arginine
    C. Isoleucine and leucine
    D. Acetoacetic acid and β-hydroxybutyric acid

41. A 13-year-old ice skater is having her routine physical before the school year. Her first morning urinalysis results include color = straw; appearance = hazy; pH = 6.0; protein = trace; a representative microscopic high-power field is shown in Color Plate 50■. All other chemical results were normal. The major formed elements are _____ and suggest _____.
    A. Hyaline casts and waxy casts; nephrotic syndrome
    B. Mucus and fibers; no pathology
    C. Granular casts and red blood cells; glomerulonephritis
    D. Hyaline casts and mucus; normal sediment

42. Phenylketonuria may be characterized by which of the following statements?
    A. It may cause brain damage if untreated.
    B. It is caused by the absence of the enzyme, phenylalanine oxidase.
    C. Phenylpyruvic acid excess appears in the blood.
    D. Excess tyrosine accumulates in the blood.

43. What condition is suggested by the number of the formed element that predominates in the high-power field of Color Plate 51■?
    A. Glomerulonephritis
    B. Improperly collected specimen
    C. Pyelonephritis
    D. Normal sample

44. Xanthochromia of cerebrospinal fluid (CSF) samples may be due to increased levels of which of the following?
    A. Chloride
    B. Protein
    C. Glucose
    D. Magnesium

45. Which of the following will be characterized by an increased number of the urinary component seen in Color Plate 52■?
    A. Acute glomerulonephritis
    B. Biliary tract obstruction
    C. Contamination from vaginal discharge
    D. Nephrotic syndrome

46. To determine amniotic fluid contamination with maternal urine, which of the following measurements could be used?
    A. Creatinine concentration
    B. Delta absorbance at 410 nm
    C. Albumin/globulin ratio
    D. Lactate dehydrogenase

47. With the development of fetal lung maturity, which of the following phospholipid concentrations in amniotic fluid significantly and consistently increases?
    A. Sphingomyelin
    B. Phosphatidyl ethanolamine
    C. Phosphatidyl inositol
    D. Phosphatidyl choline

48. A patient has been diagnosed with an upper gastrointestinal bleed. Which of the following would be characteristic for this condition?
    A. Brown stool with streaks of bright red
    B. Stool with lack of brown color ("clay-colored")
    C. Stool with a much darker brown/black color
    D. Yellow stool with increased mucus

49. A pleural effusion is found to have 3000 white blood cells per microliter and 5 g/dL total protein. From this it can be determined that the patient's effusion is
    A. A transudate
    B. An exudate
    C. Noninflammatory
    D. Hemorrhagic

50. Patients with diabetes insipidus tend to produce urine in _____ volume with _____ specific gravity.
    A. Increased; decreased
    B. Increased; increased
    C. Decreased; decreased
    D. Decreased; increased

51. The estimation of hyaluronic acid concentration by measurement of viscosity is useful in evaluating which type of fluid?
    A. Spinal
    B. Peritoneal
    C. Pleural
    D. Synovial

52. Which of the following is characteristic of an exudate effusion?
    A. Leukocyte count >1000/μL
    B. Clear appearance
    C. Protein concentration <3.0 g/dL
    D. Absence of fibrinogen

53. Which of the following systems utilizes polyelectrolytes to determine the specific gravity of urine?
    A. Refractometer
    B. Osmometer
    C. TS meter
    D. Reagent strip

54. Which methods may be used to quantify protein in both cerebrospinal fluid and urine specimens?
    A. Trichloroacetic acid and bromcresol green
    B. Ponceau S and Coomassie brilliant blue
    C. Bromcresol green and Coomassie brilliant blue
    D. Coomassie brilliant blue and trichloroacetic acid

55. Which of the following characteristics is *true* of the primary urinary components shown in Color Plate 53■?
    A. Consist of uromodulin protein
    B. Presence always indicates a disease process
    C. Can be observed with polarized microscopy
    D. Appear yellowish in brightfield microscopy

56. A characteristic of substances normally found dissolved in the urine is that they are all
    A. Water soluble
    B. Inorganic
    C. Organic
    D. Waste products

57. Which of the following statements applies to the proper collection and handling of CSF?
    A. The second tube collected should be used for chemistry analyses.
    B. The third tube collected should be used for bacteriologic studies.
    C. CSF collected in the evening should be refrigerated and assays performed only by day-shift personnel.
    D. With low-volume specimens, a culture is performed first, before cell counts are done.

58. Which of the following characteristics is *true* for the urinary components shown in Color Plate 54■?
    A. Never should appear in a freshly collected sample
    B. Can also resemble cysteine crystals
    C. Appear insoluble in alkaline urine
    D. Presence indicates an inborn error of metabolism

59. A patient sends the following question to an online consumer health Web site: "I am a 22-year-old female who experienced increasing headaches, thirst, and decreasing energy. I was studying in the library when I felt lightheaded and passed out. I was taken to a hospital emergency department and they told me that my serum Acetest® was 40 mg/dL and urine glucose was 500 mg/dL. What does this mean?" How would you reply?
    A. Your lab results pattern suggests diabetes mellitus.
    B. You probably have been crash dieting recently.
    C. The two results do not fit any disease pattern.
    D. The tests need to be repeated because they could not possibly occur together.

60. Which urinalysis reagent strip test will never be reported out as "negative"?
    A. Protein
    B. Urobilinogen
    C. Bilirubin
    D. Nitrite

61. The following urinalysis results were obtained on a 40-year-old white male whose skin appeared yellowish during the clinical examination. Color and clarity—dark brown, clear; protein—negative; glucose—negative; blood—negative; ketones—negative; bilirubin— moderate; urobilinogen—0.2 mg/dL. These results are clinically significant in which of the following conditions?
    A. Bile duct obstruction
    B. Cirrhosis
    C. Hepatitis
    D. Hemolytic anemia

62. Compared to the fecal occult blood test, which of the following is a disadvantage of performing a DNA-based test to detect colon cancer?
    A. The DNA test is more invasive.
    B. The DNA test is less sensitive.
    C. The DNA test is more expensive.
    D. Additional diet restrictions are needed for the DNA test.

63. Which of the following may be associated with morphologic examination of spermatozoa?
    A. Evaluation should include assessment of 1000 spermatozoa.
    B. A small number of sperm should have normal morphologic characteristics.
    C. Papanicolaou stain may be used.
    D. Presence of red or white cells and epithelial cells need not be noted.

64. Which condition is characterized by increased levels of immunoglobulins in the cerebrospinal fluid, originating from within the central nervous system and not from the general blood circulation?
    A. Gout
    B. Erythroblastosis fetalis
    C. Multiple myeloma
    D. Multiple sclerosis

65. Which of the following statements pertains to screening methods used to determine pregnancy?
    A. Immunoassays will use reagent anti-hCG to react with patient hCG.
    B. A random urine specimen is the preferred specimen for pregnancy screening tests.
    C. Internal controls provided within the kit will assess if the patient's specimen was collected correctly.
    D. External quality control is not needed with these methods.

66. The following urinalysis biochemical results were obtained from a 4-month-old infant who experienced vomiting and diarrhea after milk ingestion and failed to gain weight: pH—6; protein—negative; glucose—negative; ketone—negative; bilirubin—negative; Clinitest®—2+. These results are clinically significant in which of the following disorders?
    A. Diabetes mellitus
    B. Ketosis
    C. Starvation
    D. Galactosemia

67. Which of the following is a *true* statement?
    A. Renal tubular cells originate from the renal pelvis.
    B. Red blood cells in acid urine (pH 4.5) will usually be crenated because of the acidity.
    C. Bacteria introduced into a urine specimen at the time of the collection will have no immediate effect on the level of nitrite in the specimen.
    D. Pilocarpine iontophoresis is the method of choice for the collection of pericardial fluid.

answers & rationales

1.

D. The first-voided morning urine specimen is the most desirable for chemical and microscopic analysis because it is the most concentrated specimen of the day. Protein and nitrite testing is better performed on a concentrated specimen, as are the specific gravity determination and the examination of urinary sediment. However, because of the lack of food and fluid intake during the night, glucose metabolism may be better assessed on the basis of a postprandial specimen.

2.

D. Urochrome, a yellow-brown pigment derived from urobilin, is principally responsible for the yellow coloration of normal urine. Urochrome is excreted at a constant rate, showing no diurnal variation. Therefore, the color of normal urine, which may range from straw to deep amber, is dependent on the concentrating ability of the kidney and the volume of urine excreted.

3.

C. Melanin, a substance derived from tyrosine, is responsible for the pigmentation of the eyes, skin, and hair. In some malignancies, known as melanomas, the tumor or mole takes on a darkly pigmented appearance because of the melanin present. In cases of metastatic melanoma, melanogen, which is a colorless precursor of melanin, is excreted in the urine. If the urine is allowed to stand at room temperature for 24 hours, the melanogen is oxidized to melanin, imparting a dark brown or black coloration to the specimen. Qualitative screening tests for the detection of melanin in urine use ferric chloride or sodium nitroprusside as the oxidation reagent systems.

4.

D. pH is a representative symbol for the hydrogen ion concentration. The kidney plays an important role in the maintenance of the acid-base balance of body fluids by either excreting or retaining hydrogen ions. A normally functioning kidney will excrete urine with a pH between 4.5 and 8.0, depending on the overall acid-base needs of the body.

5.

B. At room temperature, the amount of bacteria present in a urine sample will increase. The bacteria are capable of metabolizing the urinary urea to ammonia. The ammonia formed through this process will cause an alkalinization of the urine.

6.

C. On the average, a normal adult excretes 1200–1500 mL of urine daily. "Polyuria" is a term used to describe the excretion of a urine volume in excess of 2000 mL/day. In oliguria, the daily urine excretion is less than 500 mL, and in anuria the urine formation is completely suppressed. Hypersthenuria refers to urines of any volume containing increased levels of dissolved solute.

7.

A. In healthy individuals the amount of protein excreted in the urine should not exceed 150 mg/24 hr. When protein is present in the urine, the colorimetric reagent test strips change color, indicating a semiquantification of the amount of protein present. Serum proteins are classified as being albumin or globulin in nature, and the type of protein excreted in the urine is dependent on the disorder present. Although the strip test is a rapid screening method for the detection of urinary protein, it must be noted that this method is more sensitive to the presence of albumin in the specimen than to the presence of globulin, Bence Jones protein, or mucoprotein.

8.

C. Normal urine does not foam on being shaken. However, urine containing bilirubin will exhibit yellow foaming when the specimen is shaken. In fact, the foam test was actually the first test for bilirubin, before the development of the chemical tests. If the shaken specimen shows a white foam, increased urine protein can be suspected.

9.

C. For quality control of reagent test strips, it is recommended that both positive and negative controls be used daily. It is necessary that any deterioration of the strips be detected in order to avoid false-positive or false-negative results. The use of positive and negative controls will act as a check on the reagents, on the technique employed, and on the interpretive ability of the person or instrument performing the test.

10.

D. The principle of the reagent strip method for the detection of protein in urine is based on a color change in an indicator system, such as tetrabromophenol blue, that is buffered to pH 3. The buffering capacity of the strip is sufficient provided that the urine pH does not exceed 8.0. Within the normal urine pH range of 4.5–8.0, a change in color in the reagent strip is an indication of the presence of protein in the urine. With a urine pH greater than 8, the buffering capacity of the strip may be exceeded, and a false-positive color change in the impregnated area will reflect the pH of the urine rather than the presence of protein. The presence of vitamin C, uric acid, or glucose in urine will not affect the test for protein.

11.

D. "Isosthenuria" is a term applied to a series of urine specimens that exhibit a fixed specific gravity of approximately 1.010. In isosthenuria there is little, if any, variation of the specific gravity between urine specimens from the same patient. This condition is abnormal and denotes the presence of severe renal damage in which both the diluting ability and the concentrating ability of the kidneys have been severely affected.

12.

C. When globulin, mucoprotein, or Bence Jones protein is present in a urine specimen, the reagent strip test may give a negative result because the strip is more sensitive to the presence of albumin than to the presence of other proteins in urine. However, the sulfosalicylic acid (SSA) test is able to detect not only albumin but also globulin, mucoprotein, and Bence Jones protein in a specimen. Therefore, it can be seen that a negative reagent strip test result for protein but a positive sulfosalicylic acid test result is possible when the protein present is some protein other than albumin. For this reason the sulfosalicylic acid test is run as a test for urinary protein if the presence of abnormal proteins is suspected.

13.

D. Although sodium is the major inorganic molecule found in urine, urea is the major organic molecule excreted. Urea is a waste product of protein/amino acid metabolism. Its level in a normal 24-hour urine with a glomerular filtration rate of 125 mL/min would be 400 mmol/day. Glucose excretion will average less than 1 mmol/day. The excretion of the inorganic molecules sodium and potassium would be 130 and 70 mmol/day, respectively.

14.

A. Preanalytical components of laboratory testing include all variables that can affect the integrity or acceptability of the patient specimen prior to analysis, such as correct collection technique. Analytical factors affect the actual analysis of the specimen (temperature, condition of equipment, timing, presence of interfering substances). Postanalytical factors affect the final handling of the results generated (reporting units, critical values, acceptability of quality control).

15.

B. Under normal metabolic conditions, the body metabolizes fat to carbon dioxide and water. With inadequate carbohydrate intake, as with dieting and starvation, or with inadequate carbohydrate metabolism, as with diabetes mellitus, there is an increased utilization of fat. Because of this increased fat metabolism, the body is unable to completely degrade the fat, resulting in a buildup of intermediary products known as ketone bodies. The term "ketone bodies" is used collectively to denote the presence of acetoacetic acid, beta-hydroxybutyric acid, and acetone. Reagent test strips impregnated with sodium nitroprusside are able to detect the presence of acetoacetic acid and acetone in urine specimens. Although beta-hydroxybutyric acid accounts for approximately 78% of the total ketones, it is not detected by the sodium nitroprusside test.

16.

C. Although a positive result on a urine test for ketones is most commonly associated with increased urinary glucose levels, as in diabetes mellitus, other conditions may cause the urine ketone test to show positive results while the urine glucose test shows negative results. In young children, a negative glucose reaction accompanied by a positive ketone reaction is sometimes seen. Ketones in the urine may be seen when a child is suffering from an acute febrile disease or toxic condition that is accompanied by vomiting or diarrhea. In these cases, because of either decreased food intake or decreased intestinal absorption, fat catabolism is increased to such an extent that the intermediary products, known as ketone bodies, are formed and excreted in the urine.

17.

C. The colorimetric reagent strip test for the detection of hemoglobin in urine utilizes a buffered test zone impregnated with a dye and organic peroxide. The peroxidase activity of hemoglobin catalyzes the oxidation of the dye with peroxide to form a colored compound. Like hemoglobin, myoglobin also has a peroxidase activity and, when present in a urine specimen, myoglobin will react, yielding false-positive results. In the presence of large amounts of ascorbic acid, antibiotics containing ascorbic acid as a preservative, formaldehyde, or nitrite, the urine reaction may be inhibited, causing false-negative results.

18.

D. Bilirubin is a compound that is formed as a result of hemoglobin breakdown. The majority of bilirubin in the blood is bound to albumin and is known as unconjugated bilirubin. Because unconjugated bilirubin is not water soluble, it may not be excreted in the urine. The remainder of the bilirubin in the blood has been processed by the liver. In the liver, the bilirubin is conjugated with glucuronic acid or sulfuric acid. This conjugated bilirubin is water soluble, and it is this portion that is excreted in increased amounts in the urine in some hepatic and obstructive biliary tract diseases. The presence of conjugated bilirubin in a urine specimen may be detected by use of the reagent test strips. The test strips are impregnated with a diazonium salt, such as diazotized 2,4-dichloroaniline, which forms a purplish azobilirubin compound with bilirubin.

19.

B. Only dissolved solutes affect specific gravity (e.g., glucose). Cells, mucus, crystals, or any other formed elements will have no effect, regardless of concentration. If the reagent strip method is used, it should be noted that only dissolved ions will contribute to specific gravity results. Thus glucose would not affect reagent strip results at any concentration. In such instances as diabetes mellitus, with urine glucose levels over 2 g/dL, there may be a discrepancy between specific gravity results obtained with a reagent strip method versus using a refractometer, because such glucose levels are known to increase refractometer results, thus requiring correction.

20.

D. The majority of renal and urinary tract diseases are characterized by an increased number of neutrophilic leukocytes in the urine. To identify correctly any white blood cells present in a urine specimen, it is necessary to examine the specimen as soon as possible after collection. This is necessary because leukocytes tend to lyse easily when exposed to either hypotonic or alkaline urine.

21.

A. The ratio of urine albumin to creatinine in a random specimen is commonly used to evaluate microalbuminuria, especially in patients with diabetes mellitus. This patient's ratio is 114 mg albumin per gram creatinine. The American Diabetes Association defines microalbuminuria as between 30 and 299 mg/g. Values greater than 299 mg/g would be "macroalbuminuria." Nephrotic syndrome is characterized by excretion of albumin in excess of 3.5 grams per day. Patients with obstructive jaundice will usually not experience proteinuria.

22.

B. To better diagnose renal and urinary tract diseases, it is necessary to examine urinary sediment carefully by the most appropriate microscopic method available. Formed elements in the urine, such as cells and casts, are more easily differentiated by the use of phase-contrast microscopy. This is especially true for the identification of the more translucent elements such as the hyaline casts. Phase microscopy tends to enhance the outline of the formed elements, allowing them to stand out and be more easily distinguished.

23.

A. Fatty materials in urinary sediment may be identified by means of staining techniques using Sudan III and oil red O or by means of polarized microscopy. Polarized microscopy is especially useful when the composition of fatty casts, fatty droplets, or oval fat bodies is primarily cholesterol. When cholesterol molecules are exposed to polarized microscopy, the effect is such that a Maltese cross formation becomes visible, simplifying the identification process. Casts and red blood cells may be better visualized using phase-contrast microscopy. Ketone bodies will be soluble and, therefore, not seen in a urine sediment.

24.

D. When neutrophils are exposed to hypotonic urine, their physical appearance becomes altered. Under hypotonic conditions, the neutrophils tend to swell and the cytoplasmic granules contained within the cells exhibit Brownian movement. This Brownian movement of the granules causes the neutrophilic contents to refract in such a way that the cells appear to glitter—thus the name "glitter cells."

25.

D. Waxy casts represent the final phase of granular cast degeneration. As the fine granules of the granular casts lyse, highly refractive, smooth, blunt-ended waxy casts are formed. When waxy casts are found in the urine sediment, the implication is that there is nephron obstruction caused by tubular inflammation and degeneration.

26.

C. Nephrotic syndrome is suggested by the increased urine protein (with serum albumin significantly decreased), the hyaline and fatty casts, and the presence of oval fat bodies. The patient's symptoms of periorbital swelling and edema reflect the loss of oncotic pressure because of the excretion of albumin. Its loss from the vascular compartment will induce plasma water movement into the tissue spaces. Glomerulonephritis will have many more red blood cells, including red blood cell casts. Multiple myeloma will not show increased urine albumin but rather immunoglobulin light chains. Chronic renal failure will have multiple types of casts present (hyaline, granular, cellular, waxy, fatty).

27.

B. There are minimal bacteria present in Color Plate 46■. Both budding yeast and white blood cells predominate this microscopic field. Patients with diabetes mellitus are prone to such yeast infections because of the increased glucose in their urine. The epithelial cells visualized in this field are transitional and not squamous. They can be distinguished by their size (about 15–20 μm), less cytoplasm than a squamous cell would have, and their central nucleus. Increased squamous epithelial cells would suggest improper collection, whereas transitional cells, if greater than 5 cells/hpf, would indicate pathology. There are fewer than 5 red blood cells in this field, and that would be below the sensitivity of the blood pad on the reagent strip.

28.

D. Alkaptonuria is a rare hereditary disease that is characterized by excessive urinary excretion of homogentisic acid. This acid, the product of phenylalanine and tyrosine metabolism, accumulates in urine because of a deficiency in the enzyme homogentisic acid oxidase, which normally catalyzes the oxidation of homogentisic acid to maleyl acetoacetic acid. Urine containing homogentisic acid turns black on standing because of an oxidative process; thus the screening test for alkaptonuria consists of the detection of a black coloration in urine that is left standing at room temperature for 24 hours.

29.

C. This student has a lower urinary tract infection (UTI), also known as cystitis. The major distinguishing features between upper and lower UTI include the presence of protein and casts in an upper UTI and not in a lower UTI. This is because both urine protein excretion and cast formation reflect what is happening within the kidney itself. The most common source of either upper or lower UTIs is contamination by enteric gram-negative bacteria. Their presence will not be found in glomerulonephritis or with urinary stones (nephro = "kidney" + lith = "stone").

30.

D. The intestinal enterochromaffin cells, sometimes called the argentaffin cells, produce a substance known as serotonin from the amino acid tryptophan. In cases of metastatic carcinoid tumors, excessive amounts of serotonin are produced. Serotonin may then undergo oxidative deamination to form the metabolite 5-hydroxyindole acetic acid (5-HIAA), which is excreted in the urine. It is the quantification of 5-HIAA that is diagnostically significant because it reflects serotonin production.

31.

B. Because of increased lipid metabolism in long-term, intensive dieting, ketone body formation will increase. Blood glucose levels in such patients will be normal or decreased. In nephrotic syndrome, the large amounts of albumin excreted will be detectable by both reagent strip and SSA methods. In multiple myeloma, however, the increased globulin light chains (Bence Jones proteins) excreted will only be detectable by SSA because the reagent strip is more sensitive to albumin. Cystitis is a lower urinary tract infection affecting the bladder but not the kidney itself. This infection will not exhibit increased protein, whereas an upper urinary tract infection will.

32.

C. Bacteria of the *Enterobacter, Citrobacter, Escherichia, Proteus, Klebsiella,* and *Pseudomonas* species produce enzymes that catalyze the reduction of nitrate, a substance normally found in urine, to nitrite. Reagent test strips have been developed that are able to detect nitrite in urine. Therefore, a positive nitrite test result is an indirect indication of the presence of bacteria in the urine specimen.

33.

C. Cerebrospinal fluid (CSF) is a clear, colorless liquid that may be described as a modified ultrafiltrate of blood. Both active transport and passive diffusion are involved in the passage of glucose from the blood into the CSF. Normally, fasting CSF glucose levels range between 50 and 80 mg/dL, representing approximately 60–70% of the blood glucose level. In hyperglycemia with plasma glucose levels of 300 mg/dL, the active transport mechanism reaches a point of maximum response, so that CSF glucose levels reflect approximately 30% of the plasma glucose level. Decreased CSF glucose levels are associated with hypoglycemia, a faulty active transport mechanism, and excess utilization of glucose by microorganisms, red or white blood cells, or the central nervous system.

34.

C. In addition to hemoglobin, the muscle protein myoglobin can cause a positive blood result in chemical reagent strip testing. Both hemoglobin and myoglobin possess pseudoperoxidase activity detected by the "blood" chemistry test. Muscle-crushing injuries (rhabdomyolysis) will release myoglobin from the muscle. The myoglobin, being a small molecule, is readily excreted by the kidneys. Myoglobinuria can lead to acute renal failure. Myoglobin can be distinguished from hemoglobin in urine by an ammonium sulfate screening test. Myoglobin will remain soluble in 80% ammonium sulfate and give a positive filtrate blood reaction after the precipitation of hemoglobin. The patient's pathology involves neither red blood cells nor bilirubin. The severe stress may be responsible for both the urine protein and granular cast results.

35.

D. The eGFR calculation is based on the "modification of diet and renal disease" formula recommended by the American Kidney Foundation. It does not use a urine sample at all, but instead requires only a serum creatinine and the patient's age, gender, and race. Values less than 60 mL/min/1.73 m^2 are considered abnormal and need to be followed up. This patient's value places him in stage 3 kidney damage (35–59 mL/min/1.73 m^2). This calculation is considered more accurate than the Cockgroft-Gault formula, but there are limitations based on the standardization of the creatinine method used.

36.

A. Normal alkaline (or neutral) urine may contain triple phosphate crystals, as seen in Color Plate 47■. These crystals can be identified by the characteristic "coffin lid" appearance. They usually do not indicate any pathology.

37.

D. Refer to Color Plate 48■. The fringed appearance at the one end of the major formed element strongly suggests that this is a fiber artifact, most likely placed in the sample at the time of collection. Casts, taking the shape of the tubule within which they are formed, will not have such a fringed end.

38.

C. The distal convoluted tubule and collecting duct provide water reabsorption through the action of antidiuretic hormone (vasopressin). The renin-angiotensin-aldosterone system is responsible for sodium reabsorption by the distal and collecting tubules. Decreased plasma volume leads to pressure alterations detected by receptors located in the kidney's juxtaglomerular apparatus and the right atrium of the heart. These changes trigger the production of renin and antidiuretic hormone, respectively.

39.

A. Only freshly voided urine specimens should be used for urinalysis testing. If the specimen cannot be examined within 1 hour after collection, it should be refrigerated to help preserve the integrity of the specimen. When urine is left standing at room temperature for an excessive period, multiplication of bacteria will occur. The bacteria are capable of converting urea in the urine to ammonia, causing the urine to become more alkaline. Loss of carbon dioxide from the specimen will also contribute to the alkalinization of the urine. Constituents such as glucose, bilirubin, and urobilinogen will also be lost from the specimen.

40.

B. The presence of cystine crystals in a patient sample is always a cause for immediate notification of the physician. Cystinuria is an autosomal recessive disorder characterized by the inability to reabsorb the amino acids cystine, lysine, arginine, and ornithine in either the renal tubules or the intestine. Cystine will crystallize in acid pH more readily than the other amino acids. Tyrosine forms needle-shaped crystals whereas leucine will appear round and oily with concentric rings. Isoleucine and phenylalanine will not form crystals in the urine. Acetoacetate and β-hydroxybutyric acid are two ketone bodies that will be soluble in the sample and give a positive reaction with nitroprusside.

41.

D. The major formed elements in Color Plate 50■ are hyaline casts and mucus fibers, which are normal in the numbers shown in this field. Waxy casts will appear yellowish with characteristic serrated edges. There are no obvious granules in the casts shown, and red blood cells are not present.

42.

A. Phenylketonuria is inherited as an autosomal recessive trait that manifests itself in the homozygous form. The basis for the disease lies in the fact that the enzyme phenylalanine hydroxylase, which is needed for the conversion of phenylalanine to tyrosine, is absent. Because of this enzyme deficiency, phenylalanine levels rise in the blood, with increased amounts of phenylpyruvic acid and other derivatives being excreted in the urine. If the disease is detected at an early stage, mental retardation may be avoided by restricting the dietary intake of phenylalanine.

43.

D. Color Plate 51■ demonstrates sperm and calcium oxalate crystals. Both formed elements are found in correctly collected normal urines from either gender. Calcium oxalate seen here is the dehydrate form. The monohydrate form will appear oval or dumbbell shaped. Neither formed element is usually associated with pathology.

44.

B. A variety of substances in CSF specimens have been associated with a xanthochromic appearance. Among those substances are oxyhemoglobin, carotenoids, bilirubin, and protein. The appearance of the specimen by itself is not usually specific for a particular disease state, but it may provide useful information in comparison with other findings. Glucose, magnesium, and chloride do not contribute to the color of the specimen.

45.

A. Refer to Color Plate 52■. Erythrocytes or red blood cells (RBCs) occur in small numbers (0–2/hpf) in a normal urine. Using brightfield microscopy, unstained RBCs appear as colorless discs with an average size of 7 μm in diameter. Increased or large numbers of RBCs are commonly seen with acute glomerulonephritis, renal calculi, acute infections, and menstrual contamination. The nephrotic syndrome is characterized by heavy proteinuria, oval fat bodies, renal tubular epithelial cells, casts, and waxy and fatty casts. Biliary tract obstruction will show pale-colored stools, whereas vaginal discharge contamination may introduce increased numbers of white blood cells.

46.

A. Because there may be technical problems associated with amniocentesis, contamination with maternal urine should be considered in evaluating specimens submitted for amniotic fluid analysis. Urinary concentrations of creatinine and urea nitrogen are anywhere from 10 to 50 times the amniotic fluid concentrations, and an increased concentration of either in the amniotic fluid would be sensitive indicators of urinary contamination. Measurements of albumin, total protein, or lactate dehydrogenase would be of little use for this purpose because their relative concentrations in urine and amniotic fluid are not predictably different. A delta absorbance at 410 nm would be used to assess the presence of bilirubin in the sample, as in assessment of erythroblastosis fetalis.

47.

D. The alveolar concentrations of the various phospholipids (surfactants) change during fetal lung development, and because these changes are reflected directly in the amniotic fluid, a number of investigations have shown that analysis of the fluid can provide good predictive information for the development of respiratory distress syndrome in the newborn. The concentrations of sphingomyelin and phosphatidyl inositol increase until about 32–34 weeks of gestation and then decline. Conversely, lecithin (phosphatidyl choline) and phosphatidyl glycerol concentrations increase rapidly after 32–34 weeks of gestation, and their concentrations relative to those of the other phospholipids are useful in assessing the development of fetal lung maturity.

48.

C. The hemoglobin released from red blood cells in an upper gastrointestinal bleed will have time to become denatured and oxidized as it travels the entire intestinal tract. This will make the stool become much darker in color by the time the hemoglobin is excreted. Clay-colored stools will result from an obstruction of the biliary duct, preventing bilirubin from entering the intestines to be converted into urobilinogen and then into urobilin. Stools with red streaks are more likely to result from a lower gastrointestinal bleed, as from the colon. Increased mucus will be associated with intestinal inflammation.

49.

B. Effusions can be transudates or exudates, and the distinguishing characteristics are cell number and total protein. Transudates, being noninflammatory, will have low numbers of cells and less than 3 g/dL protein. This patient's results suggest she has an exudate due to the high number of cells and large amount of protein.

50.

A. Diabetes insipidus is caused by a deficiency in antidiuretic hormone. Such deficiencies will result in the kidney's inability to reabsorb water at the distal and collecting tubules. This affects only water reabsorption and not the reabsorption of other urinary solutes. Excreted solute amounts will be the same, but the water volume into which they are excreted will be larger. This results in high urine volumes and low final solute concentrations. The low solute will lead to low specific gravities in these patients' specimens.

51.

D. Synovial fluid is a form of plasma ultrafiltrate with added hyaluronic acid. Decreased viscosity and poor mucin clot formation are indications of the decreased hyaluronate concentration of synovial fluid. Either of these findings is usually an indication of inflammation. Because the viscosity of synovial fluid is normally very high, it can be estimated by the length of string formed when the fluid drops from a syringe. The term "mucin" in the mucin clot test is a misnomer, because mucin is not present in synovial fluid.

52.

A. Effusions result from an imbalance of the flow of body fluids. Effusions are classified as exudates or transudates on the basis of certain characteristics. Exudates are generally formed in response to inflammation or infection with concomitant capillary wall damage. Exudates are characterized by protein levels greater than 3.0 g/dL, leukocyte counts greater than 1000/μL, and the presence of a sufficient amount of fibrinogen to cause clotting. In contrast, transudates are characterized by protein levels less than 3.0 g/dL, leukocyte counts less than 300/μL, and the absence of fibrinogen. Transudates are generally formed as the result of noninflammatory processes, including alterations in plasma oncotic pressure, pleural capillary hydrostatic pressure, or intrapleural pressure.

53.

D. A clinically useful test for assessing the concentrating and diluting ability of the kidneys is the determination of urine specific gravity. The specific gravity is a measure of the proportion of dissolved solids in a given volume of solvent. Polyelectrolytes are incorporated into urinalysis reagent strips. A dye also present in the strips will change color because of a pKa change in the polyelectrolytes. The pKa varies with the ionic concentration of the urine. The color obtained is compared with a set of standard colors, each color correlating with a different specific gravity concentration. The TS meter is a specific type of refractometer that utilizes the close correlation of a solution's refractive index with its solute concentration to determine the specific gravity of urine. The refractive index is the ratio of the velocity of light in air to the velocity of light in a solution, this being comparable to the number of dissolved particles in that solution. An osmometer measures the concentration of dissolved solute in the sample, usually through its relationship to freezing point depression.

54.

D. Trichloroacetic acid is a turbidimetric method used to quantify small amounts of protein, less than 100 mg/dL, in cerebrospinal fluid (CSF) and urine specimens. Coomassie brilliant blue is a colorimetric dye binding method in which protein complexes with the dye, forming a soluble blue complex. This method also exhibits the necessary sensitivity for detecting small quantities of protein. Bromcresol green is selective for albumin and is used to quantify albumin in serum. Ponceau S is used in serum protein electrophoresis methods to stain both albumin and globulins.

55.

A. As seen in Color Plate 53■, hyaline casts are the most commonly observed cast, and they consist completely of uromodulin (Tamm-Horsfall) protein. A reference urine may contain 0–2 hyaline casts per low-power field. Hyaline casts appear translucent using brightfield microscopy because they have a refractive index similar to urine. Phase-contrast microscopy may be used to visualize the casts better.

56.

A. To be found in urine, a solute must be water soluble. Solutes can be inorganic (e.g., sodium) or organic (e.g., urea). Excreted waste products, meaning end products of metabolism, are creatinine, urea, and uric acid. Some excreted solutes, however, are not present as waste but as overload, such as glucose or sodium.

57.

D. Cerebrospinal fluid (CSF) must be collected in sterile tubes. The first tube is generally used for chemistry and serology studies, the second tube is employed for bacteriologic examination, and the third tube is used for cell counts. Tubes used for chemistry and bacteriologic studies should be centrifuged before use. CSF should remain uncentrifuged for cell counts. Low-volume specimens need to be cultured first (to ensure sterility) before any other test is performed. Because the analysis of CSF should be performed immediately, it is critical that personnel on all shifts be able to perform the necessary testing.

58.

B. Uric acid crystals, as seen in Color Plate 54■, are commonly encountered in normal acidic urine but may be observed in neutral urine and rarely in an alkaline urine, because uric acid is soluble at alkaline pH. Using brightfield microscopy, uric acid crystals appear as diamonds, cubes, barrels, rosettes, and may even have six sides and be confused with cysteine. Because they are a reflection of the excretion of purine waste products, they may be pathologically increased in cases of gout and after chemotherapy. They show birefringence (multiple colors) under plane polarized light.

59.

A. A positive urine glucose plus a positive serum ketone strongly suggest uncontrolled diabetes mellitus. There is an increased rate of fatty acid oxidation occurring in light of the inaccessibility of the glucose, especially to skeletal muscle. If the patient had only been dieting, the glucose would be negative.

60.

B. The sensitivity of a method is the lowest concentration of the analyte that will result in a detectable reaction signal. The protein, bilirubin, and nitrite readout color scales each have a color associated with analyte concentrations less than the method's sensitivity, called "negative." Urobilinogen's readout color scale begins with its lowest reportable value, but there is no pad associated with concentrations less than this.

61.

A. In the hepatic phase of bilirubin metabolism, bilirubin is conjugated with glucuronic acid to form water-soluble conjugated bilirubin. The conjugated bilirubin passes into the bile duct and on to the intestinal tract. In the intestine, it is reduced by intestinal bacteria to form urobilinogen. Bile duct obstruction is characterized by an obstruction of the flow of conjugated bilirubin into the intestinal tract to complete its metabolism. The conjugated bilirubin, which is water soluble, will be excreted by the kidney. Because bilirubin is not entering the intestines, the normal production of urobilinogen is decreased. Therefore, the urine biochemical test will indicate a positive reagent strip test for bilirubin, positive Ictotest, and "normal" (0.2 mg/dL) urobilinogen (because there is no reagent strip pad for "negative" urobilinogen).

62.

C. DNA-based tests for detecting mutations within colon cells are generally more expensive than the fecal occult blood (FOB) methods using the pseudoperoxidase property of hemoglobin. Advantages, however, include no diet or medication restrictions prior to the testing, use of a single stool sample, and a sensitivity (50–73%) that far exceeds that of the FOB test (13–35%) for detecting colon cancer. It is, however, less sensitive than colonoscopy, which can detect 95% of colon cancers. Gene mutations associated with colorectal cancer include APC (adenomatous polyposis coli) on chromosome 5; K-ras, a mutation that often occurs after APC mutation; and p53, with mutations occurring later in the process, associated with larger adenomas.

63.

C. The morphologic characteristics of spermatozoa are best evaluated by means of smears stained with Papanicolaou stain. Other stains used include Kernechtrot, Giemsa, basic fuchsin, crystal violet, and hematoxylin. When oil immersion is used, a minimum of 200 spermatozoa should be evaluated for morphologic characteristics. Although sources differ as to the exact number, it is generally established that at least 60% of the sperm should have normal morphologic features. When this microscopic analysis is performed, the presence of erythrocytes, leukocytes, epithelial cells, and microorganisms should be indicated.

64.

D. Immunoglobulins (IgGs) are normally present at less than 1 mg/dL in the CSF. Increased CSF IgG can result from increased CSF production (e.g., multiple sclerosis) or from increased transport from the blood plasma (compromised blood-brain barrier). Neither gout, erythoblastosis fetalis (isoimmunization syndrome), nor multiple myeloma produces increased CSF IgG levels.

65.

A. Many simplified yet immunologically sophisticated methods exist currently for determining pregnancy. All are based on the reaction between patient human chorionic gonadotropin (hCG) and anti-hCG. Most kits will use an antibody recognizing one subunit of hCG (alpha or beta), whereas other kits may use both anti-α-hCG and anti-β-hCG. Internal controls in these kits will only check if the procedural steps were performed correctly. They cannot detect problems with any preanalytical variables, like specimen handling or appropriateness. In addition, internal quality control cannot be used to assess the kit's accuracy in distinguishing "positive" from "negative" specimens. Only the use of external quality control specimens can accomplish this. Because the first morning specimen is the most concentrated of the day, it is the preferred specimen for such screenings. Use of a random urine may be too dilute to detect low levels of patient hCG, thus giving a false negative.

66.

D. Galactosemia, an inborn error of metabolism, is characterized by the inability to metabolize galactose, a monosaccharide that is contained in milk as a constituent of the disaccharide lactose. Thus galactose appears in elevated levels in the blood and urine. The condition may result in liver disease, mental retardation, and cataract formation if not treated or controlled. In the biochemical analysis of the urine, the conflicting results for the two glucose tests may be explained as follows: The glucose oxidase reagent strip test is specific for glucose; therefore, the glucose will be negative. The Clinitest®, a modification of the Benedict's test procedure, detects most reducing substances. Because galactose is present in the urine and is a reducing substance, the Clinitest® is positive.

67.

C. Renal tubular cells originate from the renal medulla or cortex. Red blood cell crenation is a phenomenon reflecting increased solute concentration (hyperosmolality) and is not caused by urine pH. Red cells will, however, lyse at high alkaline pH. The nitrite reaction requires (a) a sufficient dietary source of nitrate, (b) sufficient numbers of bacteria present in the urine, and (c) sufficient incubation time (>4 hours). Bacteria introduced at collection, even in sufficient number, will not have had sufficient incubation time to convert urine nitrate to nitrite. Pilocarpine iontophoresis is the collection method for sweat.

review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.*

1. What is the molarity of a solution that contains 18.7 g of KCl in 500 mL?
   A. 0.1
   B. 0.5
   C. 1.0
   D. 5.0

2. A calcium standard solution contains 10 mg/dL of calcium. What is its concentration in millimoles per liter?
   A. 2.5 mmol/L
   B. 5.0 mmol/L
   C. 7.5 mmol/L
   D. 10.0 mmol/L

3. How much NaCl is needed to prepare 100 mL of a standard solution of concentration 135 mEq/L of Na?
   A. 0.31 g
   B. 0.79 g
   C. 1.2 g
   D. 1.8 g

4. How much 95% alcohol is required to prepare 5 L of 70% alcohol?
   A. 2.4 L
   B. 3.5 L
   C. 3.7 L
   D. 4.4 L

5. A solution of NaOH is standardized by titration with 0.100 N HCl. A total of 10.0 mL of NaOH requires 11.25 mL of HCl. What is the normality of the NaOH solution?
   A. 0.100
   B. 0.112
   C. 0.113
   D. 1.125

*Note: A periodic table of the elements is located on p. 967.

6. How many milliliters of concentrated H_2SO_4 (sp. gr. = 1.84 g/mL; assay = 97%) are required to prepare 10 L of 0.1 N H_2SO_4?
   A. 1.84
   B. 9.20
   C. 27.5
   D. 54.4

7. How many grams of $CaCl_2 \cdot 2H_2O$ must be used to prepare 500 mL of 10% anhydrous $CaCl_2$?
   A. 33.1
   B. 41.3
   C. 50.0
   D. 66.2

8. What is the osmolality of a solution containing 5.85 g NaCl and 18 g glucose in 1 kg water?
   A. 0.2
   B. 0.3
   C. 0.6
   D. 0.9

9. Physiologic saline solution is 0.85% NaCl. What is its osmolarity?
   A. 0.15
   B. 0.29
   C. 0.85
   D. 8.5

10. Convert 70°F to degrees Celsius.
    A. 7°C
    B. 12°C
    C. 20°C
    D. 21°C

11. A 5 N solution is diluted 1:4. The resulting solution is diluted 4:15. What is the concentration in normality of the final solution?
    A. 0.25
    B. 0.33
    C. 2.5
    D. 3.0

12. A colorimetric method calls for the use of 0.1 mL of serum, 5 mL of reagents, and 4.9 mL of water. What is the dilution of the serum in the final solution?
    A. 1 to 5
    B. 1 to 10
    C. 1 to 50
    D. 1 to 100

13. A solution of a colored substance that is known to follow Beer's law has an absorbance of 0.085 when measured in a cell 1 cm in length. Calculate the absorbance for a solution of twice the concentration measured in the same cell.
    A. 0.042
    B. 0.085
    C. 0.170
    D. 5.90

14. In a spectrophotometric procedure that follows Beer's law, the absorbance of a standard solution of concentration 15 mg/dL is 0.50 in a 1-cm cell. The absorbance of the sample solution is 0.62. What is the concentration?
    A. 0.62 mg/mL
    B. 6.2 mol/L
    C. 12.1 mg/dL
    D. 18.6 mg/dL

15. A stock standard solution of urea contains 10 mg/mL of urea nitrogen. How much stock (in milliliters) is needed to prepare 200 mL of a working standard containing 20 mg/dL of urea nitrogen?
    A. 1
    B. 2
    C. 4
    D. 8

16. How many milliliters of a 50%$^{v/v}$ acetic acid solution are required to prepare 1 L of 5%$^{v/v}$ acetic acid?
    A. 0.01
    B. 0.10
    C. 10
    D. 100

17. How many grams of sulfosalicylic acid (mol wt = 254) are required to prepare 1 L of a 3%$^{w/v}$ solution?
    A. 3.0
    B. 7.6
    C. 30
    D. 254

18. A quantitative protein analysis is performed on an aliquot of a 24-hour urine specimen. The test indicates the presence of 1.2% protein. If a total urine volume of 2155 mL is collected, how many grams of protein are excreted in the 24-hour specimen?
    A. 0.056
    B. 2.6
    C. 25.9
    D. 258.6

19. How many grams of H_2SO_4 are required to prepare 750 mL of a 2 M solution?
    A. 36.8
    B. 73.5
    C. 98
    D. 147

20. How many grams of NaOH are required to prepare 4 L of a 2 N solution?
    A. 40
    B. 80
    C. 160
    D. 320

21. How many milliliters of glacial acetic acid (mol wt = 60; assay = 99.7%) are required to prepare 2 L of a 5%$^{v/v}$ solution?
    A. 6
    B. 10
    C. 50
    D. 100

22. How many grams of H_2SO_4 are required to prepare 6 L of a 5 N solution?
    A. 245
    B. 294
    C. 490
    D. 1470

23. How many grams of NaOH are required to prepare 2500 mL of a 4 M solution?
    A. 40
    B. 100
    C. 160
    D. 400

24. An isotonic saline solution contains 0.85%$^{w/v}$ NaCl. How many grams of NaCl are needed to prepare 5 L of this solution?
    A. 4.25
    B. 8.5
    C. 42.5
    D. 170

25. How many grams of NaOH are required to prepare 500 mL of a 0.02 N solution?
    A. 0.4
    B. 0.8
    C. 4.0
    D. 8.0

26. What is the normality of a 30%$^{w/v}$ H_2SO_4 solution?
    A. 0.31
    B. 0.61
    C. 3.06
    D. 6.12

27. A serum chloride concentration is 369 mg/dL. What is the concentration in milliequivalents per liter?
    A. 5
    B. 10
    C. 36
    D. 104

28. A serum calcium level is 8.6 mg/dL. What is the concentration in millimoles per liter?
    A. 2.2
    B. 2.5
    C. 4.3
    D. 8.6

29. How many milliliters of concentrated HNO_3 (sp. gr. = 1.42 g/mL; assay = 70%) are required to prepare 2 L of 0.15 N HNO_3?
    A. 13.3
    B. 19.0
    C. 38.0
    D. 189.9

30. With the use of concentrated HCl (sp. gr. = 1.19 g/mL; assay = 37.5%), 3 L of 0.50 N HCl are prepared. A total of 16 mL of 0.20 N NaOH is required to titrate 7 mL of the HCl solution, indicating a lower normality of the acid solution than desired. How many milliliters of concentrated HCl must be added to the acid solution to attain an accurate 0.50 N HCl solution?
    A. 6.40
    B. 10.53
    C. 101.03
    D. 105.33

31. A serum chloride concentration is 369 mg/dL. What is the concentration in millimoles per liter?
    A. 5
    B. 10
    C. 36
    D. 104

32. How many milliliters of a stock solution of 20% $^{w/v}$ NaOH are required to prepare 800 mL of a 2.5%$^{v/v}$ solution?
    A. 20
    B. 50
    C. 100
    D. 125

33. How many milliliters of a 40% $^{w/v}$ NaOH solution are required to prepare 1.5 L of 2 N NaOH?
    A. 0.3
    B. 3.0
    C. 30
    D. 300

34. How many grams of anhydrous sodium sulfate (mol wt = 142) are required to prepare 750 mL of a 23%$^{w/v}$ solution?
    A. 2.3
    B. 106.5
    C. 172.5
    D. 230

35. With the use of concentrated HCl, 2 L of 0.20 N HCl are prepared. On titration, it is determined that the normality is actually 0.208. To correct this error, how many milliliters of deionized water must be added to the solution (10 mL used in titration process) to make an accurate 0.20 N HCl solution?
    A. 10.4
    B. 76.5
    C. 79.6
    D. 80.0

36. A serum calcium level is 8.6 mg/dL. What is the concentration in milliequivalents per liter?
    A. 2.2
    B. 2.5
    C. 4.3
    D. 8.6

37. How many milliliters of a 5 N HCl solution are required to prepare 4 L of 10%$^{w/v}$ HCl?
    A. 219
    B. 292
    C. 1460
    D. 2192

38. A serum potassium level is 19.5 mg/dL. What is the concentration in milliequivalents per liter?
    A. 0.5
    B. 5.0
    C. 10.0
    D. 19.5

39. An analysis for sodium is performed on an aliquot of a 24-hour urine specimen. A sodium value of 122.5 mmol/L is read from the instrument. What is the amount of sodium in the 24-hour urine specimen if 1540 mL of urine are collected?
    A. 79.5
    B. 188.6
    C. 1886.5
    D. 18,865

40. With the use of concentrated HNO_3, 4 L of 0.50 N HNO_3 are prepared. On titration, it is determined that the normality is actually 0.513. To correct this error, how many milliliters of deionized water must be added to the remaining 3.975 L of solution to make an accurate 0.50 N HNO_3 solution?
    A. 25.6
    B. 100.7
    C. 103.4
    D. 104.0

41. What is the pH of a 0.2 N acetic acid solution that is 1% ionized?
    A. 0.703
    B. 1.699
    C. 1.703
    D. 2.699

42. What is the ionic strength of a 0.2 M Na_2SO_4 solution?
    A. 0.4
    B. 0.5
    C. 0.6
    D. 1.2

43. A 1.0 mg/dL bilirubin standard (purity = 99.0%; mol wt = 584) is prepared by dissolving it in chloroform at 25°C. Under these conditions, the molar absorptivity at 453 nm is 60,700. What is the expected absorbance of this standard solution?
    A. 0.104
    B. 0.607
    C. 0.962
    D. 1.039

44. An enzyme assay is performed at 37°C, and absorbance readings are taken each minute for a total of 4 minutes. Given the following information, what is the enzyme activity in units per liter at 37°C?

Absorbance (*A*) Readings	Method Information
0.204 *A* at 1 minute	Reagent volume = 3.0 mL
0.406 *A* at 2 minutes	Sample volume = 200 µL
0.610 *A* at 3 minutes	Light path = 1 cm
0.813 *A* at 4 minutes	$\varepsilon_{340\ nm}$ of NADH = 6.22×10^3

    A. 490
    B. 522
    C. 525
    D. 1307

45. A patient weighs 175.5 pounds. What is the patient's weight expressed in kilograms?
    A. 8.0
    B. 38.6
    C. 79.8
    D. 87.8

46. Convert 30°C to degrees Fahrenheit.
    A. 4°F
    B. 10°F
    C. 49°F
    D. 86°F

47. A curie (Ci) is the quantity of radioactive material that exhibits
    A. 3.7×10^4 dps
    B. 3.7×10^7 dps
    C. 3.7×10^{10} dps
    D. 3.7×10^{10} dpm

48. Each radionuclide has a unique half-life associated with it. Assuming an initial activity of 100%, through how many half-life periods must a nuclide pass to bring its activity down to less than 1%?
    A. 3
    B. 7
    C. 10
    D. 100

49. ^{125}I has a half-life of 60 days. At the end of 180 days, what percent of activity would remain?
    A. 12.5
    B. 25.0
    C. 33.3
    D. 50.0

50. How many grams of NaOH are required to prepare 500 mL of a NaOH solution with a pH of 12?
    A. 0.02
    B. 0.2
    C. 0.4
    D. 2.0

51. What is the relative centrifugal force ($\times g$) of a centrifuge operating at 2500 rpm with a radius of 10 cm?
    A. 625
    B. 699
    C. 1250
    D. 6988

52. A 10-mL class A volumetric flask has an accuracy of ±0.2%. Express the ±0.2% tolerance in terms of milliliters.
    A. ±0.002
    B. ±0.01
    C. ±0.02
    D. ±0.04

53. A sample of deionized water is found to contain a lead concentration of 0.01 ppm. What is the equivalent concentration expressed as milligrams per deciliter?
    A. 0.01
    B. 0.001
    C. 0.0001
    D. 0.00001

54. Because of a malfunction, a spectrophotometer is able to show only the percent transmittance (*%T*) readings on its digital display. Convert 68.0%*T* to its corresponding absorbance.
    A. 0.109
    B. 0.168
    C. 0.320
    D. 0.495

55. Which of the following correctly states the conditions required in using a colorimetric method based on Beer's law?
    A. Incident radiation should be monochromatic.
    B. Absorption of light by the solvent must be insignificant in comparison with absorption by the solute (the analyte).
    C. The solution must be sufficiently dilute to provide a linear relation between absorption and concentration.
    D. All the above

56. When one of two variable quantities changes as a result of a change in the other, the result is frequently presented in the form of a graph. Which of the following descriptions of a graph is correct?
    A. The *x*-axis is usually used to plot the independent variable.
    B. Different scales may be used for each of the two axes.
    C. Semilog paper uses a logarithmic scale for one axis and a linear, or Cartesian, scale for the other axis.
    D. All the above

57. Primary standards used for analytical work should have what property?
    A. The substance must be available in a form not less than 99.95% pure.
    B. It should not be hygroscopic.
    C. It must be a stable substance that can be dried, preferably at 104–110°C.
    D. All the above

58. Which of the following weighs the least?
    A. 0.1 ng
    B. 0.01 g
    C. 1.0 mg
    D. 1000 pg

59. The OH^- concentration of a solution is 1×10^{-6}. What is the pH of this solution?
    A. 0.6
    B. 6
    C. 8
    D. 14

60. What amount of NaCl (mol wt = 58.5) is needed to obtain 50 mg Cl?
    A. 19.6 mg
    B. 30.3 mg
    C. 50.0 mg
    D. 82.4 mg

PERIODIC TABLE OF THE ELEMENTS

	Main groups 1A[a] 1		Transition metals										Main groups 3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1	1 **H** 1.00794	2A 2																2 **He** 4.002602
2	3 **Li** 6.941	4 **Be** 9.012182											5 **B** 10.811	6 **C** 12.0107	7 **N** 14.00674	8 **O** 15.9994	9 **F** 18.998403	10 **Ne** 20.1797
3	11 **Na** 22.989770	12 **Mg** 24.3050	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	13 **Al** 26.981538	14 **Si** 28.0855	15 **P** 30.973762	16 **S** 32.066	17 **Cl** 35.4527	18 **Ar** 39.948
4	19 **K** 39.0983	20 **Ca** 40.078	21 **Sc** 44.95591	22 **Ti** 47.867	23 **V** 50.9415	24 **Cr** 51.9961	25 **Mn** 54.938049	26 **Fe** 55.845	27 **Co** 58.933200	28 **Ni** 58.6934	29 **Cu** 63.546	30 **Zn** 65.39	31 **Ga** 69.723	32 **Ge** 72.61	33 **As** 74.92160	34 **Se** 78.96	35 **Br** 79.904	36 **Kr** 83.80
5	37 **Rb** 85.4678	38 **Sr** 87.62	39 **Y** 88.90585	40 **Zr** 91.224	41 **Nb** 92.90638	42 **Mo** 95.94	43 **Tc** [98]	44 **Ru** 101.07	45 **Rh** 102.90550	46 **Pd** 106.42	47 **Ag** 107.8682	48 **Cd** 112.411	49 **In** 114.818	50 **Sn** 118.710	51 **Sb** 121.760	52 **Te** 127.60	53 **I** 126.90447	54 **Xe** 131.29
6	55 **Cs** 132.90545	56 **Ba** 137.327	57 ***La** 138.9055	72 **Hf** 178.49	73 **Ta** 180.9479	74 **W** 183.84	75 **Re** 186.207	76 **Os** 190.23	77 **Ir** 192.217	78 **Pt** 195.078	79 **Au** 196.96655	80 **Hg** 200.59	81 **Tl** 204.3833	82 **Pb** 207.2	83 **Bi** 208.98038	84 **Po** [210]	85 **At** [210]	86 **Rn** [222]
7	87 **Fr** [223]	88 **Ra** [226]	89 **†Ac** [227]	104 **Rf** [261]	105 **Db** [262]	106 **Sg** [266]	107 **Bh** [264]	108 **Hs** [265]	109 **Mt** [268]	110 [269]	111 [272]	112 [277]	113	114 [285]	115	116 [289]	117	118 [293]

*Lanthanide series	58 **Ce** 140.116	59 **Pr** 140.90765	60 **Nd** 144.24	61 **Pm** [145]	62 **Sm** 150.36	63 **Eu** 151.964	64 **Gd** 157.25	65 **Tb** 158.92534	66 **Dy** 162.50	67 **Ho** 164.93032	68 **Er** 167.26	69 **Tm** 168.93421	70 **Yb** 173.04	71 **Lu** 174.967
†Actinide series	90 **Th** 232.0381	91 **Pa** 231.03588	92 **U** 238.0289	93 **Np** [237]	94 **Pu** [244]	95 **Am** [243]	96 **Cm** [247]	97 **Bk** [247]	98 **Cf** [251]	99 **Es** [252]	100 **Fm** [257]	101 **Md** [258]	102 **No** [259]	103 **Lr** [262]

[a] The labels on top (1A, 2A, etc.) are common American usage. The labels below these (1, 2, etc.) are those recommended by the International Union of Pure and Applied Chemistry.

The names and symbols for elements 110 and above have not yet been decided.

Atomic weights in brackets are the masses of the longest-lived or most important isotope of radioactive elements.

Further information is available at http://www.shef.ac.uk/chemistry/web-elements/

The production of elements 116 and 118 was reported in May 1999 by scientists at Lawrence Berkeley National Laboratory.

answers & rationales

1.

B. A simple and universally applicable method for solving laboratory calculation problems is to read the problem with three questions in mind:

1. What am I given?
2. What do I want?
3. What is the relation between no. 1 and no. 2?

Given: the concentration of a solution in terms of grams per 500 mL

Want: the concentration in terms of molarity (M)

Relation: molarity = moles per liter

Calculation: A molar solution is one that contains 1 gram molecular weight (usually called 1 mol) of solute per liter of solution. The gram molecular weight is the sum of the atomic weights. A table of atomic weights is used to determine that the gram molecular weight of KCl = 39 + 35.5 = 74.5 g. Express the concentration of the given solution (18.7 g/500 mL) in terms of grams per liter for easy comparison with molarity (moles per liter).

$$18.7 \text{ g}/500 \text{ mL} = 18.7 \text{ g}/0.5 \text{ L} = 37.4 \text{ g/L}$$

$$\text{moles} = \frac{\text{grams}}{\text{gram molecular weight}}$$

$$= \frac{37.4}{74.5} = 0.50$$

$$0.50 \text{ mol/L} = 0.50 \text{ M}$$

A solution that contains 18.7 g KCl/500 mL is 0.50 M KCl.

2.

A. To convert 10 mg of calcium per deciliter to millimoles per liter, proceed as follows. Calculate the milligrams of calcium per liter.

$$10 \text{ mg Ca/dL} = 10 \text{ mg/dL} \times 10 \text{ dL/L}$$

$$= 100 \text{ mg Ca/L}$$

Calculate the weight of 1 mmol of calcium.

$$\text{Mol wt of Ca} = 40 \text{ g/mol} = 40{,}000 \text{ mg/mol}$$

Thus, 1 mmol Ca = 40 mg Ca.

Then convert 100 mg Ca to mmol Ca:

$$\frac{100 \text{ mg/L}}{40 \text{ mg/mmol}} = 2.5 \text{ mmol Ca/L}$$

Therefore, a concentration of 10 mg Ca/dL = 2.5 mmol Ca/L.

3.

B. To calculate how much NaCl is needed to prepare 100 mL of a standard solution of concentration 135 mEq/L of Na, first consider how many equivalents of Na are present in a mole of NaCl. Find the molecular weight of NaCl. Note that the concentration is expressed in milliequivalents *per liter* (mEq/L), but the desired volume is 100 mL.

Calculation: Because sodium has a valence of 1, one mole of NaCl contains 1 Eq of Na. Molecular weight of NaCl = 23 + 35.5 = 58.5 g. It follows that if 1 Eq of Na is present in 58.5 g NaCl, then 1 mEq of Na is present in 0.0585 g NaCl. There are 135 mEq of Na present in 135 × 0.0585 = 7.90 g NaCl. Therefore, a concentration of 135 mEq/L = 7.90 g NaCl/L.

Let X equal the weight of NaCl in 100 mL of the desired solution. Then

$$\frac{X}{100\text{ mL}} = \frac{7.90\text{ g}}{1000\text{ mL}}$$

$$X = \frac{7.90\text{ g} \times 100\text{ mL}}{1000\text{ mL}} = 0.790\text{ g NaCl}$$

4.

C. Problems requiring the conversion of one concentration to another are conveniently solved by applying the formula

$$C_1 \times V_1 = C_2 \times V_2$$

where C_1 and V_1 are the concentration and volume of the starting solution; C_2 and V_2 are the concentration and volume of the final solution. Thus to prepare 5 L of 70% alcohol from 95% alcohol, let C_1 and V_1 refer to the 95% alcohol, and let C_2 and V_2 refer to the 70% alcohol.

$$95\% \times V_1 = 70\% \times 5\text{ L}$$

$$V_1 = \frac{70 \times 5\text{ L}}{95} = 3.7\text{ L}$$

5.

B. To solve questions involving standardization of an acid or a base by titration, bear in mind that at the endpoint of the titration, the number of equivalents of acid used will be the same as the number of equivalents of base used. Therefore, the following formula applies:

$$C_1 \times V_1 = C_2 \times V_2$$

where C_1 and V_1 are the concentration and volume of HCl needed to neutralize the NaOH solution. The equation is set up below.

$$(0.100\text{ N HCl})\ (11.25\text{ mL}) = C_2\ (10.0\text{ mL})$$

$$C_2 = \frac{11.25 \times 0.100\text{ N}}{10.0} = 0.112\text{ N}$$

6.

C. To calculate how many milliliters of concentrated H_2SO_4 (sp. gr. = 1.84 g/mL, assay = 97%) are required to prepare 10 L of 0.1 N H_2SO_4, first calculate the concentration of the concentrated sulfuric acid in terms of normality. Because the specific gravity is the weight of 1 mL, it follows that 1 mL of concentrated sulfuric acid weighs 1.84 grams. The assay is 97%; that is, 97% of the 1.84 g is H_2SO_4. Therefore, 1 mL of concentrated sulfuric acid contains

$$\frac{97}{100} \times 1.84 \text{ g } H_2SO_4 = 1.78 \text{ g } H_2SO_4$$

and 1 L of concentrated sulfuric acid contains 1780 g H_2SO_4. To convert this concentration (1780 g/L) to normality, recall that a normal solution contains 1 gram equivalent weight per liter. A gram equivalent weight (g Eq wt) is the weight that will combine with or replace 1 g of hydrogen. Because H_2SO_4 has two replaceable hydrogens, the

$$\text{g Eq wt of } H_2SO_4 = \frac{\text{mol wt}}{2} = \frac{98}{2} = 49 \text{ g}$$

Therefore, the concentrated sulfuric acid, which is 1780 g/L, contains

$$\frac{1780 \text{ g/L}}{49 \text{ Eq/L}} = 36.3 \text{ Eq/L}$$

Hence its concentration is 36.3 N.

Now the question becomes: How many milliliters of 36.3 N H_2SO_4 are required to prepare 10 L of 0.1 N H_2SO_4? Using the relation $N_1 \times V_1 = N_2 \times V_2$, let N_1 and V_1 be the concentration and volume of the concentrated H_2SO_4 and let N_2 and V_2 be the concentration and volume of the 0.1 N H_2SO_4.

$$N_1 \times V_1 = N_2 \times V_2$$
$$36.3 \text{ N} \times V_1 = 0.1 \text{ N} \times 10 \text{ L}$$
$$V_1 = \frac{0.1 \times 10 \text{ L}}{36.3} = 0.0275 \text{ L} = 27.5 \text{ mL}$$

Therefore, 27.5 mL of concentrated H_2SO_4 are required to prepare 10 L of 0.1 N H_2SO_4.

7.

D. The problem requires the preparation of 500 mL of 10%$^{w/v}$ $CaCl_2$ with the use of the hydrate $CaCl_2 \cdot 2H_2O$. First calculate how much $CaCl_2$ is needed: 500 mL of 10% $^{w/v}$ $CaCl_2$ contains

$$500 \text{ mL} \times \frac{10 \text{ g}}{100 \text{ mL}} = 50 \text{ g } CaCl_2$$

Then calculate how much $CaCl_2$ is present in $CaCl_2 \cdot 2H_2O$ by comparing molecular weights:

$$\text{mol wt of } CaCl_2 = 40 + 71 = 111 \text{ g}$$
$$\text{mol wt of } CaCl_2 \cdot 2H_2O = 40 + 71 + 4 + 32 = 147 \text{ g}$$

There are 111 g of $CaCl_2$ present in 147 g $CaCl_2 \cdot 2H_2O$; 50 g of $CaCl_2$ are present in

$$\frac{147}{111} \times 50 = 66.2 \text{ g } CaCl_2 \cdot 2H_2O$$

As an alternate method, let X = grams $CaCl_2 \cdot 2H_2O$ needed.

$$\frac{111}{147} = \frac{50}{X}$$
$$X = \frac{147 \times 50}{111} = 66.2 \text{ g}$$

Therefore, a total of 66.2 g $CaCl_2 \cdot 2H_2O$ are needed to prepare 500 mL of 10% $^{w/v}$ $CaCl_2$.

8.

B. *Given:* weights of NaCl and glucose dissolved in 1 kg of water
Want: osmolality of solution
Relation: osmolality = moles per kilogram of solvent times the number of particles into which solute molecules dissociate
Calculation: Express the amount of each solute in terms of moles (grams per molecular weight)

$$5.85 \text{ g NaCl} = \frac{5.85 \text{ g}}{58.5 \text{ g/mol}} = 0.1 \text{ mol NaCl}$$

$$18 \text{ g glucose} = \frac{18 \text{ g}}{180 \text{ g/mol}} = 0.1 \text{ mol glucose}$$

Consider the dissociation of each solute.

$$0.1 \text{ mol NaCl} \rightarrow 0.1 \text{ mol } Na^+ + 0.1 \text{ mol } Cl^-$$

Glucose does not dissociate appreciably.

Number of Osmols =

$0.1 \times 2 = 0.2$ osmol NaCl
$+ \underline{0.1}$ osmol glucose
Total 0.3 osmol in 1 kg H_2O
$= 0.3$ osmolal

9.

B. *Given:* concentration in grams per deciliter: 0.85% NaCl = 0.85 g/dL
Want: concentration in osmolarity
Relation: osmolarity = moles per liter times the number of particles into which the solute dissociates
Calculation: molecular weight of NaCl = 23 + 35.5 = 58.5 g/mol
In dilute solution, NaCl is assumed to be fully dissociated; therefore, each molecule of NaCl will produce two particles, a sodium ion and a chloride ion.

Osmolarity =
mol/L × No. of particles/dissociate molecule

$$= \frac{\text{g}}{\text{mol wt}}/\text{L} \times \text{No. of particles/dissociate molecule}$$

Convert to grams per liter:

$$0.85 \text{ g/dL} \times 10 \text{ dL/L} = 8.5 \text{ g/L}$$

$$\text{Osmolarity} = \frac{8.5 \text{ g/L}}{58.5 \text{ g/mol}} \times 2 = 0.29 \text{ osmol/L}$$

10.

D. To convert 70°F to degrees Celsius, compare the size of Fahrenheit and Celsius degrees. There are 212 − 32 = 180 Fahrenheit degrees between the boiling point and the freezing point of water, respectively. There are 100 − 0 = 100 Celsius degrees between the boiling point and the freezing point of water, respectively. Therefore, 180 Fahrenheit degrees equal 100 Celsius degrees. From the relationship

$$\frac{°C - 0}{°F - 32} = \frac{100}{180} = \frac{5}{9}$$

two formulas may be derived:

$$°F = \left(°C \times \frac{9}{5}\right) + 32$$

and

$$°C = (°F - 32) \times \frac{5}{9}$$

Thus using the relationship

$$1°F = \frac{100}{180} = \frac{5°C}{9}$$

it follows that

$$\begin{aligned} 70°F &= 70 - 32 \\ &= 38°F \text{ degrees above freezing} \\ &= 38 \times \frac{5°C}{9} \text{ degrees above freezing} \\ &= 21°C \end{aligned}$$

11.

B. When more than one dilution is carried out on a sample, the final concentration is the initial concentration multiplied by each dilution expressed as a fraction. If a 5 N solution is diluted 1:4 and then further 4:15, the final concentration is:

$$5\,N \times \frac{1}{4} \times \frac{4}{15} = 0.33\,N$$

The same principle applies in testing a specimen that is too concentrated to fall within the range of the test procedure. The specimen is diluted, the test repeated, and the result multiplied by the reciprocal of the dilution. Thus, if the specimen had to be diluted 1:10 (1/10) to fall within the range of the test procedure, the result would be multiplied by 10 (10/1) to give the correct value.

12.

D. To find the dilution of serum in a mixture, calculate the total volume. The total volume equals 0.1 mL serum + 5 mL reagents + 4.9 mL water = 10 mL. Therefore, the dilution of serum is 0.1 mL to 10 mL, or 0.1:10. Because dilutions are usually expressed as 1 to some number, multiply both the serum volume (0.1) and the total volume (10) by a common factor of 10. Thus, the 0.1:10 serum dilution may be expressed as 1:100.

13.

C. Beer's law states that $A = abc$; where A = absorbance, a = absorptivity, b = light path in cm, and c = concentration of the absorbing compound. If a and b are constant, then A is directly proportional to c. Therefore, if c is doubled, then A also is doubled:

$$0.085\,A \times 2 = 0.170\,A$$

14.

D. In any given photometric procedure, the length of the light path (= width of the cuvet) and the absorptivity of the analyte are constant. Hence, if the procedure follows Beer's law, absorbance (A) is proportional to the concentration (C). This can be expressed as a ratio:

$$C_{unknown}/C_{standard} = A_{unknown}/A_{standard}$$

$$C_u = \frac{A_u}{A_s} \times C_s$$

$$\frac{0.62}{0.50} \times 15 \text{ mg/dL} = 18.6 \text{ mg/dL}$$

15.

C. Problems requiring the conversion of one concentration to another can use the following formula:

$$C_1 \times V_1 = C_2 \times V_2$$

where C_1 and V_1 are the concentration and volume of the stock solution. C_2 and V_2 are the concentration and volume of the final solution. The units of concentration must be the same; therefore, first convert 20 mg/dL to 0.2 mg/mL before solving the equation.

$$(10 \text{ mg/mL})\ V_1 = (0.2 \text{ mg/mL})\ (200 \text{ mL})$$

$$V_1 = \frac{0.2 \times 200 \text{ mL}}{10} = 4 \text{ mL}$$

16.

D. When starting with a percent solution, one may prepare a percent solution of a lesser concentration by using the following formula:

$$V_1 \times \%_1 = V_2 \times \%_2$$

Substitute the given information of the problem in the preceding formula:

$$1000 \text{ mL} \times 5\% = V_2 \times 50\%$$

$$\frac{1000 \text{ mL} \times 5}{50} = V_2$$

$$100 \text{ mL} = V_2$$

Therefore, 100 mL of a 50%$^{v/v}$ acetic acid solution are required to prepare 1 L of 5% $^{v/v}$ acetic acid. Add 100 mL of 50% acetic acid to the solvent, diluting with solvent to a final volume of 1 L.

17.

C. Percent solution refers to a specific number of parts per hundred. The term "parts" refers to the weight of a solute in grams or the volume of liquid in milliliters. The term "hundred" refers to the final volume of 100 mL of solution or 100 g of solution. Thus percent solutions may be expressed as weight per volume (w/v), weight per weight (w/w), or volume per volume (v/v). Preparation of a weight per volume solution may be done as follows: 3%$^{w/v}$ sulfosalicylic acid = 3 g/dL (remember that 100 mL is equivalent to 1 dL). To find the number of grams needed to prepare 1 L, multiply by the required volume in deciliters as follows: 3 g/dL × 10 dL/L × 1 L = 30 g. To prepare 1 L of a 3% $^{w/v}$ sulfosalicylic acid solution, dissolve 30 g of sulfosalicylic acid in deionized water (solvent) and dilute to a final volume of 1 L using a volumetric flask.

18.

C. The urine sample contained 1.2% protein, which is equivalent to 1.2 g of protein per deciliter. Because the total urine volume is given in milliliters, it is necessary to express the volume in deciliters so that the units of measurement correspond. This may be done by dividing the 24-hour volume in milliliters by 100, because there are 100 mL in each deciliter. The amount of protein excreted in the 24-hour urine specimen may now be calculated.

$$\frac{\text{Protein conc. in g/dL} \times \text{urine volume in mL/24 hr}}{100 \text{ mL/dL}} = \text{g/24 hr}$$

$$\frac{1.2 \text{ g/dL} \times 2155 \text{ mL/24 hr}}{100 \text{ mL/dL}} = 25.9 \text{ g/24 hr}$$

19.

D. A 1 molar solution contains 1 mole or 1 g molecular weight of a solute in 1 L of solution. One gram molecular weight of H_2SO_4 consists of 98 g. Thus 1 L of a 1 M solution contains 98 g of H_2SO_4, and a 2 M solution contains twice this amount or 196 g of H_2SO_4 per liter.

$$1 \text{ M } H_2SO_4 = 98 \text{ g/L}$$
$$2 \text{ M } H_2SO_4 = 2 \times 98 \text{ g/L} = 196 \text{ g/L}$$

Thus to find the number of grams needed to prepare only 750 mL of a 2 M H_2SO_4 solution, multiply by the required volume expressed in liters as follows:

$$196 \text{ g/L} \times \frac{750 \text{ mL}}{1000 \text{ mL/L}} = 147 \text{ g}$$

Therefore, 147 g of H_2SO_4 are needed to prepare 750 mL of a 2 M H_2SO_4 solution.

20.

D. A one normal (1 N) solution contains 1 g equivalent weight of a solute in 1 L of solution. For a base, the equivalent weight is defined as that weight which combines with 1.008 g of replaceable hydrogen. One gram equivalent weight of NaOH consists of 40 g. Thus 1 L of a 1 N solution contains 40 g of NaOH, and a 2 N solution contains twice this amount or 80 g of NaOH per liter.

$$1 \text{ N NaOH} = 40 \text{ g/L}$$
$$2 \text{ N NaOH} = 2 \times 40 \text{ g/L} = 80 \text{ g/L}$$

Thus to find the number of grams needed to prepare 4 L of a 2 N NaOH solution, multiply by the required volume as follows: 80 g/L × 4 L = 320 g. Therefore, 320 g of NaOH are needed to prepare 4 L of a 2 N NaOH solution. Prepare this solution by dissolving the 320 g of solute in solvent and bring to a final volume of 4 L.

21.

D. "Percent solution" refers to a specific number of *parts per hundred.* For a volume per volume solution, measure the volume of liquid solute required in milliliters and add solvent to a final volume of 100 mL of solution. Preparation of a $5\%^{v/v}$ CH_3COOH solution may be done as follows: 5% glacial acetic acid = 5 mL/dL (remember, 100 mL is equivalent to 1 dL). Thus to find the number of milliliters needed to prepare 2 L of 5% CH_3COOH, multiply by the required volume in deciliters as follows: 5 mL/dL × 10 dL/L × 2 L = 100 mL. Thus to prepare 2 L of a 5% acetic acid solution, add 100 mL of glacial acetic acid to deionized water (remember—always add acid to water) and dilute using a volumetric flask to a final volume of 2 L.

22.

D. A 1 normal (1 N) solution contains 1 g equivalent weight of a solute in 1 L of solution. For an acid, the equivalent weight is defined as that weight which is equivalent to 1.008 g of replaceable hydrogen. The gram equivalent weight may be determined by dividing the gram molecular weight by the positive valence. One equivalent weight of H_2SO_4 equals 49 g, because the molecular weight of 98 g divided by the positive valence 2 is 49 g. Thus 1 L of a 1 N solution contains 49 g of H_2SO_4, and a 5 N solution contains five times this amount or 245 g of H_2SO_4 per liter.

$$1\ N\ H_2SO_4 = 49\ g/L$$
$$5\ N\ H_2SO_4 = 5 \times 49\ g/L = 245\ g/L$$

Thus to find the number of grams needed to prepare 6 L of a 5 N H_2SO_4 solution, multiply by the required volume as follows: 245 g/L × 6 L = 1470 g. Therefore, 1470 g of H_2SO_4 are needed to prepare 6 L of a 5 N H_2SO_4 solution.

23.

D. A 1 molar solution contains 1 mol or 1 gram molecular weight of a solute in 1 L of solution. One gram molecular weight of NaOH consists of 40 g. Thus 1 L of a 1 M solution contains 40 g of NaOH, and a 4 M solution contains four times this amount or 160 g of NaOH per liter.

$$1\ M\ NaOH = 40\ g/L$$
$$4\ M\ NaOH = 4 \times 40\ g/L = 160\ g/L$$

Thus to find the number of grams needed to prepare 2500 mL of a 4 M NaOH solution, multiply by the required volume expressed in liters as follows:

$$160\ g/L \times \frac{2500\ mL}{1000\ mL/L} = 400\ g$$

Therefore, 400 g of NaOH are needed to prepare 2500 mL (2.5 L) of a 4 M NaOH solution.

24.

C. A weight per volume percent solution contains a specific number of grams (equivalent to the percent indicated) of solute per 100 mL (equivalent to 1 dL) of solution. An $0.85\%^{w/v}$ isotonic saline solution contains 0.85 g of NaCl per 100 mL of solution. To find the number of grams needed to prepare 5 L of solution, multiply by the required volume in deciliters as follows:

$0.85\ g/dL \times 5\ L \times 10\ dL/L = 42.5\ g\ NaCl.$

Using a volumetric flask, prepare 5 L of an $0.85\%^{w/v}$ NaCl solution by dissolving 42.5 g of NaCl in deionized water and diluting to a final volume of 5 L.

25.

A. A 1 normal solution contains 1 g equivalent weight of a solute in 1 L of solution. One equivalent weight of NaOH contains 40 g; thus, a 0.02 N solution contains one-fiftieth of this amount, or 0.8 g of NaOH per liter.

$$1\ N\ NaOH = 40\ g/L$$
$$0.02\ N\ NaOH = 0.02 \times 40\ g/L = 0.8\ g/L$$

Thus to find the number of grams needed to prepare 500 mL (0.5 L) of a 0.02 N NaOH solution, multiply by the required volume as follows: 0.8 g/L × 0.5 L = 0.4 g or 400 mg. Therefore, 0.4 g (or 400 mg) of NaOH are needed to prepare 500 mL of a 0.02 N NaOH solution.

26.

D. When starting with a 30% $^{w/v}$ H_2SO_4 solution, the normality of the solution may be calculated by employing the following steps:

1. Determine the molecular weight of H_2SO_4: mol wt = 98 g.
2. Find the g/Eq in 1 L of 1 N H_2SO_4 solution: 98 g/L ÷ 2 Eq/L = 49 g/Eq = 49 g/L.
3. Find the corresponding percent of a 1 N H_2SO_4 solution: 49 g/L ÷ 10 dL/L = 4.9 g/dL = 4.9%$^{w/v}$ H_2SO_4.
4. Find the normality of the 30% H_2SO_4 solution. Divide the stated percent by the percent of the 1 N H_2SO_4 solution: 30% ÷ 4.9% = 6.12 N H_2SO_4.

27.

D. To convert 369 mg/dL of chloride to milliequivalents per liter, use the following formula:

$$\frac{\text{mg/dL} \times 10 \times \text{valence}}{\text{atomic mass}} = \text{mEq/L}$$

For the problem presented, let 10 represent the number of deciliters per liter; 35.5 is the atomic mass of chloride, and 1 is the valence of chloride:

$$\frac{369 \text{ mg/dL} \times 10 \times 1}{35.5} = 104 \text{ mEq/L}$$

28.

A. To convert 8.6 mg/dL of calcium to millimoles per liter, the following formula may be used:

$$\frac{\text{mg/L}}{\text{molecular mass}} = \text{mmol/L}$$

For the problem presented:

$$\frac{8.6 \text{ mg/dL} \times 10 \text{ dL/L}}{40} = 2.15 \text{ mmol/L}$$

The calculated value of 2.15 mmol/L rounded to the nearest tenth gives the answer 2.2 mmol/L.

29.

B. When starting with a concentrated HNO_3 solution, a 0.15 normal solution may be prepared by employing the following steps:

A. Calculate the normality of the concentrated HNO_3 solution.
   1. Multiply the specific gravity (expressed as grams per milliliter) by the assay (expressed as percent by weight) to find the number of grams per milliliter:

      $$1.42 \text{ g/mL} \times 0.70 = 0.994 \text{ g/mL}$$

   2. To find the number of grams per liter, multiply by 1000 mL/L:

      $$0.994 \text{ g/mL} \times 1000 \text{ mL/L} = 994 \text{ g/L}$$

   3. To find the normality (equivalents per liter) of the concentrated HNO_3, divide the grams per liter by the equivalent weight of HNO_3:

      $$1 \text{ N } HNO_3 = 63 \text{ g/Eq}$$
      $$994 \text{ g/L} \div 63 \text{ g/Eq} = 15.8 \text{ Eq/L}$$
      $$= 15.8 \text{ N } HNO_3$$

B. Calculate the number of milliliters of concentrated HNO_3 needed to prepare 2 L of 0.15 N HNO_3 using the formula $V_1 \times N_1 = V_2 \times N_2$.

$$2000 \text{ mL} \times 0.15 \text{ N} = V_2 \times 15.8 \text{ N}$$
$$\frac{2000 \text{ mL} \times 0.15}{15.8} = V_2$$
$$19.0 \text{ mL} = V_2$$

Thus 2 L of 0.15 N HNO_3 are prepared by diluting 19.0 mL of concentrated HNO_3 to a final volume of 2000 mL.

30.

B. Because titration reveals that the normality of the HCl solution is lower than the desired normality of 0.50 N, it is apparent that concentrated HCl must be added. Use the following formula to determine the actual normality of the acid solution:

$$V_1 \times N_1 = V_2 \times N_2$$

where V_1 = 7 mL of the HCl solution, N_1 = actual normality of the HCl solution, V_2 = 16 mL of the 0.20 N NaOH solution, and N_2 = 0.20 N NaOH solution.

1. Find the actual normality of the acid solution:

$$7 \text{ mL} \times N_1 = 16 \text{ mL} \times 0.20 \text{ N}$$

$$N_1 = \frac{16 \text{ mL} \times 0.20}{7 \text{ mL}}$$

$$N_1 = 0.457 \text{ actual N of HCl solution}$$

2. Find the number of milliequivalents per milliliter lacking in the initial HCl solution: 0.500 mEq/mL – 0.457 mEq/mL = 0.043 mEq/mL lacking.
3. Find the number of milliliters of the initial HCl solution remaining: 3000 mL – 7 mL = 2993 mL remaining.
4. Find the number of milliequivalents of HCl that must be added to the remaining solution: 2993 mL × 0.043 mEq/mL = 128.7 mEq required.
5. Find the number of grams per milliliter of HCl in concentrated HCl: 1.19 g/mL (sp. gr.) × 0.375 (assay) = 0.446 g HCl/mL of concentrated acid. It follows that there are 446 g HCl/L.
6. Find the normality of concentrated HCl (Eq wt = 36.5 g): 446 g/L ÷ 36.5 g/Eq = 12.22 Eq/L, or 12.22 N, or 12.22 mEq/mL.
7. Find the number of milliliters of concentrated HCl that must be added:

$$\frac{128.7 \text{ (mEq required)}}{12.22 \text{ (mEq/mL in conc. HCl)}} = 10.53 \text{ mL}$$

Thus, 10.53 mL of concentrated HCl must be added to the remaining volume of 2993 mL for a total volume of 3003.53 mL to prepare an accurate 0.50 N HCl solution.

31.

D. When the concentrations of the four serum electrolytes—sodium, potassium, chloride, and carbon dioxide—are expressed, the preferred terminology is an expression of the unit designation as millimoles per liter (mmol/L). To convert from milligrams per deciliter to millimoles per liter, the following formula may be used:

$$\frac{\text{mg/L}}{\text{molecular mass}} = \text{mmol/L}$$

$$\frac{369 \text{ mg/dL} \times 10 \text{ dL/L}}{35.5} = 104 \text{ mmol/L}$$

32.

C. When a dilute solution is prepared from a concentrated solution, the following equation may be used:

$$V_1 \times C_1(\%) = V_2 \times C_2(\%)$$

For the stated problem let V_1 = milliliters of the 20% NaOH solution, C_1 = the 20% NaOH solution, V_2 = 800 mL of the desired 2.5% solution, and C_2 = the 2.5% NaOH solution. Using this information in the above equation:

$$V_1 \times 20\% = 800 \text{ mL} \times 2.5\%$$

$$V_1 = \frac{800 \text{ mL} \times 2.5\%}{20\%}$$

$$V_1 = 100 \text{ mL}$$

Thus, 100 mL of the 20%$^{w/v}$ NaOH solution are required to prepare a total volume of 800 mL of 2.5%$^{v/v}$ NaOH. This may be done by diluting 100 mL of 20% NaOH to a final volume of 800 mL.

33.

D. When one starts with a percent solution, a normal solution may be prepared by employing the following steps.

A. Calculate the normality of the 40%$^{w/v}$ NaOH solution.

1. Find the number of grams in 1 equivalent weight of a 1 N NaOH solution:

$$1 \text{ N NaOH} = 40 \text{ g/Eq} = 40 \text{ g/L}$$

2. Find the grams per liter in the 40%$^{w/v}$ NaOH solution:

$$40 \text{ g}/100 \text{ mL} \times 1000 \text{ mL/L} = 400 \text{ g/L}$$

3. Find the normality of the 40% NaOH solution by dividing the number of grams per liter in the 40% NaOH solution by the equivalent weight of NaOH:

$$400 \text{ g/L} \div 40 \text{ g/Eq} = 10 \text{ Eq/L}$$

$$= 10 \text{ N NaOH}$$

B. Calculate the number of milliliters of 40%$^{w/v}$ NaOH needed to prepare 1.5 L of 2 N NaOH using the formula:

$$V_1 \times N_1 = V_2 \times N_2$$

$$1500 \text{ mL} \times 2 \text{ N} = V_2 \times 10 \text{ N}$$

$$\frac{1500 \text{ mL} \times 2}{10} = V_2$$

$$300 \text{ mL} = V_2$$

Therefore, 300 mL of 40%$^{w/v}$ NaOH are required to prepare 1.5 L of 2 N NaOH. Add 300 mL of 40% NaOH to the solvent, diluting with solvent to a final volume of 1.5 L.

34.

C. A weight per volume percent solution contains a specific number of grams of solute per 100 mL of solution. Preparation of a 23%$^{w/v}$ Na_2SO_4 solution may be done as follows: 23%$^{w/v}$ Na_2SO_4 = 23 g/dL. Thus to find the number of grams needed to prepare 750 mL, multiply by the required volume in deciliters, as follows:

$$23 \text{ g/dL} \times 10 \text{ dL/L} \times \frac{750 \text{ mL}}{1000 \text{ mL/L}} = 172.5 \text{ g}$$

To prepare 750 mL of a 23%$^{w/v}$ Na_2SO_4 solution, dissolve 172.5 g of Na_2SO_4 in solvent and dilute to a final volume of 750 mL.

35.

C. Because the normality of the HCl solution is greater than the desired normality, this problem can be considered similar to that of preparing a dilute solution from a concentrated solution. Thus the following equation may be used to determine the new total volume required, and indirectly the amount of deionized water that must be added to the remaining volume, to achieve the desired normality:

$$V_1 \times N_1 = V_2 \times N_2$$

where V_1 = mL of the desired 0.20 N HCl solution, N_1 = 0.20 N HCl solution, V_2 = 1990 mL remaining of the 0.208 N HCl solution, and N_2 = 0.208 N HCl solution. Using this information in the above equation:

$$V_1 \times 0.20 \text{ N} = 1990 \text{ mL} \times 0.208 \text{ N}$$

$$V_1 = \frac{1990 \text{ mL} \times 0.208 \text{ N}}{0.20 \text{ N}}$$

$$V_1 = 2069.6 \text{ mL}$$

Thus a new total volume of 2069.6 mL is required to make an accurate 0.20 N HCl solution. To determine the amount of deionized water that must be added to the remaining 1990 mL volume, find the difference between the new total volume and the remaining volume: 2069.6 mL – 1990 mL = 79.6 mL of deionized water that must be added to the remaining volume of solution.

36.

C. To convert milligrams per deciliter to milliequivalents per liter, use the following formula:

$$\frac{\text{mg/dL} \times 10 \times \text{valence}}{\text{atomic mass}} = \text{mEq/L}$$

For the problem presented, let 10 represent the number of deciliters per liter; 40 is the atomic mass of calcium, and 2 is the valence of calcium.

$$\frac{8.6 \text{ mg/dL} \times 10 \times 2}{40} = 4.3 \text{ mEq/L}$$

37.

D. When one starts with a 5 N HCl solution, 4 L of a 10%$^{w/v}$ HCl solution may be prepared by employing the following steps:

A. Calculate the normality of the 10%$^{w/v}$ HCl solution.
   1. Find the number of grams in 1 equivalent weight of a 1 N HCl solution:
      1 N HCl = 36.5 g/Eq.
   2. Find the number of grams per liter in the 10%$^{w/v}$ HCl solution:
      10%$^{w/v}$ HCl = 10 g/dL = 100 g/L.
   3. Find the normality of the 10%$^{w/v}$ HCl solution. Divide the number of grams per liter in the 10% solution by the equivalent weight of HCl:
      100 g/L ÷ 36.5 g/Eq = 2.74 N HCl.

B. Calculate the number of milliliters of 5 N HCl needed to prepare 4 L of 10%$^{w/v}$ HCl using the following formula:

$$V_1 \times N_1 = V_2 \times N_2$$
$$4000 \text{ mL} \times 2.74 \text{ N} = V_2 \times 5 \text{ N}$$
$$\frac{4000 \text{ mL} \times 2.74 \text{ N}}{5 \text{ N}} = V_2$$
$$2192 \text{ mL} = V_2$$

Therefore, 2192 mL of 5 N HCl are required to prepare 4 L of 10% $^{w/v}$ HCl. Add 2192 mL of 5 N HCl to the solvent, diluting with solvent to a final volume of 4 L.

38.

B. To convert milligrams per deciliter to milliequivalents per liter, use the following formula:

$$\frac{\text{mg/dL} \times 10 \times \text{valence}}{\text{atomic mass}} = \text{mEq/L}$$

For the problem presented, let 10 represent the number of deciliters per liter; 39 is the atomic mass of potassium, and 1 is the valence of potassium.

$$\frac{19.5 \text{ mg/dL} \times 10 \times 1}{39} = 5.0 \text{ mEq/L}$$

39.

B. The urine sample contained 122.5 mmol/L of sodium. Because the total urine volume is given in milliliters, it is necessary to express the volume in liters so that the units of measurement correspond. This may be done by dividing the 24-hour volume in milliliters by 1000, because each liter has 1000 mL. The amount of sodium excreted in the 24-hour urine specimen may now be calculated.

$$\frac{\text{Sodium mmol/L} \times \text{urine volume mL/24 hr}}{1000 \text{ mL/L}} = \text{mmol/24 hr}$$

$$\frac{122.5 \text{ mmol/L} \times 1540 \text{ mL/24 hr}}{1000 \text{ mL/L}}$$
$$= 188.6 \text{ mmol/24 hr}$$

40.

C. Because titration reveals that the normality of the HNO_3 solution is higher than the desired normality, it is apparent that the solution needs to be diluted. The following equation may be used to determine the new total volume required, and indirectly the amount of deionized water that must be added to the remaining volume to make the desired normality:

$$V_1 \times N_1 = V_2 \times N_2$$

where V_1 = milliliters of the desired 0.50 N HNO_3 solution, N_1 = 0.50 N HNO_3 solution, V_2 = 3975 mL remaining of the 0.513 N HNO_3 solution, and N_2 = 0.513 N HNO_3 solution. Using this information in the above equation:

$$V_1 \times 0.50 \text{ N} = 3975 \text{ mL} \times 0.513 \text{ N}$$

$$V_1 = \frac{3975 \text{ mL} \times 0.513 \text{ N}}{0.50 \text{ N}}$$

$$V_1 = 4078.4 \text{ mL}$$

Thus, a new total volume of 4078.4 mL is required to make an accurate 0.50 N HNO_3 solution. To determine the amount of deionized water that must be added to the remaining 3975 mL volume, find the difference between the new total volume and the remaining volume:

4078.4 mL − 3975 mL = 103.4 mL of deionized water that must be added to the remaining volume of solution.

41.

D. The degree to which an acid solution dissociates determines the hydrogen ion concentration and thus the strength of the acid solution. Because the dissociation of acetic acid is only 1%, it is considered a weak acid. To calculate the pH of a 0.2 N acetic acid solution that is 1% ionized, proceed as follows:

1. Find the hydrogen ion concentration with the following formula:

$$[H^+] = \text{N} \times \% \text{ ionized}$$

$$[H^+] = 0.2 \times 0.01 = 0.002 \text{ g } H^+/\text{L}$$

2. Find the pH of this solution with the following formula:

$$\text{pH} = \log \frac{1}{[H^+]}$$

$$\text{pH} = \log \frac{1}{0.002}$$

$$\text{pH} = 2.699$$

42.

C. A solution of Na_2SO_4 dissociates into two Na^+ ions and one SO_4^{2-} ion. The ionic strength is governed by the concentration (moles per liter) of each ion present in solution and the ionic charges associated with each ion. To calculate the ionic strength, derive half the sum for all ions present when the concentration of each ion present is multiplied by its valence squared. For a 0.2 M Na_2SO_4 solution, proceed as follows:

$$\text{Ionic strength} = \frac{\left[0.2 \times 2 \times (1)^2\right] + \left[0.2 \times (2)^2\right]}{2}$$

$$\text{Ionic strength} = 0.6$$

43.

D. The equation $A = abc$, where A represents the absorbance of a substance at a specified wavelength, a represents the absorptivity, b represents the length of the light path in centimeters, and c represents the concentration of the absorbing compound, is commonly referred to as Beer's law. When a 1-cm light path is used and the concentration is expressed in moles per liter, the constant a is replaced by the symbol ε (epsilon), which is referred to as the molar absorptivity. The molar absorptivity is generally defined as the absorbance determined, at a specified wavelength using a 1-cm light path, for a 1 M solution of a pure substance. The equation becomes $A = \varepsilon bc$. Use this formula to determine the absorbance of the 1.0 mg/dL (0.01 g/L) bilirubin standard solution. The molecular weight of bilirubin is 584 and ε is 60,700 at 453 nm using a 1-cm light path.

$$A = \varepsilon bc$$
$$A = (60{,}700)\ (1)\ (0.01 \div 584)$$
$$A = 1.039$$

Thus, a 1 mg/dL bilirubin standard solution should have an absorbance reading of 1.039.

44.

B. Over the years, numerous units have been used to express enzyme activity, including the Bodansky unit, the Gutman-Gutman unit, and the Bowers-McComb unit. The introduction of the international unit (IU or U) brought a common unit of comparison to enzyme assays. The International Unit is defined as the amount of enzyme activity that converts 1 μmol of substrate in 1 minute under standard conditions. The following formula is used to calculate enzyme activity.

$$\frac{\Delta A/\text{min} \times 1000 \times TV \times 1000 \times Tf}{6.22 \times 10^3 \times LP \times SV} = \text{U/L}$$

where ΔA/min is the average absorbance change per minute and 1000 converts milliliters to liters; TV is the total reaction volume and 1000 converts millimoles to micromoles; Tf is the temperature factor (1.0 at 37°C); 6.22×10^3 is the molar absorptivity of reduced nicotinamide-adenine dinucleotide (NADH) at 340 nm; LP is the light path in centimeters and SV is the sample volume. For the problem presented, determine the average ΔA/min and then substitute the given information into the equation.

Absorbance (A) Readings	**ΔA/min**	**Average ΔA/min**
0.204 A at 1 min		
	0.202	
0.406 A at 2 min		
	0.204	0.203
0.610 A at 3 min		
	0.203	
0.813 A at 4 min		

$$\frac{0.203\ \Delta A/\text{min} \times 1000 \times 3.2\ \text{mL} \times 1000 \times 1}{6.22 \times 10^3 \times 1.0 \times 0.2\ \text{mL}} = 522\ \text{U/L}$$

45.

C. A kilogram is approximately equal to 2.2 pounds. A weight expressed in pounds may be converted to kilograms by dividing the weight in pounds by the conversion factor 2.2 as follows:

$$175.5 \text{ lb} \div 2.2 \text{ lb/kg} = 79.8 \text{ kg}$$

46.

D. When converting Celsius to Fahrenheit degrees, remember that 1°C equals 9/5°F. Thus multiplying the temperature in degrees Celsius by 9/5 is necessary to express the temperature in the degrees Fahrenheit unit, and the addition of 32 allows for adjustment to the Fahrenheit scale's zero point. To convert 30°C to degrees Fahrenheit, the following formula may be used:

$$°F = (°C \times 9/5) + 32$$
$$°F = (30 \times 9/5) + 32$$
$$°F = 86$$

47.

C. Unlike most units of measurement used in chemistry, the curie (Ci) is a unit of measure for radioactivity that is independent of weight. The curie is defined as the quantity of radioactive material that exhibits 3.7×10^{10} disintegrations per second (dps). Other terms frequently used refer to units of radioactivity of smaller dimensions than the curie. These are the millicurie (mCi), which exhibits 3.7×10^{7} dps, and the microcurie (μCi), which exhibits 3.7×10^{4} dps.

48.

B. Each radionuclide has a unique half-life associated with it: ^{14}C = 5730 years; ^{3}H = 12.3 years; ^{125}I = 60 days; and ^{131}I = 8.1 days. Because half-life refers to the percent of activity remaining after a specified time, it will be necessary for all radionuclides to pass through seven half-life periods in order to reduce the initial activity of 100% to less than 1%. The following example expresses this phenomenon:

$$A_n = A_0\left(\tfrac{1}{2}\right)^7$$
$$A_7 = 100\left(\tfrac{1}{2}\right)^7$$
$$A_7 = 0.78\%$$

49.

A. The term "half-life" refers to the time required for the activity of a known amount of radioactive material to decrease to half of the initial activity. In a radioimmunoassay (RIA), this loss of activity is critical to the sensitivity of the assay. As activity decreases with time, the sensitivity of the assay will also decrease. In the problem presented, assume 100% activity originally. Because ^{125}I has a half-life of 60 days, the lapse of 180 days represents three half-life periods (3 $t_{1/2}$), 180 days ÷ 60 days/$t_{1/2}$ = 3 $t_{1/2}$. Use the following formula, where A_n is the activity at the specified number of half-life periods. A_0 is the initial activity, and n is the number of half-life periods.

$$A_n = A_0\left(\tfrac{1}{2}\right)^n$$
$$A_3 = 100\left(\tfrac{1}{2}\right)^3$$
$$A_3 = 12.5\%$$

Therefore, 12.5% of the initial activity remains at the end of 3 $t_{1/2}$.

50.

B. For a strong base such as NaOH, dissociation into Na^+ and OH^- is complete. Thus in terms of molarity, the concentration of sodium ions equals the concentration of hydroxyl ions, which in turn equals the molar concentration of NaOH originally present. Proceed with the problem for a solution of NaOH with a pH of 12 by finding the molar concentration of the hydroxyl ions.

$$\begin{aligned} pH + pOH &= 14 \\ 12 + pOH &= 14 \\ pOH &= 2 \\ pOH &= -\log[OH^-] \\ 2 &= -\log[OH^-] \\ [OH^-] &= 1 \times 10^{-2} = 0.01 \text{ mol/L} \end{aligned}$$

Because 0.01 mol/L is the molar concentration of hydroxyl ions, it follows that the molar concentration of NaOH is 0.01 mol/L. The following formula is used to find the number of grams required to prepare a 0.01 M NaOH solution (molecular weight of NaOH = 40).

$$\begin{aligned} \frac{\text{g/L}}{\text{mol wt}} &= \text{M} \\ \frac{X}{40} &= 0.01 \\ X &= 0.4 \text{ g/L} = 0.2 \text{ g/500 mL} \end{aligned}$$

Thus, 0.2 g of NaOH is required to prepare 500 mL of a solution of NaOH with a pH of 12.

51.

B. By use of centrifugal force, a centrifuge effects the separation of substances of different densities. The most common use of a centrifuge in the clinical laboratory is the separation of serum or plasma from the blood cells. The use of the proper amount of centrifugal force with serum separator tubes is especially important. In order for a thixotropic, silicone gel to form a barrier between the serum and the cell clot, it is critical that the tube be centrifuged for a specified time and with the specified centrifugal force. The following formula is used to calculate the relative centrifugal force (RCF) in terms of gravities (g), where 1.118×10^{-5} represents a constant, r represents the rotating radius in centimeters, and rpm represents the rotating speed in revolutions per minute:

$$\begin{aligned} \text{RCF} &= 1.118 \times 10^{-5} \times \text{r} \times (\text{rpm})^2 \\ &= 1.118 \times 10^{-5} \times 10 \times (2500)^2 \\ &= 698.75 \\ \text{RCF} &= 699 \times g \end{aligned}$$

52.

C. Class A volumetric flasks are calibrated at 20°C. Glassware that is designated as class A must meet the requirements of the National Institute of Standards and Technology. The College of American Pathologists (CAP) requires that CAP-approved clinical laboratories use only class A glassware. A 10 mL class A volumetric flask that is accurate to ±0.2% has a tolerance of ±0.02 mL; 10 mL × 0.2% = 0.02 mL = ±0.02 mL. Thus the capacity of a 10 mL flask is within the range of 9.98 to 10.02 mL.

53.

B. The term "parts per million" (ppm) is a unit of concentration that describes the number of parts of a substance that are contained in 1 million parts of the solution. "Parts per million" refers to the number of grams of a substance in 1 million grams of solution. To convert from parts per million to concentration, the following formula is used:

$$\frac{\text{g}}{X\ \text{mL}} \times 1{,}000{,}000 = \text{ppm}$$

In referring to parts per million, it is important to remember that the unit of measure may vary (e.g., milligrams, micrograms, or nanograms), provided that the relationship of some number of parts in 1 million parts is maintained. Therefore, it follows that to convert 0.01 ppm to mg/dL:

$$\frac{\text{mg}}{X\ \text{mL}} \times 1000 = \text{ppm}$$

$$\frac{\text{mg}}{100} \times 1000 = 0.01$$

$$\text{mg} = \frac{0.01}{1000} \times 100$$

$$\text{mg} = 0.001$$

Thus 0.01 ppm of lead is equivalent to 0.001 mg/dL or 0.001 mg/100 mL.

54.

B. To convert 68.0 $\%T$ to absorbance (A), use the following formula:

$$A = -\log T = \log\frac{1}{T} = \log\frac{100\%}{\%T}$$

$$A = \log 100 - \log \%T$$

$$A = 2 - \log \%T$$

$$A = 2 - \log 68$$

$$A = 2 - 1.832 = 0.168$$

After determining absorbance values, they may be used in the Beer's law equation, $A = abc$, to determine concentration values. Absorbance and percent transmittance values may both be used to construct standard curves to determine concentration values of unknown samples.

55.

D. Spectrophotometric analysis is based on Beer's law, which states that under appropriate conditions the concentration of a colored substance in solution is directly proportional to the amount of light absorbed and inversely proportional to the logarithm of the transmitted light. The narrower the band of wavelengths of light used, the more closely is Beer's law followed. Because Beer's law applies only to absorption of light by the analyte, absorption by any other substance, such as the solvent, would make the law inapplicable. Beer's law is applicable only to dilute solutions. Fluorescing compounds would add light to the system. A chemical reaction would change the concentration of the analyte. In either case, Beer's law would be inapplicable.

56.

D. When one of the two variable quantities changes as a result of changing the other, the result is frequently presented in the form of a graph. It is essential that the scale chosen for each axis of the graph be used consistently. For example, when using Cartesian graph paper where both axes are linear to plot spectrophotometric data (concentration versus absorbance), if 1 cm represents a concentration of 10 mg/dL on the ordinate or horizontal x-axis, 1 cm cannot represent 100 mg/dL toward the end of the scale. Similarly, for the abscissa or vertical y-axis a uniform scale must be used (e.g., 1 cm represents 0.1 absorbance units), but it need not be, and generally it is not, the same as the scale used for the horizontal axis, as illustrated by this example.

57.

D. Primary standards are substances that react quantitatively with other analytes. The caliber of the analysis depends on the caliber of the primary standard. It is therefore essential that the primary standard available be at least 99.95% pure. The primary standard must not be hygroscopic to avoid errors in weighing caused by absorption of water. For accurate weighing, the primary standard must be dry. Dryness is obtained in an oven at a temperature slightly above the boiling point of water. Hence the primary standard must be stable at 110°C. Because the primary standard should have a large equivalent weight and hence a large molecular weight, a relatively large amount will be needed to react with the analyte. The error inherent in weighing very small quantities is thereby avoided.

58.

A. To solve this problem, all the values must be converted to a common unit. In metric measurement the gram is the primary unit for weight. The value 0.1 ng is equal to 1×10^{-10} g, which is the lightest weight given. The gram relationships for the other values stated are 0.01 g $= 1 \times 10^{-2}$ g, 1.0 mg $= 1 \times 10^{-3}$ g, and 1000 pg $= 1 \times 10^{-9}$ g.

59.

C. A solution with an OH^- concentration of 1×10^{-6} has a pOH of 6. The pH + pOH = 14; therefore the pH is 8. The H^+ concentration of pure water is 10^{-7} and has a pH of 7.

60.

D. The molecular weight of NaCl is 58.5 and the atomic weight of Cl is 35.5. To change 50 mg Cl into an equivalent amount of NaCl, the answer must be greater than the amount of Cl. The solution follows:

$$50 \text{ mg Cl} \times \frac{58.5 \text{ mg NaCl}}{35.5 \text{ mg Cl}} = 82.4 \text{ mg NaCl}$$

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review questions

INSTRUCTIONS Each of the questions or incomplete statements that follows is comprised of four suggested responses. Select the *best* answer or completion statement in each case.

General Laboratory Principles

1. Which of the following is an alumina-silicate glass that is at least six times stronger than borosilicate and is resistant to alkaline etching and scratching?
   A. Kimax
   B. Pyrex
   C. Corning boron free
   D. Corex

2. The National Institute of Standards and Technology (NIST) requires that volumetric pipettes and flasks be certified as
   A. Class A
   B. Class B
   C. Class C
   D. Class D

3. Which of the following desiccants is the most hygroscopic?
   A. Silica gel
   B. Alumina
   C. Barium oxide
   D. Magnesium perchlorate

4. Lavender top blood collection tubes are used for complete blood counts. What is the anticoagulant in these tubes?
   A. EDTA
   B. Heparin
   C. Sodium citrate
   D. Sodium oxalate

5. SI units are the designated units employed by the International System of Units. The unit class that encompasses the seven fundamental quantities of measurement is
   A. Base
   B. Primary
   C. Derived
   D. Elemental

6. Which of the following parameters does *not* have an effect on centrifugal force?
   A. Temperature of the centrifuge
   B. Mass of the material being centrifuged
   C. Speed of rotation
   D. Radius of the centrifuge

7. What class of weights should be used to check the calibration of analytical balances?
   A. J
   B. M
   C. P
   D. S

8. "To deliver" (TD) pipettes are identified by
   A. Two etched bands near the top
   B. Self-draining capacity
   C. Dual-purpose pipette labels
   D. Blue graduation levels

9. If a laboratory needs to keep certain chemical materials dry, the apparatus used will be a
   A. Buret
   B. Desiccator
   C. Separatory funnel
   D. Vacuum

10. Which of the following is an advantage of the angle-head centrifuge over the horizontal-head centrifuge?
    A. Less air friction
    B. Smaller increase in sample temperature during centrifugation
    C. Can be operated at a higher speed
    D. All the above

11. The type of water desired for use in test methods requiring maximum accuracy and precision is
    A. Distilled
    B. Pure grade
    C. Reagent grade
    D. Special reagent water

12. The speed of a centrifuge should be checked at least once every 3 months with a(n)
    A. Tachometer
    B. Wiper
    C. Potentiometer
    D. Ergometer

13. The type of balance that uses an electromagnetic force to counterbalance the load placed on the pan is a(n)
    A. Trip balance
    B. Class A balance
    C. Class S balance
    D. Electronic balance

14. Which of the following blood collection tube tops would contain a serum separator gel?
    A. Blue
    B. Grey
    C. Lavender
    D. Speckled

15. When performing a venipuncture, which of the following is the proper order of draw for evacuated blood collection tubes?
    A. Sterile specimen, light blue top, and plain red top
    B. Light blue top, plain red top, and sterile specimen
    C. Lavender top, light blue top, plain red top
    D. Green top, sterile specimen, and plain red top

Laboratory Quality Assessment

16. What term applies to the sum of all the values in a set of numbers divided by the number of values in that set?
    A. Median
    B. Mode
    C. Arithmetic mean
    D. Geometric mean

17. Calculate the coefficient of variation (percent) for a set of data where the mean $(\bar{x})$ = 89 mg/dL and 2 standard deviations (s) = 14 mg/dL.
    A. 7.8
    B. 7.9
    C. 15.7
    D. 15.8

18. What does the preparation of a Levey-Jennings quality control chart for any single constituent of serum require?
    A. Analysis of control serum over a period of 20 consecutive days
    B. 20 to 30 analyses of the control serum, on 1 day, in one batch
    C. Analyses consistently performed by one person
    D. Weekly analyses of the control serum for 1 month

19. A batch of test results is out of control. What should you do first?
    A. Report the results to the physician first, and then look for the trouble.
    B. Follow the "out-of-control" procedure specified for the test method.
    C. Repeat the tests with a new lot of standards (calibrators).
    D. Repeat the tests with a new lot of reagents.

20. In addition to utilizing Levey-Jennings charts, what other criteria should be applied to interpret internal quality control data?
    A. Westgard multirule
    B. Cusum
    C. Linear regression
    D. Youden

21. A new standard (calibrator) has been prepared in error at a lower concentration than that required for the test. How would such an error appear on a quality control chart?
    A. Upward trend
    B. Downward trend
    C. Upward shift
    D. Downward shift

22. The ±2 standard deviation (±2 *s*) range of acceptable values for a digoxin control is established as 2.0–2.6 ng/mL. On the average, the expectation that a value will be greater than 2.6 ng/mL is 1 in
    A. 10
    B. 20
    C. 40
    D. 100

23. What is the purpose of a Youden plot?
    A. Compares results on two control specimens, low and high controls, for the same analyte analyzed by several laboratories
    B. Evaluates the validity of daily results on a single control specimen over a period of 30 days
    C. Compares results on a single control specimen by two different methods for the same analyte
    D. Evaluates the validity of daily results of two control specimens within a single laboratory

24. If the therapeutic range for the gentamicin assay is a trough level of less than 2 μg/mL and a peak level of 5–8 μg/mL, what would be appropriate mean values for two control levels (in micrograms per milliliter) used to monitor the system?
    A. 1 and 2
    B. 1 and 3
    C. 1.5 and 6
    D. 5 and 6

25. Which of the following monitoring factors would *not* be included in a laboratory's quality assessment program?
    A. Scheduling of staff
    B. Specimen collection and identification
    C. Accuracy and precision of analyses
    D. Preventive maintenance of instruments

26. On a quality control chart, when would a statistical out-of-control situation requiring corrective action be suspected?
    A. Six successive plots fall above and below the mean within ± 1 *s*.
    B. Six successive plots fall above and below the mean within ± 2 *s*.
    C. One plot falls within the area of ± 2 *s* to 3 *s* within a 20-consecutive-day span.
    D. One plot falls outside the area of ± 3 *s* within a 20-consecutive-day span.

27. Which of the following would result in a sudden shift in daily values on a quality control chart?
    A. Recalibrating the instrument when changing reagent lot numbers during an analytical run
    B. Replacing the instrument's sample aspiration probe
    C. Changing the spectrophotometer lamp in the middle of a sample run
    D. Changing personnel operating the instrument

28. Which of the following terms refers to the measure of scatter of experimental data around the mean of a Gaussian (normal) distribution curve?
    A. Median
    B. Mode
    C. Coefficient of variation
    D. Standard deviation

29. The percentage of individuals without a specific disease who are correctly identified or predicted by the test as not having the disease describes
    A. Sensitivity
    B. Specificity
    C. Positive predictive value
    D. Negative predictive value

30. Which of the following terms refers to deviation from the true value caused by indeterminate errors inherent in every laboratory measurement?
    A. Random error
    B. Standard error of the mean
    C. Parametric analysis
    D. Nonparametric analysis

31. What is the following formula used to calculate? $\sqrt{\frac{\Sigma(x - \bar{x})^2}{n-1}}$
    A. Coefficient of variation
    B. Variance
    C. Confidence limits
    D. Standard deviation

32. Which of the following terms refers to the closeness with which the measured value agrees with the true value?
    A. Random error
    B. Precision
    C. Accuracy
    D. Variance

33. What percentage of values will fall between ± 2 *s* in a Gaussian (normal) distribution?
    A. 34.13%
    B. 68.26%
    C. 95.45%
    D. 99.74%

34. Which of the following terms refers to a measure of dispersion or spread of values around a central value?
    A. Range
    B. Validity
    C. Variance
    D. Coefficient of variation

35. Which of the following describes the ability of an analytical method to maintain both accuracy and precision over an extended period of time?
    A. Reliability
    B. Validity
    C. Probability
    D. Sensitivity

36. What is the following formula used to calculate? $\frac{\Sigma(x - \bar{x})^2}{n - 1}$
    A. Coefficient of variation
    B. Variance
    C. Confidence limits
    D. Standard deviation

37. Which of the following does *not* pertain to the characteristics and use of assayed control material?
    A. Has physical and chemical properties resembling test specimen
    B. Contains preanalyzed concentrations of analytes being measured
    C. Can be interchanged in terms of use with primary standards or calibrators
    D. Concentrations of analytes should be in normal and abnormal ranges.

38. A group of physicians consistently complains that they are not receiving stat patient results quickly enough. The supervisor is likely to refer to which quality assessment variable?
    A. Specimen separation and aliquoting
    B. Test utilization
    C. Analytical methodology
    D. Turnaround time

39. To provide independent validation of internal quality control programs, external surveys have been developed. Which of the following is a representative survey program?
    A. CLSI (Clinical and Laboratory Standards Institute)
    B. ASCLS (American Society for Clinical Laboratory Science)
    C. ASCP (American Society for Clinical Pathology)
    D. CAP (College of American Pathologists)

40. A tech is scheduled to perform a specialized test that she/he is familiar with, but is not *exactly* certain of the steps required. What is the best course of action to take?
    A. Ask another tech to perform the test.
    B. Consult the procedure manual and notify the supervisor.
    C. Run the test as best as possible, being careful to note control values.
    D. Reject the specimen.

41. A tech has completed the first run of morning specimens. She/he notices that the one control being used is outside $\pm 3\ s$. What course of action should be taken?
    A. Release the results.
    B. Repeat the control only, and if it comes in, release results.
    C. Check equipment and reagents to determine source of error; repeat the entire analysis, including the control and patients; if the control value is within $\pm 2\ s$, release results.
    D. Repeat the control; if the same thing happens, attribute the cause to random error; release results.

42. Which of the following describes the Westgard multirule 2_{2s}?
    A. Two control data points are within ± 2 *s*.
    B. One control data point falls outside $+2$ *s* and a second point falls outside -2 *s*.
    C. Two consecutive data points fall outside $+2$ *s* or fall outside -2 *s*.
    D. Two consecutive data points fall outside $+2$ *s*.

43. Which Westgard multirule applies to a situation where one control point exceeds the mean by $+2$ *s* and a second control point exceeds the mean by -2 *s*?
    A. 1_{2s}
    B. 2_{2s}
    C. 4_{1s}
    D. R_{4s}

44. Upon admission to the hospital, a chemistry profile is performed on a patient. The patient has a total bilirubin of 2.0 mg/dL. The next day a second chemistry profile is done, and the patient's total bilirubin is 6.2 mg/dL. What should be done in regard to these results because the normal and abnormal controls are within acceptable limits?
    A. Immediately call the physician to alert him/her to the second abnormal result.
    B. Immediately send the second result to the patient's floor for charting.
    C. Repeat the entire second run of patient specimens because there must be an error.
    D. Perform a delta check and, if warranted, look for possible sources of error.

45. When comparing a potential new test with a comparative method in order to bring a new method into the laboratory, one observes error that is consistently affecting results in one direction. What is this type of error known as?
    A. Systematic error
    B. Random error
    C. Constant systematic error
    D. Proportional systematic error

46. When establishing a reference interval for a new test being introduced into the laboratory, what is the preferred number of subjects that should participate?
    A. 30
    B. 50
    C. 75
    D. 120

47. A small laboratory has collected blood samples from 20 individuals as part of a reference interval study for a new test being introduced into the laboratory. Of the test results, four are outside the reference interval published by the manufacturer. How should you proceed?
    A. Delete the four results and only use the 16 within the range to establish the lab's reference interval.
    B. Use all 20 results when calculating the ± 2 *s* range because outliers are to be expected.
    C. Run four additional samples and if within the manufacturer's range, add them to the original 16 for statistical analysis.
    D. Obtain an additional 20 samples for testing, and if two or less are outside the suggested range, then the manufacturer's reference interval can be accepted.

48. Which of the following parameters is *not* needed to determine the predictive value of a positive test?
    A. Sensitivity
    B. Specificity
    C. Disease prevalence
    D. Precision

49. Which of the following must be known in order to determine the sensitivity of a test?
    A. True positives and false negatives
    B. True negatives and false positives
    C. True positives and false positives
    D. True negatives and false negatives

50. A new test to assess for the presence of malignancy has been developed. By testing a group of benign individuals, it is determined that 45 of 50 subjects test negative for the new marker. What is the specificity of this new assay?
    A. 10%
    B. 11%
    C. 90%
    D. 100%

51. Which of the following must be known in order to determine the predictive value of a negative test; that is, the percentage of individuals who test negative and are not diseased?
    A. True negatives and false negatives
    B. True positives and false positives
    C. True positives and false negatives
    D. True negatives and false positives

52. Which of the following statements is *false* about proficiency-testing programs?
    A. Participation is mandated by the Centers for Medicare and Medicaid Services under CLIA '88.
    B. College of American Pathologists and the American Association of Bioanalysts are two major providers of these programs.
    C. Samples of unknown concentrations are periodically sent to labs participating in the program.
    D. Acceptable ranges are provided with the samples so labs can determine if it is necessary to repeat the assay.

Laboratory Safety

53. Which of the following is a reactive chemical that has the potential to become shock sensitive if stored for a prolonged period of time?
    A. Xylene
    B. Picric acid
    C. Chloroform
    D. Phenol

54. A fire extinguisher used in the event of an electrical fire should include which of the following classifications?
    A. Type A
    B. Type B
    C. Type C
    D. Type D

55. In the National Fire Protection Association identification system, four color-coded, diamond-shaped symbols are arranged to form a larger diamond shape. What type of hazard does the blue diamond identify?
    A. Flammable
    B. Health
    C. Reactivity
    D. Contact

56. Xylene, ethanol, methanol, and acetone would be in which hazard class?
    A. Corrosive
    B. Flammable
    C. Oxidizer
    D. Carcinogen

57. Which of the following terms is used to identify a chemical that causes cancer?
    A. Mutagen
    B. Teratogen
    C. Carcinogen
    D. Reactive

58. A Biosafety Level 2 (BSL-2) laboratory is designed to work with microorganisms that are
    A. Not associated with disease in healthy adult humans
    B. Associated with serious or lethal human disease for which preventative or therapeutic interventions may be available
    C. Likely to cause serious or lethal human disease for which preventative or therapeutic interventions are not usually available
    D. Associated with human disease that is rarely serious and for which preventative or therapeutic interventions are often available

59. The flash point of a liquid may be defined as the
    A. Minimum temperature at which self-sustained ignition will occur
    B. Maximum vapor pressure at which spontaneous ignition will occur
    C. Temperature at which an adequate amount of vapor is produced, forming an ignitable mixture with air at the liquid's surface
    D. Temperature that is 10°C greater than the liquid's boiling point

60. A corrosive material was spilled onto the hand of a laboratorian. After diluting the material under running cold water, what should be done next?
    A. Consult the material safety data sheet.
    B. Wipe up the spill with paper towels.
    C. Dilute the spill with water and remove it in a biohazard bag.
    D. Go to the nearest hospital emergency department.

61. The federal agency charged with the enforcement of safety and health legislation is the
    A. College of American Pathologists
    B. International Air and Transportation Association
    C. Joint Commission
    D. Occupational Safety and Health Administration

62. The maximum chemical exposure allowable for an employee during one 8-hour day is the
    A. Flash point
    B. HEPA standard
    C. Threshold limit value
    D. Limit established by the United Nations

63. The air from a chemical fume hood is
    A. Recirculated through a HEPA filter
    B. Totally exhausted from the building
    C. Totally exhausted from the building through a HEPA filter
    D. Recirculated through a HEPA filter (30%) and 70% is exhausted to the outside

64. The air-handling system for a microbiology laboratory should
    A. Maintain negative pressure with respect to the administrative areas
    B. Maintain positive pressure with respect to the administrative areas
    C. Have no particular requirement
    D. Have a HEPA filter

65. Precautions such as using a chemical fume hood, wearing rubber gloves, donning a respirator, and cleaning glassware with a strong acid or organic solvent are consistent with working with
    A. Corrosives
    B. Carcinogens
    C. Azides
    D. Acids

66. A laboratorian, properly dressed in white pants, laboratory coat, and shoes, prepares to leave the laboratory for lunch. In addition to washing his hands, he should
    A. Put on safety goggles
    B. Remove his laboratory coat
    C. Wipe the bench with water
    D. Remove polyvinyl gloves and place them into labcoat pocket for future use

67. Which of the following may be a potentially hazardous biological situation?
    A. Handling specimens collected from patients in isolation according to standard precautions
    B. Keeping the centrifuge lid closed until the system has stopped completely
    C. Discarding sharp objects, including broken glass, in a puncture-proof container
    D. Discarding disposable blood collection needles in the patient's wastebasket

68. Which of the following is associated with proper storage of chemicals?
    A. All chemicals should be stored in alphabetical order for ease of handling.
    B. Flammable chemicals should be stored in a chemical fume hood.
    C. Large containers of liquid chemicals should be stored on a top shelf to allow easy visibility from below.
    D. Volatile solvents should be stored in small amounts in an explosion-proof refrigerator.

69. A laboratorian spills a bottle of concentrated sulfuric acid and slips in the fluid, exposing the lower length of her body to the burning fluid. What would be the most advisable action for a coworker to take?
    A. Call security.
    B. Put the person under the safety shower.
    C. Take the injured person to the nearest hospital emergency department.
    D. Pour concentrated base on the person to neutralize the acid.

70. A stat procedure requiring a corrosive reagent (organic acid) is requested. To transport this reagent to the work area under the chemical fume hood a laboratorian should
    A. Employ a rubber carrier with handles
    B. Pour an amount near the storage site and transport it
    C. Pipette the required volume and carry the pipette to the work area
    D. Carry the brown bottle by the loop with one hand under the bottom of the container

71. Which of the following statements pertains to the safe handling of compressed gases?
    A. Large cylinders should be loosely placed on a hand cart when being transported.
    B. Cylinders must be secured to a wall or bench when in use.
    C. Cylinders should be stored along with flammable liquids because both are combustible.
    D. Large cylinders should be ordered to avoid frequent movement in and out of stock.

72. The major job-related disease hazard in clinical laboratories is
    A. Tularemia
    B. Salmonella
    C. Tuberculosis
    D. Hepatitis

73. The responsibility to implement and maintain a safety program in a clinical laboroatory belongs to the
    A. State public health laboratory
    B. Chemical hygiene officer
    C. Laboratory safety officer
    D. Chief pathologist

74. Based on the chemical properties of azides, which of the following factors has motivated laboratories to monitor their use?
    A. The buildup of salts can lead to explosions.
    B. They are corrosive to pipes even when diluted.
    C. They are extremely volatile.
    D. They are flammable and dangerous near an open flame.

75. A biological safety cabinet that forces HEPA filtered air downward at the front of the cabinet where the laboratorian is working describes a
    A. Class 1 cabinet
    B. Class 2 cabinet
    C. Class 3 cabinet
    D. Class 4 cabinet

76. The phrase "Standard Precautions" refers to a concept of bloodborne disease control that requires all human blood and other potentially infectious materials
    A. Be treated as if known to be infectious for bloodborne pathogens regardless of the perceived "low risk" of a patient population
    B. Be treated as if it is not infectious unless it is known to be infectious
    C. Must be handled using a respirator for aerosol exposure
    D. Need not be treated with caution unless there is a cut on your hand

77. The biological safety cabinet is the single most useful safety device in the microbiology laboratory. How do Class 2A cabinets differ from Class 2B?
    A. Class 2A exhausts HEPA filtered air into the room.
    B. Class 2B exhausts HEPA filtered air into the room.
    C. Class 2B cabinets are larger.
    D. Class 2A cabinets contain gas jets for a Bunsen burner.

78. Chlorine is most often used in the form of sodium hypochlorite (NaOCl), found in household bleach, for a disinfectant. What dilution of household bleach is recommended by the Centers for Disease Control and Prevention to clean up blood spills?
    A. 1:1
    B. 1:10
    C. 1:20
    D. 1:100

79. The clinical hematology laboratory just received a new disinfectant to use in place of the one normally used. Never having used this particular disinfectant before, how should the lab professional proceed?
    A. Use it full strength; you can always be sure if you do this.
    B. Read the manufacturer's package insert and prepare the product according to directions.
    C. Make the concentration 10% higher than the manufacturer's recommendations.
    D. Put the new disinfectant under the sink for storage.

80. What written plan of specific measures must laboratories have in place to minimize the risk of exposure to bloodborne pathogens?
    A. Chemical hygiene plan
    B. Exposure control plan
    C. Material safety data sheets
    D. Infection control plan

81. Which of the following information is *not* found on a material safety data sheet?
    A. Health hazard data
    B. Fire and explosion hazard data
    C. First aid measures
    D. Cost of chemical

82. Work is being done with *Mycobacterium tuberculosis* in the microbiology laboratory. It is important that you enter this laboratory while work is being done with positive samples. What is the most important personal protective equipment you should don before entering this laboratory?
    A. Carbon cartridge respirator
    B. Mask
    C. Gloves
    D. N95 HEPA filter respirator

83. When working with chemicals, the selection of your gloves depends on
    A. The chemical hazard and the tasks involved
    B. How far you must transport the chemical
    C. Whether or not you use a chemical fume hood
    D. Whatever is available in the laboratory

84. What is the maximum capacity of a chemical storage safety can used in the laboratory?
    A. 1 gallon
    B. 5 gallons
    C. 20 gallons
    D. 55 gallons

answers & rationales

General Laboratory Principles

1.

D. Several types of glassware are commonly used in the laboratory, each having its specific purpose. Corex glass is used in the manufacture of centrifuge tubes and thermometers. Pipettes, beakers, and flasks are generally made from Pyrex or Kimax borosilicate glass.

2.

A. The National Institute of Standards and Technology (NIST) and the College of American Pathologists (CAP) state that volumetric pipettes and flasks must be of certified accuracy. Class A glassware meets federal guidelines and fulfills the CAP requirements. All non–class A glassware must be recalibrated periodically by an acceptable verification procedure.

3.

D. Desiccants are drying agents employed to keep some chemicals, thin-layer chromatography plates, and gases used in gas chromatography from combining with water and becoming hydrated. The most effective desiccant is magnesium perchlorate; one of the least hygroscopic is silica gel. Some desiccants can be regenerated for repeated use by heating them at a high temperature for several hours.

4.

A. Lavender top blood collection tubes contain EDTA as an anticoagulant. The EDTA chelates calcium, which is required for the coagulation cascade. Heparin is the anticoagulant in green top tubes, sodium citrate in blue top tubes, and potassium oxalate in grey top tubes.

5.

A. The Système Internationale d'Unités was established to facilitate a uniform system of measurement. SI units may be classified as base, derived, or supplemental units. Base units were established for each of the seven fundamental quantities of measurement: length (meter), mass (kilogram), time (second), electric current (ampere), amount of substance (mole), temperature (kelvin), catalytic amount (katal), and luminous intensity (candela). Derived units are mathematically calculated from more than one base unit.

6.

A. Centrifugal force depends on the mass and on the speed and radius of rotation. Because most materials being centrifuged in the clinical laboratory have specific gravity close to 1.00, only the speed and radius need be considered. The relative centrifugal force (RCF) is calculated by the formula:

$$RCF = 1.118 \times 10^{-5} \times r \times (rpm)^2$$

where r = radius in centimeters and rpm = the number of revolutions per minute. The RCF is expressed as a number times the force of gravity (or the number $\times g$). The radius is measured from the center of the centrifuge shaft to the inside bottom of the centrifuge cup. The number of revolutions per minute (rpm) is measured by a tachometer. The centrifugal force is not influenced by temperature.

7.

D. The National Institute of Standards and Technology recognizes the use of five specific types of calibrating weights. Analytical balances should be checked monthly and before accurate analytical work for proper calibration using Class S weights. To calibrate the weights themselves, Class M weights are used to check other weights, because their quality is that of a primary standard. Class P weights are allowed greater tolerance levels than Class S-1, with the latter being used for routine analytical work. Class J weights are used for microanalytical work.

8.

A. Serologic pipettes are "to deliver" (TD) types and are not rinsed out. These pipettes are not self-draining but must be blown out to deliver their entire contents, as is indicated by the two etched rings at the top of the pipette. "To contain" (TC) pipettes, such as Sahli pipettes, must be rinsed out to deliver their entire contents.

9.

B. Desiccators provide a dry environment for chemical materials. A shelf is placed on top of the desiccant on which the material to be stored can be set. A heavy glass cover closes the system. An airtight seal is provided by placing stopcock grease around the ground glass joints between the desiccator and the lid.

10.

D. In an angle-head centrifuge, the cups are rigidly supported in the head at a fixed angle to the shaft, and they are fully enclosed within the head. In a horizontal-head centrifuge, the cups hang down in a vertical position when the centrifuge is at rest and swing out to a horizontal position when the centrifuge is rotating. Because cups in the angle-head centrifuge are enclosed in a head specifically designed to reduce wind resistance, there is less air friction and, consequently, less of an increase in sample temperature during centrifugation. Because of the reduced wind resistance, angle-head centrifuges can provide a force of over 9000 $\times g$, whereas horizontal-head centrifuges provide about 1650 $\times g$.

11.

C. Reagent grade water should be used when a high degree of accuracy is desired, as in quantitative chemistry assays and in the preparation of standards and buffers. Reagent grade water requires deionization through acidic and basic ion-exchange columns, removal of organic materials by activated charcoal adsorption, and semipermeable membrane filtration for the removal of microorganisms and other particulate material. Special reagent water is for specific uses and, depending upon the use, may need to be sterilized.

12.

A. Centrifugal force may be determined by knowing the mass of the solution and the speed and radius of the centrifuge. With aqueous solutions having a specific gravity near 1.0, the specific mass need not be known. To determine the speed, use either a strobe light, positioned over the revolving centrifuge head, or a tachometer to establish the revolutions per minute (rpm).

13.

D. An electronic balance is a single-pan balance that uses an electromagnetic force to counterbalance the load placed on the pan. These balances are top-loading in design and permit weighings to be made quickly. The Mettler Instrument Corporation makes a representative electronic balance.

14.

D. Speckled top tubes, without an anticoagulant, contain a separation gel and are referred to as serum separator tubes. Plasma separator tubes contain the separation gel and an anticoagulant, such as lithium heparin; they may have a speckled or solid colored top representing the anticoagulant. During centrifugation, the gel forms a barrier between the blood cells and the serum or plasma. The gel barrier helps to maintain the integrity of the analytes in the liquid portion, and it facilitates automatic sampling or removal of the serum or plasma from the tube.

15.

A. To prevent anticoagulant carryover and contamination when using a needle and needle holder for blood collection, evacuated blood collection tubes should be collected in a proper order. Sterile specimens should be drawn first to minimize risk of contamination for blood cultures. After sterile collections, the following order should be used: light blue top, plain red top or serum separator tube, green top (heparin or heparin PST), and lavender top.

Laboratory Quality Assessment

16.

C. The arithmetic mean of a set of numbers is obtained by adding all the numbers in the set and dividing the sum by the number of values in that set. It is a precise way of expressing what is often called the average. It is not to be confused with the mode, which is the value that occurs most frequently in the set. The geometric mean is the antilogarithm of the sum of the logarithms of all the values divided by the number of values. The median is the middle value in a set of numbers that are arranged according to their magnitude.

17.

B. The coefficient of variation is calculated from the formula:

$$CV = \frac{s}{\bar{x}} \times 100\%$$

where CV = coefficient of variation, s = 1 standard deviation, and $\bar{x}$ = mean. Given that the mean = 89 mg/dL and 2 s = 14 mg/dL,

$$CV = \frac{7}{89} \times 100\% = 7.86\%$$

Because there are only two significant figures in each of the given numbers, there can be only two figures in the answer. Therefore, when rounded to the nearest tenth, the answer is 7.9, not 7.8.

18.

A. Any analytical result has some degree of uncertainty because of unavoidable random errors in the procedure. A Levey-Jennings quality control chart is a graphic representation of the acceptable limits of variation in the results of an analytical method. To prepare such a chart, it is first necessary to obtain a large-enough batch of normal and abnormal pooled serum to last for a minimum of 12 months. Analyses of aliquots of the pools are done in duplicate over a period of 20 days, preferably by all workers who will subsequently be using the controls. The data thus collected are statistically analyzed to determine the mean and standard deviation. Any results falling above or below the mean ± 3 *s* are discarded. The mean and standard deviation are then recalculated. The acceptable range is assigned, usually the mean ± 2 *s*. The data thus developed are used to prepare the Levey-Jennings quality control chart. A similar protocol may also be followed when setting up standard deviation parameters for new lots of assayed control materials.

19.

B. The purpose of a quality control chart is to facilitate the identification of analytical problems that are not otherwise apparent. A quality control program must include clearly written instructions for the steps that are to be taken when a control serum value is out of control. These instructions must be used whenever a set of test results is out of the established control limits. Usually the procedures will include a visual inspection of the equipment, reagents, and instruments used, and a check of calculations. The next step might be to rerun the batch of tests with a fresh aliquot of control serum. Additional steps to take include preparing newly reconstituted controls, recalibrating the instrument, and using a fresh bottle of reagent. Out-of-control results should never be reported to the physician.

20.

A. When assessing daily, internal quality control, the Westgard multirule procedure aids in interpretation of control data. A chart similar to the Levey-Jennings chart is constructed with control limits drawn at the mean as well as ± 1 *s*, ± 2 *s*, ± 3 *s*, and even ± 4 *s*. The Westgard multirules are then applied to the graphical representation, giving a more structured approach to data interpretation.

21.

C. On a quality control chart, when the control values change abruptly and on several consecutive days are consistently on one side of the mean, although within the ± 2 *s* limits, this is called a shift. An upward shift could be produced by changing to a new standard (calibrator) that was prepared in error at a lower concentration than specified. A downward shift could be caused by the use of too concentrated a standard (calibrator) than what is specified. A gradual change observed over the course of several days is called a trend. It may be upward or downward. Its presence suggests gradual deterioration of one of the reagents or instrument components.

22.

C. If the range of acceptable values for a quality control material is based on the ± 2 *s* intervals on either side of the expected mean value for the control, then about 19 of every 20 values obtained for the control are expected to fall within the acceptable range. Conversely, about 1 of every 20 values obtained are expected to fall outside the ± 2 *s* range with about 1 of every 40 values above the upper acceptable limit and the same number below the acceptable limit. For example, if a digoxin control is established to have an acceptable range from 2.0 to 2.6 ng/mL, about 1 value in 40 would be expected above 2.6 ng/mL.

23.

A. A Youden plot is a type of quality control chart that is used to compare results obtained on a high and low control serum by several different laboratories. It is particularly useful for interlaboratory quality control programs. The Youden plot displays the results of the analyses by plotting the mean values for one specimen on the ordinate and the other specimen on the abscissa. It is desirable for a laboratory to have its point fall at the center of the plot.

24.

C. The choice of appropriate concentrations for control materials is important in implementing a quality control program. The concentrations chosen should be sensitive to assay variability in the clinically significant region of the particular compound being measured. For instance, the therapeutic range for the drug gentamicin in many laboratories is a trough level that is less than 2 μg/mL and a peak level of 5–8 μg/mL. Control levels of 1.5 and 6 μg/mL would appropriately monitor both trough and peak regions of the standard curve. If both controls are above 5 or below 4 μg/mL, only one region of the curve would be monitored.

25.

A. Scheduling of staff is not involved in quality assessment programs. Quality assessment in a broad sense includes monitoring every aspect of laboratory work from collection and identification of specimens to delivery of valid results to the physician. Laboratory protocol should include procedures that ensure that the correct specimen is collected and that the specimen is correctly labeled. The quality of the laboratory supplies, including the reagent grade water (CLRW) and reagents, should meet the specifications of the appropriate professional or governmental agency (i.e., the College of American Pathologists, the American Chemical Society, or the National Institute of Standards and Technology). The accuracy and precision of the analyses should be ensured through the use of appropriate standards (calibrators) and controls. Also, quality control charts that define the acceptable range of results for control specimens and participation in external proficiency testing programs are part of any quality assessment program. Because most analyses are done with instruments, the performance of the instruments must be monitored. Establishment of a regularly scheduled preventive maintenance program provides for optimum performance of the instruments as well as continuous monitoring of their performance.

26.

D. On a quality control chart the acceptable range generally encompasses the mean ±2 *s*. The control values should occur randomly in this area, falling to both sides of the mean. If more than five successive plots occur at a constant level in one area (e.g., near the ±2 *s* line), an out-of-control situation should be considered. One plot falling outside the mean ±2 *s* in 20 successive days is expected statistically. Although one should be alert to the possibility of a potential problem, it does not necessarily imply an out-of-control situation. However, the occurrence of a value outside the area of ±3 *s* would require corrective action.

27.

C. When a quality control chart shows a sudden shift in daily values, there are several possible causes. Use of a new batch of reagents or reference standards (calibrators) that have been improperly made can cause such a shift. Another cause for a sudden shift in daily values might be a change in one of the components of the instrument, such as a new lamp in a spectrophotometer. Whenever an instrument component is changed, the instrument must be recalibrated. A change in operating personnel should not cause any change in quality control values.

28.

D. The standard deviation reflects how much the data values vary around the mean. The mean is the arithmetic average of the data and is a measure of the location of the distribution. The median describes the middle value; half of the observations are greater than the median, half are less than the median. The mode is the most frequently obtained value. The coefficient of variation expresses random variation of analytical methods in units independent of methodology, because it is a percentage comparison of the standard deviation divided by the mean.

29.

B. Specificity is the percentage of individuals without the specific disease who are correctly identified or predicted by the test as not having the disease. The sensitivity of a test is the percentage of individuals with a specific disease who are correctly identified or predicted by the test as having the disease. The positive predictive value is the percentage of people who tested positive with the test who have the disease, and the negative predictive value is the percentage of people who tested negative with the test who do not have the disease.

30.

A. Random errors are deviations from the true value caused by unavoidable errors inherent in laboratory measurements. The standard error of the mean is a statistical concept reflecting sampling variation. It is the standard deviation of the entire population. Parametric statistics refer to a Gaussian (normal) distribution of data. Nonparametric statistics are more general and require no assumptions.

31.

D. To determine the standard deviation, compute the difference between each value and the mean ($x - \bar{x}$), square the differences, and add all the squared differences [$\Sigma(x-\bar{x})^2$]. Then divide the sum by one less than the number of values ($n - 1$) and take the square root. The standard deviation is the estimated random error.

32.

C. The accuracy of an analytical result is the closeness with which the measured value agrees with the true value. Precision is reproducibility. Accuracy and precision are independent, but it is the goal of the clinical laboratory to design methods that are both precise and accurate.

33.

C. The normal distribution is a symmetric distribution about the mean. In a normal distribution, 95.45% of the values will be within an area enclosed by the mean ±2 *s* and approximately 5% will normally fall outside; 68.26% will lie within ±1 *s*; 99.74% will lie within ±3 *s*. The ±2 *s* (95.45%) interval forms the basis of statistical quality control in the laboratory.

34.

C. Variance is one way in which members of a group are dispersed about the mean. It is a square of the standard deviation (s^2). Both standard deviation and variance are measures that describe how observed values vary.

35.

A. The reliability of an analytical procedure is its ability to maintain accuracy and precision over an extended period of time during which supplies, equipment, and personnel in the laboratory may change. It is often used interchangeably with the term "consistency." It is the goal of every clinical laboratory to produce reliable results.

36.

B. The variance is the measure of dispersion, which is the square of the standard deviation. To determine the variance, find the difference between each value and the mean, square this difference, add the differences, and divide by one less than the number of values. The variance reflects scattering about the mean.

37.

C. A control is a specimen of known concentration with physical and chemical properties closely resembling the test specimen. The control is generally composed of the same matrix as the sample. The concentrations of analytes in different control materials should be in the normal and abnormal ranges. Both primary standards and calibrators are used to calculate the concentration of specimens being analyzed. Calibrators are considered secondary standards and must meet specific criteria as outlined by the National Institute of Standards and Technology. These criteria need not be met for controls and thus their uses cannot be interchanged.

38.

D. Maintaining quality assessment includes control of preanalytical, analytical, and postanalytical factors. One variable to assess is turnaround time. It is the total amount of time required to procure the specimen, prepare the specimen, run the test, and relate the results. The supervisor should refer to the turnaround time of the stated procedure and relate to the techs the need to work within the stated limits.

39.

D. The College of American Pathologists comprehensive survey involves thousands of participating clinical chemistry laboratories. This survey and others have been established to provide independent validation of quality control programs. A CAP survey provides unknown samples for analysis. The program, when properly used, gives valid estimation of the inherent accuracy of a system. CLSI develops laboratory standards to improve the quality of medical care. ASCLS and ASCP are organizations to which medical laboratory personnel may apply for professional membership.

40.

B. Whenever one needs to review the details of a procedure, one should review the procedure manual and notify the supervisor so that guidance can be received. The procedure manual is one way in which a laboratory can document analytical protocols. This leads to consistency in test results regardless of which person is performing the analysis.

41.

C. When checking control results that fall outside acceptable limits, one can apply the Westgard multirule procedure, specifically 1_{3s}. Anytime only one control is used and it exceeds the mean $\pm 3\ s$, you must reject the test run. You should check out the instrument and reagent system to locate the problem if possible. A new control along with the patient specimens should be analyzed. No results should be reported until the control is within the limits of $\pm 2\ s$ from the mean.

42.

C. Westgard multirule 2_{2s} describes an out-of-control situation where two consecutive data points fall outside the same mean $+2\ s$ or fall outside the same mean $-2\ s$. This is an example of systematic error. The test run would be rejected, and all samples would need to be retested.

43.

D. When one control point exceeds the mean by +2 *s* and a second control point exceeds the mean by –2 *s*, the R_{4s} multirule will apply. In this case the out-of-control problem is most likely due to random error. The test run would be rejected, and all samples would need to be retested.

44.

D. When the same test is ordered on a patient more than once, a delta check can be performed to compare consecutive test results. Bilirubin results obtained on two consecutive days on an adult should not vary by more than 50%. If the results vary by greater than 50%, it is most likely that an error has occurred or an acute change has taken place. One of the first things to check is proper identification of the patient's specimen. As part of a quality assessment program, one should also check patient results based on the clinical correlation of laboratory test results.

45.

A. When comparing a potential new test with a comparative method in order to bring a new method into the laboratory, linear regression analysis should be performed using the results of the two methods. By calculating the slope and *y* intercept, the presence of systematic error can be identified. Unlike random error that is due to chance and can occur in either direction, systematic error consistently affects results in only one direction. Specific types of systematic error are termed constant and proportional, but the stated question did not give sufficient information to differentiate between the types.

46.

D. Although some laboratories may use the reference interval recommended by the manufacturer or ranges published in medical books, it is preferred that laboratories establish their own limits. When subjects are not easily available, a laboratory should use at least 20 individuals to verify a published range. Whenever possible, a minimum of 120 subjects with representatives from each age and sex group should be included in the reference interval study.

47.

D. When the population served by a laboratory is similar to that described by a manufacturer, then the reference interval published by the manufacturer can be adopted provided the laboratory successfully completes a small study. Such a study need only include 20 individuals. If two or less subjects tested have test values that fall outside the suggested range, then the manufacturer's reference interval can be used. If three or more subjects have test values that fall outside the range, then an additional 20 subjects need to be tested. Provided that two or less are outside the range on this second attempt, then the manufacturer's reference interval can be accepted. In the event that this second attempt fails, the laboratory should assess what differences there may be between their population and that of the manufacturer. If differences cannot be determined, then a complete reference interval study using 120 subjects should be completed by the laboratory.

48.

D. To determine the percent of individuals with a positive test result who actually have the disease, the predictive value of a positive test is calculated. The sensitivity, specificity, and disease prevalence must be known to calculate the predictive value. Sensitivity in this case refers to the percent of individuals having the disease who test positively. Specificity refers to the percent of individuals who do not have the disease and test negatively. Precision is not a component of the formula.

49.

A. The *sensitivity* of a test is the percentage of individuals with a specific disease that are correctly identified or predicted by the test as having the disease. To determine the sensitivity of an assay, the true positives, represented by the number of individuals correctly identified by the test as having the disease, and the false negatives, represented by the number of diseased individuals not correctly identified by the test, must be established for the assay in question. The formula for determining sensitivity follows, where TP = true positives and FN = false negatives.

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100$$

50.

C. The *specificity* of a test is the percentage of individuals without the specific disease that are correctly identified or predicted by the test as not having the disease. To determine the specificity of an assay, the true negatives, represented by the number of individuals correctly identified by the test as not having the disease, and the false positives, represented by the number of nondiseased individuals not correctly identified by the test, must be established for the assay in question. The formula for determining specificity follows, where TN = true negatives and FP = false positives.

$$\begin{aligned}\text{Specificity} &= \frac{\text{TN}}{\text{FP} + \text{TN}} \times 100 \\ &= \frac{45}{5 + 45} \times 100 \\ &= \frac{45}{50} \times 100 \\ \text{Specificity} &= 90\%\end{aligned}$$

51.

A. The predictive value of a test utilizes the parameters of test sensitivity and specificity as well as disease prevalence. The predictive value of a negative test (i.e., the percentage of individuals who test negative and are not diseased) may be determined by knowing the number of true negatives and false negatives. The formula for determining the predictive value of a negative test (PV^-) follows, where TN = true negatives and FN = false negatives.

$$PV^- = \frac{\text{TN}}{\text{TN} + \text{FN}} \times 100$$

52.

D. A proficiency testing program is part of external quality control that aids a lab in assessing the quality of its testing methods. Samples of unknown concentrations are purchased through a recognized professional agency such as the College of American Pathologists or the American Association of Bioanalysts. Samples are periodically sent to labs participating in the program. Following analysis of the samples, the lab sends its results to the agency for review. If significant problems are detected, the laboratory needs to take corrective action. Participation is mandated by the Centers for Medicare and Medicaid Services under CLIA '88.

Laboratory Safety

53.

B. Picric acid has the potential to form peroxides if stored for a long period of time and not used. If this happens, the bottle can become shock sensitive with the potential to explode if knocked. All other chemicals listed do not pose this risk.

54.

C. A fire extinguisher is classified and labeled for the type of fire on which it should be used. An ABC fire extinguisher is commonly found in laboratories. Type A extinguishers are used on fires of ordinary combustibles such as paper, cloth, wood, rubber, and plastics. Type B extinguishers are used on fires of flammable liquids including oils, gasoline, and solvents. Type C extinguishers are used on electrical equipment fires. Type D extinguishers are used on fires involving combustible metals (e.g., magnesium, sodium).

55.

B. The National Fire Protection Association developed the 704-M Identification System to provide common, recognizable warning signals for chemical hazards. The system consists of four color-coded, diamond-shaped symbols arranged to form a larger diamond shape. The blue diamond symbol located to the left identifies potential health hazards. The diamond symbol located at the top of the larger diamond is color-coded red, indicating a flammability hazard. The yellow diamond symbol to the right represents reactivity-stability hazards. The white diamond symbol located at the bottom provides information on special precautions. Contained within each color-coded diamond is a number ranging from 0 to 4, indicating the severity of the respective hazard (0 = none and 4 = extreme). A number of chemical manufacturers have adopted this warning system for their labels.

56.

B. All hazardous chemicals in the workplace must be identified and clearly marked with a National Fire Protection Association label. All the chemicals listed are flammable. Corrosive chemicals are harmful to mucous membranes, skin, eyes, or tissues.

57.

C. A carcinogen is defined as a substance or agent producing or inciting cancer. Mutagens cause changes in DNA. Teratogens cause birth defects.

58.

D. The National Institutes of Health guidelines describe Biosafety Level 2 laboratories as those laboratories that work with microorganisms associated with human disease that is rarely serious and for which preventive or therapeutic interventions are often available. Biosafety Level 1 laboratories handle agents that have no known potential for infecting healthy people. Biosafety Level 3 is recommended for materials that may contain viruses not normally encountered in a clinical laboratory.

59.

C. Both flammable and combustible liquids are commonly used in the laboratory. These two categories are differentiated on the basis of their flash points—that is, the temperature at which a liquid forms an adequate amount of vapor to produce an ignitable mixture with the air at the liquid's surface. The flash point of flammables is designated as less than 100°F and that of combustibles as greater than 100°F.

60.

A. Manufacturers of chemicals, reagents, and kits provide material safety data sheets (MSDSs) for all products. These sheets must be available to all laboratorians in case of emergency. When an individual goes to the emergency department, he or she should have the MSDS to give to the physician in order to get prompt, correct treatment. A laboratory professional must be confident to report any accident and take appropriate measures to clean it up.

61.

D. The Occupational Safety and Health Administration is the federal agency charged with the enforcement of safety and health legislation. The College of American Pathologists and The Joint Commission are private organizations that issue standards and offer accreditation. The International Air and Transportation Association develops regulations for the shipment of hazardous materials.

62.

C. The threshold limit value (TLV) is the exposure allowable for an employee during one 8-hour day. The TLV will be listed in the material safety data sheet for each chemical. More toxic chemicals will have a smaller TLV.

63.

B. A chemical fume hood is an engineering control to provide protection from chemicals. Correct airflow is critical in containing fumes. All air in a chemical fume hood is exhausted out of the building. Chemicals should not be used in a biological safety cabinet unless a Class 2B cabinet can be used, in which case all the air is also exhausted to the outside but through a HEPA filter.

64.

A. Laboratory areas should maintain negative pressure with respect to the administrative areas to prevent toxic or pathogenic materials used in laboratory work areas from escaping and injuring humans or contaminating the environment. The amount of air provided to the negative pressure laboratory should be equal to 85% of the air exhausted from the area. Positive pressure is maintained in the office areas.

65.

B. Some of the precautions that should be followed when working with carcinogenic chemicals include performing the procedure in a chemical fume hood, wearing rubber gloves and proper protective clothing, and wearing a respirator when working with organic vapors and dust-producing materials. If possible, use disposable glassware. All other glassware should be washed with a strong acid before being processed in the general wash cycle.

66.

B. Safe practices in the laboratory are essential to the well-being of all employees. Each laboratorian should disinfect his/her work area daily. Pens and pencils placed on laboratory bench tops may be contaminated and should never be placed near one's mouth. Laboratory coats should never be worn in the cafeteria because they may be contaminated. Laboratory personnel should never smoke, eat, or drink in the laboratory, nor should food be placed in a refrigerator used for storage of reagents or biologic specimens. Cosmetics should not be applied in the laboratory because of potential contamination. Personnel should always wash their hands before leaving the laboratory, discarding their used gloves in a biohazard receptacle.

67.

D. It should be remembered that all body fluids from patients are potentially hazardous to one's health. Specimens collected from infectious patients in isolation should be handled according to standard precautions. Blood specimens need to be centrifuged, and inhalation of aerosols is prevented by never raising centrifuge lids prematurely. All sharp objects, including broken glass and needles, should be disposed of in a puncture-proof container. Blood collection needles should never be discarded in a wastebasket in a patient's room, because housekeeping personnel or others may easily be injured and infected.

68.

D. Chemicals should not be stored in alphabetical order because some chemicals are incompatible with others and will react adversely. Large containers of chemicals should always be stored on a shelf as close to the floor as possible to avoid severe injury in the event of breakage. Flammable chemicals should be stored in a fire-safety cabinet. Although flammables should be used in a chemical fume hood, the hood is not a proper storage area.

69.

B. Emergency showers must be available to anyone working with corrosive materials. The victim should be removed from the area as rapidly as possible and showered with water. No attempt should be made to neutralize the acid on the person's skin.

70.

A. Bottles of chemicals and solutions should be handled carefully. Chemical containers made of glass should be transported in rubber or plastic holders that will protect them from breakage. In the event of breakage, the plastic holders will contain the spill.

71.

B. Compressed gas cylinders should be stored in a vertical position in a ventilated, fire-resistant location. Gas cylinders must never be stored in the same area as flammable liquids, because both are highly combustible. Because of their shape, gas cylinders may easily fall, causing the regulator valve to rupture. To prevent such an occurrence, cylinders must always be fastened when stored, transported, and used in the laboratory.

72.

D. Viral hepatitis is the major job-related disease hazard in all clinical laboratories. All laboratorians who handle blood or body fluids are at risk. The modes of transmission include ingestion and injection. Thus, it is crucial that the laboratorian follows proper safety practices at all times.

73.

C. The responsibility to implement and maintain a safety program belongs to the laboratory safety officer. The responsibility of the chemical hygeine officer is to provide technical guidance in the development and implementation of the chemical hygiene plan. Accreditation agencies typically require laboratories to have both a safety officer and a chemical hygiene officer.

74.

A. Although now considered a carcinogen, sodium azide has been used as a preservative in some laboratory reagents. When disposal of this reagent is made in the sewer, a build-up of copper and iron salts of azide may occur. These metallic salts are explosive, especially when subjected to mechanical shock.

75.

B. The single most useful safety device used in a clinical microbiology laboratory is the Class 2 biological safety cabinet. This engineering device is designed with inward airflow at a velocity to protect personnel. In addition, it is constructed with HEPA-filtered vertical laminar flow for protection of laboratorians.

76.

A. In 1987, the Centers for Disease Control and Prevention (CDC) established guidelines for universal precautions. These guidelines were established to lower the risk of hepatitis B virus and human immunodeficiency virus transmission in clinical laboratories and blood banks. In 1996, the CDC published new guidelines, called standard precautions, for isolation precautions in hospitals. Standard precautions (updated in 2007) synthesize the major features of body substance isolation and universal precautions to prevent transmission of a variety of organisms.

77.

A. Class 2A biological cabinets force HEPA-filtered air downward at the front of the cabinet where the laboratorian is working. This provides a barrier between the worker and the infectious material. Class 2A biological cabinets exhaust HEPA-filtered air into the building, whereas Class 2B cabinets exhaust air out of the building.

78.

B. Halogens, especially chlorine and iodine, are frequently used as disinfectants. Chlorine is most often used in the form of sodium hypochlorite (NaOCl), the compound known as household bleach. The Centers for Disease Control and Prevention recommends that counter tops be cleaned following blood spills with a 1:10 dilution of bleach.

79.

B. The most important point to remember when working with biocides or disinfectants is to prepare a working solution of the compound exactly according to the manufacturer's package insert. Many people think they will get a stronger product if they use a more concentrated dilution. The ratio of water to active ingredient may be critical, and if sufficient water is not added, the free chemical for surface disinfection may not be released.

80.

B. Each employer having an employee with occupational exposure to human blood or any other infectious materials including bloodborne pathogens must establish a written exposure control plan designed to eliminate or minimize employee exposure. The plan identifies tasks that are hazardous and promotes employee safety. The plan incorporates education, proper disposal of hazardous waste, engineering controls, use of personal protective equipment, and a post-exposure plan.

81.

D. Material safety data sheets (MSDSs) will specifically include chemical identity as it appears on the label, chemical name and common name, physical and chemical characteristics, signs and symptoms of exposure, routes of entry, exposure limits, carcinogenic potential, safe handling procedures, spill cleanup procedures, and emergency first-aid. MSDSs are provided by the manufacturers for every chemical. MSDSs contain information on the nature of the chemical, precautions if spilled, and disposal recommendations.

82.

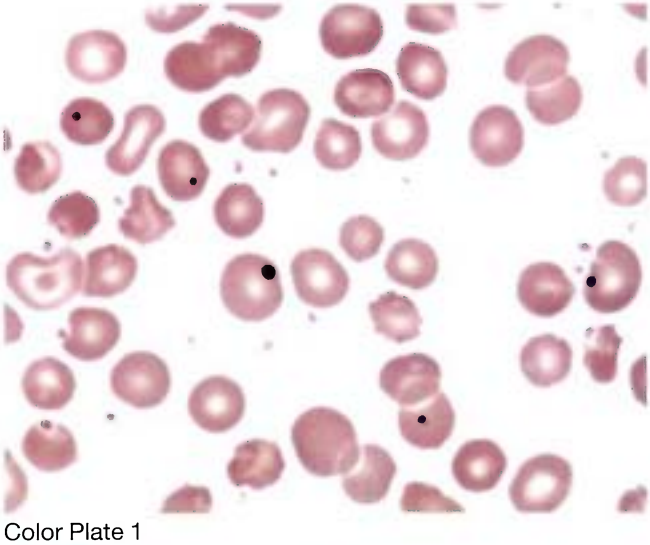
D. *Mycobacterium tuberculosis* is spread by the aerosol route. The risk of inhalation of infectious materials can occur in the laboratory environment and poses a significant potential health hazard to the employees. The proper personal protective equipment is extremely important when working with particular infectious materials. The N95 HEPA filter respirator is a high-energy particulate air filter and is used for microorganisms spread via the aerosol route.

83.

A. Hands are more likely to contact chemicals than any other part of the body. Gloves made of appropriate materials can effectively protect the hands from exposure if they are worn during routine handling of chemicals. Selection of protective gloves is based on chemical hazard and the tasks involved. The glove fabric must have an acceptable slow breakthrough time and permeation rate for the chemical of interest.

84.

B. A safety can is used to store used chemicals until the material is removed from the laboratory. An approved container has a capacity of no more than 5 gallons. A safety can has a spring-closing lid and spout cover, and it is designed to safely relieve pressure buildup within the container.



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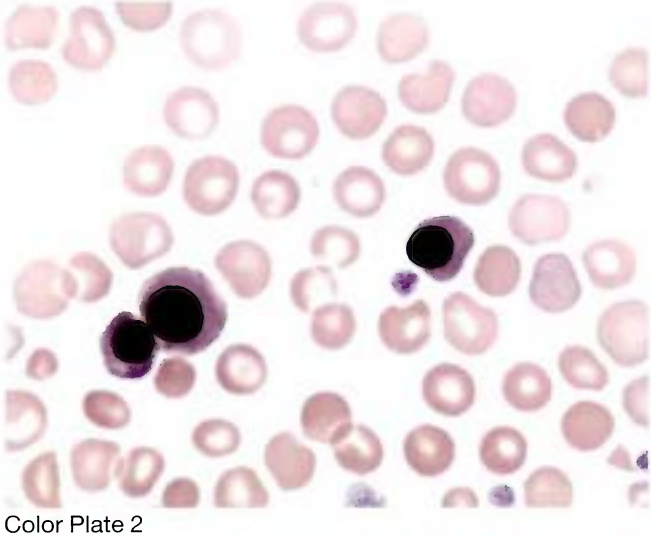

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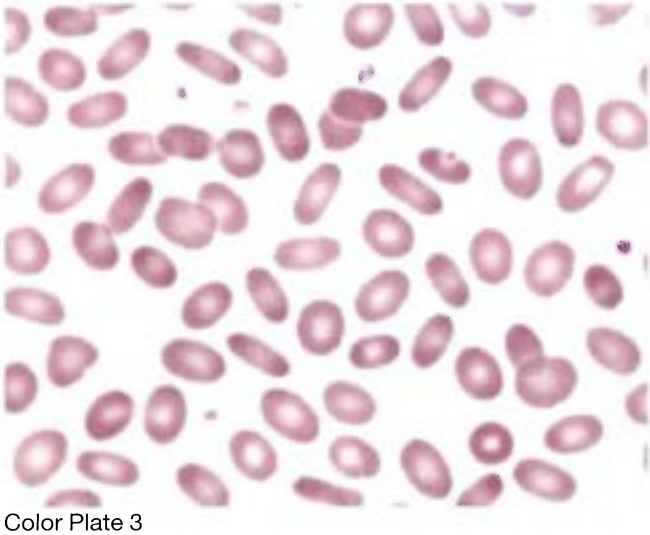

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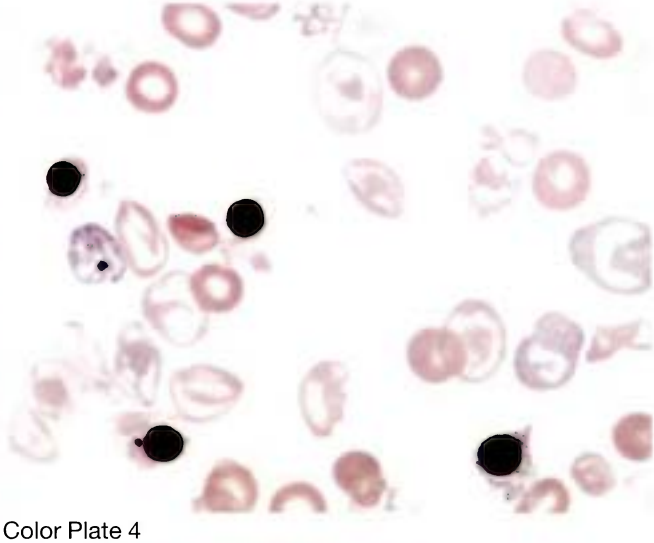

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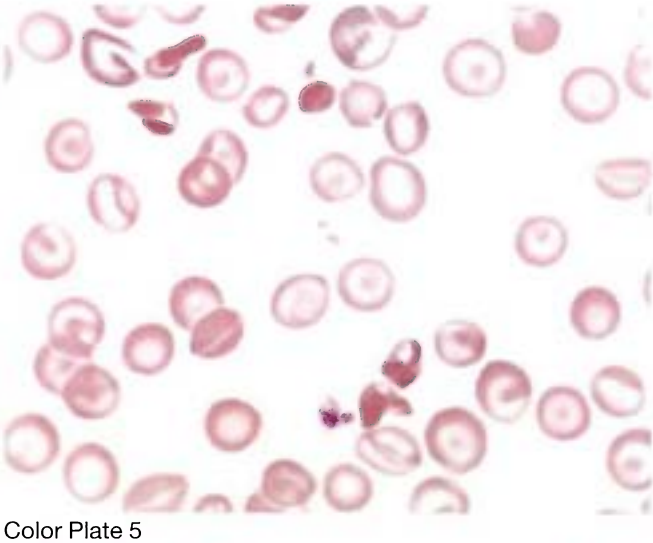

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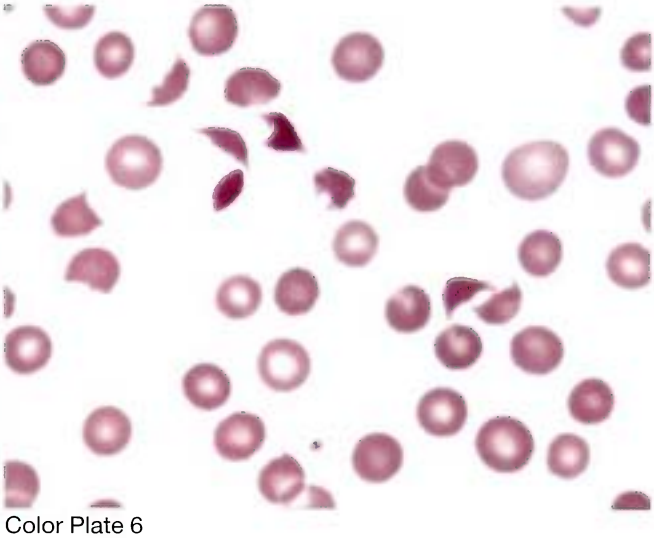

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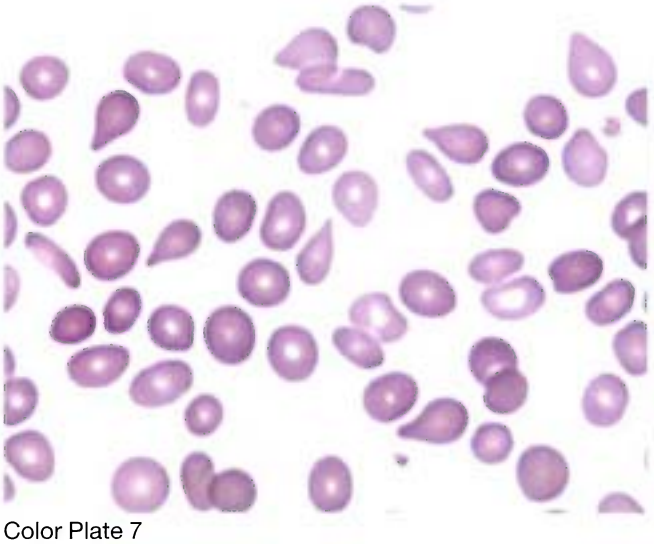

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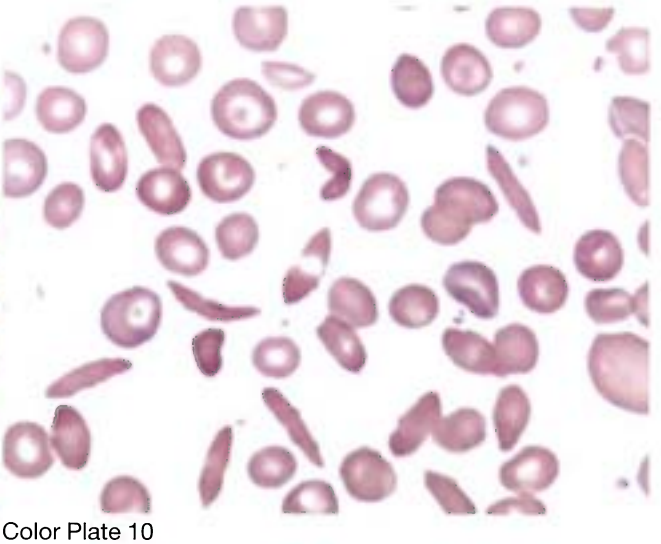

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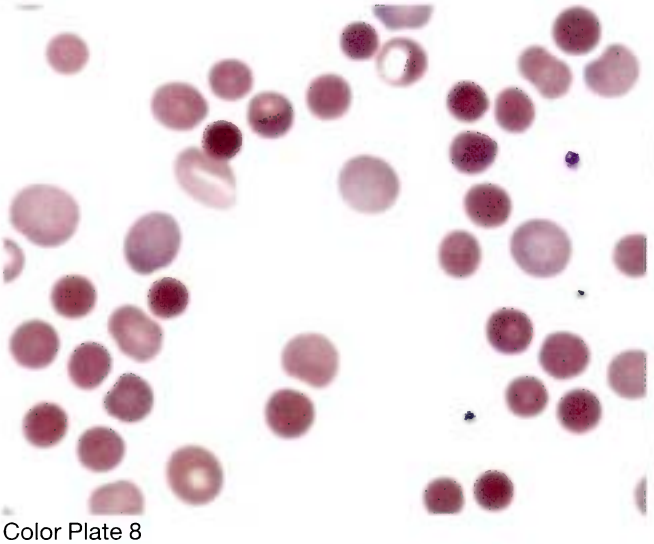

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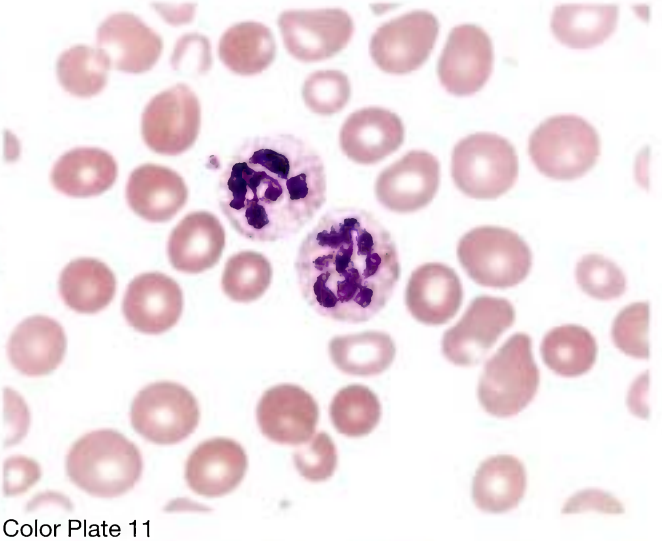

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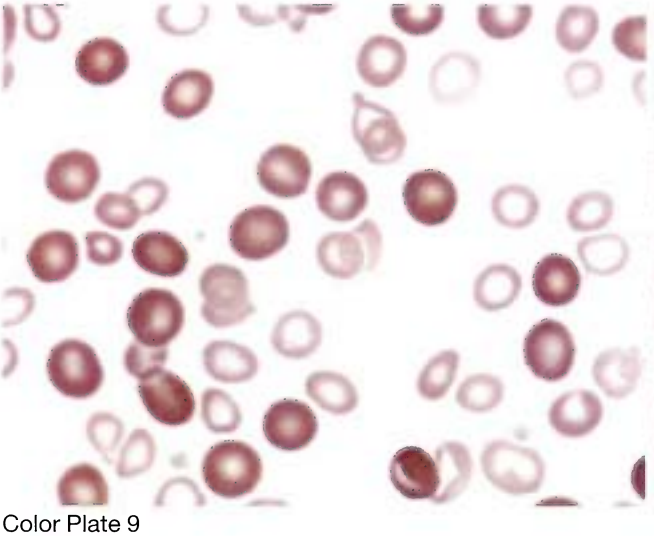

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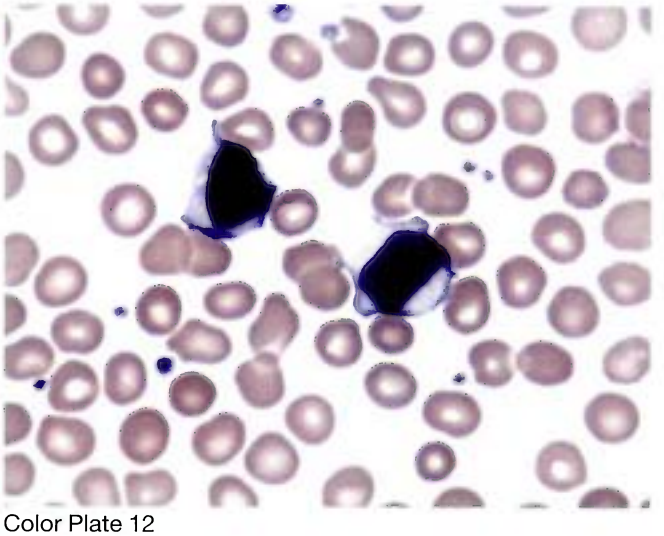

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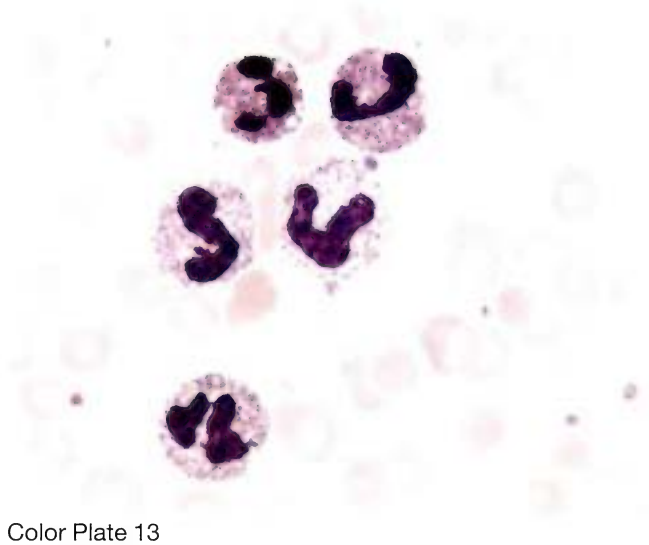

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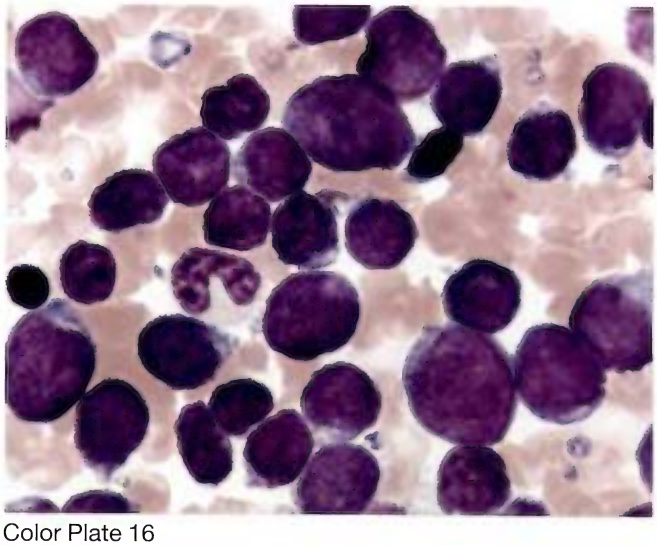

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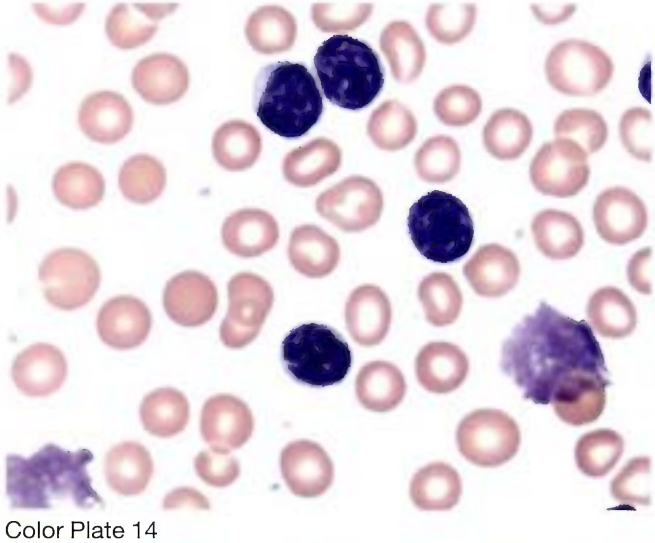

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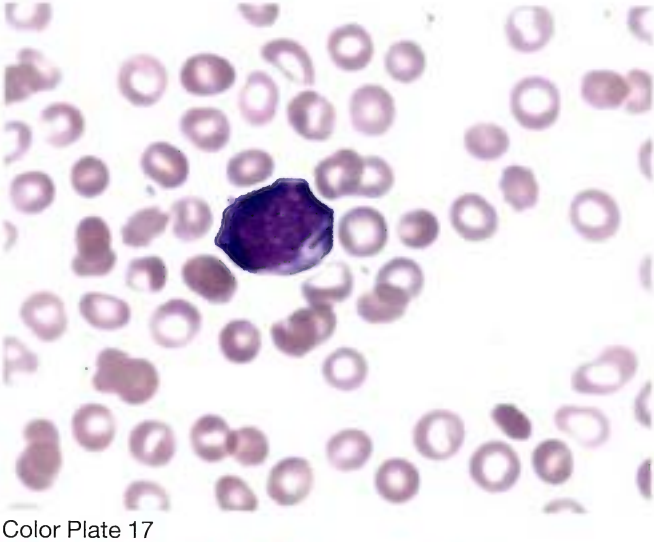

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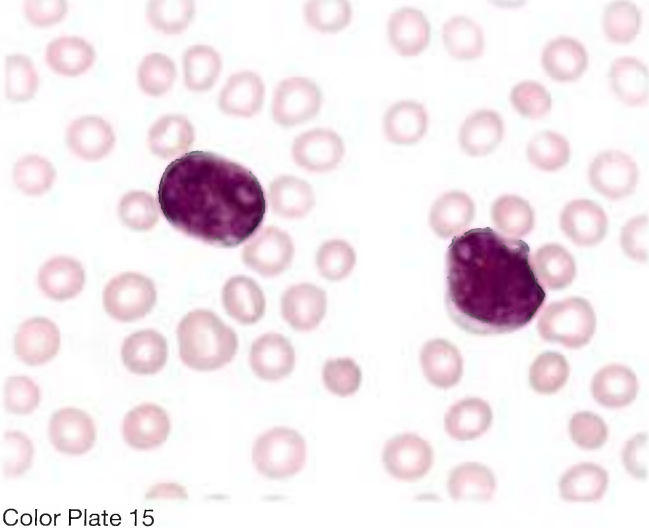

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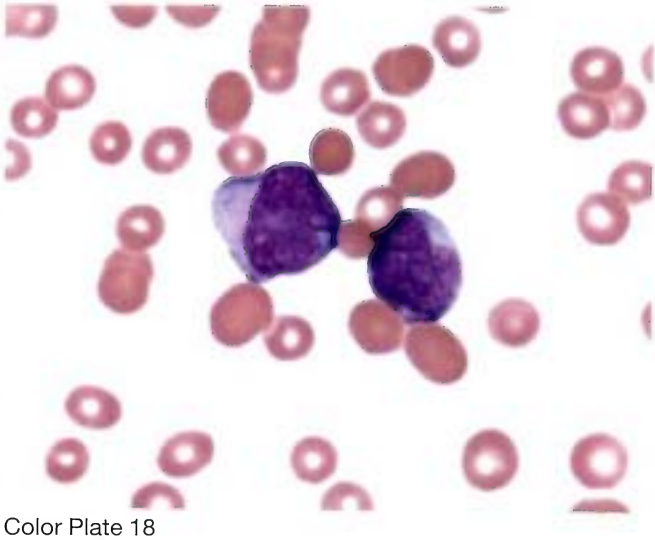

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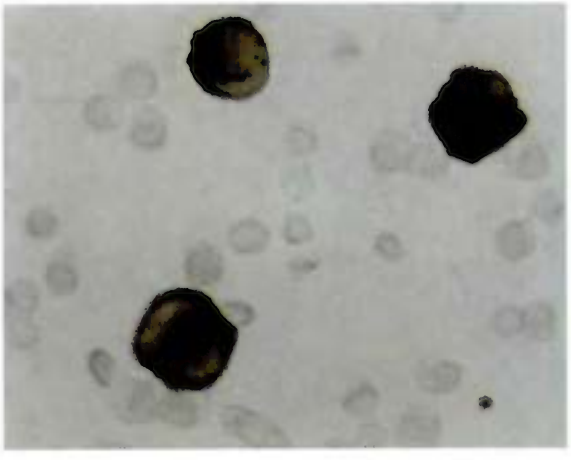

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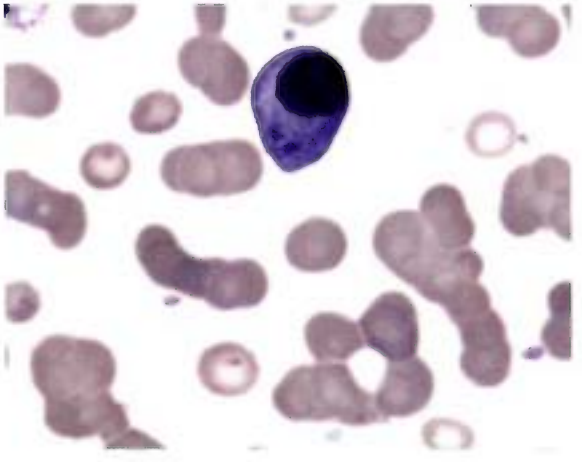

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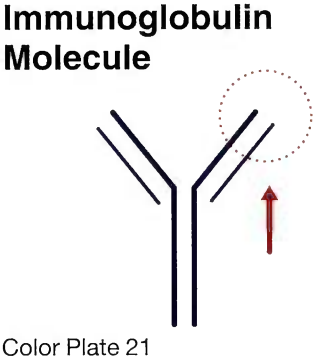

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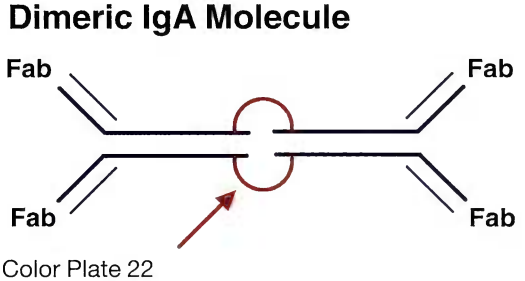

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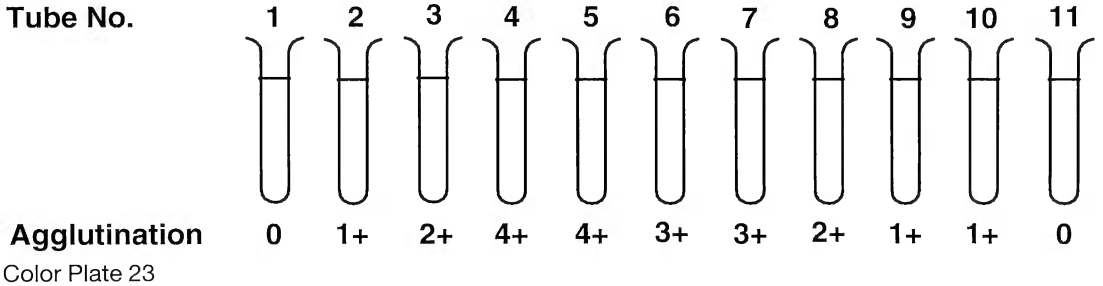

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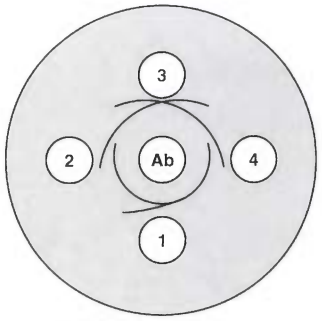

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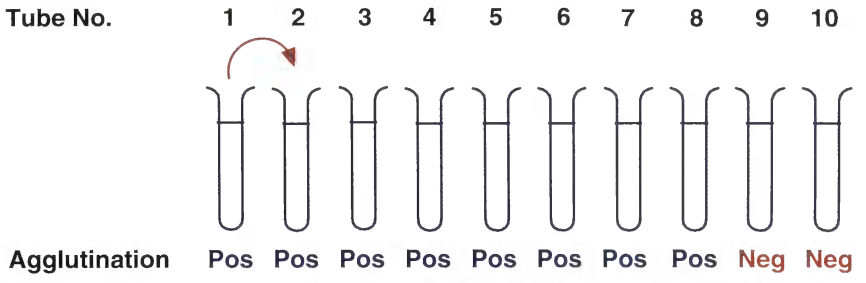

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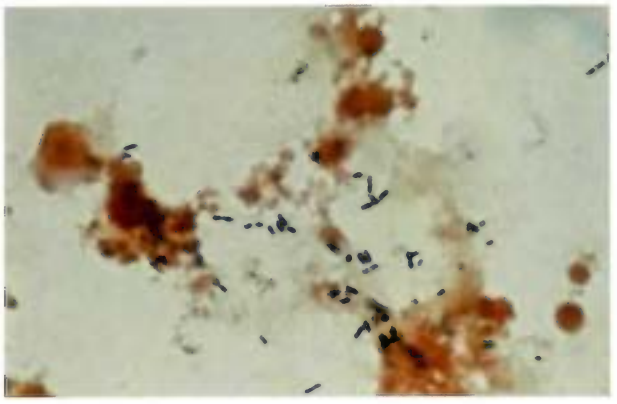

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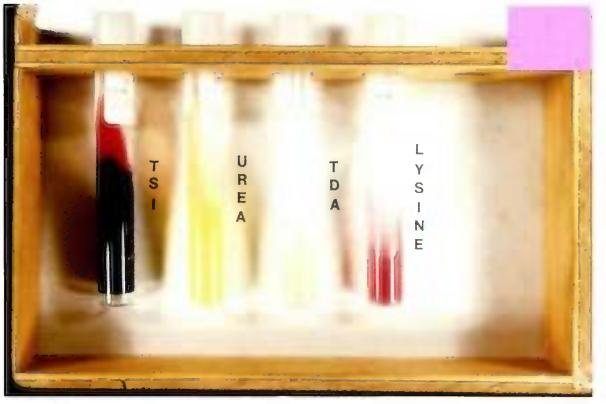

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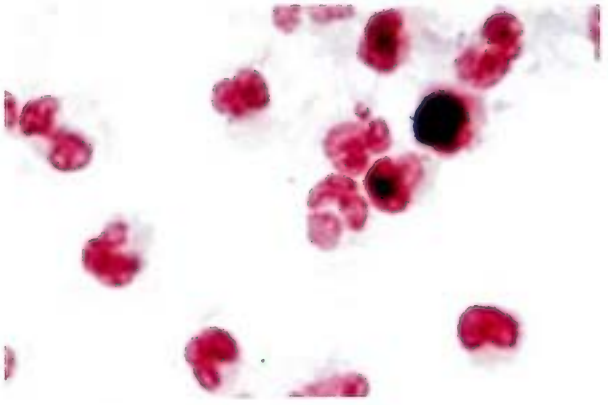

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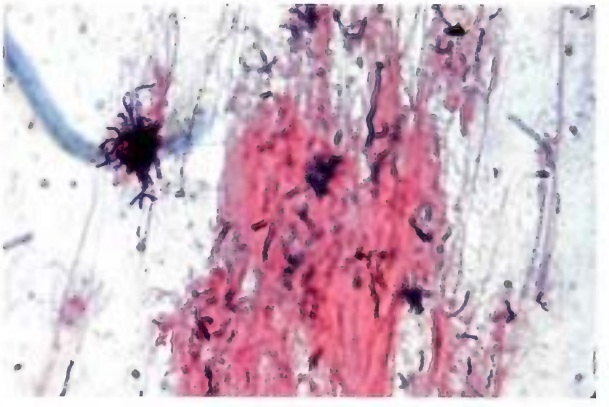

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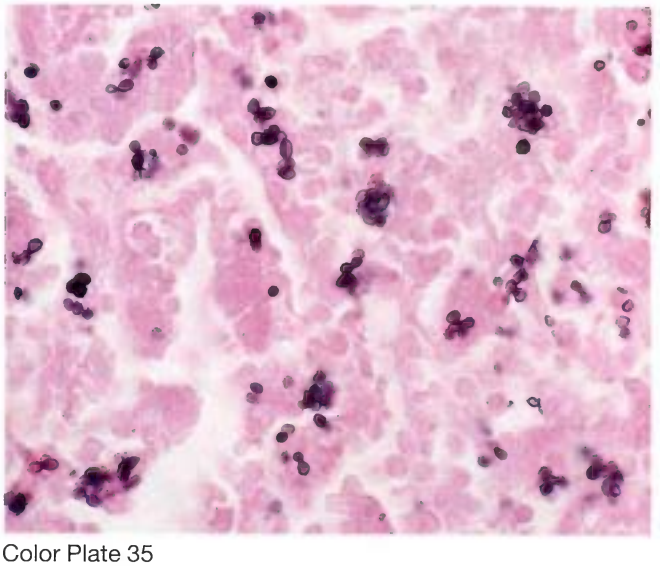

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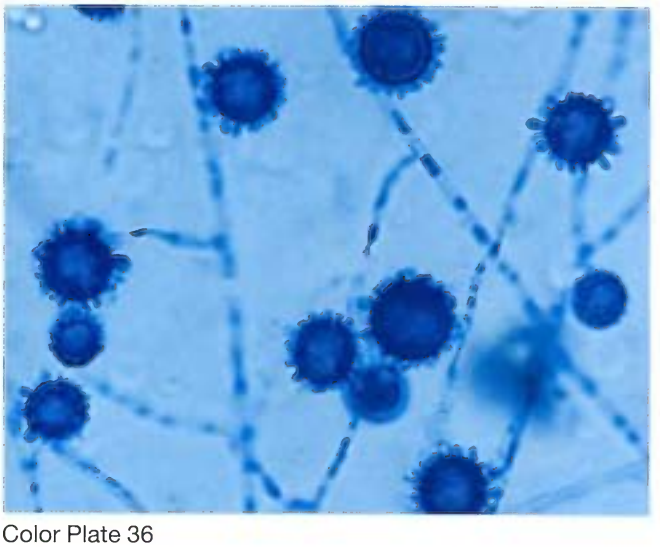

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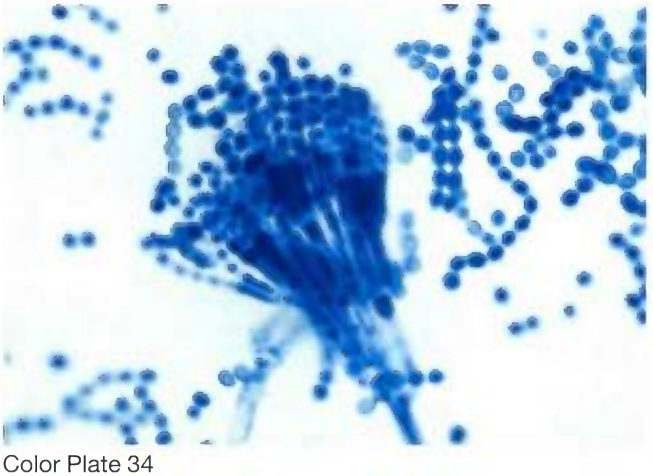

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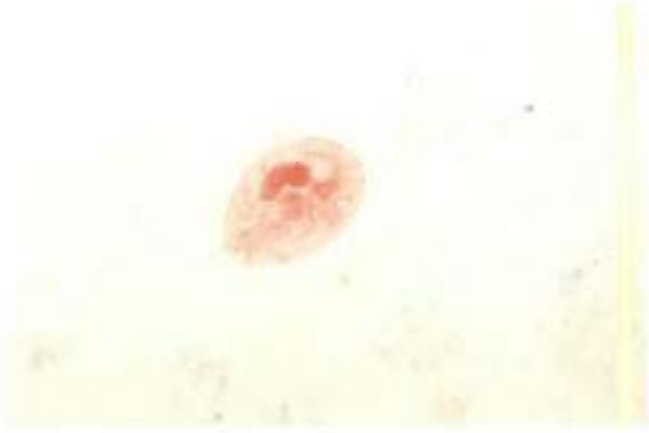

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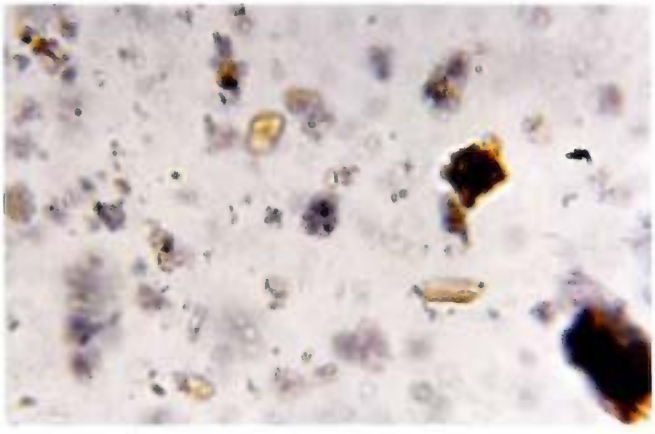

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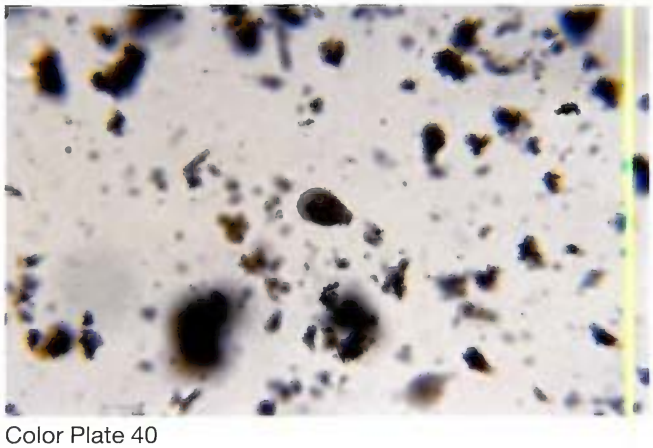

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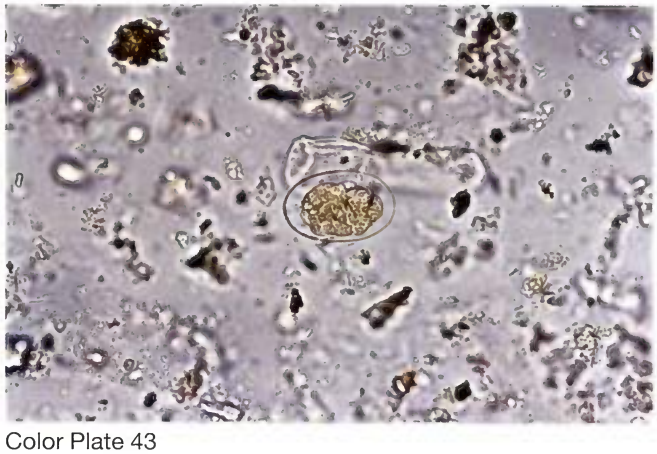

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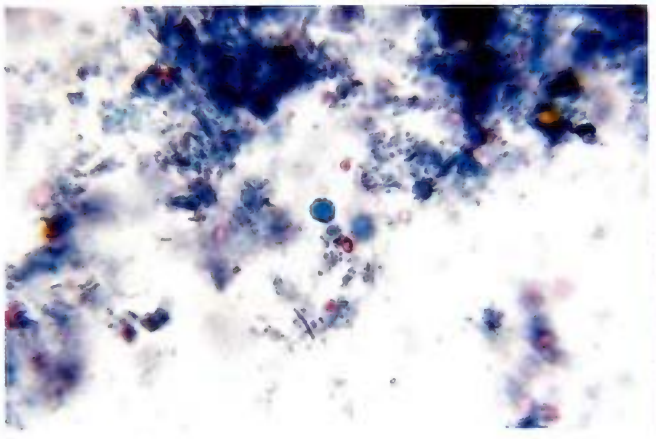

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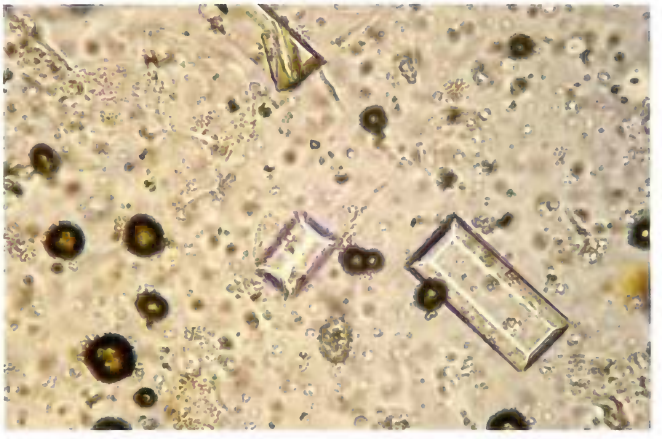

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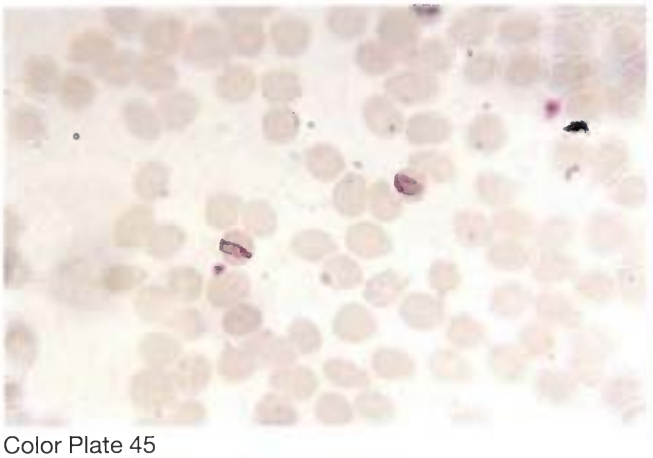

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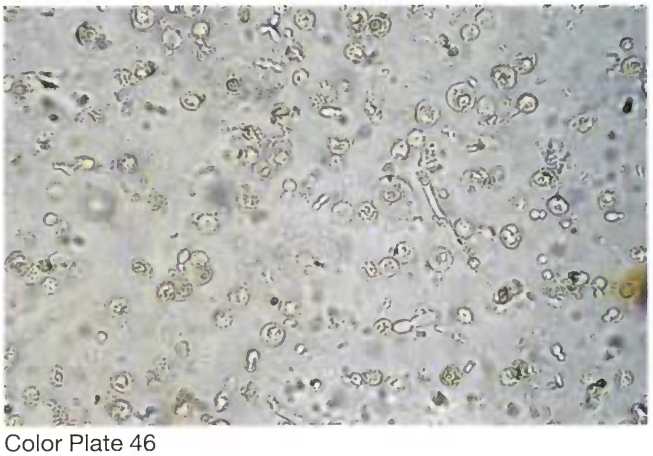

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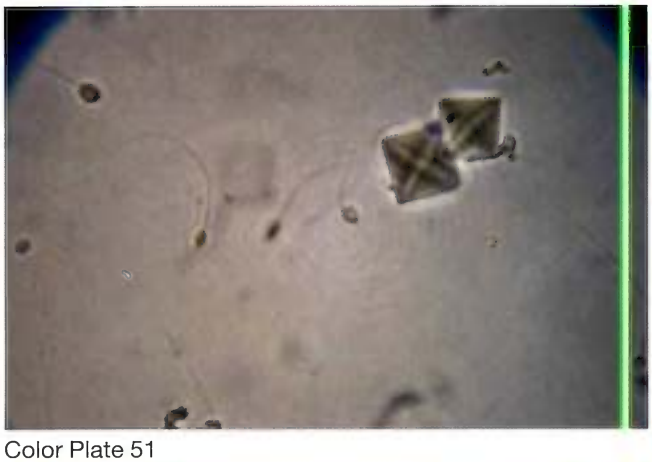
Color Plate 49

Color Plate 50



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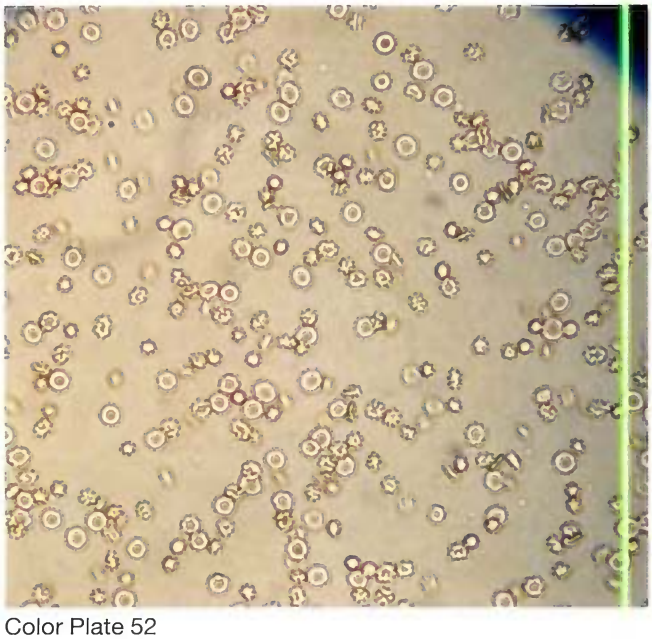

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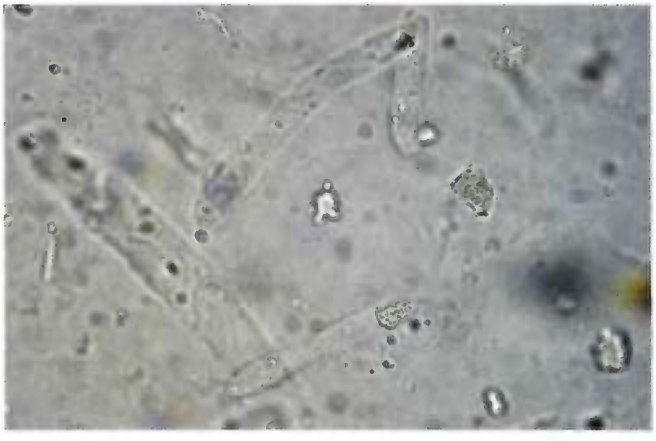

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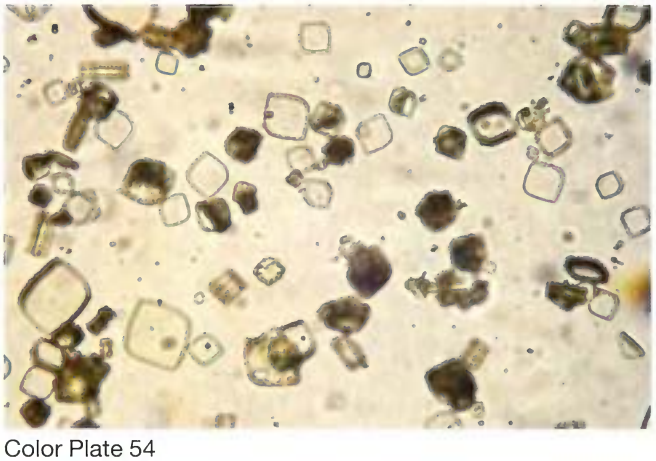

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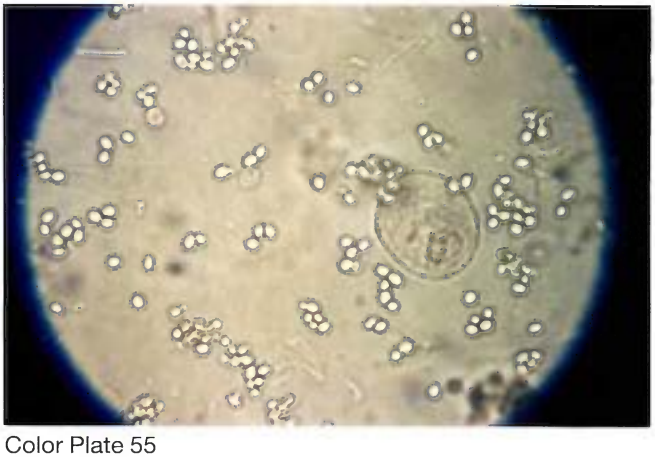

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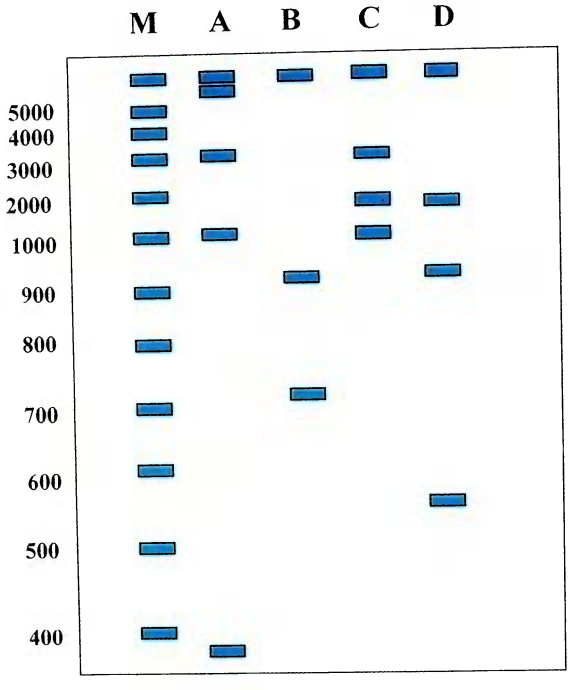
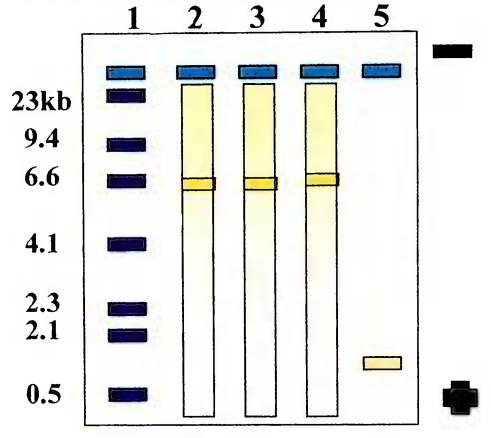


Color Plate 56

Digest DNA with restriction enzyme
Separate by electrophoresis

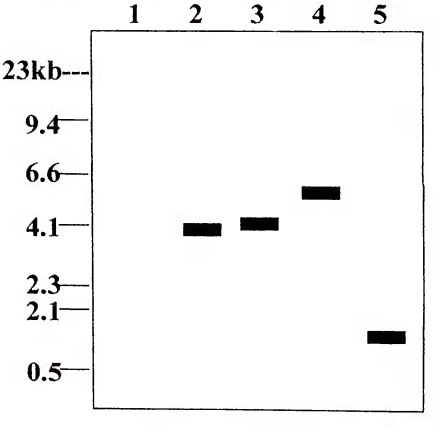


Chemical depurination
Denaturation
Neutralization

Transfer to membrane

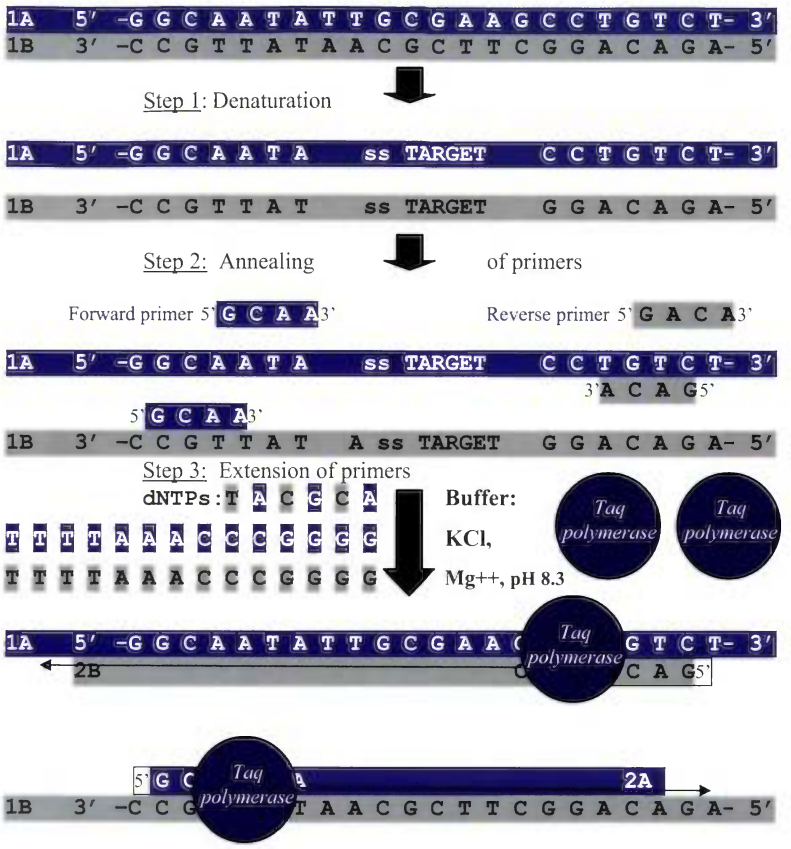
Bind DNA to membrane

Hybridize membrane with ^{32}P labeled probe
Wash off excess probe
Expose to X-ray film
Develop autoradiogram



Color Plate 57

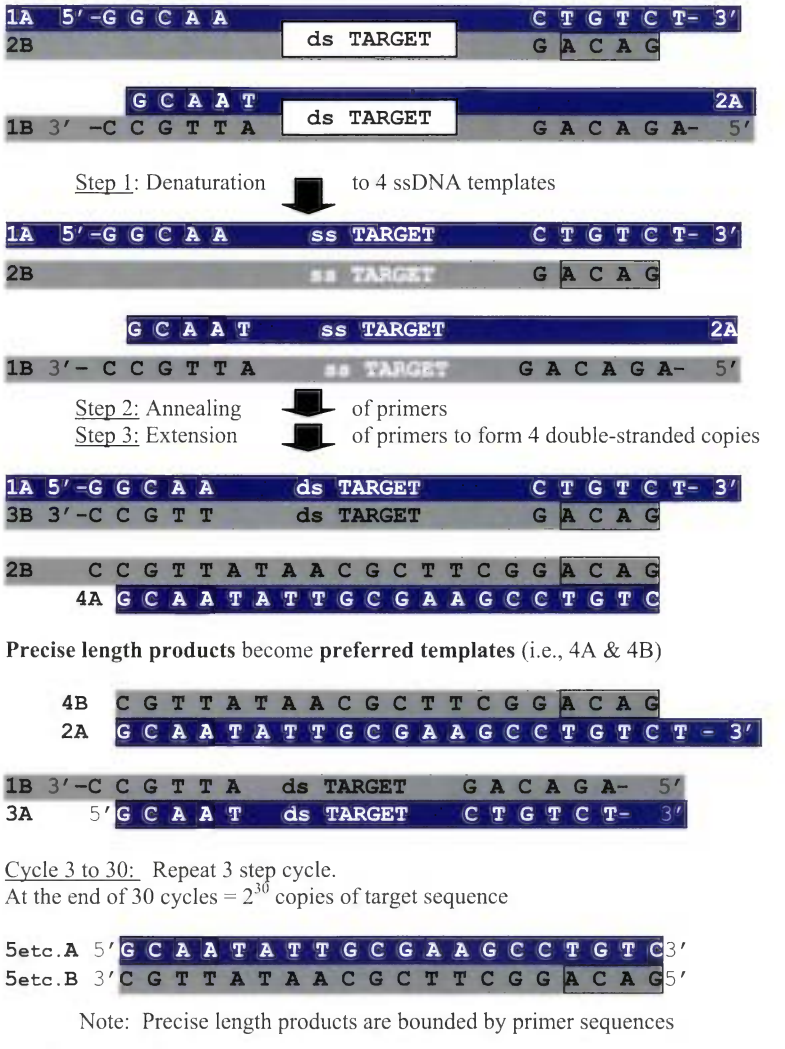
Cycle 1: Start with (1) Double-stranded DNA TEMPLATE



End of cycle 1: **2 double stranded DNA products**

Color Plate 58a

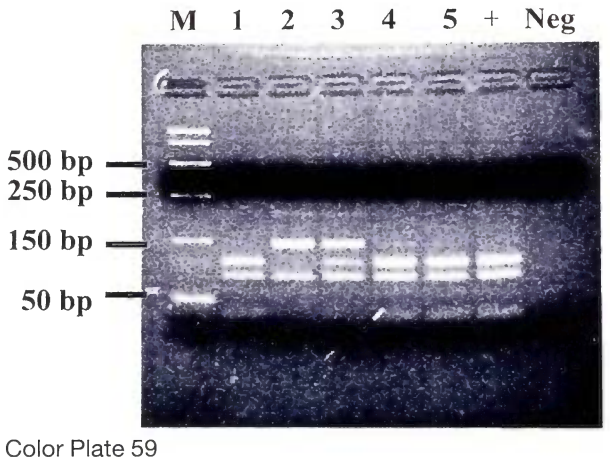
Cycle 2: Start with 2 double-stranded cDNA intermediates containing target sequence



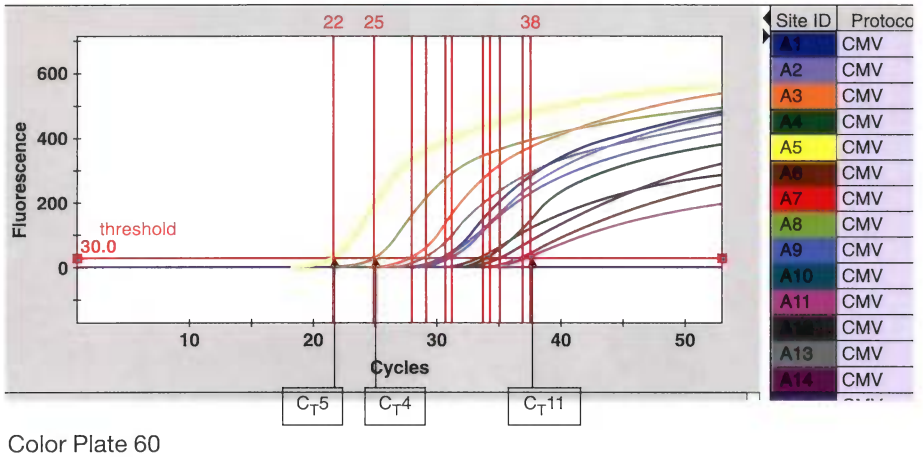
Cycle 3 to 30: Repeat 3 step cycle.
At the end of 30 cycles = 2^{30} copies of target sequence

Note: Precise length products are bounded by primer sequences

Color Plate 58b



Color Plate 59



Color Plate 60